ATARI®

XL ADDENDUM

ATARI HOME COMPUTER SYSTEM

OPERATING SYSTEM MANUAL

Supplement to ATARI 400/800™ Technical Reference Notes

ATARI®

Printed in U.S.A.

TABLE OF CONTENTS

1.0 INTRODUCTION

This manual is designed to serve as a supplement to the ATARI 400™ and ATARI 800™ OPERATING SYSTEM MANUAL.

The 1200XL, as shown in sections 3-5, is a technical upgrade of the A800. The operating system for the 1200XL has been written to maintain, as much as possible, compatibility with application programs which have already been developed for the A400/800.

Since the basic hardware which controls the user interface and the display is, for the most part, compatible with the earlier designs, the operating system, except for the enhancements or changes described here, has remained largely the same. Therefore the data contained in the OS manual for the A400/800 is still valid.

This manual has been written to provide the user with data regarding usage of the added features of the 1200XL operating system, with some details about the characteristics of the peripheral devices with which it will operate. Programmers or peripheral developers who require a greater level of detail regarding the handling of peripheral devices should refer to the documents referenced in item 2 of section 2 below.

2.0 APPLICABLE DOCUMENTS

1. ATARI Home Computer Operating Systems Manual.

   Describes the OS for the A400 and A800, which is the basis for the enhancements described in this manual.

2. ATARI Home Computer Hardware Manual and 1200XL Supplement.

   The Hardware Manual covers the hardware registers which control the various functions of the A400 and A800. The supplement to the hardware manual covers the added features for control of the 1200XL Home Computer. Details that are appropriate to the OS handling of such hardware registers are contained in this OS manual. The user who has need for other hardware-related data should refer to the hardware manual for more information.

3. DE RE ATARI

   This document provides the user with an introduction to the effective use of the ATARI Home Computer hardware. Although written to cover the A400/800, the data contained therein is valid for the 1200XL as well.

3.0 HOW THE 1200XL COMPARES TO THE A400/800

The following is a list of the features and functions which will be discussed in this chapter. Each will be explained in a separate section.

In this chapter, you will learn about:

1. The HELP Key
2. The Function Keys
3. How key codes are redefined and which ones cannot be redefined
4. How to alter the key repeat rate
5. The action of the Caps/Lowr Key
6. How the OS initializes the LED's on the keyboard
7. What happens when a cartridge is installed or removed
8. What happens during power-on self-test
9. What the option jumper assignments mean
10. What new screen modes the 1200XL can use
11. How to enable fine scrolling of the text screen
12. How the disk handler has been changed for improved operation
13. What kind of display is now produced at power-up
14. What features have been deleted as compared to the A400 or A800

Each of the items enumerated above corresponds to the paragraph number in this section which follows. For example, item 1 above is covered in paragraph 3.1, item 2 in paragraph 3.2 and so forth.

3.1 The HELP Key

The operating system, while watching the keyboard, will recognize the pressing of the HELP key as a request to set a flag in the OS database. This flag can be read by whichever application program is in control at the time and react accordingly.

The OS treats the help flag in the same way as the BREAK key in that no ATASCII code is produced but a database variable is set. Therefore, if your program is expecting the HELP key to be pressed, you must not only read the keyboard FIFO (hex location O2FC) for incoming ATASCII codes other than Help, but also occasionally check ("poll") the contents of the HELPFG (help flag) database variable to see if Help was requested.

After reading the database location, and deciding what to do, you must "clear" it for the next time the key will be pressed. The OS does not clear it for you. The Help Flag is cleared by storing a zero in its database variable.

The location of this variable is $O2DC. The conditions to which it responds are listed below, along with the codes which will be stored in HELPFG:

Hex value	Condition represented
OO	The Help flag is cleared. This flag is cleared at initial power-up reset and subsequently, if set, must be cleared by the application program.
11	HELP key alone was pressed.
51	SHIFT-HELP key combination was pressed.
91	CTRL-HELP key combination was pressed.

The HELP key can be used during the power-on display and during the self test feature. See those sections for more information.

3.2 What The FUNCTION Keys Do

NOTE: This section only applies to XL computers with function keys.

The 1200XL is provided with a set of four function keys. You may redefine the ATASCII values which these keys produce if you desire. As a matter of fact, the entire keyboard ATASCII output may be redefined as will be seen later. This section shows the normal definition of the F1-F4 keys, their functions and the ATASCII codes which they produce (if any) as a result of the power-on reset assignment. All values in the table below are given in hexadecimal.

FUNCTION KEY ASSIGNMENT SUMMARY

Key	If pressed alone
F1	Produces the Cursor-up function, returns ATASCII 1C
F2	Produces the Cursor-down function, returns ATASCII 1D
F3	Produces the Cursor-left function, returns ATASCII 1E
F4	Produces the Cursor-right function, returns ATASCII 1F

Key	If pressed with SHIFT
F1	See HOME CURSOR below
F2	See CURSOR TO LOWER LEFT CORNER below
F3	See CURSOR TO BEGINNING OF PHYSICAL LINE below
F4	See CURSOR TO FAR RIGHT OF PHYSICAL LINE below

Key	If pressed with CTRL
F1	See KEYBOARD ENABLE/DISABLE below
F2	See SCREEN DMA ENABLE/DISABLE below
F3	See KEY-CLICK ENABLE/DISABLE below
F4	See DOMESTIC/INTERNATIONAL CHARACTER SET below

Key	If pressed with CTRL and SHIFT
F1	Ignored
F2	Ignored
F3	Ignored
F4	Ignored

HOME CURSOR FUNCTION

SHIFT-F1 causes the cursor to move to the home position of the screen as well as producing the default ATASCII code 1C. The default function is reassignable.

CURSOR TO LOWER LEFT CORNER

SHIFT-F2 causes the cursor to move to the lower left corner of the screen as well as producing the default ATASCII code 1D. The default function is reassignable.

CURSOR TO BEGINNING OF PHYSICAL LINE

SHIFT-F3 causes the cursor to move to the far left of the physical line on which it is located (note, not the logical line which, in the screen editor, could be as many as 3 physical lines). This function is performed by the screen editor as well as generating the default ATASCII code 1E. The default function is reassignable.

CURSOR TO FAR RIGHT WITHIN PHYSICAL LINE

SHIFT-F4 causes the cursor to move to the far right side of the physical line on which it is located. This function is performed by the screen editor as well as generating the default ATASCII code 1F. The default function is reassignable.

KEYBOARD ENABLE/DISABLE

CTRL-F1 controls the keyboard enable/disable function. It produces no ATASCII code. This key combination affects the operating system handling of the keyboard and is not reassignable.

CTRL-F1 disables and re-enables all keyboard functions except for the following:

RESET is the 6502 RESET key, and cannot be disabled

OPTION
START
SELECT keys are not controlled by the operating system

Each time you press CTRL-F1, the operating system changes the enabled/disabled status to the opposite of what it was when you pressed this combination. In other words, if the OS had disabled the keyboard, LED 1 would be on. If, at that time, you press CTRL-F1, the OS would re-enable the keyboard and turn LED 1 off. The second press of this combination would reverse the process, disabling the keyboard again.

You may monitor or control the keyboard enable or disable function under software control by reading or writing the OS database variable called KEYDIS (hex location 026D). A value of 0 in this location means the keyboard is enabled, and a value of hex FF here means the keyboard is disabled.

SCREEN DMA ENABLE/DISABLE

CTRL-F2 controls the Screen Enable/Disable Direct Memory Access (DMA). It produces no ATASCII code. This key combination affects the operating system handling of the display function. This key combination is not reassignable.

The 1200XL, on power-up, always enables the screen DMA. What this means is that the system will always initialize itself to display anything which has been defined for the screen display during power up. This same screen DMA enable will also occur if you touch any keyboard key other than the CTRL-F2 combination.

Various types of programs which you write may be heavily involved in arithmetic computations. To speed up the processing in the A400 or A800, you may disable the screen DMA. When it is disabled, the ANTIC processor does not steal memory cycles from the 6502 to get its data for the screen. Therefore during disable mode, the screen remains blank. When it is enabled, the full display which you have defined is visible; however, the processor is slowed down by anywhere from 10 to 40 percent as explained in the section on ANTIC DMA in the Atari Hardware Manual.

On the 1200XL, to start the higher speed/ no display function, press the CTRL-F2 key combination. The display will go blank. To restore the display again at any time, you can press any other key.

During your arithmetic calculations, you may be in continuous process of updating the memory area where the display data is contained. You can then get a status of the operation in process at any time simply by pressing any key other than CTRL-F2, then again press CTRL-F2 to re-enter the higher speed mode.

Your program, then, on completion of the calculation, could exercise direct program control over the ANTIC DMA variable to restore the display when the arithmetic intensive part is over.

The DMA control database variable SMDCTL contains status bits for display list memory access as well as player missile data access. When the combination CTRL-F2 is pressed, the OS will save this value, (if it is not already zero) in database variable location DMASAV$02DD). Then the variable SMDCTL will be set to zero. When the combination is pressed again, the original value is restored to SMDCTL from DMASAV, thereby restoring the display. Your program could perform the same process.

KEY-CLICK ENABLE/DISABLE

CTRL-F3 controls the Key-Click enable/disable function. If pressed once, it disables the audible feedback on keystrokes. Pressed again reenables it. This function only affects an OS database variable and produces no ATASCII code. It is not reassignable.

You may control the key click enable/disable from your program. All that needs to be done is to change the same flag which the operating system uses to indicate whether a key click is required. This flag is called NOCLIK. It is one of the OS database variables, contained at location $02DB.

On power up and reset, the operating system initializes this variable to a value of 00, meaning that key click is enabled. This location, when it contains the value $FF, indicates that no key click is desired. The key combination CTRL-F3 toggles it between the values 00 and FF.

DOMESTIC/INTERNATIONAL CHARACTER SELECTION

CTRL-F4 controls the domestic/international character selection. Default is domestic. It affects an OS database variable only and produces no ATASCII code. It is not reassignable. It toggles the display of character sets, changing between the two each time the key combination is pressed. When the international character set is selected, LED number 2 will be lit.

The international version of the character set is located in the ROM beginning at location $CC00 You can cause the international character set to be selected by storing the constant $CC to location $02F4. This is the location CHBAS. The normal character set is located in the ROM starting at $E000. If a program stores $E0 to CHBAS, it selects the display of the normal characters.

If you have defined your own character set, however, pressing CTRL-F4 will display the international character set. This is because the operating system will test CHBAS and find that the value $CC is not there. Therefore $CC must be the next value which is to be used (selects int'l. set). When it tests CHBAS and finds $CC stored there, it knows that $E0 is the next value to use during the toggle between character sets.

Two variables are used to control the character set selection: CHBAS (02F4) and CHSALT (026B). The Screen Editor (E:) and the Display Handler (S:) initialize variables CHBAS and CHSALT at every OPEN command which you issue to either one. CHBAS is initialized to a value of hex E0 and CHSALT is initialized to a value of hex CC.

When you press CTRL-F4, the operating system swaps the values of CHBAS and CHSALT using the OS variable TEMP as the temporary holding point. Once it completes the swap, if CHBAS is equal to CC, it will light LED 2, indicating that the international character set is selected.

3.3 KEY REDEFINITION

You may redefine most of the 1200XL console keys if desired. The redefinition process consists of setting up a pair of tables which can be referenced by the operating system when it translates your keystroke into an ATASCII value.

The two tables are the KEY Definition Table and the Function Key Definition Table. The operating system has a pair of data tables from which the normal definitions are made. You may define your own set of tables however, then simply tell the operating systen where they are located in memory.

One such use of key redefinition might be to experiment with other, possibly more efficient keyboard layouts, such as perhaps the Dvorak keyboard. An example is given in Appendix A of a keyboard redefinition to allow you to do such an experiment. (Over the years, the QWERTY key layout has been the accepted standard, though many people have found DVORAK to be more efficient. This would allow you to try it for yourself.)

CONTENTS OF THE KEY DEFINITION TABLE

This table allows most of the keys of the 1200XL to generate any desired ATASCII code or special internal function. The exceptions to this are listed at the end of this section. To redefine the keys, it is necessary first to define an area in memory where a 192 byte table may be stored.

Into this table, you will store the definitions of the keys which you desire. Later you will tell the operating system where this table is located so that future references may be made to it instead of the standard definition table.

The organization of this table is as follows:

Lower case convert. Group of 64 bytes	KEYTABLE _ START (Starts at user defined address) Table of lower case conversions
Shift plus key Group of 64 bytes	Table of uppercase conversions
CTRL plus key Group of 64 bytes	Table of control key conversions
	KEYTABLE _ START + 191.

The reason that each of the subdivisions of the table has 64 bytes in it is that the hardware can generate a total of 64 hardware keycodes. These codes, numbered 00-63 decimal (00-3F hexadecimal) are used to index directly into one of the three keycode tables. Which table is referenced depends on whether the CTRL or SHIFT keys is pressed.

Note that there is no table for the combination of both CTRL and SHIFT. This combination is invalid and is ignored by the operating system.

Each of the three 64 byte subsections of the table has the form:

00 code	Byte 0	contains conversion for key code 00 for key alone, key with CTRL, or key plus SHIFT. Depends on which table is accessed per which keys pressed.
01 code	Byte 1	contains conversion for key code 01
3F code	Byte 3F	contains conversion for key code 3F

The codes which you place in your table will either generate an ATASCII code (for direct character translation) or they will tell the system to perform a specific function. Specifically any code in the range of 80 to 91 hexadecimal will be treated as special by the system. This is illustrated in the table below.

CODES AND THEIR EFFECT ON THE SYSTEM AFTER TRANSLATION

CODE	EFFECT (if any)
00 thru 7F	Used as the ATASCII code only.
92 thru FF	Used as the ATASCII code only.
80	Ignore, invalid key combination.
81	Invert the video output to the screen.
82	Alpha lock/Lower case toggle.
83	Alpha lock
84	Control Lock
85	End of file
86	ATASCII code
87	ATASCII code
88	ATASCII code
89	Key click on/off
8A	Function 1 *
8B	Function 2 *
8C	Function 3 *
8D	Function 4 *

* NOTE: When it sees these keycode translations, it is told to DO the function which is described in the Function Key descriptions. The ATASCII coded generation for the normal and shifted function keys is handled in a different table, whose description follows that for the keycode hardware translate table.

8E	Cursor to home
8F	Cursor to bottom
90	Cursor to the left margin
91	Cursor to the right margin

The table below shows the key cap corresponding to each key code. The physical position of each key switch within the table determines the hardware code which it will generate. To determine what code it is, take the row address of the cap, and add it to the column address. The result is the hexadecimal value returned to the operating system (range 00-3F) for use in the table lookup for that key.

KEYCODE DEFINITIONS TABLE

	0	1	2	3	4	5	6	7
00	L	J	;	F1	F2	K	+	*
08	O		P	U	RET	I	—	=
10	V	HLP	C	F3	F4	B	X	Z
18	4		3	6	ESC	5	2	1
20	,	SPACE	.	N		M	/	) | (
28	R		E	Y	TAB	T	W	Q
30	9		O	7	BACKS	8		
38	F	H	D		CAPS	G	S	A

As an example the key cap "C" is in the table in row 10, column 2. This means that the hardware generates a hardware code 10 + 2 or 12 hexadecimal. Therefore, in the translation tables shown above, the function code or ATASCII code for this character will be stored in the key definition table position $12 for each of the three types of "C" which are valid (c alone, Shifted C, or Control C). You may cause each of these to perform a separate function or generate a separate ATASCII code by revising the tables.

When you have decided on how you want your keys to be redefined, you tell the operating system where it may find the definitions by storing the address of those definitions in locations 79 and 7A hexadecimal. The low byte of the hexadecimal address where you have stored the keys should be placed in location 79, the high byte is location 7A. This is defined as one of the system vectors, called KEYDEF. It will point to the default, or original key definition table at power-on reset time.

REASSIGNMENT OF THE FUNCTION KEYS ONLY

There may be times when you only want to redefine the function keys and not redefine the rest of the keyboard. The 1200XL operating system allows you to redefine only the function keys by setting up an 8-byte table in place of the 192 byte table which would have otherwise been required. The format of this table is as follows:

F1	← Lowest memory location of the table
F2	
F3	
F4	
SHIFT-F1	
SHIFT-F2	
SHIFT-F3	
SHIFT-F4	← Highest memory location of the table

When you have decided what functions each combination must perform and have built the table, change the system vector FKDEF to point to the lowest address of your table. This vector is located at memory locations 60 and 61 hexadecimal. Location 60 gets the low byte of the hex address, location 61 gets the high byte.

The same codes described in the section titled "CODES AND THEIR EFFECT ON THE SYSTEM AFTER TRANSLATION" are used in this table. However, DO NOT assign codes 8A through 8D to the same function as the key itself. In other words, do not specify that the key F1 should perform function F1, etc. since this would result in an infinite loop. (F1 sensed by the OS sends it to the function key table, which tells it to look up and perform the F1 function, which sends it to the table, and so on, with no possible exit.)

NON-REASSIGNABLE KEYS AND KEY COMBINATIONS

The following keys or key combinations are either specifically wired for special functions or are subjected to special handling by the operating system.

Even though there might be a hardware-generated key code shown in the table above, and a corresponding space in the translate tables, there is no way to reassign these functions. This is because the operating system traps the hardware code directly to perform the specified function and it never gets to the translate mode. These keys or combinations are as follows:

BREAK	—	This function is fixed as a special case in the operating system. It is sensed by the hardware.
SHIFT	—	This key is an integral part of the hardware encoding of any key function.
CTRL	—	This key in an integral part of the hardware encoding of any key function.
OPTION SELECT START	—	All of these are directly wired to and are sensed by the GTIA circuitry.
RESET	—	Directly wired to the 6502 reset line.
HELP	—	Function is fixed by the operating system. The help function handling is described elsewhere in this manual.
CTRL-1		Controls the screen output start/stop function.
CTRL-F1		See KEYBOARD ENABLE/DISABLE above. As noted there, this function is not reassignable.
CTRL-F2		See SCREEN DMA CONTROL above. As noted there, this function is not reassignable.
CTRL-F3		See KEY-CLICK ENABLE/DISABLE above. As noted there, this function is not reassignable.
CTRL-F4		See DOMESTIC/INTERNATIONAL CHARACTER SET above.

3.4 USER-ALTERABLE KEY AUTO-REPEAT RATE

The 1200XL operating system allows you to control the rate at which a key, continuously held down, will repeat its entry to the system. This change can be done by modifying the OS database variable KEYREP, located at hex address O2DA.

This variable determines the repetition rate by counting the number of VBLANK (vertical blanking) intervals which occur. For the NTSC (60 Hz) system, the initial value of this variable is 6; for PAL systems, the value is 5. This assures a uniform repeat rate of 10 characters per second for either system. The key repeat rate equals the VBLANK rate (60 or 50 per second) divided by the KEYREP value.

Under control of this variable, the maximum "controllable" key repeat rate would be 50 characters per second on the PAL and 60 characters per second on the NTSC (screen refresh rate). This would occur with a value of 1 in this variable.

You may control the rate at which occurs before the key repeat starts. The OS database variable which controls this is called KRPDEL. Its hex address is O2D9.

It controls the number of VBLANKs which must occur between the sensing of the key pressed until the first repeat occurs. From that time on, the repeat rate is controlled as described above. The initial values used by the OS provide a 0.8 second initial delay for either NTSC (count = 48) or PAL (count = 40) systems.

3.5 CAPS/LOWR KEY TOGGLE ACTION

The CAPS/LOWR key on the 1200XL functions as shown in the chart below:

KEY COMBINATION	CURRENT STATE	NEW STATE
CAPS	Control Lock	Lower Case
CAPS	Alpha Lock	Lower Case
CAPS	Lower Case	Alpha Lock
SHIFT-CAPS	— any —	Alpha Lock
CTRL-CAPS	— any —	Control Lock
CTRL-SHIFT-CAPS	— any —	— no change —

The meaning of the terms is as follows:

Lower Case — All key caps respond in lower case mode

Alpha Lock — All alphabetic keys (A-Z) respond in upper case mode, all others lower case

Control Lock — All alphabetic keys (A-Z) respond as though the control key is being held down as well as the selected key

3.6 LED INITIALIZATION

The 1200XL has two LED's on the front panel, called LED 1, and LED 2. LED 1, when lit, indicates that the Keyboard is disabled. LED 2, when lit, indicates that the international character set is selected. The operating system enables the keyboard and selects the domestic character set on power up and reset. Therefore these LED's will both be off.

3.7 POWER-ON SELF-TEST

During the initial power-on, the 1200XL operating system will perform the following quick check of the integrity of the system RAM and ROM:

a. Is it possible to write $FF (all ones) to all RAM locations?

b. Is it possible to write $00 (all zeros) to all RAM locations?

c. Does a checksum of the two ROM's compare to that stored within each ROM?

If any of these tests fail, the operating system will transfer control to the self-test memory test routine. Here a more thorough test of both RAM and ROM can take place.

3.8 OPTION JUMPERS

The 1200XL is provided with a set of four hardware jumpers which are designed to tell the operating system how the system is configured. As of the date of this writing, only one of the four jumpers has been assigned, specifically J1. This is specified in the table below. During the power-on sequence, the 1200XL operating system reads the state of these jumpers and stores this state in the OS database variable JMPERS, location 030E.

The bit assignments for each of the four jumpers is as specified below. The bits are all active low, meaning that if a line reads a digital zero, the jumper is installed.

BIT	FUNCTION	HARDWARE NAME
0	Self test enable (will run self test if low)	J1 (pot 4)
1-3	Reserved for future use	
4-7	Unused	

3.9 ADDITIONAL HARDWARE SCREEN MODES

The 1200XL adds direct access to the remaining special purpose display processor operating modes. The table below shows the current mapping which has been provided for the A400 and A800. The table which follows thereafter shows the added modes and the numbers which the software can use to access the extra modes.

Mode mapping common to A400/A800:

Software Mode		ANTIC MODE		GTIA MODE
0	($00)	2	($02)	0
1	($01)	6	($06)	0
2	($02)	7	($07)	0
3	($03)	8	($08)	0
4	($04)	9	($09)	0
5	($05)	10	($0A)	0
6	($06)	11	($0B)	0
7	($07)	13	($0D)	0
8	($08)	15	($0F)	0
9	($09)	15	($0F)	1
10	($0A)	15	($0F)	2
11	($0B)	15	($0F)	3

Mode mapping for 1200XL (additional):

Software Mode		ANTIC MODE		GTIA MODE
12	($0C)	4	($04)	0 (note 1)
13	($0D)	5	($05)	0 (note 1)
14	($0E)	12	($0C)	0
15	($0F)	14	($0E)	0

Note 1: The existing character sets will not provide recognizable characters for these new modes. Therefore you will have to provide the character set if you use these modes. This is done by defining the full character set, then modifying the OS database variable CHBAS to point to the most significant byte of the address at which the character set starts. CHBAS is located at $2F4.

Appendix B of this manual contains some suggestions on the method for designing a new character set to support those added modes.

3.10 TEXT SCREEN FINE SCROLLING

The screen editor (E:) supports fine scrolling of the text screen data as an option. This fine scrolling option will be enabled if the database variable FINE (hex location 026E) is set nonzero prior to issuing the OPEN command to the screen editor. Likewise, the feature will be disabled if this location is set to 00 before issuing the OPEN.

There are only two allowed values for FINE = 0 and hex FF. Other values may produce undesirable results.

During an OPEN command to the Screen Editor (E:), if FINE (026E) is hex FF, then a fine scrolling display list is created. This display list will be one byte larger than a coarse scrolling display list. In addition, the OS places the address of a display list interrupt routine into the display list vector VDSLST (0200) replacing an other vector which you might have already stored there.

When fine scrolling is enabled, the Screen Editor's display list interrupt service routine modifies the content of color register COLPF1 (D017) for the very last visible line of the screen.

When a CLOSE command is issued for the Screen Editor, if FINE is hex FF, then the address of an RTI is placed into the display list vector VDSLST (0200). For OS versions 11 and beyond, FINE is set to zero again, and the screen is reopened with a coarse scrolling display list.

The recommended manner for enabling and disabling fine scrolling is shown below:

a. Set FINE to hex FF

b. OPEN E: using an IOCB number

c. Use E: as usual; fine scrolling is enabled

d. CLOSE E:

e. If the IOCB is now open, then you are finished, otherwise continue with the next step

f. Set FINE to zero

g. OPEN E:

3.11 DISK COMMUNICATIONS ENHANCEMENTS

The 1200XL adds the capability for the resident disk handler to read and write disk sectors having variable length from 1 to 65536 bytes. The default length, as is used on the A400 and A800 currently, is 128 bytes. Both at power-on and RESET (warm start), the 128 byte sector length is established. Your program can alter this length by modifying the OS database variable DSCTLN. The location of this two-byte variable is 02D5 and 02D6 (lo byte in 02D5, hi in 02D6).

In addition to the capability to read and write variable length sectors, the 1200XL also adds the capability to write a sector to the disk without a read-verify operation always following it. This is the command 'P' which was specifically excluded in the previous releases of the operating system.

With this capability added, you have a choice of either using the verify, for system integrity (always read after write). Or you can take a chance of writing a bad sector on rare occasions but increasing your average speed of disk usage by some value related to the verify time. You may want to experiment with some of your programs with and without verify to see the results.

3.12 POWER-ON DISPLAY ENHANCEMENT

In place of the original power-on memo pad display used by the A400 and A800 (in the absence of a cartridge or disk), the 1200XL displays a dynamic ATARI rainbow. If you press the HELP key while the rainbow is displayed, the 1200XL will enter the self-test mode.

4.0 MEMORY MAP OF THE 1200XL

The following table shows how the 6502 processor perceives the various address spaces which it can access. The maximum allowable address range, with the 16 bit address of the 6502 is hexadecimal 0000-FFFF. This address range is split, by the hardware memory management circuitry, as follows:

(Note: The 1200XL uses 64K RAM's as the main system writeable memory. Addresses within those RAM's, which would normally have filled the entire memory access space of 0000-FFFF of the processor, are prevented from access by the memory manager. This allows ROM's, cartridge memory, and peripherals to occupy a part of the memory space as is noted below.)

1200XL MEMORY MAP

HEX ADDRESS	WHAT IS ACCESSED THERE	NOTES
FFFF-D800	OS-ROM or RAM if ROM disabled	1
D7FF-D000	The special purpose chips respond to the address ranges shown in the listing below D000-D0FF GTIA D200-D2FF POKEY D300-D3FF PIA D400-D4FF ANTIC D500-D5FF Any read or write to an address in this range enables the cartridge control line CCNTL on the cartridge interface (same as A400/A800. D100-D1FF, D600-D6FF, and D700-D7FF are reserved for future use. OS-ROM physically present, but cannot be accessed here.	2
CFFF-C000	OS-ROM or RAM if ROM is disabled	1
BFFF-A000	RAM, or cartridge interface	3
9FFF-8000	RAM, or cartridge interface	3
7FFF-5800	RAM	
57FF-5000	RAM, unless in self-test mode	2
4FFF-0000	RAM	

NOTES: 1. Access to the OS ROM may be disabled by wirting a zero to port B of the PIA, bit O. Access is normally enabled, with a 1 present in this bit. (When changing this bit in the register, other bits should not be changed.)

2. The self-test ROM code is physically present in the OS ROM at actual address DOOO-D7FF. However, this area is used for the access to the memory mapped I/O devices. When the self-test feature is invoked, the RAM located from 5OOO-57FF is disabled. The memory manager re-maps the memory access such that the OS ROM physical addresses DOOO-D7FF are accessed at 5OOO-57FF. The memory manager uses port B of the PIA, bit to determine whether to access RAM or ROM in the region 5OOO-57F. If bit 7 is high, RAM is accessed. If bit 7 is low, the OS-ROM is accessed instead. (When changing this bit in the register, other bits should not be changed.)

(Port B was used in the A4OO/8OO to service the game ports 3 and 4. The use of the remaining bits of ths port are specified in Section 6 of this manual.)

3. ROM will be selected in these regions if control lines RD4 or RD5 are pulled up to +5V by the cartridge. RD4 controls ROM select in the region 8OOO-9FFF. RD5 controls ROM select in the region AOOO-BFFF.

5.0 ENHANCEMENTS TO THE A400/800 REV. B OPERATING SYSTEM INCORPORATED IN THE 1200XL

This section describes a set of enhancements which include new methods of handling peripheral products and, in a separate section, improvements in basic operations of the system.The latter might be referred to as "bug fixes".

PERIPHERAL HANDLER ADDITIONS

To accommodate a new class of peripheral devices, the operating system now includes a relocating loader, used to upload peripheral handlers through the serial (I/O interface).

In the A400/800, device handlers for the peripherals were uploaded as fixed location (absolute) object code. These handlers were loaded using a set of device inquires, or polls, known as types 0, 1 and 2. Information on types 0, 1 and 2 Poll Commands is available from Atari Customer Service.

The 1200XL adds two other types of polls to its operating system. One poll, known as type 3, is issued at power-on or reset time. The other, type 4, can be issued as a result of an OPEN command by an application program.

Type 3 Poll Command

The type 3 poll command itself is used as an "Are You There?" type of command. Associated with the type 3 poll are two other types, specifically the:

a) Poll Reset

and b) Null Poll

Poll Reset consists of the following SIO command byte sequence (refer to the SIO document for further explanation of the byte types):

Byte Position	Value (hex)
Device Address	4F
Command Byte	40
AUX1	4F
AUX2	4F
Command Checksum	Normal (checked by peripheral)

The 4F in AUX1 and AUX2 define this sequence to all peripherals as a poll reset.

After responding to a type 3 poll by sending a handler to the system, a peripheral is not supposed to respond again to a type 3 poll. The Poll Reset command, at power-up, resets all type 3 peripherals, freeing them to respond to the poll request. However, no serial bus device sends back any data as a result of a poll reset command.

Type 3 Poll (Are you there?)

There may be several types of peripherals which can respond to a type 3 poll. In types 0, 1 and 2, the device address sent on the serial line specifies which exact device is being called. In the type 3 poll processing, however, the address remains fixed (4F) and the devices each respond after a specific number of poll 3 retries. In other words, during poll 3 operations, the computer doesn't know which peripherals are actually attached, but will keep asking "is anybody there" until it has reached its last retry and no peripheral has responded.

Each peripheral which does respond to the type 3 poll must be designed to count the number of retries of type 3 polls, then to respond as described below on its own specified retry slot. Each time it sees a command other than a type 3 poll, these peripherals must reset their retry counters. This allows the computer to load the handler for each peripheral which responds, then restart its poll 3 sequence (original retry number restored) to look for another poll 3 response from the next peripheral (if any).

Since each peripheral responds only once (after a poll reset), a second request at a specific retry slot causes no peripheral response and allows the next retry slot to be polled.

This poll ("are you there?") is sent as follows:

Byte Position	Value (hex)
Device address	4F
Command Byte	40
AUX1	00
AUX2	00
Command checksum	Normal (checked by peripheral)

When, after checking the retry count, it is a peripheral's turn to respond, it sends back the following data to the computer on the serial interface:

a) An ACK response byte, and

b) 1. Low byte of handler size in bytes (must be EVEN)
   2. High byte of handler size
   3. Device Serial I/O Address to be used for loading
   4. Peripheral Revision Number

These four bytes, if sent by the peripheral, will be stored in OS variables DVSTAT (02EA hex) through DVSTAT+3. If there is a successful return to the OS (not a timeout or other problem), it indicates that there is a handler to be loaded. The loading is performed, then the type 3 poll is repeated until all retries are exhausted and no peripheral responds.

Once the device address data is received from the peripheral during this type 3 poll, it can thereafter be referenced directly on the serial bus by its address in place of the original poll address 4F.

Specific details of the actions taken by the OS after receiving an answer from a peripheral may be found in Appendix C.

Null Poll Command

This command is used as a serial bus no-operation. If any error should occur during loading of a peripheral handler or by the relocator, the system should be free to "back out" of the linking of the faulty loader and tell the peripherals that it is ready for the next one to be loaded. Since this null poll is a non-type-3 poll, all peripherals will have reset their retry counters and should be ready for another sequence of retries, looking for their own response retry slot. This maintains synchronization between the computer and the peripherals.

The structure of the Null Poll is as follows:

Byte Position	Value (hex)
Device Address	4F
Command Byte	40
AUX1	4E
AUX2	4E
Command Checksum	Normal (peripherals check it)

Type 4 Polling

This type of poll is sent out on the serial bus as a result of an application initiated request. During an OPEN command, a device which responds to a type 4 poll may conditionally or unconditionally be polled to determine if it is online and may or may not have its handler uploaded and linked to the system under control of the OS. Detailed information regarding the handling of the device under various operating conditions may be found in Appendix C.

The Type 4 Poll is a serial port command structured as follows:

- Device address of 4F hex (peripherals looking for Type 4 Poll may ignore the device address and look only for the poll command '@'; however, the device address will always be 4F hex and the peripheral may check this);
- Command is '@' (40 hex) (peripherals looking for this poll will always look for the '@' command);
- AUX1 contains the device name, which is an ATASCII upper-case letter (range 41 hex through 5A hex) (the peripheral must be assigned that device name in order to legally answer the poll);
- AUX2 contains the device number, which is an ATASCII digit (range ATASCII 1 through 9, 31 hex through 39 hex) (the peripheral may optionally use this information in deciding whethe or not to answer the poll);
- Standard command checksum (peripheral checks this).

This poll differs from the Type 3 Poll in that the device name and number is included in the poll. Therefore the peripheral need not count retries of the type 4 poll and should answer the poll as soon as the poll command is recognized. There is no limitation on the type 4 poll; the peripheral should answer its type 4 poll each time it is issued.

The peripheral response to a type 4 poll is the same as for the type 3 poll. The four response bytes are placed, by the computer, into DVSTAT through DVSTAT+3 (02EA through 02ED hex).

GENERAL ENHANCEMENTS TO THE REV. B OS FUNCTIONS

The following functions which are supported by the A400/800 Rev. B Operating System have been further enhanced by the addition of the following features:

Printer CLOSE with data in the buffer —

The printer handler will insert an EOL(end-of-line) character in the printer buffer, if one is not there, before sending the buffer to the printer on a CLOSE. This assures that the last line will be printed immediately rather than having the printer forced offline to output the final line.

Printer Unit Number Handling —

The printer handler has been changed so that it will process the unit number in the IOCB, allowing separate addressing for printers P1 through P8.

CIO Handling of Truncated Records on Read —

The CIO now places an EOL in the user's input buffer on the occurrence of either a record longer than the buffer being read or an EOF being encountered during the read attempt. This assures that all records are accessible, even if the user has not provided a sufficient buffer size, he will at least get as much of the record as he has provided for.

CIO Error Handling With Zero Length Buffer —

The CIO will return a buffer length of zero (in the 6502 A-register) when there is a handler error while effecting a zero length buffer transfer. (See CIO section in the OS manual.)

Display Handler Cursor Handling —

The display handler now accepts a screen clear code no matter what value is in the cursor X and Y coordinates.

Display Handler/Screen Editor Memory Clearing —

The Display handler and Screen editor will not clear memory beyond the end of memory as indicated by RAMTOP. Now it is possible for the user to specify the top of memory to be used by the system and to store device handlers or personal machine code in the memory area above the display. Changing display graphics modes, then, will not erase any data which has been placed in the RAM area above that assigned for use by the display or screen editor.

Rework of the Floating Point Package —

The 1200XL operating system corrects a bug in the Rev. B OS. It now produces an error status when an attempt is made to calculate the LOG or LOG10 of zero.

New ROM Vectors —

The following fixed entry point vectors have been added to the 1200XL ROM set:

E480	JMP PUPDIS	entry to power-on display
E483	JMP SLFTST	entry to the self-test pgm.
E486	JMP PHENTR	entry to uploaded handler enter.
E489	JMP PHULNK	entry to uploaded handler unlink.
E48C	JMP PHINIS	entry to uploaded handler inti.

6.0 OTHER CHANGES/GENERAL INFORMATION

This section deals with items which involve operating system changes, but which do not easily fit into any other category.

IMPROVED HANDLING OF OS DATABASE VARIABLES

During normal power-on sequence (cold start), the OS database variables from $03ED-$03FF are set to zero. During a RESET (warm start), they are NOT changed by the OS. This means that an enhanced version of the operating system in the future will be able to make use of these locations without reloading them after any RESET operation.

These bytes are all reserved for use in future OS revisions.

NTSC/PAL VERSION TIMING PROVISIONS

There are various timing differences between the NTSC (60 hz) and the PAL (50 hz) versions. To eliminate the necessity for providing a special operating system ROM set for each one, the specific timing adjustment values are handled within the single ROM set.

To determine which type of system the ROM is operating on, the operating system checks a flag within the GTIA chip and adjusts all timings accordingly. This was possible because the GTIA must be different to handle the modified display format for the 50 Hz version. By making certain timings a function of the state of this flag, it was possible to make external timings independent of the NTSC or PAL system itself.

The timing values relate to the handling of the 115 Volt cassette player (Atari 410) and the console auto-repeat rate as shown in the table below:

CASSETTE TIMINGS NOW INDEPENDENT	TIMING
Write Inter-record gap (long)	3.0 sec.
Read IRG delay (long)	2.0 sec.
Write IRG (short)	0.25 sec.
Read IRG delay (short)	0.16 sec.
Write File leader	19.2 sec.
Read Leader delay	9.6 sec.
Beep cue duration	0.5 sec.
Beep cue separation	0.16 sec.

AUTO-REPEAT FUNCTIONS NOW INDEPENDENT	TIMING
Initial delay for auto-repeat	0.8 sec.
Repeat rate	10.0 char/sec.

1200XL OS ROM IDENTIFICATION AND CHECKSUM DATA

Each of the two ROM's in which the 1200XL operating system is contained has a capacity of 64K bits organized as 8K by 8. Within each of the ROM's is a block of data organized as shown in the diagram below, to identify the ROM and to give its checksum. The checksum is tested by the operating system as part of the power up sequence.

The format of the block for the C000-DFFF ROM is as follows:

Contents		Address	Description
ROM Cksum (lo)		C000	Checksum which is the sum of all bytes in ROM except checksum bytes themselves.
ROM Cksum (hi)		C001	
D1	D1	C002	Revision date having the form DDMMYY where D = day digit M = month digit. Y = year digit. Each a 4 bit BCD digit. Reserved, contains $00 for the 1200XL.
M1	M2	C003	
Y1	Y2	C004	
Option byte		C005	
A1		C006	Part number having the form AANNNNNN, where A's represent ASCII characters, N are BCD digits.
A2		C007	
N1	N2	C008	
N3	N4	C009	
N5	N6	C00A	
Revision No.		C00B	

The format of the identification block for the E000-FFFF ROM is as follows:

Contents		Address	Description
D1	D2	FFEE	Revision date having the form DDMMYY where D = day digit M = month digit. Y = year digit Each a 4 bit BCD digit.
M1	M2	FFEF	
Y1	Y2	FFF0	
Option byte		FFF1	Hardware product identifier, will be used by Atari to identify Home Computer products, for 1200XL = $01.
A1		FFF2	Part number having the form AANNNNNN, where A's represent ASCII characters, N are BCD digits.
A2		FFF3	
N 1	N 2	FFF4	
N3	N4	FFF5	
N5	N6	FFF6	
Revision No.		FFF7	
ROM Cksum (lo)		FFF8	Checksum which is the sum of all bytes in ROM except for checksum bytes themselves.
ROM Cksum (hi)		FFF9	
vector table for for NMI, RES and IRQ		FFFA-FFFF	This area reserved for power-on reset vectors, NMI and IRQ vectors.

PORT B CHANGES

Port B of the PIA is a read/write port which no longer is connected to game I/O ports. Instead, its bits control various functions which include control of LED 1, LED 2, read enable of the the OS ROM's and other functions. To change only one single bit at a time within that port, the following technique should be used.

```
Clear A Bit (bit b)
    LDA PORTB
    AND # $FF-b
    STA PORTB ;clears only bit b in the port
Set A Bit (bit b)
    LDA PORTB
    ORA # b
    STA PORTB ;sets only bit b in the port
```

XL PORTB ($D3O1) BIT ASSIGNMENTS		
BIT	VALUE	USE
O	O	OS ROM DISABLED, RAM ENABLED
	1	OS ROM ENABLED
		The memory region mapped to the OS ROM is from $COOO to $FFFF except for the region from $DOOO to $D7FF which is always mapped to the hardware I/O chips (GTIA, POKEY, PIA, ANTIC).
1	O	BASIC ENABLED
	1	BASIC DISABLED, RAM ENABLED
		The memory region mapped to BASIC is from $BFFF.
2	O	LED #1 ON
	1	LED #1 OFF
3	O	LED #2 ON
	1	LED #2 OFF
4		RESERVED FOR FUTURE USE
5		RESERVED FOR FUTURE USE
6		RESERVED FOR FUTURE USE
7	O	SELF TEST ROM ENABLED
	1	SELF TEST ROM DISABLED, RAM ENABLED
		The memory region mapped to the self test ROM is from $5OOO to $57FF.

NOTE: The OS VBLANK process copies the port A joystick and paddle values into the Port B shadows. Thus, stick O affects both O and 2, stick 1 affects both 1 and 3.

REV-LEVEL DETERMINATION

To allow program products to determine which Atari Home Computer and Operating System Revision level it is operating with, the following tests are recommended:

If location \$FCD8 = \$A2, then product is an A400/A800 wherein:

- If location \$FFF8 = \$DD and \$FFF9 = \$57
  then OS is NTSC rev A.
- If location \$FFF8 = \$D6 and \$FFF9 = \$57
  then OS is PAL rev A.
- If location \$FFF8 = \$F3 and \$FFF9 = \$E6
  then OS is NTSC rev B.
- If location \$FFF8 = \$22 and \$FFF9 = \$58
  then OS is PAL rev B.
- Otherwise, it is some future A400/A800 OS.

If location \$FCD8 not \$A2, then product is a 1200XL or other future home computer product, wherein:

If location \$FFF1 = \$01, then OS is 1200XL
and location \$FFF7 will be the
internal rev number for the 1200XL OS.

Otherwise, location \$FFF1 = product code for
future Atari Home Computer product
and location \$FFF7 contains OS rev
level for this product.

APPENDIX A — AN EXAMPLE OF KEYBOARD REASSIGNMENT

As suggested earlier in this document, the keyboard functions may be reassigned. The table below gives the corresponding keys for the Dvorak (also known as the American Simplified) Keyboard. When the typewriter was first invented in 1867, Christopher L. Sholes chose a layout for the keys which would slow down the good typists of his day and thereby prevent his machine from jamming. This keyboard has endured to this day.

In 1932, August Dvorak invented this key layout which places the most often used characters, including the vowels, on the "home" key line and also redistributes the keystrokes from a 60-70% left-hand activity to an almost 50/50 activity. Certain manufacturers currently offer this key layout as an option. Now you can try it for yourself if you wish. Only the list of key correspondence is given here. It is left to the reader to compose the key function table using the data contained earlier in this manual.

TOP ROW OF KEYBOARD		CENTER ROW		BOTTOM ROW	
Current	Dvorak	Current	Dvorak	Current	Dvorak
Q	?	A	A	Z	:
q	/			z	;
W	,	S	O	X	Q
w	,				
E	.	D	E	C	J
e	.				
R	P	F	U	V	K
T	Y	G	I	B	X
Y	F	H	D	N	B
U	G	J	H	M	M
I	C	K	T	"	W
				"	w
O	R	L	N	.	V
				.	v
P	L	:	S	?	Z
		;	s	/	z
¼	"	"	_(underline)		
½	'	'	—		

APPENDIX B — SUGGESTIONS FOR THE CONSTRUCTION OF A NEW CHARACTER SET FOR THE NEW GRAPHICS MODES

This appendix covers the new graphics modes 12, 13, 14 and 15 now provided on the 1200XL. Modes 14 and 15 are pure graphics modes with resolutions of 160 by 20 and 160 by 40 respectively. Since these are not character modes, the discussion below will be limited only to modes 12 and 13.

Graphics 12 and 13 do not produce recognizable characters, for the most part, using the standard character set. One will understand why this is true by examining the following comparison between Graphics mode 0 to 12 and 13.

> Mode 0 is a 40 character mode. Each character is formed from an 8 wide by 8 high pixel matrix. Each pixel is one bit wide in memory and is ½ of a color clock wide on the screen.
>
> Modes 12 and 13 are also 40 character modes. However, each character is formed from a 4 wide by 8 high pixel matrix, with each pixel 2 bits wide in memory and one color clock wide on the screen. This forces the character to be the same width as that used in Graphics mode 0, but cannot convey the same information within 4 pixels as with 8 as far as character recognition is concerned. (It is difficult to form a recognizable character in a four by eight dot matrix).

Let's examine how the 4-pixel character is formed, again comparing the way the 8-pixel character is formed in mode 0:

> Mode 0 has a choice of two colors for each pixel (the hardware manual says 1 ½ colors, but it is actually either the hue and luminance of playfield 2 if there is a zero bit in the selected pixel position, or the hue from playfield 2 with the luminance of playfield 1 if there is a 1 bit in the selected pixel position. Therefore, each single bit in the character definition byte for a given line occupies a single ½-color-clock-wide pixel position. The character set built into the OS defines the characters in an 8 by 8 matrix.
>
> Mode 12 also uses 8 scan lines per character. However, it uses the character bytes in a different manner. Each of the character bytes retrieved by the ANTIC is treated as a set of four two-bit quantities, where each bit pair describes the color which is to be applied to one of the 4 single color-clock-wide pixels which are part of the character. Mode 13 is the same in its treatment of the data bytes, but each of the characters is double-height (16 scan lines instead of 8) and each data byte is used twice which effectively doubles the height of the character.

Let's look at a typical character, for example a W. The bits which form a W in the default character set are similar to the following:

1	0	0	0	0	0	0	1
1	0	0	0	0	0	0	1
1	0	0	1	1	0	0	1
1	0	0	1	1	0	0	1
1	0	1	0	0	1	0	1
1	1	0	0	0	0	1	1
1	1	0	0	0	0	1	1
1	0	0	0	0	0	0	1

display:

(**NOTE:** This is not the exact representation, but is used as an example of correct interpretation in mode 0 and incorrect interpretation in modes 12 and 13.)

If you view the sample set of bytes, each at consecutive addresses within the defined character set, it actually looks like a W when you trace the outline formed by the 1's in the byte set, as shown in the display example to the right of the byte representation.

In this mode 0 display, each of the 1's would be one color, and each of the zeros would be another color, assuring a readable display.

For the modes 12 and 13, the four (not 8) pixels are controlled as follows:

If two-bit value is:	**Then the pixel color is:**
00	the background color
01	the playfield 0 color
10	the playfield 1 color
11	the playfield 2 color (if bit 7 of char = 0)
11	the playfield 3 color (if bit 7 of char = 1)

For the example shown, then, the 4th line from the bottom would display a 10 10 01 01 or 4 pixels of playfield colors 1, 1, 0, 0 in a row, if the standard character set is used. And the bottom-most line would display playfield colors 1, BAK, BAK, 0 in a row. As may be imagined, it is difficult to recognize such a character. (This character is a mirror image left to right — nonsymmetric characters would be even more difficult to recognize.)

To build a character set for these modes 12 and 13, then, it is suggested that you build each character as double wide, to allow a total of 8 pixels (by 8 lines) to define the character. This would also mean assigning two character set locations for each character and treating each character printed in these modes as two characters to be printed. For the example of the W, the character set might look like this:

Byte set 1:

10	00	00	00
10	00	00	00
10	00	00	10
10	00	00	10
10	00	10	00
10	10	00	00
10	10	00	00
10	00	00	00

Byte set 2:

00	00	00	10
00	00	00	10
10	00	00	10
10	00	00	10
00	10	00	10
00	00	10	10
00	00	10	10
00	00	00	10

Byte set 1 may represent ATASCII value hex 57 within the new character set table, and set 2 may be at ATASCII value hex D7 (hex 57 plus hex 80) if desired. You may feel free, of course, to assign your character sets in any manner you desire.

Therefore, if you would print these two characters side by side on the screen, it would become effectively a 20 character per line mode, with the resultant 10-combination treated as the 1-bit in the mode 0 example and the 00-combination as the 0-bit in the mode 0 example, forming a recognizable W in the process.

Note also that you may want to design these new character sets in a 7 by 7 matrix starting the upper left hand corner of the bit-pair set to allow at least one blank row and column between each of the new characters. (This was not done in the example.)

Thus many combinations of colorful characters may be formed using this technique, allowing the user of the 1200XL additional program flexibility.

MEMORY REQUIREMENTS FOR NEW SCREEN MODES

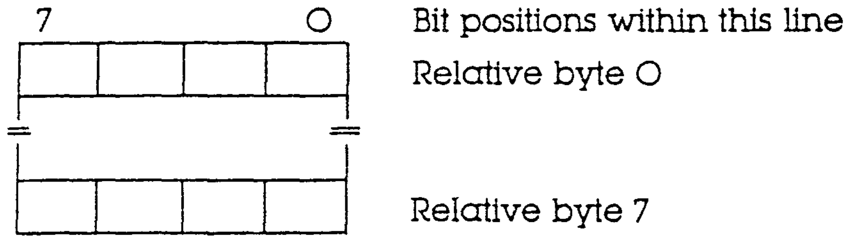
The following table summarizes the memory utilization for the new modes 12 through 15.

Mode No.	Horiz. Posit.	Vert. w/o split screen	Vert. with split screen	Colors	Data Value range	Color Reg. used	Memory Required (split screen)	Memory Required (full screen)
12	40	24	20	5	00-7F	*	1154	1152
13	40	12	10	5	00-7F	*	664	660
14	160	192	160	2	0 1	BAK PF0	4270	4296
15	160	192	160	4	0 1 2 3	BAK PF0 PF1 PF2	8112	8138

***Note:** See character definition format for modes 12 and 13.

CHARACTER DEFINITION FORMAT FOR MODES 12 AND 13

The following chart shows the layout for a single character of the character set which would be used for forming characters in modes 12 and 13. As explained above, the value of each of the bit pairs specifies what color will appear on a full-width color clock when this character mode is selected.



Each 2 bit color specification in the character definition maps to the color registers as follows:

If the bits have the value:	Then the color register used to select the color of the pixel is:
O	BAK
1	PFO
2	PF1
3	PF2 if Bit-7 (the color modifier) equals a O, or PF3 if Bit-7 (the color modifier) equals a 1.

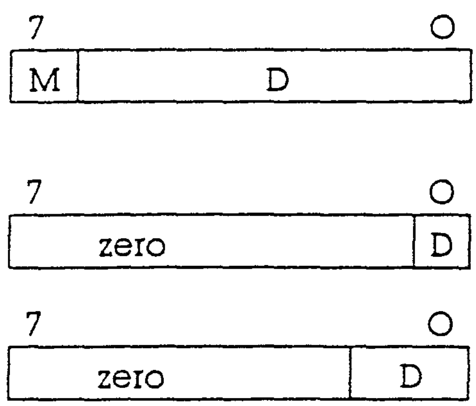
The meaning of the color modifier is shown in the following tables, which show the formats for the data bytes which are used to produce the display itself. As a reminder, the data which is to be displayed on the screen is located somewhere in memory. The data is located by the address provided in the display list. The data bytes themselves in these locations will be interpreted according to the following table.

TABLE of DATA FORMATS used for GET CHARACTER/PUT CHARACTER for MODES 12 through 15.

Modes 12, 13 — M = color modifier bit; D = truncated ATASCII

Mode 14 — D = color

Mode 15 — D = color



APPENDIX C — DATA BASE CHANGES FROM REV. B TO 1200

This appendix lists the difference in memory usage between the Rev. B operating system of the A400/800 and the operating system for the 1200XL.

LOCATION	REV. B USE	1200XL USE		
0000	reserved	LNFLG	—	for inhouse debugger
0001	reserved	NGFLAG	—	for power-up self test
001C	PTIMOT moved (0314)	ABUFPT	—	reserved
001D	PBPNT moved (02DE)	ABUFPT	—	reserved
001E	PBUFSZ moved (02DF)	ABUFPT	—	reserved
001F	PTEMP (deleted)	ABUFPT	—	reserved
0036	CRETRY moved (029C)	LTEMP	—	loader temp.
0037	DRETRY moved (02BD)	LTEMP	—	loader temp.
004A	CKEY moved (03E9)	ZCHAIN	—	handler loader temp.
004B	CASSBT moved (03E9)	ZCHAIN	—	handler loader temp.
0060	NEWROW-moved (02F5)	FKDEF	—	func. key def. ptr.
0061	NEWCOL moved (02F6)	FKDEF	—	func. key def. ptr.
0062	NEWCOL moved (02F7)	PALNTS	—	PAL/NTSC flag.
0079	ROWINC moved (02F8)	KEYDEF	—	key def. pointer
007A	COLINC moved (02F9	KEYDEF	—	key def. pointer
0233	reserved	LCOUNT	—	loader temp.
0238-0239	reserved	RELADR	—	loader
0245	reserved	RECLEN	—	loader
0247	LINBUF (deleted)	reserved		
0248-026A	LINBUF (deleted)	reserved		
026B	LINBUF (deleted)	CHSALT	—	character set ptr.
026C	LINBUF (deleted)	VSFLAG	—	fine scroll temp.
026D	LINBUF (deleted)	KEYDIS	—	keyboard disable
026E	LINBUF (deleted)	FINE	—	fine scrolling flag
0288	CSTAT (deleted)	HIBYTE	—	loader
028E	reserved	NEWADR	—	loader
029C	TMPX1 (deleted)	CRETRY	—	from 0036
02BD	HOLD5 (deleted)	DRETRY	—	from0037
02C9-02CA	reserved	RUNADR	—	loader
02CB-02CC	reserved	HIUSED	—	loader
02CD-02CE	reserved	ZHIUSE	—	loader
02CF-02D0	reserved	GBYTEA	—	loader
02D1-02D2	reserved	LOADAD	—	loader
02D3-02D4	reserved	ZLOADA	—	loader
02D5-02D6	reserved	DSCTLN	—	disk sector size
02D7-02D8	reserved	ACMISR	—	reserved
02D9	reserved	KRPDEL	—	auto key delay
02DA	reserved	KEYREP	—	auto key rate
02DB	reserved	NOCLIK	—	key click disable
02DC	reserved	HELPFG	—	HELP key flag
02DD	reserved	DMASAV	—	DMA state save
02DE	reserved	PBPNT	—	from 001D
02DF	reserved	PBUFSZ	—	from 001E

O2E9	reserved
O2F5	reserved
O2F6-O2F7	reserved
O2F8	reserved
O2F9	reserved
O3OE	ADDCOR (deleted)
O314	TEMP2 moved (O313)
O33D	reserved
O33E	reserved
O33F	reserved
O3E8	reserved
O3E9	reserved
O3EA	reserved
O3EB	reserved
O3ED-O3F8	reserved
O3F9	reserved
O3FA	reserved
O3FB-O3FC	reserved

HNDLOD	—	handler loader flag
NEWROW	—	from OO6O
NEWCOL	—	from OO61
ROWINC	—	from OO79
COLINC	—	from OO7A
JMPERS	—	option jumpers
PTIMOT	—	from OO1C
PUPBT1	—	power-up/reset
PUPBT2	—	power-up/reset
PUPBT3	—	power-up/reset
SUPERF	—	screen editor
CKEY	—	from OO4A
CASSBT	—	from OO4B
CARTCK	—	cart checksum
ACMVAR	—	reserved
MINTLK	—	reserved
GINTLK	—	cart interlock
CHLINK	—	handler chain

247-26A = free ram
CB65-CBFF = free os

```
0000                    LIST    -F,-M


              ***       Copyright 1984 ATARI.  Unauthorized reproduction, a:
              *         distribution, performance or display of this comput:
              *         or the associated audiovisual work is strictly proh:



              ***       OS - Operating System
              *
              *         NOTES
              *                 This represents an attempt to bring the OS :
              *                 into conformance with the Atari Internal So:
              *                 Standards as defined in the Software Develo:
              *                 Committee Report on Procedures and Standard:
              *                 (10/27/81).  Due to time constraints, the e:
              *                 source could not be brought up to the stand:
              *                 particularly in the area of subroutine head:
              *                 documentation (ENTRY, EXIT, CHANGES and CAL:
              *                 More complete and consistent conformance to:
              *                 standard is planned for the next revision o:
              *                 Operating System (Revision 3).
              *
              *         MODS
              *                 Revision A (400/800)
              *                 D. Crane/A. Miller/L. Kaplan/R. Whitehead
              *
              *                 Revision B (400/800)
              *                 Fix several problems.
              *                 M. Mahar/R. S. Scheiman
              *
              *                 Revision 10 (1200XL)
              *                 Support 1200XL, add new features.
              *                 H. Stewart/L. Winner/R. S. Scheiman/
              *                 Y. M. Chen/M. W. Colburn        10/26/82
              *
              *                 Revision 11 (1200XL)
              *                 Fix several problems.
              *                 R. S. Scheiman  12/23/82
              *
              *                 Revision 1 (600XL/800XL)             CO62024
              *                 Support PBI and on-board BASIC.
              *                 R. S. Scheiman/R. K. Nordin/Y. M. Chen  03/:
              *
              *                 Revision 2 (600XL/800XL)  C061598 B
              *                 Fix several problems.
              *                 R. S. Scheiman  05/10/83
              *                 Bring closer to Coding Standard (object unc:
              *                 R. K. Nordin    11/01/83
```

```
**      Program Structure
*
*       The sections of the OS appear in the following orde:
*       corresponding subtitles:
*
*       Equates and Definitions
*
*               System Symbol Equates
*               System Address Equates
*               Miscellaneous Address Equates
*               Macro Definitions
*
*       Code and Data
*
*               First 8K ROM Identification and Checksum
*
*               Interrupt Handler
*               Initialization
*               Disk Input/Ouput
*               Relocating Loader
*               Self-test, Part 1
*               Parallel Input/Output
*               Peripheral Handler Loading Facility, Part 1
*               Self-test, Part 2
*               Peripheral Handler Loading Facility, Part 2
*
*               International Character Set
*
*               Self-test, Part 3
*               Floating Point Package
*
*               Domestic Character Set
*
*               Device Handler Vector Tables
*               Jump vectors
*               Generic Parallel Device Handler Vector Tabl:
*
*               $E4C0 Patch
*               Central Input/Output
*               Peripheral Handler Loading Facility, Part 3
*               $E912 Patch
*               Peripheral Handler Loading Facility, Part 4
*               $E959 Patch
*               Serial Input/Output
*               Keyboard, Editor and Screen Handler, Part 1
*               Peripheral Handler Loading Facility, Part 5
*               $EF6B Patch
*               Keyboard, Editor and Screen Handler, Part 2
*               $F223 Patch
*               Keyboard, Editor and Screen Handler, Part 3
*               $FCD8 Patch
*               Cassette Handler
*               Printer Handler
*               Self-test, Part 4
*
*               Second 8K ROM Identification and Checksum
*               6502 Machine Vectors
```

```
0000                    **      Assembly Option Equates


     = 0000     FALSE   EQU     0
     = FFFF     TRUE    EQU     not FALSE

     = FFFF     VGC     SET     TRUE    ;virtual game controllers
     = 0000     RAMSYS  SET     FALSE   ;not RAM based system
     = 0000     LNBUG   SET     FALSE   ;no LNBUG interface
     = 0000     ACM1    SET     FALSE   ;no asynchronous communications mod:



                **      Identification Equates


     = 0002     IDREV   EQU     $02     ;identification revision number
     = 0010     IDDAY   EQU     $10     ;identification day
     = 0005     IDMON   EQU     $05     ;identification month
     = 0083     IDYEAR  EQU     $83     ;identification year
     = 0002     IDCPU   EQU     $02     ;identification CPU series
     = 0042     IDPN1   EQU     'B'     ;identification part number field 1
     = 0042     IDPN2   EQU     'B'     ;identification part number field 2
     = 0000     IDPN3   EQU     $00     ;identification part number field 3
     = 0000     IDPN4   EQU     $00     ;identification part number field 4
     = 0001     IDPN5   EQU     $01     ;identification part number field 5



                **      Configuration Equates
                *
                *       NOTES
                *               Problem: last byte of HATABS (as defined by:
                *               overlaps first power-up validation byte.


     = 0021     MAXDEV  EQU     33      ;offset to last possible entry of H:
     = 0010     IOCBSZ  EQU     16      ;length of IOCB

     = 0000     SEIOCB  EQU     0*IOCBSZ        ;screen editor IOCB index
     = 0080     MAXIOC  EQU     8*IOCBSZ        ;first invalid IOCB index

     = 0080     DSCTSZ  EQU     128     ;disk sector size

     = 0002     LEDGE   EQU     2       ;left edge
     = 0027     REDGE   EQU     39      ;right edge

     = 0700     INIML   EQU     $0700   ;initial MEMLO

     = CC00     ICSORG  EQU     $CC00   ;international character set origin
     = E000     DCSORG  EQU     $E000   ;domestic character set origin
```

```
                ** IOCB Command Code Equates

= 0003          OPEN    EQU     $03     ;open
= 0005          GETREC  EQU     $05     ;get record
= 0007          GETCHR  EQU     $07     ;get character(s)
= 0009          PUTREC  EQU     $09     ;put record
= 000B          PUTCHR  EQU     $0B     ;put character(s)
= 000C          CLOSE   EQU     $0C     ;close
= 000D          STATIS  EQU     $0D     ;status
= 000E          SPECIL  EQU     $0E     ;special


                ** Special Entry Command Equates

                ; Screen Commands

= 0011          DRAWLN  EQU     $11     ;draw line
= 0012          FILLIN  EQU     $12     ;draw line with right fill


                ** ICAX1 Auxiliary Byte 1 Equates

= 0001          APPEND  EQU     $01     ;open write append (D:) or screen r:
= 0002          DIRECT  EQU     $02     ;open for directory access (D:)
= 0004          OPNIN   EQU     $04     ;open for input (all devices)
= 0008          OPNOT   EQU     $08     ;open for output (all devices)
= 0010          MXDMOD  EQU     $10     ;open for mixed mode (E:, S:)
= 0020          INSCLR  EQU     $20     ;open for input without clearing sc:


                ** Device Code Equates

= 0043          CASSET  EQU     'C'     ;cassette
= 0044          DISK    EQU     'D'     ;disk
= 0045          SCREDT  EQU     'E'     ;screen editor
= 004B          KBD     EQU     'K'     ;keyboard
= 0050          PRINTR  EQU     'P'     ;printer
= 0053          DISPLY  EQU     'S'     ;screen display
```

```
                **      Character and Key Code Equates


 = 007D         CLS     EQU     $7D     ;clear screen
 = 009B         EOL     EQU     $9B     ;end of line (RETURN)

 = 0011         HELP    EQU     $11     ;key code for HELP
 = 0083         CNTLF1  EQU     $83     ;key code for CTRL-F1
 = 0084         CNTLF2  EQU     $84     ;key code for CTRL-F2
 = 0093         CNTLF3  EQU     $93     ;key code for CTRL-F3
 = 0094         CNTLF4  EQU     $94     ;key code for CTRL-F4
 = 009F         CNTL1   EQU     $9F     ;key code for CTRL-1



                **      Status Code Equates


 = 0001         SUCCES  EQU     1       ;successful operation

 = 0080         BRKABT  EQU     128     ;BREAK key abort
 = 0081         PRVOPN  EQU     129     ;IOCB already open error
 = 0082         NONDEV  EQU     130     ;nonexistent device error
 = 0083         WRONLY  EQU     131     ;IOCB opened for write only error
 = 0084         NVALID  EQU     132     ;invalid command error
 = 0085         NOTOPN  EQU     133     ;device/file not open error
 = 0086         BADIOC  EQU     134     ;invalid IOCB index error
 = 0087         RDONLY  EQU     135     ;IOCB opened for read only error
 = 0088         EOFERR  EQU     136     ;end of file error
 = 0089         TRNRCD  EQU     137     ;truncated record error
 = 008A         TIMOUT  EQU     138     ;peripheral device timeout error
 = 008B         DNACK   EQU     139     ;device does not acknowledge comman:
 = 008C         FRMERR  EQU     140     ;serial bus framing error
 = 008D         CRSROR  EQU     141     ;cursor overrange error
 = 008E         OVRRUN  EQU     142     ;serial bus data overrun error
 = 008F         CHKERR  EQU     143     ;serial bus checksum error
 = 0090         DERROR  EQU     144     ;device done (operation incomplete):
 = 0091         BADMOD  EQU     145     ;bad screen mode number error
 = 0092         FNCNOT  EQU     146     ;function not implemented in handle:
 = 0093         SCRMEM  EQU     147     ;insufficient memory for screen mod:

                **      DCB Device Bus ID Equates


 = 0031         DISKID  EQU     $31     ;disk bus ID
 = 0040         PDEVN   EQU     $40     ;printer bus ID
 = 0060         CASET   EQU     $60     ;cassette bus ID
```

```
                 **      Bus Command Equates

   = 0021        FOMAT   EQU     '!'     ;format command
   = 0050        PUTSEC  EQU     'P'     ;put sector command
   = 0052        READ    EQU     'R'     ;read command
   = 0053        STATC   EQU     'S'     ;status command
   = 0057        WRITE   EQU     'W'     ;write command


                 **      Command Auxiliary Byte Equates

   = 0044        DOUBLE  EQU     'D'     ;print 20 characters double width
   = 004E        NORMAL  EQU     'N'     ;print 40 characters normally
   = 0050        PLOT    EQU     'P'     ;plot
   = 0053        SIDWAY  EQU     'S'     ;print 16 characters sideways


                 **      Bus Response Equates

   = 0041        ACK     EQU     'A'     ;device acknowledged
   = 0043        COMPLT  EQU     'C'     ;device successfully completed oper:
   = 0045        ERROR   EQU     'E'     ;device incurred error in attempted:
   = 004E        NACK    EQU     'N'     ;device did not understand


                 **      Floating Point Package Miscellaneous Equates

   = 0006        FPREC   EQU     6       ;precision

   = 0005        FMPREC  EQU     FPREC-1 ;length of mantissa


                 **      Cassette Record Type Equates

   = 00FB        HDR     EQU     $FB     ;header
   = 00FC        DTA     EQU     $FC     ;data record
   = 00FA        DT1     EQU     $FA     ;last data record
   = 00FE        EOT     EQU     $FE     ;end of tape (file)

   = 0002        TONE1   EQU     2       ;record
   = 0001        TONE2   EQU     1       ;playback
```

```
                    **      Cassette Timing Equates


   = 0480           WLEADN  EQU     1152    ;NTSC 19.2 second WRITE file leader
   = 0240           RLEADN  EQU     576     ;NTSC 9.6 second READ file leader
   = 00B4           WIRGLN  EQU     180     ;NTSC 3.0 second WRITE IRG
   = 0078           RIRGLN  EQU     120     ;NTSC 2.0 second READ IRG
   = 000F           WSIRGN  EQU     15      ;NTSC 0.25 second WRITE short IRG
   = 000A           RSIRGN  EQU     10      ;NTSC 0.16 second READ short IRG
   = 001E           BEEPNN  EQU     30      ;NTSC 0.5 second beep duration
   = 000A           BEEPFN  EQU     10      ;NTSC 0.16 second beep separation

   = 03C0           WLEADP  EQU     960     ;PAL 19.2 second WRITE file leader
   = 01E0           RLEADP  EQU     480     ;PAL 9.6 second READ file leader
   = 0096           WIRGLP  EQU     150     ;PAL 3.0 second WRITE IRG
   = 0064           RIRGLP  EQU     100     ;PAL 2.0 second READ IRG
   = 000D           WSIRGP  EQU     13      ;PAL 0.25 second WRITE short IRG
   = 0008           RSIRGP  EQU     8       ;PAL 0.16 second READ short IRG
   = 0019           BEEPNP  EQU     25      ;PAL 0.5 second beep duration
   = 0008           BEEPFP  EQU     8       ;PAL 0.16 second beep separation

   = 0000           WIRGHI  EQU     0       ;high WRITE IRG
   = 0000           RIRGHI  EQU     0       ;high READ IRG



                    **      Power-up Validation Byte Value Equates


   = 005C           PUPVL1  EQU     $5C     ;power-up validation value 1
   = 0093           PUPVL2  EQU     $93     ;power-up validation value 2
   = 0025           PUPVL3  EQU     $25     ;power-up validation value 3



                    **      Relocating Loader Miscellaneous Equates


   = 009C           DATAER  EQU     156     ;end of record appears before END r:
   = 009D           MEMERR  EQU     157     ;memory insufficient for load error
```

```
                    **      Miscellaneous Equates


        = 00FF      IOCFRE  EQU     $FF       ;IOCB free indicator

        = 0028      B19200  EQU     $0028     ;19200 baud POKEY counter value
        = 05CC      B00600  EQU     $05CC     ;600 baud POKEY counter value

        = 0005      HITONE  EQU     $05       ;FSK high freq. POKEY counter value:
        = 0007      LOTONE  EQU     $07       ;FSK low freq. POKEY counter value :

        = 0034      NCOMLO  EQU     $34       ;PIA lower NOT COMMAND line command
        = 003C      NCOMHI  EQU     $3C       ;PIA raise NOT COMMAND line command

        = 0034      MOTRGO  EQU     $34       ;PIA cassette motor ON command
        = 003C      MOTRST  EQU     $3C       ;PIA cassette motor OFF command

        = 0000      NODAT   EQU     $00       ;SIO immediate operation
        = 0040      GETDAT  EQU     $40       ;SIO read data frame
        = 0080      PUTDAT  EQU     $80       ;SIO write data frame

        = 000D      CRETRI  EQU     13        ;number of command frame retries
        = 0001      DRETRI  EQU     1         ;number of device retries
        = 0002      CTIM    EQU     2         ;command frame ACK timeout

        = 0028      NBUFSZ  EQU     40        ;print normal buffer size
        = 0014      DBUFSZ  EQU     20        ;print double buffer size
        = 001D      SBUFSZ  EQU     29        ;print sideways buffer size
```

```
0000            **      Page Zero Address Equates

     = 0000     LNFLG   EQU     $0000   ;1-byte LNBUG flag (0 = not LNBUG)
     = 0001     NGFLAG  EQU     $0001   ;1-byte memory status (0 = failure)

                ;       Not Cleared

     = 0002     CASINI  EQU     $0002   ;2-byte cassette program initializa:
     = 0004     RAMLO   EQU     $0004   ;2-byte RAM address for memory test
     = 0006     TRAMSZ  EQU     $0006   ;1-byte RAM size temporary
     = 0007     CMCMD   EQU     $0007   ;1-byte command communications

                ;       Cleared upon Coldstart Only

     = 0008     WARMST  EQU     $0008   ;1-byte warmstart flag (0 = coldsta:
     = 0009     BOOT?   EQU     $0009   ;1-byte successful boot flags
     = 000A     DOSVEC  EQU     $000A   ;2-byte disk program start vector
     = 000C     DOSINI  EQU     $000C   ;2-byte disk program initialization:
     = 000E     APPMHI  EQU     $000E   ;2-byte applications memory high li:

                ;       Cleared upon Coldstart or Warmstart

     = 0010     INTZBS  EQU     $0010   ;first page zero location to clear

     = 0010     POKMSK  EQU     $0010   ;1-byte IRQEN shadow
     = 0011     BRKKEY  EQU     $0011   ;1-byte BREAK key flag (0 = no BREA:
     = 0012     RTCLOK  EQU     $0012   ;3-byte real time clock (16 millise:
     = 0015     BUFADR  EQU     $0015   ;2-byte disk interface buffer addre:
     = 0017     ICCOMT  EQU     $0017   ;1-byte CIO command table index
     = 0018     DSKFMS  EQU     $0018   ;2-byte DOS File Management System :
     = 001A     DSKUTL  EQU     $001A   ;2-byte DOS utility pointer
     = 001C     ABUFPT  EQU     $001C   ;4-byte ACMI buffer pointer area

     = 0020     ZIOCB   EQU     $0020   ;address of page zero IOCB
     = 0020     IOCBAS  EQU     $0020   ;16-byte page zero IOCB
     = 0020     ICHIDZ  EQU     $0020   ;1-byte handler ID ($FF = IOCB free:
     = 0021     ICDNOZ  EQU     $0021   ;1-byte device number
     = 0022     ICCOMZ  EQU     $0022   ;1-byte command code
     = 0023     ICSTAZ  EQU     $0023   ;1-byte status of last action
     = 0024     ICBALZ  EQU     $0024   ;1-byte low buffer address
     = 0025     ICBAHZ  EQU     $0025   ;1-byte high buffer address
     = 0026     ICPTLZ  EQU     $0026   ;1-byte low PUT-BYTE routine addres:
     = 0027     ICPTHZ  EQU     $0027   ;1-byte high PUT-BYTE routine addre:
     = 0028     ICBLLZ  EQU     $0028   ;1-byte low buffer length
     = 0029     ICBLHZ  EQU     $0029   ;1-byte high buffer length
     = 002A     ICAX1Z  EQU     $002A   ;1-byte first auxiliary information
     = 002B     ICAX2Z  EQU     $002B   ;1-byte second auxiliary informatio:
     = 002C     ICSPRZ  EQU     $002C   ;4-byte spares

     = 002C     ENTVEC  EQU     $002C   ;2-byte (not used)
     = 002E     ICIDNO  EQU     $002E   ;1-byte IOCB index (IOCB number tim:
     = 002F     CIOCHR  EQU     $002F   ;1-byte character for current CIO o:

     = 0030     STATUS  EQU     $0030   ;1-byte SIO operation status
     = 0031     CHKSUM  EQU     $0031   ;1-byte checksum (single byte sum w:
     = 0032     BUFRLO  EQU     $0032   ;1-byte low data buffer address
```

```
= 0033     BUFRHI   EQU     $0033    ;1-byte high data buffer address
= 0034     BFENLO   EQU     $0034    ;1-byte low data buffer end address
= 0035     BFENHI   EQU     $0035    ;1-byte high data buffer end addres:
= 0036     LTEMP    EQU     $0036    ;2-byte relocating loader temporary
= 0038     BUFRFL   EQU     $0038    ;1-byte data buffer full flag (0 = :
= 0039     RECVDN   EQU     $0039    ;1-byte receive-frame done flag (0 :
= 003A     XMTDON   EQU     $003A    ;1-byte transmit-frame done flag (0:
= 003B     CHKSNT   EQU     $003B    ;1-byte checksum sent flag (0 = not:
= 003C     NOCKSM   EQU     $003C    ;1-byte no checksum follows data fl:
= 003D     BPTR     EQU     $003D    ;1-byte cassette buffer pointer
= 003E     FTYPE    EQU     $003E    ;1-byte cassette IRG type (neg. = c:
= 003F     FEOF     EQU     $003F    ;1-byte cassette EOF flag (0 = no E:
= 0040     FREQ     EQU     $0040    ;1-byte cassette beep counter
= 0041     SOUNDR   EQU     $0041    ;1-byte noisy I/O flag (0 = quiet)

= 0042     CRITIC   EQU     $0042    ;1-byte critical section flag (0 = :

= 0043     FMSZPG   EQU     $0043    ;7-byte reserved for DOS File Manag:

= 004A     ZCHAIN   EQU     $004A    ;2-byte handler linkage chain point:
= 004C     DSTAT    EQU     $004C    ;1-byte display status
= 004D     ATRACT   EQU     $004D    ;1-byte attract-mode timer and flag
= 004E     DRKMSK   EQU     $004E    ;1-byte attract-mode dark (luminanc:
= 004F     COLRSH   EQU     $004F    ;1-byte attract-mode color shift
= 0050     TMPCHR   EQU     $0050    ;1-byte temporary character
= 0051     HOLD1    EQU     $0051    ;1-byte temporary
= 0052     LMARGN   EQU *   $0052    ;1-byte text column left margin
= 0053     RMARGN   EQU *   $0053    ;1-byte text column right margin
= 0054     ROWCRS   EQU *   $0054    ;1-byte cursor row
= 0055     COLCRS   EQU *   $0055    ;2-byte cursor column
= 0057     DINDEX   EQU *   $0057    ;1-byte display mode
= 0058     SAVMSC   EQU *   $0058    ;2-byte saved memory scan counter - leave inase ML u
= 005A     OLDROW   EQU *   $005A    ;1-byte prior row
= 005B     OLDCOL   EQU *   $005B    ;2-byte prior column
= 005D     OLDCHR   EQU *   $005D    ;1-byte saved character under curso:
= 005E     OLDADR   EQU *   $005E    ;2-byte saved cursor memory address
= 0060    -FKDEF    EQU *   $0060    ;2-byte function key definition tab:
= 0062    -PALNTS   EQU     $0062    ;1-byte PAL/NTSC indicator (0 = NTS:
= 0063     LOGCOL   EQU *   $0063    ;1-byte logical line cursor column
= 0064     ADRESS   EQU *   $0064    ;2-byte temporary address

= 0066     MLTTMP   EQU     $0066    ;1-byte temporary
= 0066     OPNTMP   EQU     $0066    ;1-byte open temporary
= 0066     TOADR    EQU     $0066    ;2-byte destination address   temps

= 0068     SAVADR   EQU     $0068    ;2-byte saved address
= 0068     FRMADR   EQU     $0068    ;2-byte source address

= 006A    -RAMTOP   EQU     $006A    ;1-byte RAM size
= 006B     BUFCNT   EQU *   $006B    ;1-byte buffer count (logical line :
= 006C     BUFSTR   EQU *   $006C    ;2-byte buffer start pointer
= 006E     BITMSK   EQU *   $006E    ;1-byte bit mask for bit map operat:
= 006F     SHFAMT   EQU *   $006F    ;1-byte shift amount for pixel just:
= 0070     ROWAC    EQU *   $0070    ;2-byte draw working row
= 0072     COLAC    EQU *   $0072    ;2-byte draw working column
= 0074     ENDPT    EQU *   $0074    ;2-byte end point
= 0076     DELTAR   EQU *   $0076    ;1-byte row difference
```

```
= 0077          DELTAC  EQU     $0077   ;2-byte column difference
= 0079          KEYDEF  EQU     $0079   ;2-byte key definition table addres:
= 007B          SWPFLG  EQU     $007B   ;1-byte split screen swap flag (0 =:
= 007C          HOLDCH  EQU     $007C   ;1-byte temporary character
= 007D          INSDAT  EQU     $007D   ;1-byte temporary
= 007E          COUNTR  EQU     $007E   ;2-byte draw iteration count

                ;       Reserved for Application and Floating Point Package

                ;       EQU     $0080   ;128 bytes reserved for application:


                **      Floating Point Package Page Zero Address Equates


= 00D4          FR0     EQU     $00D4   ;6-byte register 0
= 00D5          FR0M    EQU     $00D5   ;5-byte register 0 mantissa
= 00D9          QTEMP   EQU     $00D9   ;1-byte temporary

= 00DA          FRE     EQU     $00DA   ;6-byte (internal) register E

= 00E0          FR1     EQU     $00E0   ;6-byte register 1
= 00E1          FR1M    EQU     $00E1   ;5-byte register 1 mantissa

= 00E6          FR2     EQU     $00E6   ;6-byte (internal) register 2

= 00EC          FRX     EQU     $00EC   ;1-byte temporary

= 00ED          EEXP    EQU     $00ED   ;1-byte value of exponent

= 00EE          FRSIGN  EQU     $00EE   ;1-byte floating point sign
= 00EE          NSIGN   EQU     $00EE   ;1-byte sign of number

= 00EF          PLYCNT  EQU     $00EF   ;1-byte polynomial degree
= 00EF          ESIGN   EQU     $00EF   ;1-byte sign of exponent

= 00F0          SGNFLG  EQU     $00F0   ;1-byte sign flag
= 00F0          FCHFLG  EQU     $00F0   ;1-byte first character flag

= 00F1          XFMFLG  EQU     $00F1   ;1-byte transform flag
= 00F1          DIGRT   EQU     $00F1   ;1-byte number of digits after deci:

= 00F2          CIX     EQU     $00F2   ;1-byte current input index
= 00F3          INBUFF  EQU     $00F3   ;2-byte line input buffer

= 00F5          ZTEMP1  EQU     $00F5   ;2-byte temporary
= 00F7          ZTEMP4  EQU     $00F7   ;2-byte temporary
= 00F9          ZTEMP3  EQU     $00F9   ;2-byte temporary

= 00FC          FLPTR   EQU     $00FC   ;2-byte floating point number point:
= 00FE          FPTR2   EQU     $00FE   ;2-byte floating point number point:
```

```
                    **      Page One (Stack) Address Equates


                    ;       EQU     $0100   ;256-byte stack



                    **      Page Two Address Equates


      = 0200        INTABS  EQU     $0200   ;42-byte interrupt handler table

      = 0200        VDSLST  EQU     $0200   ;2-byte display list NMI vector
      = 0202        VPRCED  EQU     $0202   ;2-byte serial I/O proceed line IRQ:
      = 0204        VINTER  EQU     $0204   ;2-byte serial I/O interrupt line I:
      = 0206        VBREAK  EQU     $0206   ;2-byte BRK instruction IRQ vector
      = 0208        VKEYBD  EQU     $0208   ;2-byte keyboard IRQ vector
      = 020A        VSERIN  EQU     $020A   ;2-byte serial input ready IRQ vect:
      = 020C        VSEROR  EQU     $020C   ;2-byte serial output ready IRQ vec:
      = 020E        VSEROC  EQU     $020E   ;2-byte serial output complete IRQ :
      = 0210        VTIMR1  EQU     $0210   ;2-byte POKEY timer 1 IRQ vector
      = 0212        VTIMR2  EQU     $0212   ;2-byte POKEY timer 2 IRQ vector
      = 0214        VTIMR4  EQU     $0214   ;2-byte POKEY timer 4 IRQ vector
      = 0216        VIMIRQ  EQU     $0216   ;2-byte immediate IRQ vector
      = 0218        CDTMV1  EQU     $0218   ;2-byte countdown timer 1 value
      = 021A        CDTMV2  EQU     $021A   ;2-byte countdown timer 2 value
      = 021C        CDTMV3  EQU     $021C   ;2-byte countdown timer 3 value
      = 021E        CDTMV4  EQU     $021E   ;2-byte countdown timer 4 value
      = 0220        CDTMV5  EQU     $0220   ;2-byte countdown timer 5 value
      = 0222        VVBLKI  EQU     $0222   ;2-byte immediate VBLANK NMI vector
      = 0224        VVBLKD  EQU     $0224   ;2-byte deferred VBLANK NMI vector
      = 0226        CDTMA1  EQU     $0226   ;2-byte countdown timer 1 vector
      = 0228        CDTMA2  EQU     $0228   ;2-byte countdown timer 2 vector

      = 022A        CDTMF3  EQU     $022A   ;1-byte countdown timer 3 flag (0 =:
      = 022B        SRTIMR  EQU     $022B   ;1-byte software key repeat timer
      = 022C        CDTMF4  EQU     $022C   ;1-byte countdown timer 4 flag (0 =:
      = 022D        INTEMP  EQU     $022D   ;1-byte temporary
      = 022E        CDTMF5  EQU     $022E   ;1-byte countdown timer 5 flag (0 =:
      = 022F        SDMCTL  EQU     $022F   ;1-byte DMACTL shadow
      = 0230        SDLSTL  EQU     $0230   ;1-byte DLISTL shadow
      = 0231        SDLSTH  EQU     $0231   ;1-byte DLISTH shadow
      = 0232        SSKCTL  EQU     $0232   ;1-byte SKCTL shadow
      = 0233        LCOUNT  EQU     $0233   ;1-byte relocating loader record le:
      = 0234        LPENH   EQU     $0234   ;1-byte light pen horizontal value
      = 0235        LPENV   EQU     $0235   ;1-byte light pen vertical value
      = 0236        BRKKY   EQU     $0236   ;2-byte BREAK key vector
      = 0238        VPIRQ   EQU     $0238   ;2-byte parallel device IRQ vector
      = 023A        CDEVIC  EQU     $023A   ;1-byte command frame device ID
      = 023B        CCOMND  EQU     $023B   ;1-byte command frame command
      = 023C        CAUX1   EQU     $023C   ;1-byte command auxiliary 1
      = 023D        CAUX2   EQU     $023D   ;1-byte command auxiliary 2

      = 023E        TEMP    EQU     $023E   ;1-byte temporary

                            ASSERT  low TEMP<>$FF   ;may not be the last word o:
```

```
 = 023F     ERRFLG  EQU     $023F   ;1-byte I/O error flag (0 = no erro:

                    ASSERT  low ERRFLG<>$FF ;may not be the last word o:

 = 0240     DFLAGS  EQU     $0240   ;1-byte disk flags from sector 1
 = 0241     DBSECT  EQU     $0241   ;1-byte disk boot sector count
 = 0242     BOOTAD  EQU     $0242   ;2-byte disk boot memory address
 = 0244     COLDST  EQU     $0244   ;1-byte coldstart flag (0 = complet:
 = 0245     RECLEN  EQU     $0245   ;1-byte relocating loader record le:
 = 0246     DSKTIM  EQU     $0246   ;1-byte disk format timeout
 = 0247     PDVMSK  EQU     $0247   ;1-byte parallel device selection m:
 = 0248     SHPDVS  EQU     $0248   ;1-byte PDVS (parallel device selec:
 = 0249     PDIMSK  EQU     $0249   ;1-byte parallel device IRQ selecti:
 = 024A     RELADR  EQU     $024A   ;2-byte relocating loader relative :
 = 024C     PPTMPA  EQU     $024C   ;1-byte parallel device handler tem:
 = 024D     PPTMPX  EQU     $024D   ;1-byte parallel device handler tem:

            ;       EQU     $024E   ;6 bytes reserved for Atari

            ;       EQU     $0254   ;23 bytes reserved for Atari

 = 026B     CHSALT  EQU     $026B   ;1-byte character set alternate
 = 026C     VSFLAG  EQU     $026C   ;1-byte fine vertical scroll count
 = 026D     KEYDIS  EQU     $026D   ;1-byte keyboard disable
 = 026E     FINE    EQU     $026E   ;1-byte fine scrolling mode
 = 026F     GPRIOR  EQU     $026F   ;1-byte PRIOR shadow

 = 0270     PADDL0  EQU     $0270   ;1-byte potentiometer 0
 = 0271     PADDL1  EQU     $0271   ;1-byte potentiometer 1
 = 0272     PADDL2  EQU     $0272   ;1-byte potentiometer 2
 = 0273     PADDL3  EQU     $0273   ;1-byte potentiometer 3
 = 0274     PADDL4  EQU     $0274   ;1-byte potentiometer 4
 = 0275     PADDL5  EQU     $0275   ;1-byte potentiometer 5
 = 0276     PADDL6  EQU     $0276   ;1-byte potentiometer 6
 = 0277     PADDL7  EQU     $0277   ;1-byte potentiometer 7

 = 0278     STICK0  EQU     $0278   ;1-byte joystick 0
 = 0279     STICK1  EQU     $0279   ;1-byte joystick 1
 = 027A     STICK2  EQU     $027A   ;1-byte joystick 2
 = 027B     STICK3  EQU     $027B   ;1-byte joystick 3

 = 027C     PTRIG0  EQU     $027C   ;1-byte paddle trigger 0
 = 027D     PTRIG1  EQU     $027D   ;1-byte paddle trigger 1
 = 027E     PTRIG2  EQU     $027E   ;1-byte paddle trigger 2
 = 027F     PTRIG3  EQU     $027F   ;1-byte paddle trigger 3
 = 0280     PTRIG4  EQU     $0280   ;1-byte paddle trigger 4
 = 0281     PTRIG5  EQU     $0281   ;1-byte paddle trigger 5
 = 0282     PTRIG6  EQU     $0282   ;1-byte paddle trigger 6
 = 0283     PTRIG7  EQU     $0283   ;1-byte paddle trigger 7

 = 0284     STRIG0  EQU     $0284   ;1-byte joystick trigger 0
 = 0285     STRIG1  EQU     $0285   ;1-byte joystick trigger 1
 = 0286     STRIG2  EQU     $0286   ;1-byte joystick trigger 2
 = 0287     STRIG3  EQU     $0287   ;1-byte joystick trigger 3

 = 0288     HIBYTE  EQU     $0288   ;1-byte relocating loader high byte:
```

```
= 0289   WMODE    EQU   $0289   ;1-byte cassette WRITE mode ($80 = :
= 028A   BLIM     EQU   $028A   ;1-byte cassette buffer limit
= 028B   IMASK    EQU   $028B   ;1-byte (not used)
= 028C   JVECK    EQU   $028C   ;2-byte jump vector or temporary
= 028E   NEWADR   EQU   $028E   ;2-byte relocating address
= 0290   TXTROW   EQU   $0290   ;1-byte split screen text cursor ro:
= 0291   TXTCOL   EQU   $0291   ;2-byte split screen text cursor co:
= 0293   TINDEX   EQU   $0293   ;1-byte split scree text mode
= 0294   TXTMSC   EQU   $0294   ;2-byte split screen memory scan co:
= 0296   TXTOLD   EQU   $0296   ;6-byte OLDROW, OLDCOL, OLDCHR, OLD:
= 029C   CRETRY   EQU   $029C   ;1-byte number of command frame ret:
= 029D   HOLD3    EQU   $029D   ;1-byte temporary
= 029E   SUBTMP   EQU   $029E   ;1-byte temporary
= 029F   HOLD2    EQU   $029F   ;1-byte (not used)
= 02A0   DMASK    EQU   $02A0   ;1-byte display (pixel location) ma:
= 02A1   TMPLBT   EQU   $02A1   ;1-byte (not used)                    prev flag.
= 02A2   ESCFLG   EQU   $02A2   ;1-byte escape flag ($80 = ESC dete:
= 02A3   TABMAP   EQU   $02A3   ;15-byte (120-bit) tab stop bit map     19 bytes
= 02B2   LOGMAP   EQU   $02B2   ;8-byte (32-bit) logical line bit m:
= 02B6   INVFLG   EQU   $02B6   ;1-byte inverse video flag ($80 = i: (keyboard)
= 02B7   FILFLG   EQU   $02B7   ;1-byte right fill flag (0 = no fil:
= 02B8   TMPROW   EQU   $02B8   ;1-byte temporary row
= 02B9   TMPCOL   EQU   $02B9   ;2-byte temporary column
= 02BB   SCRFLG   EQU   $02BB   ;1-byte scroll occurence flag (0 = :
= 02BC   HOLD4    EQU   $02BC   ;1-byte temporary
= 02BD   DRETRY   EQU   $02BD   ;1-byte number of device retries
= 02BE   SHFLOK   EQU   $02BE   ;1-byte shift/control lock flags
= 02BF   BOTSCR   EQU   $02BF   ;1-byte screen bottom (24 = normal,:

= 02C0   PCOLR0   EQU   $02C0   ;1-byte player-missle 0 color/lumin:
= 02C1   PCOLR1   EQU   $02C1   ;1-byte player-missle 1 color/lumin:
= 02C2   PCOLR2   EQU   $02C2   ;1-byte player-missle 2 color/lumin:
= 02C3   PCOLR3   EQU   $02C3   ;1-byte player-missle 3 color/lumin:

= 02C4   COLOR0   EQU   $02C4   ;1-byte playfield 0 color/luminance
= 02C5   COLOR1   EQU   $02C5   ;1-byte playfield 1 color/luminance
= 02C6   COLOR2   EQU   $02C6   ;1-byte playfield 2 color/luminance
= 02C7   COLOR3   EQU   $02C7   ;1-byte playfield 3 color/luminance

= 02C8   COLOR4   EQU   $02C8   ;1-byte background color/luminance

= 02C9   PARMBL   EQU   $02C9   ;6-byte relocating loader parameter:
= 02C9   RUNADR   EQU   $02C9   ;2-byte run address
= 02CB   HIUSED   EQU   $02CB   ;2-byte highest non-zero page addre:
= 02CD   ZHIUSE   EQU   $02CD   ;2-byte highest zero page address

= 02CF   OLDPAR   EQU   $02CF   ;6-byte relocating loader parameter:
= 02CF   GBYTEA   EQU   $02CF   ;2-byte GET-BYTE routine address
= 02D1   LOADAD   EQU   $02D1   ;2-byte non-zero page load address
= 02D3   ZLOADA   EQU   $02D3   ;2-byte zero page load address

= 02D5   DSCTLN   EQU   $02D5   ;2-byte disk sector length
= 02D7   ACMISR   EQU   $02D7   ;2-byte ACMI interrupt service rout:
= 02D9   KRPDEL   EQU   $02D9   ;1-byte auto-repeat delay
= 02DA   KEYREP   EQU   $02DA   ;1-byte auto-repeat rate
= 02DB   NOCLIK   EQU   $02DB   ;1-byte key click disable
= 02DC   HELPFG   EQU   $02DC   ;1-byte HELP key flag (0 = no HELP)
```

```
= 02DD   DMASAV  EQU   $02DD   ;1-byte SDMCTL save/restore
= 02DE   PBPNT   EQU   $02DE   ;1-byte printer buffer pointer
= 02DF   PBUFSZ  EQU   $02DF   ;1-byte printer buffer size

         ;       EQU   $02E0   ;4 bytes reserved for DOS

= 02E4   RAMSIZ  EQU   $02E4   ;1-byte high RAM size
= 02E5   MEMTOP  EQU   $02E5   ;2-byte top of available user memor;
= 02E7   MEMLO   EQU   $02E7   ;2-byte bottom of available user me;
= 02E9   HNDLOD  EQU   $02E9   ;1-byte user load flag (0 = no hand;
= 02EA   DVSTAT  EQU   $02EA   ;4-byte device status buffer
= 02EE   CBAUDL  EQU   $02EE   ;1-byte low cassette baud rate
= 02EF   CBAUDH  EQU   $02EF   ;1-byte high cassette baud rate
= 02F0   CRSINH  EQU   $02F0   ;1-byte cursor inhibit (0 = cursor ;
= 02F1   KEYDEL  EQU   $02F1   ;1-byte key debounce delay timer
= 02F2   CH1     EQU   $02F2   ;1-byte prior keyboard character
= 02F3   CHACT   EQU   $02F3   ;1-byte CHACTL shadow
= 02F4   CHBAS   EQU   $02F4   ;1-byte CHBASE shadow

= 02F5   NEWROW  EQU   $02F5   ;1-byte draw destination row
= 02F6   NEWCOL  EQU   $02F6   ;2-byte draw destination column
= 02F8   ROWINC  EQU   $02F8   ;1-byte draw row increment
= 02F9   COLINC  EQU   $02F9   ;1-byte draw column increment

= 02FA   CHAR    EQU   $02FA   ;1-byte internal character
= 02FB   ATACHR  EQU   $02FB   ;1-byte ATASCII character or plot p;
= 02FC   CH      EQU   $02FC   ;1-byte keyboard code (buffer)
= 02FD   FILDAT  EQU   $02FD   ;1-byte right fill data
= 02FE   DSPFLG  EQU   $02FE   ;1-byte control character display f;
= 02FF   SSFLAG  EQU   $02FF   ;1-byte start/stop flag (0 = not st;


         **      Page Three Address Equates

= 0300   DCB     EQU   $0300   ;12-byte device control block
= 0300   DDEVIC  EQU   $0300   ;1-byte unit 1 bus ID
= 0301   DUNIT   EQU   $0301   ;1-byte unit number
= 0302   DCOMND  EQU   $0302   ;1-byte bus command
= 0303   DSTATS  EQU   $0303   ;1-byte command type/status return
= 0304   DBUFLO  EQU   $0304   ;1-byte low data buffer address
= 0305   DBUFHI  EQU   $0305   ;1-byte high data buffer address
= 0306   DTIMLO  EQU   $0306   ;1-byte timeout (seconds)
= 0307   DUNUSE  EQU   $0307   ;1-byte (not used)
= 0308   DBYTLO  EQU   $0308   ;1-byte low number of bytes to tran;
= 0309   DBYTHI  EQU   $0309   ;1-byte high number of bytes to tra;
= 030A   DAUX1   EQU   $030A   ;1-byte first command auxiliary
= 030B   DAUX2   EQU   $030B   ;1-byte second command auxiliary

= 030C   TIMER1  EQU   $030C   ;2-byte initial baud rate timer val;
= 030E   JMPERS  EQU   $030E   ;1-byte jumper options
= 030F   CASFLG  EQU   $030F   ;1-byte cassette I/O flag (0 = not ;
= 0310   TIMER2  EQU   $0310   ;2-byte final baud rate timer value
= 0312   TEMP1   EQU   $0312   ;2-byte temporary
= 0313   TEMP2   EQU   $0313   ;1-byte temporary
```

```
        = 0314     PTIMOT  EQU     $0314    ;1-byte printer timeout
        = 0315     TEMP3   EQU     $0315    ;1-byte temporary
        = 0316     SAVIO   EQU     $0316    ;1-byte saved serial data input ind:
        = 0317     TIMFLG  EQU     $0317    ;1-byte timeout flag (0 = timeout)
        = 0318     STACKP  EQU     $0318    ;1-byte SIO saved stack pointer
        = 0319     TSTAT   EQU     $0319    ;1-byte temporary status

        = 031A     HATABS  EQU     $031A    ;35-byte handler address table

        = 033D     PUPBT1  EQU     $033D    ;1-byte power-up validation byte 1
        = 033E     PUPBT2  EQU     $033E    ;1-byte power-up validation byte 2
        = 033F     PUPBT3  EQU     $033F    ;1-byte power-up validation byte 3

        = 0340     IOCB    EQU     $0340    ;128-byte I/O control blocks area
        = 0340     ICHID   EQU     $0340    ;1-byte handler ID ($FF = free)
        = 0341     ICDNO   EQU     $0341    ;1-byte device number
        = 0342     ICCOM   EQU     $0342    ;1-byte command code
        = 0343     ICSTA   EQU     $0343    ;1-byte status of last action
        = 0344     ICBAL   EQU     $0344    ;1-byte low buffer address
        = 0345     ICBAH   EQU     $0345    ;1-byte high buffer address
        = 0346     ICPTL   EQU     $0346    ;1-byte low PUT-BYTE routine addres:
        = 0347     ICPTH   EQU     $0347    ;1-byte high PUT-BYTE routine addre:
        = 0348     ICBLL   EQU     $0348    ;1-byte low buffer length
        = 0349     ICBLH   EQU     $0349    ;1-byte high buffer length
        = 034A     ICAX1   EQU     $034A    ;1-byte first auxiliary information
        = 034B     ICAX2   EQU     $034B    ;1-byte second auxiliary informatio:
        = 034C     ICSPR   EQU     $034C    ;4-byte work area

        = 03C0     PRNBUF  EQU     $03C0    ;40-byte printer buffer
        = 03E8     SUPERF  EQU     $03E8    ;1-byte editor super function flag :
        = 03E9     CKEY    EQU     $03E9    ;1-byte cassette boot request flag :
        = 03EA     CASSBT  EQU     $03EA    ;1-byte cassette boot flag (0 = not:
        = 03EB     CARTCK  EQU     $03EB    ;1-byte cartridge equivalence check:
        = 03EC     DERRF   EQU     $03EC    ;1-byte screen OPEN error flag (0 =:

                   ;       Remainder of Page Three Not Cleared upon Reset

        = 03ED     ACMVAR  EQU     $03ED    ;11 bytes reserved for ACMI
        = 03F8     BASICF  EQU     $03F8    ;1-byte BASIC switch flag (0 = BASI:
        = 03F9     MINTLK  EQU     $03F9    ;1-byte ACMI module interlock
        = 03FA     GINTLK  EQU     $03FA    ;1-byte cartridge interlock
        = 03FB     CHLINK  EQU     $03FB    ;2-byte loaded handler chain link
        = 03FD     CASBUF  EQU     $03FD    ;3-byte first 3 bytes of cassette b:
```

```
                **      Page Four Address Equates


                ;       EQU     $0400   ;128-byte remainder of cassette buf:

                ;       Reserved for Application

   = 0480       USAREA  EQU     $0480   ;128 bytes reserved for application



                **      Page Five Address Equates


                ;       Reserved for Application and Floating Point Package

                ;       EQU     $0500   ;256 bytes reserved for application:



                **      Floating Point Package Address Equates


   = 057E       LBPR1   EQU     $057E   ;1-byte LBUFF preamble
   = 057F       LBPR2   EQU     $057F   ;1-byte LBUFF preamble
   = 0580       LBUFF   EQU     $0580   ;128-byte line buffer

   = 05E0       PLYARG  EQU     $05E0   ;6-byte floating point polynomial a:
   = 05E6       FPSCR   EQU     $05E6   ;6-byte floating point temporary
   = 05EC       FPSCR1  EQU     $05EC   ;6-byte floating point temporary



                **      Page Six Address Equates


                ;       Reserved for Application

                ;       EQU     $0600   ;256 bytes reserved for application
```

```
                    **      LNBUG Address Equates

        = 0000      LNBUG   IF      LNBUG
                    LNBUG   ENDIF


                    **      Cartridge Address Equates

        = BFFA      CARTCS  EQU     $BFFA   ;2-byte cartridge coldstart address
        = BFFC      CART    EQU     $BFFC   ;1-byte cartridge present indicator
        = BFFD      CARTFG  EQU     $BFFD   ;1-byte cartridge flags
        = BFFE      CARTAD  EQU     $BFFE   ;2-byte cartridge start vector


                    **      CTIA/GTIA Address Equates

        = D000      CTIA    EQU     $D000   ;CTIA/GTIA area

                    ;       Read/Write Addresses

        = D01F      CONSOL  EQU     $D01F   ;console switches and speaker contr:

                    ;       Read Addresses

        = D000      M0PF    EQU     $D000   ;missle 0 and playfield collision
        = D001      M1PF    EQU     $D001   ;missle 1 and playfield collision
        = D002      M2PF    EQU     $D002   ;missle 2 and playfield collision
        = D003      M3PF    EQU     $D003   ;missle 3 and playfield collision

        = D004      P0PF    EQU     $D004   ;player 0 and playfield collision
        = D005      P1PF    EQU     $D005   ;player 1 and playfield collision
        = D006      P2PF    EQU     $D006   ;player 2 and playfield collision
        = D007      P3PF    EQU     $D007   ;player 3 and playfield collision

        = D008      M0PL    EQU     $D008   ;missle 0 and player collision
        = D009      M1PL    EQU     $D009   ;missle 1 and player collision
        = D00A      M2PL    EQU     $D00A   ;missle 2 and player collision
        = D00B      M3PL    EQU     $D00B   ;missle 3 and player collision

        = D00C      P0PL    EQU     $D00C   ;player 0 and player collision
        = D00D      P1PL    EQU     $D00D   ;player 1 and player collision
        = D00E      P2PL    EQU     $D00E   ;player 2 and player collision
        = D00F      P3PL    EQU     $D00F   ;player 3 and player collision

        = D010      TRIG0   EQU     $D010   ;joystick trigger 0
        = D011      TRIG1   EQU     $D011   ;joystick trigger 1

        = D012      TRIG2   EQU     $D012   ;cartridge interlock
        = D013      TRIG3   EQU     $D013   ;ACMI module interlock
```

```
  = D014        PAL      EQU     $D014    ;PAL/NTSC indicator

                ;        Write Addresses

  = D000        HPOSP0   EQU     $D000    ;player 0 horizontal position
  = D001        HPOSP1   EQU     $D001    ;player 1 horizontal position
  = D002        HPOSP2   EQU     $D002    ;player 2 horizontal position
  = D003        HPOSP3   EQU     $D003    ;player 3 horizontal position

  = D004        HPOSM0   EQU     $D004    ;missle 0 horizontal position
  = D005        HPOSM1   EQU     $D005    ;missle 1 horizontal position
  = D006        HPOSM2   EQU     $D006    ;missle 2 horizontal position
  = D007        HPOSM3   EQU     $D007    ;missle 3 horizontal position

  = D008        SIZEP0   EQU     $D008    ;player 0 size
  = D009        SIZEP1   EQU     $D009    ;player 1 size
  = D00A        SIZEP2   EQU     $D00A    ;player 2 size
  = D00B        SIZEP3   EQU     $D00B    ;player 3 size

  = D00C        SIZEM    EQU     $D00C    ;missle sizes

  = D00D        GRAFP0   EQU     $D00D    ;player 0 graphics
  = D00E        GRAFP1   EQU     $D00E    ;player 1 graphics
  = D00F        GRAFP2   EQU     $D00F    ;player 2 graphics
  = D010        GRAFP3   EQU     $D010    ;player 3 graphics

  = D011        GRAFM    EQU     $D011    ;missle graphics

  = D012        COLPM0   EQU     $D012    ;player-missle 0 color/luminance
  = D013        COLPM1   EQU     $D013    ;player-missle 1 color/luminance
  = D014        COLPM2   EQU     $D014    ;player-missle 2 color/luminance
  = D015        COLPM3   EQU     $D015    ;player-missle 3 color/luminance

  = D016        COLPF0   EQU     $D016    ;playfield 0 color/luminance
  = D017        COLPF1   EQU     $D017    ;playfield 1 color/luminance
  = D018        COLPF2   EQU     $D018    ;playfield 2 color/luminance
  = D019        COLPF3   EQU     $D019    ;playfield 3 color/luminance

  = D01A        COLBK    EQU     $D01A    ;background color/luminance

  = D01B        PRIOR    EQU     $D01B    ;priority select
  = D01C        VDELAY   EQU     $D01C    ;vertical delay
  = D01D        GRACTL   EQU     $D01D    ;graphic control
  = D01E        HITCLR   EQU     $D01E    ;collision clear
```

```
                    **      PBI Address Equates

    = D100          PBI     EQU     $D100   ;parallel bus interface area

                    ;       Read Addresses

    = D1FF          PDVI    EQU     $D1FF   ;parallel device IRQ status

                    ;       Write Addresses

    = D1FF          PDVS    EQU     $D1FF   ;parallel device select


                    **      POKEY Address Equates

    = D200          POKEY   EQU     $D200   ;POKEY area

                    ;       Read Addresses

    = D200          POT0    EQU     $D200   ;potentiometer 0
    = D201          POT1    EQU     $D201   ;potentiometer 1
    = D202          POT2    EQU     $D202   ;potentiometer 2
    = D203          POT3    EQU     $D203   ;potentiometer 3
    = D204          POT4    EQU     $D204   ;potentiometer 4
    = D205          POT5    EQU     $D205   ;potentiometer 5
    = D206          POT6    EQU     $D206   ;potentiometer 6
    = D207          POT7    EQU     $D207   ;potentiometer 7

    = D208          ALLPOT  EQU     $D208   ;potentiometer port state
    = D209          KBCODE  EQU     $D209   ;keyboard code
    = D20A          RANDOM  EQU     $D20A   ;random number generator
    = D20D          SERIN   EQU     $D20D   ;serial port input
    = D20E          IRQST   EQU     $D20E   ;IRQ interrupt status
    = D20F          SKSTAT  EQU     $D20F   ;serial port and keyboard status

                    ;       Write Addresses

    = D200          AUDF1   EQU     $D200   ;channel 1 audio frequency
    = D201          AUDC1   EQU     $D201   ;channel 1 audio control

    = D202          AUDF2   EQU     $D202   ;channel 2 audio frequency
    = D203          AUDC2   EQU     $D203   ;channel 2 audio control

    = D204          AUDF3   EQU     $D204   ;channel 3 audio frequency
    = D205          AUDC3   EQU     $D205   ;channel 3 audio control

    = D206          AUDF4   EQU     $D206   ;channel 4 audio frequency
    = D207          AUDC4   EQU     $D207   ;channel 4 audio control

    = D208          AUDCTL  EQU     $D208   ;audio control
    = D209          STIMER  EQU     $D209   ;start timers
    = D20A          SKRES   EQU     $D20A   ;reset SKSTAT status
    = D20B          POTGO   EQU     $D20B   ;start potentiometer scan sequence
```

```
  = D20D        SEROUT  EQU     $D20D   ;serial port output
  = D20E        IRQEN   EQU     $D20E   ;IRQ interrupt enable
  = D20F        SKCTL   EQU     $D20F   ;serial port and keyboard control



                **      PIA Address Equates


  = D300        PIA     EQU     $D300   ;PIA area

                ;       Read/Write Addresses

  = D300        PORTA   EQU     $D300   ;port A direction register or jacks:
  = D301        PORTB   EQU     $D301   ;port B direction register or memor:

  = D302        PACTL   EQU     $D302   ;port A control
  = D303        PBCTL   EQU     $D303   ;port B control



                **      ANTIC Address Equates


  = D400        ANTIC   EQU     $D400   ;ANTIC area

                ;       Read Addresses

  = D40B        VCOUNT  EQU     $D40B   ;vertical line counter
  = D40C        PENH    EQU     $D40C   ;light pen horizontal position
  = D40D        PENV    EQU     $D40D   ;light pen vertical position
  = D40F        NMIST   EQU     $D40F   ;NMI interrupt status

                ;       Write Addresses

  = D400        DMACTL  EQU     $D400   ;DMA control
  = D401        CHACTL  EQU     $D401   ;character control
  = D402        DLISTL  EQU     $D402   ;low display list address
  = D403        DLISTH  EQU     $D403   ;high disply list address
  = D404        HSCROL  EQU     $D404   ;horizontal scroll
  = D405        VSCROL  EQU     $D405   ;vertical scroll
  = D407        PMBASE  EQU     $D407   ;player-missle base address
  = D409        CHBASE  EQU     $D409   ;character base address
  = D40A        WSYNC   EQU     $D40A   ;wait for HBLANK synchronization
  = D40E        NMIEN   EQU     $D40E   ;NMI enable
  = D40F        NMIRES  EQU     $D40F   ;NMI interrupt status reset
```

```
                  **      PBI RAM Address Equates


       = D600     PBIRAM  EQU     $D600    ;parallel bus interface RAM area



                  **      ACMI Address Equates


       = 0000     ACMI    IF      ACMI
                  ACMI    ENDIF



                  **      Floating Point Package Address Equates

       = D800     AFP     EQU     $D800    ;convert ASCII to floating point
       = D8E6     FASC    EQU     $D8E6    ;convert floating point to ASCII
       = D9AA     IFP     EQU     $D9AA    ;convert integer to floating point
       = D9D2     FPI     EQU     $D9D2    ;convert floating point to integer
       = DA44     ZFR0    EQU     $DA44    ;zero FR0
       = DA46     ZF1     EQU     $DA46    ;zero floating point number
       = DA60     FSUB    EQU     $DA60    ;subtract floating point numbers
       = DA66     FADD    EQU     $DA66    ;add floating point numbers
       = DADB     FMUL    EQU     $DADB    ;multiply floating point numbers
       = DB28     FDIV    EQU     $DB28    ;divide floating point numbers
       = DD40     PLYEVL  EQU     $DD40    ;evaluate floating point polynomial
       = DD89     FLD0R   EQU     $DD89    ;load floating point number
       = DD8D     FLD0P   EQU     $DD8D    ;load floating point number
       = DD98     FLD1R   EQU     $DD98    ;load floating point number
       = DD9C     FLD1P   EQU     $DD9C    ;load floating point number
       = DDA7     FST0R   EQU     $DDA7    ;store floating point number
       = DDAB     FST0P   EQU     $DDAB    ;store floating point number
       = DDB6     FMOVE   EQU     $DDB6    ;move floating point number
       = DECD     LOG     EQU     $DECD    ;calculate floating point logarithm
       = DED1     LOG10   EQU     $DED1    ;calculate floating point base 10 l:
       = DDC0     EXP     EQU     $DDC0    ;calculate floating point exponenti:
       = DDCC     EXP10   EQU     $DDCC    ;calculate floating point base 10 e:
```

```
                  **      Parallel Device Address Equates

      = D803      PDID1    EQU      $D803    ;parallel device ID 1
      = D805      PDIOV    EQU      $D805    ;parallel device I/O vector
      = D808      PDIRQV   EQU      $D808    ;parallel device IRQ vector
      = D80B      PDID2    EQU      $D80B    ;parallel device ID 2
      = D80D      PDVV     EQU      $D80D    ;parallel device vector table


                  **      Device Handler Vector Table Address Equates

      = E400      EDITRV   EQU      $E400    ;editor handler vector table
      = E410      SCRENV   EQU      $E410    ;screen handler vector table
      = E420      KEYBDV   EQU      $E420    ;keyboard handler vector table
      = E430      PRINTV   EQU      $E430    ;printer handler vector table
      = E440      CASETV   EQU      $E440    ;cassette handler vector table


                  **      Jump Vector Address Equates

      = E450      DINITV   EQU      $E450    ;vector to initialize DIO
      = E453      DSKINV   EQU      $E453    ;vector to DIO
      = E456      CIOV     EQU      $E456    ;vector to CIO
      = E459      SIOV     EQU      $E459    ;vector to SIO
      = E45C      SETVBV   EQU      $E45C    ;vector to set VBLANK parameters
      = E45F      SYSVBV   EQU      $E45F    ;vector to process immediate VBLANK:
      = E462      XITVBV   EQU      $E462    ;vector to process deferred VBLANK :
      = E465      SIOINV   EQU      $E465    ;vector to initialize SIO
      = E468      SENDEV   EQU      $E468    ;vector to enable SEND
      = E46B      INTINV   EQU      $E46B    ;vector to initialize interrupt han:
      = E46E      CIOINV   EQU      $E46E    ;vector to initialize CIO
      = E471      BLKBDV   EQU      $E471    ;vector to power-up display (former:
      = E474      WARMSV   EQU      $E474    ;vector to warmstart
      = E477      COLDSV   EQU      $E477    ;vector to coldstart
      = E47A      RBLOKV   EQU      $E47A    ;vector to read cassette block
      = E47D      CSOPIV   EQU      $E47D    ;vector to open cassette for input
      = E480      PUPDIV   EQU      $E480    ;vector to power-up display
      = E483      SLFTSV   EQU      $E483    ;vector to self-test
      = E486      PHENTV   EQU      $E486    ;vector to enter peripheral handler
      = E489      PHUNLV   EQU      $E489    ;vector to unlink peripheral handle:
      = E48C      PHINIV   EQU      $E48C    ;vector to initialize peripheral ha:
```

```
                    **      Generic Parallel Device Handler Vector Table Addres:

       = E48F       GPDVV   EQU     $E48F   ;generic parallel device handler ve:
```

```
0000                    **      Self-test Page Zero Address Equates

        = 0080          STTIME  EQU     $0080   ;2-byte main screen timeout timer
        = 0082          STAUT   EQU     $0082   ;1-byte auto-mode flag
        = 0083          STJMP   EQU     $0083   ;3-byte ANTIC jump instruction
        = 0086          STSEL   EQU     $0086   ;1-byte selection
        = 0087          STPASS  EQU     $0087   ;1-byte pass
        = 0088          STSPP   EQU     $0088   ;1-byte SELECT previously pressed f:
                        ;       EQU     $0089   ;1-byte (not used)
        = 008A          STKST   EQU     $008A   ;1-byte keyboard self-test flag (0 :
        = 008B          STCHK   EQU     $008B   ;2-byte checksum
        = 008D          STSMM   EQU     $008D   ;1-byte screen memory mask
        = 008E          STSMP   EQU     $008E   ;1-byte screen memory pointer
        = 008F          ST1K    EQU     $008F   ;1-byte current 1K of memory to tes:
        = 0090          STPAG   EQU     $0090   ;2-byte current page to test
        = 0092          STPC    EQU     $0092   ;1-byte page count
        = 0093          STMVAL  EQU     $0093   ;1-byte correct value for memory te:
        = 0094          STSKP   EQU     $0094   ;1-byte simulated keypress index
        = 0095          STTMP1  EQU     $0095   ;2-byte temporary
        = 0097          STVOC   EQU     $0097   ;1-byte current voice indicator
        = 0098          STNOT   EQU     $0098   ;1-byte current note.counter
        = 0099          STCDI   EQU     $0099   ;1-byte cleft display pointer
        = 009A          STCDA   EQU     $009A   ;1-byte cleft data pointer
        = 009B          STTMP2  EQU     $009B   ;2-byte temporary
        = 009D          STTMP3  EQU     $009D   ;1-byte temporary
        = 009E          STADR1  EQU     $009E   ;2-byte temporary address
        = 00A0          STADR2  EQU     $00A0   ;2-byte temporary address
        = 00A2          STBL    EQU     $00A2   ;1-byte blink counter
        = 00A3          STTMP4  EQU     $00A3   ;1-byte temporary
        = 00A4          STLM    EQU     $00A4   ;1-byte LED mask
        = 00A5          STTMP5  EQU     $00A5   ;1-byte temporary


                        **      Self-test Address Equates

        = 3000          ST3000  EQU     $3000   ;screen memory
        = 3002          ST3002  EQU     $3002   ;cleft display
        = 3004          ST3004  EQU     $3004   ;"VOICE #" text display
        = 300B          ST300B  EQU     $300B   ;voice number display
        = 301C          ST301C  EQU     $301C   ;START key display
        = 301E          ST301E  EQU     $301E   ;SELECT key display
        = 3020          ST3020  EQU     $3020   ;OPTION key display, first 8K ROM d:
        = 3021          ST3021  EQU     $3021   ;keyboard character display
        = 3022          ST3022  EQU     $3022   ;keyboard text display
        = 3024          ST3024  EQU     $3024   ;second 8K ROM display
        = 3028          ST3028  EQU     $3028   ;"RAM" text display
        = 3038          ST3038  EQU     $3038   ;RAM display
        = 303C          ST303C  EQU     $303C   ;fifth note display
        = 304C          ST304C  EQU     $304C   ;"B S" text display
        = 3052          ST3052  EQU     $3052   ;tab key display
        = 3062          ST3062  EQU     $3062   ;cleft display
        = 306D          ST306D  EQU     $306D   ;return key display
        = 3072          ST3072  EQU     $3072   ;control key display
```

Miscellaneous Address Equates

```
= 3092      ST3092  EQU     $3092   ;"SH" text display
= 309E      ST309E  EQU     $309E   ;sixth note display
= 30AB      ST30AB  EQU     $30AB   ;"SH" text display
= 30B7      ST30B7  EQU     $30B7   ;"S P A C E   B A R" text display
= 30C1      ST30C1  EQU     $30C1   ;cleft display
= 30C2      ST30C2  EQU     $30C2   ;cleft display
= 30C7      ST30C7  EQU     $30C7   ;third note display
= 30CA      ST30CA  EQU     $30CA   ;fourth note display
= 30F8      ST30F8  EQU     $30F8   ;third note display
= 3100      ST3100  EQU     $3100   ;screen memory
= 3121      ST3121  EQU     $3121   ;cleft display
= 3122      ST3122  EQU     $3122   ;cleft display
= 313C      ST313C  EQU     $313C   ;fifth note display
= 3150      ST3150  EQU     $3150   ;first line of staff display
= 3154      ST3154  EQU     $3154   ;first note display
= 3181      ST3181  EQU     $3181   ;cleft display
= 3182      ST3182  EQU     $3182   ;cleft display
= 3186      ST3186  EQU     $3186   ;second note display
= 318C      ST318C  EQU     $318C   ;fifth note display
= 31B0      ST31B0  EQU     $31B0   ;second line of staff display
= 31C2      ST31C2  EQU     $31C2   ;cleft display
= 31CA      ST31CA  EQU     $31CA   ;fourth note display
= 31EE      ST31EE  EQU     $31EE   ;sixth note display
= 31F1      ST31F1  EQU     $31F1   ;cleft display
= 3210      ST3210  EQU     $3210   ;third line of staff display
= 321A      ST321A  EQU     $321A   ;fourth note display
= 3248      ST3248  EQU     $3248   ;third note display
= 3270      ST3270  EQU     $3270   ;fourth line of staff display
= 32D0      ST32D0  EQU     $32D0   ;fifth line of staff display
```

```
0000            **      FIX - Fix Address
                *
                *       FIX sets the origin counter to the value specified :
                *       argument.  If the current origin counter is less th:
                *       argument, FIX fills the intervening bytes with zero:
                *       issues a message to document the location and numbe:
                *       bytes that are zero filled.
                *
                *       ENTRY   FIX     address
                *
                *
                *       EXIT
                *               Origin counter set to specified address.
                *               Message issued if zero fill required.
                *
                *       CHANGES
                *               -none-
                *
                *       CALLS
                *               -none-
                *
                *       NOTES
                *               Due to ECHO limitiation of 255 iterations, :
                *               recursive.
                *               If the current origin counter value is beyo:
                *               argument, FIX generates an error.
                *
                *       MODS
                *               R. K. Nordin    11/01/83

                FIX     MACRO   address
                        IF      %1 <> *0
                        IF      %1 > *0
                        IF      %1 - *0 < 256
                        MSG     '$',%1 - *0,' free bytes from $',*0,' to $':
                        ECHO    %1 - *0
                        DB      0
                        ENDM
                        ELSE
                        FIX     *0 + 255
                        FIX     %1
                        ENDIF
                        ELSE
                        ERR             '%1 precedes current origin counter:
                        ENDIF
                        ENDIF
                        ORG     *0
                        ENDM
```

```
 `00  = C000            ORG     $C000


                **      First 8K ROM Identification and Checksum

C000  0000              DW      $0000                           ;reserved f:
C002  100583            DB      IDDAY,IDMON,IDYEAR              ;date (day,:
C005  00                DB      $00                             ;not used
C006  4242000001        DB      IDPN1,IDPN2,IDPN3,IDPN4,IDPN5   ;part numbe:
C008  02                DB      IDREV                           ;revision n:
```

```
C00C                 **      IIH - Initialize Interrupt Handler
                     *
                     *       ENTRY   JSR     IIH
                     *               TRIG3 = ACMI module interlock
                     *               TRIG2 = cartridge interlock
                     *
                     *       MODS
                     *               Original Author Unknown
                     *               1. Bring closer to Coding Standard (object :
                     *                  R. K. Nordin 11/01/83

       = C00C        IIH     =       *       ;entry

C00C  A940                   LDA     #$40
C00E  8D0ED4                 STA     NMIEN   ;disable DLI and enable VBLANK NMI

C011  AD13D0                 LDA     TRIG3   ;cartridge interlock
C014  8DFA03                 STA     GINTLK  ;cartridge interlock status

      = 0000         ACMI    IF      ACMI
                     ACMI    ENDIF

C017  60                     RTS             ;return


                     **      NMI - Process NMI
                     *
                     *       ENTRY   JMP     NMI
                     *
                     *       EXIT
                     *               Exits via appropriate vector to process NMI
                     *
                     *       MODS
                     *               Original Author Unknown
                     *               1. Bring closer to Coding Standard (object :
                     *                  R. K. Nordin 11/01/83

      = C018         NMI     =       *               ;entry

                             ASSERT  $C0=high NMI    ;for compatibility with LNB:

                     ;       Check for display list NMI.

C018  2C0FD4                 BIT     NMIST
C01B  1003 ^C020             BPL     NMI1            ;if not display list NMI

C01D  6C0002                 JMP     (VDSLST)        ;process display list NMI, :

                     ;       Initialize.

C020  D8             NMI1    CLD

                     ;       Save registers.
```

```
  21   48                 PHA                             ;save A
C022   8A                 TXA
C023   48                 PHA                             ;save X
C024   98                 TYA
C025   48                 PHA                             ;save Y

                ;         Reset NMI status.

C026   8D0FD4             STA      NMIRES                 ;reset NMI status

                ;         Process NMI.

       = 0000   LNBUG     IF       LNBUG
                LNBUG     ELSE
C029   6C2202             JMP      (VVBLKI)               ;process immediate VBLANK N:
                LNBUG     ENDIF


                **        IRQ - Process IRQ
                *
                *         ENTRY    JMP      IRQ
                *
                *         EXIT
                *                  Exits via VIMIRQ vector
                *
                *         MODS
                *                  Original Author Unknown
                *                  1. Bring closer to Coding Standard (object :
                *                     R. K. Nordin-11/01/83

       = C02C   IRQ       =        *                      ;entry

                ;         Initialize.

C02C   D8                 CLD

                ;         Process IRQ.

       = 0000   LNBUG     IF       LNBUG
                LNBUG     ELSE
C02D   6C1602             JMP      (VIMIRQ)               ;process immediate IRQ, ret:
                LNBUG     ENDIF
```

```
                        **      IIR - Process Immediate IRQ
                        *
                        *       ENTRY   JMP     IIR
                        *
                        *       EXIT
                        *               Exits via appropriate vector to process IRQ
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

         = C030         IIR     =       *               ;entry

                        ;       Initialize.

 C030  48                       PHA                     ;save A

                        ;       Check for serial input ready IRQ.

 C031  AD0ED2                   LDA     IRQST           ;IRQ status
 C034  2920                     AND     #$20            ;serial input ready
 C036  D00D ^C045               BNE     IIR1            ;if not serial input ready

                        ;       Process serial input IRQ.

 C038  A9DF                     LDA     #not $20        ;all other interrupts
 C03A  8D0ED2                   STA     IRQEN           ;enable all other interrupt:
 C03D  A510                     LDA     POKMSK
 C03F  8D0ED2                   STA     IRQEN
 C042  6C0A02                   JMP     (VSERIN)        ;process serial input ready:

                        ;       Process possible ACMI IRQ.

 C045                   IIR1
         = 0000         ACMI    IF      ACMI
                        ACMI    ENDIF

                        ;       Initialize further.

 C045  8A                       TXA
 C046  48                       PHA                     ;save X

                        ;       Check for parallel device IRQ.

 C047  ADFFD1                   LDA     PDVI            ;parallel device IRQ status:
 C04A  2D4902                   AND     PDIMSK          ;select desired IRQ statuse:
 C04D  F003 ^C052               BEQ     IIR2            ;if no desired IRQ

                        ;       Process parallel device IRQ.

 C04F  6C3802                   JMP     (VPIRQ)         ;process parallel device IR:

                        ;       Check other types of IRQ.

 C052  A206             IIR2    LDX     #TIRQL-1-1      ;offset to next to last ent:
```

```
C054  BDCFC0       IIR3   LDA   TIRQ,X          ;IRQ type
C057  E005                CPX   #5              ;offset to serial out compl:
C059  D004 ^C05F          BNE   IIR4            ;if not serial out complete

C05B  2510                AND   POKMSK          ;and with POKEY IRQ enable
C05D  F005 ^C064          BEQ   IIR5            ;if serial out complete not:

C05F  2C0ED2       IIR4   BIT   IRQST           ;IRQ interrupt status
C062  F006 ^C06A          BEQ   IIR6            ;if interrupt found

C064  CA           IIR5   DEX
C065  10ED ^C054          BPL   IIR3            ;if not done

                   ;      Coninue IRQ processing.

C067  4CA0C0              JMP   CIR             ;continue IRQ processing, r:

                   ;      Enable other interrupts.

C06A  49FF         IIR6   EOR   #$FF            ;complement mask
C06C  8D0ED2              STA   IRQEN           ;enable all others
C06F  A510                LDA   POKMSK          ;POKEY IRQ mask
C071  8D0ED2              STA   IRQEN           ;enable indicated IRQ's

                   ;      Check for BREAK key IRQ.

C074  E000                CPX   #0
C076  D005 ^C07D          BNE   IIR7            ;if not BREAK key IRQ

                   ;      Check for keyboard disabled.

C078  AD6D02              LDA   KEYDIS
C07B  D023 ^C0A0          BNE   CIR             ;if keyboard disabled, cont:

                   ;      Process IRQ.

C07D  BDD7C0       IIR7   LDA   TOIH,X          ;offset to interrupt handle:
C080  AA                  TAX
C081  BD0002              LDA   INTABS,X        ;interrupt handler address
C084  8D8C02              STA   JVECK
C087  BD0102              LDA   INTABS+1,X
C08A  8D8D02              STA   JVECK+1
C08D  68                  PLA
C08E  AA                  TAX                   ;restore X
C08F  6C8C02              JMP   (JVECK)         ;process interrupt, return
```

```
                        **      BIR - Process BREAK Key IRQ
                        *
                        *       ENTRY   JMP     BIR
                        *
                        *       EXIT
                        *               Exits via RTI
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C092          BIR     =       *               ;entry

                        ;       Process BREAK.

 C092   A900                    LDA     #0
 C094   8511                    STA     BRKKEY  ;clear BREAK key flag
 C096   8DFF02                  STA     SSFLAG  ;clear start/stop flag
 C099   8DF002                  STA     CRSINH  ;enable cursor
 C09C   854D                    STA     ATRACT  ;turn off attract-mode

                        ;       Exit.

 C09E   68              BIR1    PLA             ;restore A
 C09F   40                      RTI             ;return



                        **      CIR - Continue IRQ Processing
                        *
                        *       ENTRY   JMP     CIR
                        *
                        *       EXIT
                        *               Exits via appropriate vector to process IRQ:
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C0A0          CIR     =       *                       ;entry

                        ;       Initialize.

 C0A0   68                      PLA
 C0A1   AA                      TAX

                        ;       Check for port A interrupt.

 C0A2   2C02D3                  BIT     PACTL           ;port A control
 C0A5   1006 ^C0AD              BPL     CIR1            ;if not port A interrupt

                        ;       Process proceed line IRQ.
```

```
C0A7  AD00D3                LDA    PORTA           ;clear interrupt status bit
C0AA  6C0202                JMP    (VPRCED)        ;process proceed line IRQ, :

                     ;      Check for port B interrupt.

C0AD  2C03D3        CIR1    BIT    PBCTL           ;port B control
C0B0  1006 ^C0B8            BPL    CIR2            ;if not port B interrupt

                     ;      Process interrupt line IRQ.

C0B2  AD01D3                LDA    PORTB           ;clear interrupt status bit
C0B5  6C0402                JMP    (VINTER)        ;process interrupt line IRQ:

                     ;      Check for BRK instruction IRQ.

C0B8  68            CIR2    PLA
C0B9  8D8C02                STA    JVECK

C0BC  68                    PLA                    ;saved P
C0BD  48                    PHA                    ;resave P
C0BE  2910                  AND    #$10            ;B bit of P register
C0C0  F007 ^C0C9            BEQ    CIR3            ;if not BRK instruction IRQ

                     ;      Process BRK instruction IRQ.

C0C2  AD8C02                LDA    JVECK
C0C5  48                    PHA
C0C6  6C0602                JMP    (VBREAK)        ;process BRK instruction IR:

                     ;      Exit IRQ processing.

C0C9  AD8C02        CIR3    LDA    JVECK
C0CC  48                    PHA
                     ;      JMP    XIR             ;exit IRQ processing, retur:


                     **     XIR - Exit IRQ Processing
                     *
                     *      ENTRY  JMP    XIR
                     *
                     *      EXIT
                     *             Exits to RIR
                     *
                     *      MODS
                     *             Original Author Unknown
                     *             1. Bring closer to Coding Standard (object :
                     *                R. K. Nordin 11/01/83


      = C0CD        XIR     =      *      ;entry
C0CD  68                    PLA           ;restore A
                     ;      JMP    RIR    ;return from interrupt
```

```
                        **      RIR - Return from Interrupt
                        *
                        *       ENTRY   JMP     RIR
                        *
                        *       EXIT
                        *               Exits via RTI
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C0CE          RIR     =       *               ;entry
C0CE    40                      RTI                     ;return


                        **      AIR - Process ACMI IRQ
                        *
                        *       ENTRY   JSR     AIR
                        *
                        *       EXIT    Exits via ACMISR vector
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = 0000          ACMI    IF      -ACMI
                        ACMI    ENDIF


                        **      TIRQ - Table of IRQ Types
                        *
                        *       Entry n is the interrupt indicator of priority n (0:
                        *
                        *       NOTES
                        *               Problem: entry 7 (serial input ready) not u:

C0CF    80              TIRQ    DB      $80     ;0 - BREAK key IRQ
C0D0    40                      DB      $40     ;1 - keyboard IRQ
C0D1    04                      DB      $04     ;2 - timer 4 IRQ
C0D2    02                      DB      $02     ;3 - timer 2 IRQ
C0D3    01                      DB      $01     ;4 - timer 1 IRQ
C0D4    08                      DB      $08     ;5 - serial output complete IRQ
C0D5    10                      DB      $10     ;6 - serial output ready IRQ
C0D6    20                      DB      $20     ;7 - serial input ready IRQ

        = 0008          TIRQL   =       *-TIRQ  ;length
```

```
                    **      TOIH - Table of Offsets to Interrupt Handlers
                    *
                    *       Entry n is the offset to the interrupt handler vect:
                    *       corresponding to entry n of TIRQ.
                    *
                    *       NOTES
                    *               Problem: entry 7 (serial input ready) not u:


C0D7  36            TOIH    DB      BRKKY-INTABS    ;0 - BREAK key IRQ
C0D8  08                    DB      VKEYBD-INTABS   ;1 - keyboard IRQ
C0D9  14                    DB      VTIMR4-INTABS   ;2 - timer 4 IRQ
C0DA  12                    DB      VTIMR2-INTABS   ;3 - timer 2 IRQ
C0DB  10                    DB      VTIMR1-INTABS   ;4 - timer 1 IRQ
C0DC  0E                    DB      VSEROC-INTABS   ;5 - serial output complete:
C0DD  0C                    DB      VSEROR-INTABS   ;6 - serial output ready IR:
C0DE  0A                    DB      VSERIN-INTABS   ;7 - serial input ready IRQ


                    **      WFR - Wait for RESET
                    *
                    *       WFR loops forever.
                    *
                    *       ENTRY   JMP     WFR
                    *
                    *       EXIT
                    *               Does not exit
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = C0DF        WFR     =       *       ;entry

                    ;       Loop forever, waiting for RESET.

C0DF  4CDFC0        WFR1    JMP     WFR1    ;loop


                    **      IVNM - Process Immediate VBLANK NMI
                    *
                    *       ENTRY   JMP     IVNM
                    *
                    *       EXIT
                    *               Exits to DVNM or via VVBLKD vector
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83
```

```
        = C0E2          IVNM    =       *               ;entry

                        ;       Increment frame counter and attract-mode counter.

 C0E2   E614                    INC     RTCLOK+2        ;increment low frame counte:
 C0E4   D008 ^C0EE              BNE     IVN1            ;if low counter not zero

 C0E6   E64D                    INC     ATRACT          ;increment attract-mode cou:
 C0E8   E613                    INC     RTCLOK+1        ;increment middle frame cou:
 C0EA   D002 ^C0EE              BNE     IVN1            ;if middle counter not zero

 C0EC   E612                    INC     RTCLOK          ;increment high frame count:

                        ;       Set attract-mode effects.

 C0EE   A9FE            IVN1    LDA     #$FE            ;select no luminance change
 C0F0   A200                    LDX     #0              ;select no color shift
 C0F2   A44D                    LDY     ATRACT          ;attract-mode timer/flag
 C0F4   1006 ^C0FC              BPL     IVN2            ;if not attract-mode

 C0F6   854D                    STA     ATRACT          ;ensure continued attract-m:
 C0F8   A613                    LDX     RTCLOK+1        ;select color shift
 C0FA   A9F6                    LDA     #$F6            ;select lower luminance

 C0FC   854E            IVN2    STA     DRKMSK          ;attract-mode luminance
 C0FE   864F                    STX     COLRSH          ;attract-mode color shift

                        ;       Update COLPF1 (in case fine scrolling and critical :

 C100   ADC502                  LDA     COLOR1          ;playfield 1 color
 C103   454F                    EOR     COLRSH          ;modify color with attract-:
 C105   254E                    AND     DRKMSK          ;modify with attract-mode l:
 C107   8D17D0                  STA     COLPF1          ;set playfield 1 color/lumi:

                        ;       Process countdown timer 1.

 C10A   A200                    LDX     #0              ;indicate countdown timer 1
 C10C   2055C2                  JSR     DCT             ;decrement countdown timer
 C10F   D003 ^C114              BNE     IVN3            ;if timer not expired

 C111   204FC2                  JSR     PTO             ;process countdown timer 1 :

                        ;       Check for critical sction.

 C114   A542            IVN3    LDA     CRITIC
 C116   D008 ^C120              BNE     IVN4            ;if critical section

                        ;       Check for IRQ enabled.

 C118   BA                      TSX                     ;stack pointer
 C119   BD0401                  LDA     $0104,X         ;stacked P
 C11C   2904                    AND     #$04            ;I (IRQ disable) bit
 C11E   F003 ^C123              BEQ     IVN5            ;if IRQ enabled

                        ;       Exit.
```

```
C120  4C8AC2        IVN4   JMP    DVNM            ;process deferred VBLANK NM:

                    ;      Process IRQ enabled non-critical section.

C123                IVN5

                    ;      Check for ACMI module change.

      = 0000        ACMI   IF     ACMI
                    ACMI   ENDIF

                    ;      Check for cartridge change.

C123  AD13D0               LDA    TRIG3           ;cartridge interlock
C126  CDFA03               CMP    GINTLK          ;previous cartridge interlo:
C129  D0B4 ^C0DF           BNE    WFR             ;if cartridge change, wait :

                    ;      Set hardware registers from shadows.

C12B  AD0DD4               LDA    PENV
C12E  8D3502               STA    LPENV           ;light pen vertical positio:
C131  AD0CD4               LDA    PENH
C134  8D3402               STA    LPENH           ;light pen vertical positio:
C137  AD3102               LDA    SDLSTH
C13A  8D03D4               STA    DLISTH          ;high display list address
C13D  AD3002               LDA    SDLSTL
C140  8D02D4               STA    DLISTL          ;low display list address
C143  AD2F02               LDA    SDMCTL
C146  8D00D4               STA    DMACTL          ;DMA control
C149  AD6F02               LDA    GPRIOR
C14C  8D1BD0               STA    PRIOR           ;prioritty select

                    ;      Check for vertical scroll enabled.

C14F  AD6C02               LDA    VSFLAG          ;vertical scroll count
C152  F00E ^C162           BEQ    IVN6            ;if vertical scroll not ena:

                    ;      Scroll one line.

C154  CE6C02               DEC    VSFLAG          ;decrement vertical scroll :
C157  A908                 LDA    #8              ;scroll one line
C159  38                   SEC
C15A  ED6C02               SBC    VSFLAG          ;subtract vertical scroll c:
C15D  2907                 AND    #07
C15F  8D05D4               STA    VSCROL          ;set vertical scroll

                    ;      Turn off speaker.

C162  A208          IVN6   LDX    #$08            ;speaker off
C164  8E1FD0               STX    CONSOL          ;set speaker control

                    ;      Set color registers from shadows.

                    ;      LDX    #8              ;offset to background color

C167  58            IVN7   CLI
C168  BDC002               LDA    PCOLR0,X        ;color register shadow
```

```
C16B  454F                EOR    COLRSH          ;modify with attract-mode c:
C16D  254E                AND    DRKMSK          ;modify with attract-mode l:
C16F  9D12D0              STA    COLPM0,X        ;set color register
C172  CA                  DEX
C173  10F2 ^C167          BPL    IVN7            ;if not done

                    ;     Set character set control.

C175  ADF402              LDA    CHBAS
C178  8D09D4              STA    CHBASE
C17B  ADF302              LDA    CHACT
C17E  8D01D4              STA    CHACTL

                    ;     Process countdown timer 2.

C181  A202                LDX    #2              ;indicate countdown timer 2
C183  2055C2              JSR    DCT             ;decrement countdown timer
C186  D003 ^C18B          BNE    IVN8            ;if timer not expired

C188  2052C2              JSR    PTT             ;process countdown timer 2 :

                    ;     Process timers 3, 4 and 5.

C18B  A202          IVN8  LDX    #2              ;preset offset to timer 2

C18D  E8            IVN9  INX
C18E  E8                  INX                    ;offset to countdown timer
C18F  BD1802              LDA    CDTMV3-4,X      ;countdown timer
C192  1D1902              ORA    CDTMV3+1-4,X
C195  F006 ^C19D          BEQ    IVN10           ;if countdown timer already:

C197  2055C2              JSR    DCT             ;decrement countdown timer
C19A  9D2602              STA    CDTMF3-4,X      ;indicate timer expiration :

C19D  E008          IVN10 CPX    #8              ;offset to timer 5
C19F  D0EC ^C18D          BNE    IVN9            ;if all timers not done

                    ;     Check debounce counter.

C1A1  AD0FD2              LDA    SKSTAT          ;keyboard status
C1A4  2904                AND    #$04            ;key down indicator
C1A6  F008 ^C1B0          BEQ    IVN11           ;if key down

                    ;     Process key up.

C1A8  ADF102              LDA    KEYDEL          ;key delay counter
C1AB  F003 ^C1B0          BEQ    IVN11           ;if counted down already

C1AD  CEF102              DEC    KEYDEL          ;decrement key delay counte:

                    ;     Check software key repeat timer.

C1B0  AD2B02        IVN11 LDA    SRTIMR          ;key repeat timer
C1B3  F03E ^C1F3          BEQ    IVN13           ;if key repeat timer expire:

C1B5  AD0FD2              LDA    SKSTAT          ;keyboard status
C1B8  2904                AND    #$04            ;key down indicator
```

```
C1BA  D032 ^C1EE          BNE    IVN12           ;if key no longer down

C1BC  CE2B02              DEC    SRTIMR          ;decrement key repeat timer
C1BF  D032 ^C1F3          BNE    IVN13           ;if key repeat timer not ex:

                   ;      Process key repeat timer expiration.

C1C1  AD6D02              LDA    KEYDIS          ;keyboard disable flag
C1C4  D02D ^C1F3          BNE    IVN13           ;if keyboard disabled, no r:

C1C6  ADDA02              LDA    KEYREP          ;initial timer value
C1C9  8D2B02              STA    SRTIMR          ;reset key repeat timer
C1CC  AD09D2              LDA    KBCODE          ;key code

                   ;      Check for hidden codes.

C1CF  C99F                CMP    #CNTL1
C1D1  F020 ^C1F3          BEQ    IVN13           ;if CTRL-1

C1D3  C983                CMP    #CNTLF1
C1D5  F01C ^C1F3          BEQ    IVN13           ;if CTRL-F1

C1D7  C984                CMP    #CNTLF2
C1D9  F018 ^C1F3          BEQ    IVN13           ;if CTRL-F2

C1DB  C994                CMP    #CNTLF4
C1DD  F014 ^C1F3          BEQ    IVN13           ;if CTRL-F4

C1DF  293F                AND    #$3F
C1E1  C911                CMP    #HELP
C1E3  F00E ^C1F3          BEQ    IVN13           ;if HELP

                   ;      Set key code.

C1E5  AD09D2              LDA    KBCODE          ;key code
C1E8  8DFC02              STA    CH              ;set key code
C1EB  4CF3C1              JMP    IVN13           ;continue

                   ;      Zero key repeat timer.

C1EE  A900         IVN12  LDA    #0
C1F0  8D2B02              STA    SRTIMR          ;zero key repeat timer

                   ;      Read joysticks.

C1F3  AD00D3       IVN13  LDA    PORTA           ;joystick readings
C1F6  4A                  LSR    A
C1F7  4A                  LSR    A
C1F8  4A                  LSR    A
C1F9  4A                  LSR    A               ;joystick 1 reading
C1FA  8D7902              STA    STICK1          ;set joystick 1 reading
      = FFFF       VGC    IF     VGC
C1FD  8D7B02              STA    STICK3          ;simulate joystick 3 readin:
                   VGC    ENDIF
C200  AD00D3              LDA    PORTA           ;joystick readings
C203  290F                AND    #$0F            ;joystick 0 reading
C205  8D7802              STA    STICK0          ;set joystick 0 reading
```

```
         = FFFF      VGC     IF      VGC
 C208    8D7A02              STA     STICK2          ;simulate joystick 2 readin:
                     VGC     ENDIF

                     ;       Read joystick triggers.

 C20B    AD10D0              LDA     TRIG0           ;trigger 0 indicator
 C20E    8D8402              STA     STRIG0          ;set trigger 0 indicator
         = FFFF      VGC     IF      VGC
 C211    8D8602              STA     STRIG2          ;simulate trigger 2 indicat:
                     VGC     ENDIF
 C214    AD11D0              LDA     TRIG1           ;trigger 1 indicator
 C217    8D8502              STA     STRIG1          ;set trigger 1 indicator
         = FFFF      VGC     IF      VGC
 C21A    8D8702              STA     STRIG3          ;simulate trigger 3 indicat:
                     VGC     ENDIF

                     ;       Read potentiometers.

 C21D    A203                LDX     #3              ;offset to last potentiomet:

 C21F    BD00D2      IVN14   LDA     POT0,X          ;potentiometer reading
 C222    9D7002              STA     PADDL0,X        ;set potentiometer reading
         = FFFF      VGC     IF      VGC
 C225    9D7402              STA     PADDL4,X        ;simulate potentiometer rea:
                     VGC     ENDIF
 C228    CA                  DEX
 C229    10F4 ^C21F          BPL     IVN14           ;if not done

                     ;       Start potentiometers for next time.

 C22B    8D0BD2              STA     POTGO           ;start potentiometers

                     ;       Read paddle triggers.

 C22E    A202                LDX     #2              ;offset to paddle trigger r:
 C230    A001                LDY     #1              ;offset to joystick reading

 C232    B97802      IVN15   LDA     STICK0,Y        ;joystick reading
 C235    4A                  LSR     A
 C236    4A                  LSR     A
 C237    4A                  LSR     A               ;paddle trigger reading
 C238    9D7D02              STA     PTRIG1,X        ;set paddle trigger reading
         = FFFF      VGC     IF      VGC
 C23B    9D8102              STA     PTRIG5,X        ;simulate paddle trigger re:
                     VGC     ENDIF

 C23E    A900                LDA     #0
 C240    2A                  ROL     A               ;paddle trigger reading
 C241    9D7C02              STA     PTRIG0,X        ;set paddle trigger reading
         = FFFF      VGC     IF      VGC
 C244    9D8002              STA     PTRIG4,X        ;simulate paddle trigger re:
                     VGC     ENDIF
 C247    CA                  DEX
 C248    CA                  DEX
 C249    88                  DEY
 C24A    10E6 ^C232          BPL     IVN15           ;if not done
```

```
                                *       Exit.
  C24C  6C2402                          JMP     (VVBLKD)        ;process deferred VBLANK NMI


                                **      PTO - Process Countdown Timer One Expiration
                                *
                                *       ENTRY   JSR     PTO
                                *
                                *       MODS
                                *               Original Author Unknown
                                *               1. Bring closer to Coding Standard (object :
                                *                  R. K. Nordin 11/01/83

          = C24F        PTO     =       *               ;entry
  C24F  6C2602                  JMP     (CDTMA1)        ;process countdown timer 1 :


                                **      PTT - Process Countdown Timer Two Expiration
                                *
                                *       ENTRY   JSR     PTT
                                *
                                *       MODS
                                *               Original Author Unknown
                                *               1. Bring closer to Coding Standard (object :
                                *                  R. K. Nordin 11/01/83

          = C252        PTT     =       *               ;entry
  C252  6C2802                  JMP     (CDTMA2)        ;process countdown timer 2 :


                                **      DCT - Decrement Countdown Timer
                                *
                                *       ENTRY   JSR     DCT
                                *               X = offset to timer value
                                *
                                *       EXIT
                                *               A = 0, if timer expired
                                *                 = $FF, if timer did not expire
                                *
                                *       MODS
                                *               Original Author Unknown
                                *               1. Bring closer to Coding Standard (object :
                                *                  R. K. Nordin 11/01/83

          = C255        DCT     =       *               ;entry
  C255  BC1802                  LDY     CDTMV1,X        ;low timer value
```

```
C258  D008 ^C262           BNE   DCT1           ;if low timer value not zer:

C25A  BC1902               LDY   CDTMV1+1,X     ;high timer value
C25D  F010 ^C26F           BEQ   DCT2           ;if timer value zero, exit

C25F  DE1902               DEC   CDTMV1+1,X     ;decrement high timer value

C262  DE1802       DCT1    DEC   CDTMV1,X       ;decrement low timer value
C265  D008 ^C26F           BNE   DCT2           ;if low timer value not zer:

C267  BC1902               LDY   CDTMV1+1,X     ;high timer value
C26A  D003 ^C26F           BNE   DCT2           ;if high timer value not ze:

C26C  A900                 LDA   #0             ;indicate timer expired
C26E  60                   RTS                  ;return

C26F  A9FF         DCT2    LDA   #$FF           ;indicate timer did not exp:
C271  60                   RTS                  ;return


                   **      SVP - Set Vertical Blank Parameters
                   *
                   *       SVP sets countdown timers and VBLANK vectors.
                   *
                   *       ENTRY   JSR     SVP
                   *               X = high initial timer value or high vector:
                   *               Y = low initial timer value or low vector a:
                   *               A = 1, if timer 1 value
                   *                   2, if timer 2 value
                   *                   3, if timer 3 value
                   *                   4, if timer 4 value
                   *                   5, if timer 5 value
                   *                   6, if immediate VBLANK vector
                   *                   7, if deferred VBLANK vector
                   *
                   *       MODS
                   *               Original Author Unknown
                   *               1. Bring closer to Coding Standard (object :
                   *                  R. K. Nordin 11/01/83

        = C272     SVP     =     *              ;entry

                   ;       Initialize.

C272  0A                   ASL   A              ;compute offset+2 to value :
C273  8D2D02               STA   INTEMP         ;offset+2 to value or vecto:
C276  8A                   TXA                  ;high timer value or high v:

                   ;       Ensure no VBLANK in progress by delaying after HBLA:

C277  A205                 LDX   #5             ;20 CPU cycles
C279  8D0AD4               STA   WSYNC          ;wait for HBLANK synchroniz:

C27C  CA           SVP1    DEX
```

```
C27D  D0FD ^C27C          BNE     SVP1             ;if not done delaying

                    ;     Set timer value or vector address.

C27F  AE2D02              LDX     INTEMP           ;offset+2 to value or vecto:
C282  9D1702              STA     CDTMV1-2+1,X     ;high timer value or high v:
C285  98                  TYA
C286  9D1602              STA     CDTMV1-2,X       ;low timer value or low vec:
C289  60                  RTS                      ;return



                    **    DVNM - Process Deferred VBLANK NMI
                    *
                    *     ENTRY   JMP     DVNM
                    *
                    *     EXIT
                    *             Exits via RTI
                    *
                    *     MODS
                    *             Original Author Unknown.
                    *             1. Bring closer to Coding Standard (object :
                    *                R. K. Nordin 11/01/83

      = C28A        DVNM  =       *        ;entry
C28A  68                  PLA
C28B  A8                  TAY              ;restore Y
C28C  68                  PLA
C28D  AA                  TAX              ;restore X
C28E  68                  PLA              ;restore A
C28F  40                  RTI              ;return
```

```
C290                    **      PWS - Perform Warmstart
                        *
                        *       ENTRY   JMP     PWS
                        *
                        *       EXIT
                        *               Exits to PCS or PRS
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C290          PWS     =       *       ;entry

                        ;       Initialize.

C290  78                        SEI

                        ;       Check for cartridge change.

C291  AD13D0                    LDA     TRIG3   ;cartridge interlock
C294  CDFA03                    CMP     GINTLK  ;previous cartridge interlock statu:
C297  D02F ^C2C8                BNE     PCS     ;if cartridge changed, perform cold:

                        ;       Check for cartridge.

C299  6A                        ROR     A
C29A  9005 ^C2A1                BCC     PWS1    ;if no cartridge

                        ;       Verify no change in cartridge.

C29C  20C9C4                    JSR     CCE     ;check cartridge equivalence
C29F  D027 ^C2C8                BNE     PCS     ;if different cartridge, coldstart

                        ;       Check coldstart status.

C2A1  AD4402            PWS1    LDA     COLDST  ;coldstart status
C2A4  D022 ^C2C8                BNE     PCS     ;if coldstart was in progress, perf:

                        ;       Perform warmstart.

C2A6  A9FF                      LDA     #$FF    ;indicate warmstart
C2A8  D020 ^C2CA                BNE     PRS     ;preset memory, return
```

```
                        **      RES - Process RESET
                        *
                        *       ENTRY   JMP     RES
                        *
                        *       EXIT
                        *               Exits to PCS, if coldstart, or PWS, if warm:
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C2AA          RES     =       *               ;entry

                        ;       Initialize.

 C2AA   78                      SEI

                        ;       Delay 0.1 second for RESET bounce.

 C2AB   A28C                    LDX     #140            ;0.1 second delay

 C2AD   88              RES1    DEY
 C2AE   D0FD ^C2AD              BNE     RES1            ;if inner loop not done

 C2B0   CA                      DEX
 C2B1   D0FA ^C2AD              BNE     RES1            ;if outer loop not done

                        ;       Check power-up validation bytes.

 C2B3   AD3D03                  LDA     PUPBT1
 C2B6   C95C                    CMP     #PUPVL1
 C2B8   D00E ^C2C8              BNE     PCS             ;if validation byte 1 differs, cold:

 C2BA   AD3E03                  LDA     PUPBT2
 C2BD   C993                    CMP     #PUPVL2
 C2BF   D007 ^C2C8              BNE     PCS             ;if validation byte 2 differs, cold:

 C2C1   AD3F03                  LDA     PUPBT3
 C2C4   C925                    CMP     #PUPVL3
 C2C6   F0C8 ^C290              BEQ     PWS             ;if all bytes validated, perform wa:

                        ;       JMP     PCS             ;perform coldstart, return
```

```
                        **      PCS - Perform Coldstart
                        *
                        *       ENTRY   JMP     PCS
                        *
                        *       EXIT
                        *               Exits to PRS
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C2C8          PCS     =       *       ;entry
 C2C8   A900                    LDA     #0      ;indicate coldstart
                        ;       JMP     PRS     ;preset memory, return


                        **      PRS - Preset Memory
                        *
                        *       ENTRY   JMP     PRS
                        *
                        *       EXIT
                        *               Exits via CARTCS vector or DOSVEC vector
                        *
                        *       NOTES
                        *               Problem: in the CRASS65 version, APPMHI was:
                        *               zero-page.
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C2CA          PRS     =       *               ;entry

                        ;       Update warmstart flag.

 C2CA   8508                    STA     WARMST          ;update warmstart flag

                        ;       Set initial conditions.

 C2CC   78                      SEI
 C2CD   D8                      CLD
 C2CE   A2FF                    LDX     #$FF
 C2D0   9A                      TXS                     ;set stack pointer

                        ;       Initialize LNBUG flag, if necessary.

        = 0000          LNBUG   IF      LNBUG
                        LNBUG   ENDIF

                        ;       Perform miscellaneous initialization.
```

```
C2D1  2071C4             JSR   PMI            ;perform miscellaneous init:

                   ;     Initialize memory status.

C2D4  A901               LDA   #1             ;no failure indicator
C2D6  8501               STA   NGFLAG         ;memory status flag

                   ;     Check type.

C2D8  A508               LDA   WARMST         ;warmstart flag
C2DA  D052 ^C32E         BNE   PRS8           ;if warmstart

                   ;     Zero all RAM (except beginning of page zero).

C2DC  A900               LDA   #0
C2DE  A008               LDY   #WARMST        ;initial offset into page z:
C2E0  8504               STA   RAMLO
C2E2  8505               STA   RAMLO+1        ;initialize RAM pointer

C2E4  A9FF         PRS3  LDA   #$FF
C2E6  9104               STA   (RAMLO),Y      ;attempt to store $FF
C2E8  D104               CMP   (RAMLO),Y
C2EA  F002 ^C2EE         BEQ   PRS4           ;if $FF stored successfully

C2EC  4601               LSR   NGFLAG         ;indicate memory failure

C2EE  A900         PRS4  LDA   #$00
C2F0  9104               STA   (RAMLO),Y      ;attempt to store $00
C2F2  D104               CMP   (RAMLO),Y
C2F4  F002 ^C2F8         BEQ   PRS5           ;if $00 stored successfully

C2F6  4601               LSR   NGFLAG         ;indicate memory failure

C2F8  C8           PRS5  INY
C2F9  D0E9 ^C2E4         BNE   PRS3           ;if not end of page

                   ;     Advance to next page and check for completion.

C2FB  E605               INC   RAMLO+1        ;advance RAM pointer to nex:
C2FD  A605               LDX   RAMLO+1
C2FF  E406               CPX   TRAMSZ         ;RAM size
C301  D0E1 ^C2E4         BNE   PRS3           ;if not at end of RAM

                   ;     Initialize DOSVEC.

C303  A923               LDA   #low PPD       ;power-up display routine a:
C305  850A               STA   DOSVEC         ;initialize DOS vector
C307  A9F2               LDA   #high PPD
C309  850B               STA   DOSVEC+1

                   ;     Verify ROM checksums.

C30B  AD01D3             LDA   PORTB
C30E  297F               AND   #$7F           ;select self-test ROM
C310  8D01D3             STA   PORTB          ;port B memory control

C313  2073FF             JSR   VFR            ;verify first 8K ROM
```

```
C316  B005 ^C31D         BCS   PRS6           ;if first 8K ROM bad

C318  2092FF             JSR   VSR            ;verify second 8K ROM
C31B  9002 ^C31F         BCC   PRS7           ;if seond 8K ROM good

C31D  4601       PRS6    LSR   NGFLAG         ;indicate memory bad

C31F  AD01D3     PRS7    LDA   PORTB
C322  0980               ORA   #$80           ;disable self-test ROM
C324  8D01D3             STA   PORTB          ;update port B memory contr:

                 ;       Indicate coldstart in progress.

C327  A9FF               LDA   #$FF
C329  8D4402             STA   COLDST         ;indicate coldstart in prog:
C32C  D022 ^C350         BNE   PRS12          ;continue with coldstart pr:

                 ;       Perform warmstart procedures.

C32E  A200       PRS8    LDX   #0

C330  ADEC03             LDA   DERRF          ;screen OPEN error flag
C333  F007 ^C33C         BEQ   PRS9           ;if in screen OPEN

                 ;       Clean up APPMHI.

                 ;       STX   APPMHI
C335  8E0E00             VFD   8\$8E,8\low APPMHI,8\high APPMHI
                 ;       STX   APPMHI+1
C338  8E0F00             VFD   8\$8E,8\low [APPMHI+1],8\high [APPMHI+1]

C33B  8A                 TXA

                 ;       Clear page 2 and part of page 3.

C33C  9D0002     PRS9    STA   $0200,X        ;clear byte of page 2

C33F  E0ED               CPX   #low ACMVAR    ;start of page 3 locations :
C341  B003 ^C346         BCS   PRS10          ;if not to clear this page :

C343  9D0003             STA   $0300,X        ;clear byte of page 3

C346  CA         PRS10   DEX
C347  D0F3 ^C33C         BNE   PRS9           ;if not done

                 ;       Clear part of page 0.

C349  A210               LDX   #INTZBS        ;offset to first page 0 byt:

C34B  9500       PRS11   STA   $0000,X        ;clear byte of page 0
C34D  E8                 INX
C34E  10FB ^C34B         BPL   PRS11          ;if not done

                 ;       Record BASIC status.

C350  A200       PRS12   LDX   #0             ;initially assume BASIC ena:
C352  AD01D3             LDA   PORTB          ;port B memory control
```

```
C355  2902                  AND     #$02            ;BASIC enabled indicator
C357  F001 ^C35A            BEQ     PRS13           ;if BASIC enabled

C359  E8                    INX                     ;indicate BASIC disabled

C35A  8EF803        PRS13   STX     BASICF          ;BASIC flag

                    ;       Establish power-up validation bytes.

C35D  A95C                  LDA     #PUPVL1
C35F  8D3D03                STA     PUPBT1          ;validation byte 1
C362  A993                  LDA     #PUPVL2
C364  8D3E03                STA     PUPBT2          ;validation byte 2
C367  A925                  LDA     #PUPVL3
C369  8D3F03                STA     PUPBT3          ;validation byte 3

                    ;       Establish screen margins.

C36C  A902                  LDA     #LEDGE
C36E  8552                  STA     LMARGN          ;left margin
C370  A927                  LDA     #REDGE
C372  8553                  STA     RMARGN          ;right margin

                    ;       Establish parameters for NTSC or PAL.

C374  AD14D0                LDA     PAL             ;GTIA flag bits
C377  290E                  AND     #$0E            ;PAL/NTSC indicator
C379  D008 ^C383            BNE     PRS14           ;if NTSC

C37B  A905                  LDA     #5              ;PAL key repeat delay
C37D  A201                  LDX     #1              ;PAL indicator
C37F  A028                  LDY     #40             ;PAL key repeat initial del:
C381  D006 ^C389            BNE     PRS15           ;set parameters

C383  A906          PRS14   LDA     #6              ;NTSC key repeat delay
C385  A200                  LDX     #0              ;NTSC indicator
C387  A030                  LDY     #48             ;NTSC key repeat initial de:

C389  8DDA02        PRS15   STA     KEYREP          ;set key repeat rate
C38C  8662                  STX     PALNTS          ;set PAL/NTSC status
C38E  8CD902                STY     KRPDEL          ;set key repeat initial del:

                    ;       Initialize missing controller ports, if not simulat:

      = 0000        VGC     IF      not VGC
                    VGC     ENDIF

                    ;       Copy interrupt vector table from ROM to RAM.

C391  A225                  LDX     #TIHVL-1        ;offset to last byte of tab:

C393  BD4BC4        PRS17   LDA     TIHV,X          ;byte of table of interrupt:
C396  9D0002                STA     INTABS,X        ;byte of RAM table
C399  CA                    DEX
C39A  10F7 ^C393            BPL     PRS17           ;if not done

                    ;       Copy handler vector table from ROM to RAM.
```

```
C39C  A20E                  LDX     #THAVL-1          ;offset to last byte of tab:

C39E  BD2EC4        PRS18   LDA     THAV,X            ;byte of handler vector tab:
C3A1  9D1A03                STA     HATABS,X          ;byte of RAM table
C3A4  CA                    DEX
C3A5  10F7 ^C39E            BPL     PRS18             ;if not done

                    ;       Initialize software.

C3A7  2035C5                JSR     ISW               ;initialize software

                    ;       Initialize ACMI module, if present.

      = 0000        ACMI    IF      ACMI
                    ACMI    ENDIF

                    ;       Enable IRQ interrupts.

C3AA  58                    CLI

                    ;       Check for memory problems.

C3AB  A501                  LDA     NGFLAG            ;memory status
C3AD  D015 ^C3C4            BNE     PRS21             ;if memory good

                    ;       Perform memory self-test on bad memory.

C3AF  AD01D3                LDA     PORTB
C3B2  297F                  AND     #$7F              ;enable self-test ROM
C3B4  8D01D3                STA     PORTB             ;update port B memory contr:
C3B7  A902                  LDA     #2
C3B9  8DF302                STA     CHACT             ;CHACTL (character control):
C3BC  A9E0                  LDA     #high DCSORG      ;high domestic character se:
C3BE  8DF402                STA     CHBAS             ;CHBASE (character base) sh:
C3C1  4C0350                JMP     EMS               ;execute memory self-test

                    ;       Check for cartridge.

C3C4  A200          PRS21   LDX     #0
C3C6  8606                  STX     TRAMSZ            ;clear cartridge flag

C3C8  AEE402                LDX     RAMSIZ            ;RAM size
C3CB  E0B0                  CPX     #high $B000       ;start of cartridge area
C3CD  B00D ^C3DC            BCS     PRS22             ;if RAM in cartridge area

C3CF  AEFCBF                LDX     CART
C3D2  D008 ^C3DC            BNE     PRS22             ;if no cartridge

C3D4  E606                  INC     TRAMSZ            ;set cartridge flag
C3D6  20C9C4                JSR     CCE               ;check cartridge equivalenc:
C3D9  2029C4                JSR     ICS               ;initialize cartridge softw:

                    ;       Open screen editor.

C3DC  A903          PRS22   LDA     #OPEN
C3DE  A200                  LDX     #SEIOCB           ;screen editor IOCB index
```

```
C3E0  9D4203              STA   ICCOM,X         ;command
C3E3  A948                LDA   #low SEDS       ;screen editor device speci:
C3E5  9D4403              STA   ICBAL,X         ;buffer address
C3E8  A9C4                LDA   #high SEDS
C3EA  9D4503              STA   ICBAH,X
C3ED  A90C                LDA   #OPNIN+OPNOT    ;open for input/output
C3EF  9D4A03              STA   ICAX1,X         ;auxiliary informatin 1
C3F2  2056E4              JSR   CIOV            ;vector to CIO
C3F5  1003 ^C3FA          BPL   PRS23           ;if no error

                    ;     Process error (which should never happen).

C3F7  4CAAC2              JMP   RES             ;retry power-up

                    ;     Delay, ensuring VBLANK.

C3FA  E8          PRS23   INX
C3FB  D0FD ^C3FA          BNE   PRS23           ;if inner loop not done

C3FD  C8                  INY
C3FE  10FA ^C3FA          BPL   PRS23           ;if outer loop not done

                    ;     Attempt cassette boot.

C400  206EC6              JSR   ACB             ;attempt cassette boot

                    ;     Check cartridge for disk boot.

C403  A506                LDA   TRAMSZ
C405  F006 ^C40D          BEQ   PRS24           ;if no cartridge

C407  ADFDBF              LDA   CARTFG          ;cartridge mode flags
C40A  6A                  ROR   A
C40B  9006 ^C413          BCC   PRS25           ;if disk boot not desired

                    ;     Attempt disk boot.

C40D  208BC5      PRS24   JSR   ADB             ;attempt disk boot

                    ;     Initialize periperhal handler loading facility.

C410  2039E7              JSR   PHR             ;poll, load, relocate, init:

                    ;     Indicate coldstart complete.

C413  A900        PRS25   LDA   #0
C415  8D4402              STA   COLDST          ;indicate coldstart complet:

                    ;     Check cartridge for execution.

C418  A506                LDA   TRAMSZ
C41A  F00A ^C426          BEQ   PRS26           ;if no cartridge

C41C  ADFDBF              LDA   CARTFG          ;cartridge mode flags
C41F  2904                AND   #$04
C421  F003 ^C426          BEQ   PRS26           ;if execution not desired
```

```
                        ;       Execute cartridge.

 C423  6CFABF                   JMP     (CARTCS)        ;execute cartridge

                        ;       Exit to power-up display or booted program.

 C426  6C0A00           PRS26   JMP     (DOSVEC)        ;vector to booted program



                        **      ICS - Initialize Cartridge Software
                        *
                        *       ENTRY   JSR     ICS
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

       = C429           ICS     =       *               ;entry
 C429  6CFEBF                   JMP     (CARTAD)        ;initialize cartridge softw:



                        **      PAI - Process ACMI Interrupt
                        *
                        *       PAI does nothing.
                        *
                        *       ENTRY   JSR     PAI
                        *
                        *       NOTES
                        *               Problem: this code is unneeded unless ACMI :
                        *               option is selected.
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

       = C42C           PAI     =       *               ;entry
 C42C  18                       CLC
 C42D  60                       RTS                     ;return
```

```
                          **      THAV - Table of Handler Vectors
                          *
                          *       NOTES
                          *               THAV is moved to RAM table HATABS.


C42E  50            THAV  DB      PRINTR  ;printer device code
C42F  30E4                DW      PRINTV  ;printer handler vector table

C431  43                  DB      CASSET  ;cassette device code
C432  40E4                DW      CASETV  ;cassette handler vector table

C434  45                  DB      SCREDT  ;editor device code
C435  00E4                DW      EDITRV  ;editor handler vector table

C437  53                  DB      DISPLY  ;screen device code
C438  10E4                DW      SCRENV  ;screen handler vector table

C43A  4B                  DB      KBD     ;keyboard device code
C43B  20E4                DW      KEYBDV  ;keyboard handler vector table

      = 000F        THAVL =       *-THAV  ;length



                          **      BMSG - Boot Error Message


C43D  424F4F5420    BMSG  DB      'BOOT ERROR',EOL



                          **      Screen Editor Device Specification


C448  453A9B        SEDS  DB      'E:',EOL



                          **      TIHV - Table of Interrupt Handler Vectors
                          *
                          *       NOTES
                          *               TIHV is moved to RAM table INTABS.


C44B  CEC0          TIHV  DW      RIR     ;VDSLST - display list NMI vector
C44D  CDC0                DW      XIR     ;VPRCED - proceed line IRQ vector
C44F  CDC0                DW      XIR     ;VINTER - interrupt line IRQ vector
C451  CDC0                DW      XIR     ;VBREAK - BRK instruction IRQ vecto:
C453  19FC                DW      KIR     ;VKEYBD - keyboard IRQ vector
C455  2CEB                DW      IRIR    ;VSERIN - serial input ready IRQ ve:
C457  ADEA                DW      ORIR    ;VSEROR - serial output ready IRQ v:
C459  ECEA                DW      OCIR    ;VSEROC - serial output complete IR:
C45B  CDC0                DW      XIR     ;VTIMR1 - POKEY timer 1 IRQ vector
```

```
C45D  CDC0                DW      XIR      ;VTIMR2 - POKEY timer 2 IRQ vector
C45F  CDC0                DW      XIR      ;VTIMR4 - POKEY timer 4 IRQ vector
C461  30C0                DW      IIR      ;VIMIRQ - immediate IRQ vector
C463  0000                DW      0        ;CDTMV1 - countdown timer 1 vector
C465  0000                DW      0        ;CDTMV2 - countdown timer 2 vector
C467  0000                DW      0        ;CDTMV3 - countdown timer 3 vector
C469  0000                DW      0        ;CDTMV4 - countdown timer 4 vector
C46B  0000                DW      0        ;CDTMV5 - countdown timer 5 vector
C46D  E2C0                DW      IVNM     ;VVBLKI - immediate VBLANK NMI vect:
C46F  8AC2                DW      DVNM     ;VVBLKD - deferred VBLANK NMI vecto:

      = 0026      TIHVL   =       *-TIHV   ;length


                  **      PMI - Perform Miscellaneous Initialization
                  *
                  *       ENTRY   JSR      PMI
                  *
                  *       NOTES
                  *               Problem: initial address for sizing RAM sho:
                  *               $4000 (16K) instead of $2800.
                  *
                  *       MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83


      = C471      PMI     =       *        ;entry

                  ;       Check for cartridge special execution case.

C471  AD13D0              LDA     TRIG3
C474  6A                  ROR     A
C475  900D ^C484          BCC     PMI1     ;if cartridge not inserted

C477  ADFCBF              LDA     CART
C47A  D008 ^C484          BNE     PMI1     ;if not cartridge

C47C  ADFDBF              LDA     CARTFG   ;cartridge flags
C47F  1003 ^C484          BPL     PMI1     ;if special execution not desired

                  ;       Execute cartridge.

C481  6CFEBF              JMP     (CARTAD)         ;execute cartridge

                  ;       Initialize hardware.

C484  20DAC4      PMI1    JSR     IHW      ;initialize hardware

                  ;       Disable BASIC.

C487  AD01D3              LDA     PORTB
C48A  0902                ORA     #$02     ;disable BASIC
C48C  8D01D3              STA     PORTB    ;update port B memory control
```

```
                        ;       If warmstart, check previous BASIC status.

C48F  A508                      LDA     WARMST
C491  F007 ^C49A                BEQ     PMI2    ;if coldstart

C493  ADF803                    LDA     BASICF  ;BASIC flag
C496  D011 ^C4A9                BNE     PMI4    ;if BASIC not previously enabled

C498  F007 ^C4A1                BEQ     PMI3    ;enable BASIC

                        ;       Check OPTION key.

C49A  AD1FD0            PMI2    LDA     CONSOL  ;console switches
C49D  2904                      AND     #$04    ;OPTION key indicator
C49F  F008 ^C4A9                BEQ     PMI4    ;if OPTION key pressed, do not enab:

                        ;       Enable BASIC.

C4A1  AD01D3            PMI3    LDA     PORTB
C4A4  29FD                      AND     #$FD    ;disable BASIC
C4A6  8D01D3                    STA     PORTB   ;update port B memory control

                        ;       Determine size of RAM.

      = 0000            RAMSYS  IF      RAMSYS
                        RAMSYS  ELSE

C4A9  A900              PMI4    LDA     #low $2800      ;initial low address
C4AB  A8                        TAY                     ;offset to first byte of pa:
C4AC  8505                      STA     TRAMSZ-1        ;set initial low address

C4AE  A928                      LDA     #high $2800     ;initial RAM size
C4B0  8506                      STA     TRAMSZ          ;set initial RAM size (high:

C4B2  B105              PMI5    LDA     (TRAMSZ-1),Y    ;first byte of page
C4B4  49FF                      EOR     #$FF            ;complement
C4B6  9105                      STA     (TRAMSZ-1),Y    ;attempt to store complemen:
C4B8  D105                      CMP     (TRAMSZ-1),Y
C4BA  D00C ^C4C8                BNE     PMI6            ;if complement not stored

C4BC  49FF                      EOR     #$FF            ;original value
C4BE  9105                      STA     (TRAMSZ-1),Y    ;attempt to store original :
C4C0  D105                      CMP     (TRAMSZ-1),Y
C4C2  D004 ^C4C8                BNE     PMI6            ;if original value not stor:

C4C4  E606                      INC     TRAMSZ          ;increment high address
C4C6  D0EA ^C4B2                BNE     PMI5            ;continue

                        ;       Exit.

C4C8  60                PMI6    RTS                     ;return
                        RAMSYS  ENDIF
```

```
                        **      CCE - Check Cartridge Equivalence
                        *
                        *       ENTRY   JSR     CCE
                        *
                        *       NOTES
                        *               Problem: this code checksums $BFF0 - $C0EF;:
                        *               checksum $BF00 - $BFFF.
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

     = C4C9             CCE     =       *       ;entry

                        ;       Initialize.

C4C9  A900                      LDA     #0      ;initial sum
C4CB  AA                        TAX             ;offset to first byte
C4CC  18                        CLC

                        ;       Checksum 256 bytes of cartridge area.

C4CD  7DF0BF            CCE1    ADC     $BFF0,X ;add in byte
C4D0  E8                        INX
C4D1  D0FA ^C4CD                BNE     CCE1    ;if not done

                        ;       Exit.

C4D3  CDEB03                    CMP     CARTCK  ;previous checksum
C4D6  8DEB03                    STA     CARTCK  ;new checksum
C4D9  60                        RTS             ;return


                        **      IHW - Initialize Hardware
                        *
                        *       ENTRY   JSR     IHW
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

     = C4DA             IHW     =       *       ;entry

                        ;       Initialize CTIA, ANTIC and POKEY areas.

C4DA  A900                      LDA     #0              ;initialization value
C4DC  AA                        TAX                     ;initial offset
C4DD  8D03D3                    STA     PBCTL           ;set for direction register:

C4E0  9D00D0            IHW1    STA     CTIA,X          ;initialize CTIA/GTIA area :
C4E3  9D00D4                    STA     ANTIC,X         ;initialize ANTIC area regi:
```

```
C4E6  9D00D2                STA     POKEY,X         ;initialize POKEY area regi:
C4E9  E001                  CPX     #low PORTB
C4EB  F003 ^C4F0            BEQ     IHW2            ;if port B, don't initializ:

C4ED  9D00D3                STA     PIA,X           ;initialize PIA area regist:

C4F0  E8             IHW2   INX
C4F1  D0ED ^C4E0            BNE     IHW1            ;if not done

                     ;      Initialize PIA.

C4F3  A93C                  LDA     #$3C
C4F5  8D03D3                STA     PBCTL   ;precondition port B outputs
C4F8  A9FF                  LDA     #$FF
C4FA  8D01D3                STA     PORTB   ;all high
C4FD  A938                  LDA     #$38
C4FF  8D02D3                STA     PACTL   ;select data direction register
C502  8D03D3                STA     PBCTL   ;select data direction register
C505  A900                  LDA     #$00
C507  8D00D3                STA     PORTA   ;all inputs
C50A  A9FF                  LDA     #$FF
C50C  8D01D3                STA     PORTB   ;all outputs
C50F  A93C                  LDA     #$3C
C511  8D02D3                STA     PACTL   ;back to port
C514  8D03D3                STA     PBCTL   ;back to port
C517  AD01D3                LDA     PORTB   ;clear interrupts
C51A  AD00D3                LDA     PORTA   ;clear interrupts

                     ;      Initialize POKEY.

C51D  A922                  LDA     #$22    ;get POKEY out of initialize mode a:
C51F  8D0FD2                STA     SKCTL   ;set serial port control

C522  A9A0                  LDA     #$A0    ;pure tone, no volume
C524  8D05D2                STA     AUDC3   ;turn off channel 3
C527  8D07D2                STA     AUDC4   ;turn off channel 4

C52A  A928                  LDA     #$28    ;clock ch. 3 with 1.79 MHz, ch. 4 w:
C52C  8D08D2                STA     AUDCTL  ;set audio control

C52F  A9FF                  LDA     #$FF
C531  8D0DD2                STA     SEROUT  ;start bit only

C534  60                    RTS             ;return
```

```
                        **      ISW - Initialize Software
                        *
                        *       ENTRY   JSR     ISW
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

         = C535         ISW     =       *                  ;entry

                        ;       Initialize BREAK key handling.

C535  C611                      DEC     BRKKEY             ;turn off BREAK key flag

C537  A992                      LDA     #low BIR
C539  8D3602                    STA     BRKKY              ;set BREAK key IRQ routine :
C53C  A9C0                      LDA     #high BIR
C53E  8D3702                    STA     BRKKY+1

                        ;       Initialize RAMSIZ and MEMTOP.

C541  A506                      LDA     TRAMSZ             ;determined size of RAM
C543  8DE402                    STA     RAMSIZ             ;size of RAM
C546  8DE602                    STA     MEMTOP+1           ;high top of memory
C549  A900                      LDA     #$00
C54B  8DE502                    STA     MEMTOP             ;low top of memory

                        ;       Initialize MEMLO.

C54E  A900                      LDA     #low INIML         ;initial MEMLO address
C550  8DE702                    STA     MEMLO
C553  A907                      LDA     #high INIML
C555  8DE802                    STA     MEMLO+1

                        ;       Initialize device handlers.

C558  200CE4                    JSR     EDITRV+12          ;initialize editor handler
C55B  201CE4                    JSR     SCRENV+12          ;initialize screen handler
C55E  202CE4                    JSR     KEYBDV+12          ;initialize keyboard handle:
C561  203CE4                    JSR     PRINTV+12          ;initialize printer handler
C564  204CE4                    JSR     CASETV+12          ;initialize cassette handle:

                        ;       Initialize various routines.

C567  206EE4                    JSR     CIOINV             ;initialize CIO
C56A  2065E4                    JSR     SIOINV             ;initialize SIO
C56D  206BE4                    JSR     INTINV             ;initialize interrupt handl:
C570  2050E4                    JSR     DINITV             ;initialize DIO

                        ;       Initialize generic parallel device handler.

C573  A96E                      LDA     #low PIR
C575  8D3802                    STA     VPIRQ              ;parallel device IRQ routin:
C578  A9C9                      LDA     #high PIR
C57A  8D3902                    STA     VPIRQ+1
```

```
C57D  209BE4               JSR    GPDVV+12        ;initialize parallel device:

                     ;     Set status of START key.

C580  AD1FD0               LDA    CONSOL          ;console switches
C583  2901                 AND    #$01            ;START key indicator
C585  4901                 EOR    #$01            ;START key status
C587  8DE903               STA    CKEY            ;cassette boot request flag

C58A  60                   RTS                    ;return


                     **    ADB - Attempt Disk Boot
                     *
                     *     ENTRY  JSR    ADB
                     *
                     *     MODS
                     *            Original Author Unknown
                     *            1. Bring closer to Coding Standard (object :
                     *               R. K. Nordin 11/01/83

      = C58B         ADB   =             *    ;entry

                     ;     Check type of reset.

C58B  A508                 LDA    WARMST
C58D  F009 ^C598           BEQ    ADB1     ;if not warmstart

                     ;     Process warmstart.

C58F  A509                 LDA    BOOT?    ;successful boot flags
C591  2901                 AND    #$01     ;successful disk boot indicator
C593  F033 ^C5C8           BEQ    BAI2     ;if disk boot not successful, retur:

                     ;     Initialize disk booted software.

C595  4C3BC6               JMP    IBS      ;initialize booted software

                     ;     Process coldstart.

C598  A901           ADB1  LDA    #1
C59A  8D0103               STA    DUNIT    ;disk unit number
C59D  A953                 LDA    #STATC   ;status
C59F  8D0203               STA    DCOMND   ;command
C5A2  2053E4               JSR    DSKINV   ;issue command
C5A5  3021 ^C5C8           BMI    BAI2     ;if error, return

                     ;     Boot.

                     ;     JMP    ABI      ;attempt boot and initialize
```

```
                       **      ABI - Attempt Boot and Initialize
                       *
                       *       ENTRY   JSR     ABI
                       *
                       *       MODS
                       *               Original Author Unknown
                       *               1. Bring closer to Coding Standard (object :
                       *                  R. K. Nordin 11/01/83


     = C5A7            ABI     =       *               ;entry

C5A7  A900                     LDA     #high 1
C5A9  8D0B03                   STA     DAUX2
C5AC  A901                     LDA     #low 1          ;sector number
C5AE  8D0A03                   STA     DAUX1

C5B1  A900                     LDA     #low [CASBUF+3] ;buffer address
C5B3  8D0403                   STA     DBUFLO
C5B6  A904                     LDA     #high [CASBUF+3]
C5B8  8D0503                   STA     DBUFHI

                       ;       JMP     BAI             ;boot and initialize



                       **      BAI - Boot and Initialize
                       *
                       *       ENTRY   JSR     BAI
                       *
                       *       MODS
                       *               Original Author Unknown
                       *               1. Bring closer to Coding Standard (object :
                       *                  R. K. Nordin 11/01/83


     = C5BB            BAI     =       *       ;entry

                       ;       Read first sector.

C5BB  2059C6                   JSR     GNS     ;get next sector
C5BE  1009 ^C5C9               BPL     CBI     ;if no error, complete boot and ini:

                       ;       Process error.

C5C0  203EC6           BAI1    JSR     DBE     ;display boot error message

C5C3  ADEA03                   LDA     CASSBT
C5C6  F0DF ^C5A7               BEQ     ABI     ;if not cassette boot, try again

                       ;       Exit.

C5C8  60               BAI2    RTS             ;return
```

```
                        **      CBI - Complete Boot and Initialize
                        *
                        *       ENTRY   JSR     CBI
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C5C9          CBI     =       *               ;entry

                        ;       Transfer flags.

C5C9    A203                    LDX     #3

C5CB    BD0004          CBI1    LDA     CASBUF+3,X      ;byte from buffer
C5CE    9D4002                  STA     DFLAGS,X        ;flag byte
C5D1    CA                      DEX
C5D2    10F7 ^C5CB              BPL     CBI1            ;if not done

                        ;       Transfer sector.

C5D4    AD4202                  LDA     BOOTAD
C5D7    8504                    STA     RAMLO           ;set boot address
C5D9    AD4302                  LDA     BOOTAD+1
C5DC    8505                    STA     RAMLO+1

C5DE    AD0404                  LDA     CASBUF+7
C5E1    850C                    STA     DOSINI          ;establish initializtion ad:
C5E3    AD0504                  LDA     CASBUF+8
C5E6    850D                    STA     DOSINI+1

C5E8    A07F            CBI2    LDY     #127            ;offset to last byte of sec:

C5EA    B90004          CBI3    LDA     CASBUF+3,Y      ;byte of sector buffer
C5ED    9104                    STA     (RAMLO),Y       ;byte of boot program
C5EF    88                      DEY
C5F0    10F8 ^C5EA              BPL     CBI3            ;if not done

                        ;       Increment loader buffer pointer.

C5F2    18                      CLC
C5F3    A504                    LDA     RAMLO
C5F5    6980                    ADC     #$80
C5F7    8504                    STA     RAMLO
C5F9    A505                    LDA     RAMLO+1
C5FB    6900                    ADC     #0
C5FD    8505                    STA     RAMLO+1         ;increment boot loader buff:

                        ;       Decrement and check number of sectors.

C5FF    CE4102                  DEC     DBSECT          ;decrement number of sector:
C602    F012 ^C616              BEQ     CBI5            ;if no more sectors

                        ;       Get next sector.
```

```
C604  EE0A03                 INC     DAUX1            ;increment sector number

C607  2059C6        CBI4     JSR     GNS              ;get next sector
C60A  10DC ^C5E8             BPL     CBI2             ;if status OK

                    ;        Process error.

C60C  203EC6                 JSR     DBE              ;display boot error message
C60F  ADEA03                 LDA     CASSBT
C612  D0AC ^C5C0             BNE     BAI1             ;if cassette, start over

C614  F0F1 ^C607             BEQ     CBI4             ;try sector again

                    ;        Clean up.

C616  ADEA03        CBI5     LDA     CASSBT
C619  F003 ^C61E             BEQ     CBI6             ;if not cassette boot

C61B  2059C6                 JSR     GNS              ;get EOF record (but do not:

                    ;        Execute boot loader.

C61E  2029C6        CBI6     JSR     EBL              ;execute boot loader
C621  B09D ^C5C0             BCS     BAI1             ;if bad boot, try again

                    ;        Initialize booted software.

C623  203BC6                 JSR     IBS              ;initialize booted software
C626  E609                   INC     BOOT?            ;indicate boot success
C628  60                     RTS                      ;return


                    **       EBL - Execute Boot Loader
                    *
                    *        ENTRY   JSR     EBL
                    *
                    *        MODS
                    *                Original Author Unknown
                    *                1. Bring closer to Coding Standard (object :
                    *                   R. K. Nordin 11/01/83

      = C629        EBL      =       *                ;entry

                    ;        Move boot loader start address to RAMLO.

C629  18                     CLC
C62A  AD4202                 LDA     BOOTAD
C62D  6906                   ADC     #6
C62F  8504                   STA     RAMLO            ;boot loader start address
C631  AD4302                 LDA     BOOTAD+1
C634  6900                   ADC     #0
C636  8505                   STA     RAMLO+1

                    ;        Execute boot loader.
```

```
C638  6C0400              JMP     (RAMLO)           ;execute boot loader


                    **      IBS - Initialize Booted Software
                    *
                    *       ENTRY   JSR     IBS
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = C63B        IBS     =       *                 ;entry
C63B  6C0C00              JMP     (DOSINI)          ;initialize booted software


                    **      DBE - Display Boot Error Message
                    *
                    *       ENTRY   JSR     DBE
                    *
                    *       NOTES
                    *               Problem: bytes wasted by LDX/TXA and LDY/TY:
                    *               combinations.
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = C63E        DBE     =       *                 ;entry

                    ;       Set up IOCB.

C63E  A23D                LDX     #low BMSG         ;boot error message
C640  A0C4                LDY     #high BMSG
C642  8A                  TXA
C643  A200                LDX     #SEIOCB           ;screen editor IOCB index
C645  9D4403              STA     ICBAL,X           ;low buffer address
C648  98                  TYA
C649  9D4503              STA     ICBAH,X           ;high buffer address
C64C  A909                LDA     #PUTREC
C64E  9D4203              STA     ICCOM,X           ;command
C651  A9FF                LDA     #$FF
C653  9D4803              STA     ICBLL,X           ;buffer length

                    ;       Perform CIO.

C656  4C56E4              JMP     CIOV              ;vector to CIO, return
```

```
                        **      GNS - Get Next Sector
                        *
                        *       ENTRY   JSR     GNS
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


        = C659          GNS     =       *       ;entry

                        ;       Check type of boot.

C659  ADEA03                    LDA     CASSBT
C65C  F003 ^C661                BEQ     GNS1    ;if not cassette boot

                        ;       Read block from cassette.

C65E  4C7AE4                    JMP     RBLOKV  ;vector to read cassette block rout:

                        ;       Read sector from disk.

C661  A952              GNS1    LDA     #READ
C663  8D0203                    STA     DCOMND  ;command
C666  A901                      LDA     #1      ;drive number 1
C668  8D0103                    STA     DUNIT   ;set drive number
C66B  4C53E4                    JMP     DSKINV  ;vector to DIO, return



                        **      ACB - Attempt Cassette Boot
                        *
                        *       ENTRY   JSR     ACB
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


        = C66E          ACB     =       *               ;entry

                        ;       Check type.

C66E  A508                      LDA     WARMST          ;warmstart flag
C670  F009 ^C67B                BEQ     ACB1            ;if coldstart

                        ;       Perform warmstart procedures.

C672  A509                      LDA     BOOT?           ;successful boot flags
C674  2902                      AND     #$02            ;successful cassette boot i:
C676  F027 ^C69F                BEQ     ACB2            ;if cassette boot not succe:

C678  4CA0C6                    JMP     ACB3            ;initialize cassette
```

```
                        ;       Perform coldstart procedures.

C67B  ADE903      ACB1  LDA     CKEY            ;cassette boot request flag
C67E  F01F ^C69F        BEQ     ACB2            ;if cassette boot not reque:

                        ;       Boot cassette.

C680  A980              LDA     #$80
C682  853E              STA     FTYPE           ;set long IRG type
C684  EEEA03            INC     CASSBT          ;set cassette boot flag
C687  207DE4            JSR     CSOPIV          ;open cassette for input
C68A  208BC5            JSR     BAI             ;boot and initialize
C68D  A900              LDA     #0
C68F  8DEA03            STA     CASSBT          ;clear cassette boot flag
C692  8DE903            STA     CKEY            ;clear cassette boot reques:
C695  0609              ASL     BOOT?           ;indicate successful casset:

C697  A50C              LDA     DOSINI
C699  8502              STA     CASINI          ;cassette initialization ad:
C69B  A50D              LDA     DOSINI+1
C69D  8503              STA     CASINI+1

                        ;       Exit.

C69F  60          ACB2  RTS                     ;return

                        ;       Initialize cassette booted program.

C6A0  6C0200      ACB3  JMP     (CASINI)        ;initialize cassette booted:
```

```
C6A3                    **      IDIO - Initialize DIO
                        *
                        *       ENTRY   JSR     IDIO
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

     = C6A3             IDIO    =       *               ;entry
C6A3  A9A0                      LDA     #160            ;160 second timeout
C6A5  8D4602                    STA     DSKTIM          ;set initial disk timeout
C6A8  A980                      LDA     #low DSCTSZ     ;disk sector size
C6AA  8DD502                    STA     DSCTLN
C6AD  A900                      LDA     #high DSCTSZ
C6AF  8DD602                    STA     DSCTLN+1
C6B2  60                        RTS                     ;return


                        **      DIO - Disk I/O
                        *
                        *       ENTRY   JSR     DIO
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

     = C6B3             DIO     =       *               ;entry

                        ;       Initialize.

C6B3  A931                      LDA     #DISKID         ;disk bus ID
C6B5  8D0003                    STA     DDEVIC          ;device bus ID
C6B8  AD4602                    LDA     DSKTIM          ;timeout
C6BB  AE0203                    LDX     DCOMND          ;command
C6BE  E021                      CPX     #FOMAT
C6C0  F002 ^C6C4                BEQ     DIO1            ;if FORMAT command

C6C2  A907                      LDA     #7              ;set timeout to 7 seconds

C6C4  8D0603            DIO1    STA     DTIMLO          ;timeout

                        ;       Set SIO command.

C6C7  A240                      LDX     #GETDAT         ;assume GET DATA

C6C9  AD0203                    LDA     DCOMND          ;command
C6CC  C950                      CMP     #PUTSEC
C6CE  F004 ^C6D4                BEQ     DIO2            ;if PUT SECTOR command

C6D0  C957                      CMP     #WRITE
C6D2  D002 ^C6D6                BNE     DIO3            ;if not WRITE command
```

```
C6D4  A280          DIO2    LDX     #PUTDAT         ;select PUT DATA

                    ;       Check command.

C6D6  C953          DIO3    CMP     #STATC
C6D8  D010 ^C6EA            BNE     DIO4            ;if not STATUS command

                    ;       Set up STATUS command.

C6DA  A9EA                  LDA     #low DVSTAT
C6DC  8D0403                STA     DBUFLO          ;buffer address
C6DF  A902                  LDA     #high DVSTAT
C6E1  8D0503                STA     DBUFHI
C6E4  A004                  LDY     #low 4          ;low byte count
C6E6  A900                  LDA     #high 4         ;high byte count
C6E8  F006 ^C6F0            BEQ     DIO5            ;perform SIO

                    ;       Set up other commands.

C6EA  ACD502        DIO4    LDY     DSCTLN          ;low byte count
C6ED  ADD602                LDA     DSCTLN+1        ;high byte count

                    ;       Perform SIO.

C6F0  8E0303        DIO5    STX     DSTATS          ;SIO command
C6F3  8C0803                STY     DBYTLO          ;low byte count
C6F6  8D0903                STA     DBYTHI          ;high byte count
C6F9  2059E4                JSR     SIOV            ;vector to SIO
C6FC  1001 ^C6FF            BPL     DIO6            ;if no error

                    ;       Process error.

C6FE  60                    RTS                     ;return

                    ;       Process successful operation.

C6FF  AD0203        DIO6    LDA     DCOMND          ;command
C702  C953                  CMP     #STATC
C704  D00A ^C710            BNE     DIO7            ;if not STATUS command

C706  203AC7                JSR     SBA             ;set buffer address
C709  A002                  LDY     #2
C70B  B115                  LDA     (BUFADR),Y      ;timeout status
C70D  8D4602                STA     DSKTIM          ;disk timeout

                    ;       Set byte count.

C710  AD0203        DIO7    LDA     DCOMND
C713  C921                  CMP     #FOMAT
C715  D01F ^C736            BNE     DIO10           ;if not FORMAT command

C717  203AC7                JSR     SBA             ;set buffer address
C71A  A0FE                  LDY     #$FE            ;initial buffer pointer

C71C  C8            DIO8    INY                     ;increment buffer pointer
C71D  C8                    INY                     ;increment buffer pointer
```

```
C71E  B115           DIO9    LDA     (BUFADR),Y        ;low bad sector data
C720  C9FF                   CMP     #$FF
C722  D0F8 ^C71C             BNE     DIO8              ;if low not $FF

C724  C8                     INY
C725  B115                   LDA     (BUFADR),Y        ;high bad sector data
C727  C8                     INY
C728  C9FF                   CMP     #$FF
C72A  D0F2 ^C71E             BNE     DIO9              ;if high not $FF

C72C  88                     DEY
C72D  88                     DEY
C72E  8C0803                 STY     DBYTLO            ;low bad sector byte count
C731  A900                   LDA     #0
C733  8D0903                 STA     DBYTHI            ;high bad sector byte count

                     ;       Exit.

C736  AC0303         DIO10   LDY     DSTATS            ;status
C739  60                     RTS                       ;return


                     **      SBA - Set Buffer Address
                     *
                     *       ENTRY   JSR     SBA
                     *
                     *       MODS
                     *               Original Author Unknown
                     *               1. Bring closer to Coding Standard (object :
                     *                  R. K. Nordin 11/01/83

      = C73A         SBA     =       *                 ;entry
C73A  AD0403                 LDA     DBUFLO
C73D  8515                   STA     BUFADR            ;buffer address
C73F  AD0503                 LDA     DBUFHI
C742  8516                   STA     BUFADR+1
C744  60                     RTS                       ;return
```

```
C745                    **      RLR - Relocate Routine
                        *
                        *       RLR reloactes a relocatable routine which is assemb:
                        *       origin 0.
                        *
                        *       ENTRY   JSR     RLR
                        *               GBYTEA = GBYTEA+1 = address of get-byte rou:
                        *
                        *       MODS
                        *               Y. M. Chen      04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

     = C745             RLR     =       *               ;entry

                        ;       Clear parameter block.

C745  A205                      LDX     #5              ;offset to last parameter

C747  A900              RLR1    LDA     #0
C749  9DC902                    STA     PARMBL,X        ;clear byte of parameter bl:
C74C  CA                        DEX
C74D  10F8 ^C747                BPL     RLR1            ;if not done

                        ;       Get a new record type and set the subroutine vector:

C74F  A900              RLR2    LDA     #0
C751  8D3302                    STA     LCOUNT          ;process 0th byte of a reco:
C754  20CFC7                    JSR     GBY             ;get type ID
C757  A09C                      LDY     #DATAER
C759  B039 ^C794                BCS     RLR4            ;if EOF before END record

C75B  8D8802                    STA     HIBYTE          ;save type ID
C75E  20CFC7                    JSR     GBY             ;get record length
C761  A09C                      LDY     #DATAER
C763  B02F ^C794                BCS     RLR4            ;if EOF before END record

C765  8D4502                    STA     RECLEN
C768  AD8802                    LDA     HIBYTE          ;get type ID
C76B  C90B                      CMP     #$0B            ;END record
C76D  F026 ^C795                BEQ     END             ;if END record

C76F  2A                        ROL     A               ;set subroutine vectors
C770  AA                        TAX
C771  BDE4C8                    LDA     TRPR,X
C774  8DC902                    STA     RUNADR
C777  BDE5C8                    LDA     TRPR+1,X
C77A  8DCA02                    STA     RUNADR+1

C77D  AD4502            RLR3    LDA     RECLEN
C780  CD3302                    CMP     LCOUNT
C783  F0CA ^C74F                BEQ     RLR2            ;if LCOUNT=RECLEN, get new :

C785  20CFC7                    JSR     GBY             ;get next byte
C788  A09C                      LDY     #DATAER
C78A  B008 ^C794                BCS     RLR4            ;if EOF before END record
```

```
C78C  20D2C7                 JSR     CAL             ;call record subroutine
C78F  EE3302                 INC     LCOUNT
C792  D0E9 ^C77D             BNE     RLR3            ;continue

C794  60            RLR4     RTS                     ;return



                    **       END - Handle END Record
                    *
                    *        END handles record type of
                    *        1.End Record
                    *
                    *        Record format:
                    *        Byte 0          Type ID
                    *        Byte 1          Self-start flag
                    *        Bytes 2 - 3     Run address
                    *
                    *        Process formula
                    *
                    *        RUNADR+LOADAD ==> Start Execution Address n Loader-:
                    *        parameter block.
                    *
                    *        End record calculates the start execution address b:
                    *        RUNADR with LOADAD, and returns to the Caller with :
                    *        block and a status byte in the Y register. Y=1 mean:
                    *        successful, else is a data structure error.
                    *
                    *        ENTRY   JSR     END
                    *
                    *        MODS
                    *                Y. M. Chen      04/01/82
                    *                1. Bring closer to Coding Standard (object :
                    *                   R. K. Nordin 11/01/83


      = C795        END      =       *       ;entry
C795  20CFC7                 JSR     GBY     ;get low byte of the RUNADR
C798  A09C                   LDY     #DATAER
C79A  B02C ^C7C8             BCS     END3    ;if EOF before END record

C79C  8DC902                 STA     RUNADR
C79F  20CFC7                 JSR     GBY     ;get high byte of the RUNADR
C7A2  A09C                   LDY     #DATAER
C7A4  B022 ^C7C8             BCS     END3    ;if EOF before END record

C7A6  8DCA02                 STA     RUNADR+1
C7A9  AD4502                 LDA     RECLEN  ;RECLEN here is self-start flag
C7AC  C901                   CMP     #1
C7AE  F016 ^C7C6             BEQ     END2    ;if 1, an absolute RUNADR, no fixup

C7B0  9017 ^C7C9             BCC     END4    ;if 0, this is not a self-start pro:

                    ;        Process relative start.
```

```
C7B2  18                    CLC
C7B3  ADC902                LDA     RUNADR          ;execution address, needs f:
C7B6  6DD102                ADC     LOADAD
C7B9  A8                    TAY
C7BA  ADCA02                LDA     RUNADR+1
C7BD  6DD202                ADC     LOADAD+1        ;A= high byte, Y=low byte

C7C0  8CC902       END1     STY     RUNADR          ;set up Loader-Caller param:
C7C3  8DCA02                STA     RUNADR+1

C7C6  A001         END2     LDY     #SUCCES         ;Y=1 successful operation

C7C8  60           END3     RTS                     ;return

C7C9  A000         END4     LDY     #0              ;fill self-start parameter :
C7CB  A900                  LDA     #0              ;for non-self start program
C7CD  F0F1 ^C7C0            BEQ     END1            ;continue


                   **       GBY - Get Byte
                   *
                   *        ENTRY   JSR     GBY
                   *
                   *        MODS
                   *                Y. M. Chen      04/01/82
                   *                1. Bring closer to Coding Standard (object :
                   *                   R. K. Nordin 11/01/83

      = C7CF       GBY      =       *               ;entry
C7CF  6CCF02                JMP     (GBYTEA)        ;get byte, return


                   **       CAL - Execute at Run Address
                   *
                   *        ENTRY   JSR     CAL
                   *
                   *        MODS
                   *                Y. M. Chen      04/01/82
                   *                1. Bring closer to Coding Standard (object :
                   *                   R. K. Nordin 11/01/83

      = C7D2       CAL      =       *               ;entry
C7D2  6CC902                JMP     (RUNADR)        ;process record, return
```

```
                        **      TEX - Handle Text Record
                        *
                        *       TEX handles record types of
                        *
                        *       1.Non-zero page relocatable text
                        *       2.Zero page relocatable text
                        *       3.Absolute text
                        *
                        *       Record format
                        *
                        *       |Type   |Length         |Relative addr. |text   |
                        *       |ID     |(RECLEN)       |(RELADR)       |       |
                        *
                        *       Process formula
                        *       A register ===> (NEWADR+LCOUNT)
                        *
                        *       Relocate object text into fixed address of NEWADR+L:
                        *
                        *       ENTRY   JSR     TEX
                        *
                        *       NOTES
                        *
                        *       1.The relocating address (NEWADR) for absolute text:
                        *       relative address (RELADR), relocating address fixup:
                        *       needed.
                        *       2.There is no need to compare MEMTOP for processing:
                        *       text.
                        *       3.X register is used as an indexing to zero page va:
                        *       or non-zero page variables.    X=0 means pointing :
                        *       page variable, whereas X=2 means pointing to zero p:
                        *       variables.
                        *       4.Each byte of the object text comes in A register.
                        *
                        *       MODS
                        *               Y. M. Chen      04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

     = C7D5     TEX     =       *       ;entry
C7D5  AC3302            LDY     LCOUNT  ;A register=data coming in
C7D8  C001              CPY     #$01
C7DA  F00A ^C7E6        BEQ     TEX1    ;if 1, process highest used address

C7DC  B073 ^C851        BCS     FTX     ;if 2 or greater, relocate object t:

C7DE  8D4A02            STA     RELADR
C7E1  8D8E02            STA     NEWADR  ;for absolute text NEWADR=RELADR
C7E4  906A ^C850        BCC     TEX8

                        ;       Set highest used address.

C7E6  8D4B02    TEX1    STA     RELADR+1        ;save high byte of RELADR
C7E9  8D8F02            STA     NEWADR+1        ;for absolute text NEWADR=R:
C7EC  A200              LDX     #0              ;X=an index to non-zero or :
C7EE  AD8802            LDA     HIBYTE          ;HIBYTE=Type ID
C7F1  F006 ^C7F9        BEQ     TEX2            ;if 0, process non-zero pag:
```

```
C7F3  C90A                CMP     #$0A
C7F5  F015 ^C80C          BEQ     TEX3               ;if $0A, needs no relative :

C7F7  A202                LDX     #2                 ;X=2 for zero page text rec:

C7F9  18          TEX2    CLC                        ;fix relocating addr. for n:
C7FA  AD4A02              LDA     RELADR             ;text & zero page text
C7FD  7D0102              ADC     LOADAD,X           ;NEWADR=RELADR+LOADAD
C800  8D8E02              STA     NEWADR
C803  AD4B02              LDA     RELADR+1
C806  7D0202              ADC     LOADAD+1,X
C809  8D8F02              STA     NEWADR+1           ;Loader start relocating

C80C  18          TEX3    CLC
C80D  AD8E02              LDA     NEWADR  ;NEWADR+RECLEN is the last used mem:
C810  6D4502              ADC     RECLEN  ;for this particular record
C813  48                  PHA
C814  A900                LDA     #0      ;A=high byte, S=low byte
C816  6D8F02              ADC     NEWADR+1
C819  A8                  TAY             ;high byte
C81A  68                  PLA             ;low byte
C81B  38                  SEC
C81C  E902                SBC     #2      ;skip unwanted 2 bytes of relative :
C81E  B001 ^C821          BCS     TEX4

C820  88                  DEY

C821  48          TEX4    PHA
C822  98                  TYA
C823  DDCC02              CMP     HIUSED+1,X         ;HIUSED stores the highest :
C826  68                  PLA
C827  9010 ^C839          BCC     TEX6               ;if HIUSED>(NEWADR+RECLEN),:

C829  D005 ^C830          BNE     TEX5               ;if HIUSED<=(NEWADR+RECLEN)

C82B  DDCB02              CMP     HIUSED,X
C82E  9009 ^C839          BCC     TEX6

                  ;       Update HIUSED.

C830  9DCB02      TEX5    STA     HIUSED,X           ;update HIUSED
C833  48                  PHA
C834  98                  TYA
C835  9DCC02              STA     HIUSED+1,X
C838  68                  PLA

C839  AE8802      TEX6    LDX     HIBYTE
C83C  E001                CPX     #$01
C83E  F010 ^C850          BEQ     TEX8    ;if zero page text

                  ;       Check MEMTOP.

C840  CCE602              CPY     MEMTOP+1           ;MEMTOP>HIUSED, OK
C843  9008 ^C850          BCC     TEX8

C845  D005 ^C84C          BNE     TEX7
```

```
C847  CDE502              CMP     MEMTOP
C84A  9004 ^C850          BCC     TEX8

C84C  68          TEX7    PLA                         ;MEMTOP<=HIUSED then error
C84D  68                  PLA                         ;do a force return to calle:
C84E  A09D                LDY     #MEMERR             ;set memory insufficient fl:

C850  60          TEX8    RTS                         ;return


                  **      FTX - Relocate Text into Memory
                  *
                  *       ENTRY   JSR      FTX
                  *
                  *       NOTES
                  *               Problem: bytes wasted by JMP to RTS.
                  *
                  *       MODS
                  *               Y. M. Chen       04/01/82
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83

      = C851      FTX     =       *       ;entry
C851  38                  SEC
C852  48                  PHA             ;A register has object text
C853  AD3302              LDA     LCOUNT  ;LCOUNT counts 2 bytes of relative :
C856  E902                SBC     #2      ;-2 is the total bytes of object te:
C858  18                  CLC
C859  6D8E02              ADC     NEWADR
C85C  8536                STA     LTEMP   ;A ===>(NEWADR+LCOUNT-2)
C85E  A900                LDA     #0
C860  6D8F02              ADC     NEWADR+1
C863  8537                STA     LTEMP+1
C865  68                  PLA
C866  A000                LDY     #0
C868  9136                STA     (LTEMP),Y
C86A  4C50C8              JMP     TEX8    ;return


                  **      WOR - Handle Word Reference Record Type
                  *
                  *       WOR handles record types of
                  *
                  *       1.Non-zero page word references to non-zero page.
                  *       2.Zero page word references to non-zero page.
                  *
                  *       Record format
                  *
                  *       |Type    |Length             |Offset1|Offset2|Offsetn|
                  *       | ID     |(RECLEN)           |A Reg. |       |       |
                  *
                  *       Process formula
```

```
                        *
                        *       (A register +NEWADR)W +LOADAD ===> (NEWADR+ A regis:
                        *
                        *       Count, the offset from the start relocating address:
                        *       low byte
                        *       of a word needing to be fixed.  The fixup process i:
                        *       content of the word and add loading address, then r:
                        *       fixed word.
                        *
                        *       Offset information comes in A register.
                        *
                        *       ENTRY   JSR     WOR
                        *
                        *       MODS
                        *               Y. M. Chen      04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                 R. K. Nordin 11/01/83


      = C86D            WOR     =       *               ;entry
C86D  18                        CLC
C86E  6D8E02                    ADC     NEWADR          ;offset in A register
C871  8536                      STA     LTEMP
C873  A900                      LDA     #0
C875  6D8F02                    ADC     NEWADR+1
C878  8537                      STA     LTEMP+1         ;offset +NEWADR= LTEMP
C87A  A000                      LDY     #0
C87C  B136                      LDA     (LTEMP),Y       ;get low byte content of wh:
C87E  18                        CLC
C87F  6D0102                    ADC     LOADAD          ;fix low byte of a word
C882  9136                      STA     (LTEMP),Y
C884  E636                      INC     LTEMP           ;increment LTEMP pointer by:
C886  D002 ^C88A                BNE     WOR1            ;if low not zero

C888  E637                      INC     LTEMP+1         ;increment high

C88A  B136              WOR1    LDA     (LTEMP),Y       ;fix high byte of a word
C88C  6D0202                    ADC     LOADAD+1
C88F  9136                      STA     (LTEMP),Y       ;restore processed content
C891  60                        RTS                     ;return



                        **      LOO - Handle Low Byte and One Byte Record Types
                        *
                        *       LOO handles record types of
                        *
                        *       1.Non-zero page low byte references to non-zero ppa:
                        *       2.Zero page low-byte references to non-zero page.
                        *       3.Non-zero page one byte references to zero page.
                        *       4.Zero page one byte references to zero page.
                        *
                        *       Record format
                        *
                        *       |Type   |Length         |Offset1|Offset2|Offsetn|
                        *       |ID     |(RECLEN)       |A Reg. |A Reg. |       |
```

```
                        *
                        *       The process formula for non-zero page low byte refe:
                        *       non-zero page record and zero page low byte referen:
                        *       non-zero page record is
                        *
                        *       (offset + NEWADR)+LOADAD ===> (offset +NEWADR)
                        *
                        *       The process formula for non-zero page one byte refe:
                        *       zero
                        *       page record and zero page one byte references to ze:
                        *       record
                        *       is
                        *
                        *       (offset + NEWADR)+LOADADZ ===> (offset + NEWADR)
                        *
                        *       Count from the offset from the start relocating add:
                        *       low byte or one byte need to be fixed. Get the cont:
                        *       low byte or one byteand add either LOADAD or LOADAD:
                        *       page loading address), then restore the value.
                        *
                        *       The offset comes in A register.
                        *
                        *       The X register for this routine points to either no:
                        *       variables or zero page variables. Record type 2 & 3:
                        *       non-zero page variable, type 4 & 5 needs zero page :
                        *
                        *       X=2 points to zero page variable.
                        *
                        *       ENTRY   JSR     L00
                        *
                        *       MODS
                        *               Y. M. Chen      04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

       = C892           L00     =       *       ;entry
  C892  A200                    LDX     #0      ;X=0 points to non-zero page variab:
  C894  AC8802                  LDY     HIBYTE  ;HIBYTE has Type ID
  C897  C004                    CPY     #4      ;type 4 & 5 needs zero page variabl:
  C899  9002 ^C89D              BCC     L001    ;if type 2 or 3, need non-zero page:

  C89B  A202                    LDX     #2      ;point to zero page variable

  C89D  18              L001    CLC             ;offset is in A register
  C89E  6D8E02                  ADC     NEWADR  ;offset+NEWADR=the byte needs fixup
  C8A1  8536                    STA     LTEMP
  C8A3  A900                    LDA     #0
  C8A5  6D8F02                  ADC     NEWADR+1
  C8A8  8537                    STA     LTEMP+1
  C8AA  A000                    LDY     #0
  C8AC  B136                    LDA     (LTEMP),Y       ;get the content of offset+:
  C8AE  18                      CLC
  C8AF  7D0102                  ADC     LOADAD,X        ;do relocating fixup
  C8B2  9136                    STA     (LTEMP),Y       ;restore the being fixed va:
  C8B4  60                      RTS                     ;return
```

```
                        **      HIG - Handle High Byte Record Types
                        *
                        *       HIG handles record types of
                        *
                        *       1.Non-zero page high bytes references to non-zero p:
                        *       2.Zero page high bytes references to non-zero page.
                        *
                        *       Record format
                        *
                        *       |Type   |Length         |Offset1|Low Byte|Offset2|L:
                        *       |ID     |(RECLEN)       |HIBYTE |A Reg.  | (HIBYTE):
                        *
                        *       Process formula
                        *
                        *       (HIBYTE+NEWADR)+[[LOADAD+A]/256] ==> (HIBYTE+NEWADR:
                        *
                        *       Count the offset from the start relocating address :
                        *       byte needs to be fixed. Get the low byte informatio:
                        *       A register, then add the low byte with LOADAD and s:
                        *       flag depending on the calculation. Next do an addit:
                        *       high byte, NEWADR and the C flag. Restore the addit:
                        *       back to the high byte location.in memory.
                        *
                        *       HIBYTE is not Type ID here. HIBYTE is used to store:
                        *       byte value.
                        *
                        *       ENTRY   JSR     HIG
                        *
                        *       NOTES
                        *               Problem: many instances of jumping to RTS i:
                        *               wastes bytes.
                        *
                        *       MODS
                        *               Y. M. Chen      04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C8B5          HIG     =       *               ;entry

                        ;       Initialize.

C8B5    48                      PHA                     ;save offset pointing to hi:

                        ;       Check LCOUNT odd or even.

C8B6    AD3302                  LDA     LCOUNT
C8B9    6A                      ROR     A
C8BA    68                      PLA
C8BB    B015 ^C8D2              BCS     HIG2            ;if even number, process lo:

                        ;       Process high byte.

C8BD    18                      CLC
C8BE    6D3E02                  ADC     NEWADR
C8C1    8536                    STA     LTEMP           ;get high byte value
C8C3    A900                    LDA     #0
```

```
C8C5  6D8F02                 ADC     NEWADR+1
C8C8  8537                   STA     LTEMP+1
C8CA  A000                   LDY     #0
C8CC  B136                   LDA     (LTEMP),Y
C8CE  8D8802                 STA     HIBYTE          ;save high byte content

C8D1  60            HIG1     RTS                     ;return

                    ;        Process low byte.

C8D2  18            HIG2     CLC
C8D3  6DD102                 ADC     LOADAD          ;add low byte with LOADAD
C8D6  A900                   LDA     #0
C8D8  6DD202                 ADC     LOADAD+1
C8DB  6D8802                 ADC     HIBYTE          ;C flag+LOADAD(high byte)+H:
C8DE  A000                   LDY     #0
C8E0  9136                   STA     (LTEMP),Y       ;store being fixed high byt:
C8E2  F0ED ^C8D1             BEQ     HIG1


                    **       TRPR - Table of Record Processing Routines


C8E4  D5C7          TRPR     DW      TEX     ;0 - non-zero page relocatable text
C8E6  D5C7                   DW      TEX     ;1 - zero page relocatable text
C8E8  92C8                   DW      LOO     ;2 - non-zero page low byte to non-:
C8EA  92C8                   DW      LOO     ;3 - zero page low byte to non-zero:
C8EC  92C8                   DW      LOO     ;4 - non-zero page one byte to zero:
C8EE  92C8                   DW      LOO     ;5 - zero page one byte to zero pag:
C8F0  6DC8                   DW      WOR     ;6 - non-zero page word to non-zero:
C8F2  6DC8                   DW      WOR     ;7 - zero page word to non-zero pag:
C8F4  85C8                   DW      HIG     ;8 - non-zero page high byte to non:
C8F6  85C8                   DW      HIG     ;9 - zero page high byte to non-zer:
C8F8  D5C7                   DW      TEX     ;10 - absolute text
C8FA  95C7                   DW      END     ;11 - end record
```

```
C8FC                    **      SES - Select and Execute Self-test
                        *
                        *       SES selects the self-test ROM and executes the self:
                        *
                        *       ENTRY   JSR     SES
                        *
                        *       NOTES
                        *               Problem: this could be contiguous with othe:
                        *               self-test code (near TST0).
                        *
                        *       MODS
                        *               M. W. Colburn   10/26/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


        = C8FC          SES     =       *       ;entry

C8FC    A9FF                    LDA     #$FF
C8FE    8D4402                  STA     COLDST  ;force coldstart on RESET

C901    AD01D3                  LDA     PORTB
C904    297F                    AND     #$7F    ;enable self-test ROM
C906    8D01D3                  STA     PORTB   ;update port B memory control

C909    4C83E4                  JMP     SLFTSV  ;vector to self-test
```

```
C90C                  **      GIN - Initialize Generic Parallel Device
                      *
                      *       ENTRY   JSR     GIN
                      *
                      *       MODS
                      *               Y. M. Chen      02/18/83
                      *               1. Bring closer to Coding Standard (object :
                      *                  R. K. Nordin 11/01/83


      = C90C          GIN     =       *       ;entry

                      ;       Initialize.

C90C  A901                    LDA     #$01    ;initially select device 0
C90E  8D4802                  STA     SHPDVS  ;device select shadow

                      ;       For each potential device, initialize if device pre:

C911  AD4802          GIN1    LDA     SHPDVS  ;device select shadow
C914  8DFFD1                  STA     PDVS    ;device select

C917  AD03D8                  LDA     PDID1   ;first ID
C91A  C980                    CMP     #$80    ;required value
C91C  D00A ^C928              BNE     GIN2    ;if first ID not verified

C91E  AD0BD8                  LDA     PDID2   ;second ID
C921  C991                    CMP     #$91    ;required value
C923  D003 ^C928              BNE     GIN2    ;if second ID not verified

C925  2019D8                  JSR     PDVV+12 ;initialize parallel device handler

C928  0E4802          GIN2    ASL     SHPDVS  ;advance to next device
C92B  D0E4 ^C911              BNE     GIN1    ;if devices remain

                      ;       Exit.

C92D  A900                    LDA     #$00    ;select FPP (deselect device)
                      ;       STA     SHPDVS  ;device select shadow
C92F  8DFFD1                  STA     PDVS    ;device select
C932  60                      RTS             ;return



                      **      PIO - Parallel Input/Output
                      *
                      *       ENTRY   JSR     PIO
                      *
                      *       NOTES
                      *               Problem: in the CRASS65 version, CRITIC was:
                      *               zero-page.
                      *
                      *       MODS
                      *               Y. M. Chen      02/18/83
                      *               1. Bring closer to Coding Standard (object :
                      *                  R. K. Nordin 11/01/83
```

```
       = C933      PIO      =       *        ;entry

                   ;        Initialize.

C933   A901                 LDA     #1
                   ;        STA     CRITIC   ;indicate critical section
C935   8D4200               VFD     8\$8D,8\low CRITIC,8\high CRITIC
C938   AD0103               LDA     DUNIT    ;device unit number
C93B   48                   PHA              ;save device unit number
C93C   AD4702               LDA     PDVMSK   ;device selection mask
C93F   F01A ^C95B           BEQ     PIO2     ;if no device to select

                   ;        For each device, pass request to device I/O routine:

C941   A208                 LDX     #TPDSL   ;offset to first byte beyond table

C943   20AFC9      PIO1     JSR     SNP      ;select next parallel device
C946   F013 ^C95B           BEQ     PIO2     ;if no device selected

C948   8A                   TXA
C949   48                   PHA              ;save offset
C94A   2005D8               JSR     PDIOV    ;perform parallel device I/O
C94D   68                   PLA              ;saved offset
C94E   AA                   TAX              ;restore offset
C94F   90F2 ^C943           BCC     PIO1     ;if device did not field request

                   ;        Restore Floating Point Package.

C951   A900                 LDA     #$00     ;select FPP (deselect device)
C953   8D4802               STA     SHPDVS   ;device select shadow
C956   8DFFD1               STA     PDVS     ;device select
C959   F003 ^C95E           BEQ     PIO3     ;exit

                   ;        Perform SIO.

C95B   2071E9      PIO2     JSR     SIO      ;perform SIO

                   ;        Exit.

C95E   68          PIO3     PLA              ;saved device unit number
C95F   8D0103               STA     DUNIT    ;restore device unit number
C962   A900                 LDA     #0
                   ;        STA     CRITIC   ;indicate non-critical section
C964   8D4200               VFD     8\$8D,8\low CRITIC,8\high CRITIC
C967   8C0303               STY     DSTATS
C96A   AC0303               LDY     DSTATS   ;status (re-establish N)
C96D   60                   RTS              ;return
```

```
                        **      PIR - Handle Parallel Device IRQ
                        *
                        *       ENTRY   JMP     PIR
                        *
                        *       EXIT
                        *               Exits via RTI
                        *
                        *       MODS
                        *               Y. M. Chen      02/18/83
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = C96E            PIR     =       *       ;entry

                        ;       Determine which device made IRQ, in order of priori:

C96E  A208                      LDX     #TPDSL  ;offset to first byte beyond table

C970  6A                PIR1    ROR     A
C971  B003 ^C976                BCS     PIR2    ;if IRQ of that device

C973  CA                        DEX
C974  D0FA ^C970                BNE     PIR1    ;if devices remain

                        ;       Select device and process IRQ.

C976  AD4802            PIR2    LDA     SHPDVS          ;current device selection
C979  48                        PHA                     ;save current device select:
C97A  BD20CA                    LDA     TPDS-1,X        ;device selection desired
C97D  8D4802                    STA     SHPDVS          ;device select shadow
C980  8DFFD1                    STA     PDVS            ;device select
C983  2008D8                    JSR     PDIRQV          ;process IRQ

                        ;       Exit.

C986  68                        PLA                     ;saved device selection
C987  8D4802                    STA     SHPDVS          ;restore device select shad:
C98A  8DFFD1                    STA     PDVS            ;device select
C98D  68                        PLA                     ;saved X
C98E  AA                        TAX                     ;restore X
C98F  68                        PLA                     ;restore A
C990  40                        RTI                     ;return
```

```
                        **      GOP - Perform Generic Parallel Device OPEN
                        *
                        *       ENTRY   JSR     GOP
                        *
                        *       MODS
                        *               Y. M. Chen      02/18/83
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C991          GOP     =       *       ;entry
C991  A001                      LDY     #1      ;offset for OPEN
C993  4CDCC9                    JMP     EPC     ;execute parallel device handler co:


                        **      GCL - Perform Generic Parallel Device CLOSE
                        *
                        *       ENTRY   JSR     GCL
                        *
                        *       MODS
                        *               Y. M. Chen      02/18/83
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C996          GCL     =       *       ;entry
C996  A003                      LDY     #3      ;offset for CLOSE
C998  4CDCC9                    JMP     EPC     ;execute parallel device handler co:


                        **      GGB - Perform Generic Parallel Device GET-BYTE
                        *
                        *       ENTRY   JSR     GGB
                        *
                        *       MODS
                        *               Y. M. Chen      02/18/83
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C99B          GGB     =       *       ;entry
C99B  A005                      LDY     #5      ;offset for GET-BYTE
C99D  4CDCC9                    JMP     EPC     ;execute parallel device handler co:
```

```
                        **      GPB - Perform Generic Parallel Device PUT-BYTE
                        *
                        *       ENTRY   JSR     GPB
                        *
                        *       MODS
                        *               Y. M. Chen      02/18/83
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C9A0          GPB     =       *       ;entry
C9A0    A007                    LDY     #7      ;offset for PUT-BYTE
C9A2    4CDCC9                  JMP     EPC     ;execute parallel device handler co:


                        **      GST - Perform Generic Parallel Device STATUS
                        *
                        *       ENTRY   JSR     GST
                        *
                        *       MODS
                        *               Y. M. Chen      02/18/83
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C9A5          GST     =       *       ;entry
C9A5    A009                    LDY     #9      ;offset for STATUS
C9A7    4CDCC9                  JMP     EPC     ;execute parallel device handler co:


                        **      GSP - Perform Generic Parallel Device SPECIAL
                        *
                        *       ENTRY   JSR     GSP
                        *
                        *       MODS
                        *               Y. M. Chen      02/18/83
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = C9AA          GSP     =       *       ;entry
C9AA    A00B                    LDY     #11     ;offset for SPECIAL
C9AC    4CDCC9                  JMP     EPC     ;execute parallel device handler co:
```

```
                    **      SNP - Select Next Parallel Device
                    *
                    *       ENTRY   JSR     SNP
                    *
                    *       MODS
                    *               Y. M. Chen      02/18/83
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


        = C9AF      SNP     =       *       ;entry

                    ;       Decrement and check offset.

C9AF  CA            SNP1    DEX             ;decrement offset
C9B0  1009 ^C9BB            BPL     SNP2    ;if devices remain

                    ;       Exit.

C9B2  A900                  LDA     #$00    ;select FPP (deselect device)
C9B4  8D4802                STA     SHPDVS  ;device select shadow
C9B7  8DFFD1                STA     PDVS    ;device select.
C9BA  60                    RTS             ;return

                    ;       Ensure device is indicated by selection mask.

C9BB  AD4702        SNP2    LDA     PDVMSK  ;device selection mask
C9BE  3D21CA                AND     TPDS,X  ;device select
C9C1  F0EC ^C9AF            BEQ     SNP1    ;if device not indicated for select:

                    ;       Select device.

C9C3  8D4802                STA     SHPDVS  ;device select shadow
C9C6  8DFFD1                STA     PDVS    ;device select
C9C9  60                    RTS             ;return



                    **      IPH - Invoke Parallel Device Handler
                    *
                    *       ENTRY   JSR     IPH
                    *               Y = offset into parallel device vector tabl:
                    *               PPTMPA = original A value
                    *               PPTMPX = original X value
                    *
                    *       NOTES
                    *               Problem: wasted byte for DEY.
                    *
                    *       MODS
                    *               Y. M. Chen      02/18/83
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


        = C9CA      IPH     =       *       ;entry
C9CA  B90DD8                LDA     PDVV,Y  ;high routine address-1
```

```
 C9CD  48                    PHA               ;place on stack
 C9CE  88                    DEY
 C9CF  B900D8                LDA     PDVV,Y    ;low routine address-1
 C9D2  48                    PHA               ;place on stack
 C9D3  AD4C02                LDA     PPTMPA    ;restore A for handler
 C9D6  AE4D02                LDX     PPTMPX    ;restore X for handler
 C9D9  A092                  LDY     #FNCNOT   ;preset status
 C9DB  60                    RTS               ;invoke handler routine (address on:


                     **      EPC - Execute Parallel Device Handler Command
                     *
                     *       ENTRY   JSR     EPC
                     *
                     *       NOTES
                     *               Problem: in the CRASS65 version, CRITIC was:
                     *               zero-page.
                     *
                     *       MODS
                     *               Y. M. Chen        02/18/83
                     *               1. Bring closer to Coding Standard (object :
                     *                  R. K. Nordin 11/01/83

       = C9DC        EPC     =       *         ;entry

                     ;       Initialize.

 C9DC  8D4C02                STA     PPTMPA    ;save data byte
 C9DF  8E4D02                STX     PPTMPX    ;save X
                     ;       LDA     CRITIC
 C9E2  AD4200                VFD     8\$AD,8\low CRITIC,8\high CRITIC
 C9E5  48                    PHA               ;save critical section status
 C9E6  A901                  LDA     #1
                     ;       STA     CRITIC    ;indicate critical section
 C9E8  8D4200                VFD     8\$8D,8\low CRITIC,8\high CRITIC

                     ;       For each device, pass request to device handler.

 C9EB  A208                  LDX     #TPDSL    ;offset to first byte beyond table

 C9ED  20AFC9        EPC1    JSR     SNP       ;select next device
 C9F0  F011 ^CA03            BEQ     EPC2      ;if no device selected, return erro:

 C9F2  8A                    TXA
 C9F3  48                    PHA               ;save offset
 C9F4  98                    TYA
 C9F5  48                    PHA               ;save Y
 C9F6  20CAC9                JSR     IPH       ;invoke parallel device handler
 C9F9  9020 ^CA1B            BCC     EPC4      ;if device did not field, try next :

                     ;       Clean up.

 C9FB  8D4C02                STA     PPTMPA    ;save possible data byte
 C9FE  68                    PLA               ;clean stack
```

```
C9FF  68                  PLA
CA00  4C05CA              JMP   EPC3     ;exit

                  ;       Return Nonexistent Device error.

CA03  A082        EPC2    LDY   #NONDEV

                  ;       Restore Floating Point Package.

CA05  A900        EPC3    LDA   #$00     ;select FPP (deselect device)
CA07  8D4802              STA   SHPDVS   ;device select shadow
CA0A  8DFFD1              STA   PDVS     ;device select
CA0D  68                  PLA            ;saved critical section status
                  ;       STA   CRITIC   ;restore critical section status
CA0E  8D4200              VFD   8\$8D,8\low CRITIC,8\high CRITIC
CA11  AD4C02              LDA   PPTMPA   ;restore possible data byte
CA14  8C4D02              STY   PPTMPX
CA17  AC4D02              LDY   PPTMPX   ;status (re-establish N)
CA1A  60                  RTS            ;return

                  ;       Prepare to try next device.

CA1B  68          EPC4    PLA
CA1C  A8                  TAY            ;restore Y
CA1D  68                  PLA
CA1E  AA                  TAX            ;restore X
CA1F  90CC ^C9ED          BCC   EPC1     ;try next device



                  **      TPDS - Table of Parallel Device Selects
                  *
                  *       NOTES
                  *             Problem: bytes wasted by replication of this
                  *             elsewhere.

CA21  80          TPDS    DB    $80      ;0 - device 7 (lowest priority)
CA22  40                  DB    $40      ;1 - device 6
CA23  20                  DB    $20      ;2 - device 5
CA24  10                  DB    $10      ;3 - device 4
CA25  08                  DB    $08      ;4 - device 3
CA26  04                  DB    $04      ;5 - device 2
CA27  02                  DB    $02      ;6 - device 1
CA28  01                  DB    $01      ;7 - device 0 (highest priority)

      = 0008      TPDSL   =     *-TPDS   ;length
```

```
CA29                  **      PHL - Load and Initialize Peripheral Handler
                      *
                      *       Subroutine to load, relocate, initialize and open a
                      *       "provisionally" opened IOCB. This routine is called
                      *       upon first I/O attempt following provisional open.
                      *       It does the final opening by simulating the first
                      *       part of a normal CIO OPEN and then finishing with
                      *       code which is in CIO.
                      *
                      *       Input parameters:
                      *       ICIDNO (specifies which IOCB);
                      *       various values in the provisionally-opened IOCB:
                      *               ICSPR (handler name)
                      *               ICSPR+1 (serial address for loading);
                      *       whatever the called subroutines require.
                      *
                      *       Output parameters:
                      *       None. (Error returns are all handled by called subr:
                      *               in fact, all returns are handled by called :
                      *
                      *       Modified:
                      *       ICHID in both calling IOCB and ZIOCB (part of compl:
                      *       ICCOMT (a CIO variable);
                      *       Registers not saved.
                      *
                      *       Subroutines called:
                      *       LPH (does the loading);
                      *       PHC (initializes the loaded handler);
                      *       FDH (a CIO entry--finds handler table entry of
                      *               newly loaded/initialized handler);
                      *       IIO (a CIO entry--finishes full proper opening of I:
                      *               including calling handler OPEN entry--IIO r:
                      *               to PHL's caller);
                      *       IND (a CIO entry--returns with error to PHL's calle:
                      *
                      *       ENTRY   JSR     PHL
                      *
                      *       NOTES
                      *               Problem: in the CRASS65 version, ICIDNO was:
                      *               zero-page.
                      *
                      *       MODS
                      *               R. S. Scheiman  04/01/82
                      *               1. Bring closer to Coding Standard (object :
                      *                  R. K. Nordin 11/01/83

      = CA29          PHL     =       *                 ;entry

                      ;       Load peripheral handler.

                      ;       LDX     ICIDNO            ;IOCB index
CA29  AE2E00                  VFD     8\$AE,8\low ICIDNO,8\high ICIDNO
CA2C  BD4D03                  LDA     ICSPR+1,X
CA2F  20DEE7                  JSR     LPH               ;load peripheral handler
CA32  B020 ^CA54              BCS     PHL1              ;if error
```

```
                        ;       Initialize peripheral handler.

CA34  18                        CLC                     ;indicate zero handler size
CA35  209EE8                    JSR     PHC             ;initialize peripheral hand:
CA38  B01A ^CA54                BCS     PHL1            ;if error

                        ;       Find device handler.

                        ;       LDX     ICIDNO          ;IOCB index
CA3A  AE2E00                    VFD     8\$AE,8\low ICIDNO,8\high ICIDNO
CA3D  BD4C03                    LDA     ICSPR,X
CA40  2016E7                    JSR     FDH             ;find device handler
CA43  B00F ^CA54                BCS     PHL1            ;if not found

                        ;       Set handler ID.

                        ;       LDX     ICIDNO          ;IOCB index
CA45  AE2E00                    VFD     8\$AE,8\low ICIDNO,8\high ICIDNO
CA48  9D4003                    STA     ICHID,X         ;handler ID
CA4B  8520                      STA     ICHIDZ

                        ;       Simulate initial CIO OPEN processing.

CA4D  A903                      LDA     #OPEN   ;OPEN command
CA4F  8517                      STA     ICCOMT  ;command
CA51  4C5CE5                    JMP     IIO     ;initialize IOCB for OPEN, return

                        ;       Indicate nonexistent device error.

CA54  4C10E5            PHL1    JMP     IND     ;indicate nonexistent device error,:
```

```
CA57                  **      TSTO - Table of Self-test Text Offsets


CA57  00              TSTO    DB      TXT0-TTXT        ;0 - offset to "MEMORY TEST:
CA58  13                      DB      TXT1-TTXT        ;1 - offset to "RAM" text
CA59  16                      DB      TXT2-TTXT        ;2 - offset to "KEYBOARD TE:
CA5A  D1                      DB      TXT3-TTXT        ;3 - offset to "S P A C E  :
CA5B  E4                      DB      TXT4-TTXT        ;4 - offset to "SH" text
CA5C  E4                      DB      TXT5-TTXT        ;5 - offset to "SH" text
CA5D  E8                      DB      TXT6-TTXT        ;6 - offset to "B S" text
CA5E  29                      DB      TXT7-TTXT        ;7 - offset to keyboard tex:
CA5F  EB                      DB      TXT8-TTXT        ;8 - offset to control key :
CA60  EE                      DB      TXT9-TTXT        ;9 - offset to "VOICE #" te:



                      **      TTXT - Table of Text Sequences


      = CA61          TTXT    =       *



                      **      TXT0 - "MEMORY TEST   ROM" Text


CA61  0000            TXT0    DB      $00,$00
CA63  2D252D2F32              DB      $2D,$25,$2D,$2F,$32,$39 ;"MEMORY"
CA69  00                      DB      $00
CA6A  34253334                DB      $34,$25,$33,$34         ;"TEST"
CA6E  000000                  DB      $00,$00,$00
CA71  322F2D                  DB      $32,$2F,$2D             ;"ROM"

      = 0013          TXT0L   =       *-TXT0   ;length



                      **      TXT1 - "RAM" Text


CA74  32212D          TXT1    DB      $32,$21,$2D             ;"RAM"

      = 0003          TXT1L   =       *-TXT1   ;length
```

```
                        **      TXT2 - "KEYBOARD TEST" Text

CA77  0000              TXT2    DB      $00,$00
CA79  2B2539222F                DB      $2B,$25,$39,$22,$2F,$21,$32,$24 ;"KEYBOARD"
CA81  00                        DB      $00
CA82  34253334                  DB      $34,$25,$33,$34                 ;"TEST"
CA86  000000                    DB      $00,$00,$00
CA89  B2                        DB      $B2

      = 0013            TXT2L   =       *-TXT2  ;length


                        **      TXT7 - Keyboard

      = CA8A            TXT7    =       *

                        ;       First Row (Function Keys)

CA8A  91                        DB      $91             ;"1"
CA8B  00                        DB      $00
CA8C  92                        DB      $92             ;"2"
CA8D  00                        DB      $00
CA8E  93                        DB      $93             ;"3"
CA8F  00                        DB      $00
CA90  94                        DB      $94             ;"4"
CA91  00                        DB      $00
CA92  A8                        DB      $A8             ;"H"
CA93  00                        DB      $00
CA94  A1                        DB      $A1             ;"A"
CA95  00                        DB      $00
CA96  A2                        DB      $A2             ;"B"
CA97  000000                    DB      $00,$00,$00

                        ;       Second Row ("1 2 3 4 5 6 7 8 9 0 < >")

CA9A  5B                        DB      $5B
CA9B  00                        DB      $00
CA9C  11                        DB      $11             ;"1"
CA9D  00                        DB      $00
CA9E  12                        DB      $12             ;"2"
CA9F  00                        DB      $00
CAA0  13                        DB      $13             ;"3"
CAA1  00                        DB      $00
CAA2  14                        DB      $14             ;"4"
CAA3  00                        DB      $00
CAA4  15                        DB      $15             ;"5"
CAA5  00                        DB      $00
CAA6  16                        DB      $16             ;"6"
CAA7  00                        DB      $00
CAA8  17                        DB      $17             ;"7"
CAA9  00                        DB      $00
CAAA  18                        DB      $18             ;"8"
CAAB  00                        DB      $00
```

```
CAAC  19                    DB      $19             ;"9"
CAAD  00                    DB      $00
CAAE  10                    DB      $10             ;"0"
CAAF  00                    DB      $00
CAB0  1C                    DB      $1C             ;"<"
CAB1  00                    DB      $00
CAB2  1E                    DB      $1E             ;">"
CAB3  00                    DB      $00
CAB4  A2                    DB      $A2             ;"B"
CAB5  80                    DB      $80
CAB6  B3                    DB      $B3             ;"S"
CAB7  000000                DB      $00,$00,$00

                  ;         Third Row ("Q W E R T Y U I O P - =")

CABA  FF                    DB      $FF
CABB  FF                    DB      $FF
CABC  00                    DB      $00
CABD  31                    DB      $31             ;"Q"
CABE  00                    DB      $00
CABF  37                    DB      $37             ;"W"
CAC0  00                    DB      $00
CAC1  25                    DB      $25             ;"E"
CAC2  00                    DB      $00
CAC3  32                    DB      $32             ;"R"
CAC4  00                    DB      $00
CAC5  34                    DB      $34             ;"T"
CAC6  00                    DB      $00
CAC7  39                    DB      $39             ;"Y"
CAC8  00                    DB      $00
CAC9  35                    DB      $35             ;"U"
CACA  00                    DB      $00
CACB  29                    DB      $29             ;"I"
CACC  00                    DB      $00
CACD  2F                    DB      $2F             ;"O"
CACE  00                    DB      $00
CACF  30                    DB      $30             ;"P"
CAD0  00                    DB      $00
CAD1  0D                    DB      $0D             ;"-"
CAD2  00                    DB      $00
CAD3  1D                    DB      $1D             ;"="
CAD4  00                    DB      $00
CAD5  B2                    DB      $B2             ;"R"
CAD6  B4                    DB      $B4             ;"T"
CAD7  000000                DB      $00,$00,$00

                  ;         Fourth Row ("A S D F G H J K L ; + *")

CADA  80                    DB      $80
CADB  DC                    DB      $DC
CADC  80                    DB      $80
CADD  00                    DB      $00
CADE  21                    DB      $21             ;"A"
CADF  00                    DB      $00
CAE0  33                    DB      $33             ;"S"
CAE1  00                    DB      $00
CAE2  24                    DB      $24             ;"D"
```

```
CAE3  00                DB      $00
CAE4  26                DB      $26             ;"F"
CAE5  00                DB      $00
CAE6  27                DB      $27             ;"G"
CAE7  00                DB      $00
CAE8  28                DB      $28             ;"H"
CAE9  00                DB      $00
CAEA  2A                DB      $2A             ;"J"
CAEB  00                DB      $00
CAEC  2B                DB      $2B             ;"K"
CAED  00                DB      $00
CAEE  2C                DB      $2C             ;"L"
CAEF  00                DB      $00
CAF0  18                DB      $18             ;";"
CAF1  00                DB      $00
CAF2  0B                DB      $0B             ;"+"
CAF3  00                DB      $00
CAF4  0A                DB      $0A             ;"*"
CAF5  00                DB      $00
CAF6  A3                DB      $A3             ;"C"
CAF7  000000            DB      $00,$00,$00

                ;       Fifth Row ("Z X C V B N M , . /")

CAFA  80                DB      $80
CAFB  B3                DB      $B3             ;"S"
CAFC  A8                DB      $A8             ;"H"
CAFD  80                DB      $80
CAFE  00                DB      $00
CAFF  3A                DB      $3A             ;"Z"
CB00  00                DB      $00
CB01  38                DB      $38             ;"X"
CB02  00                DB      $00
CB03  23                DB      $23             ;"C"
CB04  00                DB      $00
CB05  36                DB      $36             ;"V"
CB06  00                DB      $00
CB07  22                DB      $22             ;"B"
CB08  00                DB      $00
CB09  2E                DB      $2E             ;"N"
CB0A  00                DB      $00
CB0B  2D                DB      $2D             ;"M"
CB0C  00                DB      $00
CB0D  0C                DB      $0C             ;","
CB0E  00                DB      $00
CB0F  0E                DB      $0E             ;"."
CB10  00                DB      $00
CB11  0F                DB      $0F             ;"/"
CB12  00                DB      $00
CB13  80                DB      $80
CB14  B3                DB      $B3             ;"S"
CB15  A8                DB      $A8             ;"H"
CB16  80                DB      $80
CB17  000000            DB      $00,$00,$00

                ;       Sixth Row (Space Bar)
```

```
CB1A  0000000000          DB      $00,$00,$00,$00,$00
CB1F  80                  DB      $80
CB20  B3                  DB      $B3             ;"S"
CB21  80                  DB      $80
CB22  B0                  DB      $B0             ;"P"
CB23  80                  DB      $80
CB24  A1                  DB      $A1             ;"A"
CB25  80                  DB      $80
CB26  A3                  DB      $A3             ;"C"
CB27  80                  DB      $80
CB28  A5                  DB      $A5             ;"E"
CB29  80                  DB      $80
CB2A  80                  DB      $80
CB2B  80                  DB      $80
CB2C  A2                  DB      $A2             ;"B"
CB2D  80                  DB      $80
CB2E  A1                  DB      $A1             ;"A"
CB2F  80                  DB      $80
CB30  B2                  DB      $B2             ;"R"
CB31  80                  DB      $80

      = 00A8      TXT7L   =       *-TXT7  ;length


            **      TXT3 - "S P A C E   B A R" Text


CB32  00          TXT3    DB      $00
CB33  33                  DB      $33             ;"S"
CB34  00                  DB      $00
CB35  30                  DB      $30             ;"P"
CB36  00                  DB      $00
CB37  21                  DB      $21             ;"A"
CB38  00                  DB      $00
CB39  23                  DB      $23             ;"C"
CB3A  00                  DB      $00
CB3B  25                  DB      $25             ;"E"
CB3C  00                  DB      $00
CB3D  00                  DB      $00
CB3E  00                  DB      $00
CB3F  22                  DB      $22             ;"B"
CB40  00                  DB      $00
CB41  21                  DB      $21             ;"A"
CB42  00                  DB      $00
CB43  32                  DB      $32             ;"R"
CB44  00                  DB      $00

      = 0013      TXT3L   =       *-TXT3  ;length
```

```
                        **      TXT4 - "SH" Text

CB45  00                TXT4    DB      $00
CB46  3328                      DB      $33,$28         ;"SH"
CB48  00                        DB      $00

      = 0004            TXT4L   =       *-TXT4  ;length


                        **      TXT5 - "SH" Text

      = CB45            TXT5    =       TXT4

      = 0004            TXT5L   =       TXT4L   ;length


                        **      TXT6 - "B S" Text

CB49  22                TXT6    DB      $22     ;"B"
CB4A  00                        DB      $00
CB4B  33                        DB      $33     ;"S"

      = 0003            TXT6L   =       *-TXT6  ;length


                        **      TXT8 - Control Key

CB4C  00                TXT8    DB      $00
CB4D  5C                        DB      $5C
CB4E  00                        DB      $00

      = 0003            TXT8L   =       *-TXT8  ;length
```

```
                  **      TXT9 - "VOICE #" Text

CB4F  362F292325  TXT9    DB      $36,$2F,$29,$23,$25     ;"VOICE"
CB54  00                  DB      $00
CB55  03                  DB      $03                     ;"#"

      = 0007      TXT9L   =       *-TXT9  ;length
```

```
CB56                    **      CLT - Checksum Linkage Table
                        *
                        *       ENTRY   JSR     CLT
                        *               ZCHAIN - ZCHAIN+1 = address of linkage tabl:
                        *
                        *       EXIT
                        *               A = checksum of linkage table
                        *
                        *       CHANGES
                        *               Y
                        *
                        *       CALLS
                        *               -none-
                        *
                        *       MODS
                        *               R. S. Scheiman  04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

     = CB56          CLT        =       *               ;entry

CB56  A011                      LDY     #17             ;offset to last byte to sum
CB58  A900                      LDA     #0              ;initial sum
CB5A  18                        CLC

CB5B  714A           CLT1       ADC     (ZCHAIN),Y      ;add byte
CB5D  88                        DEY
CB5E  10FB ^CB5B                BPL     CLT1            ;if not done

CB60  6900                      ADC     #0              ;add final carry
CB62  49FF                      EOR     #$FF            ;complement
CB64  60                        RTS                     ;return
```

```
CB65                          FIX     ICSORG



                  **          International Character Set

CC00  0000000000              DB      $00,$00,$00,$00,$00,$00,$00,$00 ;$00 - spac:
CC08  0018181818              DB      $00,$18,$18,$18,$18,$00,$18,$00 ;$01 - !
CC10  0066666600              DB      $00,$66,$66,$66,$00,$00,$00,$00 ;$02 - "
CC18  0066FF6666              DB      $00,$66,$FF,$66,$66,$FF,$66,$00 ;$03 - #
CC20  183E603C06              DB      $18,$3E,$60,$3C,$06,$7C,$18,$00 ;$04 - $
CC28  00666C1830              DB      $00,$66,$6C,$18,$30,$66,$46,$00 ;$05 - %
CC30  1C361C386F              DB      $1C,$36,$1C,$38,$6F,$66,$3B,$00 ;$06 - &
CC38  0018181800              DB      $00,$18,$18,$18,$00,$00,$00,$00 ;$07 - '
CC40  000E1C1818              DB      $00,$0E,$1C,$18,$18,$1C,$0E,$00 ;$08 - (
CC48  0070381818              DB      $00,$70,$38,$18,$18,$38,$70,$00 ;$09 - )
CC50  00663CFF3C              DB      $00,$66,$3C,$FF,$3C,$66,$00,$00 ;$0A - aste:
CC58  0018187E18              DB      $00,$18,$18,$7E,$18,$18,$00,$00 ;$0B - plus
CC60  0000000000              DB      $00,$00,$00,$00,$00,$18,$18,$30 ;$0C - comm:
CC68  0000007E00              DB      $00,$00,$00,$7E,$00,$00,$00,$00 ;$0D - minu:
CC70  0000000000              DB      $00,$00,$00,$00,$00,$18,$18,$00 ;$0E - peri:
CC78  00060C1830              DB      $00,$06,$0C,$18,$30,$60,$40,$00 ;$0F - /

CC80  003C666E76              DB      $00,$3C,$66,$6E,$76,$66,$3C,$00 ;$10 - 0
CC88  0018381818              DB      $00,$18,$38,$18,$18,$18,$7E,$00 ;$11 - 1
CC90  003C660C18              DB      $00,$3C,$66,$0C,$18,$30,$7E,$00 ;$12 - 2
CC98  007E0C180C              DB      $00,$7E,$0C,$18,$0C,$66,$3C,$00 ;$13 - 3
CCA0  000C1C3C6C              DB      $00,$0C,$1C,$3C,$6C,$7E,$0C,$00 ;$14 - 4
CCA8  007E607C06              DB      $00,$7E,$60,$7C,$06,$66,$3C,$00 ;$15 - 5
CCB0  003C607C66              DB      $00,$3C,$60,$7C,$66,$66,$3C,$00 ;$16 - 6
CCB8  007E060C18              DB      $00,$7E,$06,$0C,$18,$30,$30,$00 ;$17 - 7
CCC0  003C663C66              DB      $00,$3C,$66,$3C,$66,$66,$3C,$00 ;$18 - 8
CCC8  003C663E06              DB      $00,$3C,$66,$3E,$06,$0C,$38,$00 ;$19 - 9
CCD0  0000181800              DB      $00,$00,$18,$18,$00,$18,$18,$00 ;$1A - colo:
CCD8  0000181800              DB      $00,$00,$18,$18,$00,$18,$18,$30 ;$1B - semi:
CCE0  060C183018              DB      $06,$0C,$18,$30,$18,$0C,$06,$00 ;$1C - <
CCE8  00007E0000              DB      $00,$00,$7E,$00,$00,$7E,$00,$00 ;$1D - =
CCF0  6030180C18              DB      $60,$30,$18,$0C,$18,$30,$60,$00 ;$1E - >
CCF8  003C660C18              DB      $00,$3C,$66,$0C,$18,$00,$18,$00 ;$1F - ?

CD00  003C666E6E              DB      $00,$3C,$66,$6E,$6E,$60,$3E,$00 ;$20 - @
CD08  00183C6666              DB      $00,$18,$3C,$66,$66,$7E,$66,$00 ;$21 - A
CD10  007C667C66              DB      $00,$7C,$66,$7C,$66,$66,$7C,$00 ;$22 - B
CD18  003C666060              DB      $00,$3C,$66,$60,$60,$66,$3C,$00 ;$23 - C
CD20  00786C6666              DB      $00,$78,$6C,$66,$66,$6C,$78,$00 ;$24 - D
CD28  007E607C60              DB      $00,$7E,$60,$7C,$60,$60,$7E,$00 ;$25 - E
CD30  007E607C60              DB      $00,$7E,$60,$7C,$60,$60,$60,$00 ;$26 - F
CD38  003E60606E              DB      $00,$3E,$60,$60,$6E,$66,$3E,$00 ;$27 - G
CD40  0066667E66              DB      $00,$66,$66,$7E,$66,$66,$66,$00 ;$28 - H
CD48  007E181818              DB      $00,$7E,$18,$18,$18,$18,$7E,$00 ;$29 - I
CD50  0006060606              DB      $00,$06,$06,$06,$06,$66,$3C,$00 ;$2A - J
CD58  00666C7878              DB      $00,$66,$6C,$78,$78,$6C,$66,$00 ;$2B - K
CD60  0060606060              DB      $00,$60,$60,$60,$60,$60,$7E,$00 ;$2C - L
CD68  0063777F6B              DB      $00,$63,$77,$7F,$6B,$63,$63,$00 ;$2D - M
CD70  0066767E7E              DB      $00,$66,$76,$7E,$7E,$6E,$66,$00 ;$2E - N
```

```
CD78  003C666666        DB      $00,$3C,$66,$66,$66,$66,$3C,$00 ;$2F - O

CD80  007C66667C        DB      $00,$7C,$66,$66,$7C,$60,$60,$00 ;$30 - P
CD88  003C666666        DB      $00,$3C,$66,$66,$66,$6C,$36,$00 ;$31 - Q
CD90  007C66667C        DB      $00,$7C,$66,$66,$7C,$6C,$66,$00 ;$32 - R
CD98  003C603C06        DB      $00,$3C,$60,$3C,$06,$06,$3C,$00 ;$33 - S
CDA0  007E181818        DB      $00,$7E,$18,$18,$18,$18,$18,$00 ;$34 - T
CDA8  0066666666        DB      $00,$66,$66,$66,$66,$66,$7E,$00 ;$35 - U
CDB0  0066666666        DB      $00,$66,$66,$66,$66,$3C,$18,$00 ;$36 - V
CDB8  0063636B7F        DB      $00,$63,$63,$6B,$7F,$77,$63,$00 ;$37 - W
CDC0  0066663C3C        DB      $00,$66,$66,$3C,$3C,$66,$66,$00 ;$38 - X
CDC8  0066663C18        DB      $00,$66,$66,$3C,$18,$18,$18,$00 ;$39 - Y
CDD0  007E0C1830        DB      $00,$7E,$0C,$18,$30,$60,$7E,$00 ;$3A - Z
CDD8  001E181818        DB      $00,$1E,$18,$18,$18,$18,$1E,$00 ;$3B - [
CDE0  0040603018        DB      $00,$40,$60,$30,$18,$0C,$06,$00 ;$3C - \
CDE8  0078181818        DB      $00,$78,$18,$18,$18,$18,$78,$00 ;$3D - ]
CDF0  00081C3663        DB      $00,$08,$1C,$36,$63,$00,$00,$00 ;$3E - ^
CDF8  0000000000        DB      $00,$00,$00,$00,$00,$00,$FF,$00 ;$3F - unde:

CE00  0C183C063E        DB      $0C,$18,$3C,$06,$3E,$66,$3E,$00 ;$40 - acut:
CE08  3018006666        DB      $30,$18,$00,$66,$66,$66,$3E,$00 ;$41 - acut:
CE10  366C007676        DB      $36,$6C,$00,$76,$76,$7E,$6E,$00 ;$42 - tild:
CE18  0C187E607C        DB      $0C,$18,$7E,$60,$7C,$60,$7E,$00 ;$43 - acut:
CE20  00003C6060        DB      $00,$00,$3C,$60,$60,$3C,$18,$30 ;$44 - cedi:
CE28  3C66003C66        DB      $3C,$66,$00,$3C,$66,$66,$3C,$00 ;$45 - circ:
CE30  3018003C66        DB      $30,$18,$00,$3C,$66,$66,$3C,$00 ;$46 - grav:
CE38  3018003818        DB      $30,$18,$00,$38,$18,$18,$3C,$00 ;$47 - grav:
CE40  1C30307830        DB      $1C,$30,$30,$78,$30,$30,$7E,$00 ;$48 - U.K.:
CE48  0066003818        DB      $00,$66,$00,$38,$18,$18,$3C,$00 ;$49 - diae:
CE50  0066006666        DB      $00,$66,$00,$66,$66,$66,$3E,$00 ;$4A - umla:
CE58  36003C063E        DB      $36,$00,$3C,$06,$3E,$66,$3E,$00 ;$4B - umla:
CE60  66003C6666        DB      $66,$00,$3C,$66,$66,$66,$3C,$00 ;$4C - umla:
CE68  0C18006666        DB      $0C,$18,$00,$66,$66,$66,$3E,$00 ;$4D - grav:
CE70  0C18003C66        DB      $0C,$18,$00,$3C,$66,$66,$3C,$00 ;$4E - acut:
CE78  0066003C66        DB      $00,$66,$00,$3C,$66,$66,$3C,$00 ;$4F - umla:

CE80  6600666666        DB      $66,$00,$66,$66,$66,$66,$7E,$00 ;$50 - umla:
CE88  3C661C063E        DB      $3C,$66,$1C,$06,$3E,$66,$3E,$00 ;$51 - circ:
CE90  3C66006666        DB      $3C,$66,$00,$66,$66,$66,$3E,$00 ;$52 - circ:
CE98  3C66003818        DB      $3C,$66,$00,$38,$18,$18,$3C,$00 ;$53 - circ:
CEA0  0C183C667E        DB      $0C,$18,$3C,$66,$7E,$60,$3C,$00 ;$54 - acut:
CEA8  30183C667E        DB      $30,$18,$3C,$66,$7E,$60,$3C,$00 ;$55 - grav:
CEB0  366C007C66        DB      $36,$6C,$00,$7C,$66,$66,$66,$00 ;$56 - tild:
CEB8  3CC33C667E        DB      $3C,$C3,$3C,$66,$7E,$60,$3C,$00 ;$57 - circ:
CEC0  18003C063E        DB      $18,$00,$3C,$06,$3E,$66,$3E,$00 ;$58 - ring:
CEC8  30183C063E        DB      $30,$18,$3C,$06,$3E,$66,$3E,$00 ;$59 - grav:
CED0  1800183C66        DB      $18,$00,$18,$3C,$66,$7E,$66,$00 ;$5A - ring:
CED8  786078607E        DB      $78,$60,$78,$60,$7E,$18,$1E,$00 ;$5B - disp:
CEE0  00183C7E18        DB      $00,$18,$3C,$7E,$18,$18,$18,$00 ;$5C - up a:
CEE8  001818187E        DB      $00,$18,$18,$18,$7E,$3C,$18,$00 ;$5D - down:
CEF0  0018307E30        DB      $00,$18,$30,$7E,$30,$18,$00,$00 ;$5E - left:
CEF8  00180C7E0C        DB      $00,$18,$0C,$7E,$0C,$18,$00,$00 ;$5F - righ:

CF00  1800181818        DB      $18,$00,$18,$18,$18,$18,$18,$00 ;$60 - Span:
CF08  00003C063E        DB      $00,$00,$3C,$06,$3E,$66,$3E,$00 ;$61 - a
CF10  0060607C66        DB      $00,$60,$60,$7C,$66,$66,$7C,$00 ;$62 - b
CF18  00003C6060        DB      $00,$00,$3C,$60,$60,$60,$3C,$00 ;$63 - c
```

```
CF20  0006063E66      DB      $00,$06,$06,$3E,$66,$66,$3E,$00 ;$64 - d
CF28  00003C667E      DB      $00,$00,$3C,$66,$7E,$60,$3C,$00 ;$65 - e
CF30  000E183E18      DB      $00,$0E,$18,$3E,$18,$18,$18,$00 ;$66 - f
CF38  00003E6666      DB      $00,$00,$3E,$66,$66,$3E,$06,$7C ;$67 - g
CF40  0060607C66      DB      $00,$60,$60,$7C,$66,$66,$66,$00 ;$68 - h
CF48  0018003818      DB      $00,$18,$00,$38,$18,$18,$3C,$00 ;$69 - i
CF50  0006000606      DB      $00,$06,$00,$06,$06,$06,$06,$3C ;$6A - j
CF58  0060606C78      DB      $00,$60,$60,$6C,$78,$6C,$66,$00 ;$6B - k
CF60  0038181818      DB      $00,$38,$18,$18,$18,$18,$3C,$00 ;$6C - l
CF68  0000667F7F      DB      $00,$00,$66,$7F,$7F,$6B,$63,$00 ;$6D - m
CF70  00007C6666      DB      $00,$00,$7C,$66,$66,$66,$66,$00 ;$6E - n
CF78  00003C6666      DB      $00,$00,$3C,$66,$66,$66,$3C,$00 ;$6F - o

CF80  00007C6666      DB      $00,$00,$7C,$66,$66,$7C,$60,$60 ;$70 - p
CF88  00003E6666      DB      $00,$00,$3E,$66,$66,$3E,$06,$06 ;$71 - q
CF90  00007C6660      DB      $00,$00,$7C,$66,$60,$60,$60,$00 ;$72 - r
CF98  00003E603C      DB      $00,$00,$3E,$60,$3C,$06,$7C,$00 ;$73 - s
CFA0  00187E1818      DB      $00,$18,$7E,$18,$18,$18,$0E,$00 ;$74 - t
CFA8  0000666666      DB      $00,$00,$66,$66,$66,$66,$3E,$00 ;$75 - u
CFB0  0000666666      DB      $00,$00,$66,$66,$66,$3C,$18,$00 ;$76 - v
CFB8  000063687F      DB      $00,$00,$63,$6B,$7F,$3E,$36,$00 ;$77 - w
CFC0  0000663C18      DB      $00,$00,$66,$3C,$18,$3C,$66,$00 ;$78 - x
CFC8  0000666666      DB      $00,$00,$66,$66,$66,$3E,$0C,$78 ;$79 - y
CFD0  00007E0C18      DB      $00,$00,$7E,$0C,$18,$30,$7E,$00 ;$7A - z
CFD8  6666183C66      DB      $66,$66,$18,$3C,$66,$7E,$66,$00 ;$7B - umla:
CFE0  1818181818      DB      $18,$18,$18,$18,$18,$18,$18,$18 ;$7C - |
CFE8  007E787C6E      DB      $00,$7E,$78,$7C,$6E,$66,$06,$00 ;$7D - disp:
CFF0  0818387838      DB      $08,$18,$38,$78,$38,$18,$08,$00 ;$7E - disp:
CFF8  10181C1E1C      DB      $10,$18,$1C,$1E,$1C,$18,$10,$00 ;$7F - disp:
```

```
D000                         FIX     $D000
D000   = 5000#               LOC     $5000    ;$D000 - $D7FF mapped to $5000 - $5:


                   **        STH - Self-test Hardware
                   *
                   *         ENTRY   JSR      STH
                   *
                   *         NOTES
                   *                 Problem: this is superfluous; SLFTSV could :
                   *                 EST.
                   *
                   *         MODS
                   *                 M. W. Colburn   10/26/82
                   *                 1. Bring closer to Coding Standard (object :
                   *                    R. K. Nordin 11/01/83

     = 5000        STH       =       *        ;entry
5000# 4C0950                 JMP     EST      ;execute self-test


                   **        EMS - Execute Memory Self-test
                   *
                   *         ENTRY   JSR      EMS
                   *
                   *         MODS
                   *                 M. W. Colburn   10/26/82
                   *                 1. Bring closer to Coding Standard (object :
                   *                    R. K. Nordin 11/01/83

     = 5003        EMS       =       *        ;entry
5003# 208650                 JSR     IST      ;initialize self-test
5006# 4C9152                 JMP     STM      ;self-test memory


                   **        EST - Execute Self-test
                   *
                   *         ENTRY   JSR      EST
                   *
                   *         MODS
                   *                 M. W. Colburn   10/26/82
                   *                 1. Bring closer to Coding Standard (object :
                   *                    R. K. Nordin 11/01/83

     = 5009        EST       =       *        ;entry
5009# 208650                 JSR     IST      ;initialize self-test
                   ;         JMP     SEL      ;self-test
```

```
                        **      SEL - Self-test
                        *
                        *       ENTRY   JSR     SEL
                        *
                        *       MODS
                        *               M. W. Colburn   10/26/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

         = 500C         SEL     =       *               ;entry

                        ;       Initialize.

500C# A900                      LDA     #0
500E# 8580                      STA     STTIME          ;clear main screen timeout :
5010# 8581                      STA     STTIME+1
5012# 8582                      STA     STAUT           ;clear auto-mode flag
5014# 8D08D2                    STA     AUDCTL          ;initialize audio control r:
5017# A903                      LDA     #$03            ;initialize POKEY
5019# 8D0FD2                    STA     SKCTL           ;serial port control
501C# 201055                    JSR     SAS             ;silence all sounds
501F# A940                      LDA     #$40            ;disable DLI
5021# 8D0ED4                    STA     NMIEN           ;NMI enable
5024# A200                      LDX     #0              ;main screen colors
5026# 207357                    JSR     SUC             ;set up colors
5029# A23A                      LDX     #low DISL1      ;display list for main scre:
502B# A051                      LDY     #high DISL1
502D# 209E50                    JSR     SDL             ;set up display list
5030# A9D0                      LDA     #low PMD        ;process main screen DLI ro:
5032# 8D0002                    STA     VDSLST          ;display list NMI address
5035# A950                      LDA     #high PMD
5037# 8D0102                    STA     VDSLST+1
503A# A20C                      LDX     #3*4            ;main screen bold lines
503C# A9AA                      LDA     #$AA            ;color 1
503E# 202A57                    JSR     SVR             ;set value in range

                        ;       Wait for all screen DLI's to clear and for VBLANK.

5041# A200                      LDX     #0

5043# 8E0AD4            SEL1    STX     WSYNC           ;wait for HBLANK synchroniz:
5046# E8                        INX
5047# D0FA ^5043                BNE     SEL1            ;if not done waiting

                        ;       Wait until beam close to top (main screen DLI near):

5049# AD0BD4            SEL2    LDA     VCOUNT
504C# C918                      CMP     #24
504E# B0F9 ^5049                BCS     SEL2            ;if not done waiting

                        ;       Preset for self-test type determination.

5050# A910                      LDA     #$10            ;initially select memory te:
5052# 8587                      STA     STPASS          ;pass indicator
5054# A9C0                      LDA     #$C0            ;enable DLI
5056# 8D0ED4                    STA     NMIEN
```

```
                         ;       Determine type of self-test.

5059# AD1FD0        SEL3    LDA     CONSOL          ;console switches
505C# 2901                  AND     #$01            ;START key indicator
505E# D0F9 ^5059            BNE     SEL3            ;if START key not pressed

5060# A9FF                  LDA     #$FF            ;clear character
5062# 8DFC02                STA     CH

5065# A586                  LDA     STSEL           ;selection
5067# 290F                  AND     #$0F            ;selection
5069# C901                  CMP     #$01            ;memory test indicator
506B# F010 ^507D            BEQ     SEL5            ;if memory test

506D# C902                  CMP     #$02
506F# F00F ^5080            BEQ     SEL6            ;if audio-visual test

5071# C904                  CMP     #$04
5073# F00E ^5083            BEQ     SEL7            ;if keyboard test

                         ;       Self-test all.

5075# A988          SEL4    LDA     #$88            ;indicate all tests
5077# 8586                  STA     STSEL           ;selection
5079# A9FF                  LDA     #$FF            ;auto-mode indicator
507B# 8582                  STA     STAUT           ;auto-mode flag

                         ;       Self-test memory.

507D# 4C9152        SEL5    JMP     STM             ;self-test memory

                         ;       Self-test audio-visual.

5080# 4C5755        SEL6    JMP     STV             ;self-test audio-visual

                         ;       Self-test keyboard.

5083# 4C5054        SEL7    JMP     STK             ;self-test keyboard



                         **      IST - Initialize Self-test
                         *
                         *       ENTRY   JSR     IST
                         *
                         *       MODS
                         *               M. W. Colburn   10/26/82
                         *               1. Bring closer to Coding Standard (object :
                         *                  R. K. Nordin 11/01/83


      = 5086        IST     =       *       ;entry
5086# A911                  LDA     #$11    ;indicate memory test
5088# 8586                  STA     STSEL   ;selection
508A# A921                  LDA     #$21
```

```
508C# 8D2F02            STA     SDMCTL  ;select small size playfield
508F# A9C0              LDA     #$C0
5091# 8D0ED4            STA     NMIEN   ;enable DLI
5094# A941              LDA     #$41
5096# 8583              STA     STJMP   ;ANTIC jump instruction
5098# A9FF              LDA     #$FF    ;clear code indicator
509A# 8DFC02            STA     CH      ;key code
509D# 60                RTS             ;return


                **      SDL - Set Up Display List
                *
                *       ENTRY   JSR     SDL
                *
                *       MODS
                *               M. W. Colburn   10/26/82
                *               1. Bring closer to Coding Standard (object :
                *                  R. K. Nordin 11/01/83

      = 509E    SDL     =       *       ;entry

509E# 858A              STA     STKST           ;keyboard self-test flag
50A0# 98                TYA
50A1# 48                PHA                     ;save high address
50A2# 8A                TXA
50A3# 48                PHA                     ;save low address
50A4# A900              LDA     #0
50A6# 8D2F02            STA     SDMCTL          ;DMACTL (DMA control) shado:
50A9# 8DDC02            STA     HELPFG          ;HELP key flag
50AC# A9DA              LDA     #low POD        ;process DLI routine
50AE# 8D0002            STA     VDSLST
50B1# A953              LDA     #high POD
50B3# 8D0102            STA     VDSLST+1
50B6# A200              LDX     #0*4            ;screen memory
50B8# 8A                TXA                     ;value is 0
50B9# 202A57            JSR     SVR             ;set value in range
50BC# 68                PLA                     ;saved low address
50BD# AA                TAX
50BE# 68                PLA                     ;saved high address
50BF# A8                TAY
50C0# 8E3002            STX     SDLSTL          ;low display list address
50C3# 8684              STX     STJMP+1         ;low display list address
50C5# 8C3102            STY     SDLSTH          ;high display list address
50C8# 8485              STY     STJMP+2         ;high display list address
50CA# A921              LDA     #$21
50CC# 8D2F02            STA     SDMCTL
50CF# 60                RTS             ;return
```

```
                    **      PMD - Process Main Screen DLI
                    *
                    *       1) IF MAIN SCREEN IS ON FOR MORE than FIVE MINUTES
                    *       THEN 'ALL TESTS' SELECTION IS SELECTED AND EXECUTED
                    *       2) COLORS FOR CURRENTLY SELECTED CHOICE AND THE
                    *       NON-SELECTED CHOICES ARE DISPLAYED ON FLY
                    *       3) SELECTION PROCESS IS HANDLED
                    *
                    *       ENTRY   JMP     PMD
                    *
                    *       EXIT
                    *               Exits via RTI
                    *
                    *       MODS
                    *               M. W. Colburn   10/26/82
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


       = 50D0       PMD     =       *       ;entry

                    ;       Initialize.

50D0# 48                    PHA             ;save A
50D1# 8A                    TXA
50D2# 48                    PHA             ;save X

                    ;       Check for 4th time.

50D3# A27A          PMD1    LDX     #$7A    ;assume non-selected color
50D5# A587                  LDA     STPASS  ;pass indicator
50D7# C901                  CMP     #$01    ;4th time indicator
50D9# F01F ^50FA            BEQ     PMD3    ;if 4th time

                    ;       Check for selection.

50DB# 2901                  AND     #$01    ;selection indicatorn
50DD# F00A ^50E9            BEQ     PMD2    ;if selected

                    ;       Increment and check blink counter.

50DF# E6A2                  INC     STBL    ;increment blink counter
50E1# A5A2                  LDA     STBL    ;blink counter
50E3# 2920                  AND     #$20    ;blink indicator
50E5# F002 ^50E9            BEQ     PMD2    ;if not to blink

50E7# A22C                  LDX     #$2C    ;use selected color

                    ;       Set color.

50E9# 8E0AD4        PMD2    STX     WSYNC   ;wait for HBLANK synchronization
50EC# 8E16D0                STX     COLPF0  ;playfield 0 color
50EF# 18                    CLC
50F0# 6687                  ROR     STPASS  ;advance pass indicator
50F2# A900                  LDA     #0
50F4# 854D                  STA     ATRACT
```

```
                        ;       Exit.

50F6# 68                        PLA
50F7# AA                        TAX             ;restore X
50F8# 68                        PLA             ;restore A
50F9# 40                        RTI             ;return

                        ;       Check for SELECT previously pressed.

50FA# A588              PMD3    LDA     STSPP   ;SELECT previously pressed  flag
50FC# D016 ^5114                BNE     PMD4    ;if SELECT previously pressed

                        ;       Check for SELECT pressed.

50FE# AD1FD0                    LDA     CONSOL  ;console switches
5101# 2902                      AND     #$02    ;SELECT key indicator
5103# D01A ^511F                BNE     PMD5    ;if SELECT not pressed, exit

                        ;       Process SELECT pressed.

5105# A586                      LDA     STSEL   ;current selection
5107# 2A                        ROL     A
5108# 2686                      ROL     STSEL   ;next selection
510A# A920                      LDA     #$20    ;blink indicator
510C# 85A2                      STA     STBL    ;blink counter
510E# A9FF                      LDA     #$FF    ;SELECT previously pressed indicato:
5110# 8588                      STA     STSPP   ;SELECT previously pressed flag
5112# D00B ^511F                BNE     PMD5

                        ;       Process SELECT previously pressed.

5114# AD1FD0            PMD4    LDA     CONSOL  ;console switches
5117# 2902                      AND     #$02    ;SELECT key indicator
5119# F004 ^511F                BEQ     PMD5    ;if SELECT still pressed

511B# A900                      LDA     #0      ;SELECT not previously pressed indi:
511D# 8588                      STA     STSPP   ;SELECT previously pressed flag

511F# A586              PMD5    LDA     STSEL   ;selection
5121# 290F                      AND     #$0F
5123# 0910                      ORA     #$10    ;reset indicate memory test
5125# 8587                      STA     STPASS  ;pass indicator

                        ;       Advance main screen timer.

5127# E680                      INC     STTIME
5129# D002 ^512D                BNE     PMD6    ;if low not zero

512B# E681                      INC     STTIME+1

                        ;       Check main screen timer.

512D# A581              PMD6    LDA     STTIME+1
512F# C9FA                      CMP     #250    ;main screen timeout
5131# D004 ^5137                BNE     PMD7    ;if main screen timed out

                        ;       Process main screen timeout.
```

```
5133# 58                      CLI
5134# 4C7550                  JMP     SEL4      ;self-test all

                     ;        Continue.

5137# 4CD350         PMD7     JMP     PMD1      ;continue



                     **       DISL1 - Display List for Main Screen

513A# 7070707070     DISL1    DB      $70,$70,$70,$70,$70
513F# 47                      DB      $47
5140# 6151                    DW      SMEM1
5142# 707070                  DB      $70,$70,$70
5145# 4E                      DB      $4E
5146# 0030                    DW      ST3000
5148# 70                      DB      $70
5149# F0                      DB      $F0
514A# C6                      DB      $C6
514B# 7151                    DW      SMEM2
514D# 7086                    DB      $70,$86
514F# 7086                    DB      $70,$86
5151# 7006                    DB      $70,$06
5153# 7070                    DB      $70,$70
5155# 4E                      DB      $4E
5156# 0030                    DW      ST3000
5158# 707070                  DB      $70,$70,$70
515B# 42                      DB      $42
515C# 8151                    DW      SMEM3
515E# 41                      DB      $41
515F# 3A51                    DW      DISL1



                     **       SMEM1 - "SELF TEST" Text

5161# 00000000       SMEM1    DB      $00,$00,$00,$00
5165# 33252C26                DB      $33,$25,$2C,$26          ;"SELF"
5169# 00                      DB      $00
516A# 34253334                DB      $34,$25,$33,$34          ;"TEST"
516E# 000000                  DB      $00,$00,$00
```

```
                        **      SMEM2 - "MEMORY AUDIO-VISUAL KEYBOARD ALL TESTS" Te:

5171# 0000          SMEM2   DB      $00,$00
5173# 2D252D2F32            DB      $2D,$25,$2D,$2F,$32,$39                 ;"M:
5179# 0000000000            DB      $00,$00,$00,$00,$00
517E# 0000000000            DB      $00,$00,$00,$00,$00
5183# 213524292F            DB      $21,$35,$24,$29,$2F                     ;"A:
5188# 0D                    DB      $0D                                     ;"-:
5189# 3629333521            DB      $36,$29,$33,$35,$21,$2C                 ;"V:
518F# 00000000              DB      $00,$00,$00,$00
5193# 2B2539222F            DB      $2B,$25,$39,$22,$2F,$21,$32,$24         ;"K:
519B# 0000000000            DB      $00,$00,$00,$00,$00,$00,$00,$00
51A3# 212C2C                DB      $21,$2C,$2C                             ;"A:
51A6# 00                    DB      $00
51A7# 3425333433            DB      $34,$25,$33,$34,$33                     ;"T:
51AC# 0000000000            DB      $00,$00,$00,$00,$00


                        **      SMEM3 - "SELECT,START OR RESET" Text

51B1# 00000000      SMEM3   DB      $00,$00,$00,$00
51B5# 42                    DB      $42
51B6# B3A5ACA5A3            DB      $B3,$A5,$AC,$A5,$A3,$B4         ;"SELECT"
51BC# 56                    DB      $56
51BD# 0C                    DB      $0C                             ;","
51BE# 42                    DB      $42
51BF# B3B4A1B2B4            DB      $B3,$B4,$A1,$B2,$B4             ;"START"
51C4# 56                    DB      $56
51C5# 2F32                  DB      $2F,$32                         ;"OR"
51C7# 42                    DB      $42
51C8# B2A5B3A5B4            DB      $B2,$A5,$B3,$A5,$B4             ;"RESET"
51CD# 56                    DB      $56
51CE# 000000                DB      $00,$00,$00


                        **      DISL2 - Display List for Memory Test

51D1# 707070        DISL2   DB      $70,$70,$70
51D4# 46                    DB      $46
51D5# 0030                  DW      ST3000
51D7# 70                    DB      $70
51D8# 7006                  DB      $70,$06
51DA# 7008                  DB      $70,$08
51DC# 70                    DB      $70
51DD# 7006                  DB      $70,$06
51DF# 7008                  DB      $70,$08
51E1# 7008                  DB      $70,$08
51E3# 7008                  DB      $70,$08
51E5# 7008                  DB      $70,$08
51E7# 707070                DB      $70,$70,$70
```

```
51EA# 01                        DB      $01
51EB# ED51                      DW      DISL3



                        **      DISL3 - Display List for Exit Text


51ED# A040              DISL3   DB      $A0,$40
51EF# 42                        DB      $42
51F0# F551                      DW      SMEM4
51F2# 01                        DB      $01
51F3# 8300                      DW      STJMP



                        **      SMEM4 - "RESET OR HELP TO EXIT" Text


51F5# 0000000000        SMEM4   DB      $00,$00,$00,$00,$00
51FA# 42                        DB      $42
51FB# B2A5B3A5B4                DB      $B2,$A5,$B3,$A5,$B4     ;"RESET"
5200# 56                        DB      $56
5201# 2F32                      DB      $2F,$32                 ;"OR"
5203# 42                        DB      $42
5204# A8A5ACB0                  DB      $A8,$A5,$AC,$B0         ;"HELP"
5208# 56                        DB      $56
5209# 342F                      DB      $34,$2F                 ;"TO"
520B# 00                        DB      $00
520C# 25382934                  DB      $25,$38,$29,$34         ;"EXIT"
5210# 0000000000                DB      $00,$00,$00,$00,$00



                        **      DISL4 - Display List for Keyboard Test


5215# 70707070          DISL4   DB      $70,$70,$70,$70
5219# 46                        DB      $46
521A# 0030                      DW      ST3000
521C# 707070                    DB      $70,$70,$70
521F# 7002                      DB      $70,$02
5221# 70                        DB      $70
5222# 7002                      DB      $70,$02
5224# 7002                      DB      $70,$02
5226# 7002                      DB      $70,$02
5228# 7002                      DB      $70,$02
522A# 7002                      DB      $70,$02
522C# 7070                      DB      $70,$70
522E# 01                        DB      $01
522F# ED51                      DW      DISL3
```

```
                          **      DISL5 - Display List for Audio-visual Test


5231# 70707070     DISL5    DB       $70,$70,$70,$70
5235# 46                    DB       $46
5236# 7152                  DW       SMEM5
5238# 7006                  DB       $70,$06
523A# 7070                  DB       $70,$70
523C# 48                    DB       $48
523D# 0031                  DW       ST3100
523F# 0B0B0B0B0B0B          DB       $0B,$0B,$0B,$0B,$0B,$0B,$0B,$0B
5247# 0B0B0B0B0B0B          DB       $0B,$0B,$0B,$0B,$0B,$0B,$0B,$0B
524F# 0B0B0B0B0B0B          DB       $0B,$0B,$0B,$0B,$0B,$0B,$0B,$0B
5257# 0B0B0B0B0B0B          DB       $0B,$0B,$0B,$0B,$0B,$0B,$0B,$0B
525F# 0B0B0B0B0B0B          DB       $0B,$0B,$0B,$0B,$0B,$0B,$0B,$0B
5267# 0B0B                  DB       $0B,$0B
5269# 70                    DB       $70
526A# 46                    DB       $46
526B# 0030                  DW       ST3000
526D# 70                    DB       $70
526E# 01                    DB       $01
526F# ED51                  DW       DISL3


                          **      SMEM5 - "AUDIO-VISUAL TEST" Text


5271# 0000         SMEM5    DB       $00,$00
5273# 213524292F            DB       $21,$35,$24,$29,$2F                ;"AUDIO"
5278# 0D                    DB       $0D                                ;"-"
5279# 362933352l            DB       $36,$29,$33,$35,$21,$2C            ;"VISUAL"
527F# 00000000              DB       $00,$00,$00,$00
5283# 00000000              DB       $00,$00,$00,$00
5287# 34253334              DB       $34,$25,$33,$34                    ;"TEST"
528B# 0000000000            DB       $00,$00,$00,$00,$00,$00



                          **      STM - Self-test Memory
                          *
                          *       STM verifies ROM and RAM by verifying the ROM check:
                          *       writing and reading all possible values to each byt:
                          *
                          *       ENTRY   JSR     STM
                          *
                          *       NOTES
                          *               Problem: searches beyond end of TMNT.
                          *
                          *       MODS
                          *               M. W. Colburn   10/26/82
                          *               1. Bring closer to Coding Standard (object :
                          *                  R. K. Nordin 11/01/83
```

```
       = 5291       STM     =       *               ;entry

                    ;       Initialize.

5291# A201                  LDX     #low DISL2      ;memory test display list
5293# A051                  LDY     #high DISL2
5295# A900                  LDA     #0              ;indicate not keyboard self:
5297# 209E50                JSR     SDL             ;set up display list
529A# A201                  LDX     #1              ;memory test colors
529C# 207357                JSR     SUC             ;set up colors
529F# A200                  LDX     #0              ;offset to "MEMORY TEST   R:
52A1# 205957                JSR     SSM             ;set screen memory
52A4# A201                  LDX     #1              ;offset to "RAM" text
52A6# 205957                JSR     SSM             ;set screen memory

                    ;       Test first 8K ROM.

52A9# AD2030        STM1    LDA     ST3020
52AC# C9AA                  CMP     #$AA            ;color 1 for failure
52AE# F017 ^52C7            BEQ     STM4            ;if first 8K ROM already fa:

52B0# A955                  LDA     #$55            ;color 0 for test
52B2# 208E53                JSR     DFS             ;display first ROM status
52B5# 208153                JSR     DMW             ;delay a middling while
52B8# 2073FF                JSR     VFR             ;verify first 8K ROM
52BB# B005 ^52C2            BCS     STM2            ;if ROM failed

52BD# A9FF                  LDA     #$FF            ;color 2 for success
52BF# 4CC452                JMP     STM3

52C2# A9AA          STM2    LDA     #$AA            ;color 1 for failure

52C4# 208E53        STM3    JSR     DFS             ;display first ROM status

                    ;       Test second 8K ROM.

52C7# AD2430        STM4    LDA     ST3024
52CA# C9AA                  CMP     #$AA            ;color 1 for failure
52CC# F017 ^52E5            BEQ     STM7            ;if second 8K ROM already f:

52CE# A955                  LDA     #$55            ;color 0 for test
52D0# 209953                JSR     DSS             ;display second ROM status
52D3# 208153                JSR     DMW             ;delay a middling while
52D6# 2092FF                JSR     VSR             ;verify second 8K ROM
52D9# B005 ^52E0            BCS     STM5            ;if ROM failed

52DB# A9FF                  LDA     #$FF            ;color 2 for success
52DD# 4CE252                JMP     STM6

52E0# A9AA          STM5    LDA     #$AA            ;color 1 for failure

52E2# 209953        STM6    JSR     DSS             ;display second ROM status

                    ;       Test RAM.

52E5# A9C0          STM7    LDA     #$C0            ;mask for left side of a sc:
```

```
52E7# 858D                  STA     STSMM
52E9# A904                  LDA     #$04           ;initially select LED 1 off
52EB# 85A4                  STA     STLM           ;LED mask
52ED# A900                  LDA     #0
52EF# 858E                  STA     STSMP
52F1# 8590                  STA     STPAG          ;initialize current page
52F3# 8591                  STA     STPAG+1
52F5# 858F                  STA     ST1K           ;initialize current 1K to t:

                     ;      Test 1K of RAM.

52F7# A68E           STM8   LDX     STSMP          ;screen memory pointer
52F9# BD3830                LDA     ST3038,X
52FC# 258D                  AND     STSMM
52FE# C980                  CMP     #$80
5300# F05C ^535E            BEQ     STM17          ;if already failed

5302# C908                  CMP     #$08
5304# F058 ^535E            BEQ     STM17          ;if already failed

5306# A944                  LDA     #$44           ;color 0 for test
5308# 20C353                JSR     DRS            ;display RAM block status
530B# A5A4                  LDA     STLM           ;LED mask
530D# 20A453                JSR     SLD            ;set LED's
5310# A5A4                  LDA     STLM           ;current LED mask
5312# 490C                  EOR     #$0C           ;complement LED's selected
5314# 85A4                  STA     STLM           ;update LED mask

                     ;      Check for memory not to test.

5316# A207                  LDX     #TMNTL-1+2     ;2 bytes beyond last byte o:

5318# BD4A54         STM9   LDA     TMNT,X         ;range to test
531B# C591                  CMP     STPAG+1        ;high current page
531D# F037 ^5356            BEQ     STM15          ;if not to test, indicate s:

531F# CA                    DEX
5320# 10F6 ^5318            BPL     STM9           ;if not done

                     ;      Test 1K of RAM.

5322# A904                  LDA     #4             ;number of pages to test
5324# 8592                  STA     STPC           ;page count

                     ;      Write initial list to page.

5326# A200           STM10  LDX     #0             ;initial value to write

                     ;      Write list to page.

5328# A000           STM11  LDY     #0             ;offset to first byte of pa:

532A# 8A             STM12  TXA
532B# 9190                  STA     (STPAG),Y      ;byte of page
532D# E8                    INX
532E# C8                    INY
532F# D0F9 ^532A            BNE     STM12          ;if not done writing page
```

```
                        ;       Verify list written to page.

5331# 8693                      STX     STMVAL          ;first correct value to tes:
5333# A000                      LDY     #0              ;offset to first byte of pa:

5335# B190              STM13   LDA     (STPAG),Y       ;byte of page
5337# C593                      CMP     STMVAL          ;correct value
5339# D010 ^534B                BNE     STM14           ;if not correct value

533B# E693                      INC     STMVAL          ;increment value to test
533D# C8                        INY
533E# D0F5 ^5335                BNE     STM13           ;if not done verifying page

                        ;       Increment and test initial value to write.

5340# E8                        INX                     ;increment initial value to:
5341# D0E5 ^5328                BNE     STM11           ;if not done, write another:

                        ;       Decrement and test page counter.

5343# E691                      INC     STPAG+1         ;increment high current pag:
5345# C692                      DEC     STPC            ;decrement page count
5347# D0DD ^5326                BNE     STM10           ;if not done testing pages

5349# F00E ^5359                BEQ     STM16           ;indicate success

                        ;       Display failure.

534B# 208153            STM14   JSR     DMW             ;delay a middling while
534E# A988                      LDA     #$88            ;color 1 for failure
5350# 20C353                    JSR     DRS             ;display RAM block status
5353# 4C5E53                    JMP     STM17

                        ;       Delay for simulating test of memory not to test.

5356# 208553            STM15   JSR     DLW             ;delay a long while

                        ;       Display success.

5359# A9CC              STM16   LDA     #$CC            ;color 2 for success
535B# 20C353                    JSR     DRS             ;display RAM block status

535E# A58D              STM17   LDA     STSMM
5360# 3026 ^5388                BMI     STM20

5362# A9C0                      LDA     #$C0
5364# 858D                      STA     STSMM
5366# E68E                      INC     STSMP           ;increment screen memory po:

5368# 18                STM18   CLC
5369# A58F                      LDA     ST1K            ;current 1K to test
536B# 6904                      ADC     #high $0400     ;add 1K
536D# 8591                      STA     STPAG+1         ;high current page
536F# 858F                      STA     ST1K            ;update current 1K to test
5371# CDE402                    CMP     RAMSIZ          ;RAM size
5374# D081 ^52F7                BNE     STM8            ;if not done testing RAM
```

```
                        ;       Check for auto-mode.

5376# A582                      LDA     STAUT           ;auto-mode flag
5378# D003 ^537D                BNE     STM19           ;if auto-mode, perform audi:

                        ;       Test memory again.

537A# 4CA952                    JMP     STM1            ;test memory again

                        ;       Process auto-mode.

537D# A90C              STM19   LDA     #$0C    ;indicate LED 1 and 2 off
537F# 20A453                    JSR     SLD     ;set LED's
5382# 208553                    JSR     DLW     ;delay a long while
5385# 4C5755                    JMP     STV     ;self-test audio-visual

5388# A90C              STM20   LDA     #$0C
538A# 858D                      STA     STSMM
538C# D0DA ^5368                BNE     STM18



                        **      DFS - Display First ROM Status
                        *
                        *       ENTRY   JSR     DFS
                        *
                        *       MODS
                        *               M. W. Colburn   10/26/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = 538E            DFS     =       *               ;entry
538E# A204                      LDX     #1*4            ;first 8K ROM display
5390# 202A57                    JSR     SVR             ;set value in range
5393# 29FC                      AND     #$FC
5395# 8D2330                    STA     ST3020+3
5398# 60                        RTS                     ;return



                        **      DSS - Display Second ROM Status
                        *
                        *       ENTRY   JSR     DSS
                        *
                        *       MODS
                        *               M. W. Colburn   10/26/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = 5399            DSS     =       *               ;entry
5399# A208                      LDX     #2*4            ;second 8K ROM display
539B# 202A57                    JSR     SVR             ;set value in range
```

```
539E# 29FC              AND     #$FC
53A0# 8D2730            STA     ST3024+3
53A3# 60                RTS                     ;return



                **      SLD - Set LED's
                *
                *       ENTRY   JSR     SLD
                *               A = LED mask (bit 3 - LED 2, bit 2 - LED 1)
                *
                *       MODS
                *               M. W. Colburn   10/26/82
                *               1. Bring closer to Coding Standard (object :
                *                  R. K. Nordin 11/01/83

     = 53A4     SLD     =       *               ;entry
53A4# 85A5              STA     STTMP5          ;save LED mask
53A6# AD01D3            LDA     PORTB
53A9# 29F3              AND     #$F3            ;clear.LED control
53AB# 05A5              ORA     STTMP5          ;set LED control according :
53AD# 8D01D3            STA     PORTB           ;update port B memory contr:
53B0# 60                RTS                     ;return



                **      DMW - Delay a Middling While
                *
                *       ENTRY   JSR     DMW
                *
                *       MODS
                *               M. W. Colburn   10/26/82
                *               1. Bring closer to Coding Standard (object :
                *                  R. K. Nordin 11/01/83

     = 53B1     DMW     =       *               ;entry
53B1# A23C              LDX     #60             ;60-VBLANK delay
53B3# D002 ^53B7        BNE     DAW             ;delay a while
```

```
                        **      DLW - Delay a Long While
                        *
                        *       ENTRY   JSR     DLW
                        *
                        *       MODS
                        *               M. W. Colburn   10/26/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = 53B5          DLW     =       *               ;entry
53B5# A296                      LDX     #150            ;150-VBLANK delay
                        ;       JMP     DAW             ;delay a while, return


                        **      DAW - Delay a While
                        *
                        *       ENTRY   JSR     DAW
                        *
                        *       MODS
                        *               M. W. Colburn   10/26/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = 53B7          DAW     =       *               ;entry

53B7# A0FF              DAW1    LDY     #$FF            ;initialize inner loop coun:

53B9# 8C0AD4            DAW2    STY     WSYNC           ;wait for HBLANK synchroniz:
53BC# 88                        DEY
53BD# D0FA ^53B9                BNE     DAW2            ;if inner loop not done

53BF# CA                        DEX
53C0# D0F5 ^53B7                BNE     DAW1            ;if outer loop not done

53C2# 60                        RTS                     ;return


                        **      DRS - Display RAM Block Status
                        *
                        *       ENTRY   JSR     DRS
                        *
                        *       MODS
                        *               M. W. Colburn   10/26/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = 53C3          DRS     =       *               ;entry
53C3# 48                        PHA                     ;save color
53C4# A68E                      LDX     STSMP
53C6# A58D                      LDA     STSMM
```

```
53C8# 49FF              EOR     #$FF            ;complement
53CA# 3D3830            AND     ST3038,X
53CD# 9D3830            STA     ST3038,X
53D0# 68                PLA                     ;saved color
53D1# 258D              AND     STSMM
53D3# 1D3630            ORA     ST3038,X
53D6# 9D3830            STA     ST3038,X
53D9# 60                RTS                     ;return


                **      POD - Process Other DLI's
                *
                *       POD turns the last line on the screen into white on:
                *       handles keyboard self-test display of console switc:
                *       HELP key for exit, and ensures no attract-mode.
                *
                *       ENTRY   JMP     POD
                *
                *       EXIT
                *               Exits via RTI
                *
                *       MODS
                *               M. W. Colburn   10/26/82
                *               1. Bring closer to Coding Standard (object :
                *                  R. K. Nordin 11/01/83


      = 53DA    POD     =       *       ;entry

                ;       Initialize.

53DA# 48                PHA             ;save A

                ;       Select colors.

53DB# A90C              LDA     #$0C    ;white color
53DD# 8D17D0            STA     COLPF1  ;playfield 1 color
53E0# ADC802            LDA     COLOR4  ;background color
53E3# 8D18D0            STA     COLPF2  ;playfield 2 color

                ;       Ensure no attract-mode.

53E6# A900              LDA     #0      ;no attract-mode
53E8# 854D              STA     ATRACT  ;attract-mode timer/flag

                ;       Check HELP key.

53EA# ADDC02            LDA     HELPFG  ;HELP key flag
53ED# F00E ^53FD        BEQ     POD1    ;if HELP not pressed

                ;       Process HELP key.

53EF# A900              LDA     #0      ;HELP key not pressed indicator
53F1# 8DDC02            STA     HELPFG  ;HELP key flag
53F4# A90C              LDA     #$0C    ;LED's off
```

```
53F6# 20A453                JSR     SLD         ;set LED's
53F9# 58                    CLI
53FA# 4C0C50                JMP     SEL         ;start over with main screen

                    ;       Check for keyboard self-test.

53FD# A58A          POD1    LDA     STKST       ;keyboard self-test flag
53FF# F047 ^5448            BEQ     POD10       ;if not keyboard self-test, exit

                    ;       Set display of console switches pressed.

5401# AD1FD0                LDA     CONSOL      ;console switches
5404# 2901                  AND     #$01        ;START key indicator
5406# F004 ^540C            BEQ     POD2        ;if START key pressed

5408# A9B3                  LDA     #$B3
540A# D002 ^540E            BNE     POD3        ;set display

540C# A933          POD2    LDA     #$33

540E# 8D1C30        POD3    STA     ST301C      ;set START key display

5411# AD1FD0                LDA     CONSOL      ;console switches
5414# 2902                  AND     #$02        ;SELECT key indicator
5416# F004 ^541C            BEQ     POD4        ;if SELECT key pressed

5418# A9F3                  LDA     #$F3
541A# D002 ^541E            BNE     POD5        ;set display

541C# A973          POD4    LDA     #$73

541E# 8D1E30        POD5    STA     ST301E      ;set SELECT key display

5421# AD1FD0                LDA     CONSOL      ;console switches
5424# 2904                  AND     #$04        ;OPTION key indicator
5426# F004 ^542C            BEQ     POD6        ;if OPTION key pressed

5428# A9AF                  LDA     #$AF
542A# D002 ^542E            BNE     POD7        ;set display

542C# A92F          POD6    LDA     #$2F

542E# 8D2030        POD7    STA     ST3020      ;set OPTION key display

                    ;       Sound tone if console switches pressed.

5431# AD1FD0                LDA     CONSOL      ;console switches
5434# 2907                  AND     #$07        ;key indicators
5436# C907                  CMP     #$07        ;no keys pressed
5438# F009 ^5443            BEQ     POD8        ;if no keys pressed

543A# A964                  LDA     #100        ;frequency
543C# 8D02D2                STA     AUDF2       ;set frequency of voice 2
543F# A9A8                  LDA     #$A8        ;pure tone, half volume
5441# D002 ^5445            BNE     POD9        ;set control of voice 2

5443# A900          POD8    LDA     #0          ;zero volume
```

```
5445# 8D03D2        POD9    STA     AUDC2   ;set control of voice 2

                    ;       Exit.

5448# 68            POD10   PLA             ;restore A
5449# 40                    RTI             ;return



                    **      TMNT - Table of Memory Not to Test
                    *
                    *       NOTES
                    *               Problem: bytes wasted by redundant entries.

544A# 00            TMNT    DB      high $0000      ;$0000 - $03FF, zero page a:
544B# 50                    DB      high $5000      ;$5000 - $53FF, self-test R:
544C# 54                    DB      high $5400      ;$5400 - $57FF, self-test R:
544D# 30                    DB      high ST3000     ;ST3000 - ST3000+$03FF, scr:
544E# 30                    DB      high ST3000     ;ST3000 - ST3000+$03FF, scr:
544F# 30                    DB      high ST3000     ;ST3000 - ST3000+$03FF, scr:

      = 0006        TMNTL   =       *-TMNT  ;length



                    **      STK - Self-test Keyboard
                    *
                    *       STK verifies the operation of the keyboard by displ:
                    *       keys as they are pressed.  In auto-mode, the verifi:
                    *       is simulated.
                    *
                    *       ENTRY   JSR     STK
                    *
                    *       NOTES
                    *               Problem: one too many bytes taken from TSKP:
                    *               Problem: wasted bytes for extra LDA CH.
                    *               Problem: logic is convoluted (due to SBT an:
                    *               subroutines appearing in the middle of STK):
                    *
                    *       MODS
                    *               M. W. Colburn   10/26/82
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = 5450        STK     =       *               ;entry

                    ;       Initialize.

5450# A200                  LDX     #0
5452# 8694                  STX     STSKP           ;initialize simulated keypr:
5454# A203                  LDX     #3              ;keyboard test colors
```

```
5456# 207357            JSR     SUC             ;set up colors
5459# A215              LDX     #low DISL4      ;keyboard display list
545B# A052              LDY     #high DISL4
545D# A9FF              LDA     #$FF            ;indicate keyboard self-tes:
545F# 209E50            JSR     SDL             ;set up display list

                ;       Test keyboard.

5462# A202      STK1    LDX     #2              ;offset to "KEYBOARD TEST" :
5464# 205957            JSR     SSM             ;set screen memory
5467# A207              LDX     #7              ;offset to keyboard text
5469# 205957            JSR     SSM             ;set screen memory

                ;       Check auto-mode.

546C# A582              LDA     STAUT           ;auto-mode flag
546E# F013 ^5483        BEQ     STK3            ;if not auto-mode

                ;       Simulate keypress.

5470# A694      STK2    LDX     STSKP           ;offset to next simulated k:
5472# BD4555            LDA     TSKP,X          ;simulated keypress
5475# E694              INC     STSKP           ;advance offset to simulate:
5477# A694              LDX     STSKP           ;offset to simulated keypre:
5479# E013              CPX     #TSKPL+1        ;last offset+1+1
547B# D014 ^5491        BNE     STK4            ;if last keypress not proce:

                ;       Self-test memory.

547D# 208553            JSR     DLW             ;delay a long while
5480# 4C9152            JMP     STM             ;self-test memory

                ;       Get a keypress.

5483# ADFC02    STK3    LDA     CH              ;key code
5486# C9FF              CMP     #$FF            ;clear code indicator
5488# F0F9 ^5483        BEQ     STK3            ;if no key pressed

548A# C9C0              CMP     #$C0
548C# B0F5 ^5483        BCS     STK3

548E# ADFC02            LDA     CH              ;key code

                ;       Process keypress.

5491# A2FF      STK4    LDX     #$FF            ;clear code indicator
5493# 8EFC02            STX     CH              ;key code
5496# 48                PHA                     ;save key code
5497# 2980              AND     #$80
5499# F005 ^54A0        BEQ     STK5            ;if not CTRL

549B# A208              LDX     #8              ;offset to control key text
549D# 205957            JSR     SSM             ;set screen memory

                ;       Check for shift key.

54A0# 68        STK5    PLA                     ;saved key code
```

```
54A1# 48                      PHA                       ;save key code
54A2# 2940                    AND     #$40
54A4# F00A ^54B0              BEQ     STK6              ;if not shift key

                      ;       Process keyboard shift key display.

54A6# A205                    LDX     #5                ;offset to "SH"
54A8# 205957                  JSR     SSM               ;set screen memory
54AB# A204                    LDX     #4                ;offset to "SH"
54AD# 205957                  JSR     SSM               ;set screen memory

                      ;       Check for special keys.

54B0# 68              STK6    PLA                       ;saved key code
54B1# 293F                    AND     #$3F
54B3# C921                    CMP     #$21
54B5# F068 ^551F              BEQ     KSB               ;if space bar, process disp:

54B7# C92C                    CMP     #$2C
54B9# F074 ^552F              BEQ     KTK               ;if tab key, process displa:

54BB# C934                    CMP     #$34
54BD# F068 ^5527              BEQ     KBK               ;if backspace key, process :

54BF# C90C                    CMP     #$0C
54C1# F076 ^5539              BEQ     KRK               ;if return key, process dis:

                      ;       Process other key displays.

54C3# AA                      TAX                       ;key code
54C4# BD9C57                  LDA     TSMC,X            ;display character
54C7# 48                      PHA                       ;save display character

54C8# A921                    LDA     #low ST3021
54CA# 8595                    STA     STTMP1            ;screen pointer
54CC# A930                    LDA     #high ST3021
54CE# 8596                    STA     STTMP1+1

                      ;       Find display character in screen memory.

54D0# 68                      PLA                       ;saved display character
54D1# A0FF                    LDY     #$FF              ;preset offset

54D3# C8              STK7    INY
54D4# D195                    CMP     (STTMP1),Y
54D6# D0FB ^54D3              BNE     STK7              ;if not found

                      ;       Display inverse video.

54D8# B195                    LDA     (STTMP1),Y
54DA# 4980                    EOR     #$80              ;invert video
54DC# 9195                    STA     (STTMP1),Y

                      ;       Check auto-mode.

54DE# A582            STK8    LDA     STAUT             ;auto-mode flag
54E0# F013 ^54F5              BEQ     STK9              ;if not auto-mode
```

```
                    ;       Process auto-mode.

54E2# 200555                JSR     SBT                 ;sound beep tone
54E5# A214                  LDX     #20                 ;20-VBLANK delay
54E7# 208753                JSR     DAW                 ;delay a while
54EA# 201055                JSR     SAS                 ;silence all sounds
54ED# A20A                  LDX     #10                 ;10-VBLANK delay
54EF# 208753                JSR     DAW                 ;delay a while
54F2# 4C6254                JMP     STK1                ;get next simulated keypres:

                    ;       Process manual mode.

54F5# 200555        STK9    JSR     SBT                 ;sound beep tone

54F8# AD0FD2        STK10   LDA     SKSTAT
54FB# 2904                  AND     #$04
54FD# F0F9 ^54F8            BEQ     STK10

54FF# 201055                JSR     SAS                 ;silence all sounds
5502# 4C6254                JMP     STK1                ;get next keypress


                    **      SBT - Sound Beep Tone
                    *
                    *       ENTRY   JSR     SBT
                    *
                    *       MODS
                    *               M. W. Colburn   10/26/82
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = 5505        SBT     =       *       ;entry
5505# A964                  LDA     #$64    ;frequency
5507# 8D00D2                STA     AUDF1   ;set frequency
550A# A9A8                  LDA     #$A8    ;pure tone, half volume
550C# 8D01D2                STA     AUDC1   ;set control
550F# 60                    RTS             ;return


                    **      SAS - Silence All Sounds
                    *
                    *       ENTRY   JSR     SAS
                    *
                    *       MODS
                    *               M. W. Colburn   10/26/82
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = 5510        SAS     =       *       ;entry
5510# A900                  LDA     #0      ;volume 0
```

```
5512# 8D01D2            STA     AUDC1   ;silence voice 1
5515# 8D03D2            STA     AUDC2   ;silence voice 2
5518# 8D05D2            STA     AUDC3   ;silence voice 3
551B# 8D07D2            STA     AUDC4   ;silence voice 4
551E# 60                RTS             ;return


                **      KSB - Process Keyboard Space Bar Display
                *
                *       ENTRY   JSR     KSB
                *
                *       MODS
                *               M. W. Colburn   10/26/82
                *               1. Bring closer to Coding Standard (object :
                *                  R. K. Nordin 11/01/83


     = 551F     KSB     =       *       ;entry
551F# A203              LDX     #3      ;offset to "S P A C E   B A R" text
5521# 205957            JSR     SSM     ;set screen memory
5524# 4CDE54            JMP     STK8    ;continue


                **      KBK - Process Keyboard Backspace Key Display
                *
                *       ENTRY   JSR     KBK
                *
                *       MODS
                *               M. W. Colburn   10/26/82
                *               1. Bring closer to Coding Standard (object :
                *                  R. K. Nordin 11/01/83


     = 5527     KBK     =       *       ;entry
5527# A206              LDX     #6      ;offset to "B S" text
5529# 205957            JSR     SSM     ;set screen memory
552C# 4CDE54            JMP     STK8    ;continue


                **      KTK - Process Keyboard Tab Key Display
                *
                *       ENTRY   JSR     KTK
                *
                *       MODS
                *               M. W. Colburn   10/26/82
                *               1. Bring closer to Coding Standard (object :
                *                  R. K. Nordin 11/01/83


     = 552F     KTK     =       *       ;entry
552F# A97F              LDA     #$7F
```

```
5531# 8D5230                STA     ST3052
5534# 8D5330                STA     ST3052+1
5537# D0A5 ^54DE            BNE     STK8     ;continue



                      **    KRK - Process Keyboard Return Key Display
                      *
                      *     ENTRY   JSR     KRK
                      *
                      *     MODS
                      *             M. W. Colburn   10/26/82
                      *             1. Bring closer to Coding Standard (object :
                      *                R. K. Nordin 11/01/83

      = 5539          KRK   =       *        ;entry
5539# A932                  LDA     #$32
553B# 8D6D30                STA     ST306D
553E# A934                  LDA     #$34
5540# 8D6E30                STA     ST306D+1
5543# D099 ^54DE            BNE     STK8     ;continue



                      **    TSKP - Table of Simulated Keypresses

5545# 52080A2B28      TSKP  DB      $52,$08,$0A,$2B,$28,$0D,$3D,$39,$2D     ;"C:
554E# 1F30351A              DB      $1F,$30,$35,$1A                         ;"1:
5552# 7F2D3F280D            DB      $7F,$2D,$3F,$28,$0D                     ;"A:

      = 0012          TSKPL =       *-TSKP   ;length



                      **    STV - Self-test Audio-visual
                      *
                      *     STV verifies the operation of the display and voice:
                      *     displaying and playing a tune.
                      *
                      *     ENTRY   JSR     STV
                      *
                      *     MODS    M. W. Colburn   10/26/82
                      *             1. Bring closer to Coding Standard (object :
                      *                R. K. Nordin 11/01/83

      = 5557          STV   =       *                 ;entry

                      ;     Initialize.

5557# A202                  LDX     #2                ;audio-visual test colors
```

```
5559# 207357            JSR     SUC             ;set up colors

                ;       Test audio-visual.

555C# A900      STV1    LDA     #0
555E# 8597              STA     STVOC           ;initialize voice indicator

                ;       Test voice.

5560# A900      STV2    LDA     #0
5562# 8598              STA     STNOT           ;initialize note counter
5564# A231              LDX     #low DISL5      ;audio-visual display list
5566# A052              LDY     #high DISL5
5568# A900              LDA     #0              ;indicate not keyboard self:
556A# 209E50            JSR     SDL             ;set up display list

                ;       Display voice number.

556D# A209              LDX     #9              ;offset to "VOICE #" text
556F# 205957            JSR     SSM             ;set screen memory
5572# A597              LDA     STVOC           ;voice indicator
5574# 4A                LSR     A               ;voice number
5575# 18                CLC
5576# 6911              ADC     #$11            ;adjust for screen memory
5578# 8D0B30            STA     ST300B          ;voice number display

                ;       Display staff.

557B# A20F              LDX     #$0F            ;offset to last byte of sta:

557D# A9FF      STV3    LDA     #$FF            ;color 2
557F# 9D5031            STA     ST3150,X        ;byte of first line of staf:
5582# 9DB031            STA     ST31B0,X        ;byte of second line of sta:
5585# 9D1032            STA     ST3210,X        ;byte of third line of staf:
5588# 9D7032            STA     ST3270,X        ;byte of fourth line of sta:
558B# 9DD032            STA     ST32D0,X        ;byte of fifth line of staf:
558E# CA                DEX
558F# 10EC ^557D        BPL     STV3            ;if not done

                ;       Display cleft.

5591# A900              LDA     #0              ;offset to first cleft disp:
5593# 8599              STA     STCDI           ;cleft display pointer
5595# A90C              LDA     #2*6
5597# 859A              STA     STCDA           ;cleft data pointer

5599# A699      STV4    LDX     STCDI           ;cleft display pointer
559B# BD1757            LDA     TCDA+1,X        ;high address of cleft disp:
559E# A8                TAY
559F# BD1657            LDA     TCDA,X          ;low address of cleft displ:
55A2# AA                TAX
55A3# A59A              LDA     STCDA           ;cleft data pointer
55A5# 208556            JSR     DVN
55A8# 18                CLC
55A9# A59A              LDA     STCDA           ;cleft data pointer
55AB# 6906              ADC     #6
55AD# 859A              STA     STCDA           ;update cleft data pointer
```

```
55AF# E699                  INC     STCDI           ;increment cleft display po;
55B1# E699                  INC     STCDI
55B3# A599                  LDA     STCDI           ;cleft display pointer
55B5# C914                  CMP     #TCDAL          ;length of cleft display ta;
55B7# D0E0 ^5599            BNE     STV4            ;if not done

                      ;     Delay.

55B9# 20B153                JSR     DMW             ;delay a middling while

                      ;     Display and play first note.

55BC# A254                  LDX     #low ST3154
55BE# A031                  LDY     #high ST3154
55C0# A900                  LDA     #0*6
55C2# 208556                JSR     DVN

55C5# A951                  LDA     #$51            ;first note frequency
55C7# 206C56                JSR     SVN

                      ;     Display and play second note.

55CA# A286                  LDX     #low ST3186
55CC# A031                  LDY     #high ST3186
55CE# A900                  LDA     #0*6
55D0# 208556                JSR     DVN

55D3# A95B                  LDA     #$5B            ;second note frequency
55D5# 206C56                JSR     SVN

                      ;     Display and play third note.

55D8# A2F8                  LDX     #low ST30F8
55DA# A030                  LDY     #high ST30F8
55DC# A948                  LDA     #12*6
55DE# 208556                JSR     DVN
55E1# A2C7                  LDX     #low ST30C7
55E3# A030                  LDY     #high ST30C7
55E5# A954                  LDA     #14*6
55E7# 208556                JSR     DVN
55EA# A248                  LDX     #low ST3248
55EC# A032                  LDY     #high ST3248
55EE# A94E                  LDA     #13*6
55F0# 208556                JSR     DVN

55F3# A944                  LDA     #$44            ;third note frequency
55F5# 206C56                JSR     SVN

                      ;     Display and play fourth note.

55F8# A2CA                  LDX     #low ST30CA
55FA# A030                  LDY     #high ST30CA
55FC# A948                  LDA     #12*6
55FE# 208556                JSR     DVN
5601# A21A                  LDX     #low ST321A
5603# A032                  LDY     #high ST321A
5605# A94E                  LDA     #13*6
```

```
5607# 208556          JSR     DVN
560A# A2CA            LDX     #low ST31CA
560C# A031            LDY     #high ST31CA
560E# A906            LDA     #1*6
5610# 208556          JSR     DVN

5613# A93C            LDA     #$3C          ;fourth note frequency
5615# 206C56          JSR     SVN

              ;       Display and play fifth note.

5618# A23C            LDX     #low ST303C
561A# A030            LDY     #high ST303C
561C# A948            LDA     #12*6
561E# 208556          JSR     DVN
5621# A28C            LDX     #low ST318C
5623# A031            LDY     #high ST318C
5625# A94E            LDA     #13*6
5627# 208556          JSR     DVN
562A# A23C            LDX     #low ST313C
562C# A031            LDY     #high ST313C
562E# A906            LDA     #1*6
5630# 208556          JSR     DVN

5633# A92D            LDA     #$2D          ;fifth note frequency
5635# 206C56          JSR     SVN

              ;       Display and play sixth note.

5638# A29E            LDX     #low ST309E
563A# A030            LDY     #high ST309E
563C# A948            LDA     #12*6
563E# 208556          JSR     DVN
5641# A2EE            LDX     #low ST31EE
5643# A031            LDY     #high ST31EE
5645# A94E            LDA     #13*6
5647# 208556          JSR     DVN

564A# A935            LDA     #$35          ;sixth note frequency
564C# 206C56          JSR     SVN

              ;       Delay.

564F# 208553          JSR     DLW           ;delay a long while

              ;       Advance to next voice.

5652# E697            INC     STVOC         ;increment voice indicator
5654# E697            INC     STVOC
5656# A597            LDA     STVOC         ;voice indicator
5658# C908            CMP     #8            ;last voice indicator
565A# D007 ^5663      BNE     STV5          ;if all voices not processe:

              ;       Process test completion.

565C# A582            LDA     STAUT         ;auto-mode flag
565E# D006 ^5666      BNE     STV6          ;if auto-mode, perform keyb:
```

```
5660# 4C5C55                JMP     STV1            ;repeat audio-visual test

                    ;       Test next voice.

5663# 4C6055        STV5    JMP     STV2            ;test next voice

                    ;       Self-test keyboard.

5666# 208553        STV6    JSR     DLW             ;delay a long while
5669# 4C5054                JMP     STK             ;self-test keyboard


                    **      SVN - Sound Tone
                    *
                    *       ENTRY   JSR     SVN
                    *
                    *       MODS
                    *               M. W. Colburn   10/26/82
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

        = 566C      SVN     =       *               ;entry

                    ;       Sound note.

566C# A497                  LDY     STVOC           ;current voice indicator
566E# 9900D2                STA     AUDF1,Y         ;set frequency
5671# A9A8                  LDA     #$A8            ;pure tone, half volume
5673# 9901D2                STA     AUDC1,Y         ;set control

                    ;       Delay a while.

5676# A698                  LDX     STNOT           ;current note
5678# BDB656                LDA     TNDD,X          ;delay time
567B# AA                    TAX
567C# 208753                JSR     DAW             ;delay a while

                    ;       Increment note counter.

567F# E698                  INC     STNOT           ;increment note counter

                    ;       Exit.

5681# 201055                JSR     SAS             ;silence all sounds
5684# 60                    RTS                     ;return
```

```
                        **      DVN - Display
                        *
                        *       ENTRY   JSR     DVN
                        *
                        *       MODS
                        *               M. W. Colburn   10/26/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = 5685          DVN     =       *               ;entry
5685# 8698                      STX     STTMP2
5687# 849C                      STY     STTMP2+1
5689# AA                        TAX
568A# A000                      LDY     #0
568C# A910                      LDA     #16
568E# 859D                      STA     STTMP3
5690# A906                      LDA     #6
5692# 85A3                      STA     STTMP4

5694# BDBC56            DVN1    LDA     TAVD,X
5697# 119B                      ORA     (STTMP2),Y
5699# 919B                      STA     (STTMP2),Y
569B# 20AA56                    JSR     AST             ;add 16
569E# C69D                      DEC     STTMP3
56A0# D0F2 ^5694                BNE     DVN1

56A2# E69D                      INC     STTMP3
56A4# E8                        INX
56A5# C6A3                      DEC     STTMP4
56A7# D0EB ^5694                BNE     DVN1

56A9# 60                        RTS                     ;return


                        **      AST - Add Sixteen
                        *
                        *       ENTRY   JSR     AST
                        *
                        *       MODS
                        *               M. W. Colburn   10/26/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = 56AA          AST     =       *               ;entry
56AA# 18                        CLC
56AB# A59B                      LDA     STTMP2          ;current low value
56AD# 6910                      ADC     #16             ;add 16
56AF# 859B                      STA     STTMP2          ;new low value
56B1# 9002 ^56B5                BCC     AST1            ;if no carry

56B3# E69C                      INC     STTMP2+1        ;adjust high value

56B5# 60                AST1    RTS                     ;return
```

```
                    **      TNDD - Table of Note Duration Delays


56B6# 20            TNDD    DB        32        ;0 - first note
56B7# 20                    DB        32        ;1 - second note
56B8# 20                    DB        32        ;2 - third note
56B9# 10                    DB        16        ;3 - fourth note
56BA# 10                    DB        16        ;4 - fifth note
56BB# 20                    DB        32        ;5 - sixth note



                    **      TAVD - Table of Audio-visual Test Display Data


56BC# 011F3F7F3E    TAVD    DB        $01,$1F,$3F,$7F,$3E,$1C        ;0
56C2# 0041424C70            DB        $00,$41,$42,$4C,$70,$40        ;1
56C8# 0001020408            DB        $00,$01,$02,$04,$08,$10        ;2
56CE# 0043444848            DB        $00,$43,$44,$48,$48,$48        ;3
56D4# 0044221008            DB        $00,$44,$22,$10,$08,$07        ;4
56DA# 0004080502            DB        $00,$04,$08,$05,$02,$00        ;5
56E0# 0030488884            DB        $00,$30,$48,$88,$84,$84        ;6
56E6# 00888890A0            DB        $00,$88,$88,$90,$A0,$C0        ;7
56EC# 00F0888482            DB        $00,$F0,$88,$84,$82,$82        ;8
56F2# 0082828488            DB        $00,$82,$82,$84,$88,$F0        ;9
56F8# 0000000000            DB        $00,$00,$00,$00,$00,$80        ;10
56FE# 8080808080            DB        $80,$80,$80,$80,$80,$80        ;11
5704# 001C3E7F7E            DB        $00,$1C,$3E,$7F,$7E,$7C        ;12
570A# 4000000000            DB        $40,$00,$00,$00,$00,$00        ;13
5710# 0004040605            DB        $00,$04,$04,$06,$05,$06        ;14



                    **      TCDA - Table of Cleft Display Addresses


5716# C130          TCDA    DW        ST30C1    ;0
5718# 2131                  DW        ST3121    ;1
571A# 8131                  DW        ST3181    ;2
571C# F131                  DW        ST31F1    ;3
571E# 0230                  DW        ST3002    ;4
5720# 6230                  DW        ST3062    ;5
5722# 2231                  DW        ST3122    ;6
5724# 8231                  DW        ST3182    ;7
5726# C230                  DW        ST30C2    ;8
5728# C231                  DW        ST31C2    ;9

      = 0014        TCDAL   =         *-TCDA              ;length
```

```
                    **      SVR - Set Value in Range
                    *
                    *       ENTRY   JSR     SVR
                    *               A = value to set
                    *               X = offset to TARS range
                    *
                    *       EXIT
                    *               A = value set
                    *
                    *       MODS
                    *               M. W. Colburn   10/26/82
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = 572A        SVR     =       *               ;entry

                    ;       Initialize.

572A# 48                    PHA                     ;save value

                    ;       Set address range.

572B# BDDC57                LDA     TARS,X          ;start of range
572E# 859E                  STA     STADR1
5730# BDDD57                LDA     TARS+1,X
5733# 859F                  STA     STADR1+1
5735# BDDE57                LDA     TARS+2,X        ;end of range
5738# 85A0                  STA     STADR2
573A# BDDF57                LDA     TARS+3,X
573D# 85A1                  STA     STADR2+1

                    ;       Set value in range.

573F# A000                  LDY     #0              ;offset to first byte

5741# 68            SVR1    PLA                     ;saved value
5742# 919E                  STA     (STADR1),Y      ;byte of range
5744# E69E                  INC     STADR1          ;increment low address
5746# D002 ^574A            BNE     SVR2            ;if no carry

5748# E69F                  INC     STADR1+1        ;adjust high address

574A# 48            SVR2    PHA                     ;save value
574B# A59E                  LDA     STADR1          ;low current address
574D# C5A0                  CMP     STADR2          ;low end of range
574F# D0F0 ^5741            BNE     SVR1            ;if definitely not done

5751# A59F                  LDA     STADR1+1        ;high current address
5753# C5A1                  CMP     STADR2+1        ;high end of range
5755# D0EA ^5741            BNE     SVR1            ;if not done

                    ;       Exit.

5757# 68                    PLA                     ;restore value
5758# 60                    RTS                     ;return
```

```
                        **      SSM - Set Screen Memory
                        *
                        *       ENTRY   JSR     SSM
                        *
                        *       MODS
                        *               M. W. Colburn   10/26/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = 5759            SSM     =       *               ;entry
 5759# BD57CA                   LDA     TSTO,X          ;offset to source
 575C# A8                       TAY
 575D# BDEC57                   LDA     TSTL,X          ;length of source
 5760# 859E                     STA     STADR1          ;length
 5762# BDF657                   LDA     TSTD,X          ;offset to destination
 5765# AA                       TAX

 5766# B961CA           SSM1    LDA     TTXT,Y          ;byte of source
 5769# 9D0030                   STA     ST3000,X        ;byte of destination
 576C# C8                       INY
 576D# E8                       INX
 576E# C69E                     DEC     STADR1          ;decrement length
 5770# D0F4 ^5766               BNE     SSM1            ;if not done

 5772# 60                       RTS                     ;return



                        **      SUC - Set Up Colors
                        *
                        *       ENTRY   JSR     SUC
                        *               X = 0, if main screen colors
                        *                 = 1, if memory test colors
                        *                 = 2, if keyboard test colors
                        *                 = 3, if audio-visual test colors
                        *
                        *       EXIT
                        *               COLOR0, COLOR1, COLOR2 and COLOR4 set.
                        *
                        *       CHANGES
                        *               A
                        *
                        *       CALLS
                        *               -none-
                        *
                        *       MODS
                        *               M. W. Colburn   10/26/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = 5773            SUC     =       *               ;entry

 5773# BD8C57                   LDA     SUCA,X
 5776# 8DC402                   STA     COLOR0  ;playfield 0 color
```

```
5779# BD9057              LDA     SUCB,X
577C# 8DC502              STA     COLOR1  ;playfield 1 color

577F# BD9457              LDA     SUCC,X
5782# 8DC602              STA     COLOR2  ;playfield 2 color

5785# BD9857              LDA     SUCD,X
5788# 8DC802              STA     COLOR4  ;background color

578B# 60                  RTS             ;return


578C# 2C          SUCA    DB      $2C     ;0 - main screen playfield 0 color
578D# 0C                  DB      $0C     ;1 - memory test playfield 0 color
578E# 2A                  DB      $2A     ;2 - keyboard test playfield 0 colo:
578F# 18                  DB      $18     ;3 - audio-visual test playfield 0 :

5790# 0F          SUCB    DB      $0F     ;0 - main screen playfield 1 color
5791# 32                  DB      $32     ;1 - memory test playfield 1 color
5792# 0C                  DB      $0C     ;2 - keyboard test playfield 1 colo:
5793# 0E                  DB      $0E     ;3 - audio-visual test playfield 1 :

5794# D2          SUCC    DB      $D2     ;0 - main screen playfield 2 color
5795# D6                  DB      $D6     ;1 - memory test playfield 2 color
5796# 00                  DB      $00     ;2 - keyboard test playfield 2 colo:
5797# 84                  DB      $84     ;3 - audio-visual test playfield 2 :

5798# D2          SUCD    DB      $D2     ;0 - main screen background color
5799# A0                  DB      $A0     ;1 - memory test background color
579A# 30                  DB      $30     ;2 - keyboard test background color
579B# 84                  DB      $84     ;3 - audio-visual test background c:


                  **      TSMC - Table of Screen Memory Character Codes
                  *
                  *       Entry n is the screen memory character code for key:


579C# 2C          TSMC    DB      $2C     ;$00 - L key
579D# 2A                  DB      $2A     ;$01 - J key
579E# 1B                  DB      $1B     ;$02 - semicolon key
579F# 91                  DB      $91     ;$03
57A0# 92                  DB      $92     ;$04
57A1# 2B                  DB      $2B     ;$05 - K key
57A2# 0B                  DB      $0B     ;$06 - plus key
57A3# 0A                  DB      $0A     ;$07 - asterisk key
57A4# 2F                  DB      $2F     ;$08 - O key
57A5# 00                  DB      $00     ;$09
57A6# 30                  DB      $30     ;$0A - P key
57A7# 35                  DB      $35     ;$0B - U key
57A8# B2                  DB      $B2     ;$0C - RETURN key
57A9# 29                  DB      $29     ;$0D - I key
57AA# 0D                  DB      $0D     ;$0E - minus key
57AB# 1D                  DB      $1D     ;$0F - = key
```

```
57AC# 36                DB      $36     ;$10 - V key
57AD# A8                DB      $A8     ;$11
57AE# 23                DB      $23     ;$12 - C key
57AF# 93                DB      $93     ;$13
57B0# 94                DB      $94     ;$14
57B1# 22                DB      $22     ;$15 - B key
57B2# 38                DB      $38     ;$16 - X key
57B3# 3A                DB      $3A     ;$17 - Z key
57B4# 14                DB      $14     ;$18 - 4 key
57B5# 00                DB      $00     ;$19
57B6# 13                DB      $13     ;$1A - 3 key
57B7# 16                DB      $16     ;$1B - 6 key
57B8# 5B                DB      $5B     ;$1C - ESC key
57B9# 15                DB      $15     ;$1D - 5 key
57BA# 12                DB      $12     ;$1E - 2 key
57BB# 11                DB      $11     ;$1F - 1 key

57BC# 0C                DB      $0C     ;$20 - comma key
57BD# 00                DB      $00     ;$21 - space key
57BE# 0E                DB      $0E     ;$22 - period key
57BF# 2E                DB      $2E     ;$23 - N key
57C0# 00                DB      $00     ;$24
57C1# 2D                DB      $2D     ;$25 - M key
57C2# 0F                DB      $0F     ;$26 - / key
57C3# A1                DB      $A1     ;$27 - inverse video key
57C4# 32                DB      $32     ;$28 - R key
57C5# 00                DB      $00     ;$29
57C6# 25                DB      $25     ;$2A - E key
57C7# 39                DB      $39     ;$2B - Y key
57C8# FF                DB      $FF     ;$2C - TAB key
57C9# 34                DB      $34     ;$2D - T key
57CA# 37                DB      $37     ;$2E - W key
57CB# 31                DB      $31     ;$2F - Q key

57CC# 19                DB      $19     ;$30 - 9 key
57CD# 00                DB      $00     ;$31
57CE# 10                DB      $10     ;$32 - 0 key
57CF# 17                DB      $17     ;$33 - 7 key
57D0# A2                DB      $A2     ;$34 - backspace key
57D1# 18                DB      $18     ;$35 - 8 key
57D2# 1C                DB      $1C     ;$36 - < key
57D3# 1E                DB      $1E     ;$37 - > key
57D4# 26                DB      $26     ;$38 - F key
57D5# 28                DB      $28     ;$39 - H key
57D6# 24                DB      $24     ;$3A - D key
57D7# 00                DB      $00     ;$3B
57D8# A3                DB      $A3     ;$3C - CAPS key
57D9# 27                DB      $27     ;$3D - G key
57DA# 33                DB      $33     ;$3E - S key
57DB# 21                DB      $21     ;$3F - A key
```

```
                        **      TARS - Table of Address Ranges to Set

57DC# 0030FF3E          TARS    DW      ST3000,ST3000+$0EFF     ;0 - screen memory
57E0# 20302430                  DW      ST3020,ST3020+4         ;1 - memory test fi:
57E4# 24302830                  DW      ST3024,ST3024+4         ;2 - memory test se:
57E8# 00302030                  DW      ST3000,ST3000+32        ;3 - main screen bo:


                        **      TSTL - Table of Self-test Text Lengths

57EC# 13                TSTL    DB      TXT0L   ;0 - length of "MEMORY TEST   ROM" :
57ED# 03                        DB      TXT1L   ;1 - length of "RAM" text
57EE# 13                        DB      TXT2L   ;2 - length of "KEYBOARD TEST" text
57EF# 13                        DB      TXT3L   ;3 - length of "S P A C E   B A R" :
57F0# 04                        DB      TXT4L   ;4 - length of "SH" text
57F1# 04                        DB      TXT5L   ;5 - length of "SH" text
57F2# 03                        DB      TXT6L   ;6 - length of "B S" text
57F3# A8                        DB      TXT7L   ;7 - length of keyboard text
57F4# 03                        DB      TXT8L   ;8 - length of control key text
57F5# 07                        DB      TXT9L   ;9 - length of "VOICE #" text


                        **      TSTD - Table of Self-test Text Destination Offsets

57F6# 00                TSTD    DB      ST3000-ST3000   ;0 - offset to "MEMORY TEST:
57F7# 28                        DB      ST3028-ST3000   ;1 - offset to "RAM" text
57F8# 00                        DB      ST3000-ST3000   ;2 - offset to "KEYBOARD TE:
57F9# 87                        DB      ST3087-ST3000   ;3 - offset to "S P A C E  :
57FA# 92                        DB      ST3092-ST3000   ;4 - offset to "SH" text
57FB# AB                        DB      ST30AB-ST3000   ;5 - offset to "SH" text
57FC# 4C                        DB      ST304C-ST3000   ;6 - offset to "B S" text
57FD# 22                        DB      ST3022-ST3000   ;7 - offset to keyboard tex:
57FE# 72                        DB      ST3072-ST3000   ;8 - offset to control key :
57FF# 04                        DB      ST3004-ST3000   ;9 - offset to "VOICE #" te:
```

```
5800#            ***     (C) Copyright 1978 Shepardson Microsystems, Inc.


5800#                    FIX     $D800


                 ***     FPP - Floating Point Package
                 *
                 *       FPP is a collection of routines for floating point
                 *       computations.  A floating point number is represent:
                 *       in 6 bytes:
                 *
                 *       Byte 0
                 *               Bit 7           Sign of mantissa
                 *               Bits 0 - 6      BCD exponent, biased by $40
                 *
                 *       Bytes 1 - 5             BCD mantissa
                 *
                 *       MODS
                 *               Shepardson Microsystems
                 *
                 *               Produce 2K version.
                 *               M. Lorenzen     09/06/81


D800                     FIX     AFP


                 **      AFP - Convert ASCII to Floating Point
                 *
                 *       ENTRY   JSR     AFP
                 *               INBUFF = line buffer pointer
                 *               CIX = offset to first byte of number
                 *
                 *       EXIT
                 *               C clear, if valid number
                 *               C set, if invalid number
                 *
                 *       NOTES
                 *               Problem: bytes wasted by check for "-", nea:
                 *
                 *       MODS
                 *               Original Author Unknown
                 *               1. Bring closer to Coding Standard (object :
                 *                  R. K. Nordin 11/01/83

                 ;AFP    =       *               ;entry
```

```
                        ;       Initialize.

D800  20A1DB                    JSR     SLB     ;skip leading blanks

                        ;       Check for number.

D803  20B8DB                    JSR     TVN     ;test for valid number character
D806  B039 ^D841                BCS     AFP5    ;if not number character

                        ;       Set initial values.

D808  A2ED                      LDX     #EEXP   ;exponent
D80A  A004                      LDY     #4      ;indicate 4 bytes to clear
D80C  2048DA                    JSR     ZXLY
D80F  A2FF                      LDX     #$FF
D811  86F1                      STX     DIGRT   ;number of digits after decimal poi:
D813  2044DA                    JSR     ZFR0    ;zero FR0
D816  F004 ^D81C                BEQ     AFP2    ;get first character

                        ;       Indicate not first character.

D818  A9FF              AFP1    LDA     #$FF    ;indicate not first character
D81A  85F0                      STA     FCHFLG  ;first character flag

                        ;       Get next character.

D81C  2094DB            AFP2    JSR     GNC     ;get next character
D81F  B021 ^D842                BCS     AFP6    ;if character not numeric

                        ;       Process numeric character.

D821  48                        PHA                     ;save digit
D822  A6D5                      LDX     FR0M            ;first byte
D824  D011 ^D837                BNE     AFP3            ;if not zero

D826  20E8DB                    JSR     S0L             ;shift FR0 left 1 digit
D829  68                        PLA                     ;saved digit
D82A  05D9                      ORA     FR0M+FMPREC-1   ;insert into last byte
D82C  85D9                      STA     FR0M+FMPREC-1   ;update last byte

                        ;       Check for decimal point.

D82E  A6F1                      LDX     DIGRT   ;number of digits after decimal poi:
D830  30E6 ^D818                BMI     AFP1    ;if no decimal point, process next :

                        ;       Increment number of digits after decimal point.

D832  E8                        INX             ;increment number of digits
D833  86F1                      STX     DIGRT   ;number of digits after decimal poi:
D835  D0E1 ^D818                BNE     AFP1    ;process next character

                        ;       Increment exponent, if necessary.

D837  68                AFP3    PLA             ;clean stack
D838  A6F1                      LDX     DIGRT   ;number of digits after decimal poi:
D83A  1002 ^D83E                BPL     AFP4    ;if already have decimal point
```

```
D83C  E6ED                INC    EEXP    ;increment number of digits more th:

                    ;     Process next character.

D83E  4C18D8        AFP4  JMP    AFP1    ;process next character

                    ;     Exit.

D841  60            AFP5  RTS            ;return

                    ;     Process non-numeric character.

D842  C92E          AFP6  CMP    #'.'
D844  F014 ^D85A          BEQ    AFP8    ;if ".", process decimal point

D846  C945                CMP    #'E'
D848  F019 ^D863          BEQ    AFP9    ;if "E", process exponent

D84A  A6F0                LDX    FCHFLG  ;first character flag
D84C  D068 ^D8B6          BNE    AFP16   ;if not first character, process en:

D84E  C92B                CMP    #'+'
D850  F0C6 ^D818          BEQ    AFP1    ;if "+", process next character

D852  C92D                CMP    #'-'
D854  F000 ^D856          BEQ    AFP7    ;if "-", process negative sign

                    ;     Process negative sign.

D856  85EE          AFP7  STA    NSIGN   ;sign of number
D858  F0BE ^D818          BEQ    AFP1    ;process next character

                    ;     Process decimal point.

D85A  A6F1          AFP8  LDX    DIGRT   ;number of digits after decimal poi:
D85C  1058 ^D8B6          BPL    AFP16   ;if already have decimal point

D85E  E8                  INX            ;zero
D85F  86F1                STX    DIGRT   ;number of digits after decimal poi:
D861  F0B5 ^D818          BEQ    AFP1    ;process next character

                    ;     Process exponent.

D863  A5F2          AFP9  LDA    CIX     ;offset to character
D865  85EC                STA    FRX     ;save offset to character
D867  2094D8              JSR    GNC     ;get next character
D86A  B037 ^D8A3          BCS    AFP13   ;if not numeric

                    ;     Process numeric character in exponent.

D86C  AA            AFP10 TAX            ;first character of exponent
D86D  A5ED                LDA    EEXP    ;number of digits more than 9
D86F  48                  PHA            ;save number of digits more than 9
D870  86ED                STX    EEXP    ;first character of exponent

                    ;     Process second character of exponent.
```

```
D872  2094D8              JSR    GNC      ;get next character
D875  B017 ^D88E          BCS    AFP11    ;if not numeric, no second digit

D877  48                  PHA             ;save second digit
D878  A5ED                LDA    EEXP     ;first digit
D87A  0A                  ASL    A        ;2 times first digit
D87B  85ED                STA    EEXP     ;2 times first digit
D87D  0A                  ASL    A        ;4 times first digit
D87E  0A                  ASL    A        ;8 times first digit
D87F  65ED                ADC    EEXP     ;add 2 times first digit
D881  85ED                STA    EEXP     ;save 10 times first digit
D883  68                  PLA             ;saved second digit
D884  18                  CLC
D885  65ED                ADC    EEXP     ;insert in exponent
D887  85ED                STA    EEXP     ;update exponent

                 ;        Process third character of exponent.

D889  A4F2                LDY    CIX      ;offset to third character
D88B  209DD8              JSR    ICX      ;increment offset

D88E  A5EF       AFP11    LDA    ESIGN    ;sign of exponent
D890  F009 ^D89B          BEQ    AFP12    ;if no sign on exponent

                 ;        Process negative exponent.

D892  A5ED                LDA    EEXP     ;exponent
D894  49FF                EOR    #$FF     ;complement exponent
D896  18                  CLC
D897  6901                ADC    #1       ;add 1 for 2's complement
D899  85ED                STA    EEXP     ;update exponent

                 ;        Add in number of digits more than 9.

D89B  68         AFP12    PLA             ;saved number of digits more than 9
D89C  18                  CLC
D89D  65ED                ADC    EEXP     ;add exponent
D89F  85ED                STA    EEXP     ;update exponent
D8A1  D013 ^D8B6          BNE    AFP16    ;process end of input

                 ;        Process non-numeric in exponent.

D8A3  C92B       AFP13    CMP    #'+'
D8A5  F006 ^D8AD          BEQ    AFP14    ;if "+", process next character

D8A7  C92D                CMP    #'-'
D8A9  D007 ^D8B2          BNE    AFP15

D8AB  85EF                STA    ESIGN    ;save sign of exponent

                 ;        Process next character.

D8AD  2094D8     AFP14    JSR    GNC      ;get next character
D8B0  90BA ^D86C          BCC    AFP10    ;if numeric, process numeric charac:

                 ;        Process other non-numeric in exponent.
```

```
D8B2  A5EC          AFP15   LDA   FRX      ;saved offset
D8B4  85F2                  STA   CIX      ;restore offset

                    ;       Process end of input.

D8B6  C6F2          AFP16   DEC   CIX      ;decrement offset
D8B8  A5ED                  LDA   EEXP     ;exponent
D8BA  A6F1                  LDX   DIGRT    ;number of digits after decimal poi:
D8BC  3005 ^D8C3            BMI   AFP17    ;if no decimal point

D8BE  F003 ^D8C3            BEQ   AFP17    ;if no digits after decimal point

D8C0  38                    SEC
D8C1  E5F1                  SBC   DIGRT    ;subtract number of digits after de:

D8C3  48            AFP17   PHA            ;save adjusted exponent
D8C4  2A                    ROL   A        ;set C with sign of exponent
D8C5  68                    PLA            ;saved adjusted exponent
D8C6  6A                    ROR   A        ;shift right
D8C7  85ED                  STA   EEXP     ;save power of 100
D8C9  9003 ^D8CE            BCC   AFP18    ;if no carry, process even number

D8CB  20E8D8                JSR   SOL      ;shift FR0 left 1 digit

D8CE  A5ED          AFP18   LDA   EEXP     ;exponent
D8D0  18                    CLC
D8D1  6944                  ADC   #$40+4   ;add bias plus 4 for normalization
D8D3  85D4                  STA   FR0      ;save exponent

D8D5  2000DC                JSR   NORM     ;normalize number
D8D8  B008 ^D8E5            BCS   AFP20    ;if error

                    ;       Check sign of number.

D8DA  A6EE                  LDX   NSIGN    ;sign of number
D8DC  F006 ^D8E4            BEQ   AFP19    ;if sign of number not negative

                    ;       Process negative number.

D8DE  A5D4                  LDA   FR0      ;first byte of mantissa
D8E0  0980                  ORA   #$80     ;indicate negative
D8E2  85D4                  STA   FR0      ;update first byte of mantissa

                    ;       Exit.

D8E4  18            AFP19   CLC            ;indicate valid number

D8E5  60            AFP20   RTS            ;return
```

```
D8E6                            FIX     FASC


                        **      FASC - Convert Floating Point Number to ASCII
                        *
                        *       ENTRY   JSR     FASC
                        *               FR0 - FR0+5 = number to convert
                        *
                        *       EXIT
                        *               INBUFF = pointer to start of number
                        *               High order bit of last charecter set
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

                        ;FASC   =       *       ;entry

                        ;       Initialize.

D8E6  2051DA                    JSR     ILP     ;initialize line buffer pointer
D8E9  A930                      LDA     #'0'
D8EB  8D7F05                    STA     LBPR2   ;put "0" in front of line buffer

                        ;       Check for E format required.

D8EE  A5D4                      LDA     FR0     ;exponent
D8F0  F028 ^D91A                BEQ     FASC2   ;if exponent zero, number zero

D8F2  297F                      AND     #$7F    ;clear sign
D8F4  C93F                      CMP     #$40-1  ;bias-1
D8F6  9028 ^D920                BCC     FASC3   ;if exponent < bias-1, E format req:

D8F8  C945                      CMP     #$40+5  ;bias+5
D8FA  B024 ^D920                BCS     FASC3   ;if >= bias+5, E format required

                        ;       Process E format not required.

D8FC  38                        SEC
D8FD  E93F                      SBC     #$40-1  ;subtract bias-1, yielding decimal :
D8FF  2070DC                    JSR     COA     ;convert FR0 to ASCII
D902  20A4DC                    JSR     FNZ     ;find last non-zero character
D905  0980                      ORA     #$80    ;set high order bit
D907  9D8005                    STA     LBUFF,X ;update last character
D90A  AD8005                    LDA     LBUFF   ;first character
D90D  C92E                      CMP     #'.'
D90F  F003 ^D914                BEQ     FASC1   ;if decimal point

D911  4C88D9                    JMP     FASC10

D914  20C1DC            FASC1   JSR     DLP     ;decrement line buffer pointer
D917  4C9CD9                    JMP     FASC11  ;perform final adjustment
```

```
                        ;       Process zero.

D91A  A980      FASC2   LDA     #$80+'0'        ;"0" with high order bit se:
D91C  8D8005            STA     LBUFF           ;put zero character in line:
D91F  60                RTS                     ;return

                        ;       Process E format required.

D920  A901      FASC3   LDA     #1
D922  2070DC            JSR     COA     ;convert FR0 to ASCII
D925  20A4DC            JSR     FNZ     ;find last non-zero character
D928  E8                INX             ;increment offset to last character
D929  86F2              STX     CIX     ;save offset to last character

                        ;       Adjust exponent.

D92B  A5D4              LDA     FR0     ;exponent
D92D  0A                ASL     A       ;double exponent
D92E  38                SEC
D92F  E980              SBC     #$40*2  ;subtract 2 times bias

                        ;       Check first character for "0"..

D931  AE8005            LDX     LBUFF   ;first character
D934  E030              CPX     #'0'
D936  F017 ^D94F        BEQ     FASC5   ;if "0"

                        ;       Put decimal after first character.

D938  AE8105            LDX     LBUFF+1 ;second character
D93B  AC8205            LDY     LBUFF+2 ;decimal point
D93E  8E8205            STX     LBUFF+2 ;decimal point
D941  8C8105            STY     LBUFF+1 ;third character
D944  A6F2              LDX     CIX     ;offset
D946  E002              CPX     #2      ;former offset to decimal point
D948  D002 ^D94C        BNE     FASC4   ;if offset pointed to second charac:

D94A  E6F2              INC     CIX     ;increment offset

D94C  18        FASC4   CLC
D94D  6901              ADC     #1      ;adjust exponent for movement of de:

                        ;       Convert exponent to ASCII.

D94F  85ED      FASC5   STA     EEXP    ;exponent
D951  A945              LDA     #'E'
D953  A4F2              LDY     CIX     ;offset
D955  209FDC            JSR     SAL     ;store ASCII character in line buff:
D958  84F2              STY     CIX     ;save offset
D95A  A5ED              LDA     EEXP    ;exponent
D95C  1008 ^D969        BPL     FASC6   ;if exponent positive

D95E  A900              LDA     #0
D960  38                SEC
D961  E5ED              SBC     EEXP    ;complement exponent
D963  85ED              STA     EEXP    ;update exponent
D965  A920              LDA     #'-'
```

```
D967  D002 ^D96B           BNE     FASC7    ;store "-"

D969  A92B        FASC6    LDA     #'+'

D96B  209FDC      FASC7    JSR     SAL      ;store ASCII character in line buff:
D96E  A200                 LDX     #0       ;initial number of 10's
D970  A5ED                 LDA     EEXP     ;exponent

D972  38          FASC8    SEC
D973  E90A                 SBC     #10      ;subtract 10
D975  9003 ^D97A           BCC     FASC9    ;if < 0, done

D977  E8                   INX              ;increment number of 10's
D978  D0F8 ^D972           BNE     FASC8    ;continue

D97A  18          FASC9    CLC
D97B  690A                 ADC     #10      ;add back 10
D97D  48                   PHA              ;save remainder
D97E  8A                   TXA              ;number of 10's
D97F  209DDC               JSR     SNL      ;store number in line buffer
D982  68                   PLA              ;saved remainder
D983  0980                 ORA     #$80     ;set high order bit
D985  209DDC               JSR     SNL      ;store number in line buffer

                  ;        Perform final adjustment.

D988  AD8005      FASC10   LDA     LBUFF    ;first character
D98B  C930                 CMP     #'0'
D98D  D00D ^D99C           BNE     FASC11

                  ;        Increment pointer to point to non-zero character.

D98F  18                   CLC
D990  A5F3                 LDA     INBUFF           ;line buffer pointer
D992  6901                 ADC     #1               ;add 1
D994  85F3                 STA     INBUFF           ;update line buffer pointer
D996  A5F4                 LDA     INBUFF+1
D998  6900                 ADC     #0
D99A  85F4                 STA     INBUFF+1

                  ;        Check for positive exponent.

D99C  A5D4        FASC11   LDA     FR0              ;exponent
D99E  1009 ^D9A9           BPL     FASC12           ;if exponent positive, exit

                  ;        Process negative exponent.

D9A0  20C1DC               JSR     DLP              ;decrement line buffer poin:
D9A3  A000                 LDY     #0               ;offset to first character
D9A5  A92D                 LDA     #'-'
D9A7  91F3                 STA     (INBUFF),Y       ;put "-" in line buffer

                  ;        Exit.

D9A9  60          FASC12   RTS                      ;return
```

```
D9AA                          FIX     IFP



                        **    IFP - Convert Integer to Floating Point Number
                        *
                        *     ENTRY   JSR     IFP
                        *             FR0 - FR0+1 = integer to convert
                        *
                        *     EXIT
                        *             FR0 - FR0+5 = floating point number
                        *
                        *     MODS
                        *             Original Author Unknown
                        *             1. Bring closer to Coding Standard (object :
                        *                R. K. Nordin 11/01/83


                        ;IFP  =       *         ;entry

                        ;     Initialize.

D9AA  A5D4                    LDA     FR0               ;low integer
D9AC  85F8                    STA     ZTEMP4+1          ;save low integer
D9AE  A5D5                    LDA     FR0+1             ;high integer
D9B0  85F7                    STA     ZTEMP4            ;save high integer
D9B2  2044DA                  JSR     ZFR0              ;zero FR0

                        ;     Convert to floating point.

D9B5  F8                      SED
D9B6  A010                    LDY     #16               ;number of bits in integer

D9B8  06F8              IFP1  ASL     ZTEMP4+1          ;shift integer
D9BA  26F7                    ROL     ZTEMP4            ;shift integer, setting C i:

D9BC  A203                    LDX     #3                ;offset to last possible by:

D9BE  B5D4              IFP2  LDA     FR0,X             ;byte of number
D9C0  75D4                    ADC     FR0,X             ;double byte, adding in car:
D9C2  95D4                    STA     FR0,X             ;update byte of number
D9C4  CA                      DEX
D9C5  D0F7 ^D9BE              BNE     IFP2              ;if not done

D9C7  88                      DEY                       ;decrement count of integer:
D9C8  D0EE ^D9B8              BNE     IFP1              ;if not done

D9CA  D8                      CLD

                        ;     Set exponent.

D9CB  A942                    LDA     #$40+2            ;indicate decimal after las:
D9CD  85D4                    STA     FR0               ;exponent

                        ;     Exit.
```

```
D9CF  4C00DC                 JMP    NORM              ;normalize, return



D9D2                         FIX    FPI



                     **      FPI - Convert Floating Point Number to Integer
                     *
                     *       ENTRY  JSR    FPI
                     *              FR0 - FR0+5 = floating point number
                     *
                     *       EXIT
                     *              C set, if error
                     *              C clear, if no error
                     *              FR0 - FR0+1 = integer
                     *
                     *       MODS
                     *              Original Author Unknown
                     *              1. Bring closer to Coding Standard (object :
                     *                 R. K. Nordin 11/01/83

                     ;FPI    =      *                 ;entry

                     ;       Initialize.

D9D2  A900                   LDA    #0
D9D4  85F7                   STA    ZTEMP4            ;zero integer
D9D6  85F8                   STA    ZTEMP4+1

                     ;       Check exponent.

D9D8  A5D4                   LDA    FR0               ;exponent
D9DA  3066 ^DA42             BMI    FPI4              ;if sign of exponent is neg:

D9DC  C943                   CMP    #$40+3            ;bias+3
D9DE  B062 ^DA42             BCS    FPI4              ;if number too big, error

D9E0  38                     SEC
D9E1  E940                   SBC    #$40              ;subtract bias
D9E3  903F ^DA24             BCC    FPI2              ;if number less than 1, tes:

                     ;       Compute number of digits to convert.

D9E5  6900                   ADC    #0                ;add carry
D9E7  0A                     ASL    A                 ;2 times exponent-$40+1
D9E8  85F5                   STA    ZTEMP1            ;number of digits to conver:

                     ;       Convert.

D9EA  205ADA         FPI1    JSR    SIL               ;shift integer left
D9ED  B053 ^DA42             BCS    FPI4              ;if number too big, error
```

```
D9EF  A5F7                 LDA   ZTEMP4          ;2 times integer
D9F1  85F9                 STA   ZTEMP3          ;save 2 times integer
D9F3  A5F8                 LDA   ZTEMP4+1
D9F5  85FA                 STA   ZTEMP3+1
D9F7  205ADA               JSR   SIL             ;shift integer left
D9FA  B046 ^DA42           BCS   FPI4            ;if number too big, error

D9FC  205ADA               JSR   SIL             ;shift integer left
D9FF  B041 ^DA42           BCS   FPI4            ;if number too big, error

DA01  18                   CLC
DA02  A5F8                 LDA   ZTEMP4+1        ;8 times integer
DA04  65FA                 ADC   ZTEMP3+1        ;add 2 times integer
DA06  85F8                 STA   ZTEMP4+1        ;10 times integer
DA08  A5F7                 LDA   ZTEMP4
DA0A  65F9                 ADC   ZTEMP3
DA0C  85F7                 STA   ZTEMP4
DA0E  B032 ^DA42           BCS   FPI4            ;if overflow, error

DA10  20B9DC               JSR   GND             ;get next digit
DA13  18                   CLC
DA14  65F8                 ADC   ZTEMP4+1        ;insert digit
DA16  85F8                 STA   ZTEMP4+1
DA18  A5F7                 LDA   ZTEMP4
DA1A  6900                 ADC   #0              ;add carry
DA1C  B024 ^DA42           BCS   FPI4            ;if overflow, error

DA1E  85F7                 STA   ZTEMP4
DA20  C6F5                 DEC   ZTEMP1          ;decrement count of digits :
DA22  D0C6 ^D9EA           BNE   FPI1            ;if not done

                    ;      Check for round required.

DA24  20B9DC        FPI2   JSR   GND             ;get next digit
DA27  C905                 CMP   #5
DA29  900D ^DA38           BCC   FPI3            ;if digit less than 5, do n:

                    ;      Round.

DA2B  18                   CLC
DA2C  A5F8                 LDA   ZTEMP4+1
DA2E  6901                 ADC   #1              ;add 1 to round
DA30  85F8                 STA   ZTEMP4+1
DA32  A5F7                 LDA   ZTEMP4
DA34  6900                 ADC   #0
DA36  85F7                 STA   ZTEMP4

                    ;      Return integer.

DA38  A5F8          FPI3   LDA   ZTEMP4+1        ;low integer
DA3A  85D4                 STA   FR0             ;low integer result
DA3C  A5F7                 LDA   ZTEMP4          ;high integer
DA3E  85D5                 STA   FR0+1           ;high integer result
DA40  18                   CLC                   ;indicate success
DA41  60                   RTS                   ;return

                    ;      Return error.
```

```
DA42  38          FPI4    SEC                        ;indicate error
DA43  60                  RTS                        ;return



DA44                      FIX     ZFR0



                  **      ZFR0 - Zero FR0
                  *
                  *       ENTRY   JSR     ZFR0
                  *
                  *       MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83

                  ;ZFR0   =       *       ;entry

DA44  A2D4                LDX     #FR0    ;indicate zero FR0
                  ;       JMP     ZF1     ;zero floating point number, return



DA46                      FIX     ZF1



                  **      ZF1 - Zero Floating Point Number
                  *
                  *       ENTRY   JSR     ZF1
                  *               X = offset to register
                  *
                  *       MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83

                  ;ZF1    =       *       ;entry

DA46  A006                LDY     #6      ;number of bytes to zero
                  ;       JMP     ZXLY    ;zero bytes, return
```

```
                    **      ZXLY - Zero Page Zero Location X for Length Y
                    *
                    *       ENTRY   JSR     ZXLY
                    *               X = offset
                    *               Y = length
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = DA48        ZXLY    =       *       ;entry

DA48  A900                  LDA     #0

DA4A  9500          ZXLY1   STA     $0000,X ;zero byte
DA4C  E8                    INX
DA4D  88                    DEY
DA4E  D0FA ^DA4A            BNE     ZXLY1   ;if not done

DA50  60                    RTS             ;return



                    **      ILP - Initialize Line Buffer Pointer
                    *
                    *       ENTRY   JSR     ILP
                    *
                    *       EXIT
                    *               INBUFF - INBUFF+1 = line buffer address
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = DA51        ILP     =       *               ;entry
DA51  A905                  LDA     #high LBUFF     ;high buffer address
DA53  85F4                  STA     INBUFF+1        ;high line buffer pointer
DA55  A980                  LDA     #low LBUFF      ;low buffer address
DA57  85F3                  STA     INBUFF          ;low line buffer pointer
DA59  60                    RTS                     ;return
```

```
                        **      SIL - Shift Integer Left
                        *
                        *       ENTRY   JSR     SIL
                        *               ZTEMP4 - ZTEMP4+1 = number (high, low) to s:
                        *
                        *       EXIT
                        *               ZTEMP4 - ZTEMP4+1 shifted left 1
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

         = DA5A         SIL     =       *               ;entry
DA5A  18                        CLC
DA5B  26F8                      ROL     ZTEMP4+1        ;shift low
DA5D  26F7                      ROL     ZTEMP4          ;shift high
DA5F  60                        RTS                     ;return


DA60                            FIX     FSUB


                        **      FSUB - Perform Floating Point Subtract
                        *
                        *       FSUB subtracts FR1 from FR0.
                        *
                        *       ENTRY   JSR     FSUB
                        *               FR0 - FR0+5 = minuend
                        *               FR1 - FR1+5 = subtrahend
                        *
                        *       EXIT
                        *               C set, if error
                        *               C clear, if no error
                        *               FR0 - FR0+5 = difference
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

                        ;FSUB   =       *       ;entry

                        ;       Complement sign of subtrahend and add.

DA60  A5E0                      LDA     FR1     ;subtrahend exponent
DA62  4980                      EOR     #$80    ;complement sign of subtrahend
DA64  85E0                      STA     FR1     ;update subtrahend exponent
                        ;       JMP     FADD    ;perform add, return
```

```
DA66                            FIX     FADD



                        **      FADD - Perform Floating Point Add
                        *
                        *       ENTRY   JSR     FADD
                        *               FR0 - FR0+5 = augend
                        *               FR1 - FR1+5 = addend
                        *
                        *       EXIT
                        *               C set, if error
                        *               C clear, if no error
                        *               FR0 - FR0+5 = sum
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


                        ;FADD   =       *       ;entry

                        ;       Initialize.

DA66  A5E0              FADD1   LDA     FR1     ;exponent of addend
DA68  297F                      AND     #$7F    ;clear sign of addend mantissa
DA6A  85F7                      STA     ZTEMP4  ;save addend exponent
DA6C  A5D4                      LDA     FR0     ;exponent of augend
DA6E  297F                      AND     #$7F    ;clear sign of augend mantissa
DA70  38                        SEC
DA71  E5F7                      SBC     ZTEMP4  ;subtract addend exponent
DA73  1010 ^DA85                BPL     FADD3   ;if augend exponent >= addend expon:

                        ;       Swap augend and addend.

DA75  A205                      LDX     #FPREC-1        ;offset to last byte

DA77  B5D4              FADD2   LDA     FR0,X           ;byte of augend
DA79  B4E0                      LDY     FR1,X           ;byte of addend
DA7B  95E0                      STA     FR1,X           ;move byte of augend to add:
DA7D  98                        TYA
DA7E  95D4                      STA     FR0,X           ;move byte of addend to aug:
DA80  CA                        DEX
DA81  10F4 ^DA77                BPL     FADD2           ;if not done

DA83  30E1 ^DA66                BMI     FADD1           ;re-initialize

                        ;       Check alignment.

DA85  F007 ^DA8E        FADD3   BEQ     FADD4   ;if exponent difference zero, alrea:

DA87  C905                      CMP     #FMPREC ;mantissa precision
DA89  B019 ^DAA4                BCS     FADD6   ;if exponent difference < mantissa :

                        ;       Align.
```

```
DA8B  203EDC                JSR     S1R       ;shift FR1 right

                  ;         Check for like signs of mantissas.

DA8E  F8          FADD4     SED
DA8F  A5D4                  LDA     FR0       ;augend exponent
DA91  45E0                  EOR     FR1       ;EOR with addend exponent
DA93  301E ^DAB3            BMI     FADD8     ;if signs differ, subtract

                  ;         Add.

DA95  A204                  LDX     #FMPREC-1         ;offset to last byte of man:
DA97  18                    CLC

DA98  B5D5        FADD5     LDA     FR0M,X            ;byte of augend mantissa
DA9A  75E1                  ADC     FR1M,X            ;add byte of addend mantiss:
DA9C  95D5                  STA     FR0M,X            ;update byte of result mant:
DA9E  CA                    DEX
DA9F  10F7 ^DA98            BPL     FADD5             ;if not done

DAA1  D8                    CLD
DAA2  B003 ^DAA7            BCS     FADD7             ;if carry, process carry

                  ;         Exit.

DAA4  4C00DC      FADD6     JMP     NORM              ;normalize, return

                  ;         Process carry.

DAA7  A901        FADD7     LDA     #1                ;indicate shift 1
DAA9  203ADC                JSR     S0R               ;shift FR0 right
DAAC  A901                  LDA     #1                ;carry
DAAE  85D5                  STA     FR0M              ;set carry in result

                  ;         Exit.

DAB0  4C00DC                JMP     NORM              ;normalize, return

                  ;         Subtract.

DAB3  A204        FADD8     LDX     #FMPREC-1         ;offset to last byte of man:
DAB5  38                    SEC

DAB6  B5D5        FADD9     LDA     FR0M,X            ;byte of augend mantissa
DAB8  F5E1                  SBC     FR1M,X            ;subtract byte of addend ma:
DABA  95D5                  STA     FR0M,X            ;update byte of result mant:
DABC  CA                    DEX
DABD  10F7 ^DAB6            BPL     FADD9             ;if not done

DABF  9004 ^DAC5            BCC     FADD10            ;if borrow, process borrow

                  ;         Exit.

DAC1  D8                    CLD
DAC2  4C00DC                JMP     NORM              ;normalize, return
```

```
                        ;       Process borrow.

DAC5  A5D4      FADD10  LDA     FR0             ;result exponent
DAC7  4980              EOR     #$80            ;complement sign of result
DAC9  85D4              STA     FR0             ;update result exponent

DACB  38                SEC
DACC  A204              LDX     #FMPREC-1       ;offset to last byte of man:

DACE  A900      FADD11  LDA     #0
DAD0  F5D5              SBC     FROM,X          ;complement byte of result :
DAD2  95D5              STA     FROM,X          ;update byte of result mant:
DAD4  CA                DEX
DAD5  10F7 ^DACE        BPL     FADD11          ;if not done

                        ;       Exit.

DAD7  D8                CLD
DAD8  4C00DC            JMP     NORM            ;normalize, return


DADB                    FIX     FMUL



                        **      FMUL - Perform Floating Point Multiply
                        *
                        *       ENTRY   JSR     FMUL
                        *               FR0 - FR0+5 = multiplicand
                        *               FR1 - FR1+5 = multiplier
                        *
                        *       EXIT
                        *               C set, if error
                        *               C clear, if no error
                        *               FR0 - FR0+5 = product
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


                        ;FMUL   =       *       ;entry

                        ;       Check for zero multiplicand.

DADB  A5D4              LDA     FR0     ;multiplicand exponent
DADD  F045 ^DB24        BEQ     FMUL8   ;if multiplicand exponent zero, res:

                        ;       Check for zero multiplier.

DADF  A5E0              LDA     FR1     ;multiplier exponent
DAE1  F03E ^DB21        BEQ     FMUL7   ;if multiplier exponent zero, resul:
```

```
DAE3  20CFDC                 JSR     SUE      ;set up exponent
DAE6  38                     SEC
DAE7  E940                   SBC     #$40     ;subtract bias
DAE9  38                     SEC              ;add 1
DAEA  65E0                   ADC     FR1      ;add multiplier exponent
DAEC  3038 ^DB26             BMI     FMUL9    ;if overflow, error

                     ;       Set up.

DAEE  20E0DC                 JSR     SUP      ;set up

                     ;       Compute number of times to add multiplicand.

DAF1  A5DF           FMUL1   LDA     FRE+FPREC-1     ;last byte of FRE
DAF3  290F                   AND     #$0F            ;extract low order digit
DAF5  85F6                   STA     ZTEMP1+1

                     ;       Check for completion.

DAF7  C6F6           FMUL2   DEC     ZTEMP1+1        ;decrement counter
DAF9  3006 ^Dd01             BMI     FMUL3           ;if done

DAFB  2001DD                 JSR     FRA10           ;add FR1 to FR0
DAFE  4CF7DA                 JMP     FMUL2           ;continue

                     ;       Compute number of times to add 10 times multiplicand

DB01  A5DF           FMUL3   LDA     FRE+FPREC-1     ;last byte of FRE
DB03  4A                     LSR     A
DB04  4A                     LSR     A
DB05  4A                     LSR     A
DB06  4A                     LSR     A               ;high order digit
DB07  85F6                   STA     ZTEMP1+1

                     ;       Check for completion.

DB09  C6F6           FMUL4   DEC     ZTEMP1+1        ;decrement counter
DB0B  3006 ^Dd13             BMI     FMUL5           ;if done

DB0D  2005DD                 JSR     FRA20           ;add FR2 to FR0
DB10  4C09DB                 JMP     FMUL4           ;continue

                     ;       Set up for next set of adds.

DB13  2062DC         FMUL5   JSR     S0ER            ;shift FR0/FRE right

                     ;       Decrement counter and test for completion.

DB16  C6F5                   DEC     ZTEMP1          ;decrement
DB18  D0D7 ^DAF1             BNE     FMUL1           ;if not done

                     ;       Set exponent.

DB1A  A5ED           FMUL6   LDA     EEXP            ;exponent
DB1C  85D4                   STA     FR0             ;result exponent
DB1E  4C04DC                 JMP     NOE             ;normalize, return
```

```
                        ;       Return zero result.

DB21  2044DA            FMUL7   JSR     ZFR0            ;zero FR0

                        ;       Return no error.

DB24  18                FMUL8   CLC                     ;indicate no error
DB25  60                        RTS                     ;return

                        ;       Return error.

DB26  38                FMUL9   SEC                     ;indicate error
DB27  60                        RTS                     ;return



DB28                            FIX     FDIV



                        **      FDIV - Perform Floating Point Divide
                        *
                        *       ENTRY   JSR     FDIV
                        *               FR0 - FR0+5 = dividend
                        *               FR1 - FR1+5 = divisor
                        *
                        *       EXIT
                        *               C clear, if no error
                        *               C set, if error
                        *               FR0 - FR0+5 = quotient
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


                        ;FDIV   =       *       ;entry

                        ;       Check for zero divisor.

DB28  A5E0                      LDA     FR1     ;divisor exponent
DB2A  F0FA ^DB26                BEQ     FMUL9   ;if divisor exponent zero, error

                        ;       Check for zero dividend.

DB2C  A5D4                      LDA     FR0     ;dividend exponent
DB2E  F0F4 ^DB24                BEQ     FMUL8   ;if dividend exponent zero, result :

DB30  20CFDC                    JSR     SUE     ;set up exponent
DB33  38                        SEC
DB34  E5E0                      SBC     FR1     ;subtract divisor exponent
DB36  18                        CLC
DB37  6940                      ADC     #$40    ;add bias
DB39  30EB ^DB26                BMI     FMUL9   ;if overflow, error
```

```
DB3B  20E0DC            JSR   SUP      ;set up
DB3E  E6F5              INC   ZTEMP1   ;divide requires extra pass
DB40  4C4EDB            JMP   FDIV3    ;skip shift

                  ;     Shift FR0/FRE left one byte.

DB43  A200       FDIV1  LDX   #0              ;offset to first byte to sh:

DB45  B5D5       FDIV2  LDA   FR0+1,X         ;byte to shift
DB47  95D4              STA   FR0,X           ;byte of destination
DB49  E8                INX
DB4A  E00C              CPX   #FMPREC*2+2     ;number of bytes to shift
DB4C  D0F7 ^DB45        BNE   FDIV2           ;if not done

                  ;     Subtract 2 times divisor from dividend.

DB4E  A005       FDIV3  LDY   #FPREC-1        ;offset to last byte
DB50  38                SEC
DB51  F8                SED

DB52  B9DA00     FDIV4  LDA   FRE,Y           ;byte of dividend
DB55  F9E600            SBC   FR2,Y           ;subtract byte of 2*divisor
DB58  99DA00            STA   FRE,Y           ;update byte of dividend
DB5B  88                DEY
DB5C  10F4 ^DB52        BPL   FDIV4           ;if not done

DB5E  D8                CLD
DB5F  9004 ^DB65        BCC   FDIV5           ;if difference < 0

DB61  E6D9              INC   QTEMP           ;increment
DB63  D0E9 ^DB4E        BNE   FDIV3           ;continue

                  ;     Adjust.

DB65  200FDD     FDIV5  JSR   FRA2E    ;add FR2 to FR0

                  ;     Shift last byte of quotient left one digit.

DB68  06D9              ASL   QTEMP
DB6A  06D9              ASL   QTEMP
DB6C  06D9              ASL   QTEMP
DB6E  06D9              ASL   QTEMP

                  ;     Subtract divisor from dividend.

DB70  A005       FDIV6  LDY   #FPREC-1        ;offset to last byte
DB72  38                SEC
DB73  F8                SED

DB74  B9DA00     FDIV7  LDA   FRE,Y           ;byte of dividend
DB77  F9E000            SBC   FR1,Y           ;subtract byte of divisor
DB7A  99DA00            STA   FRE,Y           ;update byte of dividend
DB7D  88                DEY
DB7E  10F4 ^DB74        BPL   FDIV7           ;if not done

DB80  D8                CLD
```

```
DB81  9004 ^DB87            BCC     FDIV8           ;if difference < 0

DB83  E6D9                  INC     QTEMP           ;increment
DB85  D0E9 ^DB70            BNE     FDIV6           ;continue

                      ;     Adjust.

DB87  2009DD        FDIV8   JSR     FRA1E   ;add FR1 to FR0
DB8A  C6F5                  DEC     ZTEMP1  ;decrement
DB8C  D0B5 ^DB43            BNE     FDIV1   ;if not done

                      ;     Clear exponent.

DB8E  2062DC                JSR     S0ER    ;shift  FR0/FRE right

                      ;     Exit.

DB91  4C1ADB                JMP     FMUL6


                      **    GNC - Get Next Character
                      *
                      *     ENTRY   JSR     GNC
                      *             INBUFF - INBUFF+1 = line buffer pointer
                      *             CIX = offset to character
                      *
                      *     EXIT
                      *             C set, if character not numeric
                      *             A = non-numeric character
                      *             C clear, if character numeric
                      *             CIX = offset to next character
                      *
                      *     MODS
                      *             Original Author Unknown
                      *             1. Bring closer to Coding Standard (object :
                      *                R. K. Nordin 11/01/83


      = DB94          GNC   =       *               ;entry
DB94  20AFDB                JSR     TNC             ;test for numeric character
DB97  A4F2                  LDY     CIX             ;offset
DB99  9002 ^DB9D            BCC     ICX             ;if numeric, increment offs:

DB9B  B1F3                  LDA     (INBUFF),Y      ;character
                      ;     JMP     ICX             ;increment offset, return
```

```
                        **      ICX - Increment Character Offset
                        *
                        *       ENTRY   JSR     ICX
                        *               Y = offset
                        *
                        *       EXIT
                        *               CIX = offset to next character
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = DB9D          ICX     =       *       ;entry
DB9D    C8                      INY             ;increment offset
DB9E    84F2                    STY     CIX     ;offset
DBA0    60                      RTS             ;return


                        **      SLB - Skip Leading Blanks
                        *
                        *       ENTRY   JSR     SLB
                        *               INBUFF - INBUFF+1 = line buffer pointer
                        *               CIX = offset
                        *
                        *       EXIT
                        *               CIX = offset to first non-blank character
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = DBA1          SLB     =       *               ;entry

                        ;       Initialize.

DBA1    A4F2                    LDY     CIX             ;offset to character
DBA3    A920                    LDA     #' '

                        ;       Search for first non-blank character.

DBA5    D1F3            SLB1    CMP     (INBUFF),Y      ;character
DBA7    D003 ^DBAC              BNE     SLB2            ;if non-blank character

DBA9    C8                      INY
DBAA    D0F9 ^DBA5              BNE     SLB1            ;if not done

                        ;       Exit.

DBAC    84F2            SLB2    STY     CIX             ;offset to first non-blank :
DBAE    60                      RTS                     ;return
```

```
                        **      TNC - Test for Numeric Character
                        *
                        *       ENTRY   JSR     TNC
                        *               INBUFF - INBUFF+1 = line buffer pointer
                        *               CIX = offset
                        *
                        *       EXIT
                        *               C set, if numeric
                        *               C clear if non-numeric
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = DBAF            TNC     =       *               ;entry
DBAF  A4F2                      LDY     CIX             ;offset
DBB1  B1F3                      LDA     (INBUFF),Y      ;character
DBB3  38                        SEC
DBB4  E930                      SBC     #'0'
DBB6  9018 ^DBD0                BCC     TVN2            ;if < "0", return failure

DBB8  C90A                      CMP     #'9'-'0'+1      ;return success or failure
DBBA  60                        RTS                     ;return



                        **      TVN - Test for Valid Number Character
                        *
                        *       ENTRY   JSR     TVN
                        *
                        *       EXIT
                        *               C set, if not number
                        *               C clear, if number
                        *
                        *       NOTES
                        *               Problem: bytes wasted by BCC TVN5.
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = DBBB            TVN     =       *       ;entry

                        ;       Initialize.

DBBB  A5F2                      LDA     CIX     ;offset
DBBD  48                        PHA             ;save offset

                        ;       Check next character.

DBBE  2094DB                    JSR     GNC     ;get next character
DBC1  901F ^DBE2                BCC     TVN5    ;if numeric, return success
```

```
DBC3  C92E                    CMP     #'.'
DBC5  F014 ^DBDB              BEQ     TVN4     ;if ".", check next character

DBC7  C92B                    CMP     #'+'
DBC9  F007 ^DBD2              BEQ     TVN3     ;if "+", check next character

DBCB  C92D                    CMP     #'-'
DBCD  F003 ^DBD2              BEQ     TVN3     ;if "-", check next character

                      ;       Clean stack.

DBCF  68              TVN1    PLA              ;clean stack

                      ;       Return failure.

DBD0  38              TVN2    SEC              ;indicate failure
DBD1  60                      RTS              ;return

                      ;       Check character after "+" or "-".

DBD2  2094DB          TVN3    JSR     GNC      ;get next character
DBD5  900B ^DBE2              BCC     TVN5     ;if numeric, return success

DBD7  C92E                    CMP     #'.'
DBD9  D0F4 ^DBCF              BNE     TVN1     ;if not ".", return failure

                      ;       Check character after ".".

DBDB  2094DB          TVN4    JSR     GNC      ;get next character
DBDE  9002 ^DBE2              BCC     TVN5     ;if numeric, return success

DBE0  B0ED ^DBCF              BCS     TVN1     ;return failure

                      ;       Return success.

DBE2  68              TVN5    PLA              ;saved offset
DBE3  85F2                    STA     CIX      ;restore offset
DBE5  18                      CLC              ;indicate success
DBE6  60                      RTS              ;return



                      **      S2L - Shift FR2 Left One Digit
                      *
                      *       ENTRY   JSR     S2L
                      *
                      *       MODS
                      *               Original Author Unknown
                      *               1. Bring closer to Coding Standard (object :
                      *                  R. K. Nordin 11/01/83


      = DBE7          S2L     =       *        ;entry
DBE7  A2E7                    LDX     #FR2+1   ;indicate shift of FR2 mantissa
```

```
DBE9  D002 ^DBED            BNE     SML     ;shift mantissa left 1 digit, retur:



                      **     SOL - Shift FR0 Left One Digit
                      *
                      *      ENTRY   JSR     SOL
                      *
                      *      MODS
                      *              Original Author Unknown
                      *              1. Bring closer to Coding Standard (object :
                      *                 R. K. Nordin 11/01/83

      = DBEB          SOL    =       *       ;entry
DBEB  A205                   LDX     #FR0M   ;indicate shift of FR0 mantissa
                      ;      JMP     SML     ;shift mantissa left 1 digit, retur:



                      **     SML - Shift Mantissa Left One Digit
                      *
                      *      ENTRY   JSR     SML
                      *
                      *      EXIT
                      *              FRX = excess digit
                      *
                      *      MODS
                      *              Original Author Unknown
                      *              1. Bring closer to Coding Standard (object :
                      *                 R. K. Nordin 11/01/83

      = DBED          SML    =       *       ;entry
DBED  A004                   LDY     #4      ;number of bits to shift

DBEF  18              SML2   CLC
DBF0  3604                   ROL     $0004,X ;shift 5th byte left 1 bit
DBF2  3603                   ROL     $0003,X ;shift 4th byte left 1 bit
DBF4  3602                   ROL     $0002,X ;shift 3rd byte left 1 bit
DBF6  3601                   ROL     $0001,X ;shift 2nd byte left 1 bit
DBF8  3600                   ROL     $0000,X ;shift 1st byte left 1 bit
DBFA  26EC                   ROL     FRX     ;shift excess digit left 1 bit
DBFC  88                     DEY
DBFD  D0F0 ^DBEF             BNE     SML2    ;if not done

DBFF  60                     RTS             ;return
```

```
                        **      NORM - Normalize FR0
                        *
                        *       ENTRY   JSR     NORM
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = DC00          NORM    =       *               ;entry
DC00    A200                    LDX     #0
DC02    86DA                    STX     FRE             ;byte to shift in
                        ;       JMP     N0E             ;normalize FR0/FRE, return



                        **      N0E - Normalize FR0/FRE
                        *
                        *       ENTRY   JSR     N0E
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = DC04          N0E     =       *               ;entry
DC04    A204                    LDX     #FMPREC-1       ;mantissa size
DC06    A5D4                    LDA     FR0             ;exponent
DC08    F02E ^DC38              BEQ     N0E5            ;if exponent zero, number i:

DC0A    A5D5            N0E1    LDA     FR0M            ;first byte of mantissa
DC0C    D01A ^DC28              BNE     N0E3            ;if not zero, no shift

                        ;       Shift mantissa left 1 byte.

DC0E    A000                    LDY     #0              ;offset to first byte of ma:

DC10    B9D600          N0E2    LDA     FR0M+1,Y        ;byte to shift
DC13    99D500                  STA     FR0M,Y          ;byte of destination
DC16    C8                      INY
DC17    C005                    CPY     #FMPREC         ;size of mantissa
DC19    90F5 ^DC10              BCC     N0E2            ;if not done

                        ;       Decrement exponent and check for completion.

DC1B    C6D4                    DEC     FR0             ;decrement exponent
DC1D    CA                      DEX
DC1E    D0EA ^DC0A              BNE     N0E1            ;if not done

                        ;       Check first byte of mantissa.

DC20    A5D5                    LDA     FR0M    ;first byte of mantissa
DC22    D004 ^DC26              BNE     N0E3    ;if mantissa not zero
```

```
                        ;       Zero exponent.

DC24  85D4                      STA     FR0     ;zero exponent
DC26  18                        CLC
DC27  60                        RTS             ;return

                        ;       Check for overflow.

DC28  A5D4              NOE3    LDA     FR0     ;exponent
DC2A  297F                      AND     #$7F    ;clear sign
DC2C  C971                      CMP     #$40+49 ;bias+49
DC2E  9001 ^DC31                BCC     NOE4    ;if exponent < 49, no overflow

                        ;       Return error.

                        ;       SEC             ;indicate error
DC30  60                        RTS             ;return

                        ;       Check for underflow.

DC31  C90F              NOE4    CMP     #$40-49
DC33  B003 ^DC36                BCS     NOE5    ;if exponent >= -49, no underflow

                        ;       Zero result.

DC35  2044DA                    JSR     ZFR0    ;zero FR0

                        ;       Exit.

DC38  18                NOE5    CLC             ;indicate no error
DC39  60                        RTS             ;return



                        **      S0R - Shift FR0 Right
                        *
                        *       ENTRY   JSR     S0R
                        *               A = shift count
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = DC3A            S0R     =       *       ;entry
DC3A  A2D4                      LDX     #FR0    ;indicate shift of FR0
DC3C  D002 ^DC40                BNE     SRR     ;shift register right, return
```

```
                        **      S1R - Shift FR1 Right
                        *
                        *       ENTRY   JSR     S1R
                        *               A = shift count
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

       = DC3E           S1R     =       *       ;entry
DC3E   A2E0                     LDX     #FR1    ;indicate shift of FR1
                        ;       JMP     SRR     ;shift register right, return



                        **      SRR - Shift Register Right
                        *
                        *       ENTRY   JSR     SRR
                        *               X = offset to register
                        *               A = shift count
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

       = DC40           SRR     =       *               ;entry
DC40   86F9                     STX     ZTEMP3          ;register
DC42   85F7                     STA     ZTEMP4          ;shift count
DC44   85F8                     STA     ZTEMP4+1        ;save shift count

DC46   A004             SRR1    LDY     #FMPREC-1       ;mantissa size-1

DC48   B504             SRR2    LDA     $0004,X         ;byte to shift
DC4A   9505                     STA     $0005,X         ;byte of destination
DC4C   CA                       DEX
DC4D   88                       DEY
DC4E   D0F8 ^DC48               BNE     SRR2            ;if not done

DC50   A900                     LDA     #0
DC52   9505                     STA     $0005,X         ;first byte of mantissa
DC54   A6F9                     LDX     ZTEMP3          ;register
DC56   C6F7                     DEC     ZTEMP4          ;decrement shift count
DC58   D0EC ^DC46               BNE     SRR1            ;if not done

                        ;       Adjust exponent.

DC5A   B500                     LDA     $0000,X         ;exponent
DC5C   18                       CLC
DC5D   65F8                     ADC     ZTEMP4+1        ;subtract shift count
DC5F   9500                     STA     $0000,X         ;update exponent
DC61   60                       RTS                     ;return
```

```
                             **      S0ER - Shift FR0/FRE Right
                             *
                             *       ENTRY   JSR     S0ER
                             *
                             *       MODS
                             *               Original Author Unknown
                             *               1. Bring closer to Coding Standard (object :
                             *                  R. K. Nordin 11/01/83


          = DC62             S0ER    =       *               ;entry
DC62  A20A                           LDX     #FMPREC*2       ;number of bytes to shift

DC64  B5D4                   S0ER1   LDA     FR0,X           ;byte to shift
DC66  95D5                           STA     FR0+1,X         ;byte of destination
DC68  CA                             DEX
DC69  10F9 ^DC64                     BPL     S0ER1           ;if not done

DC6B  A900                           LDA     #0
DC6D  85D4                           STA     FR0             ;shift in 0
DC6F  60                             RTS                     ;return



                             **      COA - Convert FR0 to ASCII
                             *
                             *       ENTRY   JSR     COA
                             *               A = decimal point position
                             *
                             *       MODS
                             *               Original Author Unknown
                             *               1. Bring closer to Coding Standard (object :
                             *                  R. K. Nordin 11/01/83

          = DC70             COA     =       *       ;entry

                             ;       Initialize.

DC70  85F7                           STA     ZTEMP4  ;decimal point position counter
DC72  A200                           LDX     #0      ;offset to first byte of FROM
DC74  A000                           LDY     #0      ;offset to first byte of LBUF

                             ;       Convert next byte.

DC76  2093DC                 COA1    JSR     TDP     ;test for decimal point
DC79  38                             SEC
DC7A  E901                           SBC     #1      ;decrement deciaml point position
DC7C  85F7                           STA     ZTEMP4  ;update deciaml point position coun:

                             ;       Convert first digit of next byte.

DC7E  B5D5                           LDA     FROM,X  ;byte
DC80  4A                             LSR     A
DC81  4A                             LSR     A
DC82  4A                             LSR     A
```

```
DC83  4A                LSR     A         ;first digit
DC84  209DDC            JSR     SNL       ;store number in line buffer

                  ;     Convert second digit of next byte.

DC87  B5D5              LDA     FROM,X    ;byte
DC89  290F              AND     #$0F      ;extract second digit
DC8B  209DDC            JSR     SNL       ;store number in line buffer
DC8E  E8                INX
DC8F  E005              CPX     #FMPREC   ;nuber of bytes
DC91  90E3 ^DC76        BCC     COA1      ;if not done

                  ;     Exit.

                  ;     JMP     TDP       ;test for decimal point, return



                  **    TDP - Test for Decimal Point
                  *
                  *     ENTRY   JSR     TDP
                  *             ZTEMP4 = decimal point position counter
                  *
                  *     MODS
                  *             Original Author Unknown
                  *             1. Bring closer to Coding Standard (object :
                  *                R. K. Nordin 11/01/83

      = DC93      TDP   =       *         ;entry

                  ;     Check decimal point position counter.

DC93  A5F7              LDA     ZTEMP4    ;decimal point position counter
DC95  D005 ^DC9C        BNE     TDP1      ;if not decimal point position, exi:

                  ;     Insert decimal point.

DC97  A92E              LDA     #'.'
DC99  209FDC            JSR     SAL       ;store ASCII character in line buff:

                  ;     Exit.

DC9C  60          TDP1  RTS               ;return
```

```
                        **      SNL - Store Number in Line Buffer
                        *
                        *       ENTRY   JSR     SNL
                        *               A = digit to store
                        *               Y = offset
                        *
                        *       EXIT
                        *               ASCII digit placed in line buffer
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = DC9D          SNL     =       *       ;entry
DC9D    0930                    ORA     #$30    ;convert digit to ASCII
                        ;       JMP     SAL     ;store ASCII character in line buff:


                        **      SAL - Store ASCII Character in Line Buffer
                        *
                        *       ENTRY   JSR     SAL
                        *               Y = offset
                        *               A = character
                        *
                        *       EXIT
                        *               Character placed in line buffer
                        *               Y = incremented offset
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = DC9F          SAL     =       *       ;entry
DC9F    998005                  STA     LBUFF,Y ;store character in line buffer
DCA2    C8                      INY             ;increment offset
DCA3    60                      RTS             ;return
```

```
                        **      FNZ - Find Last Non-zero Character in Line Buffer
                        *
                        *       FNZ returns the last non-zero character.  If the la:
                        *       non-zero character is ".", FNZ returns the characte:
                        *       preceding the ".".  If no other non-zero character :
                        *       encountered, FNZ returns the first character.
                        *
                        *       ENTRY   JSR     FNZ
                        *
                        *       EXIT
                        *               A = character
                        *               X = offset to character
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


        = DCA4          FNZ     =       *       ;entry

                        ;       Initialize.

DCA4    A20A                    LDX     #10     ;offset to last possible character

                        ;       Check next character.

DCA6    BD8005          FNZ1    LDA     LBUFF,X ;character
DCA9    C92E                    CMP     #'.'
DCAB    F007 ^DCB4              BEQ     FNZ2    ;if ".", return preceding character

DCAD    C930                    CMP     #'0'
DCAF    D007 ^DCB8              BNE     FNZ3    ;if not "0", exit

                        ;       Decrement offset and check for completion.

DCB1    CA                      DEX
DCB2    D0F2 ^DCA6              BNE     FNZ1    ;if not done

                        ;       Return character preceding "." or first character.

DCB4    CA              FNZ2    DEX             ;offset to character
DCB5    BD8005                  LDA     LBUFF,X ;character

                        ;       Exit.

DCB8    60              FNZ3    RTS             ;return
```

```
                        **      GND - Get Next Digit
                        *
                        *       ENTRY   JSR     GND
                        *               FR0 - FR0+5 = number
                        *
                        *       EXIT
                        *               A = digit
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = DCB9          GND     =       *       ;entry
DCB9    20EBDB                  JSR     SOL     ;shift FR0 left 1 digit
DCBC    A5EC                    LDA     FRX     ;excess digit
DCBE    290F                    AND     #$0F    ;extract low order digit
DCC0    60                      RTS             ;return



                        **      DLP - Decrement Line Buffer Pointer
                        *
                        *       ENTRY   JSR     DLP
                        *               INBUFF - INBUFF+1 = line buffer pointer
                        *
                        *       EXIT
                        *               INBUFF - INBUFF+1 = incremented line buffer:
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = DCC1          DLP     =       *               ;entry
DCC1    38                      SEC
DCC2    A5F3                    LDA     INBUFF          ;line buffer pointer
DCC4    E901                    SBC     #1              ;subtract 1
DCC6    85F3                    STA     INBUFF          ;update line buffer pointer
DCC8    A5F4                    LDA     INBUFF+1
DCCA    E900                    SBC     #0
DCCC    85F4                    STA     INBUFF+1
DCCE    60                      RTS                     ;return
```

```
                         **     SUE - Set Up Exponent for Multiply or Divide
                         *
                         *      ENTRY   JSR     SUE
                         *
                         *      EXIT
                         *              A = FR0 exponent (without sign)
                         *              FR1 = FR1 exponent (without sign)
                         *              FRSIGN = sign of result
                         *
                         *      MODS
                         *              Original Author Unknown
                         *              1. Bring closer to Coding Standard (object :
                         *                 R. K. Nordin 11/01/83


       = DCCF            SUE    =       *       ;entry
DCCF   A5D4                     LDA     FR0     ;FR0 exponent
DCD1   45E0                     EOR     FR1     ;EOR with FR1 exponent
DCD3   2980                     AND     #$80    ;extract sign
DCD5   85EE                     STA     FRSIGN  ;sign of result
DCD7   06E0                     ASL     FR1     ;shift out FR1 sign
DCD9   46E0                     LSR     FR1     ;FR1 exponent without sign
DCDB   A5D4                     LDA     FR0     ;FR0 exponent
DCDD   297F                     AND     #$7F    ;FR0 exponent without sign
DCDF   60                       RTS             ;return



                         **     SUP - Set Up for Multiply or Divide
                         *
                         *      ENTRY   JSR     SUP
                         *              A = exponent
                         *
                         *      MODS
                         *              Original Author Unknown
                         *              1. Bring closer to Coding Standard (object :
                         *                 R. K. Nordin 11/01/83


       = DCE0            SUP    =       *       ;entry
DCE0   05EE                     ORA     FRSIGN  ;place sign in exponent
DCE2   85ED                     STA     EEXP    ;exponent
DCE4   A900                     LDA     #0
DCE6   85D4                     STA     FR0     ;clear FR0 exponent
DCE8   85E0                     STA     FR1     ;clear FR0 exponent
DCEA   2028DD                   JSR     M12     ;move FR1 to FR2
DCED   20E7DB                   JSR     S2L     ;shift FR2 left 1 digit
DCF0   A5EC                     LDA     FRX     ;excess digit
DCF2   290F                     AND     #$0F    ;extract low order digit
DCF4   85E6                     STA     FR2     ;shift in low order digit
DCF6   A905                     LDA     #FMPREC ;mantissa size
DCF8   85F5                     STA     ZTEMP1  ;mantissa size
DCFA   2034DD                   JSR     M0E     ;move FR0 to FRE
DCFD   2044DA                   JSR     ZFR0    ;zero FR0
DD00   60                       RTS             ;return
```

```
                          **        FRA10 - Add FR1 to FR0
                          *
                          *         ENTRY   JSR     FRA10
                          *                 FR0 - FR0+5 = augend
                          *                 FR1 - FR1+5 = addend
                          *
                          *         EXIT
                          *                 FR0 - FR0+5 = sum
                          *
                          *         MODS
                          *                 Original Author Unknown
                          *                 1. Bring closer to Coding Standard (object :
                          *                    R. K. Nordin 11/01/83

     = DD01      FRA10    =         *                   ;entry
DD01  A209                LDX       #FR0+FPREC-1        ;offset to last byte of FR0
DD03  D006 ^DD0B          BNE       F1R



                          **        FRA20 - Add FR2 to FR0
                          *
                          *         ENTRY   JSR     FRA20
                          *                 FR0 - FR0+5 = augend
                          *                 FR2 - FR2+5 = addend
                          *
                          *         EXIT
                          *                 FR0 - FR0+5 = sum
                          *
                          *         MODS
                          *                 Original Author Unknown
                          *                 1. Bring closer to Coding Standard (object :
                          *                    R. K. Nordin 11/01/83

     = DD05      FRA20    =         *                   ;entry
DD05  A209                LDX       #FR0+FPREC-1        ;offset to last byte of FR0
DD07  D008 ^DD11          BNE       F2R



                          **        FRA1E - Add FR1 to FRE
                          *
                          *         ENTRY   JSR     FRA1E
                          *                 FRE - FRE+5 = augend
                          *                 FR1 - FR1+5 = addend
                          *
                          *         EXIT
                          *                 FRE - FRE+5 = sum
                          *
                          *         MODS
                          *                 Original Author Unknown
                          *                 1. Bring closer to Coding Standard (object :
                          *                    R. K. Nordin 11/01/83
```

```
      = DD09      FRA1E    =        *                 ;entry
DD09  A2DF                 LDX      #FRE+FPREC-1      ;offset to last byte of FRE
                  ;        JMP      F1R               ;add FR1 to register, retur:



                  **       F1R - Add FR1 to Register
                  *
                  *        ENTRY    JSR      F1R
                  *                 X = offset to last byte of augend register
                  *                 FR1 - FR1+5 = addend
                  *
                  *        EXIT
                  *                 Sum in augend register
                  *
                  *        MODS
                  *                 Original Author Unknown
                  *                 1. Bring closer to Coding Standard (object :
                  *                    R. K. Nordin 11/01/83

      = DD0B      F1R      =        *                 ;entry
DD0B  A0E5                 LDY      #FR1+FPREC-1      ;offset to last byte of FR1
DD0D  D004 ^DD13           BNE      FARR



                  **       FRA2E - Add FR2 to FRE
                  *
                  *        ENTRY    JSR      FRA2E
                  *                 FRE - FRE+5 = augend
                  *                 FR2 - FR2+5 = addend
                  *
                  *        EXIT
                  *                 FRE - FRE+5 = sum
                  *
                  *        MODS
                  *                 Original Author Unknown
                  *                 1. Bring closer to Coding Standard (object :
                  *                    R. K. Nordin 11/01/83

      = DD0F      FRA2E    =        *                 ;entry
DD0F  A2DF                 LDX      #FRE+FPREC-1      ;offset to last byte of FRE
                  ;        JMP      F2R
```

```
                        **      F2R - Add FR2 to Register
                        *
                        *       ENTRY   JSR     F2R
                        *               X = offset to last byte of augend register
                        *               FR2 - FR2+5 = addend
                        *
                        *       EXIT
                        *               Sum in augend register
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

     = DD11             F2R     =       *                       ;entry
DD11 A0EB                       LDY     #FR2+FPREC-1            ;offset to last byte of FR2
                        ;       JMP     FARR



                        **      FARR - Add Register to Register
                        *
                        *       ENTRY   JSR     FARR
                        *               X = offset to last byte of augend register
                        *               Y = offset to last byte of addend register
                        *
                        *       EXIT
                        *               Sum in augend register
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

     = DD13             FARR    =       *                       ;entry

                        ;       Initialize.

DD13 A905                       LDA     #FPREC-1                ;floating point number size:
DD15 85F7                       STA     ZTEMP4                  ;byte count
DD17 18                         CLC
DD18 F8                         SED

                        ;       Add.

DD19 B500               FARR1   LDA     $0000,X                 ;byte of augend
DD1B 790000                     ADC     $0000,Y                 ;add byte of addend
DD1E 9500                       STA     $0000,X                 ;update byte of augend
DD20 CA                         DEX
DD21 88                         DEY
DD22 C6F7                       DEC     ZTEMP4                  ;decrement byte count
DD24 10F3 ^DD19                 BPL     FARR1                   ;if not done

                        ;       Exit.
```

```
DD26  D8                    CLD
DD27  60                    RTS                     ;return



                     **     M12 - Move FR1 to FR2
                     *
                     *      ENTRY   JSR     M12
                     *              FR1 - FR1+5 = number to move
                     *
                     *      EXIT
                     *              FR2 - FR2+5 = moved number
                     *
                     *      MODS
                     *              Original Author Unknown
                     *              1. Bring closer to Coding Standard (object :
                     *                 R. K. Nordin 11/01/83

      = DD28         M12    =       *               ;entry
DD28  A005                  LDY     #FPREC-1        ;offset to last byte

DD2A  B9E000         M121   LDA     FR1,Y           ;byte of source
DD2D  99E600                STA     FR2,Y           ;byte of destination
DD30  88                    DEY
DD31  10F7 ^DD2A            BPL     M121            ;if not done

DD33  60                    RTS                     ;return



                     **     M0E - Move FR0 to FRE
                     *
                     *      ENTRY   JSR     M0E
                     *              FR0 - FR0+5 = number to move
                     *
                     *      EXIT
                     *              FRE - FRE+5 = moved number
                     *
                     *      MODS
                     *              Original Author Unknown
                     *              1. Bring closer to Coding Standard (object :
                     *                 R. K. Nordin 11/01/83

      = DD34         M0E    =       *               ;entry
DD34  A005                  LDY     #FPREC-1        ;offset to last byte

DD36  B9D400         M0E1   LDA     FR0,Y           ;byte of source
DD39  99DA00                STA     FRE,Y           ;byte of destination
DD3C  88                    DEY
DD3D  10F7 ^DD36            BPL     M0E1            ;if not done
```

```
DD3F  60                RTS                     ;return



DD40                    FIX     PLYEVL


                **      PLYEVL - Evaluate Polynomial
                *
                *       Y = A(0)+A(1)*X+A(2)*X^2+...+A(N)*X^N
                *
                *       ENTRY   JSR     PLYEVL
                *               X = low address of coefficient table
                *               Y = high address of coefficient table
                *               FR0 - FR0+5 = X argument
                *               A = N+1
                *
                *       EXIT
                *               FR0 - FR0+5 = Y result
                *
                *       MODS
                *               Original Author Unknown
                *               1. Bring closer to Coding Standard (object :
                *                  R. K. Nordin 11/01/83


                ;PLYEVL =       *               ;entry

DD40  86FE              STX     FPTR2           ;save pointer to coefficien:
DD42  84FF              STY     FPTR2+1
DD44  85EF              STA     PLYCNT          ;degree
DD46  A2E0              LDX     #low PLYARG
DD48  A005              LDY     #high PLYARG
DD4A  20A7DD            JSR     FSTOR           ;save argument
DD4D  20B6DD            JSR     FMOVE           ;move argument to FR1
DD50  A6FE              LDX     FPTR2
DD52  A4FF              LDY     FPTR2+1
DD54  2089DD            JSR     FLD0R           ;initialize sum in FR0
DD57  C6EF              DEC     PLYCNT          ;decrement degree
DD59  F02D ^DD86        BEQ     PLY3            ;if complete, exit

DD5B  20DBDA    PLY1    JSR     FMUL            ;argument times current sum
DD5E  B028 ^DD88        BCS     PLY3            ;if overflow

DD60  18                CLC
DD61  A5FE              LDA     FPTR2           ;current low coefficient ad:
DD63  6906              ADC     #FPREC          ;add floating point number :
DD65  85FE              STA     FPTR2           ;update low coefficient add:
DD67  9006 ^DD6F        BCC     PLY2            ;if no carry

DD69  A5FF              LDA     FPTR2+1         ;current high coefficceint :
DD6B  6900              ADC     #0              ;adjust high coefficient ad:
DD6D  85FF              STA     FPTR2+1         ;update high coefficient ad:
```

```
DD6F  A6FE         PLY2    LDX     FPTR2             ;low coefficient address
DD71  A4FF                 LDY     FPTR2+1           ;high coefficient address
DD73  2098DD               JSR     FLD1R             ;get next coefficient
DD76  2066DA               JSR     FADD              ;add coefficient to argument
DD79  B00D ^DD88           BCS     PLY3              ;if overflow

DD7B  C6EF                 DEC     PLYCNT            ;decrement degree
DD7D  F009 ^DD88           BEQ     PLY3              ;if complete, exit

DD7F  A2E0                 LDX     #low PLYARG       ;low argument address
DD81  A005                 LDY     #high PLYARG      ;high argument address
DD83  2098DD               JSR     FLD1R             ;get argument
DD86  30D3 ^DD5B           BMI     PLY1              ;continue

DD88  60           PLY3    RTS                       ;return



DD89                       FIX     FLD0R



                   **      FLD0R - Load FR0
                   *
                   *       ENTRY   JSR     FLD0R
                   *               X = low pointer
                   *               Y = high pointer
                   *
                   *       EXIT
                   *               FR0 loaded
                   *
                   *       MODS
                   *               Original Author Unknown
                   *               1. Bring closer to Coding Standard (object :
                   *                  R. K. Nordin 11/01/83


                   ;FLD0R  =       *                 ;entry
DD89  86FC                 STX     FLPTR             ;low pointer
DD8B  84FD                 STY     FLPTR+1           ;high pointer
                   ;       JMP     FLD0P             ;load FR0, return
```

```
DD8D                        FIX     FLD0P



                    **      FLD0P - Load FR0
                    *
                    *       ENTRY   JSR     FLD0P
                    *               FLPTR - FLPTR+1 = pointer
                    *
                    *       EXIT
                    *               FR0 loaded
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

                    ;FLD0P  =       *                   ;entry

DD8D  A005                  LDY     #FPREC-1            ;offset to last byte

DD8F  B1FC          FLD01   LDA     (FLPTR),Y           ;byte of source
DD91  99D400                STA     FR0,Y               ;byte of destination
DD94  88                    DEY
DD95  10F8 ^DD8F            BPL     FLD01               ;if not done

DD97  60                    RTS                         ;return



DD98                        FIX     FLD1R



                    **      FLD1R - Load FR1
                    *
                    *       ENTRY   JSR     FLD1R
                    *               X = low pointer
                    *               Y = high pointer
                    *
                    *       EXIT
                    *               FR1 loaded
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

                    ;FLD1R  =       *                   ;entry
DD98  86FC                  STX     FLPTR               ;low pointer
DD9A  84FD                  STY     FLPTR+1             ;high pointer
                    ;       JMP     FLD1P               ;load FR1, return
```

```
DD9C                        FIX     FLD1P



                    **      FLD1P - Load FR1
                    *
                    *       ENTRY   JSR     FLD1P
                    *               FLPTR - FLPTR+1 = pointer
                    *
                    *       EXIT
                    *               FR1 loaded
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

                    ;FLD1P  =       *               ;entry
DD9C  A005                  LDY     #FPREC-1        ;offset to last byte

DD9E  B1FC          FLD11   LDA     (FLPTR),Y       ;byte of source
DDA0  99E000                STA     FR1,Y           ;byte of destination
DDA3  88                    DEY
DDA4  10F8 ^DD9E            BPL     FLD11           ;if not done

DDA6  60                    RTS                     ;return



DDA7                        FIX     FSTOR



                    **      FSTOR - Store FR0
                    *
                    *       ENTRY   JSR     FSTOR
                    *               FR0 - FR0+5 = number
                    *               X = low pointer
                    *               Y = high pointer
                    *
                    *       EXIT
                    *               FR0 stored
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

                    ;FSTOR  =       *               ;entry
DDA7  86FC                  STX     FLPTR           ;low pointer
DDA9  84FD                  STY     FLPTR+1         ;high pointer
```

```
                        ;       JMP     FSTOP           ;store FR0, return


DDAB                            FIX     FSTOP


                        **      FSTOP - Store FR0
                        *
                        *       ENTRY   JSR     FSTOP
                        *               FR0 - FR0+5 = number
                        *               FLPTR - FLPTR+1 = pointer
                        *
                        *       EXIT
                        *               FR0 stored
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


                        ;FSTOP  =       *               ;entry

DDAB  A005                      LDY     #FPREC-1        ;offset to last byte

DDAD  B9D400            FST01   LDA     FR0,Y           ;byte of source
DDB0  91FC                      STA     (FLPTR),Y       ;byte of destination
DDB2  88                        DEY
DDB3  10F8 ^DDAD                BPL     FST01           ;if not done

DDB5  60                        RTS                     ;return


DDB6                            FIX     FMOVE


                        **      FMOVE - Move FR0 to FR1
                        *
                        *       ENTRY   JSR     FMOVE
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

                        ;FMOVE  =       *               ;entry

DDB6  A205                      LDX     #FPREC-1        ;offset to last byte
```

```
DDB8  B5D4         FM01   LDA    FR0,X           ;byte of source
DDBA  95E0                STA    FR1,X           ;byte of destination
DDBC  CA                  DEX
DDBD  10F9 ^DDB8          BPL    FM01            ;if not done

DDBF  60                  RTS                    ;return



DDC0                      FIX    EXP



                   **     EXP - Compute Power of e
                   *
                   *      ENTRY   JSR     EXP
                   *
                   *      MODS
                   *              Original Author Unknown
                   *              1. Bring closer to Coding Standard (object :
                   *                 R. K. Nordin 11/01/83

                   ;EXP   =       *                ;entry

                   ;      Initialize.

DDC0  A289                LDX    #low LOG10E     ;base 10 logarithm of e
DDC2  A0DE                LDY    #high LOG10E
DDC4  2098DD              JSR    FLD1R           ;load FR1

                   ;      Compute X*LOG10(E).

DDC7  20DBDA              JSR    FMUL            ;multiply
DDCA  B07F ^DE4B          BCS    EXP6            ;if overflow, error

                   ;      Compute result = 10^(X*LOG10(E)).

                   ;      JMP    EXP10           ;compute power of 10, retur:
```

```
DDCC                    FIX     EXP10



                **      EXP10 - Compute Power of 10
                *
                *       ENTRY   JSR     EXP10
                *
                *       MODS
                *               Original Author Unknown
                *               1. Bring closer to Coding Standard (object :
                *                  R. K. Nordin 11/01/83

                ;EXP10  =       *               ;entry

                ;       Initialize.

DDCC  A900              LDA     #0
DDCE  85F1              STA     XFMFLG          ;zero integer part
DDD0  A5D4              LDA     FR0
DDD2  85F0              STA     SGNFLG          ;save argument sign
DDD4  297F              AND     #$7F            ;extract absolute value
DDD6  85D4              STA     FR0             ;update argument

                ;       Check for argument less than 1.

DDD8  38                SEC
DDD9  E940              SBC     #$40            ;subtract bias
DDDB  3026 ^DE03        BMI     EXP1            ;if argument < 1

                ;       Extract integer and fractional parts of exponent.

DDDD  C904              CMP     #FPREC-2
DDDF  106A ^DE4B        BPL     EXP6            ;if argument too big, error

DDE1  A2E6              LDX     #low FPSCR
DDE3  A005              LDY     #high FPSCR
DDE5  20A7DD            JSR     FSTOR           ;save argument
DDE8  20D2D9            JSR     FPI             ;convert argument to intege:
DDEB  A5D4              LDA     FR0
DDED  85F1              STA     XFMFLG          ;save interger part
DDEF  A5D5              LDA     FR0+1           ;most significant byte of i:
DDF1  D058 ^DE4B        BNE     EXP6            ;if integer part too large,:

DDF3  20AAD9            JSR     IFP             ;convert integer part to fl:
DDF6  20B6DD            JSR     FMOVE
DDF9  A2E6              LDX     #low FPSCR
DDFB  A005              LDY     #high FPSCR
DDFD  2089DD            JSR     FLDOR           ;argument
DE00  2060DA            JSR     FSUB            ;subtract to get fractional:

                ;       Compute 10 to fractional exponent.

DE03  A90A      EXP1    LDA     #NPCOEF
DE05  A24D              LDX     #low P10COF
```

```
DE07  A0DE                    LDY     #high P10COF
DE09  2040DD                  JSR     PLYEVL          ;P(X)
DE0C  2086DD                  JSR     FMOVE
DE0F  20DBDA                  JSR     FMUL            ;P(X)*P(X)

                      ;       Check integer part.

DE12  A5F1                    LDA     XFMFLG          ;integer part
DE14  F023 ^DE39              BEQ     EXP4            ;if integer part zero

                      ;       Compute 10 to integer part.

DE16  18                      CLC
DE17  6A                      ROR     A               ;integer part divided by 2
DE18  85E0                    STA     FR1             ;exponent
DE1A  A901                    LDA     #1              ;assume mantissa 1
DE1C  9002 ^DE20              BCC     EXP2            ;if integer part even

DE1E  A910                    LDA     #$10            ;substitute mantissa 10

DE20  85E1            EXP2    STA     FR1M            ;mantissa
DE22  A204                    LDX     #FMPREC-1       ;offset to last byte of man:
DE24  A900                    LDA     #0

DE26  95E2            EXP3    STA     FR1M+1,X        ;zero byte of mantissa
DE28  CA                      DEX
DE29  10FB ^DE26              BPL     EXP3            ;if not done

DE2B  A5E0                    LDA     FR1             ;exponent
DE2D  18                      CLC
DE2E  6940                    ADC     #$40            ;add bias
DE30  B019 ^DE4B              BCS     EXP6            ;if too big, error

DE32  3017 ^DE4B              BMI     EXP6            ;if underflow, error

DE34  85E0                    STA     FR1             ;10 to integer part

                      ;       Compute product of 10 to integer part and 10 to fra:

DE36  20DBDA                  JSR     FMUL            ;multiply to get result

                      ;       Invert result if argument < 0.

DE39  A5F0            EXP4    LDA     SGNFLG          ;argument sign
DE3B  100D ^DE4A              BPL     EXP5            ;if argument >= 0

DE3D  2086DD                  JSR     FMOVE
DE40  A28F                    LDX     #low FONE
DE42  A0DE                    LDY     #high FONE
DE44  2089DD                  JSR     FLD0R           ;load FR0
DE47  2028DB                  JSR     FDIV            ;divide to get result

                      ;       Exit.

DE4A  60              EXP5    RTS                     ;return

                      ;       Return error.
```

```
DE4B  38            EXP6    SEC                         ;indicate error
DE4C  60                    RTS                         ;return



                    **      P10COF - Power of 10 Coefficients


DE4D  3D17941900    P10COF  DB      $3D,$17,$94,$19,$00,$00 ;0.0000179419
DE53  3D57330500            DB      $3D,$57,$33,$05,$00,$00 ;0.0000573305
DE59  3E05547662            DB      $3E,$05,$54,$76,$62,$00 ;0.0005547662
DE5F  3E32196227            DB      $3E,$32,$19,$62,$27,$00 ;0.0032176227
DE65  3F01686030            DB      $3F,$01,$68,$60,$30,$36 ;0.0168603036
DE6B  3F07320327            DB      $3F,$07,$32,$03,$27,$41 ;0.0732032741
DE71  3F25433456            DB      $3F,$25,$43,$34,$56,$75 ;0.2543345675
DE77  3F66273730            DB      $3F,$66,$27,$37,$30,$50 ;0.6627373050
DE7D  4001151292            DB      $40,$01,$15,$12,$92,$55 ;1.15129255
DE83  3F99999999            DB      $3F,$99,$99,$99,$99,$99 ;0.9999999999

      = 000A        NPCOEF  =       [*-P10COF]/FPREC



                    **      LOG10E - Base 10 Logarithm of e


DE89  3F43429448    LOG10E  DB      $3F,$43,$42,$94,$48,$19 ;base 10 logarithm :



                    **      FONE - 1.0


DE8F  4001000000    FONE    DB      $40,$01,$00,$00,$00,$00 ;1.0



                    **      XFORM - Transform
                    *
                    *       Z = (X-C)/(X+C)
                    *
                    *       ENTRY   JSR     XFORM
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = DE95        XFORM   =       *                   ;entry
DE95  86FE                  STX     FPTR2
DE97  84FF                  STY     FPTR2+1
```

```
DE99  A2E0                LDX     #low PLYARG
DE9B  A005                LDY     #high PLYARG
DE9D  20A7DD              JSR     FSTOR              ;save argument
DEA0  A6FE                LDX     FPTR2
DEA2  A4FF                LDY     FPTR2+1
DEA4  2098DD              JSR     FLD1R              ;load FR1
DEA7  2066DA              JSR     FADD               ;X+C
DEAA  A2E6                LDX     #low FPSCR
DEAC  A005                LDY     #high FPSCR
DEAE  20A7DD              JSR     FSTOR              ;store FR0
DEB1  A2E0                LDX     #low PLYARG
DEB3  A005                LDY     #high PLYARG
DEB5  2089DD              JSR     FLD0R              ;load FR0
DEB8  A6FE                LDX     FPTR2
DEBA  A4FF                LDY     FPTR2+1
DEBC  2098DD              JSR     FLD1R              ;load FR1
DEBF  2060DA              JSR     FSUB               ;X-C
DEC2  A2E6                LDX     #low FPSCR
DEC4  A005                LDY     #high FPSCR
DEC6  2098DD              JSR     FLD1R              ;load FR1
DEC9  2028DB              JSR     FDIV               ;divide to get result
DECC  60                  RTS                        ;return


DECD                      FIX     LOG


                  **      LOG - Compute Base e Logarithm
                  *
                  *       ENTRY   JSR     LOG
                  *               FR0 - FR0+5 = argument
                  *
                  *       MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83


                  ;LOG    =       *       ;entry

DECD  A901                LDA     #1      ;indicate base e logarithm
DECF  D002 ^DED3          BNE     LOGS    ;compute logertihm, return
```

```
DED1                            FIX     LOG10



                        **      LOG10 - Compute Base 10 Logarithm
                        *
                        *       ENTRY   JSR     LOG10
                        *               FR0 - FR0+5 = argument
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

                        ;LOG10  =       *       ;entry

DED1  A900                      LDA     #0      ;indicate base 10 logartihm
                        ;       JMP     LOGS    ;compute logarithm, return



                        **      LOGS - Compute Logarithm
                        *
                        *       ENTRY   JSR     LOGS
                        *               A = 0, if base 10 logarithm
                        *                 = 1, if base e logartihm
                        *               FR0 - FR0+5 = argument
                        *
                        *       EXIT
                        *               C set, if error
                        *               C clear, if no error
                        *               FR0 - FR0+5 = result
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = DED3            LOGS    =       *       ;entry

                        ;       Initialize.

DED3  85F0                      STA     SGNFLG  ;save logarithm base indicator

                        ;       Check argument.

DED5  A5D4                      LDA     FR0     ;argument exponent
DED7  F005 ^DEDE                BEQ     LOGS1   ;if argument zero, error

DED9  3003 ^DEDE                BMI     LOGS1   ;if argument negative, error

                        ;       X = F*(10^Y), 1<F<10
                        ;       10^Y HAS SAME EXP BYTE AS X
```

```
                        ;       & MANTISSA BYTE = 1 OR 10

DEDB    4CF6DF                  JMP     LOGQ

                        ;       Return error.

DEDE    38              LOGS1   SEC                     ;indicate error
DEDF    60                      RTS                     ;return



                        **      LOGC - Complete Computation of Logarithm
                        *
                        *       ENTRY   JSR     LOGC
                        *               SGNFLG = 0, if base 10 logarithmr
                        *                      = 1, if base e logarithm
                        *
                        *       NOTES
                        *               Problem: logic is convoluted because LOGQ c:
                        *               was moved.
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = DEE0          LOGC    =       *               ;entry

                        ;       Initialize.

DEE0    E940                    SBC     #$40
DEE2    0A                      ASL     A
DEE3    85F1                    STA     XFMFLG          ;save Y
DEE5    A5D5                    LDA     FR0+1
DEE7    29F0                    AND     #$F0
DEE9    D004 ^DEEF              BNE     LOGC2

DEEB    A901                    LDA     #1              ;mantissa is 1
DEED    D004 ^DEF3              BNE     LOGC3           ;set mantissa

DEEF    E6F1            LOGC2   INC     XFMFLG          ;increment Y
DEF1    A910                    LDA     #$10            ;mantissa is 10

DEF3    85E1            LOGC3   STA     FR1M            ;mantissa
DEF5    A204                    LDX     #FMPREC-1       ;offset to last byte of man:
DEF7    A900                    LDA     #0

DEF9    95E2            LOGC4   STA     FR1M+1,X        ;zero byte of mantissa
DEFB    CA                      DEX
DEFC    10FB ^DEF9              BPL     LOGC4           ;if not done

DEFE    2028DB                  JSR     FDIV            ;X = X/(10^Y), S.B. IN (1,1:

                        ;       Compute LOG10(X), 1 <= X <= 10.
```

```
DF01  A266                LDX   #low SQR10
DF03  A0DF                LDY   #high SQR10
DF05  2095DE              JSR   XFORM          ;Z = (X-C)/(X+C); C*C = 10
DF08  A2E6                LDX   #low FPSCR
DF0A  A005                LDY   #high FPSCR
DF0C  20A7DD              JSR   FSTOR          ;SAVE Z
DF0F  20B6DD              JSR   FMOVE
DF12  20DBDA              JSR   FMUL           ;Z*Z
DF15  A90A                LDA   #NLCOEF
DF17  A272                LDX   #low LGCOEF
DF19  A0DF                LDY   #high LGCOEF
DF1B  2040DD              JSR   PLYEVL         ;P(Z*Z)
DF1E  A2E6                LDX   #low FPSCR
DF20  A005                LDY   #high FPSCR
DF22  2098DD              JSR   FLD1R          ;load FR1
DF25  20DBDA              JSR   FMUL           ;Z*P(Z*Z)
DF28  A26C                LDX   #low FHALF
DF2A  A0DF                LDY   #high FHALF
DF2C  2098DD              JSR   FLD1R
DF2F  2066DA              JSR   FADD           ;0.5 + Z*P(Z*Z)
DF32  20B6DD              JSR   FMOVE
DF35  A900                LDA   #0
DF37  85D5                STA   FR0+1
DF39  A5F1                LDA   XFMFLG
DF3B  85D4                STA   FR0
DF3D  1007 ^DF46          BPL   LOGC5

DF3F  49FF                EOR   #-$01          ;complement sign
DF41  18                  CLC
DF42  6901                ADC   #1
DF44  85D4                STA   FR0

DF46  20AAD9      LOGC5   JSR   IFP            ;convert integer to floatin;
DF49  24F1                BIT   XFMFLG
DF4B  1006 ^DF53          BPL   LOGC6

DF4D  A980                LDA   #$80
DF4F  05D4                ORA   FR0
DF51  85D4                STA   FR0            ;update exponent

DF53  2066DA      LOGC6   JSR   FADD           ;LOG(X) = LOG(X)+Y

                  ;       Check base of logarithm.

DF56  A5F0                LDA   SGNFLG         ;logarithm base indicator
DF58  F00A ^DF64          BEQ   LOGC7          ;if LOG10 (not LOG)

                  ;       Compute base e logarithm.

DF5A  A289                LDX   #low LOG10E    ;base 10 logarithm of e
DF5C  A0DE                LDY   #high LOG10E
DF5E  2098DD              JSR   FLD1R          ;load FR1
DF61  2028DB              JSR   FDIV           ;result is LOG(X) divided b:

                  ;       Exit.

DF64  18          LOGC7   CLC                  ;indicate success
```

```
DF65  60                     RTS                                 ;return



                    **       SQR10 - Square Root of 10


DF66  4003162277  SQR10      DB       $40,$03,$16,$22,$77,$66 ;square root of 10



                    **       FHALF - 0.5


DF6C  3F50000000  FHALF      DB       $3F,$50,$00,$00,$00,$00 ;0.5



                    **       LGCOEF - Logartihm Coefficients


DF72  3F49155711  LGCOEF     DB       $3F,$49,$15,$57,$11,$08 ;0.4915571108
DF78  BF51704947             DB       $BF,$51,$70,$49,$47,$08 ;-0.5170494708
DF7E  3F39205761             DB       $3F,$39,$20,$57,$61,$95 ;0.3920576195
DF84  BF04396303             DB       $BF,$04,$39,$63,$03,$55 ;-0.0439630355
DF8A  3F10093012             DB       $3F,$10,$09,$30,$12,$64 ;0.1009301264
DF90  3F09390804             DB       $3F,$09,$39,$08,$04,$60 ;0.0939080460
DF96  3F12425847             DB       $3F,$12,$42,$58,$47,$42 ;0.1242584742
DF9C  3F17371206             DB       $3F,$17,$37,$12,$06,$08 ;0.1737120608
DFA2  3F28952971             DB       $3F,$28,$95,$29,$71,$17 ;0.2895297117
DFA8  3F86858896             DB       $3F,$86,$85,$88,$96,$44 ;0.8685889644

      = 000A      NLCOEF     =        [*-LGCOEF]/FPREC



                    **       ATCOEF - Arctangent Coefficients
                    *
                    *        NOTES
                    *                 Problem: not used.


DFAE  3E16054449             DB       $3E,$16,$05,$44,$49,$00 ;0.001605444900
DFB4  BE95683845             DB       $BE,$95,$68,$38,$45,$00 ;-0.009568384500
DFBA  3F02687994             DB       $3F,$02,$68,$79,$94,$16 ;0.0268799416
DFC0  BF04927890             DB       $BF,$04,$92,$78,$90,$80 ;-0.0492789080
DFC6  3F07031520             DB       $3F,$07,$03,$15,$20,$00 ;0.0703152000
DFCC  BF08922912             DB       $BF,$08,$92,$29,$12,$44 ;-0.0892291244
DFD2  3F11084009             DB       $3F,$11,$08,$40,$09,$11 ;0.1108400911
DFD8  BF14283156             DB       $BF,$14,$28,$31,$56,$04 ;-0.1428315604
DFDE  3F19999877             DB       $3F,$19,$99,$98,$77,$44 ;0.1999987744
DFE4  BF33333331             DB       $BF,$33,$33,$33,$31,$13 ;-0.3333333113
DFEA  3F99999999             DB       $3F,$99,$99,$99,$99,$99 ;0.9999999999
```

```
DFF0  3F78539816          DB      $3F,$78,$53,$98,$16,$34 ;pi/4 = arctan 1


                    **      LOGQ - Continue Computation of Loagarithm
                    *
                    *       ENTRY   JSR     LOGQ
                    *
                    *       NOTES
                    *               Problem: logic is convoluted because this c:
                    *               moved.
                    *               Problem: for readability, this might be rel:
                    *               before tables.
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = DFF6        LOGQ    =       *       ;entry
DFF6  A5D4                  LDA     FR0
DFF8  85E0                  STA     FR1
DFFA  38                    SEC
DFFB  4CE0DE                JMP     LOGC    ;complete computation of logarithm,:
```

```
DFFE                          FIX     DCSORG


                    **        Domestic Character Set

E000  0000000000              DB      $00,$00,$00,$00,$00,$00,$00,$00 ;$00 - spac:
E008  0018181818              DB      $00,$18,$18,$18,$18,$00,$18,$00 ;$01 - !
E010  0066666600              DB      $00,$66,$66,$66,$00,$00,$00,$00 ;$02 - "
E018  0066FF6666              DB      $00,$66,$FF,$66,$66,$FF,$66,$00 ;$03 - #
E020  183E603C06              DB      $18,$3E,$60,$3C,$06,$7C,$18,$00 ;$04 - $
E028  00666C1830              DB      $00,$66,$6C,$18,$30,$66,$46,$00 ;$05 - %
E030  1C361C386F              DB      $1C,$36,$1C,$38,$6F,$66,$3B,$00 ;$06 - &
E038  0018181800              DB      $00,$18,$18,$18,$00,$00,$00,$00 ;$07 - '
E040  000E1C1818              DB      $00,$0E,$1C,$18,$18,$1C,$0E,$00 ;$08 - (
E048  0070381818              DB      $00,$70,$38,$18,$18,$38,$70,$00 ;$09 - )
E050  00663CFF3C              DB      $00,$66,$3C,$FF,$3C,$66,$00,$00 ;$0A - aste:
E058  0018187E18              DB      $00,$18,$18,$7E,$18,$18,$00,$00 ;$0B - plus
E060  0000000000              DB      $00,$00,$00,$00,$00,$18,$18,$30 ;$0C - comm:
E068  0000007E00              DB      $00,$00,$00,$7E,$00,$00,$00,$00 ;$0D - minu:
E070  0000000000              DB      $00,$00,$00,$00,$00,$18,$18,$00 ;$0E - peri:
E078  00060C1830              DB      $00,$06,$0C,$18,$30,$60,$40,$00 ;$0F - /

E080  003C666E76              DB      $00,$3C,$66,$6E,$76,$66,$3C,$00 ;$10 - 0
E088  0018381818              DB      $00,$18,$38,$18,$18,$18,$7E,$00 ;$11 - 1
E090  003C660C18              DB      $00,$3C,$66,$0C,$18,$30,$7E,$00 ;$12 - 2
E098  007E0C180C              DB      $00,$7E,$0C,$18,$0C,$66,$3C,$00 ;$13 - 3
E0A0  000C1C3C6C              DB      $00,$0C,$1C,$3C,$6C,$7E,$0C,$00 ;$14 - 4
E0A8  007E607C06              DB      $00,$7E,$60,$7C,$06,$66,$3C,$00 ;$15 - 5
E0B0  003C607C66              DB      $00,$3C,$60,$7C,$66,$66,$3C,$00 ;$16 - 6
E0B8  007E060C18              DB      $00,$7E,$06,$0C,$18,$30,$30,$00 ;$17 - 7
E0C0  003C663C66              DB      $00,$3C,$66,$3C,$66,$66,$3C,$00 ;$18 - 8
E0C8  003C663E06              DB      $00,$3C,$66,$3E,$06,$0C,$38,$00 ;$19 - 9
E0D0  0000181800              DB      $00,$00,$18,$18,$00,$18,$18,$00 ;$1A - colo:
E0D8  0000181800              DB      $00,$00,$18,$18,$00,$18,$18,$30 ;$1B - semi:
E0E0  060C183018              DB      $06,$0C,$18,$30,$18,$0C,$06,$00 ;$1C - <
E0E8  00007E0000              DB      $00,$00,$7E,$00,$00,$7E,$00,$00 ;$1D - =
E0F0  6030180C18              DB      $60,$30,$18,$0C,$18,$30,$60,$00 ;$1E - >
E0F8  003C660C18              DB      $00,$3C,$66,$0C,$18,$00,$18,$00 ;$1F - ?

E100  003C666E6E              DB      $00,$3C,$66,$6E,$6E,$60,$3E,$00 ;$20 - @
E108  00183C6666              DB      $00,$18,$3C,$66,$66,$7E,$66,$00 ;$21 - A
E110  007C667C66              DB      $00,$7C,$66,$7C,$66,$66,$7C,$00 ;$22 - B
E118  003C666060              DB      $00,$3C,$66,$60,$60,$66,$3C,$00 ;$23 - C
E120  00786C6666              DB      $00,$78,$6C,$66,$66,$6C,$78,$00 ;$24 - D
E128  007E607C60              DB      $00,$7E,$60,$7C,$60,$60,$7E,$00 ;$25 - E
E130  007E607C60              DB      $00,$7E,$60,$7C,$60,$60,$60,$00 ;$26 - F
E138  003E60606E              DB      $00,$3E,$60,$60,$6E,$66,$3E,$00 ;$27 - G
E140  0066667E66              DB      $00,$66,$66,$7E,$66,$66,$66,$00 ;$28 - H
E148  007E181818              DB      $00,$7E,$18,$18,$18,$18,$7E,$00 ;$29 - I
E150  0006060606              DB      $00,$06,$06,$06,$06,$66,$3C,$00 ;$2A - J
E158  00666C7878              DB      $00,$66,$6C,$78,$78,$6C,$66,$00 ;$2B - K
E160  0060606060              DB      $00,$60,$60,$60,$60,$60,$7E,$00 ;$2C - L
E168  0063777F6B              DB      $00,$63,$77,$7F,$6B,$63,$63,$00 ;$2D - M
E170  0066767E7E              DB      $00,$66,$76,$7E,$7E,$6E,$66,$00 ;$2E - N
```

```
E178  003C666666     DB   $00,$3C,$66,$66,$66,$66,$3C,$00 ;$2F - O

E180  007C66667C     DB   $00,$7C,$66,$66,$7C,$60,$60,$00 ;$30 - P
E188  003C666666     DB   $00,$3C,$66,$66,$66,$6C,$36,$00 ;$31 - Q
E190  007C66667C     DB   $00,$7C,$66,$66,$7C,$6C,$66,$00 ;$32 - R
E198  003C603C06     DB   $00,$3C,$60,$3C,$06,$06,$3C,$00 ;$33 - S
E1A0  007E181818     DB   $00,$7E,$18,$18,$18,$18,$18,$00 ;$34 - T
E1A8  0066666666     DB   $00,$66,$66,$66,$66,$66,$7E,$00 ;$35 - U
E1B0  0066666666     DB   $00,$66,$66,$66,$66,$3C,$18,$00 ;$36 - V
E1B8  0063636B7F     DB   $00,$63,$63,$6B,$7F,$77,$63,$00 ;$37 - W
E1C0  0066663C3C     DB   $00,$66,$66,$3C,$3C,$66,$66,$00 ;$38 - X
E1C8  0066663C18     DB   $00,$66,$66,$3C,$18,$18,$18,$00 ;$39 - Y
E1D0  007E0C1830     DB   $00,$7E,$0C,$18,$30,$60,$7E,$00 ;$3A - Z
E1D8  001E181818     DB   $00,$1E,$18,$18,$18,$18,$1E,$00 ;$3B - [
E1E0  0040603018     DB   $00,$40,$60,$30,$18,$0C,$06,$00 ;$3C - \
E1E8  0078181818     DB   $00,$78,$18,$18,$18,$18,$78,$00 ;$3D - ]
E1F0  00081C3663     DB   $00,$08,$1C,$36,$63,$00,$00,$00 ;$3E - ^
E1F8  0000000000     DB   $00,$00,$00,$00,$00,$00,$FF,$00 ;$3F - unde:

E200  00367F7F3E     DB   $00,$36,$7F,$7F,$3E,$1C,$08,$00 ;$40 - hear:
E208  1818181F1F     DB   $18,$18,$18,$1F,$1F,$18,$18,$18 ;$41 - mid :
E210  0303030303     DB   $03,$03,$03,$03,$03,$03,$03,$03 ;$42 - righ:
E218  181818F8F8     DB   $18,$18,$18,$F8,$F8,$00,$00,$00 ;$43 - low :
E220  181818F8F8     DB   $18,$18,$18,$F8,$F8,$18,$18,$18 ;$44 - mid :
E228  000000F8F8     DB   $00,$00,$00,$F8,$F8,$18,$18,$18 ;$45 - up r:
E230  03070E1C38     DB   $03,$07,$0E,$1C,$38,$70,$E0,$C0 ;$46 - righ:
E238  C0E070381C     DB   $C0,$E0,$70,$38,$1C,$0E,$07,$03 ;$47 - left:
E240  0103070F1F     DB   $01,$03,$07,$0F,$1F,$3F,$7F,$FF ;$48 - righ:
E248  000000000F     DB   $00,$00,$00,$00,$0F,$0F,$0F,$0F ;$49 - low :
E250  80C0E0F0F8     DB   $80,$C0,$E0,$F0,$F8,$FC,$FE,$FF ;$4A - left:
E258  0F0F0F0F00     DB   $0F,$0F,$0F,$0F,$00,$00,$00,$00 ;$4B - up r:
E260  F0F0F0F000     DB   $F0,$F0,$F0,$F0,$00,$00,$00,$00 ;$4C - up l:
E268  FFFF000000     DB   $FF,$FF,$00,$00,$00,$00,$00,$00 ;$4D - top :
E270  0000000000     DB   $00,$00,$00,$00,$00,$00,$FF,$FF ;$4E - bott:
E278  00000000F0     DB   $00,$00,$00,$00,$F0,$F0,$F0,$F0 ;$4F - low :

E280  001C1C7777     DB   $00,$1C,$1C,$77,$77,$08,$1C,$00 ;$50 - club:
E288  0000001F1F     DB   $00,$00,$00,$1F,$1F,$18,$18,$18 ;$51 - up l:
E290  000000FFFF     DB   $00,$00,$00,$FF,$FF,$00,$00,$00 ;$52 - mid :
E298  181818FFFF     DB   $18,$18,$18,$FF,$FF,$18,$18,$18 ;$53 - mid :
E2A0  00003C7E7E     DB   $00,$00,$3C,$7E,$7E,$7E,$3C,$00 ;$54 - soli:
E2A8  00000000FF     DB   $00,$00,$00,$00,$FF,$FF,$FF,$FF ;$55 - bott:
E2B0  C0C0C0C0C0     DB   $C0,$C0,$C0,$C0,$C0,$C0,$C0,$C0 ;$56 - left:
E2B8  000000FFFF     DB   $00,$00,$00,$FF,$FF,$18,$18,$18 ;$57 - up m:
E2C0  181818FFFF     DB   $18,$18,$18,$FF,$FF,$00,$00,$00 ;$58 - low :
E2C8  F0F0F0F0F0     DB   $F0,$F0,$F0,$F0,$F0,$F0,$F0,$F0 ;$59 - left:
E2D0  1818181F1F     DB   $18,$18,$18,$1F,$1F,$00,$00,$00 ;$5A - low :
E2D8  786078607E     DB   $78,$60,$78,$60,$7E,$18,$1E,$00 ;$5B - disp:
E2E0  00183C7E18     DB   $00,$18,$3C,$7E,$18,$18,$18,$00 ;$5C - up a:
E2E8  001818187E     DB   $00,$18,$18,$18,$7E,$3C,$18,$00 ;$5D - down:
E2F0  0018307E30     DB   $00,$18,$30,$7E,$30,$18,$00,$00 ;$5E - left:
E2F8  00180C7E0C     DB   $00,$18,$0C,$7E,$0C,$18,$00,$00 ;$5F - righ:

E300  00183C7E7E     DB   $00,$18,$3C,$7E,$7E,$3C,$18,$00 ;$60 - diam:
E308  00003C063E     DB   $00,$00,$3C,$06,$3E,$66,$3E,$00 ;$61 - a
E310  0060607C66     DB   $00,$60,$60,$7C,$66,$66,$7C,$00 ;$62 - b
E318  00003C6060     DB   $00,$00,$3C,$60,$60,$60,$3C,$00 ;$63 - c
```

Domestic Character Set

```
E320  0006063E66      DB      $00,$06,$06,$3E,$66,$66,$3E,$00 ;$64 - d
E328  00003C667E      DB      $00,$00,$3C,$66,$7E,$60,$3C,$00 ;$65 - e
E330  000E183E18      DB      $00,$0E,$18,$3E,$18,$18,$18,$00 ;$66 - f
E338  00003E6666      DB      $00,$00,$3E,$66,$66,$3E,$06,$7C ;$67 - g
E340  0060607C66      DB      $00,$60,$60,$7C,$66,$66,$66,$00 ;$68 - h
E348  0018003818      DB      $00,$18,$00,$38,$18,$18,$3C,$00 ;$69 - i
E350  0006000606      DB      $00,$06,$00,$06,$06,$06,$06,$3C ;$6A - j
E358  0060606C78      DB      $00,$60,$60,$6C,$78,$6C,$66,$00 ;$6B - k
E360  0038181818      DB      $00,$38,$18,$18,$18,$18,$3C,$00 ;$6C - l
E368  0000667F7F      DB      $00,$00,$66,$7F,$7F,$6B,$63,$00 ;$6D - m
E370  00007C6666      DB      $00,$00,$7C,$66,$66,$66,$66,$00 ;$6E - n
E378  00003C6666      DB      $00,$00,$3C,$66,$66,$66,$3C,$00 ;$6F - o

E380  00007C6666      DB      $00,$00,$7C,$66,$66,$7C,$60,$60 ;$70 - p
E388  00003E6666      DB      $00,$00,$3E,$66,$66,$3E,$06,$06 ;$71 - q
E390  00007C6660      DB      $00,$00,$7C,$66,$60,$60,$60,$00 ;$72 - r
E398  00003E603C      DB      $00,$00,$3E,$60,$3C,$06,$7C,$00 ;$73 - s
E3A0  00187E1818      DB      $00,$18,$7E,$18,$18,$18,$0E,$00 ;$74 - t
E3A8  0000666666      DB      $00,$00,$66,$66,$66,$66,$3E,$00 ;$75 - u
E3B0  0000666666      DB      $00,$00,$66,$66,$66,$3C,$18,$00 ;$76 - v
E3B8  0000636B7F      DB      $00,$00,$63,$6B,$7F,$3E,$36,$00 ;$77 - w
E3C0  0000663C18      DB      $00,$00,$66,$3C,$18,$3C,$66,$00 ;$78 - x
E3C8  0000666666      DB      $00,$00,$66,$66,$66,$3E,$0C,$78 ;$79 - y
E3D0  00007E0C18      DB      $00,$00,$7E,$0C,$18,$30,$7E,$00 ;$7A - z
E3D8  00183C7E7E      DB      $00,$18,$3C,$7E,$7E,$18,$3C,$00 ;$7B - spad;
E3E0  1818181818      DB      $18,$18,$18,$18,$18,$18,$18,$18 ;$7C - |
E3E8  007E787C6E      DB      $00,$7E,$78,$7C,$6E,$66,$06,$00 ;$7D - disp;
E3F0  0818387838      DB      $08,$18,$38,$78,$38,$18,$08,$00 ;$7E - disp;
E3F8  10181C1E1C      DB      $10,$18,$1C,$1E,$1C,$18,$10,$00 ;$7F - disp;
```

```
E400                    FIX     EDITRV



                **      EDITRV - Editor Handler Vector Table


E400  93EF              DW      EOP-1   ;perform editor OPEN
E402  2DF2              DW      ECL-1   ;perform editor CLOSE
E404  49F2              DW      EGB-1   ;perform editor GET-BYTE
E406  AFF2              DW      EPB-1   ;perform editor PUT-BYTE
E408  1DF2              DW      SST-1   ;perform editor STATUS (screen STAT:
E40A  2CF2              DW      ESP-1   ;perform editor SPECIAL
E40C  4C6EEF            JMP     SIN     ;initialize editor (initialize scre:
E40F  00                DB      0       ;reserved


E410                    FIX     SCRENV



                **      SCRENV - Screen Handler Vector Table


E410  8DEF              DW      SOP-1   ;perform screen OPEN
E412  2DF2              DW      ECL-1   ;perform screen CLOSE (editor CLOSE:
E414  7FF1              DW      SGB-1   ;perform screen GET-BYTE
E416  A3F1              DW      SPB-1   ;perform screen PUT-BYTE
E418  1DF2              DW      SST-1   ;perform screen STATUS
E41A  AEF9              DW      SSP-1   ;perform screen SPECIAL
E41C  4C6EEF            JMP     SIN     ;initialize screen
E41F  00                DB      0       ;reserved


E420                    FIX     KEYBDV



                **      KEYBDV - Keyboard Handler Vector Table


E420  1DF2              DW      SST-1   ;perform keyboard OPEN (screen STAT:
E422  1DF2              DW      SST-1   ;perform keyboard CLOSE (screen STA:
E424  FCF2              DW      KGB-1   ;perform keyboard GET-BYTE
E426  2CF2              DW      ESP-1   ;perform keyboard SPECIAL (editor S:
E428  1DF2              DW      SST-1   ;perform keyboard STATUS (screen ST:
E42A  2CF2              DW      ESP-1   ;perform keyboard SPECIAL (editor S:
E42C  4C6EEF            JMP     SIN     ;initialize keyboard (initialize sc:
E42F  00                DB      0       ;reserved
```

```
E430                       FIX     PRINTV



                **         PRINTV - Printer Handler Vector Table


E430  C1FE                 DW      POP-1     ;perform printer OPEN
E432  06FF                 DW      PCL-1     ;perform printer CLOSE
E434  C0FE                 DW      PSP-1     ;perform printer SPECIAL
E436  CAFE                 DW      PPB-1     ;perform printer PUT-BYTE
E438  A2FE                 DW      PST-1     ;perform printer STATUS
E43A  C0FE                 DW      PSP-1     ;perform printer SPECIAL
E43C  4C99FE               JMP     PIN       ;initialize printer
E43F  00                   DB      0         ;reserved



E440                       FIX     CASETV



                **         CASETV - Cassette Handler Vector Table


E440  E5FC                 DW      COP-1     ;perform cassette OPEN
E442  CEFD                 DW      CCL-1     ;perform cassette CLOSE
E444  79FD                 DW      CGB-1     ;perform cassette GET-BYTE
E446  B3FD                 DW      CPB-1     ;perform cassette PUT-BYTE
E448  C8FD                 DW      CST-1     ;perform cassette STATUS
E44A  E4FC                 DW      CSP-1     ;perform cassette SPECIAL
E44C  4CDBFC               JMP     CIN       ;initialize cassette
E44F  00                   DB      0         ;reserved
```

```
E450                    **      Jump Vectors


E450                            FIX     DINITV
E450    4CA3C6                  JMP     IDIO      ;initialize DIO

E453                            FIX     DSKINV
E453    4CB3C6                  JMP     DIO       ;perform DIO

E456                            FIX     CIOV
E456    4CDFE4                  JMP     CIO       ;perform CIO

E459                            FIX     SIOV
E459    4C33C9                  JMP     PIO       ;perform PIO

E45C                            FIX     SETVBV
E45C    4C72C2                  JMP     SVP       ;set VBLANK parameters

E45F                            FIX     SYSVBV
E45F    4CE2C0                  JMP     IVNM      ;process immediate VBLANK NMI

E462                            FIX     XITVBV
E462    4C8AC2                  JMP     DVNM      ;process deferred VBLANK NMI

E465                            FIX     SIOINV
E465    4C5CE9                  JMP     ISIO      ;initialize SIO

E468                            FIX     SENDEV
E468    4C17EC                  JMP     ESS       ;enable SIO SEND

E46B                            FIX     INTINV
E46B    4C0CC0                  JMP     IIH       ;initialize interrupt handler

E46E                            FIX     CIOINV
E46E    4CC1E4                  JMP     ICIO      ;initialize CIO

E471                            FIX     BLKBDV
E471    4C23F2                  JMP     PPD       ;perform power-up display

E474                            FIX     WARMSV
E474    4C90C2                  JMP     PWS       ;perform warmstart

E477                            FIX     COLDSV
E477    4CC8C2                  JMP     PCS       ;perform coldstart

E47A                            FIX     RBLOKV
E47A    4C8DFD                  JMP     RCB       ;read cassette block

E47D                            FIX     CSOPIV
E47D    4CF7FC                  JMP     OCI       ;open cassette for input

E480                            FIX     PUPDIV
E480    4C23F2                  JMP     PPD       ;perform power-up display

E483                            FIX     SLFTSV
E483    4C0050                  JMP     STH       ;self-test hardware
```

```
E486                    FIX     PHENTV
E486  4C8CEE            JMP     PHE       ;perform peripheral handler entry

E489                    FIX     PHUNLV
E489  4C15E9            JMP     PHU       ;perform peripheral handler unlinki:

E48C                    FIX     PHINIV
E48C  4C98E6            JMP     PHI       ;perform peripheral handler initial:
```

```
E48F                    FIX     GPDVV


                  **    GPDVV - Generic Parallel Device Handler Vector Tabl:


E48F  90C9              DW      GOP-1    ;perform generic parallel device OP:
E491  95C9              DW      GCL-1    ;perform generic parallel device CL:
E493  9AC9              DW      GGB-1    ;perform generic parallel device GE:
E495  9FC9              DW      GPB-1    ;perform generic parallel device PU:
E497  A4C9              DW      GST-1    ;perform generic parallel device ST:
E499  A9C9              DW      GSP-1    ;perform generic parallel device SP:
E49B  4C0CC9            JMP     GIN      ;initialize generic parallel device
```

```
E49E                    FIX     $E4C0



                **      E4C0 - $E4C0 Patch
                *
                *       For compatibility with OS Revision B, return.

E4C0  60                RTS             ;return
```

```
E4C1                  **      ICIO - Initialize CIO
                      *
                      *       ENTRY   JSR     ICIO
                      *
                      *       MODS
                      *               Original Author Unknown
                      *               1. Bring closer to Coding Standard (object :
                      *                  R. K. Nordin 11/01/83

      = E4C1          ICIO    =       *               ;entry

                      ;       Initialize IOCB's.

E4C1  A200                    LDX     #0              ;index of first IOCB

E4C3  A9FF            ICIO1   LDA     #IOCFRE         ;IOCB free indicator
E4C5  9D4003                  STA     ICHID,X         ;set IOCB free
E4C8  A9DB                    LDA     #low [IIN-1]
E4CA  9D4603                  STA     ICPTL,X         ;initialize PUT-BYTE routin:
E4CD  A9E4                    LDA     #high [IIN-1]
E4CF  9D4703                  STA     ICPTH,X
E4D2  8A                      TXA                     ;index of current IOCB
E4D3  18                      CLC
E4D4  6910                    ADC     #IOCBSZ         ;add IOCB size
E4D6  AA                      TAX                     ;index of next IOCB
E4D7  C980                    CMP     #MAXIOC         ;index of first invalid IOC:
E4D9  90E8 ^E4C3              BCC     ICIO1           ;if not done

E4DB  60                      RTS                     ;return



                      **      IIN - Indicate IOCB Not Open Error
                      *
                      *       ENTRY   JSR     IIN
                      *
                      *       EXIT
                      *               Y = IOCB Not Open error code
                      *
                      *       MODS
                      *               Original Author Unknown
                      *               1. Bring closer to Coding Standard (object :
                      *                  R. K. Nordin 11/01/83

      = E4DC          IIN     =       *       ;entry
E4DC  A085                    LDY     #NOTOPN ;IOCB not open error
E4DE  60                      RTS             ;return
```

```
                        **      CIO - Central Input/Output
                        *
                        *       ENTRY   JSR     CIO
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


       = E4DF           CIO     =       *       ;entry

                        ;       Initialize.

E4DF   852F                     STA     CIOCHR  ;save possible output byte value
E4E1   862E                     STX     ICIDNO  ;save IOCB index

                        ;       Check IOCB index validity.

E4E3   8A                       TXA             ;IOCB index
E4E4   290F                     AND     #$0F    ;index modulo 16
E4E6   D004 ^E4EC               BNE     CIO1    ;if IOCB not multiple of 16, error

E4E8   E080                     CPX     #MAXIOC ;index of first invalid IOCB
E4EA   9005 ^E4F1               BCC     CIO2    ;if index within range

                        ;       Indicate Invalid IOCB Index error.

E4EC   A086             CIO1    LDY     #BADIOC ;invalid IOCB index error
E4EE   4C70E6                   JMP     SSC     ;set status and complete operation,:

                        ;       Move part of IOCB to zero page IOCB.

E4F1   A000             CIO2    LDY     #0              ;offset to first byte of pa:

E4F3   BD4003           CIO3    LDA     IOCB,X          ;byte of IOCB
E4F6   992000                   STA     IOCBAS,Y        ;byte of zero page IOCB
E4F9   E8                       INX
E4FA   C8                       INY
E4FB   C00C                     CPY     #ICSPRZ-IOCBAS  ;offset to first undesired :
E4FD   90F4 ^E4F3               BCC     CIO3            ;if not done

                        ;       Check for provisionally open IOCB.

E4FF   A520                     LDA     ICHIDZ  ;handler ID
E501   C97F                     CMP     #$7F    ;provisionally open indicator
E503   D015 ^E51A               BNE     PCC     ;if not provisionally open, perform:

                        ;       Check for CLOSE command.

E505   A522                     LDA     ICCOMZ  ;command
E507   C90C                     CMP     #CLOSE
E509   F071 ^E57C               BEQ     XCL     ;if CLOSE command

                        ;       Check handler load flag.

E50B   ADE902                   LDA     HNDLOD
```

```
E50E  D005 ^E515             BNE     LHO      ;if handler load desired

                    ;        Indicate nonexistent device error.

                    ;        JMP     IND      ;indicate nonexistent device error,:



                    **       IND - Indicate Nonexistent Device Error
                    *
                    *        ENTRY   JSR      IND
                    *
                    *        MODS
                    *                Original Author Unknown
                    *                1. Bring closer to Coding Standard (object :
                    *                   R. K. Nordin 11/01/83

      = E510        IND      =       *        ;entry
E510  A082                   LDY     #NONDEV  ;nonexistent device error

E512  4C70E6        IND1     JMP     SSC      ;set status and complete operation,:



                    **       LHO - Load Peripheral Handler for OPEN
                    *
                    *        ENTRY   JSR      LHO
                    *
                    *        MODS
                    *                Original Author Unknown
                    *                1. Bring closer to Coding Standard (object :
                    *                   R. K. Nordin 11/01/83

      = E515        LHO      =       *        ;entry
E515  2029CA                 JSR     PHL      ;load and initialize peripheral han:
E518  30F8 ^E512             BMI     IND1     ;if error

                    ;        JMP     PCC      ;perform CIO command, return
```

```
                        **      PCC - Perform CIO Command
                        *
                        *       ENTRY   JSR     PCC
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = E51A      PCC   =       *       ;entry

                        ;       Check command validity.

E51A  A084                      LDY     #NVALID         ;assume invalid code
E51C  A522                      LDA     ICCOMZ          ;command
E51E  C903                      CMP     #OPEN           ;first valid command
E520  9025 ^E547                BCC     XOP1            ;if command invalid

E522  A8                        TAY                     ;command

E523  C00E                      CPY     #SPECIL         ;last valid command
E525  9002 ^E529                BCC     PCC1            ;if valid

E527  A00E                      LDY     #SPECIL         ;substitute SPECIAL command

                        ;       Obtain vector offset.

E529  8417            PCC1      STY     ICCOMT          ;save command
E52B  B92AE7                    LDA     TCVO-3,Y        ;vector offset for command
E52E  F00F ^E53F                BEQ     XOP             ;if OPEN command, process

                        ;       Perform command.

E530  C902                      CMP     #2
E532  F048 ^E57C                BEQ     XCL             ;if CLOSE command, process

E534  C908                      CMP     #8
E536  B05F ^E597                BCS     XSS             ;if STATUS or SPECIAL comma:

E538  C904                      CMP     #4
E53A  F076 ^E5B2                BEQ     XGT             ;if GET command, process

E53C  4C1EE6                    JMP     XPT             ;process PUT command, proce:
```

```
                        **      XOP - Execute OPEN Command
                        *
                        *       ENTRY   JSR     XOP
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


        = E53F          XOP     =       *       ;entry

                        ;       Check IOCB free.

E53F    A520                    LDA     ICHIDZ  ;handler ID
E541    C9FF                    CMP     #IOCFRE ;IOCB free indicator
E543    F005 ^E54A              BEQ     XOP2    ;if IOCB free

                        ;       Process error.

E545    A081                    LDY     #PRVOPN ;IOCB previously open error

E547    4C70E6          XOP1    JMP     SSC     ;set status and complete operation,:

                        ;       Check handler load.

E54A    ADE902          XOP2    LDA     HNDLOD
E54D    D027 ^E576              BNE     PPO     ;if user wants unconditional poll

                        ;       Search handler table.

E54F    20FFE6                  JSR     SHT     ;search handler table
E552    B022 ^E576              BCS     PPO     ;if not found, poll

                        ;       Initialize status.

E554    A900                    LDA     #0
E556    8DEA02                  STA     DVSTAT  ;clear status
E559    8DEB02                  STA     DVSTAT+1

                        ;       Initialize IOCB.

                        ;       JMP     IIO     ;initialize IOCB for OPEN, return
```

```
                        **      IIO - Initialize IOCB for OPEN
                        *
                        *       ENTRY   JSR     IIO
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = E55C          IIO     =       *       ;entry

                        ;       Compute handler entry point.

E55C  2095E6                    JSR     CEP     ;compute handler entry point
E55F  B0E6 ^E547                BCS     XOP1    ;if error

                        ;       Execute command.

E561  20EAE6                    JSR     EHC     ;execute handler command

                        ;       Set PUT-BYTE routine address in IOCB.

E564  A90B                      LDA     #PUTCHR
E566  8517                      STA     ICCOMT  ;command
E568  2095E6                    JSR     CEP     ;compute handler entry point
E56B  A52C                      LDA     ICSPRZ  ;PUT-BYTE routine address
E56D  8526                      STA     ICPTLZ  ;IOCB PUT-BYTE routine address
E56F  A52D                      LDA     ICSPRZ+1
E571  8527                      STA     ICPTHZ
E573  4C72E6                    JMP     CCO     ;complete CIO operation, return



                        **      PPO - Poll Peripheral for OPEN
                        *
                        *       ENTRY   JSR     PPO
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = E576          PPO     =       *       ;entry
E576  20F9EE                    JSR     PHO     ;poll
E579  4C70E6                    JMP     SSC     ;set status and complete operation,:
```

```
                    **      XCL - Execute CLOSE Command
                    *
                    *       ENTRY   JSR     XCL
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

     = E57C         XCL     =       *               ;entry

                    ;       Initialize.

E57C  A001                  LDY     #SUCCES         ;assume success
E57E  8423                  STY     ICSTAZ          ;status
E580  2095E6                JSR     CEP             ;compute handler entry poin:
E583  B003 ^E588            BCS     XCL1            ;if error

                    ;       Execute command.

E585  20EAE6                JSR     EHC             ;execute handler command

                    ;       Close IOCB.

E588  A9FF          XCL1    LDA     #IOCFRE         ;IOCB free indicator
E58A  8520                  STA     ICHIDZ          ;indicate IOCB free
E58C  A9E4                  LDA     #high [IIN-1]
E58E  8527                  STA     ICPTHZ          ;reset initial PUT-BYTE rou:
E590  A9D8                  LDA     #low [IIN-1]
E592  8526                  STA     ICPTLZ
E594  4C72E6                JMP     CCO             ;complete CIO operation, re:



                    **      XSS - Execute STATUS and SPECIAL Commands
                    *
                    *       ENTRY   JSR     XSS
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

     = E597         XSS     =       *       ;entry

                    ;       Check IOCB free.

E597  A520                  LDA     ICHIDZ  ;handler ID
E599  C9FF                  CMP     #IOCFRE
E59B  D005 ^E5A2            BNE     XSS1    ;if IOCB not free

                    ;       Open IOCB.

E59D  20FFE6                JSR     SHT     ;search handler table
```

```
E5A0  B0A5 ^E547          BCS     XOP1     ;if error

                  ;       Execute command.

E5A2  2095E6      XSS1    JSR     CEP      ;compute handler entry point
E5A5  20EAE6              JSR     EHC      ;execute handler command

                  ;       Restore handler ID, in case IOCB implicitly opened.

E5A8  A62E                LDX     ICIDNO   ;IOCB index
E5AA  BD4003              LDA     ICHID,X  ;original handler ID
E5AD  8520                STA     ICHIDZ   ;restore zero page handler ID
E5AF  4C72E6              JMP     CCO      ;complete CIO operation, return


                  **      XGT - Execute GET Command
                  *
                  *       ENTRY   JSR     XGT
                  *
                  *       MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83

      = E5B2      XGT     =       *        ;entry

                  ;       Check GET validity.

E5B2  A522                LDA     ICCOMZ   ;command
E5B4  252A                AND     ICAX1Z
E5B6  D005 ^E5BD          BNE     XGT2     ;if GET command valid

                  ;       Process error.

E5B8  A083                LDY     #WRONLY  ;IOCB opened for write only error

E5BA  4C70E6      XGT1    JMP     SSC      ;set status and complete operation,:

                  ;       Compute and check handler entry point.

E5BD  2095E6      XGT2    JSR     CEP      ;compute handler entry point
E5C0  B0F8 ^E5BA          BCS     XGT1     ;if error

                  ;       Check buffer length.

E5C2  A528                LDA     ICBLLZ           ;buffer length
E5C4  0529                ORA     ICBLLZ+1
E5C6  D008 ^E5D0          BNE     XGT3             ;if buffer length non-zero

                  ;       Get byte.

E5C8  20EAE6              JSR     EHC      ;execute handler command
E5CB  852F                STA     CIOCHR   ;data
E5CD  4C72E6              JMP     CCO      ;complete CIO operation, return
```

```
                        ;       Fill buffer.

E5D0  20EAE6     XGT3   JSR     EHC             ;execute handler command
E5D3  852F              STA     CIOCHR          ;data
E5D5  3041 ^E618        BMI     XGT7            ;if error, end transfer

E5D7  A000              LDY     #0
E5D9  9124              STA     (ICBALZ),Y      ;byte of buffer
E5DB  20D1E6            JSR     IBP             ;increment buffer pointer
E5DE  A522              LDA     ICCOMZ          ;command
E5E0  2902              AND     #$02
E5E2  D00C ^E5F0        BNE     XGT4            ;if GET RECORD command

                        ;       Check for EOL.

E5E4  A52F              LDA     CIOCHR  ;data
E5E6  C99B              CMP     #EOL
E5E8  D006 ^E5F0        BNE     XGT4    ;if not EOL

                        ;       Process EOL.

E5EA  20B8E6            JSR     DBL     ;decrement buffer length
E5ED  4C18E6            JMP     XGT7    ;clean up

                        ;       Check buffer full.

E5F0  20B8E6     XGT4   JSR     DBL     ;decrement buffer length
E5F3  D0DB ^E5D0        BNE     XGT3    ;if buffer not full, continue

                        ;       Check command.

E5F5  A522              LDA     ICCOMZ  ;command
E5F7  2902              AND     #$02
E5F9  D01D ^E618        BNE     XGT7    ;if GET CHARACTER command, clean up

                        ;       Process GET RECORD.

E5FB  20EAE6     XGT5   JSR     EHC     ;execute handler command
E5FE  852F              STA     CIOCHR  ;data
E600  300A ^E60C        BMI     XGT6    ;if error

                        ;       Check for EOL.

E602  A52F              LDA     CIOCHR  ;data
E604  C99B              CMP     #EOL
E606  D0F3 ^E5FB        BNE     XGT5    ;if not EOL, continue

                        ;       Process end of record.

E608  A989              LDA     #TRNRCD ;truncated record error
E60A  8523              STA     ICSTAZ  ;status

                        ;       Process error.

E60C  20C8E6     XGT6   JSR     DBP             ;decrement buffer pointer
E60F  A000              LDY     #0
```

```
E611  A99B                LDA    #EOL
E613  9124                STA    (ICBALZ),Y       ;set EOL in buffer
E615  20D1E6              JSR    IBP              ;increment buffer pointer

                    ;     Clean up.

E618  2008E6        XGT7  JSR    SFL     ;set final buffer length
E61B  4C72E6              JMP    CCO     ;complete CIO operation, return



                    **    XPT - Execute PUT Command
                    *
                    *     ENTRY   JSR     XPT
                    *
                    *     MODS
                    *             Original Author Unknown
                    *             1. Bring closer to Coding Standard (object :
                    *                R. K. Nordin 11/01/83


      = E61E        XPT   =       *       ;entry

                    ;     Check PUT validity.

E61E  A522                LDA    ICCOMZ  ;command
E620  252A                AND    ICAX1Z
E622  D005 ^E629          BNE    XPT2    ;if PUT command valid

                    ;     Process error.

E624  A087                LDY    #RDONLY ;IOCB opened for read only error

E626  4C70E6        XPT1  JMP    SSC     ;set status and complete operation,:

                    ;     Compute and check handler entry point.

E629  2095E6        XPT2  JSR    CEP     ;compute handler entry point
E62C  B0F8 ^E626          BCS    XPT1    ;if error

                    ;     Check buffer length.

E62E  A528                LDA    ICBLLZ  ;buffer length
E630  0529                ORA    ICBLLZ+1
E632  D006 ^E63A          BNE    XPT3    ;if buffer length non-zero

                    ;     Put byte.

E634  A52F                LDA    CIOCHR  ;data
E636  E628                INC    ICBLLZ  ;set buffer length to 1
E638  D006 ^E640          BNE    XPT4    ;transfer one byte

                    ;     Transfer data from buffer to handler.

E63A  A000          XPT3  LDY    #0
E63C  B124                LDA    (ICBALZ),Y       ;byte from buffer
```

```
E63E  852F                STA     CIOCHR          ;data

E640  20EAE6      XPT4    JSR     EHC             ;execute handler command
E643  08                  PHP                     ;save status
E644  20D1E6              JSR     IBP             ;increment buffer pointer
E647  20BBE6              JSR     DBL             ;decrement buffer length
E64A  28                  PLP                     ;status
E64B  301D ^E66A          BMI     XPT6            ;if error

                  ;       Check command.

E64D  A522                LDA     ICCOMZ  ;command
E64F  2902                AND     #$02
E651  D006 ^E659          BNE     XPT5    ;if PUT RECORD command

                  ;       Check for EOL.

E653  A52F                LDA     CIOCHR  ;data
E655  C99B                CMP     #EOL
E657  F011 ^E66A          BEQ     XPT6    ;if EOL, clean up

                  ;       Check for buffer empty.

E659  A528        XPT5    LDA     ICBLLZ          ;buffer length
E65B  0529                ORA     ICBLLZ+1
E65D  D0DB ^E63A          BNE     XPT3            ;if buffer not empty, conti:

                  ;       Check command.

E65F  A522                LDA     ICCOMZ  ;command
E661  2902                AND     #$02
E663  D005 ^E66A          BNE     XPT6    ;if PUT CHARACTER command

                  ;       Write EOL.

E665  A99B                LDA     #EOL
E667  20EAE6              JSR     EHC     ;execute handler command

                  ;       Clean up.

E66A  20D8E6      XPT6    JSR     SFL     ;set final buffer length
E66D  4C72E6              JMP     CCO     ;complete CIO operation, return
```

```
                        **      SSC - Set Status and Complete Operation
                        *
                        *       ENTRY   JSR     SSC
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = E670            SSC     =       *       ;entry
E670  8423                      STY     ICSTAZ  ;status
                        ;       JMP     CCO     ;complete CIO operation, return



                        **      CCO - Complete CIO Operation
                        *
                        *       ENTRY   JSR     CCO
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = E672            CCO     =       *                       ;entry

                        ;       Initialize.

E672  A42E                      LDY     ICIDNO                  ;IOCB index

                        ;       Restore buffer pointer.

E674  B94403                    LDA     ICBAL,Y
E677  8524                      STA     ICBALZ                  ;restore buffer pointer
E679  B94503                    LDA     ICBAH,Y
E67C  8525                      STA     ICBAHZ

                        ;       Move part of zero page IOCB to IOCB.

E67E  A200                      LDX     #0                      ;first byte of zero page IO:
E680  8EE902                    STX     HNDLOD

E683  B520              CCO1    LDA     IOCBAS,X                ;byte of zero page IOCB
E685  994003                    STA     IOCB,Y                  ;byte of IOCB
E688  E8                        INX
E689  C8                        INY
E68A  E00C                      CPX     #ICSPRZ-IOCBAS          ;offset to first undesired :
E68C  90F5 ^E683                BCC     CCO1                    ;if not done

                        ;       Restore A, X and Y.

E68E  A52F                      LDA     CIOCHR                  ;data
E690  A62E                      LDX     ICIDNO                  ;IOCB index
E692  A423                      LDY     ICSTAZ                  ;status
```

```
E694  60                    RTS                         ;return



                    **      CEP - Compute Handler Entry Point
                    *
                    *       ENTRY   JSR     CEP
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = E695        CEP     =       *                   ;entry

                    ;       Check handler ID validity.

E695  A420                  LDY     ICHIDZ              ;handler ID
E697  C022                  CPY     #MAXDEV+1           ;first invalid ID
E699  9004 ^E69F            BCC     CEP1                ;if handler ID within range

                    ;       Process error.

E69B  A085                  LDY     #NOTOPN             ;IOCB not open error
E69D  B01B ^E6BA            BCS     CEP2                ;return

                    ;       Compute entry point.

E69F  B91B03        CEP1    LDA     HATABS+1,Y          ;low address
E6A2  852C                  STA     ICSPRZ
E6A4  B91C03                LDA     HATABS+2,Y          ;high address
E6A7  852D                  STA     ICSPRZ+1
E6A9  A417                  LDY     ICCOMT              ;command
E6AB  B92AE7                LDA     TCVO-3,Y            ;vector offset for command
E6AE  A8                    TAY
E6AF  B12C                  LDA     (ICSPRZ),Y          ;low vector address
E6B1  AA                    TAX                         ;low vector address
E6B2  C8                    INY
E6B3  B12C                  LDA     (ICSPRZ),Y          ;high vector address
E6B5  852D                  STA     ICSPRZ+1            ;set high address
E6B7  862C                  STX     ICSPRZ              ;set low address
E6B9  18                    CLC                         ;indicate success

                    ;       Exit.

E6BA  60            CEP2    RTS                         ;return
```

```
                        **      DBL - Decrement Buffer Length
                        *
                        *       ENTRY   JSR     DBL
                        *
                        *       EXIT
                        *               Z set if buffer length = 0
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = E6BB      DBL     =       *               ;entry
E6BB    A528                LDA     ICBLLZ          ;low buffer length
E6BD    D002 ^E6C1          BNE     DBL1            ;if low buffer length non-z:

E6BF    C629                DEC     ICBLLZ+1        ;decrement high buffer leng:

E6C1    C628        DBL1    DEC     ICBLLZ          ;decrement low buffer lengt:
E6C3    A528                LDA     ICBLLZ
E6C5    0529                ORA     ICBLLZ+1        ;indicate buffer length sta:
E6C7    60                  RTS                     ;return


                        **      DBP - Decrement Buffer Pointer
                        *
                        *       ENTRY   JSR     DBP
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = E6C8      DBP     =       *               ;entry
E6C8    A524                LDA     ICBALZ          ;low buffer address
E6CA    D002 ^E6CE          BNE     DBP1            ;if low buffer address non-:

E6CC    C625                DEC     ICBALZ+1        ;decrement high buffer addr:

E6CE    C624        DBP1    DEC     ICBALZ          ;decrement low buffer addre:
E6D0    60                  RTS                     ;return
```

```
                        **      IBP - Increment Buffer Pointer
                        *
                        *       ENTRY   JSR     IBP
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = E6D1             IBP     =       *                       ;entry
E6D1 E624                       INC     ICBALZ                  ;increment low buffer addre:
E6D3 D002 ^E6D7                 BNE     IBP1                    ;if low buffer address non-:

E6D5 E625                       INC     ICBALZ+1                ;increment high buffer addr:

E6D7 60                 IBP1    RTS                             ;return



                        **      SFL - Set Final Buffer Length
                        *
                        *       ENTRY   JSR     SFL
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = E6D8             SFL     =       *                       ;entry
E6D8 A62E                       LDX     ICIDNO                  ;IOCB index
E6DA 38                         SEC
E6DB BD4803                     LDA     ICBLL,X                 ;initial length
E6DE E528                       SBC     ICBLLZ                  ;subtract byte count
E6E0 8528                       STA     ICBLLZ                  ;update length
E6E2 BD4903                     LDA     ICBLH,X
E6E5 E529                       SBC     ICBLLZ+1
E6E7 8529                       STA     ICBLHZ
E6E9 60                         RTS                             ;return



                        **      EHC - Execute Handler Command
                        *
                        *       ENTRY   JSR     EHC
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = E6EA             EHC     =       *                       ;entry
E6EA A092                       LDY     #FNCNOT                 ;assume function not define:
```

```
E6EC  20F4E6              JSR     IDH              ;invoke device handler
E6EF  8423                STY     ICSTAZ           ;status
E6F1  C000                CPY     #0               ;set N accordingly
E6F3  60                  RTS                      ;return


                  **      IDH - Invoke Device Handler
                  *
                  *       ENTRY   JSR     IDH
                  *
                  *       MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83

      = E6F4      IDH     =       *                ;entry
E6F4  AA                  TAX                      ;save A
E6F5  A52D                LDA     ICSPRZ+1         ;high vector
E6F7  48                  PHA                      ;put high vector on stack
E6F8  A52C                LDA     ICSPRZ           ;low vector
E6FA  48                  PHA                      ;put low vector on stack
E6FB  8A                  TXA                      ;restore A
E6FC  A62E                LDX     ICIDNO           ;IOCB index
E6FE  60                  RTS                      ;invoke handler (address on:


                  **      SHT - Search Handler Table
                  *
                  *       ENTRY   JSR     SHT
                  *
                  *       MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83

      = E6FF      SHT     =       *       ;entry

                  ;       Set device number.

E6FF  38                  SEC
E700  A001                LDY     #1
E702  B124                LDA     (ICBALZ),Y       ;device number
E704  E931                SBC     #'1'
E706  3004 ^E70C          BMI     SHT1             ;if number less than  "1"

E708  C909                CMP     #'9'-'1'+1
E70A  9002 ^E70E          BCC     SHT2             ;if number in range "1" to :

E70C  A900        SHT1    LDA     #0               ;substitute device number ":

E70E  8521        SHT2    STA     ICDNOZ           ;device number (0 through 8:
```

```
E710  E621                   INC     ICDNOZ           ;adjust number to range 1 t:

                     ;       Find device handler.

E712  A000                   LDY     #0               ;offset to device code
E714  B124                   LDA     (ICBALZ),Y       ;device code
                     ;       JMP     FDH              ;find device handler, retur:



                     **      FDH - Find Device Handler
                     *
                     *       ENTRY   JSR     FDH
                     *
                     *       MODS
                     *               Original Author Unknown
                     *               1. Bring closer to Coding Standard (object :
                     *                  R. K. Nordin 11/01/83

      = E716         FDH     =       *                ;entry

                     ;       Check device code.

E716  F00C ^E724             BEQ     FDH2             ;if device code null

                     ;       Search handler table for device.

E718  A021                   LDY     #MAXDEV          ;offset to last possible en:

E71A  D91A03         FDH1    CMP     HATABS,Y         ;device code from table
E71D  F009 ^E728             BEQ     FDH3             ;if device found

E71F  88                     DEY
E720  88                     DEY
E721  88                     DEY
E722  10F6 ^E71A             BPL     FDH1             ;if not done

                     ;       Process device not found.

E724  A082           FDH2    LDY     #NONDEV          ;nonexistent device error
E726  38                     SEC                      ;indicate error
E727  60                     RTS                      ;return

                     ;       Set handler ID.

E728  98             FDH3    TYA                      ;offset to device code in t:
E729  8520                   STA     ICHIDZ           ;set handler ID
E72B  18                     CLC                      ;indicate no error
E72C  60                     RTS                      ;return
```

```
                        **      TCVO - Table of Command Vector Offsets
                        *
                        *       Entry n is the vector offset for command n+3.

E72D  00                TCVO    DB      0       ;3 - open
E72E  04                        DB      4       ;4
E72F  04                        DB      4       ;5 - get record
E730  04                        DB      4       ;6
E731  04                        DB      4       ;7 - get byte(s)
E732  06                        DB      6       ;8
E733  06                        DB      6       ;9 - put record
E734  06                        DB      6       ;10
E735  06                        DB      6       ;11 - put byte(s)
E736  02                        DB      2       ;12 - close
E737  08                        DB      8       ;13 - status
E738  0A                        DB      10      ;14 - special
```

```
E739                    **      PHR - Perform Peripheral Handler Loading Initializa:
                        *
                        *       * Performs Power-up Polling, with Handler loading a:
                        *       and Initialization;
                        *       * Performs System Reset Re-initialization of all ha:
                        *
                        *       Input Parameters:
                        *       WARMST (used to distinguish Cold and Warm Start).
                        *
                        *       Output Parameters:
                        *       None.
                        *
                        *       Modified:
                        *       Registers are not saved;
                        *       All kinds of side effects when any handler is loade:
                        *       (potentially MEMLO, DVSTAT thru DVSTAT+3, the DCB,
                        *       CHLINK, ZCHAIN, TEMP1, TEMP2, TEMP3.    This list m:
                        *       not be complete.).
                        *
                        *       ENTRY   JSR     PHR
                        *
                        *       MODS
                        *               R. S. Scheiman  04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = E739            PHR     =       *       ;entry

                        ;       Check for coldstart.

E739  A508                      LDA     WARMST          ;warmstart flag
E73B  F025 ^E762                BEQ     PHR2            ;if coldstart

                        ;       Process warmstart.

E73D  A9E9                      LDA     #low [CHLINK-18]
E73F  854A                      STA     ZCHAIN
E741  A903                      LDA     #high [CHLINK-18]
E743  854B                      STA     ZCHAIN+1

                        ;       Check next link.

E745  A012              PHR1    LDY     #18             ;offset to link
E747  18                        CLC
E748  B14A                      LDA     (ZCHAIN),Y      ;low link
E74A  AA                        TAX
E74B  C8                        INY
E74C  714A                      ADC     (ZCHAIN),Y      ;high link
E74E  F026 ^E776                BEQ     PHR4            ;if forward link null

                        ;       Re-initialize peripheral handler.

E750  B14A                      LDA     (ZCHAIN),Y      ;high link
E752  854B                      STA     ZCHAIN+1
E754  864A                      STX     ZCHAIN
E756  2056CB                    JSR     CLT             ;checksum linkage table
```

```
E759  D01B ^E776             BNE     PHR4              ;if checksum bad

E75B  2094E8                 JSR     PHW               ;re-initialize peripheral h:
E75E  B016 ^E776             BCS     PHR4              ;if error

                     ;       Continue with next handler.

E760  90E3 ^E745             BCC     PHR1              ;continue with next handler

                     ;       Process coldstart.

E762  A900           PHR2    LDA     #0
E764  8DFB03                 STA     CHLINK            ;clear chain link
E767  8DFC03                 STA     CHLINK+1
E76A  A94F                   LDA     #$4F              ;send POLL RESET poll
E76C  D02D ^E79B             BNE     PHR7

                     ;       Perform type 3 poll.

E76E  A900           PHR3    LDA     #0
E770  A8                     TAY
E771  20BEE7                 JSR     PHP
E774  1001 ^E777             BPL     PHR5              ;if poll answered

                     ;       Exit.

E776  60             PHR4    RTS                       ;return

                     ;       Process answered poll.

E777  18             PHR5    CLC
E778  ADE702                 LDA     MEMLO
E77B  6DEA02                 ADC     DVSTAT
E77E  8D1203                 STA     TEMP1
E781  ADE802                 LDA     MEMLO+1
E784  6DEB02                 ADC     DVSTAT+1
E787  8D1303                 STA     TEMP1+1           ;(TEMP2 := MEMLO + handler :
E78A  38                     SEC
E78B  ADE502                 LDA     MEMTOP
E78E  ED1203                 SBC     TEMP1
E791  ADE602                 LDA     MEMTOP+1
E794  ED1303                 SBC     TEMP1+1           ;(subtract MEMTOP)
E797  B009 ^E7A2             BCS     PHR8              ;if room to load

                     ;       Prepare for another poll.

E799  A94E           PHR6    LDA     #$4E              ;following any load or init:
                                                       ;prepare for another Type 3:
                                                       ;sending a "special" load c:
                                                       ;serial port.

                     ;       Poll.

E79B  A8             PHR7    TAY                       ;Send either "special" load:
E79C  20BEE7                 JSR     PHP
E79F  4C6EE7                 JMP     PHR3              ;go poll again
```

```
                        ;       Load peripheral handler.

E7A2  ADEC02            PHR8    LDA     DVSTAT+2        ;call the loader
E7A5  AEE702                    LDX     MEMLO
E7A8  8EEC02                    STX     DVSTAT+2        ;(Parameter = load address)
E7AB  AEE802                    LDX     MEMLO+1
E7AE  8EED02                    STX     DVSTAT+3
E7B1  20DEE7                    JSR     LPH             ;load peripheral handler
E7B4  30E3 ^E799                BMI     PHR6            ;if load error, poll again

E7B6  38                        SEC                     ;Call for initialize new ha:
E7B7  209EE8                    JSR     PHC             ;(Parameter = add size to M:
E7BA  B0DD ^E799                BCS     PHR6            ;if init error, poll again

E7BC  90B0 ^E76E                BCC     PHR3            ;poll again normally



                        **      PHP - Perform Poll
                        *
                        *       Polling subroutine calls SIO for Type 3 or 4 Poll.
                        *
                        *       Input Parameters:
                        *       A       Value for AUX1
                        *       Y       Value for AUX2
                        *
                        *       Output Parameters:
                        *       Y       SIO status from poll
                        *       DVSTAT: Device minimum size (low), if poll answered
                        *       DVSTAT+1: Device minimum size (high), if poll answe:
                        *       DVSTAT+2: Device address for loading, if poll answe:
                        *       DVSTAT+3: Device version number, if poll answered
                        *
                        *       Modified:
                        *       The registers are not saved.
                        *
                        *       Subroutines called:
                        *       SIO (performs poll and returns to PHP's caller).
                        *
                        *       ENTRY   JSR     PHP
                        *
                        *       MODS
                        *               R. S. Scheiman  04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = E7BE            PHP     =       *               ;entry

                        ;       Initialize.

E7BE  48                        PHA                     ;save parameter

                        ;       Set up DCB.

E7BF  A209                      LDX     #PHPAL-1        ;offset to last byte of DCB:
```

```
E7C1  BDD4E7      PHP1    LDA     PHPA,X          ;byte of DCB data
E7C4  9D0003              STA     DCB,X           ;byte of DCB
E7C7  CA                  DEX
E7C8  10F7 ^E7C1          BPL     PHP1            ;if not done

                  ;       Set parameters in DCB auxiliary bytes.

E7CA  8C0B03              STY     DAUX2
E7CD  68                  PLA
E7CE  8D0A03              STA     DAUX1

                  ;       Perform SIO.

E7D1  4C59E4              JMP     SIOV            ;vector to SIO, return


                  ;       DCB Poll Request Data

E7D4  4F          PHPA    DB      $4F     ;device bus ID
E7D5  01                  DB      1       ;unit number
E7D6  40                  DB      '@'     ;type 3 or 4 poll command
E7D7  40                  DB      $40     ;I/O direction
E7D8  EA02                DW      DVSTAT  ;buffer
E7DA  1E                  DB      30      ;timeout
E7DB  00                  DB      0
E7DC  0400                DW      4       ;buffer length

      = 000A      PHPAL   =       *-PHPA  ;length


                  **      LPH - Load Peripheral Handler
                  *
                  *       This subroutine calls the relocating loader to load
                  *       a handler from a peripheral.
                  *
                  *       Input parameters:
                  *       A       Peripheral serial address for load;
                  *       DVSTAT+2: Load address (low)
                  *       DVSTAT+3: Load address (high)
                  *
                  *       Output parameters:
                  *       From the relocating loader.
                  *
                  *       Modified:
                  *       TEMP1, TEMP2, TEMP3,
                  *       DVSTAT+3, DVSTAT+3 (forced even),
                  *       Relocating loader variables and parameters,
                  *       Registers not saved.
                  *
                  *       Subroutines called:
                  *       RLR (relocating loader).
                  *
                  *       ENTRY   JSR     LPH
                  *
```

```
                        *       MODS
                        *               R. S. Scheiman  04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = E7DE            LPH     =       *       ;entry

                        ;       Initialize.

E7DE  8D1303                    STA     TEMP2   ;save peripheral address
E7E1  A200                      LDX     #0
E7E3  8E1203                    STX     TEMP1   ;set starting block number
E7E6  CA                        DEX
E7E7  8E1503                    STX     TEMP3   ;set starting byte number

                        ;       Ensure load address even.

E7EA  ADEC02                    LDA     DVSTAT+2        ;low load address
E7ED  6A                        ROR     A
E7EE  9008 ^E7F8                BCC     LPH1            ;if even

E7F0  EEEC02                    INC     DVSTAT+2        ;increment low load address
E7F3  D003 ^E7F8                BNE     LPH1            ;if no carry

E7F5  EEED02                    INC     DVSTAT+3        ;increment high load addres:

                        ;       Set up relocating loader parameters.

E7F8  ADEC02            LPH1    LDA     DVSTAT+2        ;load address
E7FB  8DD102                    STA     LOADAD
E7FE  ADED02                    LDA     DVSTAT+3
E801  8DD202                    STA     LOADAD+1
E804  A916                      LDA     #low PHG        ;get-byte routine address
E806  8DCF02                    STA     GBYTEA
E809  A9E8                      LDA     #high PHG
E80B  8DD002                    STA     GBYTEA+1
E80E  A980                      LDA     #$80            ;loader page zero load addr:
E810  8DD302                    STA     ZLOADA

                        ;       Reloacte routine.

E813  4C45C7                    JMP     RLR             ;relocate routine, return
```

```
                        **      PHG - Perform Peripheral Handler GET-BYTE
                        *
                        *       Get a byte subroutine for relocating loader passes
                        *       bytes from peripheral to relocating loader via
                        *       cassette buffer. Calls GNL each time new
                        *       buffer is needed.
                        *
                        *       Input parameters:
                        *       TEMP1: Next block number;
                        *       TEMP2: Peripheral address (for GNL);
                        *       TEMP3: Next byte number (index to CASBUF).
                        *
                        *       Output parameters (for relocating loader):
                        *       Carry bit on indicates error;
                        *       A       Next byte, if no error.
                        *
                        *       Modified:
                        *       Cassette buffer CASBUF;
                        *       TEMP3;
                        *       X, Y not saved.
                        *
                        *       Subroutines called:
                        *       GNL, which calls SIO to get load records.
                        *
                        *       ENTRY   JSR     PHG
                        *
                        *       MODS
                        *               R. S. Scheiman  04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *.                 R. K. Nordin 11/01/83

      = E816            PHG     =       *                       ;entry

                        ;       Check for another byte in buffer.

E816  AE1503                    LDX     TEMP3
E819  E8                        INX
E81A  8E1503                    STX     TEMP3
E81D  F008 ^E827                BEQ     PHG2                    ;if empty, load next block

                        ;       Retrieve next byte.

E81F  AE1503            PHG1    LDX     TEMP3
E822  BD7D03                    LDA     CASBUF-$80,X    ;byte
E825  18                        CLC                             ;indicate no error
E826  60                        RTS                             ;return

                        ;       Load next block and retrieve next byte.

E827  A980              PHG2    LDA     #-128   ;offset to first byte
E829  8D1503                    STA     TEMP3
E82C  2033E8                    JSR     GNL     ;get next load block
E82F  10EE ^E81F                BPL     PHG1    ;if no error, retrieve next byte

                        ;       Process error.
```

```
E831  38                        SEC                   ;indicate error
E832  60                        RTS                   ;return



                        **      GNL - Get Next Load Block
                        *
                        *       Subroutine to get a load block from the peripheral.
                        *
                        *       Input parameters:
                        *       TEMP1: Block number.
                        *
                        *       Output parameters:
                        *       Negative bit is set by SIO if I/O error occurs.
                        *
                        *       Modified:
                        *       TEMP1;
                        *       the DCB (SIO);
                        *       Registers not saved.
                        *
                        *       Subroutines called:
                        *       SIO.
                        *
                        *       ENTRY   JSR     GNL
                        *
                        *       MODS
                        *               R. S. Scheiman  04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = E833            GNL     =       *                   ;entry

                        ;       Set up DCB.

E833  A20B                      LDX     #GNLAL-1            ;offset to last DCB data by:

E835  BD51E8            GNL1    LDA     GNLA,X              ;byte of DCB data
E838  9D0003                    STA     DCB,X               ;byte of DCB
E83B  CA                        DEX
E83C  10F7 ^E835                BPL     GNL1                ;if not done

                        ;       Set DCB parameters.

E83E  AE1203                    LDX     TEMP1               ;block number
E841  8E0A03                    STX     DAUX1               ;auxiliary 1
E844  E8                        INX
E845  8E1203                    STX     TEMP1               ;next block number
E848  AD1303                    LDA     TEMP2               ;device address
E84B  8D0003                    STA     DDEVIC              ;device bus ID

                        ;       Perform SIO.

E84E  4C59E4                    JMP     SIOV                ;vector to SIO, return
```

```
                        ;       DCB Data

E851  00                GNLA    DB      $00     ;dummy device bus ID
E852  01                        DB      1       ;dummy unit number
E853  26                        DB      '&'     ;load command
E854  40                        DB      $40     ;I/O direction
E855  FD03                      DW      CASBUF  ;buffer
E857  1E                        DB      30      ;timeout
E858  00                        DB      0
E859  8000                      DW      128     ;buffer length
E85B  00                        DB      0       ;auxiliary 1
E85C  00                        DB      0       ;auxiliary 2

      = 000C            GNLAL   =       *-GNLA  ;length


                        **      SHC - Search Handler Chain
                        *
                        *       Forward chain search searches for pointer to handle:
                        *       table whose address matches caller's parameter. If :
                        *       parameter is zero, this routine looks for the point:
                        *       the final linkage table since this table's forward :
                        *       is zero (null).
                        *
                        *       Input parameters:
                        *       A       Linkage table address to match (High)
                        *       Y       Linkage table address to match (Low)
                        *
                        *       Output parameters:
                        *       ZCHAIN points to linkage table whose forward pointe:
                        *               contains the match (if match is found);
                        *               if the match is found just following the li:
                        *               chain base CHLINK, then ZCHAIN points to CH:
                        *               minus 18;
                        *       If match successful, A (High) and X (Low) contain
                        *               matched address (equiv. to A and Y parms.);
                        *       Carry bit is set to indicate no match or checksum v:
                        *               along the chain. [Note: the linkage table p:
                        *               to by ZCHAIN upon return is not checksum ch:
                        *
                        *       Modified:
                        *       TEMP1, TEMP2, ZCHAIN;
                        *       The registers are not saved.
                        *
                        *       Subroutines called:
                        *       CLT.
                        *
                        *       ENTRY   JSR     SHC
                        *
                        *       MODS
                        *               R. S. Scheiman  04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83
```

```
        = E85D         SHC        =       *          ;entry

                       ;          Initialize.

E85D    8C1203                    STY     TEMP1
E860    8D1303                    STA     TEMP1+1
E863    A9E9                      LDA     #low [CHLINK-18]
E865    854A                      STA     ZCHAIN               ;start ZCHAIN at proper off:
E867    A903                      LDA     #high [CHLINK-18]
E869    854B                      STA     ZCHAIN+1

                       ;          Check for match.

E86B    A012           SHC1       LDY     #18
E86D    B14A                      LDA     (ZCHAIN),Y
E86F    AA                        TAX                          ;low chain pointer
E870    C8                        INY
E871    B14A                      LDA     (ZCHAIN),Y           ;high chain pointer
E873    CD1303                    CMP     TEMP2                ;check for match with param:
E876    D007 ^E87F                BNE     SHC2                 ;if no match

E878    EC1203                    CPX     TEMP1
E87B    D002 ^E87F                BNE     SHC2                 ;if no match

                       ;          Exit.

E87D    18                        CLC                ;indicate match
E87E    60                        RTS                ;return

                       ;          Check for end of chain.

E87F    C900           SHC2       CMP     #0         ;end of chain indicator
E881    D006 ^E889                BNE     SHC4       ;if not end of chain

E883    E000                      CPX     #0
E885    D002 ^E889                BNE     SHC4       ;if not end of chain

                       ;          Process end of chain or checksum error.

E887    38             SHC3       SEC                ;return error (checksum or end)
E888    60                        RTS                ;return

                       ;          Set link to new linkage table.

E889    864A           SHC4       STX     ZCHAIN     ;link to new linkage table
E88B    854B                      STA     ZCHAIN+1

E88D    2056CB                    JSR     CLT        ;checksum linkage table
E890    D0F5 ^E887                BNE     SHC3       ;if error

                       ;          Continue searching chain.

E892    F0D7 ^E86B                BEQ     SHC1       ;continue searching chain
```

```
**      PHW - Perform Peripheral Handler Warmstart Initiali:
*
*       PHC is the main entry. This performs full initializ:
*               including adding the new linkage table into:
*               table chain;
*       PHW does all initialization except adding to the li:
*               table chain (intended for warm start reinit:
*       PHI is the full initialization entry for calling
*               init from outside the OS.
*
*       The code does the following:
*       1)      Links new handler to end of chain;
*       2)      Calls handler init subroutine in handler;
*       3)      If 2 failed, unlinks handler from chain,
*               and returns with carry;
*       4)      Else, conditionally zeroes handler size ent:
*               handler linkage table (per parameter);
*       5)      Adds handler size entry (possibly zeroed) t:
*       6)      If handler size entry is nonzero, MEMLO is :
*               forced even;
*       7)      Calculates and enters linkage table checksu:
*       8)      Returns with carry clear.
*
*       PHC is called by PHR when loading handlers at cold
*               initialization;and by PHL when loading a ha:
*               application request under CIO;
*       PHW is called by PHR to reinitialize a handler duri:
*               warm-start;
*       PHI is vectored by OS vector at $E49E and is intend:
*               for use by system-level applications which :
*               handlers (ie., AUTORUN.SYS handler loader, :
*
*       Input parameters:
*       PHC:
*               DVSTAT, DVSTAT+1 contain handler size (for
*               handler init, not used by this routine);
*               DVSTAT+2, DVSTAT+3 contain handler linkage :
*               address.
*       PHW:
*               DVSTAT+2, DVSTAT+3 same;
*               DVSTAT, DVSTAT+1 undefined.
*       PHI:
*               A and Y contain handler linkage table addre:
*               they are copied into DVSTAT+3 and DVSTAT+2;
*               DVSTAT, DVSTAT+1 may or may not be signific:
*               any concern about these are up to the progr:
*               of the peripheral handler init routine and :
*               is making use of the non-OS-caller entry PH:
*
*       For PHI and PHC, the Carry bit specifies whether
*               the handler size entry of the linkage table:
*               be zeroed prior to adding to MEMLO: Carry s:
*               do NOT zero this entry.
*
*       Output parameters:
*       Carry indicates error (initialization failed);
*       The registers are not saved.
```

```
                        *
                        *       Modified:
                        *       DVSTAT+2, DVSTAT+3 are modified by PHI;
                        *       ZCHAIN, TEMP1, TEMP2;
                        *       MEMLO, MEMLO+1 conditionally incremented by handler:
                        *
                        *       Subroutines called:
                        *       SHC (to find end of linkage table chain);
                        *       PHU (to unlink handler if init. error);
                        *       CLT (to insert linkage table checksum);
                        *       loaded handler's INIT entry.
                        *
                        *       ENTRY   JSR     PHW
                        *
                        *       MODS
                        *               R. S. Scheiman  04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = E894    PHW     =       *       ;entry
E894  38                SEC             ;indicate not zeroing handler size
E895  08                PHP
E896  B028 ^E8C0        BCS     PHQ     ;initialize handler and update MEML:



                **      PHI - Perfrom Peripheral Handler Initialization wit:
                *
                *       ENTRY   JSR     PHI
                *
                *       MODS
                *               R. S. Scheiman  04/01/82
                *               1. Bring closer to Coding Standard (object :
                *                  R. K. Nordin 11/01/83


      = E898    PHI     =       *               ;entry
E898  8DED02            STA     DVSTAT+3
E89B  8CEC02            STY     DVSTAT+2
                ;       JMP     PHC             ;perfrom coldstart initiali:
```

```
                    **      PHC - Perfrom Peripheral Handler Coldstart Initiali:
                    *
                    *       ENTRY   JSR     PHC
                    *
                    *       MODS
                    *               R. S. Scheiman  04/01/82
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

    = E89E          PHC     =       *       ;entry

                    ;       Initialize.

E89E  08                    PHP

                    ;       Search for end of chain.

E89F  A900                  LDA     #0      ;indicate searching for end of chai:
E8A1  A8                    TAY
E8A2  205DE8                JSR     SHC     ;search handler chain
E8A5  B027 ^E8CE            BCS     PHQ1    ;if error, exit

                    ;       Enter at end of chain.

E8A7  A012                  LDY     #18             ;offset
E8A9  ADEC02                LDA     DVSTAT+2
E8AC  914A                  STA     (ZCHAIN),Y      ;low link
E8AE  AA                    TAX
E8AF  C8                    INY
E8B0  ADED02                LDA     DVSTAT+3
E8B3  914A                  STA     (ZCHAIN),Y      ;high link
E8B5  864A                  STX     ZCHAIN          ;link to new table
E8B7  854B                  STA     ZCHAIN+1
E8B9  A900                  LDA     #0              ;indicate end of chain
E8BB  914A                  STA     (ZCHAIN),Y      ;low link
E8BD  88                    DEY
E8BE  914A                  STA     (ZCHAIN),Y      ;high link

                    ;       Initialize handler.

                    ;       JMP     PHQ             ;initialize handler, return
```

```
                        **      PHQ - Initialize Handler and Update MEMLO
                        *
                        *       ENTRY   JSR     PHQ
                        *
                        *       MODS
                        *               R. S. Scheiman  04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = E8C0            PHQ     =       *               ;entry

                        ;       Initialize handler.

E8C0  2000E9                    JSR     PHX             ;initialize handler
E8C3  900C ^E8D1                BCC     PHQ2            ;if no error

                        ;       Process error.

E8C5  ADED02                    LDA     DVSTAT+3
E8C8  ACEC02                    LDY     DVSTAT+2
E8CB  2015E9                    JSR     PHU             ;unlink handler

                        ;       Exit, indicating error.

E8CE  28                PHQ1    PLP                     ;fix stack
E8CF  38                        SEC                     ;indicate error
E8D0  60                        RTS                     ;return

                        ;       Check for zeroing handler size.

E8D1  28                PHQ2    PLP
E8D2  B009 ^E8DD                BCS     PHQ3            ;if not to zero

                        ;       Zero handler size.

E8D4  A900                      LDA     #0
E8D6  A010                      LDY     #16             ;offset
E8D8  914A                      STA     (ZCHAIN),Y      ;zero size
E8DA  C8                        INY
E8DB  914A                      STA     (ZCHAIN),Y

                        ;       Increase MEMLO by size.

E8DD  18                PHQ3    CLC
E8DE  A010                      LDY     #16             ;offset to size
E8E0  ADE702                    LDA     MEMLO
E8E3  714A                      ADC     (ZCHAIN),Y      ;add low size
E8E5  8DE702                    STA     MEMLO           ;new low MEMLO
E8E8  C8                        INY
E8E9  ADE802                    LDA     MEMLO+1
E8EC  714A                      ADC     (ZCHAIN),Y      ;add high size
E8EE  8DE802                    STA     MEMLO+1         ;new high MEMLO

                        ;       Put checksum in linkage table.

E8F1  A00F                      LDY     #15             ;offset to checksum
```

```
E8F3  A900                  LDA     #0
E8F5  914A                  STA     (ZCHAIN),Y      ;clear checksum
E8F7  2056CB                JSR     CLT             ;checksum linkage table
E8FA  A00F                  LDY     #15             ;offset to checksum
E8FC  914A                  STA     (ZCHAIN),Y      ;checksum

                    ;       Exit.

E8FE  18                    CLC                     ;indicate success
E8FF  60                    RTS                     ;return



                    **      PHX - Initialize Handler
                    *
                    *       ENTRY   JSR     PHX
                    *
                    *       MODS
                    *               R. S. Scheiman  04/01/82
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = E900        PHX     =       *               ;entry
E900  18                    CLC
E901  A54A                  LDA     ZCHAIN
E903  690C                  ADC     #12
E905  8D1203                STA     TEMP1           ;low handler initialization:
E908  A54B                  LDA     ZCHAIN+1
E90A  6900                  ADC     #0
E90C  8D1303                STA     TEMP1+1         ;high handler initializatio:
E90F  6C1203                JMP     (TEMP1)         ;initialize handler, return
```

```
E912                    FIX     $E912


                **      E912 - $E912 Patch
                *
                *       For compatibilty with OS Revision B, set VBLANK par:

E912  4C72C2            JMP     SVP      ;set VBLANK parameters, return
```

```
E915                    **      PHU - Perform Peripheral Handler Unlinking
                        *
                        *       Handler entry unlinking routine. This routine is ca:
                        *       by the OS handler initialization to unlink a handle:
                        *       initialization fails, or by the handler itself if i:
                        *       the handler unload feature.     This routine is ent:
                        *       OS vector at $E49B.
                        *
                        *       Input parameters:
                        *       A       Address of linkage table to unlink (High);
                        *       Y       Address of linkage table to unlink (Low).
                        *       COLDST: Tested to see if PHU is called during cold :
                        *               if so, chain entry is unlinked even if at M:
                        *
                        *       Output parameters:
                        *       Carry is set to indicate error;in this case,
                        *               no unlinking has occurred.
                        *
                        *       Modified:
                        *       TEMP1, TEMP2;
                        *       ZCHAIN,ZCHAIN+1;
                        *       The forward chain pointer in the predecessor of the:
                        *       table being removed is modified to point to the suc:
                        *       of the removed table if the removal is successful--
                        *       this forward chain pointer may be CHLINK,CHLINK+1.
                        *
                        *       The registers are not saved.
                        *
                        *       Subroutines called:
                        *       SHC, CLT.
                        *
                        *       ENTRY   JSR     PHU
                        *
                        *       MODS
                        *               R. S. Scheiman  04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = E915    PHU     =       *                       ;entry

                ;       Search handler chain.

E915  205DE8            JSR     SHC                     ;search handler chain
E918  B03B ^E955        BCS     PHU3                    ;if error

                ;       Perform unlinking.

E91A  A8                TAY                             ;(save return parameter)
E91B  A54A              LDA     ZCHAIN                  ;save ZCHAIN (points to pre:
E91D  48                PHA
E91E  A54B              LDA     ZCHAIN+1
E920  48                PHA
E921  864A              STX     ZCHAIN                  ;make ZCHAIN point to linka:
E923  844B              STY     ZCHAIN+1                ;to be removed
E925  AD4402            LDA     COLDST                  ;coldstart flag
E928  D00F ^E939        BNE     PHU1                    ;if coldstart, unconditiona:
```

```
E92A  A010              LDY     #16             ;check if loaded at MEMLO..:
E92C  18                CLC                     ;by checking if size is non:
E92D  B14A              LDA     (ZCHAIN),Y
E92F  C8                INY
E930  714A              ADC     (ZCHAIN),Y
E932  D01F ^E953        BNE     PHU2            ;if handler size non-zero

E934  2056C6            JSR     CLT             ;checksum linkage table
E937  D01A ^E953        BNE     PHU2            ;if checksum nonzero, bad c:

E939  A012        PHU1  LDY     #18             ;take link from table being:
E93B  B14A              LDA     (ZCHAIN),Y
E93D  AA                TAX
E93E  C8                INY
E93F  B14A              LDA     (ZCHAIN),Y
E941  A8                TAY
E942  68                PLA             ;Make ZCHAIN point to the predecess:
E943  854B              STA     ZCHAIN+1
E945  68                PLA
E946  854A              STA     ZCHAIN
E948  98                TYA             ;And put forward link from table be:
E949  A013              LDY     #19     ;removed into its predecessors link:
E94B  914A              STA     (ZCHAIN),Y
E94D  88                DEY
E94E  8A                TXA
E94F  914A              STA     (ZCHAIN),Y
E951  18                CLC             ;indicate success
E952  60                RTS             ;return

                  ;     Clean stack and process error.

E953  68          PHU2  PLA             ;Error return--restore stack
E954  68                PLA

                  ;     Process error.

E955  38          PHU3  SEC             ;indicate error
E956  60                RTS             ;return
```

```
E957                    FIX     $E959



                **      E959 - $E959 Patch
                *
                *       For compatibilty with OS Revision B, perform PIO.

E959  4C33C9            JMP     PIO     ;perform PIO, return
```

```
E95C                  **      ISIO - Initialize SIO
                      *
                      *       ENTRY   JSR     ISIO
                      *
                      *       MODS
                      *               Original Author Unknown
                      *               1. Bring closer to Coding Standard (object :
                      *                  R. K. Nordin 11/01/83

        = E95C        ISIO    =       *       ;entry

E95C  A93C                    LDA     #MOTRST
E95E  8D02D3                  STA     PACTL   ;turn off motor

E961  A93C                    LDA     #NCOMHI
E963  8D03D3                  STA     PBCTL   ;raise NOT COMMAND line

E966  A903                    LDA     #$03    ;POKEY out of initialize mode
E968  8D3202                  STA     SSKCTL  ;SKCTL shadow
E96B  8541                    STA     SOUNDR  ;select noisy I/O
E96D  8D0FD2                  STA     SKCTL

E970  60                      RTS             ;return



                      **      SIO - Serial Input/Output
                      *
                      *       ENTRY   JSR     SIO
                      *
                      *       MODS
                      *               Original Author Unknown
                      *               1. Bring closer to Coding Standard (object :
                      *                  R. K. Nordin 11/01/83

        = E971        SIO     =       *       ;entry

                      ;       Initialize.

E971  BA                      TSX
E972  8E1803                  STX     STACKP  ;save stack pointer
E975  A901                    LDA     #1      ;critical section indicator
E977  8542                    STA     CRITIC  ;indicate critical section

                      ;       Check device ID.

E979  AD0003                  LDA     DDEVIC  ;device ID
E97C  C960                    CMP     #CASET
E97E  D003 ^E983              BNE     SIO1    ;if not cassette

                      ;       Process cassette.

E980  4C9DE8                  JMP     PCI     ;process cassette I/O, return
```

```
                        ;       Process intelligent device.

E983  A900              SI01    LDA     #0
E985  8D0F03                    STA     CASFLG  ;indicate not cassette

E988  A901                      LDA     #DRETRI
E98A  8DBD02                    STA     DRETRY  ;set device retry count

E98D  A90D              SI02    LDA     #CRETRI
E98F  8D9C02                    STA     CRETRY  ;set command frame retry count

                        ;       Send command frame.

E992  A928              SI03    LDA     #low B19200
E994  8D04D2                    STA     AUDF3           ;set baud rate to 19200
E997  A900                      LDA     #high B19200
E999  8D06D2                    STA     AUDF4

                        ;       Set up command buffer.

E99C  18                        CLC
E99D  AD0003                    LDA     DDEVIC          ;device ID
E9A0  6D0103                    ADC     DUNIT           ;add unit number
E9A3  69FF                      ADC     #$FF            ;subtract 1
E9A5  8D3A02                    STA     CDEVIC          ;device bus ID
E9A8  AD0203                    LDA     DCOMND          ;command
E9AB  8D3B02                    STA     CCOMND
E9AE  AD0A03                    LDA     DAUX1           ;auxiliary information 1
E9B1  8D3C02                    STA     CAUX1
E9B4  AD0B03                    LDA     DAUX2           ;auxiliary information 2
E9B7  8D3D02                    STA     CAUX2

                        ;       Set buffer pointer to command frame buffer.

E9BA  18                        CLC
E9BB  A93A                      LDA     #low CDEVIC     ;low buffer address
E9BD  8532                      STA     BUFRLO          ;low buffer address
E9BF  6904                      ADC     #4
E9C1  8534                      STA     BFENLO          ;low buffer end address
E9C3  A902                      LDA     #high CDEVIC    ;high buffer address
E9C5  8533                      STA     BUFRHI          ;high buffer address
E9C7  8535                      STA     BFENHI          ;high buffer end address

                        ;       Send command frame to device.

E9C9  A934                      LDA     #NCOMLO
E9CB  8D03D3                    STA     PBCTL           ;lower NOT COMMAND line
E9CE  20AFEC                    JSR     SID             ;send command frame
E9D1  AD3F02                    LDA     ERRFLG          ;error flag
E9D4  D003 ^E9D9                BNE     SI04            ;if error received

E9D6  98                        TYA                     ;status
E9D7  D008 ^E9E1                BNE     SI05            ;if ACK received

                        ;       Process NAK or timeout.

E9D9  CE9C02            SI04    DEC     CRETRY  ;decrement command frame retry count
```

```
E9DC  10B4 ^E992             BPL     SI03    ;if retries not exhausted

                     ;       Process command frame retries exhausted.

E9DE  4C22EA                 JMP     SI010   ;process error

                     ;       Process ACK.

E9E1  AD0303         SI05    LDA     DSTATS
E9E4  100D ^E9F3             BPL     SI06    ;if no data to send

                     ;       Send data frame to device.

E9E6  A90D                   LDA     #CRETRI
E9E8  8D9C02                 STA     CRETRY  ;set command frame retry count
E9EB  2087EB                 JSR     SBP     ;set buffer pointers
E9EE  20AFEC                 JSR     SID     ;send data frame
E9F1  F02F ^EA22             BEQ     SI010   ;if error

                     ;       Wait for complete.

E9F3  209AEC         SI06    JSR     GTO     ;set device timeout
E9F6  A900                   LDA     #0
E9F8  8D3F02                 STA     ERRFLG  ;clear error flag
E9FB  20C0EC                 JSR     STW     ;set timer and wait
E9FE  F012 ^EA12             BEQ     SI08    ;if timeout

                     ;       Process no timeout.

EA00  2C0303                 BIT     DSTATS
EA03  7007 ^EA0C             BVS     SI07    ;if more data follows

EA05  AD3F02                 LDA     ERRFLG  ;error flag
EA08  D018 ^EA22             BNE     SI010   ;if error

                     ;       Process no error.

EA0A  F01E ^EA2A             BEQ     CSO     ;complete SIO operation

                     ;       Receive data frame from device.

EA0C  2087EB         SI07    JSR     SBP     ;set buffer pointers
EA0F  20FDEA                 JSR     REC     ;receive

                     ;       Check error flag.

EA12  AD3F02         SI08    LDA     ERRFLG  ;error flag
EA15  F005 ^EA1C             BEQ     SI09    ;if no error preceded data

                     ;       Process error.

EA17  AD1903                 LDA     TSTAT   ;temporary status
EA1A  8530                   STA     STATUS  ;status

                     ;       Check status.

EA1C  A530           SI09    LDA     STATUS  ;status
```

```
EA1E  C901                   CMP     #SUCCES
EA20  F008 ^EA2A             BEQ     CSO       ;if successful, complete operation;:

                      ;      Process error.

EA22  CEBD02          SIO10  DEC     DRETRY    ;decrement device retry count
EA25  3003 ^EA2A             BMI     CSO       ;if retries exhausted, complete, re:

                      ;      Retry.

EA27  4C8DE9                 JMP     SIO2      ;retry



                      **     CSO - Complete SIO Operation
                      *
                      *      ENTRY   JSR     CSO
                      *
                      *      MODS
                      *              Original Author Unknown
                      *              1. Bring closer to Coding Standard (object :
                      *                 R. K. Nordin 11/01/83


      = EA2A          CSO    =       *         ;entry
EA2A  2084EC                 JSR     DSR       ;disable SEND and RECEIVE
EA2D  A900                   LDA     #0        ;not critical section indicator
EA2F  8542                   STA     CRITIC    ;critical section flag
EA31  A430                   LDY     STATUS    ;status
EA33  8C0303                 STY     DSTATS    ;status
EA36  60                     RTS               ;return



                      **     WCA - Wait for Completion or ACK
                      *
                      *      ENTRY   JSR     WCA
                      *
                      *      EXIT
                      *              Y = 0, if failure
                      *                = $FF, if success
                      *
                      *      NOTES
                      *              Problem: WCA does not handle NAK correctly?:
                      *              Just before WCA3 should be removed.
                      *
                      *      MODS
                      *              Original Author Unknown
                      *              1. Bring closer to Coding Standard (object :
                      *                 R. K. Nordin 11/01/83

      = EA37          WCA    =       *                   ;entry

                      ;      Initialize.
```

```
EA37  A900                LDA   #0              ;clear error flag
EA39  8D3F02              STA   ERRFLG          ;clear error flag

                    ;     Set buffer pointer.

EA3C  18                  CLC
EA3D  A93E                LDA   #low TEMP       ;low temporary address
EA3F  8532                STA   BUFRLO          ;low buffer address
EA41  6901                ADC   #1
EA43  8534                STA   BFENLO          ;low buffer end address
EA45  A902                LDA   #high TEMP      ;high temporary address
EA47  8533                STA   BUFRHI          ;high buffer address
EA49  8535                STA   BFENHI          ;high buffer end address
EA4B  A9FF                LDA   #$FF
EA4D  853C                STA   NOCKSM          ;indicate no checksum follo:
EA4F  20FDEA              JSR   REC             ;receive
EA52  A0FF                LDY   #$FF            ;assume success
EA54  A530                LDA   STATUS          ;status
EA56  C901                CMP   #SUCCES
EA58  D019 ^EA73          BNE   WCA2            ;if failure

EA5A  AD3E02              LDA   TEMP            ;byte received
EA5D  C941                CMP   #ACK
EA5F  F021 ^EA82          BEQ   WCA4            ;if ACK, exit

EA61  C943                CMP   #COMPLT
EA63  F01D ^EA82          BEQ   WCA4            ;if complete, exit

EA65  C945                CMP   #ERROR
EA67  D006 ^EA6F          BNE   WCA1            ;if device did not send bac:

                    ;     Process unrecognized response.

EA69  A990                LDA   #DERROR
EA6B  8530                STA   STATUS          ;indicate device error
EA6D  D004 ^EA73          BNE   WCA2            ;check for timeout

                    ;     Process nothing sent back.

EA6F  A98B          WCA1  LDA   #DNACK
EA71  8530                STA   STATUS          ;indicate NAK

                    ;     Check for timeout.

EA73  A530          WCA2  LDA   STATUS          ;status
EA75  C98A                CMP   #TIMOUT
EA77  F007 ^EA80          BEQ   WCA3            ;if timeout

                    ;     Process other error.

EA79  A9FF                LDA   #$FF            ;error indicator
EA7B  8D3F02              STA   ERRFLG          ;indicate error
EA7E  D002 ^EA82          BNE   WCA4            ;exit

                    ;     Indicate failure.
```

```
EA80  A000          WCA3    LDY     #0              ;failure indicator

                    ;       Exit.

EA82  A530          WCA4    LDA     STATUS          ;status
EA84  8D1903                STA     TSTAT           ;temporary status
EA87  60                    RTS                     ;return


                    **      SEN - Send
                    *
                    *       SEN sends a buffer over the serial bus.
                    *
                    *       ENTRY   JSR     SEN
                    *
                    *       NOTES
                    *               Problem: an interrupt may occur before CHKS:
                    *               initialized, causing an incorrect checksum :
                    *               STA CHKSUM should precede STA SEROUT.
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = EA88        SEN     =       *               ;entry

                    ;       Initialize.

EA88  A901                  LDA     #SUCCES         ;assume success
EA8A  8530                  STA     STATUS          ;status
EA8C  2017EC                JSR     ESS             ;enable SIO SEND
EA8F  A000                  LDY     #0
EA91  8431                  STY     CHKSUM          ;clear checksum
EA93  843B                  STY     CHKSNT          ;clear checksum sent flag
EA95  843A                  STY     XMTDON          ;clear transmit-frame done :

                    ;       Initiate TRANSMIT.

EA97  B132                  LDA     (BUFRLO),Y      ;first byte from buffer
EA99  8D0DD2                STA     SEROUT          ;serial output register
EA9C  8531                  STA     CHKSUM          ;checksum

                    ;       Check BREAK key.

EA9E  A511          SEN1    LDA     BRKKEY
EAA0  D003 ^EAA5            BNE     SEN2            ;if BREAK key not pressed

                    ;       Process BREAK key.

EAA2  4CC7ED                JMP     PBK             ;process BREAK key, return

                    ;       Process BREAK key not pressed.
```

```
EAA5  A53A        SEN2    LDA     XMTDON            ;transmit-frame done flag
EAA7  F0F5 ^EA9E          BEQ     SEN1              ;if transmit-frame not done

                  ;       Exit.

EAA9  2084EC              JSR     DSR               ;disable SEND and RECEIVE
EAAC  60                  RTS                       ;return



                  **      ORIR - Process Serial Output Ready IRQ
                  *
                  *       ENTRY   JMP     ORIR
                  *
                  *       EXIT
                  *               Exits via RTI
                  *
                  *       MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83

     = EAAD       ORIR    =       *       ;entry

                  ;       Initialize.

EAAD  98                  TYA
EAAE  48                  PHA             ;save Y
EAAF  E632                INC     BUFRLO  ;increment low buffer pointer
EAB1  D002 ^EAB5          BNE     ORI1    ;if low buffer pointer non-zero

EAB3  E633                INC     BUFRHI  ;increment high buffer pointer

                  ;       Check end of buffer.

EAB5  A532        ORI1    LDA     BUFRLO  ;buffer address
EAB7  C534                CMP     BFENLO  ;buffer end address
EAB9  A533                LDA     BUFRHI
EABB  E535                SBC     BFENHI
EABD  901C ^EADB          BCC     ORI4    ;if not past end of buffer

                  ;       Process end of buffer.

EABF  A536                LDA     CHKSNT  ;checksum sent flag
EAC1  D008 ^EACE          BNE     ORI2    ;if checksum already sent

                  ;       Send checksum.

EAC3  A531                LDA     CHKSUM  ;checksum
EAC5  8D0DD2              STA     SEROUT  ;serial output register
EAC8  A9FF                LDA     #$FF
EACA  8538                STA     CHKSNT  ;indicate checksum sent
EACC  D009 ^EAD7          BNE     ORI3

                  ;       Enable TRANSMIT done interrupt.
```

```
EACE  A510          ORI2    LDA     POKMSK
EAD0  0908                  ORA     #$08
EAD2  8510                  STA     POKMSK
EAD4  8D0ED2                STA     IRQEN

                    ;       Exit.

EAD7  68            ORI3    PLA
EAD8  A8                    TAY             ;restore Y
EAD9  68                    PLA             ;restore A
EADA  40                    RTI             ;return

                    ;       Transmit next byte from buffer.

EADB  A000          ORI4    LDY     #0
EADD  B132                  LDA     (BUFRLO),Y      ;byte from buffer
EADF  8D0DD2                STA     SEROUT          ;serial output register
EAE2  18                    CLC
EAE3  6531                  ADC     CHKSUM          ;add byte to checksum
EAE5  6900                  ADC     #0
EAE7  8531                  STA     CHKSUM          ;update checksum
EAE9  4CD7EA                JMP     ORI3            ;exit


                    **      OCIR - Process Serial Output Complete IRQ
                    *
                    *       ENTRY   JMP     OCIR
                    *
                    *       EXIT
                    *               Exits via RTI
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = EAEC        OCIR    =       *       ;entry

                    ;       Check checksum sent.

EAEC  A53B                  LDA     CHKSNT  ;checksum sent flag
EAEE  F00B ^EAFB            BEQ     OCI1    ;if checksum not yet sent

                    ;       Process checksum sent.

EAF0  853A                  STA     XMTDON  ;indicate transmit-frame done

                    ;       Disable TRANSMIT done interrupt.

EAF2  A510                  LDA     POKMSK
EAF4  29F7                  AND     #$F7
EAF6  8510                  STA     POKMSK
EAF8  8D0ED2                STA     IRQEN
```

```
                        ;       Exit.

EAFB  68                OCI1    PLA             ;restore A
EAFC  40                        RTI             ;return



                        **      REC - Receive
                        *
                        *       ENTRY   JSR     REC
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = EAFD            REC     =       *       ;entry

                        ;       Initialize.

EAFD  A900                      LDA     #0
EAFF  AC0F03                    LDY     CASFLG
EB02  D002 ^EB06                BNE     REC1    ;if cassette

EB04  8531                      STA     CHKSUM  ;initialize checksum

EB06  8538              REC1    STA     BUFRFL  ;clear buffer full flag
EB08  8539                      STA     RECVDN  ;clear receive-frame done flag
EB0A  A901                      LDA     #SUCCES ;assume success
EB0C  8530                      STA     STATUS  ;status
EB0E  2040EC                    JSR     ESR     ;enable SIO RECEIVE
EB11  A93C                      LDA     #NCOMHI
EB13  8D03D3                    STA     PBCTL

                        ;       Check BREAK key.

EB16  A511              REC2    LDA     BRKKEY
EB18  D003 ^EB1D                BNE     REC3    ;if BREAK key not pressed

                        ;       Process BREAK key.

EB1A  4CC7ED                    JMP     PBK     ;process BREAK key, return

                        ;       Process BREAK key not pressed.

EB1D  AD1703            REC3    LDA     TIMFLG  ;timeout flag
EB20  F005 ^EB27                BEQ     ITO     ;if timeout, indicate timeout

                        ;       Process no timeout.

EB22  A539                      LDA     RECVDN  ;receive-frame done flag
EB24  F0F0 ^EB16                BEQ     REC2    ;if receive-frame done, continue

                        ;       Exit.
```

```
EB26  60                   RTS                ;return



                     **    ITO - Indicate Timeout
                     *
                     *     ENTRY   JSR     ITO
                     *
                     *     MODS
                     *             Original Author Unknown
                     *             1. Bring closer to Coding Standard (object :
                     *                R. K. Nordin 11/01/83

      = EB27         ITO   =       *       ;entry
EB27  A98A                 LDA     #TIMOUT ;timeout indicator
EB29  8530                 STA     STATUS  ;indicate timeout
EB2B  60                   RTS             ;return



                     **    IRIR - Process Serial Input Ready IRQ
                     *
                     *     ENTRY   JMP     IRIR
                     *
                     *     EXIT
                     *             Exits via RTI
                     *
                     *     MODS
                     *             Original Author Unknown
                     *             1. Bring closer to Coding Standard (object :
                     *                R. K. Nordin 11/01/83

      = EB2C         IRIR  =       *       ;entry

                     ;     Initialize.

EB2C  98                   TYA
EB2D  48                   PHA             ;save Y
EB2E  AD0FD2               LDA     SKSTAT
EB31  8D0AD2               STA     SKRES   ;reset status register

                     ;     Check for frame error.

EB34  3004 ^EB3A           BMI     IRI1    ;if no frame error

                     ;     Process frame error.

EB36  A08C                 LDY     #FRMERR ;frame error
EB38  8430                 STY     STATUS  ;indicate frame error

                     ;     Check for overrun error.
```

```
EB3A  2920          IRI1   AND    #$20
EB3C  D004 ^EB42           BNE    IRI2     ;if no overrun error

                    ;      Process overrun error.

EB3E  A08E                 LDY    #OVRRUN ;overrun error
EB40  8430                 STY    STATUS  ;indicate overrun error

                    ;      Check for buffer full.

EB42  A538          IRI2   LDA    BUFRFL
EB44  F013 ^EB59           BEQ    IRI5     ;if buffer not yet full

                    ;      Process buffer full.

EB46  AD0DD2               LDA    SERIN   ;checksum from device
EB49  C531                 CMP    CHKSUM  ;computed checksum
EB4B  F004 ^EB51           BEQ    IRI3     ;if checksums match

                    ;      Process checksum error.

EB4D  A08F                 LDY    #CHKERR ;checksum error
EB4F  8430                 STY    STATUS  ;indicate checksum error

                    ;      Indicate receive-frame done.

EB51  A9FF          IRI3   LDA    #$FF    ;receive-frame done indicator
EB53  8539                 STA    RECVDN  ;indicate receive-frame done

                    ;      Exit.

EB55  68            IRI4   PLA
EB56  A8                   TAY             ;restore Y
EB57  68                   PLA             ;restore A
EB58  40                   RTI             ;return

                    ;      Process buffer not full.

EB59  AD0DD2        IRI5   LDA    SERIN             ;serial input register
EB5C  A000                 LDY    #0
EB5E  9132                 STA    (BUFRLO),Y        ;byte of buffer
EB60  18                   CLC
EB61  6531                 ADC    CHKSUM            ;add byte to checksum
EB63  6900                 ADC    #0
EB65  8531                 STA    CHKSUM            ;update checksum
EB67  E632                 INC    BUFRLO            ;increment low buffer point:
EB69  D002 ^EB6D           BNE    IRI6              ;if low buffer pointer non-:

EB6B  E633                 INC    BUFRHI            ;increment high buffer poin:

                    ;      Check end of buffer.

EB6D  A532          IRI6   LDA    BUFRLO            ;buffer address
EB6F  C534                 CMP    BFENLO            ;buffer end address
EB71  A533                 LDA    BUFRHI
EB73  E535                 SBC    BFENHI
EB75  900E ^EB55           BCC    IRI4              ;if not past end of buffer
```

```
                        ;       Process end of buffer.

EB77  A53C                      LDA     NOCKSM          ;no checksum follows flag
EB79  F006 ^EB81                BEQ     IRI7            ;if checksum will follow

                        ;       Process no checksum will follow.

EB7B  A900                      LDA     #0
EB7D  853C                      STA     NOCKSM          ;clear no checksum follows :
EB7F  F0D0 ^EB51                BEQ     IRI3            ;indicate receive-frame don:

                        ;       Process checksum will follow.

EB81  A9FF              IRI7    LDA     #$FF
EB83  8538                      STA     BUFRFL          ;indicate buffer full
EB85  D0CE ^EB55                BNE     IRI4            ;exit



                        **      SBP - Set Buffer Pointers
                        *
                        *       ENTRY   JSR     SBP
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = EB87            SBP     =       *       ;entry
EB87  18                        CLC
EB88  AD0403                    LDA     DBUFLO
EB8B  8532                      STA     BUFRLO  ;low buffer address
EB8D  6D0803                    ADC     DBYTLO
EB90  8534                      STA     BFENLO  ;low buffer end address
EB92  AD0503                    LDA     DBUFHI
EB95  8533                      STA     BUFRHI  ;high buffer address
EB97  6D0903                    ADC     DBYTHI
EB9A  8535                      STA     BFENHI  ;high buffer end address
EB9C  60                        RTS             ;return
```

```
                        **      PCI - Process Cassette I/O
                        *
                        *       ENTRY   JSR     PCI
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

     = EB9D             PCI     =       *               ;entry

                        ;       Check command type.

EB9D AD0303                     LDA     DSTATS          ;command type
EBA0 1032 ^EBD4                 BPL     PCI3            ;if READ

                        ;       Write a record.

EBA2 A9CC                       LDA     #low B00600
EBA4 8D04D2                     STA     AUDF3           ;set 600 baud
EBA7 A905                       LDA     #high B00600
EBA9 8D06D2                     STA     AUDF4
EBAC 2017EC                     JSR     ESS             ;enable SIO SEND
EBAF A662                       LDX     PALNTS          ;PAL/NTSC offset
EBB1 BC15EE                     LDY     WSIRGX,X        ;low short WRITE IRG time
EBB4 AD0B03                     LDA     DAUX2           ;IRG type
EBB7 3003 ^EBBC                 BMI     PCI1            ;if short IRG is desired

EBB9 BC11EE                     LDY     WIRGLX,X        ;low long WRITE IRG time

EBBC A200               PCI1    LDX     #WIRGHI         ;high IRG time
EBBE 20E2ED                     JSR     SSV             ;set SIO VBLANK parameters
EBC1 A934                       LDA     #MOTRGO
EBC3 8D02D3                     STA     PACTL           ;turn on motor

EBC6 AD1703             PCI2    LDA     TIMFLG          ;timeout flag
EBC9 D0FB ^EBC6                 BNE     PCI2            ;if no timeout

EBCB 2087EB                     JSR     SBP             ;set buffer pointers
EBCE 2088EA                     JSR     SEN             ;send
EBD1 4C04EC                     JMP     PCI6            ;exit

                        ;       Read a record.

EBD4 A9FF               PCI3    LDA     #$FF            ;cassette I/O indicator
EBD6 8D0F03                     STA     CASFLG          ;cassette I/O flag

EBD9 A662                       LDX     PALNTS          ;PAL/NTSC offset
EBDB BC17EE                     LDY     RSIRGX,X        ;low short READ IRG time
EBDE AD0B03                     LDA     DAUX2           ;IRG type
EBE1 3003 ^EBE6                 BMI     PCI4            ;if short IRG desired

EBE3 BC13EE                     LDY     RIRGLX,X        ;low long READ IRG time

EBE6 A200               PCI4    LDX     #RIRGHI         ;high READ IRG time
EBE8 20E2ED                     JSR     SSV             ;set SIO VBLANK parameters
```

```
EBEB  A934                  LDA     #MOTRGO
EBED  8D02D3                STA     PACTL           ;turn on motor

EBF0  AD1703       PCI5     LDA     TIMFLG          ;timeout flag
EBF3  D0FB ^EBF0            BNE     PCI5            ;if no timeout

EBF5  2087EB                JSR     SBP             ;set buffer pointers
EBF8  209AEC                JSR     GTO             ;get device timeout
EBFB  20E2ED                JSR     SSV             ;set SIO VBLANK parameters
EBFE  203DED                JSR     SBR             ;set initial baud rate
EC01  20FDEA                JSR     REC             ;receive

                   ;        Exit.

EC04  AD0B03       PCI6     LDA     DAUX2           ;IRG type
EC07  3005 ^EC0E            BMI     PCI7            ;if doing short IRG

EC09  A93C                  LDA     #MOTRST
EC0B  8D02D3                STA     PACTL           ;turn off motor

EC0E  4C2AEA       PCI7     JMP     CSO             ;complete SIO operation, re:


                   **       PTE - Process Timer Expiration
                   *
                   *        ENTRY   JSR     PTE
                   *
                   *        MODS
                   *                Original Author Unknown
                   *                1. Bring closer to Coding Standard (object :
                   *                   R. K. Nordin 11/01/83


      = EC11       PTE      =       *       ;entry
EC11  A900                  LDA     #0      ;timeout indicator
EC13  8D1703                STA     TIMFLG  ;timeout flag
EC16  60                    RTS             ;return



                   **       ESS - Enable SIO SEND
                   *
                   *        ENTRY   JSR     ESS
                   *
                   *        MODS
                   *                Original Author Unknown
                   *                1. Bring closer to Coding Standard (object :
                   *                   R. K. Nordin 11/01/83


      = EC17       ESS      =       *       ;entry

                   ;        Initialize.
```

```
EC17  A907                  LDA     #$07     ;mask off previous serial bus contr:
EC19  2D3202                AND     SSKCTL
EC1C  0920                  ORA     #$20     ;set SEND mode

                      ;     Check device type.

EC1E  AC0003                LDY     DDEVIC
EC21  C060                  CPY     #CASET
EC23  D00C ^EC31            BNE     ESS1     ;if not cassette

                      ;     Process cassette.

EC25  0908                  ORA     #$08     ;set FSK output
EC27  A007                  LDY     #LOTONE  ;set FSK tone frequencies
EC29  8C02D2                STY     AUDF2
EC2C  A005                  LDY     #HITONE
EC2E  8C00D2                STY     AUDF1

                      ;     Set serial bus control.

EC31  8D3202          ESS1  STA     SSKCTL   ;SKCTL shadow
EC34  8D0FD2                STA     SKCTL
EC37  A9C7                  LDA     #$C7     ;mask off previous serial bus inter:
EC39  2510                  AND     POKMSK   ;and with POKEY IRQ enable
EC3B  0910                  ORA     #$10     ;enable output data needed interrup:
EC3D  4C56EC                JMP     SSR      ;set for SEND, return



                      **    ESR - Enable SIO RECEIVE
                      *
                      *     ENTRY   JSR      ESR
                      *
                      *     MODS
                      *             Original Author Unknown
                      *             1. Bring closer to Coding Standard (object :
                      *                R. K. Nordin 11/01/83

      = EC40          ESR   =       *        ;entry
EC40  A907                  LDA     #$07     ;mask off previous serial bus contr:
EC42  2D3202                AND     SSKCTL
EC45  0910                  ORA     #$10     ;set receive mode asynchronous
EC47  8D3202                STA     SSKCTL   ;SKCTL shadow
EC4A  8D0FD2                STA     SKCTL
EC4D  8D0AD2                STA     SKRES
EC50  A9C7                  LDA     #$C7     ;mask off previous serial bus inter:
EC52  2510                  AND     POKMSK   ;and with POKEY IRQ enable
EC54  0920                  ORA     #$20     ;enable RECEIVE interrupt
                      ;     JMP     SSR      ;set for RECEIVE, return
```

```
                        **      SSR - Set for SEND or RECEIVE
                        *
                        *       ENTRY   JSR     SSR
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


        = EC56          SSR     =       *       ;entry

                        ;       Initialize.

EC56  8510                      STA     POKMSK  ;update POKEY IRQ enable
EC58  8D0ED2                    STA     IRQEN   ;IRQ enable
EC5B  A928                      LDA     #$28    ;clock ch. 3 with 1.79 MHz, ch. 4 w:
EC5D  8D08D2                    STA     AUDCTL  ;set audio control

                        ;       Set voice controls.

EC60  A206                      LDX     #6      ;offset to last voice control
EC62  A9A8                      LDA     #$A8    ;pure tone, half volume
EC64  A441                      LDY     SOUNDR  ;noisy I/O flag.
EC66  D002 ^EC6A                BNE     SSR1    ;if noisy I/O desired

EC68  A9A0                      LDA     #$A0    ;pure tone, no volume

EC6A  9D01D2    SSR1            STA     AUDC1,X ;set tone and volume
EC6D  CA                        DEX
EC6E  CA                        DEX
EC6F  10F9 ^EC6A                BPL     SSR1    ;if not done

                        ;       Turn off certain voices.

EC71  A9A0                      LDA     #$A0    ;pure tone, no volume
EC73  8D05D2                    STA     AUDC3   ;turn off sound on voice 3
EC76  AC0003                    LDY     DDEVIC  ;device bus ID
EC79  C060                      CPY     #CASET  ;cassette device ID
EC7B  F006 ^EC83                BEQ     SSR2    ;if cassette device

EC7D  8D01D2                    STA     AUDC1   ;turn off sound on voice 1
EC80  8D03D2                    STA     AUDC2   ;turn off sound on voice 2

EC83  60        SSR2            RTS             ;return
```

```
                        **      DSR - Disable SEND and RECEIVE
                        *
                        *       ENTRY   JSR     DSR
                        *
                        *       NOTES
                        *               Problem: NOP may not be necessary.
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = EC84             DSR     =       *       ;entry

                        ;       Disable serial bus interrupts.

EC84 EA                         NOP
EC85 A9C7                       LDA     #$C7    ;mask to clear serial bus interrupt:
EC87 2510                       AND     POKMSK  ;and with POKEY IRQ enable
EC89 8510                       STA     POKMSK  ;update POKEY IRQ enable
EC8B 8D0ED2                     STA     IRQEN   ;IRQ enable

                        ;       Turn off audio volume.

EC8E A206                       LDX     #6      ;offset to last voice control
EC90 A900                       LDA     #$00    ;no volume

EC92 9D01D2             DSR1    STA     AUDC1,X ;turn off voice
EC95 CA                         DEX
EC96 CA                         DEX
EC97 10F9 ^EC92                 BPL     DSR1    ;if not done

EC99 60                         RTS             ;return



                        **      GTO - Get Device Timeout
                        *
                        *       ENTRY   JSR     GTO
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = EC9A             GTO     =       *       ;entry
EC9A AD0603                     LDA     DTIMLO  ;device timeout
EC9D 6A                         ROR     A
EC9E 6A                         ROR     A
EC9F A8                         TAY             ;rotated timeout
ECA0 293F                       AND     #$3F    ;lower 6 bits
ECA2 AA                         TAX             ;high timeout
ECA3 98                         TYA             ;rotated timeout
ECA4 6A                         ROR     A
```

```
ECA5  29C0                 AND     #$C0      ;upper 2 bits
ECA7  A8                   TAY               ;low timeout
ECA8  60                   RTS               ;return


                    **      TSIH - Table of SIO Interrupt Handlers
                    *
                    *       NOTES
                    *               Problem: not used.

ECA9  2CEB          TSIH   DW      IRIR      ;serial input ready IRQ
ECAB  ADEA                 DW      ORIR      ;serial output ready IRQ
ECAD  ECEA                 DW      OCIR      ;serial output complete IRQ


                    **      SID - Send to Intelligent Device
                    *
                    *       ENTRY   JSR     SID
                    *
                    *       NOTES
                    *               Problem: bytes wasted by outer delay loop.
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = ECAF        SID    =       *         ;entry

                    ;      Delay.

ECAF  A201                 LDX     #1

ECB1  A0FF          SID1   LDY     #255

ECB3  88            SID2   DEY
ECB4  D0FD ^ECB3           BNE     SID2              ;if inner loop not done

ECB6  CA                   DEX
ECB7  D0F6 ^ECB1           BNE     SID1              ;if outer loop not done

                    ;      Send data frame.

ECB9  2088EA               JSR     SEN               ;send

                    ;      Set timer and wait.

ECBC  A002                 LDY     #low CTIM         ;frame acknowledge timeout
ECBE  A200                 LDX     #high CTIM
                    ;      JMP     STW               ;set timer and wait, return
```

```
                         **      STW - Set Timer and Wait
                         *
                         *       ENTRY   JSR     STW
                         *
                         *       MODS
                         *               Original Author Unknown
                         *               1. Bring closer to Coding Standard (object :
                         *                  R. K. Nordin 11/01/83

     = ECC0          STW     =       *       ;entry
ECC0  20E2ED                 JSR     SSV     ;set SIO VBLANK parameters
ECC3  2037EA                 JSR     WCA     ;wait for completion or ACK
ECC6  98                     TYA             ;wait termination status
ECC7  60                     RTS             ;return



                         **      CBR - Compute Baud Rate
                         *
                         *       CBR computes value for POKEY frequency for the baud:
                         *       measured by an interval of the VCOUNT timer.
                         *
                         *       ENTRY   JSR     CBR
                         *
                         *       MODS
                         *               Original Author Unknown
                         *               1. Bring closer to Coding Standard (object :
                         *                  R. K. Nordin 11/01/83

     = ECC8          CBR     =       *               ;entry
ECC8  8D1003                 STA     TIMER2          ;save final timer value
ECCB  8C1103                 STY     TIMER2+1
ECCE  202EED                 JSR     AVV             ;adjust VCOUNT value
ECD1  8D1003                 STA     TIMER2          ;save adjusted timer 2 valu:
ECD4  AD0C03                 LDA     TIMER1
ECD7  202EED                 JSR     AVV             ;adjust VCOUNT value
ECDA  8D0C03                 STA     TIMER1          ;save adjusted timer 1 valu:
ECDD  AD1003                 LDA     TIMER2
ECE0  38                     SEC
ECE1  ED0C03                 SBC     TIMER1
ECE4  8D1203                 STA     TEMP1           ;save difference
ECE7  AD1103                 LDA     TIMER2+1
ECEA  38                     SEC
ECEB  ED0D03                 SBC     TIMER1+1
ECEE  A8                     TAY                     ;difference
ECEF  A662                   LDX     PALNTS
ECF1  A900                   LDA     #0
ECF3  38                     SEC
ECF4  FD19EE                 SBC     CONS1X,X

ECF7  18             CBR1    CLC
ECF8  7D19EE                 ADC     CONS1X,X        ;accumulate product
ECFB  88                     DEY
ECFC  10F9 ^ECF7             BPL     CBR1            ;if not done
```

```
ECFE  18                CLC
ECFF  6D1203            ADC     TEMP1          ;add to get total VCOUNT di:
ED02  A8                TAY                    ;total VCOUNT difference
ED03  4A                LSR     A
ED04  4A                LSR     A
ED05  4A                LSR     A
ED06  0A                ASL     A              ;interval divided by 4
ED07  38                SEC
ED08  E916              SBC     #22            ;adjust offset
ED0A  AA                TAX                    ;offset
ED0B  98                TYA                    ;total VCOUNT difference
ED0C  2907              AND     #7             ;extract lower 3 bits of in:
ED0E  A8                TAY                    ;lower 3 bits of interval
ED0F  A9F5              LDA     #-11

ED11  18          CBR2  CLC
ED12  690B              ADC     #11            ;accumulate interpolation c:
ED14  88                DEY
ED15  10FA ^ED11        BPL     CBR2           ;if done

ED17  A000              LDY     #0             ;assume no addition correct:
ED19  38                SEC
ED1A  E907              SBC     #7             ;adjust interpolation const:
ED1C  1001 ^ED1F        BPL     CBR3

ED1E  88                DEY                    ;indicate addition correcti:

ED1F  18          CBR3  CLC
ED20  7DF9ED            ADC     TPFV,X         ;add constant to table valu:
ED23  8DEE02            STA     CBAUDL         ;low POKEY frequency value
ED26  98                TYA
ED27  7DFAED            ADC     TPFV+1,X
ED2A  8DEF02            STA     CBAUDH         ;high POKEY frequency value
ED2D  60                RTS                    ;return


                  **    AVV - Adjust VCOUNT Value
                  *
                  *     ENTRY   JSR     AVV
                  *
                  *     MODS
                  *             Original Author Unknown
                  *             1. Bring closer to Coding Standard (object :
                  *                R. K. Nordin 11/01/83


      = ED2E      AVV   =       *       ;entry
ED2E  C97C              CMP     #$7C
ED30  3004 ^ED36        BMI     AVV1    ;if >= $7C

ED32  38                SEC
ED33  E97C              SBC     #$7C
ED35  60                RTS             ;return
```

```
ED36  18              AVV1    CLC
ED37  A662                    LDX     PALNTS
ED39  7D1BEE                  ADC     CONS2X,X
ED3C  60                      RTS             ;return



                      **      SBR - Set Initial Baud Rate
                      *
                      *       INITIAL BAUD RATE MEASUREMENT -- USED TO SET THE
                      *       BAUD RATE AT THE START OF A RECORD.
                      *
                      *       IT IS ASSUMED THAT THE FIRST TWO BYTES OF EVERY
                      *       RECORD ARE $AA.
                      *
                      *       ENTRY   JSR     SBR
                      *
                      *       NOTES
                      *               Problem: bytes wasted by branch around bran:
                      *
                      *       MODS
                      *               Original Author Unknown
                      *               1. Bring closer to Coding Standard (object :
                      *                  R. K. Nordin 11/01/83


      = ED3D          SBR     =       *                       ;entry

ED3D  A511            SBR1    LDA     BRKKEY
ED3F  D003 ^ED44              BNE     SBR2                    ;if BREAK key not pressed

ED41  4CC7ED                  JMP     PBK                     ;process BREAK key, return

ED44  78              SBR2    SEI
ED45  AD1703                  LDA     TIMFLG                  ;timeout flag
ED48  D002 ^ED4C              BNE     SBR3                    ;if no timeout

ED4A  F025 ^ED71              BEQ     SBR5                    ;process timeout

ED4C  AD0FD2          SBR3    LDA     SKSTAT
ED4F  2910                    AND     #$10                    ;extract start bit
ED51  D0EA ^ED3D              BNE     SBR1                    ;if start bit

ED53  8D1603                  STA     SAVIO                   ;save serial data in
ED56  AE0BD4                  LDX     VCOUNT                  ;vertical line counter
ED59  A414                    LDY     RTCLOK+2                ;low byte of VBLANK clock
ED5B  8E0C03                  STX     TIMER1
ED5E  8C0D03                  STY     TIMER1+1                ;save initial timer value
ED61  A201                    LDX     #1
ED63  8E1503                  STX     TEMP3                   ;set mode flag
ED66  A00A                    LDY     #10                     ;10 bits

ED68  A511            SBR4    LDA     BRKKEY
ED6A  F05B ^EDC7              BEQ     PBK                     ;if BREAK key pressed, proc:

ED6C  AD1703                  LDA     TIMFLG                  ;timeout flag
```

```
ED6F  D004 ^ED75          BNE    SBR6           ;if no timeout

ED71  58         SBR5     CLI
ED72  4C27EB              JMP    ITO            ;indicate timeout, return

ED75  AD0FD2     SBR6     LDA    SKSTAT
ED78  2910                AND    #$10           ;extract
ED7A  CD1603              CMP    SAVIO          ;previous serial data in
ED7D  F0E9 ^ED68          BEQ    SBR4           ;if data in not changed

ED7F  8D1603              STA    SAVIO          ;save serial data in
ED82  88                  DEY                   ;decrement bit counter
ED83  D0E3 ^ED68          BNE    SBR4           ;if not done

ED85  CE1503              DEC    TEMP3          ;decrement mode
ED88  300C ^ED96          BMI    SBR7           ;if done with both modes

ED8A  AD0BD4              LDA    VCOUNT
ED8D  A414                LDY    RTCLOK+2
ED8F  20C8EC              JSR    CBR            ;compute baud rate
ED92  A009                LDY    #9             ;9 bits
ED94  D0D2 ^ED68          BNE    SBR4           ;set bit counter

ED96  ADEE02     SBR7     LDA    CBAUDL
ED99  8D04D2              STA    AUDF3
ED9C  ADEF02              LDA    CBAUDH
ED9F  8D06D2              STA    AUDF4          ;set POKEY baud rate
EDA2  A900                LDA    #0
EDA4  8D0FD2              STA    SKSTAT
EDA7  AD3202              LDA    SSKCTL
EDAA  8D0FD2              STA    SKSTAT         ;initialize POKEY serial po:
EDAD  A955                LDA    #$55
EDAF  9132                STA    (BUFRLO),Y     ;first byte of buffer
EDB1  C8                  INY
EDB2  9132                STA    (BUFRLO),Y     ;second byte of buffer
EDB4  A9AA                LDA    #$AA           ;checksum
EDB6  8531                STA    CHKSUM         ;checksum
EDB8  18                  CLC
EDB9  A532                LDA    BUFRLO
EDBB  6902                ADC    #2             ;add 2
EDBD  8532                STA    BUFRLO         ;update low buffer pointer
EDBF  A533                LDA    BUFRHI
EDC1  6900                ADC    #0
EDC3  8533                STA    BUFRHI         ;update high buffer pointer
EDC5  58                  CLI
EDC6  60                  RTS                   ;return
```

```
                        **      PBK - Process BREAK Key
                        *
                        *       ENTRY   JSR     PBK
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = EDC7             PBK     =       *       ;entry
EDC7 2064EC                     JSR     DSR     ;disable SEND and RECEIVE
EDCA A93C                       LDA     #MOTRST
EDCC 8D02D3                     STA     PACTL   ;turn off motor
EDCF A93C                       LDA     #NCOMHI
EDD1 8D03D3                     STA     PBCTL   ;raise NOT COMMAND line
EDD4 A980                       LDA     #BRKABT ;BREAK abort error
EDD6 8530                       STA     STATUS  ;status
EDD8 AE1803                     LDX     STACKP  ;saved stack pointer
EDDB 9A                         TXS             ;restore stack pointer
EDDC C611                       DEC     BRKKEY  ;indicate BREAK
EDDE 58                         CLI
EDDF 4C2AEA                     JMP     CSO     ;complete SIO operation, return to :



                        **      SSV - Set SIO VBLANK Parameters
                        *
                        *       ENTRY   JSR     SSV
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = EDE2             SSV     =       *               ;entry
EDE2 A911                       LDA     #low PTE        ;timer expiration routine a:
EDE4 8D2602                     STA     CDTMA1
EDE7 A9EC                       LDA     #high PTE
EDE9 8D2702                     STA     CDTMA1+1
EDEC A901                       LDA     #1              ;timer 1
EDEE 78                         SEI
EDEF 205CE4                     JSR     SETVBV          ;set VBLANK parameters
EDF2 A901                       LDA     #1              ;no timeout indicator
EDF4 8D1703                     STA     TIMFLG          ;timeout flag
EDF7 58                         CLI
EDF8 60                         RTS                     ;return
```

```
                    **      TPFV - Table of POKEY Frequency Values
                    *
                    *       TPFV translates VCOUNT interval timer measurements :
                    *       frequency register values.
                    *
                    *       Table entries are AUDF+7.
                    *
                    *       Frequency-out is Frequency-in divided by 2*(AUDF+M):
                    *       Frequency-in = 1.78979 Mhz and M = 7.
                    *
                    *       AUDF+7=(11.365167)*T-out, where T-out is the number:
                    *       (127 used cd soulution) of VCOUNT for one character
                    *       time (10 bit times).

                    ;       DW      636     ;baud rate 1407, VCOUNT interval 56
                    ;       DW      727     ;baud rate 1231, VCOUNT interval 64
                    ;       DW      818     ;baud rate 1094, VCOUNT interval 72
                    ;       DW      909     ;baud rate 985, VCOUNT interval 80

EDF9  E803          TPFV    DW      1000    ;baud rate 895, VCOUNT interval 88
EDFB  4304                  DW      1091    ;baud rate 820, VCOUNT interval 96
EDFD  9E04                  DW      1182    ;baud rate 757, VCOUNT interval 104
EDFF  F904                  DW      1273    ;baud rate 703, VCOUNT interval 112
EE01  5405                  DW      1364    ;baud rate 656, VCOUNT interval 120
EE03  AF05                  DW      1455    ;baud rate 615, VCOUNT interval 128
EE05  0A06                  DW      1546    ;baud rate 579, VCOUNT interval 136
EE07  6506                  DW      1637    ;baud rate 547, VCOUNT interval 144
EE09  C006                  DW      1728    ;baud rate 518, VCOUNT interval 152
EE0B  1A07                  DW      1818    ;baud rate 492, VCOUNT interval 160
EE0D  7507                  DW      1909    ;baud rate 469, VCOUNT interval 168
EE0F  D007                  DW      2000    ;baud rate 447, VCOUNT interval 176

                    ;       DW      2091    ;baud rate 428, VCOUNT interval 184
                    ;       DW      2182    ;baud rate 410, VCOUNT interval 192
                    ;       DW      2273    ;baud rate 394, VCOUNT interval 200
                    ;       DW      2364    ;baud rate 379, VCOUNT interval 208
                    ;       DW      2455    ;baud rate 365, VCOUNT interval 216
                    ;       DW      2546    ;baud rate 352, VCOUNT interval 224
                    ;       DW      2637    ;baud rate 339, VCOUNT interval 232
                    ;       DW      2728    ;baud rate 328, VCOUNT interval 240
                    ;       DW      2819    ;baud rate 318, VCOUNT interval 248
```

```
                        **      NTSC/PAL Constant Tables

EE11  84          WIRGLX  DB      low WIRGLN       ;NTSC low long write IRG
EE12  96                  DB      low WIRGLP       ;PAL low long write IRG

EE13  78          RIRGLX  DB      low RIRGLN       ;NTSC low long read IRG
EE14  64                  DB      low RIRGLP       ;PAL low long read IRG

EE15  0F          WSIRGX  DB      low WSIRGN       ;NTSC low short write IRG
EE16  0D                  DB      low WSIRGP       ;PAL low short write IRG

EE17  0A          RSIRGX  DB      low RSIRGN       ;NTSC low short read IRG
EE18  08                  DB      low RSIRGP       ;PAL low short read IRG

EE19  83          CONS1X  DB      131              ;NTSC
EE1A  9C                  DB      156              ;PAL

EE1B  07          CONS2X  DB      7                ;NTSC
EE1C  20                  DB      32               ;PAL
```

```
EE1D                    **      TSMA - Table of Screen Memory Allocation
                        *
                        *       Entry n is the number of $40-byte blocks to allocat:
                        *       graphics mode n.
                        *
                        *       NOTES
                        *               Problem: For readability, this, and other t:
                        *               this area, could be moved closer to the oth:
                        *               the Keyboard, Editor and Screen Handler (Ju:
                        *               the EF6B patch).

EE1D  18                TSMA    DB      24      ;0
EE1E  10                        DB      16      ;1
EE1F  0A                        DB      10      ;2
EE20  0A                        DB      10      ;3
EE21  10                        DB      16      ;4
EE22  1C                        DB      28      ;5
EE23  34                        DB      52      ;6
EE24  64                        DB      100     ;7
EE25  C4                        DB      196     ;8
EE26  C4                        DB      196     ;9
EE27  C4                        DB      196     ;10
EE28  C4                        DB      196     ;11
EE29  1C                        DB      28      ;12
EE2A  10                        DB      16      ;13
EE2B  64                        DB      100     ;14
EE2C  C4                        DB      196     ;15


                        **      TDLE - Table of Display List Entry Counts
                        *
                        *       Each entry is 2 bytes.

EE2D  1717              TDLE    DB      23,23   ;0
EE2F  0B17                      DB      11,23   ;1
EE31  2F2F                      DB      47,47   ;2
EE33  5F5F                      DB      95,95   ;3
EE35  6161                      DB      97,97   ;4
EE37  6161                      DB      97,97   ;5
EE39  170B                      DB      23,11   ;6
EE3B  BF61                      DB      191,97  ;7
EE3D  1313                      DB      19,19   ;8
EE3F  0913                      DB      9,19    ;9
EE41  2727                      DB      39,39   ;10
EE43  4F4F                      DB      79,79   ;11
EE45  4141                      DB      65,65   ;12
EE47  4141                      DB      65,65   ;13
EE49  1309                      DB      19,9    ;14
EE4B  9F41                      DB      159,65  ;15
```

```
                        **      TAGM - Table of ANTIC Graphics Modes
                        *
                        *       Entry n is the ANTIC graphics mode corresponding to:
                        *       graphics mode n.

EE4D  02                TAGM    DB      $02     ;internal 0 - 40x2x8 characters
EE4E  06                        DB      $06     ;internal 1 - 20x5x8 characters
EE4F  07                        DB      $07     ;internal 2 - 20x5x16 characters
EE50  08                        DB      $08     ;internal 3 - 40x4x8 graphics
EE51  09                        DB      $09     ;internal 4 - 80x2x4 graphics
EE52  0A                        DB      $0A     ;internal 5 - 80x4x4 graphics
EE53  0B                        DB      $0B     ;internal 6 - 160x2x2 graphics
EE54  0D                        DB      $0D     ;internal 7 - 160x4x2 graphics
EE55  0F                        DB      $0F     ;internal 8 - 320x2x1 graphics
EE56  0F                        DB      $0F     ;internal 9 - 320x2x1 GTIA "lum" mo:
EE57  0F                        DB      $0F     ;internal 10 - 320x2x1 GTIA "color/:
EE58  0F                        DB      $0F     ;internal 11 - 320x2x1 GTIA "color":
EE59  04                        DB      $04     ;internal 12 - 40x5x8 characters
EE5A  05                        DB      $05     ;internal 13 - 40x5x16 characters
EE5B  0C                        DB      $0C     ;internal 14 - 160x2x1 graphics
EE5C  0E                        DB      $0E     ;internal 15 - 160x4x1 graphics


                        **      TDLV - Table of Display List Vulnerability
                        *
                        *       Entry n is non-zero if the display list for mode n :
                        *       cross a page boundary.

EE5D  00                TDLV    DB      0       ;0
EE5E  00                        DB      0       ;1
EE5F  00                        DB      0       ;2
EE60  00                        DB      0       ;3
EE61  00                        DB      0       ;4
EE62  00                        DB      0       ;5
EE63  00                        DB      0       ;6
EE64  01                        DB      1       ;7
EE65  01                        DB      1       ;8
EE66  01                        DB      1       ;9
EE67  01                        DB      1       ;10
EE68  01                        DB      1       ;11
EE69  00                        DB      0       ;12
EE6A  00                        DB      0       ;13
EE6B  01                        DB      1       ;14
EE6C  01                        DB      1       ;15
```

```
                        **      TLSC - Table of Left Shift Counts
                        *
                        *       Entry n is the NUMBER OF LEFT SHIFTS NEEDED TO MULT:
                        *       COLCRS BY # BYTES/ROW ((ROWCRS*5)/(2**TLSC)) for mo:

EE6D  03                TLSC    DB      3       ;0
EE6E  02                        DB      2       ;1
EE6F  02                        DB      2       ;2
EE70  01                        DB      1       ;3
EE71  01                        DB      1       ;4
EE72  02                        DB      2       ;5
EE73  02                        DB      2       ;6
EE74  03                        DB      3       ;7
EE75  03                        DB      3       ;8
EE76  03                        DB      3       ;9
EE77  03                        DB      3       ;10
EE78  03                        DB      3       ;11
EE79  03                        DB      3       ;12
EE7A  03                        DB      3       ;13
EE7B  02                        DB      2       ;14
EE7C  03                        DB      3       ;15


                        **      TMCC - Table of Mode Column Counts
                        *
                        *       Entry n is the low column count for mode n.

EE7D  28                TMCC    DB      low 40          ;0
EE7E  14                        DB      low 20          ;1
EE7F  14                        DB      low 20          ;2
EE80  28                        DB      low 40          ;3
EE81  50                        DB      low 80          ;4
EE82  50                        DB      low 80          ;5
EE83  A0                        DB      low 160         ;6
EE84  A0                        DB      low 160         ;7
EE85  40                        DB      low 320         ;8
EE86  50                        DB      low 80          ;9
EE87  50                        DB      low 80          ;10
EE88  50                        DB      low 80          ;11
EE89  28                        DB      low 40          ;12
EE8A  28                        DB      low 40          ;13
EE8B  A0                        DB      low 160         ;14
EE8C  A0                        DB      low 160         ;15
```

```
                        **      TMRC - Table of Mode Row Counts
                        *
                        *       Entry n is the row count for mode n.


EE8D  18                TMRC    DB      24      ;0
EE8E  18                        DB      24      ;1
EE8F  0C                        DB      12      ;2
EE90  18                        DB      24      ;3
EE91  30                        DB      48      ;4
EE92  30                        DB      48      ;5
EE93  60                        DB      96      ;6
EE94  60                        DB      96      ;7
EE95  C0                        DB      192     ;8
EE96  C0                        DB      192     ;9
EE97  C0                        DB      192     ;10
EE98  C0                        DB      192     ;11
EE99  18                        DB      24      ;12
EE9A  0C                        DB      12      ;13
EE9B  C0                        DB      192     ;14
EE9C  C0                        DB      192     ;15


                        **      TRSC - Table of Right Shift Counts
                        *
                        *       Entry n is HOW MANY RIGHT SHIFTS FOR HCRSR FOR PART;
                        *       BYTE MODES for mode n.


EE9D  00                TRSC    DB      0       ;0
EE9E  00                        DB      0       ;1
EE9F  00                        DB      0       ;2
EEA0  02                        DB      2       ;3
EEA1  03                        DB      3       ;4
EEA2  02                        DB      2       ;5
EEA3  03                        DB      3       ;6
EEA4  02                        DB      2       ;7
EEA5  03                        DB      3       ;8
EEA6  01                        DB      1       ;9
EEA7  01                        DB      1       ;10
EEA8  01                        DB      1       ;11
EEA9  00                        DB      0       ;12
EEAA  00                        DB      0       ;13
EEAB  03                        DB      3       ;14
EEAC  02                        DB      2       ;15
```

```
                        **      TDSM - Table of Display Masks
                        *
                        *       NOTES
                        *               Includes TBTM - Table of Bit Masks.

EEAD  FF                TDSM    DB      $FF     ;1
EEAE  F0                        DB      $F0     ;2
EEAF  0F                        DB      $0F     ;3
EEB0  C0                        DB      $C0     ;4
EEB1  30                        DB      $30     ;5
EEB2  0C                        DB      $0C     ;6
EEB3  03                        DB      $03     ;7

EEB4  80                TBTM    DB      $80     ;8 (0)
EEB5  40                        DB      $40     ;9 (1)
EEB6  20                        DB      $20     ;10 (2)
EEB7  10                        DB      $10     ;11 (3)
EEB8  08                        DB      $08     ;12 (4)
EEB9  04                        DB      $04     ;13 (5)
EEBA  02                        DB      $02     ;14 (6)
EEBB  01                        DB      $01     ;15 (7)
```

```
EEBC                    **      PHE - Perform Peripheral Handler Entry
                        *
                        *       PHE attempts to enter a peripheral handler in the h:
                        *
                        *       ENTRY   JSR     PHE
                        *               X = device code
                        *               A = high linkage table address
                        *               Y = low linkage table address
                        *
                        *       EXIT
                        *               Success:
                        *               C clear
                        *               Handler table entry made
                        *
                        *               Failure due to entry previously made:
                        *               C set
                        *               N clear
                        *               X = offset to second byte of duplicate entr:
                        *               A, Y unchanged
                        *
                        *               Failure due to handler table full:
                        *               C set
                        *               N set
                        *
                        *       CHANGES
                        *               A X Y
                        *
                        *       CALLS
                        *               -none-
                        *
                        *       MODS
                        *               R. S. Scheiman  04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = EEBC            PHE     =       *               ;entry

                        ;       Initialize.

EEBC  48                        PHA                     ;save high linkage table address
EEBD  98                        TYA
EEBE  48                        PHA                     ;save low linkage table address

                        ;       Search for device code in handler table.

EEBF  8A                        TXA                             ;device code
EEC0  A200                      LDX     #0                      ;offset to first entry of t:

EEC2  DD1A03            PHE1    CMP     HATABS,X                ;device code from table
EEC5  F01E ^EEE5                BEQ     PHE3                    ;if device code found

EEC7  E8                        INX
EEC8  E8                        INX
EEC9  E8                        INX
EECA  E022                      CPX     #MAXDEV+1               ;offset+1 of last possible :
EECC  30F4 ^EEC2                BMI     PHE1                    ;if not done
```

```
                        ;       Search for empty entry in handler table.

EECE  A200                      LDX     #0              ;offset to first entry of t:
EED0  A8                        TAY                     ;save device code
EED1  A900                      LDA     #0

EED3  DD1A03            PHE2    CMP     HATABS,X        ;device code from table
EED6  F013 ^EEEB                BEQ     PHE4            ;if empty entry found

EED8  E8                        INX
EED9  E8                        INX
EEDA  E8                        INX
EEDB  E022                      CPX     #MAXDEV+1       ;offset+1 of last possible :
EEDD  30F4 ^EED3                BMI     PHE2            ;if not done

                        ;       Return table full condition.

EEDF  68                        PLA             ;clean stack
EEE0  68                        PLA
EEE1  A0FF                      LDY     #$FF    ;indicate table full (set N)
EEE3  38                        SEC             ;indicate failure
EEE4  60                        RTS             ;return

                        ;       Return device code found condition.

EEE5  68                PHE3    PLA             ;saved Y
EEE6  A8                        TAY             ;restore Y
EEE7  68                        PLA             ;restore A
EEE8  E8                        INX             ;indicate device code found (clear :
EEE9  38                        SEC             ;indicate failure
EEEA  60                        RTS             ;return

                        ;       Enter handler in table.

EEEB  98                PHE4    TYA                     ;device code
EEEC  9D1A03                    STA     HATABS,X        ;enter device code
EEEF  68                        PLA                     ;saved low linkage table ad:
EEF0  9D1B03                    STA     HATABS+1,X      ;low address
EEF3  68                        PLA                     ;saved high linkage table a:
EEF4  9D1C03                    STA     HATABS+2,X      ;high address

                        ;       Return success condition.

EEF7  18                        CLC             ;indicate success
EEF8  60                        RTS             ;return
```

```
                        **      PHO - Perform Peripheral Handler Poll at OPEN
                        *
                        *       Subroutine to perform Type 4 Poll at OPEN time, and
                        *       "provisionally" open IOCB if peripheral answers.
                        *
                        *       Input parameters:
                        *       ICIDNO identifies calling IOCB;
                        *       From zero-page IOCB:
                        *               ICBALZ,ICBAHZ (buffer pointer)
                        *               ICDNOZ (device number from caller's filespe:
                        *       From caller's buffer: device name (in filespec.).
                        *
                        *       Output parameters:
                        *       "No device" error returned if Poll not answered.
                        *       If poll is answered, the calling IOCB is "Provision:
                        *               opened (and successful status is returned)-:
                        *               ICHIDZ set to mark provisional open
                        *               ICPTLZ,ICPTHZ points to PTL (special PUT-BY:
                        *               ICSPR in calling IOCB set to device name (f:
                        *               ICSPR+1 in calling IOCB set to device seria:
                        *
                        *       Modified:
                        *       Registers not saved.
                        *
                        *       Subroutines called:
                        *       PHP performs poll.
                        *
                        *       ENTRY   JSR     PHO
                        *
                        *       NOTES
                        *               Problem: in the CRASS65 version, ICIDNO was:
                        *               zero-page.
                        *
                        *       MODS
                        *               R. S. Scheiman  04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = EEF9        PHO     =       *                   ;entry
EEF9 A000                  LDY     #0                  ;Call for Type 4 Poll with
EEFB B124                  LDA     (ICBALZ),Y          ;device name from user
EEFD A421                  LDY     ICDNOZ              ;OPEN
EEFF 208EE7                JSR     PHP
EF02 1003 ^EF07            BPL     PHO1                ;if poll answered

EF04 A082                  LDY     #NONDEV             ;Return "no device" error
EF06 60                    RTS                         ;return

EF07 A97F          PHO1    LDA     #$7F                ;"Provisionally" OPEN the I:
EF09 8520                  STA     ICHIDZ              ;(Mark "provisional")
EF0B A925                  LDA     #low [PTL-1]
EF0D 8526                  STA     ICPTLZ              ;(Special put byte routine :
EF0F A9EF                  LDA     #high [PTL-1]
EF11 8527                  STA     ICPTHZ
EF13 ADEC02                LDA     DVSTAT+2            ;(Peripheral address for lo:
                   ;       LDX     ICIDNO
```

```
EF16  AE2E00            VFD     8\$AE,8\low ICIDNO,8\high ICIDNO
EF19  9D4D03            STA     ICSPR+1,X
EF1C  A000              LDY     #0
EF1E  B124              LDA     (ICBALZ),Y      ;(Device name from user)
EF20  9D4C03            STA     ICSPR,X
EF23  A001              LDY     #SUCCES         ;indicate success
EF25  60                RTS                     ;return


                **      PTL - Perform PUT-BYTE for Provisionally Open IOCB
                *
                *       Put byte entry for provisionally opened IOCB's.
                *       This routine performs load, relocation, initializat:
                *       and finishes OPEN, then calls handler's put byte en:
                *
                *       Input parameters:
                *       A       Byte to output;
                *       X       IOCB index (IOCB number times 16);
                *       Y       "Function not supported" error code $92.
                *       AUX1 and AUX2 in zero-page IOCB are copied from the:
                *               IOCB prior to the call to PTL.
                *
                *       Output parameters:
                *       Various errors may be returned if loading fails (ei:
                *               did not allow loading by setting HNDLOD fla:
                *               was a loading error or calling error);
                *       If no loading error, this routine returns nothing--:
                *               returned is returned by the loaded PUT-BYTE:
                *               is called by this routine after the handler:
                *               initialized, and opened.
                *
                *       Modified:
                *       ICIDNO (a CIO variable);
                *       all of the zero-page IOCB is copied from the callin:
                *       normal CIO open-operation variables are affected;
                *       after opening, the zero-page IOCB is copied to the :
                *       Registers not saved if error return;if handler is l:
                *               and opened properly, the caller's A and X r:
                *               passed to the loaded handler's PUT-BYTE rou:
                *               Y is passed to that routine as $92)--then r:
                *               on return is up to handler PUT-BYTE since i:
                *               directly to caller.
                *
                *       Subroutines called:
                *       PHL (does loading, initializing and opening--calls :
                *       loaded handler's INIT, OPEN, and PUT-BYTE entries a:
                *       The PUT-BYTE entry returns directly to the PTL call:
                *
                *       ENTRY   JSR     PTL
                *
                *       NOTES
                *               Problem: in the CRASS65 version, ICIDNO was:
                *               zero-page.
                *
                *       MODS
```

```
                        *               R. S. Scheiman  04/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


        = EF26          PTL     =       *       ;entry
EF26    48                      PHA             ;save byte to output
EF27    8A                      TXA             ;IOCB index
EF28    48                      PHA             ;save IOCB index
EF29    290F                    AND     #$0F    ;IOCB index modulo 16
EF2B    D010 ^EF3D              BNE     PTL2    ;if IOCB not divisble by 16, error

EF2D    E080                    CPX     #MAXIOC
EF2F    100C ^EF3D              BPL     PTL2    ;if IOCB index invalid

EF31    ADE902                  LDA     HNDLOD
EF34    D008 ^EF41              BNE     PTL3    ;if user wants loading

EF36    A082                    LDY     #NONDEV ;indicate nonexistent device error

                        ;       Return error.

EF38    68              PTL1    PLA             ;clean stack
EF39    68                      PLA
EF3A    C000                    CPY     #0      ;indicate failure (set N)
EF3C    60                      RTS             ;return

EF3D    A086            PTL2    LDY     #BADIOC ;indicate bad IOCB number error
EF3F    30F7 ^EF38              BMI     PTL1    ;return error

                        ;       Simulate beginning of CIO, since CIO bypassed.

EF41                    PTL3
                        ;       STX     ICIDNO  ;IOCB index
EF41    8E2E00                  VFD     8\$8E,8\low ICIDNO,8\high ICIDNO
EF44    A000                    LDY     #0      ;offset to first byte of page zero :

                        ;       Copy IOCB to page zero IOCB.

EF46    BD4003          PTL4    LDA     IOCB,X  ;byte of IOCB
EF49    992000                  STA     ZIOCB,Y ;byte of page zero IOCB
EF4C    E8                      INX
EF4D    C8                      INY
EF4E    C00C                    CPY     #12
EF50    30F4 ^EF46              BMI     PTL4    ;if not done

EF52    2029CA                  JSR     PHL     ;load and initialize peripheral han:
EF55    30E1 ^EF38              BMI     PTL1    ;if error

EF57    68                      PLA             ;Re-do the put byte call,
EF58    AA                      TAX             ;this time calling real handler...
EF59    68                      PLA
EF5A    A8                      TAY
EF5B    A527                    LDA     ICPTHZ
EF5D    48                      PHA             ;(Put byte entry address minus one)
EF5E    A526                    LDA     ICPTLZ
EF60    48                      PHA
```

```
EF61  98                TYA
EF62  A092              LDY     #FNCNOT
EF64  60                RTS             ;invoke handler (address on stack)
```

```
EF6B                    FIX     $EF6B


                **      EF6B - $EF6B Patch
                *
                *       For compatibility with OS Revision B, initiate cass:

EF6B  4C05FD            JMP     ICR     ;initiate cassette READ, return
```

```
EF6E                    **      SIN - Initialize Screen
                        *
                        *       ENTRY   JSR     SIN
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = EF6E          SIN     =       *               ;entry

EF6E    A9FF                    LDA     #$FF            ;clear code indicator
EF70    8DFC02                  STA     CH              ;key code

EF73    ADE402                  LDA     RAMSIZ          ;size of RAM
EF76    856A                    STA     RAMTOP          ;RAM size

EF78    A940                    LDA     #$40            ;CAPS lock indicator
EF7A    8DBE02                  STA     SHFLOK          ;shift/control lock flags

EF7D    A951                    LDA     #low TCKD       ;table of character key def:
EF7F    8579                    STA     KEYDEF          ;key definition table addre:
EF81    A9FB                    LDA     #high TCKD
EF83    857A                    STA     KEYDEF+1

EF85    A911                    LDA     #low TFKD       ;table of function key defi:
EF87    8560                    STA     FKDEF           ;function key definition ta:
EF89    A9FC                    LDA     #high TFKD
EF8B    8561                    STA     FKDEF+1

EF8D    60                      RTS                     ;return


                        **      SOP - Perform Screen OPEN
                        *
                        *       ENTRY   JSR     SOP
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = EF8E          SOP     =       *       ;entry

                        ;       Check mode.

EF8E    A52B                    LDA     ICAX2Z
EF90    290F                    AND     #$0F
EF92    D008 ^EF9C              BNE     COC     ;if not mode 0, complete OPEN comma:

                        ;       Process mode 0.

                        ;       JMP     EOP     ;perform editor OPEN, return
```

```
                        **      EOP - Perform Editor OPEN
                        *
                        *       ENTRY   JSR     EOP
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = EF94            EOP     =       *       ;entry
EF94  A52A                      LDA     ICAX1Z
EF96  290F                      AND     #$0F
EF98  852A                      STA     ICAX1Z
EF9A  A900                      LDA     #0
                        ;       JMP     COC     ;complete OPEN command, return



                        **      COC - Complete OPEN Command
                        *
                        *       ENTRY   JSR     COC
                        *               A = mode
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = EF9C            COC     =       *       ;entry

                        ;       Check mode.

EF9C  8557                      STA     DINDEX  ;save mode
EF9E  C910                      CMP     #16
EFA0  9005 ^EFA7                BCC     COC1    ;if mode within range

                        ;       Process invalid mode.

EFA2  A991                      LDA     #BADMOD
EFA4  4C54F1                    JMP     COC17

                        ;       Initialize for OPEN.

EFA7  A9E0              COC1    LDA     #high DCSORG    ;high domestic character se:
EFA9  8DF402                    STA     CHBAS           ;character set base
EFAC  A9CC                      LDA     #high ICSORG    ;high international charact:
EFAE  8D6B02                    STA     CHSALT          ;alternate character set ba:
EFB1  A902                      LDA     #2
EFB3  8DF302                    STA     CHACT
EFB6  8D2F02                    STA     SDMCTL          ;turn off DMA
EFB9  A901                      LDA     #SUCCES
EFBB  854C                      STA     DSTAT           ;clear status
EFBD  A9C0                      LDA     #$C0            ;enable IRQ
EFBF  0510                      ORA     POKMSK
```

```
EFC1  8510                   STA     POKMSK
EFC3  8D0E02                 STA     IRQEN

                    ;        Set DLI status.

EFC6  A940                   LDA     #$40             ;disable DLI
EFC8  8D0ED4                 STA     NMIEN
EFCB  2C6E02                 BIT     FINE
EFCE  100C ^EFDC             BPL     COC2             ;if not fine scrolling (VBL:

EFD0  A9C4                   LDA     #low FDL
EFD2  8D0002                 STA     VDSLST           ;DLI vector
EFD5  A9FC                   LDA     #high FDL
EFD7  8D0102                 STA     VDSLST+1
EFDA  A9C0                   LDA     #$C0

EFDC  8D0ED4        COC2     STA     NMIEN

                    ;        Clear control.

EFDF  A900                   LDA     #0
EFE1  8D9302                 STA     TINDEX           ;clear text index (must alw:
EFE4  8564                   STA     ADRESS
EFE6  857B                   STA     SWPFLG
EFE8  8DF002                 STA     CRSINH

                    ;        Set initial tab stops.

EFEB  A00E                   LDY     #14              ;offset to last byte of bit:
EFED  A901                   LDA     #$01             ;tab stop every 8 character:

EFEF  994302        COC3     STA     TABMAP,Y         ;set tab stop
EFF2  88                     DEY
EFF3  10FA ^EFEF             BPL     COC3             ;if not done

                    ;        Load initialize color register shadows.

EFF5  A204                   LDX     #4               ;offset to last color regis:

EFF7  BD08FB        COC4     LDA     TDSC,X           ;default screen color
EFFA  9DC402                 STA     COLOR0,X         ;set color register shadow
EFFD  CA                     DEX
EFFE  10F7 ^EFF7             BPL     COC4             ;if not done

                    ;        Set up.

F000  A46A                   LDY     RAMTOP           ;(high) RAM size
F002  88                     DEY                      ;decrement (high) RAM size
F003  8C9502                 STY     TXTMSC+1
F006  A960                   LDA     #low [$0000-160]         ;low RAM size - 160
F008  8D9402                 STA     TXTMSC
F00B  A657                   LDX     DINDEX           ;mode
F00D  BD4DEE                 LDA     TAGM,X           ;convert to ANTIC code
F010  8551                   STA     HOLD1            ;ANTIC code
F012  A56A                   LDA     RAMTOP           ;(high) RAM size
F014  8565                   STA     ADRESS+1
```

```
                        ;       Allocate memory.

F016  BC1DEE                    LDY     TSMA,X          ;number of 40-byte blocks t:

F019  A928          COC5        LDA     #40             ;40 bytes
F01B  207AF5                    JSR     DBS             ;perform double byte subtra:
F01E  88                        DEY
F01F  D0F8 ^F019                BNE     COC5            ;if not done

                        ;       Clear GTIA modes.

F021  AD6F02                    LDA     GPRIOR
F024  293F                      AND     #$3F            ;clear GTIA modes
F026  8567                      STA     OPNTMP+1
F028  A8                        TAY

                        ;       Determine mode.

F029  E008                      CPX     #8
F02B  901F ^F04C                BCC     COC7            ;if mode < 8

F02D  E00F                      CPX     #15
F02F  F00D ^F03E                BEQ     COC6            ;if mode 15

F031  E00C                      CPX     #12
F033  B017 ^F04C                BCS     COC7            ;if mode >= 12

                        ;       Process modes 9, 10 and 11.

F035  8A                        TXA                     ;mode
F036  6A                        ROR     A
F037  6A                        ROR     A
F038  6A                        ROR     A
F039  29C0                      AND     #$C0            ;extract 2 low bits (in 2 h:
F03B  0567                      ORA     OPNTMP+1
F03D  A8                        TAY

                        ;       Establish line boundary at X000.

F03E  A910          COC6        LDA     #16             ;subtract 16 for page bound:
F040  207AF5                    JSR     DBS             ;perform double byte subtra:

                        ;       Check for mode 11.

F043  E00B                      CPX     #11
F045  D005 ^F04C                BNE     COC7            ;if mode 11

                        ;       Set GTIA luminance.

F047  A906                      LDA     #6              ;GTIA luminance value
F049  8DC802                    STA     COLOR4          ;background color

                        ;       Set new priority.

F04C  8C6F02        COC7        STY     GPRIOR          ;new priority

                        ;       Set memory scan counter.
```

```
F04F  A564               LDA   ADRESS         ;memory scan counter
F051  8558               STA   SAVMSC         ;save memory scan counter
F053  A565               LDA   ADRESS+1
F055  8559               STA   SAVMSC+1

                   ;     Wait for VBLANK.

F057  AD0BD4      COC8   LDA   VCOUNT
F05A  C97A               CMP   #$7A
F05C  D0F9 ^F057         BNE   COC8           ;if VBLANK has not occured

                   ;     Put display list under RAM.

F05E  2078F5             JSR   DSD            ;perform double byte single:
F061  BD5DEE             LDA   TDLV,X         ;display list vulnerability
F064  F006 ^F06C         BEQ   COC9           ;if not vulnerable

F066  A9FF               LDA   #$FF
F068  8564               STA   ADRESS
F06A  C665               DEC   ADRESS+1       ;drop down 1 page

F06C  2065F5      COC9   JSR   DDD            ;perform double byte double:
F06F  A564               LDA   ADRESS         ;end of display list
F071  8568               STA   SAVADR         ;save address
F073  A565               LDA   ADRESS+1
F075  8569               STA   SAVADR+1

                   ;     Set up.

F077  A941               LDA   #$41           ;ANTIC wait for VBLANK and :
F079  2070F5             JSR   SDI            ;store data indirect
F07C  8666               STX   OPNTMP
F07E  A918               LDA   #24
F080  8DBF02             STA   BOTSCR

                   ;     Check for modes 9 ,10 and 11.

F083  A557               LDA   DINDEX         ;mode
F085  C90C               CMP   #12
F087  B004 ^F08D         BCS   COC10          ;if mode >= 12, mixed mode :

F089  C909               CMP   #9
F08B  B039 ^F0C6         BCS   COC12          ;if mode >= 9, mixed mode n:

                   ;     Check for mixed mode.

F08D  A52A        COC10  LDA   ICAX1Z
F08F  2910               AND   #MXDMOD
F091  F033 ^F0C6         BEQ   COC12          ;if not mixed mode

                   ;     Process mixed mode.

F093  A904               LDA   #4
F095  8DBF02             STA   BOTSCR
F098  A202               LDX   #2
F09A  AD6E02             LDA   FINE
```

```
F09D  F003 ^F0A2          BEQ    COC11              ;if not fine scrolling

F09F  20A0F5              JSR    SSE                ;set scrolling display list:

F0A2  A902        COC11   LDA    #$02
F0A4  2069F5              JSR    SDF                ;store data indirect for fi:
F0A7  CA                  DEX
F0A8  10F8 ^F0A2          BPL    COC11              ;if not done

                  ;       Reload MSC for text.

F0AA  A46A                LDY    RAMTOP             ;(high) RAM size
F0AC  88                  DEY                       ;decrement (high) RAM size
F0AD  98                  TYA
F0AE  2070F5              JSR    SDI                ;store data indirect
F0B1  A960                LDA    #low [$0000-160]          ;low RAM size - 160
F0B3  2070F5              JSR    SDI                ;store data indirect
F0B6  A942                LDA    #$42               ;fine scrolling
F0B8  2069F5              JSR    SDF                ;store data indirect
F0BB  18                  CLC
F0BC  A910                LDA    #MXDMOD
F0BE  6566                ADC    OPNTMP
F0C0  A8                  TAY
F0C1  BE2DEE              LDX    TDLE,Y
F0C4  D015 ^F0DB          BNE    COC13

                  ;       Check mode.

F0C6  A466        COC12   LDY    OPNTMP
F0C8  BE2DEE              LDX    TDLE,Y             ;number of display list ent:
F0CB  A557                LDA    DINDEX             ;mode
F0CD  D00C ^F0DB          BNE    COC13              ;if not mode 0

                  ;       Check for fine scrolling.

F0CF  AD6E02              LDA    FINE               ;fine scrolling flag
F0D2  F007 ^F0DB          BEQ    COC13              ;if not fine scrolling

                  ;       Process fine scrolling.

F0D4  20A0F5              JSR    SSE                ;set scrolling display list:
F0D7  A922                LDA    #$22
F0D9  8551                STA    HOLD1

                  ;       Continue.

F0DB  A551        COC13   LDA    HOLD1
F0DD  2070F5              JSR    SDI                ;store data indirect
F0E0  CA                  DEX
F0E1  D0F8 ^F0DB          BNE    COC13              ;if not done

                  ;       Determine mode.

F0E3  A557                LDA    DINDEX             ;mode
F0E5  C908                CMP    #8
F0E7  9026 ^F10F          BCC    COC16              ;if mode < 8
```

```
F0E9  C90F                 CMP     #15
F0EB  F004 ^F0F1           BEQ     COC14           ;if mode 15

F0ED  C90C                 CMP     #12
F0EF  B01E ^F10F           BCS     COC16           ;if mode >= 12

                    ;      Process modes 8, 9, 10, 11 and 15.

F0F1  A25D          COC14  LDX     #93             ;remaining number of DLE's
F0F3  A56A                 LDA     RAMTOP          ;(high) RAM size
F0F5  38                   SEC
F0F6  E910                 SBC     #high $1000     ;subtract 4K
F0F8  2070F5               JSR     SDI             ;store data indirect
F0FB  A900                 LDA     #low $0000
F0FD  2070F5               JSR     SDI             ;store data indirect
F100  A551                 LDA     HOLD1           ;ANTIC MSC code
F102  0940                 ORA     #$40
F104  2070F5               JSR     SDI             ;store data indirect

F107  A551          COC15  LDA     HOLD1           ;remaining DLE's
F109  2070F5               JSR     SDI             ;store data indirect
F10C  CA                   DEX
F10D  D0F8 ^F107           BNE     COC15           ;if DLE's remain

                    ;      Complete display list with LMS.

F10F  A559          COC16  LDA     SAVMSC+1        ;high saved memory scan cou:
F111  2070F5               JSR     SDI             ;store data indirect
F114  A558                 LDA     SAVMSC          ;low saved memory scan coun:
F116  2070F5               JSR     SDI             ;store data indirect
F119  A551                 LDA     HOLD1
F11B  0940                 ORA     #$40
F11D  2070F5               JSR     SDI             ;store data indirect
F120  A970                 LDA     #$70            ;8 blank lines
F122  2070F5               JSR     SDI             ;store data indirect
F125  A970                 LDA     #$70            ;8 blank lines
F127  2070F5               JSR     SDI             ;store data indirect
F12A  A564                 LDA     ADRESS          ;display list address
F12C  8D3002               STA     SDLSTL          ;save display list address
F12F  A565                 LDA     ADRESS+1
F131  8D3102               STA     SDLSTL+1
F134  A970                 LDA     #$70            ;8 blank lines
F136  2070F5               JSR     SDI             ;store data indirect
F139  A564                 LDA     ADRESS          ;display list address
F13B  8DE502               STA     MEMTOP          ;update top of memory
F13E  A565                 LDA     ADRESS+1
F140  8DE602               STA     MEMTOP+1
F143  A001                 LDY     #1              ;offset
F145  AD3002               LDA     SDLSTL          ;saved display list address
F148  9168                 STA     (SAVADR),Y
F14A  C8                   INY
F14B  AD3102               LDA     SDLSTL+1
F14E  9168                 STA     (SAVADR),Y

                    ;      Check status.

F150  A54C                 LDA     DSTAT           ;status
```

```
F152  1010 ^F164            BPL     COC18          ;if no error

                    ;       Process error.

F154  8DEC03        COC17   STA     DERRF          ;screen OPEN error flag
F157  2094EF                JSR     EOP            ;perform editor OPEN
F15A  ADEC03                LDA     DERRF          ;restore status
F15D  A000                  LDY     #0             ;no screen OPEN error indic:
F15F  8CEC03                STY     DERRF          ;screen OPEN error flag
F162  A8                    TAY                    ;status
F163  60                    RTS                    ;return

                    ;       Check clear inhibit.

F164  A52A          COC18   LDA     ICAX1Z
F166  2920                  AND     #$20           ;extract clear inhibit bit
F168  D00B ^F175            BNE     COC19          ;if clear inhibited

                    ;       Clear screen.

F16A  2020F4                JSR     CSC            ;clear screen
F16D  8D9002                STA     TXTROW         ;set cursor at top row
F170  A552                  LDA     LMARGN         ;left margin
F172  8D9102                STA     TXTCOL         ;set cursor at left margin

                    ;       Exit.

F175  A922          COC19   LDA     #$22           ;turn on DMA control
F177  0D2F02                ORA     SDMCTL
F17A  8D2F02                STA     SDMCTL
F17D  4C0BF2                JMP     SEC            ;set exit conditions, retur:


                    **      SGB - Perform Screen GET-BYTE
                    *
                    *       ENTRY   JSR     SGB
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = F180        SGB     =       *       ;entry
F180  20CAF6                JSR     CCR     ;check cursor range
F183  208FF1                JSR     GDC     ;get data under cursor
F186  206AF7                JSR     CIA     ;convert internal character to ATAS:
F189  200AF6                JSR     SZA     ;set zero data and advance cursor
F18C  4C1EF2                JMP     SST     ;perform screen STATUS, return
```

```
                        **      GDC - Get Data under Cursor
                        *
                        *       ENTRY   JSR     GDC
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = F18F            GDC     =       *       ;entry
F18F  20ACF5                    JSR     CCA     ;convert cursor row/column to addre:
F192  B164                      LDA     (ADRESS),Y
F194  2D4002                    AND     DMASK

F197  466F              GDC1    LSR     SHFAMT  ;shift data down to low bits
F199  B003 ^F19E                BCS     GDC2    ;if done

F19B  4A                        LSR     A
F19C  10F9 ^F197                BPL     GDC1    ;continue shifting

F19E  8DFA02            GDC2    STA     CHAR
F1A1  C900                      CMP     #0      ;retore flags
F1A3  60                        RTS             ;return



                        **      SPB - Perform Screen PUT-BYTE
                        *
                        *       ENTRY   JSR     SPB
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = F1A4            SPB     =       *       ;entry
F1A4  8DFB02                    STA     ATACHR

                        ;       JSR     ROD     ;restore old data under cursor

F1A7  C97D                      CMP     #CLS
F1A9  D006 ^F1B1                BNE     SPB1    ;if not clear screen

F1AB  2020F4                    JSR     CSC     ;clear screen
F1AE  4C06F2                    JMP     SEC     ;set exit conditions, return

F1B1  20CAF6            SPB1    JSR     CCR     ;check cursor range
                        ;       JMP     CEL     ;check EOL, return
```

```
                    **      CEL - Check End of Line
                    *
                    *       ENTRY   JSR     CEL
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


     = F1B4         CEL     =       *       ;entry
F1B4 ADFB02                 LDA     ATACHR
F1B7 C99B                   CMP     #EOL
F1B9 D006 ^F1C1             BNE     CEL1    ;if not EOL

F1BB 2061F6                 JSR     RWS     ;return with scrolling
F1BE 4C0BF2                 JMP     SEC     ;set exit conditions, return

F1C1 20CAF1         CEL1    JSR     PLO     ;plot point
F1C4 200EF6                 JSR     SEA     ;set EOL data and advance cursor
F1C7 4C0BF2                 JMP     SEC     ;set exit conditions, return



                    **      PLO - Plot Point
                    *
                    *       ENTRY   JSR     PLO
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


     = F1CA         PLO     =       *               ;entry

                    ;       Wait for start/stop flag clear.

F1CA ADFF02         PLO0    LDA     SSFLAG          ;start/stop flag
F1CD D0FB ^F1CA             BNE     PLO0            ;if start/stop flag non-zer:

                    ;       Save cursor row/column.

F1CF A202                   LDX     #2              ;offset to last byte

F1D1 B554           PLO1    LDA     ROWCRS,X        ;byte of cursor row/column
F1D3 955A                   STA     OLDROW,X        ;save byte of cursor row/co:
F1D5 CA                     DEX
F1D6 10F9 ^F1D1             BPL     PLO1            ;if not done

                    ;       Convert ATASCII character to internal.

F1D8 ADFB02                 LDA     ATACHR          ;character
F1DB A8                     TAY                     ;character
F1DC 2A                     ROL     A
F1DD 2A                     ROL     A
```

```
                        **      SEC - Set Exit Conditions
                        *
                        *       ENTRY   JSR     SEC
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


        = F208          SEC     =       *       ;entry
F208    208FF1                  JSR     GDC     ;get data under cursor
F20E    855D                    STA     OLDCHR
F210    A657                    LDX     DINDEX  ;mode
F212    D00A ^F21E              BNE     SST     ;if graphics, no cursor

F214    AEF002                  LDX     CRSINH  ;cursor inhibit flag
F217    D005 ^F21E              BNE     SST     ;if cursor inhibited

F219    4980                    EOR     #$80    ;complement most significant bit
F21B    20E9F1                  JSR     SPQ     ;display
                        ;       JMP     SST     ;perform screen status, return



                        **      SST - Perform Screen STATUS
                        *
                        *       ENTRY   JSR     SST
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


        = F21E          SST     =       *       ;entry
F21E    A44C                    LDY     DSTAT   ;status
F220    4C26F2                  JMP     SST1    ;continue
```

```
F1DE  2A                    ROL     A
F1DF  2A                    ROL     A
F1E0  2903                  AND     #3
F1E2  AA                    TAX                         ;index into TAIC
F1E3  98                    TYA                         ;character
F1E4  299F                  AND     #$9F                ;strip off column address
F1E6  1D49FB                ORA     TAIC,X              ;or in new column address
                  ;         JMP     SPQ                 ;display, return



                  **        SPQ - Display
                  *
                  *         ENTRY   JSR     SPQ
                  *
                  *         MODS
                  *                 Original Author Unknown
                  *                 1. Bring closer to Coding Standard (object :
                  *                    R. K. Nordin 11/01/83

      = F1E9      SPQ       =       *                   ;entry

                  ;         Set CHAR.

F1E9  8DFA02                STA     CHAR                ;character

                  ;         Convert cursor row/column to address.

F1EC  20ACF5                JSR     CCA                 ;convert cursor row/column :

                  ;         Shift up to proper position.

F1EF  ADFA02                LDA     CHAR                ;character

F1F2  406F        SPQ1      LSR     SHFAMT
F1F4  B004 ^F1FA            BCS     SPQ2                ;if done

F1F6  0A                    ASL     A
F1F7  4CF2F1                JMP     SPQ1                ;continue shifting

                  ;         Update data.

F1FA  2DA002      SPQ2      AND     DMASK
F1FD  8550                  STA     TMPCHR              ;save shifted data
F1FF  ADA002                LDA     DMASK               ;display mask
F202  49FF                  EOR     #$FF                ;complement display mask
F204  3164                  AND     (ADRESS),Y          ;mask off old data
F206  0550                  ORA     TMPCHR              ;or in new data
F208  9164                  STA     (ADRESS),Y          ;update data
F20A  60                    RTS                         ;return
```

```
F223                          FIX     $F223



                        **      F223 - $F223 Patch
                        *
                        *       For compatibility with OS Revision B, perform power:

      = F223      PPD     =       *       ;entry
F223  4CFCC8              JMP     SES     ;select and execute self-test
```

```
F226                    ;       Continue.

F226  A901              SST1    LDA     #SUCCES ;indicate success
F228  854C                      STA     DSTAT   ;status
F22A  ADFB02                    LDA     ATACHR  ;data
                        ;       JMP     ESP     ;return



                        **      ESP - Perform Editor SPECIAL
                        *
                        *       ESP does nothing.
                        *
                        *       ENTRY   JSR     ESP
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = F22D            ESP     =       *       ;entry
F22D  60                        RTS             ;return



                        **      ECL - Perform Editor CLOSE
                        *
                        *       ENTRY   JSR     ECL
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = F22E            ECL     =       *       ;entry

                        ;       Check for fine scrolling.

F22E  2C6E02                    BIT     FINE    ;fine scrolling flag
F231  10EB ^F21E                BPL     SST     ;if not fine scrolling, perform STA:

                        ;       Process fine scrolling.

F233  A940                      LDA     #$40
F235  8D0ED4                    STA     NMIEN           ;disable DLI
F238  A900                      LDA     #0              ;clear fine scrolling flag
F23A  8D6E02                    STA     FINE
F23D  A9CE                      LDA     #low RIR        ;return from interrupt rout:
F23F  8D0002                    STA     VDSLST          ;restore initial DLI vector:
F242  A9C0                      LDA     #high RIR
F244  8D0102                    STA     VDSLST+1
F247  4C94EF                    JMP     EOP             ;perform editor OPEN, retur:
```

```
                        **      EGB - Perform Editor GET-BYTE
                        *
                        *       ENTRY   JSR     EGB
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = F24A            EGB     =       *       ;entry

                        ;       Initialize.

F24A  2062F9                    JSR     SWA     ;swap
F24D  208CF6                    JSR     CRE     ;check cursor range for editor
F250  A56B                      LDA     BUFCNT  ;buffer count
F252  D034 ^F288                BNE     EGB4    ;if something in the buffer

                        ;       Get line.

F254  A554                      LDA     ROWCRS          ;cursor row
F256  856C                      STA     BUFSTR          ;buffer start pointer
F258  A555                      LDA     COLCRS          ;low cursor column
F25A  856D                      STA     BUFSTR+1        ;high buffer start pointer

F25C  20FDF2            EGB1    JSR     KGB     ;perform keyboard GET-BYTE
F25F  844C                      STY     DSTAT   ;status
F261  ADFB02                    LDA     ATACHR  ;ATASCII character
F264  C99B                      CMP     #EOL
F266  F012 ^F27A                BEQ     EGB3    ;if EOL

F268  208EF2                    JSR     PCH     ;process character
F26B  2062F9                    JSR     SWA     ;swap
F26E  A563                      LDA     LOGCOL  ;logical column
F270  C971                      CMP     #113    ;column near column 120
F272  D003 ^F277                BNE     EGB2    ;if not near column 120, no beep

F274  2056F5                    JSR     BEL     ;beep

F277  4C5CF2            EGB2    JMP     EGB1    ;process next character

                        ;       Process EOL.

F27A  2018F7            EGB3    JSR     ROD             ;restore old data under cur:
F27D  2081F8                    JSR     CBC             ;compute buffer count
F280  A56C                      LDA     BUFSTR          ;buffer start pointer
F282  8554                      STA     ROWCRS          ;cursor row
F284  A56D                      LDA     BUFSTR+1        ;high buffer start pointer
F286  8555                      STA     COLCRS          ;low cursor column

                        ;       Check buffer count.

F288  A56B              EGB4    LDA     BUFCNT  ;buffer count
F28A  F011 ^F29D                BEQ     EGB6    ;if buffer count zero

                        ;       Decrement and check buffer count.
```

```
F28C  C666        EGB5    DEC     BUFCNT  ;decrement buffer count
F28E  F00D ^F29D          BEQ     EGB6    ;if buffer count zero

                  ;       Check status.

F290  A54C                LDA     DSTAT   ;status
F292  30F8 ^F28C          BMI     EGB5    ;if error, continue decrementing.

                  ;       Perfrom GET-BYTE.

F294  2080F1              JSR     SGB     ;perform screen GET-BYTE
F297  8DFB02              STA     ATACHR  ;ATASCII character
F29A  4C62F9              JMP     SWA     ;swap, return

                  ;       Exit.

F29D  2061F6      EGB6    JSR     RWS     ;return with scrolling
F2A0  A99B                LDA     #EOL
F2A2  8DFB02              STA     ATACHR  ;ATASCII character
F2A5  2006F2              JSR     SEC     ;set exit conditions
F2A8  844C                STY     DSTAT   ;status
F2AA  4C62F9              JMP     SWA     ;swap, return



                  **      IRA - Invoke Routine Pointed to by ADRESS
                  *
                  *       ENTRY   JSR     IRA
                  *
                  *       MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83


   = F2AD         IRA     =       *               ;entry
F2AD  6C6400              JMP     (ADRESS)        ;execute, return



                  **      EPB - Perform Editor PUT-BYTE
                  *
                  *       ENTRY   JSR     EBP
                  *
                  *       MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83


   = F2B0         EPB     =       *       ;entry
F2B0  8DFB02              STA     ATACHR  ;ATASCII character
F2B3  2062F9              JSR     SWA     ;swap
F2B6  20BCF6              JSR     CRE     ;check cursor range for editor
```

```
F2B9  A900                  LDA     #0
F2BB  8DE803                STA     SUPERF  ;clear super function flag
                    ;       JMP     PCH     ;process character, return



                    **      PCH - Process Character
                    *
                    *       PCH displays the character or processes control cha:
                    *       super functions (shifted function keys).
                    *
                    *       ENTRY   JSR     PCH
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = F2BE        PCH     =       *       ;entry
F2BE  2018F7                JSR     ROD     ;restore old data under cursor
F2C1  203CF9                JSR     CCC     ;check for control character
F2C4  F009 ^F2CF            BEQ     PCH2    ;if control character

                    ;       Display character.

F2C6  0EA202        PCH1    ASL     ESCFLG  ;escape flag
F2C9  2084F1                JSR     CEL     ;check EOL
F2CC  4C62F9                JMP     SWA     ;swap, return

                    ;       Process control character.

F2CF  ADFE02        PCH2    LDA     DSPFLG  ;display flag
F2D2  0DA202                ORA     ESCFLG  ;escape flag
F2D5  D0EF ^F2C6            BNE     PCH1    ;if dislay or escape, display chara:

                    ;       Continue.

F2D7  0EA202                ASL     ESCFLG
F2DA  E8                    INX

                    ;       Check for super function.

F2DB  ADE803                LDA     SUPERF
F2DE  F005 ^F2E5            BEQ     PCH3            ;if not super function

                    ;       Adjust for super function.

F2E0  8A                    TXA
F2E1  18                    CLC
F2E2  6920                  ADC     #TSFR-TCCR-3
F2E4  AA                    TAX                     ;adjusted offset

                    ;       Process control character or super function.

F2E5  BD00FB        PCH3    LDA     TCCR,X          ;low routine address
```

```
F2E8  8564                  STA     ADRESS
F2EA  BD0EF3                LDA     TCCR+1,X          ;high routine address
F2ED  8565                  STA     ADRESS+1
F2EF  20ADF2                JSR     IRA               ;invoke routine pointed to :
F2F2  2008F2                JSR     SEC               ;set exit conditions
F2F5  4C62F9                JMP     SWA               ;swap, return



                  **        IGN - Ignore Character and Perform Keyboard GET-BYT:
                  *
                  *         ENTRY   JSR     IGN
                  *
                  *         EXIT
                  *                 CH = $FF
                  *
                  *         MODS
                  *                 Original Author Unknown
                  *                 1. Bring closer to Coding Standard (object :
                  *                    R. K. Nordin 11/01/83


      = F2F8      IGN       =       *       ;entry
F2F8  A9FF                  LDA     #$FF    ;clear code indicator
F2FA  8DFC02                STA     CH      ;key code
                  ;         JMP     KGB     ;perform keyborad GET-BYTE, return



                  **        KGB - Perform Keyboard GET-BYTE
                  *
                  *         ENTRY   JSR     KGB
                  *
                  *         NOTES
                  *                 Problem: byte wasted by unnecessary TAX nea:
                  *
                  *         MODS
                  *                 Original Author Unknown
                  *                 1. Bring closer to Coding Standard (object :
                  *                    R. K. Nordin 11/01/83


      = F2FD      KGB       =       *       ;entry

                  ;         Initialize.

F2FD  A900        KGB1      LDA     #0
F2FF  8DE803                STA     SUPERF  ;clear super function flag

                  ;         Check for special edit read mode.

F302  A52A                  LDA     ICAX1Z
F304  4A                    LSR     A
F305  B06F ^F376            BCS     KGB11   ;if special edit read mode
```

```
                        ;       Check for BREAK abort.

F307  A980                      LDA     #BRKABT ;assume BREAK abort
F309  A611                      LDX     BRKKEY  ;BREAK key flag
F30B  F065 ^F372                BEQ     KGB10   ;if BREAK abort

                        ;       Check for character.

F30D  ADFC02                    LDA     CH      ;key code
F310  C9FF                      CMP     #$FF    ;clear code indicator
F312  F0E9 ^F2FD                BEQ     KGB1    ;if no character

                        ;       Process character.

F314  857C                      STA     HOLDCH  ;save character
F316  A2FF                      LDX     #$FF    ;clear code indicator
F318  8EFC02                    STX     CH      ;key code

                        ;       Sound key click if desired.

F31B  AED802                    LDX     NOCLIK  ;click inhibit flag
F31E  D003 ^F323                BNE     KGB2    ;if click inhibited

F320  2083F9                    JSR     SKC     ;sound key click

                        ;       Set offset to key definition.

F323  A8                KGB2    TAY             ;save character

                        ;       Check for CTRL and SHIFT together.

F324  C0C0                      CPY     #$C0
F326  B0D0 ^F2F8                BCS     IGN     ;if CTRL and SHIFT together, ignore

                        ;       Convert to ATASCII character.

F328  B179                      LDA     (KEYDEF),Y      ;ATASCII character

                        ;       Set ATASCII character.

F32A  8DFB02            KGB3    STA     ATACHR  ;ATASCII character
F32D  AA                        TAX
F32E  3003 ^F333                BMI     KGB4    ;if special key

F330  4CB4F3                    JMP     KGB17   ;process shift/control lock

                        ;       Check for null character.

F333  C980              KGB4    CMP     #$80
F335  F0C1 ^F2F8                BEQ     IGN     ;if null, ignore

                        ;       Check for inverse video key.

F337  C981                      CMP     #$81
F339  D00A ^F345                BNE     KGB5    ;if not inverse video key

                        ;       Process inverse video key.
```

```
F33B  AD8602              LDA     INVFLG
F33E  4980                EOR     #$80
F340  8D8602              STA     INVFLG
F343  B0B3 ^F2F8          BCS     IGN      ;ignore

                  ;       Check for CAPS key.

F345  C982        KGB5    CMP     #$82
F347  D00C ^F355          BNE     KGB6     ;if not CAPS key

                  ;       Process CAPS key.

F349  ADBE02              LDA     SHFLOK   ;shift/control lock flags
F34C  F00B ^F359          BEQ     KGB7     ;if no lock, process CAPS lock

F34E  A900                LDA     #$00     ;no lock indicator
F350  8DBE02              STA     SHFLOK   ;shift/control lock flags
F353  F0A3 ^F2F8          BEQ     IGN      ;ignore

                  ;       Check for SHIFT-CAPS key.

F355  C983        KGB6    CMP     #$83
F357  D007 ^F360          BNE     KGB8     ;if not SHIFT-CAPS

                  ;       Process SHIFT-CAPS key.

F359  A940        KGB7    LDA     #$40     ;CAPS lock indicator
F35B  8DBE02              STA     SHFLOK   ;shift/control lock flags
F35E  D098 ^F2F8          BNE     IGN      ;ignore

                  ;       Check for CTRL-CAPS key.

F360  C984        KGB8    CMP     #$84
F362  D008 ^F36C          BNE     KGB9     ;if not CTRL-CAPS

                  ;       Process CTRL-CAPS key.

F364  A980                LDA     #$80     ;control lock indicator
F366  8DBE02              STA     SHFLOK   ;shift/control lock flags
F369  4CF8F2              JMP     IGN      ;ignore

                  ;       Check for CTRL-3 key.

F36C  C985        KGB9    CMP     #$85
F36E  D00B ^F37B          BNE     KGB12    ;if not CTRL-3 key.

                  ;       Process CTRL-3 key.

F370  A988                LDA     #EOFERR

                  ;       Set status and BREAK key flag.

F372  854C        KGB10   STA     DSTAT    ;status
F374  8511                STA     BRKKEY   ;BREAK key flag

                  ;       Set EOL character.
```

```
F376  A99B          KGB11   LDA    #EOL
F378  4CDAF3                JMP    KGB19    ;set ATASCII character

                    ;       Check for CTRL-F3 key.

F37B  C989          KGB12   CMP    #$89
F37D  D010 ^F38F            BNE    KGB14    ;if not CTRL-F3 key

                    ;       Process CTRL-F3 key.

F37F  ADDB02                LDA    NOCLIK   ;toggle keyclick status
F382  49FF                  EOR    #$FF
F384  8DDB02                STA    NOCLIK
F387  D003 ^F38C            BNE    KGB13    ;if click inhibited

F389  2083F9                JSR    SKC      ;sound key click

F38C  4CF8F2        KGB13   JMP    IGN      ;ignore

                    ;       Check for function key.

F38F  C98E          KGB14   CMP    #$8E
F391  B012 ^F3A5            BCS    KGB16    ;if code >= $8E, not a function key

F393  C98A                  CMP    #$8A
F395  90F5 ^F38C            BCC    KGB13    ;if code < $8A, not a function key,:

                    ;       Process function key.

F397  E98A                  SBC    #$8A             ;convert $8A - $8D TO 0 - 3
F399  067C                  ASL    HOLDCH           ;saved character
F39B  1002 ^F39F            BPL    KGB15            ;if no SHIFT

F39D  0904                  ORA    #$04             ;convert 0 - 3 to 4 - 7

F39F  A8            KGB15   TAY                     ;offset to function key def:
F3A0  B160                  LDA    (FKDEF),Y        ;function key
F3A2  4C2AF3                JMP    KGB3             ;set ATASCII character

                    ;       Check for super function.

F3A5  C992          KGB16   CMP    #$92
F3A7  B00B ^F3B4            BCS    KGB17    ;if code >= $92, process shift/cont:

F3A9  C98E                  CMP    #$8E
F3AB  90DF ^F38C            BCC    KGB13    ;if code < $8E, not super function,:

                    ;       Process super function.

F3AD  E972                  SBC    #$8E-$1C         ;convert $8E - $91 TO $1C -:
F3AF  EEE803                INC    SUPERF           ;set super function flag
F3B2  D026 ^F3DA            BNE    KGB19            ;set ATASCII character

                    ;       Process shift/control lock.

F3B4  A57C          KGB17   LDA    HOLDCH   ;saved character
```

```
F3B6  C940                    CMP     #$40
F3B8  B015 ^F3CF              BCS     KGB18    ;if not lower case

F3BA  ADFB02                  LDA     ATACHR   ;ATASCII character
F3BD  C961                    CMP     #'a'
F3BF  900E ^F3CF              BCC     KGB18    ;if < "a", do not process

F3C1  C97B                    CMP     #'z'+1
F3C3  B00A ^F3CF              BCS     KGB18    ;if > "z", do not process

F3C5  ADBE02                  LDA     SHFLOK   ;shift/control lock flags
F3C8  F005 ^F3CF              BEQ     KGB18    ;if no lock

F3CA  057C                    ORA     HOLDCH   ;modify character
F3CC  4C23F3                  JMP     KGB2     ;reprocess character

                    ;         Invert character, if necessary.

F3CF  203CF9        KGB18     JSR     CCC      ;check for control character
F3D2  F009 ^F3DD              BEQ     KGB20    ;if control character, do not invert

F3D4  ADFB02                  LDA     ATACHR   ;ATASCII character
F3D7  4DB602                  EOR     INVFLG   ;invert character

                    ;         Set ATASCII character.

F3DA  8DFB02        KGB19     STA     ATACHR   ;ATASCII character

                    ;         Exit.

F3DD  4C1EF2        KGB20     JMP     SST      ;perform screen status, return



                    **        ESC - Escape
                    *
                    *         ENTRY   JSR      ESC
                    *
                    *         MODS
                    *                 Original Author Unknown
                    *                 1. Bring closer to Coding Standard (object :
                    *                    R. K. Nordin 11/01/83


      = F3E0        ESC       =       *        ;entry
F3E0  A980                    LDA     #$80     ;indicate escape detected
F3E2  8DA202                  STA     ESCFLG   ;escape flag
F3E5  60                      RTS              ;return
```

```
                        **      CUP - Move Cursor Up
                        *
                        *       ENTRY   JSR     CUP
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = F3E6             CUP     =       *       ;entry
F3E6  C654                      DEC     ROWCRS  ;decrement cursor row
F3E8  1006 ^F3F0                BPL     CUP2    ;if row positive

F3EA  AEBF02                    LDX     BOTSCR  ;screen bottom
F3ED  CA                        DEX             ;screen bottom - 1

F3EE  8654              CUP1    STX     ROWCRS  ;update cursor row

F3F0  4C0CF9            CUP2    JMP     SBS     ;set buffer start and logical colum:



                        **      CDN - Move Cursor Down
                        *
                        *       ENTRY   JSR     CDN
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = F3F3             CDN     =       *       ;entry
F3F3  E654                      INC     ROWCRS  ;increment cursor row
F3F5  A554                      LDA     ROWCRS  ;cursor row
F3F7  CDBF02                    CMP     BOTSCR  ;screen bottom
F3FA  90F4 ^F3F0                BCC     CUP2    ;if at bottom, set buffer start, re:

F3FC  A200                      LDX     #0
F3FE  F0EE ^F3EE                BEQ     CUP1    ;update cursor row, return
```

```
                    **      CLF - Move Cursor Left
                    *
                    *       ENTRY   JSR     CLF
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


     = F400         CLF     =       *       ;entry
F400 C655                   DEC     COLCRS  ;decrement low cursor column
F402 A555                   LDA     COLCRS  ;low cursor column
F404 3004 ^F40A             BMI     CRM     ;if negative, move cursor to margin:

F406 C552                   CMP     LMARGN  ;left margin
F408 B004 ^F40E             BCS     SCC1    ;if at left margin, set logical col:

                    ;       JMP     CRM     ;move cursor to right margin, retur:



                    **      CRM - Move Cursor to Right Margin
                    *
                    *       ENTRY   JSR     CRM
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


     = F40A         CRM     =       *       ;entry
F40A A553                   LDA     RMARGN  ;right margin
                    ;       JMP     SCC     ;set cursor column, return



                    **      SCC - Set Cursor Column
                    *
                    *       ENTRY   JSR     SCC
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


     = F40C         SCC     =       *       ;entry
F40C 8555                   STA     COLCRS  ;set low cursor column

F40E 4C8EF6         SCC1    JMP     SLC     ;set logical column, return
```

```
                        **      CRT - Move Cursor Right
                        *
                        *       ENTRY   JSR     CRT
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = F411             CRT     =       *       ;entry
F411  E655                      INC     COLCRS  ;increment low cursor column
F413  A555                      LDA     COLCRS  ;low cursor column
F415  C553                      CMP     RMARGN  ;right margin
F417  90F5 ^F40E                BCC     SCC1    ;if before right margin, process, r:

F419  F0F3 ^F40E                BEQ     SCC1    ;if at right margin

                        ;       JMP     CLM     ;move cursor to left margin, return



                        **      CLM - Move Cursor to Left Margin
                        *
                        *       ENTRY   JSR     CLM
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = F41B             CLM     =       *       ;entry
F41B  A552                      LDA     LMARGN  ;left margin
F41D  4C0CF4                    JMP     SCC     ;set cursor column, return



                        **      CSC - Clear Screen
                        *
                        *       ENTRY   JSR     CSC
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = F420             CSC     =       *               ;entry

                        ;       Set memory scan counter address.

F420  20A6F9                    JSR     SMS             ;set memory scan counter ad:

                        ;       Clear address.
```

```
F423  A464                  LDY    ADRESS
F425  A900                  LDA    #0
F427  8564                  STA    ADRESS

F429  9164        CSC1      STA    (ADRESS),Y
F42B  C8                    INY
F42C  D0FB ^F429            BNE    CSC1            ;if not done with page

F42E  E665                  INC    ADRESS+1
F430  A665                  LDX    ADRESS+1
F432  E46A                  CPX    RAMTOP          ;(high) RAM size
F434  90F3 ^F429            BCC    CSC1            ;if not done

                  ;         Clean up logical line bit map.

                  ;         LDY    #0              ;offset to first byte of bi:
F436  A9FF                  LDA    #$FF

F438  99B202      CSC2      STA    LOGMAP,Y        ;byte of logical line bit m:
F43B  C8                    INY
F43C  C004                  CPY    #4              ;4 bytes
F43E  90F8 ^F438            BCC    CSC2            ;if not done

                  ;         Exit.

                  ;         JMP    CHM             ;move cursor home, return



                  **        CHM - Move Cursor Home
                  *
                  *         ENTRY  JSR    CHM
                  *
                  *         MODS
                  *                Original Author Unknown
                  *                1. Bring closer to Coding Standard (object :
                  *                   R. K. Nordin 11/01/83


      = F440      CHM       =      *               ;entry
F440  2097F9                JSR    SCL             ;set cursor at left edge
F443  8563                  STA    LOGCOL          ;logical column
F445  856D                  STA    BUFSTR+1        ;high buffer start
F447  A900                  LDA    #0
F449  8554                  STA    ROWCRS          ;cursor row
F44B  8556                  STA    COLCRS+1        ;high cursor column
F44D  856C                  STA    BUFSTR          ;low buffer start pointer
F44F  60                    RTS                    ;return
```

```
                    **      BSP - Backspace
                    *
                    *       ENTRY   JSR     BSP
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = F450        BSP     =       *       ;entry
F450  A563                  LDA     LOGCOL  ;logical column
F452  C552                  CMP     LMARGN  ;left margin
F454  F021 ^F477            BEQ     BSP3    ;if at left margin

F456  A555                  LDA     COLCRS  ;low cursor column
F458  C552                  CMP     LMARGN  ;left margin
F45A  D003 ^F45F            BNE     BSP1    ;if not at left margin

F45C  2023F9                JSR     DWQ     ;see if line should be deleted

F45F  2000F4        BSP1    JSR     CLF     ;move cursor left
F462  A555                  LDA     COLCRS  ;low cursor column
F464  C553                  CMP     RMARGN  ;right margin
F466  D007 ^F46F            BNE     BSP2    ;if not at right margin

F468  A554                  LDA     ROWCRS  ;cursor row
F46A  F003 ^F46F            BEQ     BSP2    ;if row zero  } this can't happen!  because top li
                                                                                 is start of

F46C  20E6F3                JSR     CUP     ;move cursor up

F46F  A920          BSP2    LDA     #' '
F471  8DFB02                STA     ATACHR  ;ATASCII character
F474  20CAF1                JSR     PLO     ;plot point

F477  4C8EF8        BSP3    JMP     SLC     ;set logical column, return



                    **      TAB - Tab
                    *
                    *       ENTRY   JSR     TAB
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = F47A        TAB     =       *       ;entry

F47A  2011F4        TAB1    JSR     CRT     ;move cursor right
F47D  A555                  LDA     COLCRS  ;low cursor column
F47F  C552                  CMP     LMARGN  ;left margin
F481  D008 ^F48B            BNE     TAB2    ;if not at left margin
```

```
F483  2065F6                 JSR     RET      ;return
F486  2058F7                 JSR     BLG      ;get bit from logical line bit map
F489  B007 ^F492             BCS     TAB3     ;if end of logical line

                     ;       Check for tab stop.

F48B  A563          TAB2     LDA     LOGCOL   ;logical column
F48D  205DF7                 JSR     BMG      ;get bit from bit map
F490  90E8 ^F47A             BCC     TAB1     ;if not tab stop, keep looking

                     ;       Set logical column.

F492  4C8EF8        TAB3     JMP     SLC      ;set logical column, return



                     **      STB - Set Tab
                     *
                     *       ENTRY   JSR     STB
                     *
                     *       MODS
                     *               Original Author Unknown
                     *               1. Bring closer to Coding Standard (object :
                     *                  R. K. Nordin 11/01/83

      = F495        STB      =       *        ;entry
F495  A563                   LDA     LOGCOL   ;logical column
F497  4C3EF7                 JMP     BMS      ;set bit in bit map, return



                     **      CTB - Clear Tab
                     *
                     *       ENTRY   JSR     CTB
                     *
                     *       MODS
                     *               Original Author Unknown
                     *               1. Bring closer to Coding Standard (object :
                     *                  R. K. Nordin 11/01/83

      = F49A        CTB      =       *        ;entry
F49A  A563                   LDA     LOGCOL   ;logical column
F49C  4C4AF7                 JMP     BMC      ;clear bit in bit map, return
```

```
                        **      ICH - Insert Character
                        *
                        *       ENTRY   JSR     ICH
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


        = F49F          ICH     =       *       ;entry
F49F    204CF9                  JSR     SRC     ;save row and column
F4A2    208FF1                  JSR     GDC     ;get data under cursor
F4A5    857D                    STA     INSDAT
F4A7    A900                    LDA     #0
F4A9    8DBB02                  STA     SCRFLG

F4AC    20E9F1          ICH1    JSR     SPQ     ;store data
F4AF    A563                    LDA     LOGCOL  ;logical column
F4B1    48                      PHA             ;save logical column
F4B2    2012F6                  JSR     ACC     ;advance cursor column
F4B5    68                      PLA             ;saved logical column
F4B6    C563                    CMP     LOGCOL  ;logical column
F4B8    B00C ^F4C6              BCS     ICH2    ;if saved logical column >= logical:

F4BA    A57D                    LDA     INSDAT
F4BC    48                      PHA
F4BD    208FF1                  JSR     GDC     ;get data under cursor
F4C0    857D                    STA     INSDAT
F4C2    68                      PLA
F4C3    4CACF4                  JMP     ICH1    ;continue

                        ;       Exit.

F4C6    2057F9          ICH2    JSR     RRC     ;restore row and column

F4C9    CEBB02          ICH3    DEC     SCRFLG
F4CC    3004 ^F4D2              BMI     ICH4    ;if scroll occurred

F4CE    C654                    DEC     ROWCRS  ;decrement cursor row
F4D0    D0F7 ^F4C9              BNE     ICH3    ;continue

F4D2    4C8EF6          ICH4    JMP     SLC     ;set logical column, return
```

```
                        **      DCH - Delete Character
                        *
                        *       ENTRY   JSR     DCH
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = F4D5            DCH     =       *                 ;entry

                        ;       Save row and column.

F4D5  204CF9                    JSR     SRC               ;save row and column

                        ;       Get data to the right of cursor.

F4D8  20ACF5            DCH1    JSR     CCA     ;convert cursor row/column to addre:
F4DB  A564                      LDA     ADRESS
F4DD  8568                      STA     SAVADR  ;save address
F4DF  A565                      LDA     ADRESS+1
F4E1  8569                      STA     SAVADR+1
F4E3  A563                      LDA     LOGCOL  ;logical column.
F4E5  48                        PHA             ;save logical column
F4E6  200AF6                    JSR     SZA     ;set zero data and advance cursor
F4E9  68                        PLA             ;saved logical column
F4EA  C563                      CMP     LOGCOL  ;logical column
F4EC  B010 ^F4FE                BCS     DCH2    ;if saved logical column >= logical:

F4EE  A554                      LDA     ROWCRS            ;cursor row
F4F0  CDBF02                    CMP     BOTSCR            ;screen bottom
F4F3  B009 ^F4FE                BCS     DCH2              ;if row off screen, exit

F4F5  208FF1                    JSR     GDC               ;get data under cursor
F4F8  A000                      LDY     #0
F4FA  9168                      STA     (SAVADR),Y        ;put data in previous posit:
F4FC  F0DA ^F4D8                BEQ     DCH1              ;continue

F4FE  A000              DCH2    LDY     #0
F500  98                        TYA
F501  9168                      STA     (SAVADR),Y        ;clear last position
F503  2018F9                    JSR     DQQ               ;try to delete a line
F506  2057F9                    JSR     RRC               ;restore row and column
F509  4C8EF8                    JMP     SLC               ;set logical column, return
```

```
                    **      ILN - Insert Line
                    *
                    *       ENTRY   JSR     ILN
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = F50C        ILN     =       *       ;entry
F50C  38                    SEC
                    ;       JMP     ILN1



                    **      ILN1 - Insert Line
                    *
                    *       ENTRY   JSR     ILN1
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = F50D        ILN1    =       *       ;entry
F50D  20C2F7                JSR     ELL     ;extend logical line
F510  A552                  LDA     LMARGN  ;left margin
F512  8555                  STA     COLCRS  ;low cursor column
F514  20ACF5                JSR     CCA     ;convert cursor row/column to addre:
F517  208EF7                JSR     MLN     ;move line
F51A  20E2F7                JSR     CLN     ;clear current line
F51D  4C8EF8                JMP     SLC     ;set logical column, return



                    **      DLN - Delete Line
                    *
                    *       ENTRY   JSR     DLN
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


      = F520        DLN     =       *       ;entry
F520  208EF8                JSR     SLC     ;set logical column
F523  A451                  LDY     HOLD1
F525  8454                  STY     ROWCRS  ;cursor row
                    ;       JMP     DLN1
```

```
                        **      DLN1 - Delete Line
                        *
                        *       ENTRY   JSR     DLN1
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


        = F527          DLN1    =       *               ;entry

F527    A454            DLN0    LDY     ROWCRS          ;cursor row

F529    98              DLN2    TYA
F52A    38                      SEC
F52B    205BF7                  JSR     BLG2            ;get next bit
F52E    08                      PHP
F52F    98                      TYA
F530    18                      CLC
F531    6978                    ADC     #8*[LOGMAP-TABMAP]      ;add offset for log:
F533    28                      PLP
F534    203CF7                  JSR     BMP             ;put bit in bit map
F537    C8                      INY
F538    C018                    CPY     #24
F53A    D0ED ^F529              BNE     DLN2            ;if not done

F53C    ADB402                  LDA     LOGMAP+2
F53F    0901                    ORA     #1              ;set least significant bit
F541    8DB402                  STA     LOGMAP+2        ;update logical line bit ma:
F544    A900                    LDA     #0              ;delete line of data
F546    8555                    STA     COLCRS          ;low cursor column
F548    20ACF5                  JSR     CCA             ;convert cursor row/column :
F54B    202AF8                  JSR     SSD             ;scroll screen for delete

                        ;       Check for new logical line.

F54E    205BF7                  JSR     BLG             ;get bit from logical line :
F551    90D4 ^F527              BCC     DLN0            ;if not new logical line

                        ;       Move cursor to left margin.

F553    4C1BF4                  JMP     CLM             ;move cursor to left margin:
```

```
                        **      BEL - Sound Bell
                        *
                        *       ENTRY   JSR     BEL
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

       = F556           BEL     =       *       ;entry
F556   A020                     LDY     #$20

F558   2083F9           BEL1    JSR     SKC     ;sound key click
F55B   88                       DEY
F55C   10FA ^F558               BPL     BEL1    ;if not done

F55E   60                       RTS             ;return



                        **      CBT - Move Cursor to Bottom
                        *
                        *       ENTRY   JSR     CBT
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

       = F55F           CBT     =       *       ;entry
F55F   2040F4                   JSR     CHM     ;move cursor home
F562   4CE6F3                   JMP     CUP     ;move cursor up, return



                        **      DDD - Perform Double Byte Double Decrement
                        *
                        *       ENTRY   JSR     DDD
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

       = F565           DDD     =       *       ;entry
F565   A902                     LDA     #2      ;indicate subtracting 2
F567   D011 ^F57A               BNE     DBS     ;perform double byte subtract, retu:
```

```
                    **      SDF - Store Data Indirect for Fine Scrolling
                    *
                    *       ENTRY   JSR     SDF
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83


     = F569         SDF     =       *       ;entry
F569 AC6E02                 LDY     FINE
F56C F002 ^F570             BEQ     SDI     ;if not fine scrolling

F56E 0920                   ORA     #$20    ;enable vertical scroll
                    ;       JMP     SDI     ;store data indirect, return



                    **      SDI - Store Data Indirect
                    *
                    *       ENTRY   JSR     SDI
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

     = F570         SDI     =       *       ;entry

                    ;       Check current status.

F570 A44C                   LDY     DSTAT   ;status
F572 3028 ^F59F             BMI     DBS3    ;if error, return

                    ;       Store data.

F574 A000                   LDY     #0
F576 9164                   STA     (ADRESS),Y

                    ;       Decrement.

                    ;       JMP     DSD     ;perform double byte single decreme:
```

```
                    **      DSD - Perform Double Byte Single Decrement
                    *
                    *       ENTRY   JSR     DSD
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

        = F578      DSD     =       *       ;entry
F578    A901                LDA     #1      ;indicate subtracting 1
                    ;       JMP     DBS     ;perform double byte subtract, retu:



                    **      DBS - Perform Double Byte Subtract
                    *
                    *       ENTRY   JSR     DBS
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

        = F57A      DBS     =       *       ;entry

                    ;       Initialize.

F57A    8D9E02              STA     SUBTMP

                    ;       Check current status.

F57D    A54C                LDA     DSTAT           ;status
F57F    301E ^F59F          BMI     DBS3            ;if error

                    ;       Subtract.

F581    A564                LDA     ADRESS
F583    38                  SEC
F584    ED9E02              SBC     SUBTMP
F587    8564                STA     ADRESS
F589    B002 ^F58D          BCS     DBS1            ;if no borrow

F58B    C665                DEC     ADRESS+1        ;adjust high byte

                    ;       Check for overwriting APPMHI.

F58D    A50F        DBS1    LDA     APPMHI+1
F58F    C565                CMP     ADRESS+1
F591    900C ^F59F          BCC     DBS3            ;if not overwriting APPMHI

F593    D006 ^F59B          BNE     DBS2            ;if overwriting APPMHI, err:

F595    A50E                LDA     APPMHI
```

```
F597  C564                 CMP    ADRESS
F599  9004 ^F59F           BCC    DBS3           ;if not overwriting APPMHI

                   ;       Process error.

F59B  A993         DBS2    LDA    #SCRMEM        ;indicate insufficient memo:
F59D  854C                 STA    DSTAT          ;status

                   ;       Exit.

F59F  60           DBS3    RTS           ;return



                   **      SSE - Set Scrolling Display List Entry
                   *
                   *       Store extra line in display list for fine scrolling:
                   *
                   *       ENTRY  JSR    SSE
                   *
                   *       MODS
                   *              H. Stewart      06/01/82
                   *              1. Bring closer to Coding Standard (object :
                   *                 R. K. Nordin 11/01/83


      = F5A0       SSE     =      *      ;entry
F5A0  A902                 LDA    #$02
F5A2  2070F5               JSR    SDI    ;store data indirect
F5A5  A9A2                 LDA    #$A2   ;DLI on last visible line
F5A7  2070F5               JSR    SDI    ;store data indirect
F5AA  CA                   DEX
F5AB  60                   RTS           ;return



                   **      CCA - Convert Cursor Row/Column to Address
                   *
                   *       ENTRY  JSR    CCA
                   *
                   *       MODS
                   *              L. Winner       06/01/82
                   *              1. Bring closer to Coding Standard (object :
                   *                 R. K. Nordin 11/01/83


      = F5AC       CCA     =      *              ;entry
F5AC  A201                 LDX    #1
F5AE  8666                 STX    MLTTMP         ;initialize
F5B0  CA                   DEX
F5B1  8665                 STX    ADRESS+1       ;clear high address
F5B3  A554                 LDA    ROWCRS         ;cursor row position
F5B5  0A                   ASL    A              ;2 times row position
F5B6  2665                 ROL    ADRESS+1
F5B8  0A                   ASL    A              ;4 time row position
```

```
F5B9  2665                ROL   ADRESS+1
F5BB  6554                ADC   ROWCRS       ;add to get 5 times row pos:
F5BD  8564                STA   ADRESS
F5BF  9002 ^F5C3          BCC   CCA1

F5C1  E665                INC   ADRESS+1

F5C3  A457         CCA1   LDY   DINDEX       ;mode
F5C5  BE6DEE              LDX   TLSC,Y       ;left shift count

F5C8  0664         CCA2   ASL   ADRESS       ;ADRESS = ADRESS*X
F5CA  2665                ROL   ADRESS+1     ;divide
F5CC  CA                  DEX
F5CD  D0F9 ^F5C8          BNE   CCA2

F5CF  A556                LDA   COLCRS+1     ;high cursor column
F5D1  4A                  LSR   A            ;save least significant bit
F5D2  A555                LDA   COLCRS       ;low cursor column
F5D4  BE9DEE              LDX   TRSC,Y       ;right shift count
F5D7  F006 ^F5DF          BEQ   CCA4         ;if no shift

F5D9  6A           CCA3   ROR   A            ;roll in carry
F5DA  0666                ASL   MLTTMP       ;shift index
F5DC  CA                  DEX
F5DD  D0FA ^F5D9          BNE   CCA3

F5DF  6564         CCA4   ADC   ADRESS       ;add address
F5E1  9002 ^F5E5          BCC   CCA5         ;if no carry

F5E3  E665                INC   ADRESS+1     ;adjust high address

F5E5  18           CCA5   CLC
F5E6  6558                ADC   SAVMSC       ;add saved memory scan coun:
F5E8  8564                STA   ADRESS       ;update address
F5EA  855E                STA   OLDADR       ;save address
F5EC  A565                LDA   ADRESS+1
F5EE  6559                ADC   SAVMSC+1
F5F0  8565                STA   ADRESS+1
F5F2  855F                STA   OLDADR+1

F5F4  BE9DEE              LDX   TRSC,Y
F5F7  BD04FB              LDA   TMSK,X
F5FA  2555                AND   COLCRS       ;and in low cursor column
F5FC  6566                ADC   MLTTMP
F5FE  A8                  TAY
F5FF  B9ACEE              LDA   TDSM-1,Y     ;display mask
F602  8DA002              STA   DMASK        ;display mask
F605  856F                STA   SHFAMT
F607  A000                LDY   #0

F609  60           CCA6   RTS                ;return
```

```
                    **      SZA - Set Zero Data and Advance Cursor Column
                    *
                    *       ENTRY   JSR     SZA
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = F60A        SZA     =       *       ;entry
F60A  A900                  LDA     #0
F60C  F002 ^F610            BEQ     SDA     ;set data and advance cursor


                    **      SEA - Set EOL Data and Advance Cursor Column
                    *
                    *       ENTRY   JSR     SEA
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = F60E        SEA     =       *       ;entry
F60E  A99B                  LDA     #EOL    ;special case eliminator
                    ;       JMP     SDA     ;set data and advance cursor, retur:


                    **      SDA - Set Data and Advance Cusor Column
                    *
                    *       ENTRY   JSR     SDA
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = F610        SDA     =       *       ;entry
F610  857D                  STA     INSDAT  ;set data
                    ;       JMP     ACC     ;advance cursor column, return
```

```
                        **      ACC - Advance Cursor Column
                        *
                        *       ENTRY   JSR     ACC
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = F612            ACC     =       *               ;entry
F612  E663                      INC     LOGCOL          ;increment logical column
F614  E655                      INC     COLCRS          ;increment low cursor colum:
F616  D002 ^F61A                BNE     ACC1            ;if no carry

F618  E656                      INC     COLCRS+1        ;adjust high cursor column

F61A  A555              ACC1    LDA     COLCRS          ;low cursor column
F61C  A657                      LDX     DINDEX          ;mode
F61E  DD7DEE                    CMP     TMCC,X
F621  F00A ^F62D                BEQ     ACC2            ;if equal, process EOL

F623  E000                      CPX     #0
F625  D0E2 ^F609                BNE     CCA6            ;if not mode 0, exit

F627  C553                      CMP     RMARGN          ;right margin
F629  F0DE ^F609                BEQ     CCA6            ;if at right margin, exit

F62B  90DC ^F609                BCC     CCA6            ;if before right margin, ex:

F62D  E008              ACC2    CPX     #8
F62F  D004 ^F635                BNE     ACC3            ;if not mode 8

F631  A556                      LDA     COLCRS+1        ;high cursor column
F633  F0D4 ^F609                BEQ     CCA6            ;if only at 64

F635  A557              ACC3    LDA     DINDEX          ;mode
F637  D02C ^F665                BNE     RET             ;if mode 0, exit

F639  A563                      LDA     LOGCOL          ;logical column
F63B  C951                      CMP     #81
F63D  900A ^F649                BCC     ACC4            ;if < 81, definitely not li:

F63F  A57D                      LDA     INSDAT
F641  F022 ^F665                BEQ     RET             ;if non-zero, do not do log:

F643  2061F6                    JSR     RWS             ;return with scrolling
F646  4CABF6                    JMP     RETS            ;return

F649  2065F6            ACC4    JSR     RET             ;return
F64C  A554                      LDA     ROWCRS          ;curosr row
F64E  18                        CLC
F64F  6978                      ADC     #8*[LOGMAP-TABMAP]      ;add offset for log:
F651  205DF7                    JSR     BMG             ;get bit from bit map
F654  9008 ^F65E                BCC     ACC5

F656  A57D                      LDA     INSDAT
```

```
F658  F004 ^F65E          BEQ     ACC5              ;if zero, do not extend

F65A  18                  CLC
F65B  2000F5              JSR     ILN1              ;insert line

F65E  4C8EF8      ACC5    JMP     SLC               ;set logical column, return



                  **      RWS - Return with Scrolling
                  *
                  *       ENTRY   JSR     RWS
                  *
                  *       MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83


      = F661      RWS     =       *       ;entry
F661  A99B                LDA     #EOL    ;select scrolling
F663  857D                STA     INSDAT
                  ;       JMP     RET     ;return, return .



                  **      RET - Return
                  *
                  *       ENTRY   JSR     RET
                  *
                  *       MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83


      = F665      RET     =       *                 ;entry
F665  2097F9              JSR     SCL               ;set cursor at left edge
F668  A900                LDA     #0
F66A  8556                STA     COLCRS+1          ;high cursor column
F66C  E654                INC     ROWCRS            ;increment cursor row
F66E  A657                LDX     DINDEX
F670  A018                LDY     #24               ;assume 24 lines
F672  247B                BIT     SWPFLG
F674  1005 ^F67B          BPL     RET1              ;if normal

F676  A004                LDY     #4                ;substitute 4 lines
F678  98                  TYA
F679  D003 ^F67E          BNE     RET2

F67B  BD8DEE      RET1    LDA     TMRC,X  ;mode row count

F67E  C554        RET2    CMP     ROWCRS  ;cursor row
F680  D029 ^F6AB          BNE     RET5
```

```
F682  8C9D02               STY     HOLD3
F685  8A                   TXA             ;mode
F686  D023 ^F6AB           BNE     RET5    ;if mode not 0, do not scroll

F688  A57D                 LDA     INSDAT
F68A  F01F ^F6AB           BEQ     RET5    ;if zero, do not scroll

                   ;       If EOL, roll in a 0.

F68C  C99B                 CMP     #EOL    ;to extend bottom logical line
F68E  F001 ^F691           BEQ     RET3    ;if EOL

F690  18                   CLC

F691  20F7F7       RET3    JSR     SCR
F694  EEBB02               INC     SCRFLG
F697  C66C                 DEC     BUFSTR
F699  1002 ^F69D           BPL     RET4

F69B  E66C                 INC     BUFSTR

F69D  CE9D02       RET4    DEC     HOLD3
F6A0  ADB202               LDA     LOGMAP
F6A3  38                   SEC             ;indicate for partial line
F6A4  10EB ^F691           BPL     RET3    ;if partial logical line

F6A6  AD9D02               LDA     HOLD3
F6A9  8554                 STA     ROWCRS  ;cursor row

F6AB  4C8EF3       RET5    JMP     SLC     ;set logical column, return



                   **      SEP - Subtract End Point
                   *
                   *       ENTRY   JSR     SEP
                   *               X = 0, if row or 2, if column
                   *
                   *       MODS
                   *               Original Author Unknown
                   *               1. Bring closer to Coding Standard (object :
                   *                  R. K. Nordin 11/01/83


      = F6AE       SEP     =       *                ;entry
F6AE  38                   SEC
F6AF  B570                 LDA     ROWAC,X          ;low value from which to su:
F6B1  E574                 SBC     ENDPT
F6B3  9570                 STA     ROWAC,X          ;new low value
F6B5  B571                 LDA     ROWAC+1,X        ;high value from which to s:
F6B7  E575                 SBC     ENDPT+1
F6B9  9571                 STA     ROWAC+1,X        ;new high value
F6BB  60                   RTS                      ;return
```

```
                        **      CRE - Check Cursor Range for Editor
                        *
                        *       ENTRY   JSR     CRE
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

       = F6BC           CRE     =       *       ;entry

                        ;       Check for mixed mode.

F6BC   AD8F02                   LDA     BOTSCR
F6BF   C904                     CMP     #4      ;mixed mode indicator
F6C1   F007 ^F6CA               BEQ     CCR     ;if mixed mode, check cursor range,:

                        ;       Check for mode 0.

F6C3   A557                     LDA     DINDEX  ;mode
F6C5   F003 ^F6CA               BEQ     CCR     ;if mode 0, check cursor range

                        ;       Open editor.

F6C7   2094EF                   JSR     EOP     ;perform editor OPEN
                        ;       JMP     CCR     ;check cursor range, return



                        **      CCR - Check Cursor Range
                        *
                        *       ENTRY   JSR     CCR
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

       = F6CA           CCR     =       *               ;entry
F6CA   A927                     LDA     #39
F6CC   C553                     CMP     RMARGN          ;right margin
F6CE   B002 ^F6D2               BCS     CCR1            ;if 39 >= right margin

F6D0   8553                     STA     RMARGN          ;set right margin

F6D2   A657             CCR1    LDX     DINDEX
F6D4   BD8DEE                   LDA     TMRC,X          ;mode row count
F6D7   C554                     CMP     ROWCRS          ;cursor row
F6D9   902A ^F705               BCC     CCR5            ;if count > row position, e:

F6DB   F028 ^F705               BEQ     CCR5            ;if count = row position, e:

F6DD   E008                     CPX     #8
F6DF   D00A ^F6EB               BNE     CCR2            ;if not mode 8
```

```
F6E1  A556                 LDA    COLCRS+1         ;high cursor column
F6E3  F013 ^F6F8           BEQ    CCR4             ;if high cursor column zero

F6E5  C901                 CMP    #1
F6E7  D01C ^F705           BNE    CCR5             ;if >1, bad

F6E9  F004 ^F6EF           BEQ    CCR3             ;if 1, check low

F6EB  A556         CCR2    LDA    COLCRS+1         ;high cursor column
F6ED  D016 ^F705           BNE    CCR5             ;if high cursor column non-:

F6EF  BD7DEE       CCR3    LDA    TMCC,X           ;mode column count
F6F2  C555                 CMP    COLCRS           ;low cursor column
F6F4  900F ^F705           BCC    CCR5             ;if count > column position:

F6F6  F00D ^F705           BEQ    CCR5             ;if count = column position:

F6F8  A901         CCR4    LDA    #SUCCES          ;success indicator
F6FA  854C                 STA    DSTAT            ;indicate success
F6FC  A980                 LDA    #BRKABT          ;assume BREAK abort
F6FE  A611                 LDX    BRKKEY           ;BREAK key status
F700  8511                 STA    BRKKEY           ;clear BREAK key status
F702  F006 ^F70A           BEQ    CCR6             ;if BREAK

F704  60                   RTS                     ;return

                   ;       Process range error.

F705  2040F4       CCR5    JSR    CHM              ;move cursor home
F708  A98D                 LDA    #CRSROR          ;indicate cursor overrange

                   ;       Exit.

F70A  854C         CCR6    STA    DSTAT            ;status
F70C  68                   PLA                     ;clean stack for return to :
F70D  68                   PLA
F70E  A578                 LDA    SWPFLG
F710  1003 ^F715           BPL    CCR7             ;if not swapped

F712  4C62F9               JMP    SWA              ;swap, return

F715  4C1EF2       CCR7    JMP    SST              ;return (to CIO)
```

```
                        **      ROD - Restore Old Data under Cursor
                        *
                        *       ENTRY   JSR     ROD
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = F718             ROD     =       *                       ;entry
F718  A000                      LDY     #0
F71A  A55F                      LDA     OLDADR+1
F71C  F004 ^F722                BEQ     ROD1                    ;if page zero

F71E  A55D                      LDA     OLDCHR                  ;old data
F720  915E                      STA     (OLDADR),Y

F722  60                ROD1    RTS                             ;return



                        **      BMI - Initialize for Bit Map Operation
                        *
                        *       BMI sets the bit mask in BITMSK and byte offset in :
                        *
                        *       ENTRY   JSR     BMI
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = F723             BMI     =       *               ;entry
F723  48                        PHA                     ;save logical column
F724  2907                      AND     #7              ;logical column modulo 8
F726  AA                        TAX                     ;offset to bit mask
F727  BD84EE                    LDA     TBTM,X          ;bit mask
F72A  856E                      STA     BITMSK          ;set bit mask
F72C  68                        PLA                     ;logical column
F72D  4A                        LSR     A
F72E  4A                        LSR     A
F72F  4A                        LSR     A               ;logical column divided by 8
F730  AA                        TAX                     ;offset
F731  60                        RTS                     ;return
```

```
                        **      BLR - Rotate Logical Line Bit Map Left
                        *
                        *       BLR rotates the logical line bit map left, scrollin:
                        *       logical lines up.
                        *
                        *       ENTRY   JSR     BLR
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = F732          BLR       =       *               ;entry
F732  2EB402                    ROL     LOGMAP+2
F735  2EB302                    ROL     LOGMAP+1
F738  2EB202                    ROL     LOGMAP
F73B  60                        RTS                     ;return



                        **      BMP - Put Bit in Bit Map
                        *
                        *       PUT CARRY INTO BITMAP
                        *
                        *       ENTRY   JSR     BMP
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = F73C          BMP       =       *       ;entry
F73C  900C ^F74A                BCC     BMC     ;if C clear, clear bit in bit map, :

                        ;       JMP     BMS     ;set bit in bit map, return



                        **      BMS - Set Bit in Bit Map
                        *
                        *       ENTRY   JSR     BMS
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = F73E          BMS       =       *               ;entry
F73E  2023F7                    JSR     BMI             ;initialize for bit mask op:
F741  BDA302                    LDA     TABMAP,X
F744  056E                      ORA     BITMSK          ;set bit
F746  9DA302                    STA     TABMAP,X        ;update bit map
```

```
F749  60                     RTS                        ;return



                      **       BMC - Clear Bit in Bit Map
                      *
                      *        ENTRY   JSR     BMC
                      *
                      *        MODS
                      *                Original Author Unknown
                      *                1. Bring closer to Coding Standard (object :
                      *                   R. K. Nordin 11/01/83

      = F74A          BMC      =       *                  ;entry
F74A  2023F7                   JSR     BMI                ;initialize for bit mask op:
F74D  A56E                     LDA     BITMSK
F74F  49FF                     EOR     #$FF
F751  3DA302                   AND     TABMAP,X           ;clear bit
F754  9DA302                   STA     TABMAP,X           ;update bit map
F757  60                       RTS                        ;return



                      **       BLG - Get Bit from Logical Line Bit Map
                      *
                      *        ENTRY   JSR     BLG
                      *
                      *        MODS
                      *                Original Author Unknown
                      *                1. Bring closer to Coding Standard (object :
                      *                   R. K. Nordin 11/01/83

      = F758          BLG      =       *       ;entry
F758  A554                     LDA     ROWCRS  ;cursor row
                      ;        JMP     BLG1



                      **       BLG1 - Get Bit from Logical Line Bit Map
                      *
                      *        ENTRY   JSR     BLG1
                      *
                      *        MODS
                      *                Original Author Unknown
                      *                1. Bring closer to Coding Standard (object :
                      *                   R. K. Nordin 11/01/83

      = F75A          BLG1     =       *       ;entry
F75A  18                       CLC
                      ;        JMP     BLG2
```

```
                        **      BLG2 - Get Bit from Logical Line Bit Map
                        *
                        *       ENTRY   JSR     BLG2
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = F75B            BLG2    =       *       ;entry
F75B  6978                      ADC     #8*[LOGMAP-TABMAP]      ;add offset for log:
                        ;       JMP     BMG     ;get bit from bit map, return



                        **      BMG - Get Bit from Bit Map
                        *
                        *       ENTRY   JSR     BMG
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = F75D            BMG     =       *       ;entry
F75D  2023F7                    JSR     BMI     ;initialize for bit mask operation
F760  18                        CLC
F761  BDA302                    LDA     TABMAP,X
F764  256E                      AND     BITMSK
F766  F001 ^F769                BEQ     BMG1

F768  38                        SEC

F769  60                BMG1    RTS             ;return



                        **      CIA - Convert Internal Character to ATASCII
                        *
                        *       ENTRY   JSR     CIA
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = F76A            CIA     =       *       ;entry

                        ;       Initialize.

F76A  ADFA02                    LDA     CHAR
```

```
                        ;       Check mode.

F76D  A457                      LDY     DINDEX  ;mode
F76F  C00E                      CPY     #14
F771  B017 ^F78A                BCS     CIA2    ;if mode >= 14

F773  C00C                      CPY     #12
F775  B004 ^F77B                BCS     CIA1    ;if mode 12 or 13

F777  C003                      CPY     #3
F779  B00F ^F78A                BCS     CIA2    ;if mode >= 3

                        ;       Convert internal character to ATASCII.

F77B  2A                CIA1    ROL     A
F77C  2A                        ROL     A
F77D  2A                        ROL     A
F77E  2A                        ROL     A
F77F  2903                      AND     #3
F781  AA                        TAX
F782  ADFA02                    LDA     CHAR    ;character
F785  299F                      AND     #$9F    ;strip off cloumn address
F787  1D4DFB                    ORA     TIAC,X  ;or in new column address

                        ;       Exit.

F78A  8DFB02            CIA2    STA     ATACHR  ;ATASCII character

F78D  60                CIA3    RTS             ;return



                        **      MLN - Move Line
                        *
                        *       ENTRY   JSR     MLN
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = F78E            MLN     =       *                       ;entry

                        ;       Initialize.

F78E  A66A                      LDX     RAMTOP                  ;(high) RAM size
F790  CA                        DEX                             ;decrement (high) RAM size
F791  8669                      STX     FRMADR+1                ;high source address
F793  8667                      STX     TOADR+1                 ;high destination address
F795  A9B0                      LDA     #low [$0000-80]         ;low RAM size - 80
F797  8568                      STA     FRMADR                  ;low source address
F799  A9D8                      LDA     #low [$0000-40]         ;low RAM size - 40
F79B  8566                      STA     TOADR                   ;low destination address
```

```
F79D  A654                 LDX    ROWCRS          ;cursor row

                     ;     Check for completion.

F79F  E8           MLN1    INX
F7A0  ECBF02               CPX    BOTSCR          ;screen bottom
F7A3  F0E8 ^F78D           BEQ    CIA3            ;if done, return

                     ;     Move line.

F7A5  A027                 LDY    #39             ;offset to last byte

F7A7  B168         MLN2    LDA    (FRMADR),Y      ;byte of source
F7A9  9166                 STA    (TOADR),Y       ;byte of destination
F7AB  88                   DEY
F7AC  10F9 ^F7A7           BPL    MLN2            ;if not done

                     ;     Adjust source and destination addresses.

F7AE  38                   SEC
F7AF  A568                 LDA    FRMADR          ;source address
F7B1  8566                 STA    TOADR           ;update destination address
F7B3  E928                 SBC    #low 40         ;subtract 40
F7B5  8568                 STA    FRMADR          ;update.source address
F7B7  A569                 LDA    FRMADR+1
F7B9  8567                 STA    TOADR+1
F7BB  E900                 SBC    #high 40
F7BD  8569                 STA    FRMADR+1

                     ;     Continue.

F7BF  4C9FF7               JMP    MLN1            ;continue



                     **    ELL - Extend Logical Line
                     *
                     *     ENTRY  JSR    ELL
                     *
                     *     MODS
                     *            Original Author Unknown
                     *            1. Bring closer to Coding Standard (object :
                     *               R. K. Nordin 11/01/83


      = F7C2       ELL     =      *      ;entry
F7C2  08                   PHP           ;save bit
F7C3  A016                 LDY    #22

F7C5  98           ELL1    TYA
F7C6  205AF7               JSR    BLG1
F7C9  08                   PHP
F7CA  98                   TYA
F7CB  18                   CLC
F7CC  6979                 ADC    #8*[LOGMAP-TABMAP]+1    ;add offset for log:
F7CE  28                   PLP
```

```
F7CF  203CF7               JSR     BMP      ;put bit in bit map
F7D2  88                   DEY
F7D3  3004 ^F7D9           BMI     ELL2

F7D5  C454                 CPY     ROWCRS   ;cursor row
F7D7  B0EC ^F7C5           BCS     ELL1

F7D9  A554          ELL2   LDA     ROWCRS   ;cursor row
F7DB  18                   CLC
F7DC  6978                 ADC     #8*(LOGMAP-TABMAP)      ;add offset for log:
F7DE  28                   PLP
F7DF  4C3CF7               JMP     BMP      ;put bit in bit map, return



                    **     CLN - Clear Line
                    *
                    *      ENTRY   JSR     CLN
                    *
                    *      MODS
                    *              Original Author Unknown
                    *              1. Bring closer to Coding Standard (object :
                    *                 R. K. Nordin 11/01/83


      = F7E2        CLN    =       *        ;entry
F7E2  A552                 LDA     LMARGN   ;left margin
F7E4  8555                 STA     COLCRS   ;low cursor column
F7E6  20ACF5               JSR     CCA      ;convert cursor row/column to addre:
F7E9  38                   SEC
F7EA  A553                 LDA     RMARGN   ;right margin
F7EC  E552                 SBC     LMARGN   ;subtract left margin
F7EE  A8                   TAY              ;screen width
F7EF  A900                 LDA     #0

F7F1  9164          CLN1   STA     (ADRESS),Y
F7F3  88                   DEY
F7F4  10FB ^F7F1           BPL     CLN1     ;if not done

F7F6  60                   RTS              ;return



                    **     SCR - Scroll
                    *
                    *      ENTRY   JSR     SCR
                    *
                    *      MODS
                    *              Original Author Unknown
                    *              1. Bring closer to Coding Standard (object :
                    *                 R. K. Nordin 11/01/83


      = F7F7        SCR    =       *        ;entry
```

```
                    ;       Initialize.

F7F7  2032F7                JSR     BLR     ;rotate logical line bit map left

                    ;       Check for fine scrolling.

F7FA  AD6E02                LDA     FINE
F7FD  F028 ^F827            BEQ     SCR5    ;if not fine scrolling

F7FF  AD6C02        SCR1    LDA     VSFLAG  ;vertical scroll count
F802  D0FB ^F7FF            BNE     SCR1    ;if prior scroll not yet done

F804  A908                  LDA     #8
F806  8D6C02                STA     VSFLAG  ;vertical scroll count

                    ;       Wait for scroll to complete.

F809  AD6C02        SCR2    LDA     VSFLAG  ;vertical scroll count
F80C  C901                  CMP     #1      ;start of last scan
F80E  D0F9 ^F809            BNE     SCR2    ;if not done waiting

F810  AD0BD4        SCR3    LDA     VCOUNT
F813  C940                  CMP     #$40
F815  B0F9 ^F810            BCS     SCR3    ;if not done waiting for safe place

F817  A20D                  LDX     #$0D
F819  AD8F02                LDA     BOTSCR
F81C  C904                  CMP     #4
F81E  D002 ^F822            BNE     SCR4    ;if not split screen

F820  A270                  LDX     #$70

F822  EC0BD4        SCR4    CPX     VCOUNT
F825  B0FB ^F822            BCS     SCR4    ;if not done waiting

                    ;       Exit.

F827  20A6F9        SCR5    JSR     SMS     ;set memory scan counter address
                    ;       JMP     SSD     ;scroll screen for delete, return



                    **      SSD - Scroll Screen for Delete
                    *
                    *       ENTRY   JSR     SSD
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = F82A        SSD     =       *               ;entry

                    ;       Initialize.
```

```
F82A  A564                    LDA    ADRESS          ;address
F82C  A665                    LDX    ADRESS+1

                      ;       Calculate number of bytes to move.

F82E  E8              SSD1    INX
F82F  E46A                    CPX    RAMTOP
F831  F006 ^F839              BEQ    SSD2            ;if at RAMTOP

F833  38                      SEC
F834  E910                    SBC    #$10
F836  4C2EF8                  JMP    SSD1            ;continue

F839  6927            SSD2    ADC    #39             ;(CLC and ADC #40)
F83B  D00A ^F847              BNE    SSD3            ;if byte count non-zero

F83D  A665                    LDX    ADRESS+1
F83F  E8                      INX
F840  E46A                    CPX    RAMTOP
F842  F038 ^F87C              BEQ    SSD6            ;if at RAMTOP

F844  18                      CLC
F845  6910                    ADC    #$10

                      ;       Adjust address.

F847  A8              SSD3    TAY                    ;number of bytes
F848  857E                    STA    COUNTR
F84A  38                      SEC
F84B  A564                    LDA    ADRESS
F84D  E57E                    SBC    COUNTR          ;subtract
F84F  8564                    STA    ADRESS          ;update low address
F851  B002 ^F855              BCS    SSD4            ;if no borrow

F853  C665                    DEC    ADRESS+1        ;adjust high address

                      ;       Move data down.

F855  A564            SSD4    LDA    ADRESS
F857  18                      CLC
F858  6928                    ADC    #40
F85A  857E                    STA    COUNTR          ;address + 40
F85C  A565                    LDA    ADRESS+1
F85E  6900                    ADC    #0
F860  857F                    STA    COUNTR+1

F862  B17E            SSD5    LDA    (COUNTR),Y      ;byte to move
F864  9164                    STA    (ADRESS),Y      ;move byte
F866  C8                      INY
F867  D0F9 ^F862              BNE    SSD5            ;if not done (256-16 times)

F869  A010                    LDY    #256-240
F86B  A564                    LDA    ADRESS
F86D  C9D8                    CMP    #-40
F86F  F00B ^F87C              BEQ    SSD6            ;if all done

F871  18                      CLC
```

```
F872  69F0                    ADC     #240
F874  8564                    STA     ADRESS          ;update low address
F876  900D ^F855              BCC     SSD4            ;if no carry

F878  E665                    INC     ADRESS+1        ;adjust high address
F87A  D0D9 ^F855              BNE     SSD4            ;continue

                      ;       Clear last line.

F87C  A66A            SSD6    LDX     RAMTOP
F87E  CA                      DEX
F87F  867F                    STX     COUNTR+1
F881  A2D8                    LDX     #-40
F883  867E                    STX     COUNTR
F885  A900                    LDA     #0
F887  A027                    LDY     #39

F889  917E            SSD7    STA     (COUNTR),Y      ;clear byte of last line
F88B  88                      DEY
F88C  10FB ^F889              BPL     SSD7            ;if not done

                      ;       JMP     SLC             ;set logical column, return


                      **      SLC - Set Logical Column
                      *
                      *       ENTRY   JSR     SLC
                      *
                      *       MODS
                      *               Original Author Unknown
                      *               1. Bring closer to Coding Standard (object :
                      *                . R. K. Nordin 11/01/83

      = F88E          SLC     =       *       ;entry

                      ;       Initialize.

F88E  A900                    LDA     #0
F890  8563                    STA     LOGCOL  ;initialize logical column
F892  A554                    LDA     ROWCRS  ;cursor row
F894  8551                    STA     HOLD1   ;working row

                      ;       Search for beginning of line.

F896  A551            SLC1    LDA     HOLD1   ;add in row component
F898  205AF7                  JSR     BLG1
F89B  B00C ^F8A9              BCS     SLC2    ;if beginning of line found

F89D  A563                    LDA     LOGCOL  ;logical column
F89F  18                      CLC
F8A0  6928                    ADC     #40     ;add number of characters per line
F8A2  8563                    STA     LOGCOL  ;update logical column
F8A4  C651                    DEC     HOLD1   ;decrement working row
F8A6  4C96F8                  JMP     SLC1    ;continue
```

```
                        ;       Add in cursor column.

F8A9  18                SLC2    CLC
F8AA  A563                      LDA     LOGCOL  ;logical column
F8AC  6555                      ADC     COLCRS  ;add low cursor cloumn
F8AE  8563                      STA     LOGCOL  ;update logical column
F8B0  60                        RTS             ;return



                        **      CBC - Compute Buffer Count
                        *
                        *       CBC computes the buffer count as the number of byte:
                        *       buffer start to the end of the logical line (with t:
                        *       spaces removed).
                        *
                        *       ENTRY   JSR     CBC
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83.


      = F8B1            CBC     =       *       ;entry

                        ;       Initialize.

F8B1  204CF9                    JSR     SRC             ;save row and column
F8B4  A563                      LDA     LOGCOL          ;logical column
F8B6  48                        PHA                     ;save logical column
F8B7  A56C                      LDA     BUFSTR          ;start of buffer
F8B9  8554                      STA     ROWCRS          ;cursor row
F8BB  A56D                      LDA     BUFSTR+1
F8BD  8555                      STA     COLCRS          ;low cursor column
F8BF  A901                      LDA     #1
F8C1  856B                      STA     BUFCNT          ;initialize buffer count

                        ;       Determine last line on screen.

F8C3  A217              CBC1    LDX     #23             ;normal last line on screen
F8C5  A57B                      LDA     SWPFLG
F8C7  1002 ^F8CB                BPL     CBC2            ;if not swapped

F8C9  A203                      LDX     #3              ;last line on screen

                        ;       Check for cursor on last line of screen.

F8CB  E454              CBC2    CPX     ROWCRS  ;cursor row
F8CD  D00B ^F8DA                BNE     CBC3    ;if cursor on last line

F8CF  A555                      LDA     COLCRS  ;low cursor column
F8D1  C553                      CMP     RMARGN  ;right margin
F8D3  D005 ^F8DA                BNE     CBC3    ;if not at right margin
```

```
F8D5  E66B                 INC     BUFCNT  ;fake SEA to avoid scrolling
F8D7  4CEAF8               JMP     CBC4

F8DA  200AF6        CBC3   JSR     SZA     ;set zero data and advance cursor
F8DD  E66B                 INC     BUFCNT
F8DF  A563                 LDA     LOGCOL  ;logical column
F8E1  C552                 CMP     LMARGN  ;left margin
F8E3  D0DE ^F8C3           BNE     CBC1    ;if not yet at left margin

F8E5  C654                 DEC     ROWCRS  ;decrement cursor row
F8E7  2000F4               JSR     CLF     ;move cursor left

F8EA  208FF1        CBC4   JSR     GDC     ;get data under cursor
F8ED  D017 ^F906           BNE     CBC6    ;if non-zero, quit

F8EF  C66B                 DEC     BUFCNT  ;DECREMENT COUNTER
F8F1  A563                 LDA     LOGCOL  ;logical column
F8F3  C552                 CMP     LMARGN  ;left margin
F8F5  F00F ^F906           BEQ     CBC6    ;if beginning of logical line, exit

F8F7  2000F4               JSR     CLF     ;move cursor left
F8FA  A555                 LDA     COLCRS  ;low cursor column
F8FC  C553                 CMP     RMARGN  ;right margin
F8FE  D002 ^F902           BNE     CBC5    ;if cursor column not right margin

F900  C654                 DEC     ROWCRS  ;decrement cursor row

F902  A56B          CBC5   LDA     BUFCNT
F904  D0E4 ^F8EA           BNE     CBC4    ;if BUFCNT non-zero, continue

F906  68            CBC6   PLA             ;saved logical column
F907  8563                 STA     LOGCOL  ;restore logical column
F909  4C57F9               JMP     RRC     ;restore row and column, return


                    **     SBS - Set Buffer Start and Logical Column
                    *
                    *      ENTRY   JSR     SBS
                    *
                    *      MODS
                    *              Original Author Unknown
                    *              1. Bring closer to Coding Standard (object :
                    *                 R. K. Nordin 11/01/83


      = F90C        SBS    =       *               ;entry
F90C  208EF8               JSR     SLC             ;set logical column
F90F  A551                 LDA     HOLD1
F911  856C                 STA     BUFSTR
F913  A552                 LDA     LMARGN          ;left margin
F915  856D                 STA     BUFSTR+1

F917  60            SBS1   RTS                     ;return
```

advance cursor cd

```
                        **      DQQ - Delete Line
                        *
                        *       ENTRY   JSR     DQQ
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = F918            DQQ     =       *       ;entry
F918  A563                      LDA     LOGCOL  ;logical column
F91A  C552                      CMP     LMARGN  ;left margin
F91C  D002 ^F920                BNE     DQQ1    ;if not at left margin

F91E  C654                      DEC     ROWCRS  ;decrement cursor row

F920  208EF8            DQQ1    JSR     SLC     ;set logical column
                        ;       JMP     DWQ



                        **      DWQ - Delete Line
                        *
                        *       ENTRY   JSR     DWQ
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = F923            DWQ     =       *       ;entry

                        ;       Check for left margin.

F923  A563                      LDA     LOGCOL  ;logical column
F925  C552                      CMP     LMARGN  ;left margin
F927  F0EE ^F917                BEQ     SBS1    ;if at left margin, return

F929  20ACF5                    JSR     CCA     ;convert cursor row/column to addre:
F92C  A553                      LDA     RMARGN  ;right margin
F92E  38                        SEC
F92F  E552                      SBC     LMARGN  ;subtract left margin
F931  A8                        TAY             ;offset to last byte

F932  B164              DWQ1    LDA     (ADRESS),Y
F934  D0E1 ^F917                BNE     SBS1

F936  88                        DEY
F937  10F9 ^F932                BPL     DWQ1    ;if not done

F939  4C27F5                    JMP     DLN1    ;delete line, return
```

```
                        **      CCC - Check for Control Character
                        *
                        *       ENTRY   JSR     CCC
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


        = F93C          CCC     =       *               ;entry

F93C    A220                    LDX     #TCCRL-3        ;offset to last entry

F93E    BD00FB          CCC1    LDA     TCCR,X          ;control character
F941    CDFB02                  CMP     ATACHR          ;ATASCII character
F944    F005 ^F94B              BEQ     CCC2            ;if character found, exit

F946    CA                      DEX
F947    CA                      DEX
F948    CA                      DEX
F949    10F3 ^F93E              BPL     CCC1            ;if not done, continue sear:

F94B    60              CCC2    RTS                     ;return



                        **      SRC - Save Row and Column
                        *
                        *       ENTRY   JSR     SRC
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


        = F94C          SRC     =       *               ;entry
F94C    A202                    LDX     #2              ;offset to last byte

F94E    B554            SRC1    LDA     ROWCRS,X        ;byte of cursor row/column
F950    9DB802                  STA     TMPROW,X        ;save byte of cursor row/co:
F953    CA                      DEX
F954    10F8 ^F94E              BPL     SRC1            ;if not done

F956    60                      RTS                     ;return
```

```
                    **      RRC - Restore Row and Column
                    *
                    *       ENTRY   JSR     RRC
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

     = F957         RRC     =       *                   ;entry
F957  A202                  LDX     #2                  ;offset to last byte

F959  BD8802        RRC1    LDA     TMPROW,X            ;byte of saved cursor row/c:
F95C  9554                  STA     ROWCRS,X            ;byte of cursor row/column
F95E  CA                    DEX
F95F  10F8 ^F959            BPL     RRC1                ;if not done

F961  60                    RTS                         ;return



                    **      SWA - Swap Cursor Position with Regular Cursor Posi:
                    *
                    *       ENTRY   JSR     SWA
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

     = F962         SWA     =       *                   ;entry

                    ;       Check for split screen.

F962  ADBF02                LDA     BOTSCR              ;screen bottom
F965  C918                  CMP     #24                 ;normal indicator
F967  F017 ^F980            BEQ     SWA2                ;if not split screen

                    ;       Swap cursor parameters.

F969  A20B                  LDX     #11                 ;offset to last byte

F96B  B554          SWA1    LDA     ROWCRS,X            ;destination cursor paramet:
F96D  48                    PHA                         ;save cursor parameter
F96E  BD9002                LDA     TXTROW,X            ;source cursor parameter
F971  9554                  STA     ROWCRS,X            ;update destination cursor :
F973  68                    PLA                         ;saved cursor paramater
F974  9D9002                STA     TXTROW,X            ;update source cursor param:
F977  CA                    DEX
F978  10F1 ^F96B            BPL     SWA1                ;if not done

                    ;       Complement swap flag.

F97A  A57B                  LDA     SWPFLG              ;swap flag
```

```
F97C  49FF                    EOR     #$FF            ;complement swap flag
F97E  857B                    STA     SWPFLG          ;update swap flag

                      ;       Exit.

F980  4C1EF2          SWA2    JMP     SST     ;perform screen STATUS, return



                      **      SKC - Sound Key Click
                      *
                      *       ENTRY   JSR     SKC
                      *
                      *       MODS
                      *               Original Author Unknown
                      *               1. Bring closer to Coding Standard (object :
                      *                  R. K. Nordin 11/01/83

      = F983          SKC     =       *       ;entry

                      ;       Initialize.

F983  A27E                    LDX     #2*63   ;2 times trip count
F985  48                      PHA             ;save A

                      ;       Turn loudspeaker on.

F986  8E1FD0          SKC1    STX     CONSOL  ;turn loudspeaker on

                      ;       Wait for VBLANK (loudspeaker off).

F989  AD0BD4                  LDA     VCOUNT  ;vertical line counter

F98C  CD0BD4          SKC2    CMP     VCOUNT  ;current vertical line counter
F98F  F0FB ^F98C              BEQ     SKC2    ;if vertical line not changed

                      ;       Decrement and check trip count.

F991  CA                      DEX
F992  CA                      DEX
F993  10F1 ^F986              BPL     SKC1    ;if not done

                      ;       Exit.

F995  68                      PLA             ;restore A
F996  60                      RTS             ;return
```

```
                    **      SCL - Set Cursor at Left Edge
                    *
                    *       ENTRY   JSR     SCL
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

     = F997         SCL     =       *       ;entry

F997 A900                   LDA     #0      ;assume 0

F999 A67B                   LDX     SWPFLG  ;swap flag
F99B D004 ^F9A1             BNE     SCL1    ;if not swapped

F99D A657                   LDX     DINDEX  ;mode
F99F D002 ^F9A3             BNE     SCL2    ;if not mode 0

F9A1 A552           SCL1    LDA     LMARGN  ;use left margin instead of 0

F9A3 8555           SCL2    STA     COLCRS  ;set low cursor column
F9A5 60                     RTS             ;return


                    **      SMS - Set Memory Scan Counter Address
                    *
                    *       ENTRY   JSR     SMS
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

     = F9A6         SMS     =       *               ;entry
F9A6 A558                   LDA     SAVMSC          ;saved low memory scan coun:
F9A8 8564                   STA     ADRESS          ;set low address
F9AA A559                   LDA     SAVMSC+1        ;saved high memory scan cou:
F9AC 8565                   STA     ADRESS+1        ;set high address
F9AE 60                     RTS                     ;return
```

```
                        **      SSP - Perform Screen SPECIAL
                        *
                        *       SSP draws a line from OLDROW/OLDCOL to NEWROW/NEWCO:
                        *
                        *       ENTRY   JSR     SSP
                        *
                        *       MODS
                        *               A. Miller
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = F9AF            SSP     =       *               ;entry

                        ;       Determine command.

F9AF  A200                      LDX     #0              ;assume no fill
F9B1  A522                      LDA     ICCOMZ          ;command
F9B3  C911                      CMP     #$11            ;DRAW command
F9B5  F008 ^F9BF                BEQ     SSP2            ;if DRAW command

F9B7  C912                      CMP     #$12            ;FILL command
F9B9  F003 ^F9BE                BEQ     SSP1            ;if FILL command

F9BB  A084                      LDY     #NVALID         ;invalid command error
F9BD  60                        RTS                     ;return

F9BE  E8                SSP1    INX                     ;indicate fill

F9BF  8EB702            SSP2    STX     FILFLG          ;right fill flag

                        ;       Set destination row/coulmn.

F9C2  A554                      LDA     ROWCRS          ;cursor row
F9C4  8DF502                    STA     NEWROW
F9C7  A555                      LDA     COLCRS          ;cursor column
F9C9  8DF602                    STA     NEWCOL
F9CC  A556                      LDA     COLCRS+1
F9CE  8DF702                    STA     NEWCOL+1

                        ;       Compute row increment and difference.

F9D1  A901                      LDA     #1              ;assume increment +1
F9D3  8DF802                    STA     ROWINC          ;row increment
F9D6  8DF902                    STA     COLINC          ;column increment
F9D9  38                        SEC
F9DA  ADF502                    LDA     NEWROW          ;destination row
F9DD  E55A                      SBC     OLDROW          ;subtract source row
F9DF  8576                      STA     DELTAR          ;row difference
F9E1  B00E ^F9F1                BCS     SSP3            ;if difference positive

                        ;       Set row increment to -1 and complement row differen:

F9E3  A9FF                      LDA     #$FF            ;increment -1
F9E5  8DF802                    STA     ROWINC          ;update row increment
F9E8  A576                      LDA     DELTAR          ;row difference
F9EA  49FF                      EOR     #$FF
```

```
F9EC  18                   CLC
F9ED  6901                 ADC     #1            ;add 1 for 2's complement
F9EF  8576                 STA     DELTAR        ;update row difference

                  ;        Compute column increment and difference.

F9F1  38          SSP3     SEC
F9F2  ADF602               LDA     NEWCOL        ;destination column
F9F5  E55B                 SBC     OLDCOL        ;source column
F9F7  8577                 STA     DELTAC        ;column difference
F9F9  ADF702               LDA     NEWCOL+1
F9FC  E55C                 SBC     OLDCOL+1
F9FE  8578                 STA     DELTAC+1
FA00  B017 ^FA19           BCS     SSP4          ;if difference positive

                  ;        Set column increment to -1 and complement column di:

FA02  A9FF                 LDA     #$FF          ;increment -1
FA04  8DF902               STA     COLINC        ;update column increment
FA07  A577                 LDA     DELTAC        ;column difference
FA09  49FF                 EOR     #$FF          ;absolute value of column d:
FA0B  8577                 STA     DELTAC        ;update column difference
FA0D  A578                 LDA     DELTAC+1
FA0F  49FF                 EOR     #$FF
FA11  8578                 STA     DELTAC+1
FA13  E677                 INC     DELTAC        ;add 1 for 2's complement
FA15  D002 ^FA19           BNE     SSP4          ;if no carry

FA17  E678                 INC     DELTAC+1      ;adjust for 2's complement

                  ;        Set up working row/column and cursor row/column.

FA19  A202        SSP4     LDX     #2            ;offset to last byte
FA1B  A000                 LDY     #0
FA1D  8473                 STY     COLAC+1       ;zero high working column

FA1F  98          SSP5     TYA
FA20  9570                 STA     ROWAC,X       ;zero byte of working row/c:
FA22  B55A                 LDA     OLDROW,X      ;byte of source row/column
FA24  9554                 STA     ROWCRS,X      ;byte of cursor row/column
FA26  CA                   DEX
FA27  10F6 ^FA1F           BPL     SSP5          ;if not done

                  ;        Determine difference.

FA29  A577                 LDA     DELTAC        ;low column difference
FA2B  E8                   INX                   ;offset to working row
FA2C  A8                   TAY                   ;low column difference
FA2D  A578                 LDA     DELTAC+1      ;high column difference
FA2F  857F                 STA     COUNTR+1      ;initialize high iteration :
FA31  8575                 STA     ENDPT+1       ;initialize high end point
FA33  D00B ^FA40           BNE     SSP6          ;if high column difference :

FA35  A577                 LDA     DELTAC        ;low column difference
FA37  C576                 CMP     DELTAR        ;row difference
FA39  B005 ^FA40           BCS     SSP6          ;if column difference > row:
```

```
FA3B  A576             LDA   DELTAR        ;row difference
FA3D  A202             LDX   #2            ;offset to working column
FA3F  A8               TAY                 ;row difference

FA40  98         SSP6  TYA                 ;low maximum difference
FA41  857E             STA   COUNTR        ;low iteration counter
FA43  8574             STA   ENDPT         ;low end point
FA45  48               PHA                 ;save low end point
FA46  A575             LDA   ENDPT+1       ;high end point
FA48  4A               LSR   A             ;C = LSB of high end point
FA49  68               PLA                 ;saved low end point
FA4A  6A               ROR   A
FA4B  9570             STA   ROWAC,X       ;low working row or column

                 ;     Check for iteration counter zero.

FA4D  A57E       SSP7  LDA   COUNTR        ;low iteration counter
FA4F  057F             ORA   COUNTR+1      ;or in high iteration count;
FA51  D003 ^FA56       BNE   SSP8          ;if iteration counter is no;

FA53  4C01FB           JMP   SSP19         ;exit

                 ;     Update working row.

FA56  18         SSP8  CLC
FA57  A570             LDA   ROWAC         ;working row
FA59  6576             ADC   DELTAR        ;row difference
FA5B  8570             STA   ROWAC         ;update working row
FA5D  9002 ^FA61       BCC   SSP9          ;if no carry

FA5F  E671             INC   ROWAC+1       ;adjust high working row

FA61  A571       SSP9  LDA   ROWAC+1       ;high working row
FA63  C575             CMP   ENDPT+1       ;high end point
FA65  9015 ^FA7C       BCC   SSP11         ;if high working row < high;

FA67  D006 ^FA6F       BNE   SSP10         ;if high working row > high;

FA69  A570             LDA   ROWAC         ;low working row
FA6B  C574             CMP   ENDPT         ;low end point
FA6D  900D ^FA7C       BCC   SSP11         ;if low working row < low e;

FA6F  18         SSP10 CLC
FA70  A554             LDA   ROWCRS        ;cursor row
FA72  6DF802           ADC   ROWINC        ;add row increment
FA75  8554             STA   ROWCRS        ;update cursor row
FA77  A200             LDX   #0            ;indicate subtract from wor;
FA79  20AEF6           JSR   SEP           ;subtract end pointer

FA7C  18         SSP11 CLC
FA7D  A572             LDA   COLAC         ;low working column
FA7F  6577             ADC   DELTAC        ;add column difference
FA81  8572             STA   COLAC         ;update working column
FA83  A573             LDA   COLAC+1
FA85  6578             ADC   DELTAC+1
FA87  8573             STA   COLAC+1
FA89  C575             CMP   ENDPT+1       ;high end point
```

```
FA8B  9026 ^FAB5          BCC    SSP15       ;if high working column < h:

FA8D  D006 ^FA95          BNE    SSP12       ;if high working column > h:

FA8F  A572                LDA    COLAC       ;low working column
FA91  C574                CMP    ENDPT       ;low end point
FA93  9020 ^FAB5          BCC    SSP15       ;if low working column < lo:

FA95  2CF902      SSP12   BIT    COLINC      ;column increment
FA98  1010 ^FAAA          BPL    SSP13       ;if column increment positi:

FA9A  C655                DEC    COLCRS      ;decrement low cursor colum:
FA9C  A555                LDA    COLCRS      ;low cursor column
FA9E  C9FF                CMP    #$FF
FAA0  D00E ^FAB0          BNE    SSP14

FAA2  A556                LDA    COLCRS+1    ;high cursor column
FAA4  F00A ^FAB0          BEQ    SSP14       ;if zero, do not decrement

FAA6  C656                DEC    COLCRS+1    ;decrement high cursor colu:
FAA8  1006 ^FAB0          BPL    SSP14

FAAA  E655        SSP13   INC    COLCRS      ;increment low cursor colum:
FAAC  D002 ^FAB0          BNE    SSP14       ;if no carry

FAAE  E656                INC    COLCRS+1    ;adjust high cursor column

FAB0  A202        SSP14   LDX    #2          ;indicate subtract from wor:
FAB2  20AEF6              JSR    SEP         ;subtract end pointer

                  ;       Plot point.

FAB5  20CAF6      SSP15   JSR    CCR         ;check cursor range
FAB8  20CAF1              JSR    PLO         ;plot point

                  ;       Check for right fill.

FABB  ADB702              LDA    FILFLG      ;right fill flag
FABE  F02F ^FAEF          BEQ    SSP18       ;if no right fill

                  ;       Process right fill.

FAC0  204CF9              JSR    SRC         ;save row and column
FAC3  ADFB02              LDA    ATACHR      ;plot point
FAC6  8DBC02              STA    HOLD4       ;save plot point

FAC9  A554        SSP16   LDA    ROWCRS      ;cursor row
FACB  48                  PHA                ;save cursor row
FACC  2012F6              JSR    ACC         ;advance cursor column
FACF  68                  PLA                ;saved cursor row
FAD0  8554                STA    ROWCRS      ;restore cursor row
FAD2  20CAF6              JSR    CCR         ;check cursor range
FAD5  208FF1              JSR    GDC         ;get data under cursor
FAD8  D00C ^FAE6          BNE    SSP17       ;if non-zero data encounter:

FADA  ADFD02              LDA    FILDAT      ;fill data
FADD  8DFB02              STA    ATACHR      ;plot point
```

```
FAE0  20CAF1                 JSR     PLO              ;plot point
FAE3  4CC9FA                 JMP     SSP16            ;continue

FAE6  ADBC02         SSP17   LDA     HOLD4            ;saved plot point
FAE9  8DFB02                 STA     ATACHR           ;restore plot point
FAEC  2057F9                 JSR     RRC              ;restore row and column

                     ;       Subtract 1 from iteration counter.

FAEF  38             SSP18   SEC
FAF0  A57E                   LDA     COUNTR           ;iteration counter
FAF2  E901                   SBC     #1               ;subtract 1
FAF4  857E                   STA     COUNTR           ;update iteration counter
FAF6  A57F                   LDA     COUNTR+1
FAF8  E900                   SBC     #0
FAFA  857F                   STA     COUNTR+1

                     ;       Check for completion.

FAFC  3003 ^FB01             BMI     SSP19    ;if iteration counter negative, exit

FAFE  4C4DFA                 JMP     SSP7     ;continue

                     ;       Exit.

FB01  4C1EF2         SSP19   JMP     SST      ;perform screen STATUS, return



                     **      TMSK - Table of Bit Masks


FB04  00             TMSK    DB      $00      ;0 - mask for no bits
FB05  01                     DB      $01      ;1 - mask for lower 1 bit
FB06  03                     DB      $03      ;2 - mask for lower 2 bits
FB07  07                     DB      $07      ;3 - mask for lower 3 bits



                     **      TDSC - Table of Default Screen Colors


FB08  28             TDSC    DB      $28      ;default playfield 0 color
FB09  CA                     DB      $CA      ;default playfield 1 color
FB0A  94                     DB      $94      ;default playfield 2 color
FB0B  46                     DB      $46      ;default playfield 3 color
FB0C  00                     DB      $00      ;default background color
```

```
                        **      TCCR - Table of Control Character Routines
                        *
                        *       Each entry is 3 bytes.  The first byte is the contr:
                        *       character; the second and third bytes are the addre:
                        *       the routine which processes the control character.

FB0D  1B        TCCR    DB      $1B
FB0E  E0F3              DW      ESC     ;escape

FB10  1C                DB      $1C
FB11  E6F3              DW      CUP     ;move cursor up

FB13  1D                DB      $1D
FB14  F3F3              DW      CDN     ;move cursor down

FB16  1E                DB      $1E
FB17  00F4              DW      CLF     ;move cursor left

FB19  1F                DB      $1F
FB1A  11F4              DW      CRT     ;move  cursor right

FB1C  7D                DB      $7D
FB1D  20F4              DW      CSC     ;clear screen .

FB1F  7E                DB      $7E
FB20  50F4              DW      BSP     ;backspace

FB22  7F                DB      $7F
FB23  7AF4              DW      TAB     ;tab

FB25  9B                DB      $9B
FB26  61F6              DW      RWS     ;return with scrolling

FB28  9C                DB      $9C
FB29  20F5              DW      DLN     ;delete line

FB2B  9D                DB      $9D
FB2C  0CF5              DW      ILN     ;insert line

FB2E  9E                DB      $9E
FB2F  9AF4              DW      CTB     ;clear tab

FB31  9F                DB      $9F
FB32  95F4              DW      STB     ;set tab

FB34  FD                DB      $FD
FB35  56F5              DW      BEL     ;sound bell

FB37  FE                DB      $FE
FB38  D5F4              DW      DCH     ;delete character

FB3A  FF                DB      $FF
FB3B  9FF4              DW      ICH     ;insert character

      = 0030    TCCRL   =       *-TCCR  ;length
```

```
                        **      TSFR - Table of Super Function (Shifted Function Ke:
                        *
                        *       Each entry is 3 bytes.  The first byte is the super:
                        *       character; the second and third bytes are the addre:
                        *       routine which processes the super function.


FB3D  1C                TSFR    DB      $1C
FB3E  40F4                      DW      CHM     ;move cursor home

FB40  1D                        DB      $1D
FB41  5FF5                      DW      CBT     ;move cursor to bottom

FB43  1E                        DB      $1E
FB44  1BF4                      DW      CLM     ;move cursor to left margin

FB46  1F                        DB      $1F
FB47  0AF4                      DW      CRM     ;move cursor to right margin



                        **      TAIC - Table of ATASCII to Internal Conversion Cons:


FB49  40                TAIC    DB      $40     ;0
FB4A  00                        DB      $00     ;1
FB4B  20                        DB      $20     ;2
FB4C  60                        DB      $60     ;3



                        **      TIAC - Table of Internal to ATASCII Conversion Cons:


FB4D  20                TIAC    DB      $20     ;0
FB4E  40                        DB      $40     ;1
FB4F  00                        DB      $00     ;2
FB50  60                        DB      $60     ;3



                        **      TCKD - Table of Character Key Definitions
                        *
                        *       Entry n is the ATASCII equivalent of key code n.


      = FB51            TCKD    =       *

                        ;       Lower Case Characters

FB51  6C                        DB      $6C     ;$00 - l
FB52  6A                        DB      $6A     ;$01 - j
FB53  3B                        DB      $3B     ;$02 - semicolon
FB54  8A                        DB      $8A     ;$03 - F1
```

```
FB55  8B          DB      $8B        ;$04 - F2
FB56  6B          DB      $6B        ;$05 - k
FB57  2B          DB      $2B        ;$06 - +
FB58  2A          DB      $2A        ;$07 - *
FB59  6F          DB      $6F        ;$08 - o
FB5A  80          DB      $80        ;$09 - (invalid)
FB5B  70          DB      $70        ;$0A - p
FB5C  75          DB      $75        ;$0B - u
FB5D  9B          DB      $9B        ;$0C - return
FB5E  69          DB      $69        ;$0D - i
FB5F  2D          DB      $2D        ;$0E - -
FB60  3D          DB      $3D        ;$0F - =

FB61  76          DB      $76        ;$10 - v
FB62  80          DB      $80        ;$11 - (invalid)
FB63  63          DB      $63        ;$12 - c
FB64  8C          DB      $8C        ;$13 - F3
FB65  8D          DB      $8D        ;$14 - F4
FB66  62          DB      $62        ;$15 - b
FB67  78          DB      $78        ;$16 - x
FB68  7A          DB      $7A        ;$17 - z
FB69  34          DB      $34        ;$18 - 4
FB6A  80          DB      $80        ;$19 - (invalid)
FB6B  33          DB      $33        ;$1A - 3
FB6C  36          DB      $36        ;$1B - 6
FB6D  1B          DB      $1B        ;$1C - escape
FB6E  35          DB      $35        ;$1D - 5
FB6F  32          DB      $32        ;$1E - 2
FB70  31          DB      $31        ;$1F - 1

FB71  2C          DB      $2C        ;$20 - comma
FB72  20          DB      $20        ;$21 - space
FB73  2E          DB      $2E        ;$22 - period
FB74  6E          DB      $6E        ;$23 - n
FB75  80          DB      $80        ;$24 - (invalid)
FB76  6D          DB      $6D        ;$25 - m
FB77  2F          DB      $2F        ;$26 - /
FB78  81          DB      $81        ;$27 - inverse
FB79  72          DB      $72        ;$28 - r
FB7A  80          DB      $80        ;$29 - (invalid)
FB7B  65          DB      $65        ;$2A - e
FB7C  79          DB      $79        ;$2B - y
FB7D  7F          DB      $7F        ;$2C - tab
FB7E  74          DB      $74        ;$2D - t
FB7F  77          DB      $77        ;$2E - w
FB80  71          DB      $71        ;$2F - q

FB81  39          DB      $39        ;$30 - 9
FB82  80          DB      $80        ;$31 - (invalid)
FB83  30          DB      $30        ;$32 - 0
FB84  37          DB      $37        ;$33 - 7
FB85  7E          DB      $7E        ;$34 - backspace
FB86  38          DB      $38        ;$35 - 8
FB87  3C          DB      $3C        ;$36 - <
FB88  3E          DB      $3E        ;$37 - >
FB89  66          DB      $66        ;$38 - f
FB8A  68          DB      $68        ;$39 - h
```

```
FB8B  64          DB    $64    ;$3A - d
FB8C  80          DB    $80    ;$3B - (invalid)
FB8D  82          DB    $82    ;$3C - CAPS
FB8E  67          DB    $67    ;$3D - g
FB8F  73          DB    $73    ;$3E - s
FB90  61          DB    $61    ;$3F - a

              ;   Upper Case Characters

FB91  4C          DB    $4C    ;$40 - L
FB92  4A          DB    $4A    ;$41 - J
FB93  3A          DB    $3A    ;$42 - colon
FB94  8A          DB    $8A    ;$43 - SHIFT-F1
FB95  8B          DB    $8B    ;$44 - SHIFT-F2
FB96  4B          DB    $4B    ;$45 - K
FB97  5C          DB    $5C    ;$46 - \
FB98  5E          DB    $5E    ;$47 - ^
FB99  4F          DB    $4F    ;$48 - O
FB9A  80          DB    $80    ;$49 - (invalid)
FB9B  50          DB    $50    ;$4A - P
FB9C  55          DB    $55    ;$4B - U
FB9D  9B          DB    $9B    ;$4C - SHIFT-return
FB9E  49          DB    $49    ;$4D - I
FB9F  5F          DB    $5F    ;$4E - _
FBA0  7C          DB    $7C    ;$4F - |

FBA1  56          DB    $56    ;$50 - V
FBA2  80          DB    $80    ;$51 - (invalid)
FBA3  43          DB    $43    ;$52 - C
FBA4  8C          DB    $8C    ;$53 - SHIFT-F3
FBA5  8D          DB    $8D    ;$54 - SHIFT-F4
FBA6  42          DB    $42    ;$55 - B
FBA7  58          DB    $58    ;$56 - X
FBA8  5A          DB    $5A    ;$57 - Z
FBA9  24          DB    $24    ;$58 - $
FBAA  80          DB    $80    ;$59 - (invalid)
FBAB  23          DB    $23    ;$5A - #
FBAC  26          DB    $26    ;$5B - &
FBAD  1B          DB    $1B    ;$5C - SHIFT-escape
FBAE  25          DB    $25    ;$5D - %
FBAF  22          DB    $22    ;$5E - "
FBB0  21          DB    $21    ;$5F - !

FBB1  5B          DB    $5B    ;$60 - [
FBB2  20          DB    $20    ;$61 - SHIFT-space
FBB3  5D          DB    $5D    ;$62 - ]
FBB4  4E          DB    $4E    ;$63 - N
FBB5  80          DB    $80    ;$64 - (invalid)
FBB6  4D          DB    $4D    ;$65 - M
FBB7  3F          DB    $3F    ;$66 - ?
FBB8  81          DB    $81    ;$67 - SHIFT-inverse
FBB9  52          DB    $52    ;$68 - R
FBBA  80          DB    $80    ;$69 - (invalid)
FBBB  45          DB    $45    ;$6A - E
FBBC  59          DB    $59    ;$6B - Y
FBBD  9F          DB    $9F    ;$6C - SHIFT-tab
FBBE  54          DB    $54    ;$6D - T
```

```
FBBF  57                 DB     $57     ;$6E - W
FBC0  51                 DB     $51     ;$6F - Q

FBC1  28                 DB     $28     ;$70 - (
FBC2  80                 DB     $80     ;$71 - (invalid)
FBC3  29                 DB     $29     ;$72 - )
FBC4  27                 DB     $27     ;$73 - '
FBC5  9C                 DB     $9C     ;$74 - SHIFT-delete
FBC6  40                 DB     $40     ;$75 - @
FBC7  7D                 DB     $7D     ;$76 - SHIFT-clear
FBC8  9D                 DB     $9D     ;$77 - SHIFT-insert
FBC9  46                 DB     $46     ;$78 - F
FBCA  48                 DB     $48     ;$79 - H
FBCB  44                 DB     $44     ;$7A - D
FBCC  80                 DB     $80     ;$7B - (invalid)
FBCD  83                 DB     $83     ;$7C - SHIFT-CAPS
FBCE  47                 DB     $47     ;$7D - G
FBCF  53                 DB     $53     ;$7E - S
FBD0  41                 DB     $41     ;$7F - A

               ;         Control Characters

FBD1  0C                 DB     $0C     ;$80 - CTRL-L
FBD2  0A                 DB     $0A     ;$81 - CTRL-J  .
FBD3  7B                 DB     $7B     ;$82 - CTRL-semicolon
FBD4  80                 DB     $80     ;$83 - (invalid)
FBD5  80                 DB     $80     ;$84 - (invalid)
FBD6  0B                 DB     $0B     ;$85 - CTRL-K
FBD7  1E                 DB     $1E     ;$86 - CTRL-left arrow
FBD8  1F                 DB     $1F     ;$87 - CTRL-right arrow
FBD9  0F                 DB     $0F     ;$88 - CTRL-O
FBDA  80                 DB     $80     ;$89 - (invalid)
FBDB  10                 DB     $10     ;$8A - CTRL-P
FBDC  15                 DB     $15     ;$8B - CTRL-U
FBDD  9B                 DB     $9B     ;$8C - CTRL-return
FBDE  09                 DB     $09     ;$8D - CTRL-I
FBDF  1C                 DB     $1C     ;$8E - CTRL-up arrow
FBE0  1D                 DB     $1D     ;$8F - CTRL-down arrow

FBE1  16                 DB     $16     ;$90 - CTRL-V
FBE2  80                 DB     $80     ;$91 - (invalid)
FBE3  03                 DB     $03     ;$92 - CTRL-C
FBE4  89                 DB     $89     ;$93 - CTRL-F3
FBE5  80                 DB     $80     ;$94 - (invalid)
FBE6  02                 DB     $02     ;$95 - CTRL-B
FBE7  18                 DB     $18     ;$96 - CTRL-X
FBE8  1A                 DB     $1A     ;$97 - CTRL-Z
FBE9  80                 DB     $80     ;$98 - (invalid)
FBEA  80                 DB     $80     ;$99 - (invalid)
FBEB  85                 DB     $85     ;$9A - CTRL-3
FBEC  80                 DB     $80     ;$9B - (invalid)
FBED  1B                 DB     $1B     ;$9C - CTRL-escape
FBEE  80                 DB     $80     ;$9D - (invalid)
FBEF  FD                 DB     $FD     ;$9E - CTRL-2
FBF0  80                 DB     $80     ;$9F - (invalid)

FBF1  00                 DB     $00     ;$A0 - CTRL-comma
```

```
FBF2  20               DB      $20         ;$A1 - CTRL-space
FBF3  60               DB      $60         ;$A2 - CTRL-period
FBF4  0E               DB      $0E         ;$A3 - CTRL-N
FBF5  80               DB      $80         ;$A4 - (invalid)
FBF6  0D               DB      $0D         ;$A5 - CTRL-M
FBF7  80               DB      $80         ;$A6 - (invalid)
FBF8  81               DB      $81         ;$A7 - CTRL-inverse
FBF9  12               DB      $12         ;$A8 - CTRL-R
FBFA  80               DB      $80         ;$A9 - (invalid)
FBFB  05               DB      $05         ;$AA - CTRL-E
FBFC  19               DB      $19         ;$AB - CTRL-Y
FBFD  9E               DB      $9E         ;$AC - CTRL-tab
FBFE  14               DB      $14         ;$AD - CTRL-T
FBFF  17               DB      $17         ;$AE - CTRL-W
FC00  11               DB      $11         ;$AF - CTRL-Q

FC01  80               DB      $80         ;$B0 - (invalid)
FC02  80               DB      $80         ;$B1 - (invalid)
FC03  80               DB      $80         ;$B2 - (invalid)
FC04  80               DB      $80         ;$B3 - (invalid)
FC05  FE               DB      $FE         ;$B4 - CTRL-delete
FC06  80               DB      $80         ;$B5 - (invalid)
FC07  7D               DB      $7D         ;$B6 - CTRL-clear
FC08  FF               DB      $FF         ;$B7 - CTRL-insert
FC09  06               DB      $06         ;$B8 - CTRL-F
FC0A  08               DB      $08         ;$B9 - CTRL-H
FC0B  04               DB      $04         ;$BA - CTRL-D
FC0C  80               DB      $80         ;$BB - (invalid)
FC0D  84               DB      $84         ;$BC - CTRL-CAPS
FC0E  07               DB      $07         ;$BD - CTRL-G
FC0F  13               DB      $13         ;$BE - CTRL-S
FC10  01               DB      $01         ;$BF - CTRL-A


                 **    TFKD - Table of Function Key Definitions
                 *
                 *     Entry n is the ATASCII equivalent of adjusted funct:
                 *     code n.


FC11  1C         TFKD  DB      $1C         ;0 - F1 key
FC12  1D               DB      $1D         ;1 - F2 key
FC13  1E               DB      $1E         ;2 - F3 key
FC14  1F               DB      $1F         ;3 - F4 key

FC15  8E               DB      $8E         ;4 - SHIFT-F1 key
FC16  8F               DB      $8F         ;5 - SHIFT-F2 key
FC17  90               DB      $90         ;6 - SHIFT-F3 key
FC18  91               DB      $91         ;7 - SHIFT-F4 key
```

```
                        **      KIR - Process Keyboard IRQ
                        *
                        *       ENTRY   JMP     KIR
                        *
                        *       EXIT
                        *               Exits via RTI
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


          = FC19        KIR     =       *       ;entry

                        ;       Initialize.

FC19    8A                      TXA
FC1A    48                      PHA             ;save X
FC1B    98                      TYA
FC1C    48                      PHA             ;save Y
FC1D    AC01D3                  LDY     PORTB   ;port B memory control
FC20    AD09D2                  LDA     KBCODE  ;keyboard code
FC23    CDF202                  CMP     CH1     ;last key code
FC26    D005 ^FC2D              BNE     KIR1    ;if not last key code

FC28    AEF102                  LDX     KEYDEL  ;keyboard debounce delay
FC2B    D049 ^FC76              BNE     KIR8    ;if delay not expired, treat as bou:

                        ;       Check for CTRL-F1.

FC2D    AE6D02          KIR1    LDX     KEYDIS  ;save keyboard disable flag
FC30    C983                    CMP     #CNTLF1
FC32    D013 ^FC47              BNE     KIR4    ;if not CTRL-F1

                        ;       Process CTRL-F1.

FC34    8A                      TXA             ;keyboard disable flag
FC35    49FF                    EOR     #$FF    ;complement keyboard disable flag
FC37    8D6D02                  STA     KEYDIS  ;update keyboard disable flag
FC3A    D005 ^FC41              BNE     KIR2    ;if keyboard disabled

FC3C    98                      TYA             ;port B memory control
FC3D    0904                    ORA     #$04    ;turn off LED 1
FC3F    D003 ^FC44              BNE     KIR3    ;update port B memory control

FC41    98              KIR2    TYA             ;port B memory control
FC42    29FB                    AND     #$FB    ;turn on LED 1

FC44    A8              KIR3    TAY             ;updated port B memory control
FC45    B026 ^FC6D              BCS     KIR7    ;reset keyboard controls

                        ;       Check keyboard disable.

FC47    8A              KIR4    TXA             ;keyboard disable flag
FC48    D03D ^FC87              BNE     KIR9    ;if keyboard disabled, exit
```

```
                        ;       Get character.

FC4A  AD0902                    LDA     KBCODE  ;keyboard code
FC4D  AA                        TAX             ;character

                        ;       Check for CTRL-1.

FC4E  C99F                      CMP     #CNTL1
FC50  D00A ^FC5C                BNE     KIR5    ;if not CTRL-1

                        ;       Process CTRL-1.

FC52  ADFF02                    LDA     SSFLAG  ;start/stop flag
FC55  49FF                      EOR     #$FF    ;complement start/stop flag
FC57  8DFF02                    STA     SSFLAG  ;update start/stop flag
FC5A  B011 ^FC6D                BCS     KIR7    ;make CTRL-1 invisible

                        ;       Check character.

FC5C  293F              KIR5    AND     #$3F    ;mask off shift and control bits
FC5E  C911                      CMP     #HELP
FC60  D02E ^FC90                BNE     KIR10   ;if not HELP key

                        ;       Process HELP.

FC62  8EDC02                    STX     HELPFG  ;indicate HELP key pressed
FC65  F006 ^FC6D                BEQ     KIR7    ;reset keyboard controls

                        ;       Process character.

FC67  8EFC02            KIR6    STX     CH      ;key code
FC6A  8EF202                    STX     CH1     ;reset previous key code

                        ;       Reset keyboard controls.

FC6D  A903              KIR7    LDA     #3
FC6F  8DF102                    STA     KEYDEL  ;re-initialize for debounce
FC72  A900                      LDA     #0
FC74  854D                      STA     ATRACT  ;clear attract-mode timer/flag

                        ;       Prepare to exit.

FC76  ADD902            KIR8    LDA     KRPDEL  ;auto-repeat delay
FC79  8D2802                    STA     SRTIMR  ;reset software key repeat timer
FC7C  AD2F02                    LDA     SDMCTL  ;DMA control
FC7F  D006 ^FC87                BNE     KIR9    ;if DMA not disabled, exit

FC81  ADD002                    LDA     DMASAV  ;saved DMA control
FC84  8D2F02                    STA     SDMCTL  ;DMA control

                        ;       Exit.

FC87  8C01D3            KIR9    STY     PORTB   ;update port B memory control
FC8A  68                        PLA             ;saved Y
FC8B  A8                        TAY             ;restore Y
FC8C  68                        PLA             ;saved X
FC8D  AA                        TAX             ;restore X
```

```
FC8E  68                    PLA                 ;restore A
FC8F  40                    RTI                 ;return

                    ;       Check for CTRL-F2 or CTRL-F4.

FC90  E084          KIR10   CPX     #CNTLF2
FC92  F021 ^FCB5            BEQ     KIR12   ;if CTRL-F2

FC94  E094                  CPX     #CNTLF4
FC96  D0CF ^FC67            BNE     KIR6    ;if not CTRL-F4

                    ;       Process CTRL-F4.

FC98  ADF402                LDA     CHBAS   ;character set base
FC9B  AE6B02                LDX     CHSALT  ;character set alternate
FC9E  8D6B02                STA     CHSALT  ;update character set alternate
FCA1  8EF402                STX     CHBAS   ;update character set base

FCA4  E0CC                  CPX     #high ICSORG    ;high international charact:
FCA6  F006 ^FCAE            BEQ     KIR11           ;if international character:

FCA8  98                    TYA             ;port B memory control
FCA9  0908                  ORA     #$08    ;turn off LED 2
FCAB  A8                    TAY             ;updated port B memory control
FCAC  D0BF ^FC6D            BNE     KIR7    ;reset keyboard controls

FCAE  98            KIR11   TYA             ;port B memory control
FCAF  29F7                  AND     #$F7    ;turn on LED 2
FCB1  A8                    TAY             ;updated port B memory control
FCB2  4C6DFC                JMP     KIR7    ;reset keyboard controls

                    ;       Process CTRL-F2.

FCB5  AD2F02        KIR12   LDA     SDMCTL  ;DMA control
FCB8  F0CD ^FC87            BEQ     KIR9    ;if disabled, exit

FCBA  8DDD02                STA     DMASAV  ;save DMA state
FCBD  A900                  LDA     #0      ;disable DMA
FCBF  8D2F02                STA     SDMCTL  ;DMA control
FCC2  F0C3 ^FC87            BEQ     KIR9    ;exit



                    **      FDL - Process Display List Interrupt for Fine Scrol:
                    *
                    *       ENTRY   JMP     FDL
                    *
                    *       EXIT
                    *               Exits via RTI
                    *
                    *       NOTES
                    *               Problem: in the CRASS65 version, COLRSH was:
                    *               zero-page.
                    *               Problem: in the CRASS65 version, DRKMSK was:
                    *               zero-page.
                    *
```

```
                        *       MODS
                        *               H. Stewart      06/01/82
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


        = FCC4          FDL     =       *               ;entry
FCC4    48                      PHA                     ;save A
FCC5    ADC602                  LDA     COLOR2          ;playfield 2 color
                        ;       EOR     COLRSH          ;modify with attract-mode c:
FCC8    4D4F00                  VFD     8\$4D,8\low COLRSH,8\high COLRSH
                        ;       AND     DRKMSK          ;modify with attract-mode l:
FCCB    2D4E00                  VFD     8\$2D,8\low DRKMSK,8\high COLRSH
FCCE    8D0AD4                  STA     WSYNC           ;wait for HBLANK synchroniz:
FCD1    8D17D0                  STA     COLPF1          ;playfield 1 color/luminanc:
FCD4    68                      PLA                     ;restore A
FCD5    40                      RTI                     ;return
```

```
FCD6                      FIX     $FCD8



                    **    FCD8 - $FCD8 Patch
                    *
                    *     For compatibility with OS Revision B, sound key cli:

FCD8  4C83F9              JMP     SKC     ;sound key click, return
```

```
FCDB                    **      CIN - Initialize Cassette
                        *
                        *       ENTRY   JSR     CIN
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = FCDB             CIN     =       *                       ;entry
FCDB  A9CC                      LDA     #low B00600             ;indicate 600 baud
FCDD  8DEE02                    STA     CBAUDL                  ;cassette baud rate
FCE0  A905                      LDA     #high B00600
FCE2  8DEF02                    STA     CBAUDH
                        ;       JMP     CSP                     ;return



                        **      CSP - Perform Cassette SPECIAL
                        *
                        *       CSP does nothing.
                        *
                        *       ENTRY   JSR     CSP
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = FCE5             CSP     =       *       ;entry
FCE5  60                        RTS             ;return



                        **      COP - Perform Cassette OPEN
                        *
                        *       ENTRY   JSR     COP
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


     = FCE6             COP     =       *       ;entry

                        ;       Set cassette IRG type.

FCE6  A52B                      LDA     ICAX2Z  ;second auxiliary information
FCE8  853E                      STA     FTYPE   ;cassette IRG type

                        ;       Check OPEN mode.
```

```
FCEA  A52A                 LDA     ICAX1Z  ;OPEN mode
FCEC  290C                 AND     #$0C    ;open for input and output bits
FCEE  C904                 CMP     #$04    ;open for input bit
FCF0  F005 ^FCF7           BEQ     OCI     ;if open for input, process, return

FCF2  C908                 CMP     #$08    ;open for output bit
FCF4  F03E ^FD34           BEQ     OCO     ;if open for output, process, retur:

                   ;       Exit.

FCF6  60                   RTS             ;return



                   **      OCI - Open Cassette for Input
                   *
                   *       ENTRY   JSR     OCI
                   *
                   *       MODS
                   *               Original Author Unknown
                   *               1. Bring closer to Coding Standard (object :
                   *                  R. K. Nordin 11/01/83

      = FCF7       OCI     =       *       ;entry

                   ;       Process open for input.

FCF7  A900                 LDA     #0      ;indicate reading
FCF9  8D8902               STA     WMODE   ;WRITE mode
FCFC  853F                 STA     FEOF    ;indicate no EOF yet
FCFE  A901                 LDA     #TONE2  ;tone for pressing PLAY
FD00  20FCFD               JSR     AUB     ;alert user with beep
FD03  3029 ^FD2E           BMI     PBC1    ;if error

                   ;       Initiate cassette READ.

                   ;       JMP     ICR     ;initiate cassette READ, return



                   **      ICR - Initiate Cassette READ
                   *
                   *       ENTRY   JSR     ICR
                   *
                   *       MODS
                   *               Original Author Unknown
                   *               1. Bring closer to Coding Standard (object :
                   *                  R. K. Nordin 11/01/83

      = FD05       ICR     =       *               ;entry

                   ;       Initialize.
```

```
FD05  A934                LDA    #MOTRGO          ;motor on
FD07  8D02D3              STA    PACTL            ;port A control

                    ;     Wait for leader read.

FD0A  A662                LDX    PALNTS
FD0C  BC93FE              LDY    RLEADL,X         ;low READ leader
FD0F  BD91FE              LDA    RLEADH,X         ;high READ leader
FD12  AA                  TAX
FD13  A903                LDA    #3
FD15  8D2A02              STA    CDTMF3
FD18  205CE4              JSR    SETVBV           ;set up VBLANK timer

FD1B  AD2A02        ICR1  LDA    CDTMF3
FD1E  D0FB ^FD1B          BNE    ICR1             ;if not done waiting

                    ;     Initialize.

FD20  A980                LDA    #128   ;buffer size
FD22  853D                STA    BPTR   ;initialize buffer pointer
FD24  8D8A02              STA    BLIM   ;initialize buffer limit
FD27  4C77FD              JMP    OC02   ;exit


                    **    PBC - Process BREAK for Cassette Operation
                    *
                    *     ENTRY  JSR    PBC
                    *
                    *     MODS
                    *            Original Author Unknown
                    *            1. Bring closer to Coding Standard (object :
                    *               R. K. Nordin 11/01/83


      = FD2A        PBC   =      *      ;entry
FD2A  A080                LDY    #BRKABT ;BREAK abort error
FD2C  C611                DEC    BRKKEY ;reset BREAK key flag

FD2E  A900          PBC1  LDA    #0     ;indicate reading
FD30  8D8902              STA    WMODE  ;WRITE mode
FD33  60                  RTS           ;return
```

```
                        **      OCO - Open Cassette for Output
                        *
                        *       ENTRY   JSR     OCO
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

        = FD34          OCO     =       *       ;entry

                        ;       Initialize.

FD34    A980                    LDA     #$80    ;indicate writing
FD36    8D8902                  STA     WMODE   ;WRITE mode
FD39    A902                    LDA     #TONE1
FD3B    20FCFD                  JSR     AUB     ;alert user with beep
FD3E    30EE ^FD2E              BMI     PBC1    ;if error

                        ;       Set baud rate to 600.

FD40    A9CC                    LDA     #low B00600     ;600 baud
FD42    8D04D2                  STA     AUDF3
FD45    A905                    LDA     #high B00600
FD47    8D06D2                  STA     AUDF4

                        ;       Write marks.

FD4A    A960                    LDA     #$60
FD4C    8D0003                  STA     DDEVIC
FD4F    2068E4                  JSR     SENDEV
FD52    A934                    LDA     #MOTRGO         ;write 5 second blank tape
FD54    8D02D3                  STA     PACTL

                        ;       Wait for leader written.

FD57    A662                    LDX     PALNTS
FD59    BC8FFE                  LDY     WLEADL,X
FD5C    BD8DFE                  LDA     WLEADH,X
FD5F    AA                      TAX
FD60    A903                    LDA     #3
FD62    205CE4                  JSR     SETVBV          ;set VBLANK parmaeters
FD65    A9FF                    LDA     #$FF
FD67    8D2A02                  STA     CDTMF3

FD6A    A511            OCO1    LDA     BRKKEY  ;BREAK key flag
FD6C    F0BC ^FD2A              BEQ     PBC     ;if BREAK during write leader, proc:

FD6E    AD2A02                  LDA     CDTMF3
FD71    D0F7 ^FD6A              BNE     OCO1    ;if not done waiting

                        ;       Initialize buffer pointer.

FD73    A900                    LDA     #0
FD75    853D                    STA     BPTR    ;buffer pointer
```

```
                       ;        Indicate success.

FD77  A001      0C02   LDY     #SUCCES ;indicate success
FD79  60               RTS             ;return



                       **       CGB - Perform Cassette GET-BYTE
                       *
                       *        ENTRY   JSR     CGB
                       *
                       *        MODS
                       *                Original Author Unknown
                       *                1. Bring closer to Coding Standard (object :
                       *                   R. K. Nordin 11/01/83

      = FD7A    CGB    =        *               ;entry

                       ;        Check for EOF.

FD7A  A53F             LDA     FEOF            ;EOF flag
FD7C  3033 ^FDB1       BMI     RCB3            ;if at EOF already

                       ;        Check for end of buffer.

FD7E  A63D             LDX     BPTR            ;buffer pointer
FD80  EC8A02           CPX     BLIM            ;buffer limit
FD83  F008 ^FD8D       BEQ     RCB             ;if end of buffer, read blo:

                       ;        Get next byte.

FD85  BD0004           LDA     CASBUF+3,X      ;byte
FD88  E63D             INC     BPTR            ;increment pointer
FD8A  A001             LDY     #SUCCES         ;indicate success

FD8C  60        CGB1   RTS                     ;return



                       **       RCB - Read Cassette Block
                       *
                       *        ENTRY   JSR     RCB
                       *
                       *        MODS
                       *                Original Author Unknown
                       *                1. Bring closer to Coding Standard (object :
                       *                   R. K. Nordin 11/01/83

      = FD8D    RCB    =        *       ;entry

                       ;        Perform READ.

FD8D  A952             LDA     #'R'    ;read
```

```
FD8F  203FFE                JSR   SCB       ;perform SIO on cassette buffer
FD92  98                    TYA
FD93  30F7 ^FD8C            BMI   CGB1      ;if SIO error

FD95  A900                  LDA   #0
FD97  853D                  STA   BPTR      ;reset pointer
FD99  A280                  LDX   #$80      ;default number of bytes

                      ;     Check for header.

FD9B  ADFF03                LDA   CASBUF+2
FD9E  C9FE                  CMP   #EOT
FDA0  F00D ^FDAF            BEQ   RCB2      ;if header, read again

                      ;     Check for last record.

FDA2  C9FA                  CMP   #DT1
FDA4  D003 ^FDA9            BNE   RCB1              ;if not last data record

FDA6  AE7F04                LDX   CASBUF+130        ;number of bytes

                      ;     Set number of bytes.

FDA9  8E8A02          RCB1  STX   BLIM

                      ;     Perform cassette GET-BYTE.

FDAC  4C7AFD                JMP   CGB               ;perform cassette GET-BYTE,:

                      ;     Set EOF flag.

FDAF  C63F            RCB2  DEC   FEOF              ;set EOF flag

                      ;     Exit.

FDB1  A088            RCB3  LDY   #EOFERR           ;end of file indicator
FDB3  60                    RTS                     ;return



                      **    CPB - Perform Cassette PUT-BYTE
                      *
                      *     ENTRY   JSR     CPB
                      *
                      *     MODS
                      *             Original Author Unknown
                      *             1. Bring closer to Coding Standard (object :
                      *                R. K. Nordin 11/01/83

      = FDB4          CPB   =     *                 ;entry

                      ;     Move data to buffer.

FDB4  A63D                  LDX   BPTR              ;buffer pointer
FDB6  9D0004                STA   CASBUF+3,X        ;data
```

```
FDB9  E63D                 INC    BPTR          ;increment buffer pointer
FDBB  A001                 LDY    #SUCCES       ;assume success

                  ;        Check buffer full.

FDBD  E07F                 CPX    #127          ;offset to last byte of buf:
FDBF  F001 ^FDC2           BEQ    CPB1          ;if buffer full

FDC1  60                   RTS                  ;return

                  ;        Write cassette buffer.

FDC2  A9FC        CPB1     LDA    #DTA    ;indicate data record type
FDC4  207CFE               JSR    WCB     ;write cassette buffer
FDC7  A900                 LDA    #0
FDC9  853D                 STA    BPTR    ;reset buffer pointer
FDCB  60                   RTS            ;return



                  **       CST - Perform Cassette STATUS
                  *
                  *        ENTRY  JSR    CST
                  *
                  *        MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83


      = FDCC      CST      =      *       ;entry
FDCC  A001                 LDY    #SUCCES ;indicate success
FDCE  60                   RTS            ;return



                  **       CCL - Perform Cassette CLOSE
                  *
                  *        ENTRY  JSR    CCL
                  *
                  *        MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83


      = FDCF      CCL      =      *       ;entry

                  ;        Cheeck mode.

FDCF  AD8902               LDA    WMODE   ;WRITE mode
FDD2  3008 ^FDDC           BMI    CCL2    ;if writing

                  ;        Process reading.
```

```
FDD4  A001                  LDY     #SUCCES ;indicate success

                    ;       Exit.

FDD6  A93C          CCL1    LDA     #MOTRST
FDD8  8D02D3                STA     PACTL   ;stop motor
FDDB  60                    RTS             ;return

                    ;       Process writing.

FDDC  A63D          CCL2    LDX     BPTR              ;buffer pointer
FDDE  F00A ^FDEA            BEQ     CCL3              ;if no data bytes in buffer

FDE0  8E7F04                STX     CASBUF+130        ;number of bytes
FDE3  A9FA                  LDA     #DT1              ;indicate data record type
FDE5  207CFE                JSR     WCB               ;write cassette buffer
FDE8  30EC ^FDD6            BMI     CCL1              ;if error, exit

                    ;       Zero buffer.

FDEA  A27F          CCL3    LDX     #127              ;offset to last byte in buf:
FDEC  A900                  LDA     #0

FDEE  9D0004        CCL4    STA     CASBUF+3,X        ;zero byte
FDF1  CA                    DEX
FDF2  10FA ^FDEE            BPL     CCL4              ;if not done

                    ;       Write cassette buffer.

FDF4  A9FE                  LDA     #EOT    ;indicate EOT record type
FDF6  207CFE                JSR     WCB     ;write cassette buffer

                    ;       Exit.

FDF9  4CD6FD                JMP     CCL1    ;exit



                    **      AUB - Alert User with Beep
                    *
                    *       ON ENTRY A= FREQ
                    *
                    *       ENTRY   JSR     AUB
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = FDFC        AUB     =       *                 ;entry

                    ;       Initialize.

FDFC  8540                  STA     FREQ              ;frequency
```

```
                          ;       Compute termination time of beep duration.

FDFE  A514           AUB1    LDA     RTCLOK+2          ;current time
FE00  18                     CLC
FE01  A662                   LDX     PALNTS
FE03  7D95FE                 ADC     BEEPNX,X          ;add constant for 1 second :
FE06  AA                     TAX                       ;beep duration termination :

                          ;       Turn on speaker.

FE07  A9FF           AUB2    LDA     #$FF
FE09  8D1FD0                 STA     CONSOL            ;turn on speaker
FE0C  A900                   LDA     #$00

                          ;       Delay.

FE0E  A0F0                   LDY     #$F0

FE10  88             AUB3    DEY
FE11  D0FD ^FE10             BNE     AUB3              ;if not done delaying

                          ;       Turn off speaker.

FE13  8D1FD0                 STA     CONSOL            ;turn off speaker

                          ;       Delay.

FE16  A0F0                   LDY     #$F0

FE18  88             AUB4    DEY
FE19  D0FD ^FE18             BNE     AUB4              ;if not done delaying

                          ;       Check for beep duration termination time.

FE1B  E414                   CPX     RTCLOK+2          ;compare current time
FE1D  D0E8 ^FE07             BNE     AUB2              ;if termination time not re:

FE1F  C640                   DEC     FREQ              ;decrement frequency
FE21  F00E ^FE31             BEQ     AUB6              ;if all done, wait for anot:

                          ;       Compute termination time of beep separation.

FE23  8A                     TXA
FE24  18                     CLC
FE25  A662                   LDX     PALNTS
FE27  7D97FE                 ADC     BEEPFX,X          ;add constant
FE2A  AA                     TAX                       ;beep separation terminatio:

                          ;       Wait for termination of beep separation.

FE2B  E414           AUB5    CPX     RTCLOK+2          ;compare current time
FE2D  D0FC ^FE2B             BNE     AUB5              ;if termination time not re:

                          ;       Beep again.

FE2F  F0CD ^FDFE             BEQ     AUB1              ;beep again
```

```
                        ;       Wait for key.

FE31  2036FE     AUB6   JSR     WFK             ;wait for key
FE34  98                TYA                     ;status
FE35  60                RTS                     ;return



                        **      WFK - Wait for Key
                        *
                        *       ENTRY   JSR     WFK
                        *
                        *       NOTES
                        *               Problem: bytes wasted by not doing LDA #hig:
                        *               and LDA #low[KGB-1].
                        *               Problem: bytes wasted by this being a subro:
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = FE36     WFK    =       *               ;entry
FE36  AD25E4            LDA     KEYBDV+5        ;keyboard GET-BYTE routine :
FE39  48                PHA                     ;put address on stack
FE3A  AD24E4            LDA     KEYBDV+4
FE3D  48                PHA
FE3E  60                RTS                     ;invoke keyboard GET-BYTE r:



                        **      SCB - Perform SIO on Cassette Buffer
                        *
                        *       ENTRY   JSR     SCB
                        *
                        *       NOTES
                        *               Problem: byte wasted by JSR/RTS exit.
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = FE3F     SCB    =       *               ;entry
FE3F  8D0203            STA     DCOMND          ;command
FE42  A900              LDA     #high 131
FE44  8D0903            STA     DBYTHI          ;buffer length
FE47  A983              LDA     #low 131
FE49  8D0803            STA     DBYTLO
FE4C  A903              LDA     #high CASBUF
FE4E  8D0503            STA     DBUFHI          ;buffer address
FE51  A9FD              LDA     #low CASBUF
FE53  8D0403            STA     DBUFLO
```

```
FE56  A960                LDA   #$60           ;cassette bus ID
FE58  8D0003              STA   DDEVIC
FE5B  A900                LDA   #0
FE5D  8D0103              STA   DUNIT
FE60  A923                LDA   #35            ;timeout
FE62  8D0603              STA   DTIMLO
FE65  AD0203              LDA   DCOMND         ;command
FE68  A040                LDY   #GETDAT        ;assume SIO GET-DATA comman:
FE6A  C952                CMP   #READ
FE6C  F002 ^FE70          BEQ   SCB1           ;if READ command

FE6E  A080                LDY   #PUTDAT        ;SIO PUT-DATA command

FE70  8C0303      SCB1    STY   DSTATS         ;SIO command
FE73  A53E                LDA   FTYPE          ;IRG type
FE75  8D0B03              STA   DAUX2          ;second auxiliary informati:
FE78  2059E4              JSR   SIOV           ;vector to SIO
FE7B  60                  RTS                  ;return


                  **      WCB - Write Cassette Buffer
                  *
                  *       ENTRY   JSR     WCB
                  *
                  *       NOTES
                  *               Problem: byte wasted by JSR/RTS exit.
                  *
                  *       MODS
                  *               Original Author Unknown
                  *               1. Bring closer to Coding Standard (object :
                  *                  R. K. Nordin 11/01/83


      = FE7C      WCB     =     *              ;entry
FE7C  8DFF03              STA   CASBUF+2       ;record type
FE7F  A955                LDA   #$55
FE81  8DFD03              STA   CASBUF+0
FE84  8DFE03              STA   CASBUF+1
FE87  A957                LDA   #'W'           ;write
FE89  203FFE              JSR   SCB            ;perform SIO on cassette bu:
FE8C  60                  RTS                  ;return
```

```
                        **      NTSC/PAL Constant Tables

FE8D  04           WLEADH  DB      high WLEADN     ;high NTSC WRITE file leade:
FE8E  03                   DB      high WLEADP     ;high PAL WRITE file leader

FE8F  80           WLEADL  DB      low WLEADN      ;low NTSC WRITE file leader
FE90  C0                   DB      low WLEADP      ;low PAL WRITE file leader

FE91  02           RLEADH  DB      high RLEADN     ;high NTSC READ file leader
FE92  01                   DB      high RLEADP     ;high PAL READ file leader

FE93  40           RLEADL  DB      low RLEADN      ;low NTSC READ file leader
FE94  E0                   DB      low RLEADP      ;low PAL READ file leader

FE95  1E           BEEPNX  DB      BEEPNN          ;NTSC beep duration
FE96  19                   DB      BEEPNP          ;PAL beep duration

FE97  0A           BEEPFX  DB      BEEPFN          ;NTSC beep separation
FE98  08                   DB      BEEPFP          ;PAL beep separation
```

```
FE99                 **      PIN - Initialize Printer
                     *
                     *       ENTRY   JSR     PIN
                     *
                     *       MODS
                     *               Original Author Unknown
                     *               1. Bring closer to Coding Standard (object :
                     *                  R. K. Nordin 11/01/83

     = FE99          PIN     =       *       ;entry
FE99 A91E                    LDA     #30     ;30 second timeout
FE9B 8D1403                  STA     PTIMOT  ;printer timeout
FE9E 60                      RTS             ;return


                     **      Printer Handler Address Data
                     *
                     *       NOTES
                     *               Problem: bytes wasted by tables and code.  :
                     *               Immediate instructions should be used.

FE9F EA02            PSTB    DW      DVSTAT  ;status buffer address

FEA1 C003            PPRB    DW      PRNBUF  ;printer buffer address


                     **      PST - Perform Printer STATUS
                     *
                     *       ENTRY   JSR     PST
                     *
                     *       MODS
                     *               Original Author Unknown
                     *               1. Bring closer to Coding Standard (object :
                     *                  R. K. Nordin 11/01/83

     = FEA3          PST     =       *       ;entry

                     ;       Get status.

FEA3 A904                    LDA     #4      ;4 bytes for status
FEA5 8DDF02                  STA     PBUFSZ  ;buffer size
FEA8 AE9FFE                  LDX     PSTB    ;address of status buffer
FEAB ACA0FE                  LDY     PSTB+1
FEAE A953                    LDA     #STATC  ;status command
FEB0 8D0203                  STA     DCOMND  ;command
FEB3 8D0A03                  STA     DAUX1
FEB6 2014FF                  JSR     SDP     ;set up DCB for printer
FEB9 2059E4                  JSR     SIOV    ;vector to SIO
FEBC 3003 ^FEC1              BMI     PSP     ;if error, return
```

```
                    ;         Exit.

FEBE  2044FF                  JSR     STS       ;set printer timeout from status
                    ;         JMP     PSP       ;return



                    **        PSP - Perform Printer SPECIAL
                    *
                    *         PSP does nothing.
                    *
                    *         ENTRY   JSR     PSP
                    *
                    *         MODS
                    *                 Original Author Unknown
                    *                 1. Bring closer to Coding Standard (object :
                    *                    R. K. Nordin 11/01/83

      = FEC1        PSP       =       *         ;entry
FEC1  60                      RTS               ;return



                    **        POP - Perform Printer OPEN
                    *
                    *         ENTRY   JSR     POP
                    *
                    *         MODS
                    *                 Original Author Unknown
                    *                 1. Bring closer to Coding Standard (object :
                    *                    R. K. Nordin 11/01/83

      = FEC2        POP       =       *         ;entry
FEC2  20A3FE                  JSR     PST       ;perform printer STATUS
FEC5  A900                    LDA     #0
FEC7  8DDE02                  STA     PBPNT     ;clear printer buffer pointer
FECA  60                      RTS               ;return



                    **        PPB - Perform Printer PUT-BYTE
                    *
                    *         ENTRY   JSR     PPB
                    *
                    *         MODS
                    *                 Original Author Unknown
                    *                 1. Bring closer to Coding Standard (object :
                    *                    R. K. Nordin 11/01/83

      = FECB        PPB       =       *                   ;entry
```

```
                        ;       Initialize.

FECB  48                        PHA                     ;save data
FECC  BD4103                    LDA     ICDNO,X         ;device number
FECF  8521                      STA     ICDNOZ          ;device number
FED1  204BFF                    JSR     PPM             ;process print mode

                        ;       Put data in buffer.

FED4  AEDE02                    LDX     PBPNT           ;printer buffer pointer
FED7  68                        PLA                     ;saved data
FED8  9DC003                    STA     PRNBUF,X        ;put data in buffer
FEDB  E8                        INX

                        ;       Check for buffer full.

FEDC  ECDF02                    CPX     PBUFSZ          ;printer buffer size
FEDF  F015 ^FEF6                BEQ     PPP             ;if buffer full, perform PU:

                        ;       Update printer buffer pointer.

FEE1  8EDE02                    STX     PBPNT           ;printer buffer pointer

                        ;       Check for EOL.

FEE4  C99B                      CMP     #EOL
FEE6  F003 ^FEEB                BEQ     PPB1            ;if EOL, space fill

                        ;       Exit.

FEE8  A001                      LDY     #SUCCES         ;indicate success
FEEA  60                        RTS                     ;return

                        ;       Space fill buffer.

FEEB  A920              PPB1    LDA     #' '            ;indicate space fill
                        ;       JMP     FPB             ;fill printer buffer, retur:


                        **      FPB - Fill Printer Buffer
                        *
                        *       ENTRY   JSR     FPB
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83

      = FEED            FPB     =       *               ;entry

                        ;       Fill printer buffer.

FEED  9DC003            FPB1    STA     PRNBUF,X        ;byte of printer buffer
FEF0  E8                        INX
```

```
FEF1  ECDF02                CPX     PBUFSZ          ;printer buffer size
FEF4  D0F7 ^FEED            BNE     FPB1            ;if not done

                    ;       Perform printer PUT.

                    ;       JMP     PPP             ;perform printer PUT, retur:



                    **      PPP - Perform Printer PUT
                    *
                    *       ENTRY   JSR     PPP
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = FEF6        PPP     =       *       ;entry

                    ;       Clear printer buffer pointer.

FEF6  A900                  LDA     #0
FEF8  8DDE02                STA     PBPNT   ;clear printer buffer pointer

                    ;       Set up DCB.

FEFB  AEA1FE                LDX     PPRB    ;address of printer buffer
FEFE  ACA2FE                LDY     PPRB+1
FF01  2014FF                JSR     SDP     ;set up DCB for printer

                    ;       Perform PUT.

FF04  4C59E4                JMP     SIOV    ;vector to SIO, return



                    **      PCL - Perform Printer CLOSE
                    *
                    *       ENTRY   JSR     PCL
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = FF07        PCL     =       *       ;entry

                    ;       Initialize.

FF07  204BFF                JSR     PPM     ;process print mode

                    ;       Check buffer pointer.
```

```
FF0A  A99B                LDA    #EOL     ;indicate EOL fill
FF0C  AEDE02              LDX    PBPNT    ;printer buffer pointer
FF0F  D0DC ^FEED          BNE    FPB      ;if buffer pointer non-zero, fill b:

                    ;     Exit.

FF11  A001                LDY    #SUCCES  ;indicate success
FF13  60                  RTS             ;return



                    **    SDP - Set Up DCB for Printer
                    *
                    *     ENTRY  JSR     SDP
                    *
                    *     MODS
                    *            Original Author Unknown
                    *            1. Bring closer to Coding Standard (object :
                    *               R. K. Nordin 11/01/83


      = FF14        SDP   =      *        ;entry
FF14  8E0403              STX    DBUFLO   ;low buffer address
FF17  8C0503              STY    DBUFHI   ;high buffer address
FF1A  A940                LDA    #PDEVN   ;printer device bus ID
FF1C  8D0003              STA    DDEVIC   ;device bus ID
FF1F  A521                LDA    ICDNOZ   ;device number
FF21  8D0103              STA    DUNIT    ;unit number
FF24  A980                LDA    #$80     ;SIO WRITE command
FF26  AE0203              LDX    DCOMND   ;I/O direction
FF29  E053                CPX    #STATC   ;STATUS command
FF2B  D002 ^FF2F          BNE    SDP1     ;if STATUS command

FF2D  A940                LDA    #$40     ;SIO READ command

FF2F  8D0303        SDP1  STA    DSTATS   ;SIO command
FF32  ADDF02              LDA    PBUFSZ
FF35  8D0803              STA    DBYTLO   ;low buffer size
FF38  A900                LDA    #0
FF3A  8D0903              STA    DBYTHI   ;high buffer size
FF3D  AD1403              LDA    PTIMOT
FF40  8D0603              STA    DTIMLO   ;device timeout
FF43  60                  RTS             ;return
```

```
                        **      STS - Set Printer Timeout from Status
                        *
                        *       ENTRY   JSR     STS
                        *
                        *       NOTES
                        *               Problem: bytes wasted by this code's being :
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = FF44            STS     =       *                       ;entry
FF44  ADEC02                    LDA     DVSTAT+2                ;timeout
FF47  8D1403                    STA     PTIMOT                  ;set printer timeout
FF4A  60                        RTS                             ;return



                        **      PPM - Process Print Mode
                        *
                        *       PPM sets up the DCB according to the print mode.
                        *
                        *       ENTRY   JSR     PPM
                        *
                        *       MODS
                        *               Original Author Unknown
                        *               1. Bring closer to Coding Standard (object :
                        *                  R. K. Nordin 11/01/83


      = FF4B            PPM     =       *       ;entry

                        ;       Initialize.

FF4B  A057                      LDY     #WRITE  ;WRITE command
FF4D  A52B                      LDA     ICAX2Z  ;print mode

                        ;       Determine buffer size.

FF4F  C94E              PPM1    CMP     #NORMAL ;NORMAL mode
FF51  D004 ^FF57                BNE     PPM2    ;if not NORMAL mode

FF53  A228                      LDX     #NBUFSZ ;NORMAL mode buffer size
FF55  D00E ^FF65                BNE     PPM4    ;set buffer size

FF57  C944              PPM2    CMP     #DOUBLE ;DOUBLE mode
FF59  D004 ^FF5F                BNE     PPM3    ;if not DOUBLE mode

FF5B  A214                      LDX     #DBUFSZ ;DOUBLE mode buffer size
FF5D  D006 ^FF65                BNE     PPM4    ;set buffer size

FF5F  C953              PPM3    CMP     #SIDWAY ;SIDEWAYS mode
FF61  D00C ^FF6F                BNE     PPM5    ;if not SIDEWAYS mode, assume NORMA:
```

```
FF63  A21D                  LDX     #SBUFSZ ;SIDEWAYS mode buffer size

                    ;       Set buffer size.

FF65  8EDF02        PPM4    STX     PBUFSZ  ;set printer buffer size

                    ;       Set DCB command and mode.

FF68  8C0203                STY     DCOMND  ;command
FF6B  8D0A03                STA     DAUX1   ;print mode
FF6E  60                    RTS             ;return

                    ;       Assume NORMAL mode.

FF6F  A94E          PPM5    LDA     #NORMAL ;NORMAL mode
FF71  D0DC ^FF4F            BNE     PPM1    ;set buffer size
```

```
FF73                **      VFR - Verify First 8K ROM
                    *
                    *       ENTRY   JSR     VFR
                    *
                    *       EXIT
                    *               C clear, if verified
                    *                 set, if not verified
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = FF73        VFR     =       *       ;entry

                    ;       Initialize.

FF73  A200                  LDX     #0      ;offset to first region to checksum
FF75  868B                  STX     STCHK   ;initial sum is zero
FF77  868C                  STX     STCHK+1

                    ;       Checksum ROM.

FF79  20A9FF        VFR1    JSR     CRR     ;checksum region of ROM
FF7C  E00C                  CPX     #12
FF7E  D0F9 ^FF79            BNE     VFR1    ;if not done

                    ;       Compare result.

FF80  AD00C0                LDA     $C000   ;low checksum in ROM
FF83  AE01C0                LDX     $C001   ;high checksum in ROM
                    ;       JMP     VCS     ;verify checksum, return



                    **      VCS - Verify Checksum
                    *
                    *       ENTRY   JSR     VCS
                    *
                    *       MODS
                    *               Original Author Unknown
                    *               1. Bring closer to Coding Standard (object :
                    *                  R. K. Nordin 11/01/83

      = FF86        VCS     =       *       ;entry
FF86  C58B                  CMP     STCHK   ;low checksum
FF88  D006 ^FF90            BNE     VCS1    ;if low checksum bad

FF8A  E48C                  CPX     STCHK+1 ;high checksum
FF8C  D002 ^FF90            BNE     VCS1    ;if high checksum bad

FF8E  18                    CLC             ;indicate verified
FF8F  60                    RTS             ;return
```

```
FF90  38             VCS1    SEC                ;indicate not verified
FF91  60                     RTS                ;return



                     **      VSR - Verify Second 8K ROM
                     *
                     *       ENTRY   JSR     VSR
                     *
                     *       EXIT
                     *               C clear, if verified
                     *                 set, if not verified
                     *
                     *       MODS
                     *               Original Author Unknown
                     *               1. Bring closer to Coding Standard (object :
                     *                  R. K. Nordin 11/01/83


      = FF92         VSR     =       *          ;entry
FF92  A200                   LDX     #0
FF94  868B                   STX     STCHK      ;initial sum is zero
FF96  868C                   STX     STCHK+1
FF98  A20C                   LDX     #12        ;offset to first region to checksum
FF9A  20A9FF                 JSR     CRR        ;checksum region of ROM
FF9D  20A9FF                 JSR     CRR        ;checksum region of ROM
FFA0  ADF8FF                 LDA     $FFF8      ;low checksum from ROM
FFA3  AEF9FF                 LDX     $FFF9      ;high checksum from ROM
FFA6  4C86FF                 JMP     VCS        ;verify checksum, return



                     **      CRR - Checksum Region of ROM
                     *
                     *       ENTRY   JSR     CRR
                     *               X = offset
                     *
                     *       MODS
                     *               Original Author Unknown
                     *               1. Bring closer to Coding Standard (object :
                     *                  R. K. Nordin 11/01/83


      = FFA9         CRR     =       *          ;entry

                     ;       Transfer range addresses.

FFA9  A000                   LDY     #0

FFAB  BDD7FF         CRR1    LDA     TARV,X
FFAE  999E00                 STA     STADR1,Y
FFB1  E8                     INX
FFB2  C8                     INY
FFB3  C004                   CPY     #4         ;4 bytes for 2 addresses
FFB5  D0F4 ^FFAB             BNE     CRR1       ;if not done
```

```
                        ;       Checksum range.

FFB7  A000                      LDY     #0

FFB9  18            CRR2        CLC
FFBA  B19E                      LDA     (STADR1),Y
FFBC  658B                      ADC     STCHK
FFBE  858B                      STA     STCHK
FFC0  9002 ^FFC4                BCC     CRR3    ;if low value non-zero

FFC2  E68C                      INC     STCHK+1 ;adjust high value

FFC4  E69E          CRR3        INC     STADR1  ;advance address
FFC6  D002 ^FFCA                BNE     CRR4    ;if low address non-zero

FFC8  E69F                      INC     STADR1+1        ;adjust high address

FFCA  A59E          CRR4        LDA     STADR1  ;current address
FFCC  C5A0                      CMP     STADR2  ;end of range
FFCE  D0E9 ^FFB9                BNE     CRR2    ;if not done

FFD0  A59F                      LDA     STADR1+1
FFD2  C5A1                      CMP     STADR2+1
FFD4  D0E3 ^FFB9                BNE     CRR2    ;if not done

FFD6  60                        RTS             ;return


                        **      TARV - Table of Address Ranges to Verify


FFD7  02C000D0      TARV        DW      $C002,$D000     ;first 8K ROM, $C002 - $CFF:
FFDB  00500058                  DW      $5000,$5800     ;first 8K ROM, $D000 - $D7F:
FFDF  00D800E0                  DW      $D800,$E000     ;first 8K ROM, $D800 - $DFF:

FFE3  00E0F8FF                  DW      $E000,$FFF8     ;second 8K ROM, $E000 - $FF:
FFE7  FAFF0000                  DW      $FFFA,$0000     ;second 8K ROM, $FFFA - $FF:
```

```
FFEB                     FIX     $FFEE



                **       Second 8K ROM Identification and Checksum


FFEE  100583             DB      IDDAY,IDMON,IDYEAR             ;date (day,:
FFF1  02                 DB      IDCPU                          ;CPU series
FFF2  4242000001         DB      IDPN1,IDPN2,IDPN3,IDPN4,IDPN5  ;part numbe:
FFF7  02                 DB      IDREV                          ;revision n:
FFF8  0000               DW      $0000                          ;reserved f:
```

```
FFFA                          FIX     $FFFA



                    **        6502 Machine Vectors


FFFA  18C0                    DW      NMI     ;vector to process NMI
FFFC  AAC2                    DW      RES     ;vector to process RESET
FFFE  2CC0                    DW      IRQ     ;vector to process IRQ



0000                          END


no ERRORs, 1783 Labels, $0B57 free.


   ABI     C5A7     62#15   62/54
 n ABUFPT  001C     10#37
   ACB     C66E     53/29   66#47
   ACB1    C67B     66/52   67# 7
   ACB2    C69F     66/58   67/ 8   67#29
   ACB3    C6A0     66/60   67#33
   ACC     F612    301/25  311#15  336/53
   ACC1    F61A    311/18  311#22
   ACC2    F62D    311/25  311#35
   ACC3    F635    311/36  311#41
   ACC4    F649    311/46  311#54
   ACC5    F65E    311/59  312/ 5  312#10
   ACK     0041      7#31  240/28
   ACMI    0000      4#14   23/16   30/25   32/41   36/36   39/13   52/19
 n ACMISR  02D7     15#57
   ACMVAR  03ED     17#42   50/42
   ADB     C58B     53/42   61#30
   ADB1    C598     61/35   61#49
   ADRESS  0064     11#44  275/27  275/60  277/ 6  277/ 8  277/24  277/25
                   277/28  277/30  279/42  279/44  279/48  279/50  281/17
                   283/53  283/55  288/43  290/ 5  290/ 7  298/ 6  298/ 8
                   298/10  298/14  298/15  302/24  302/26  306/45  307/45
                   307/48  307/51  307/56  308/ 5  308/57  308/60  309/ 5
                   309/ 7  309/10  309/15  309/16  309/31  309/34  309/38
                   309/40  309/42  322/41  324/ 5  324/ 6  324/21  324/34
                   324/36  324/39  324/43  324/47  324/52  324/57  325/ 6
                   325/ 9  328/52  332/45  332/47
   AFP     D800     23#25  138/37
   AFP1    D818    139#26  139/48  139/54  140/ 9  140/27  140/35  140/44
   AFP10   D86C    140#55  141/58
   AFP11   D88E    141/ 6  141#26
   AFP12   D89B    141/27  141#39
   AFP13   D8A3    140/51  141#47
   AFP14   D8AD    141/48  141#57
```

```
  AFP15  D8B2    141/51  142# 5
  AFP16  D8B6    140/24  140/40  141/43  142#10
  AFP17  D8C3    142/13  142/15  142#20
  AFP18  D8CE    142/25  142#29
  AFP19  D8E4    142/40  142#50
  AFP2   D81C    139/22  139#31
  AFP20  D8E5    142/35  142#52
  AFP3   D837    139/38  139#58
  AFP4   D83E    139/60  140# 9
  AFP5   D841    139/12  140#13
  AFP6   D842    139/32  140#17
  AFP7   D856    140/30  140#34
  AFP8   D85A    140/18  140#39
  AFP9   D863    140/21  140#48
n ALLPOT D208     21#37
  ANTIC  D400     22#31   58/61
n APPEND 0001      5#34
  APPMHI 000E     10#24   50/32   50/32   50/34   50/34  307/55  307/61
  AST    56AA    131/28  131#52
  AST1   56B5    131/57  131#61
  ATACHR 02FB     16#30  281/44  282/16  282/58  286/ 9  287/33  288/17
                 288/24  288/59  291/45  294/ 8  294/26  294/31  299/37
                 320/32  329/20  336/48  336/61  337/ 9
  ATRACT 004D     11#26   34/26   38/13   38/23   38/26  107/60  119/50
                 345/43
  AUB    FDFC    350/38  352/22  356#56
  AUB1   FDFE    357# 7  357/60
  AUB2   FE07    357#15  357/40
  AUB3   FE10    357#23  357/24
  AUB4   FE18    357#34  357/35
  AUB5   FE2B    357#55  357/56
  AUB6   FE31    357/43  358# 7
  AUDC1  D201     21#47  124/44  125/ 5  130/37  251/33  251/46  252/33
  AUDC2  D203     21#50  121/ 6  125/ 6  251/47
  AUDC3  D205     21#53   59/39  125/ 7  251/41
  AUDC4  D207     21#56   59/40  125/ 8
  AUDCTL D208     21#58   59/43  104/23  251/22
  AUDF1  D200     21#46  124/42  130/35  250/21
  AUDF2  D202     21#49  120/57  250/19
  AUDF3  D204     21#52  237/19  248/25  257/29  352/28
  AUDF4  D206     21#55  237/21  248/27  257/31  352/30
  AVV    ED2E    254/40  254/43  255#54
  AVV1   ED36    255/56  256# 5
  B00600 05CC      9#11  248/24  248/26  349/16  349/18  352/27  352/29
  B19200 0028      9#10  237/18  237/20
  BADIOC 0086      6#32  201/33  270/32
  BADMOD 0091      6#43  274/46
  BAI    C5BB     62#42   67/16
  BAI1   C5C0     62#51   64/14   64/28
  BAI2   C5C8     61/41   61/54   62#58
  BASICF 03F8     17#43   51/10   57/11
  BEEPFN 000A      8#15  360/23
  BEEPFP 0008      8#24  360/24
  BEEPFX FE97    357/50  360#23
  BEEPNN 001E      8#14  360/20
  BEEPNP 0019      8#23  360/21
  BEEPNX FE95    357/10  360#20
```

```
BEL     F556     287/43   305#15   338/52
BEL1    F558     305#18   305/20
BFENHI  0035      11# 7   237/46   240/18   242/45   246/60   247/45
BFENLO  0034      11# 6   237/43   240/15   242/43   246/58   247/41
BIR     C092      34#18    60/21    60/23
BIR1    C09E      34#30
BITMSK  006E      11#56   316/45   317/60   318/22   319/36
BLG     F758     300/ 6   304/42   318#41
BLG1    F75A     318#58   321/56   325/53
BLG2    F75B     304/21   319#15
BLIM    028A      15# 6   351/25   353/33   354/28
BLKBDV  E471      24#43   196/42
BLR     F732     317#18   323/ 7
BMC     F74A     300/51   317/40   318#20
BMG     F75D     300/12   311/58   319#32
BMG1    F769     319/37   319#41
BMI     F723     316#40   317/58   318/21   319/33
BMP     F73C     304/27   317#39   322/ 5   322/16
BMS     F73E     300/34   317#57
BMSG    C43D      55#34    65/45    65/46
BOOT?   0009      10#21    61/39    64/33    66/56    67/20
BOOTAD  0242      14#12    63/28    63/30    64/54    64/57
BOTSCR  028F      15#32   277/39   277/59   295/19   295/42   302/36   314/19
                 321/10   323/31   330/42
BPTR    003D      11#14   351/24   352/60   353/32   353/39   354/10   354/60
                 355/ 5   355/20   356/15
BRKABT  0080       6#26   258/21   291/ 7   315/25   351/42
BRKKEY  0011      10#31    34/23    60/19   241/53   244/44   256/34   256/58
                 258/25   291/ 8   292/59   315/26   315/27   351/43   352/51
BRKKY   0236      13#52    37/14    60/22    60/24
BSP     F450     299#15   338/31
BSP1    F45F     299/22   299#26
BSP2    F46F     299/29   299/32   299#36
BSP3    F477     299/18   299#40
BUFADR  0015      10#33    69/48    70/ 6    70/11    70/42    70/44
BUFCNT  006B      11#54   287/21   287/58   288/ 6   326/43   327/ 5   327/ 9
                 327/20   327/32
BUFRFL  0038      11# 9   244/34   246/15   247/20
BUFRHI  0033      11# 5   237/45   240/17   242/38   242/44   246/53   246/59
                 247/43   257/46   257/48
BUFRLO  0032      10#61   237/41   240/13   241/47   242/35   242/42   243/21
                 246/45   246/50   246/57   247/39   257/37   257/39   257/43
                 257/45
BUFSTR  006C      11#55   287/27   287/29   287/51   287/53   298/49   298/53
                 313/21   313/24   326/38   326/40   327/55   327/57
C0A     DC70     143/49   144/14   166#41
C0A1    DC76     166#51   167/15
CAL     C7D2      72/ 6    73#52
CART    BFFC      19#18    52/51    56/43
CARTAD  BFFE      19#20    54/28    56/51
CARTCK  03EB      17#37    58/35    58/36
CARTCS  BFFA      19#17    54/ 7
CARTFG  BFFD      19#19    53/36    53/58    56/46
CASBUF  03FD      17#47    62/22    62/24    63/21    63/33    63/35    63/40
                 223/49   225/11   353/38   354/15   354/24   354/61   356/18
                 356/28   358/58   358/60   359/41   359/43   359/44
CASET   0060       6#52   236/55   250/12   251/43
```

CASETV	E440	24#24	55/15	60/47	195/26			
CASFLG	030F	16#58	237/ 8	244/29	248/51			
CASINI	0002	10#13	67/23	67/25	67/33			
CASSBT	03EA	17#36	62/53	64/13	64/20	66/19	67/14	67/18
CASSET	0043	5#47	55/14					
CAUX1	023C	13#56	237/33					
CAUX2	023D	13#57	237/35					
CBAUDH	02EF	16#17	255/38	257/30	349/19			
CBAUDL	02EE	16#16	255/35	257/28	349/17			
CBC	F8B1	287/50	326#31					
CBC1	F8C3	326#47	327/12					
CBC2	F8CB	326/49	326#55					
CBC3	F8DA	326/56	326/60	327# 8				
CBC4	F8EA	327/ 6	327#17	327/33				
CBC5	F902	327/28	327#32					
CBC6	F906	327/18	327/23	327#35				
CBI	C5C9	62/47	63#15					
CBI1	C5CB	63#21	63/24					
CBI2	C5E8	63#38	64/ 8					
CBI3	C5EA	63#40	63/43					
CBI4	C607	64# 7	64/16					
CBI5	C616	63/58	64#20					
CBI6	C61E	64/21	64#27					
CBR	ECC8	254#37	257/24					
CBR1	ECF7	254#56	254/61					
CBR2	ED11	255#21	255/24					
CBR3	ED1F	255/29	255#33					
CBT	F55F	305#37	339/16					
CCA	F5AC	281/16	283/35	302/23	303/36	304/37	308#53	322/34
		328/46						
CCA1	F5C3	309/ 8	309#12					
CCA2	F5C8	309#15	309/18					
CCA3	F5D9	309#26	309/29					
CCA4	F5DF	309/24	309#31					
CCA5	F5E5	309/32	309#36					
CCA6	F609	309#55	311/28	311/31	311/33	311/39		
CCC	F93C	289/27	294/23	329#15				
CCC1	F93E	329#19	329/26					
CCC2	F948	329/21	329#28					
CCE	C4C9	46/37	52/55	58#19				
CCE1	C4CD	58#29	58/31					
CCL	FDCF	195/34	355#53					
CCL1	FDD6	356# 9	356/21	356/39				
CCL2	FDDC	355/58	356#15					
CCL3	FDEA	356/16	356#25					
CCL4	FDEE	356#28	356/30					
CCO	E672	205/35	206/36	207/17	207/61	209/12	210/46	211#32
CCO1	E683	211#50	211/55					
CCOMND	0238	13#55	237/31					
CCR	F6CA	280/51	281/54	314/21	314/26	314#46	336/37	336/56
CCR1	F6D2	314/49	314#53					
CCR2	F6EB	314/61	315#14					
CCR3	F6EF	315/12	315#17					
CCR4	F6F8	315/ 7	315#23					
CCR5	F705	314/56	314/58	315/10	315/15	315/19	315/21	315#34
CCR6	F70A	315/28	315#39					
CCR7	F715	315/43	315#47					

```
  CDEVIC 023A    13#54  237/29  237/40  237/44
  CDN    F3F3   295#39  338/19
  CDTMA1 0226    13#37   43/24  258/44  258/46
  CDTMA2 0228    13#38   43/40
  CDTMF3 022A    13#40   40/37  351/15  351/18  352/49  352/54
n CDTMF4 022C    13#42
n CDTMF5 022E    13#44
  CDTMV1 0218    13#30   43/61   44/ 7   44/10   44/12   44/15   45/10
                 45/12
n CDTMV2 021A    13#31
  CDTMV3 021C    13#32   40/32   40/33
n CDTMV4 021E    13#33
n CDTMV5 0220    13#34
  CEL    F1B4   282#15  289/33
  CEL1   F1C1   282/18  282#23
  CEP    E695   205/19  205/30  206/21  207/ 9  207/48  209/43  212#20
  CEP1   E69F   212/26  212#35
  CEP2   E6BA   212/31  212#52
  CGB    FD7A   195/35  353#23  354/32
  CGB1   FD8C   353#42  354/ 7
  CH     02FC    16#31   41/40  105/13  106/11  122/39  122/46  122/51
                273/18  290/30  291/13  291/21  345/35
  CH1    02F2    16#20  344/28  345/36
  CHACT  02F3    16#21   40/15   52/37  274/56
  CHACTL D401    22#43   40/16
  CHAR   02FA    16#29  281/26  283/31  283/39  319/61  320/26
  CHBAS  02F4    16#22   40/13   52/39  274/52  346/18  346/21
  CHBASE D409    22#49   40/14
  CHKERR 008F     6#41  246/26
  CHKSNT 003B    11#12  241/42  242/50  242/58  243/49
  CHKSUM 0031    10#60  241/41  241/49  242/55  243/24  243/26  244/32
                246/21  246/47  246/49  257/41
  CHLINK 03F8    17#46  218/41  218/43  219/17  219/18  226/11  226/13
  CHM    F440   298#46  305/38  315/34  339/13
  CHSALT 026B    14#27  274/54  346/19  346/20
  CIA    F76A   280/53  319#57
  CIA1   F77B   320/13  320#20
  CIA2   F78A   320/10  320/16  320#32
  CIA3   F78D   320#34  321/11
  CIN    FCDB   195/39  349#15
  CIO    E4DF   196/15  201#15
  CIO1   E4EC   201/26  201#33
  CIO2   E4F1   201/29  201#38
  CIO3   E4F3   201#40  201/45
  CIOCHR 002F    10#57  201/19  207/60  208/ 9  208/21  208/44  208/49
                209/54  210/ 5  210/22  211/59
  CIOINV E46E    24#42   60/51  196/39
  CIOV   E456    24#34   53/12   65/59  196/15
  CIR    C0A0    33/21   33/38   34#49
  CIR1   C0AD    34/59   35#11
  CIR2   C0B8    35/12   35#21
  CIR3   C0C9    35/27   35#37
  CIX    00F2    12#49  140/48  141/23  142/ 6  142/10  144/17  144/38
                144/42  144/51  144/53  158/47  159/21  159/46  159/59
                160/22  160/55  161/42
  CKEY   03E9    17#35   61/13   67/ 7   67/19
  CLF    F400   296#15  299/26  327/15  327/25  338/22
```

```
  CLM     F41B    297#38  304/47  339/19
  CLN     F7E2    303/38  322#31
  CLN1    F7F1    322#41  322/43
  CLOSE   000C      5#13  201/56
  CLS     007D      6# 8  281/48
  CLT     C856     99#25  218/61  226/52  231/ 7  234/13
  CLT1    C85B     99#31   99/33
n CMCMD   0007     10#16
  CNTL1   009F      6#16   41/21  345/12
  CNTLF1  0083      6#12   41/24  344/37
  CNTLF2  0084      6#13   41/27  346/10
n CNTLF3  0093      6#14
  CNTLF4  0094      6#15   41/30  346/13
  COC     EF9C    273/57  274#36
  COC1    EFA7    274/42  274#51
  COC10   F08D    277/45  277#52
  COC11   F0A2    278/ 5  278# 9  278/12
  COC12   F0C6    277/48  277/54  278#33
  COC13   F0DB    278/29  278/36  278/41  278#51  278/54
  COC14   F0F1    279/ 6  279#13
  COC15   F107    279#24  279/27
  COC16   F10F    278/60  279/ 9  279#31
  COC17   F154    274/47  280# 9
  COC18   F164    280/ 5  280#19
  COC19   F175    280/21  280#32
  COC2    EFDC    275/13  275#21
  COC3    EFEF    275#36  275/38
  COC4    EFF7    275#44  275/47
  COC5    F019    276# 9  276/12
  COC6    F03E    276/27  276#44
  COC7    F04C    276/24  276/30  276/50  276#59
  COC8    F057    277#13  277/15
  COC9    F06C    277/21  277#27
  COLAC   0072     11#59  334/39  335/55  335/57  335/58  335/60  336/ 9
n COLBK   D01A     20#43
  COLCRS  0055     11#34  287/28  287/54  296/16  296/17  296/56  297/16
                  297/17  298/52  299/20  299/27  299/58  303/35  304/36
                  309/20  309/22  309/47  311/17  311/20  311/22  311/38
                  312/46  315/ 6  315/14  315/18  322/33  326/10  326/41
                  326/58  327/26  332/27  333/40  333/42  336/16  336/17
                  336/21  336/24  336/27  336/30
  COLDST  0244     14#13   46/42   50/19   53/51   81/24  233/60
  COLDSV  E477     24#45  196/48
  COLINC  02F9     16#27  333/49  334/23  336/13
  COLOR0  02C4     15#39  134/61  275/45
  COLOR1  02C5     15#40   38/35  135/ 7
  COLOR2  02C6     15#41  135/10  347/13
n COLOR3  02C7     15#42
  COLOR4  02C8     15#44  119/44  135/13  276/55
  COLPF0  D016     20#38  107/56
  COLPF1  D017     20#39   38/38  119/43  347/19
  COLPF2  D018     20#40  119/45
n COLPF3  D019     20#41
  COLPM0  D012     20#33   40/ 7
n COLPM1  D013     20#34
n COLPM2  D014     20#35
n COLPM3  D015     20#36
```

```
COLRSH 004F     11#28   38/31   38/36   40/ 5  347/15  347/15  347/17
COMPLT 0043      7#32  240/31
CONS1X EE19    254/56  254/59  260#20
CONS2X EE1B    256/ 7  260#23
CONSOL D01F     19#32   39/54   57/18   61/10  105/ 8  108/19  108/36
               120/16  120/27  120/38  120/51  331/34  357/16  357/28
COP    FCE6    195/33  349#53
COUNTR 007E     12#10  324/32  324/35  324/46  324/49  324/51  325/16
               325/18  325/22  334/54  335/10  335/21  335/22  337/15
               337/17  337/18  337/20
CPB    FDB4    195/36  354#56
CPB1   FDC2    355/11  355#17
CRE    F6BC    287/20  288/61  314#15
CRETRI 000D      9#26  237/13  238/18
CRETRY 029C     15#15  237/14  237/61  238/19
CRITIC 0042     11#20   38/50   83/13   83/13   83/50   83/50   88/38
                88/38   88/42   88/42   89/19   89/19  236/50  239/33
CRM    F40A    296/18  296#38  339/22
CRR    FFA9    368/29  369/30  369/31  369#50
CRR1   FFAB    369#56  369/61
CRR2   FFB9    370#10  370/25  370/29
CRR3   FFC4    370/14  370#18
CRR4   FFCA    370/19  370#23
CRSINH 02F0     16#18   34/25  275/29  284/21
CRSROR 008D      6#39  315/35
CRT    F411    297#15  299/57  338/25
CSC    F420    280/25  281/51  297#55  338/28
CSC1   F429    298#10  298/12  298/17
CSC2   F438    298#24  298/27
CSO    EA2A    238/42  239/ 6  239/11  239#30  249/25  258/27
CSOPIV E47D     24#47   67/15  196/54
CSP    FCE5    195/38  349#37
CST    FDCC    195/37  355#36
CTB    F49A    300#49  338/46
CTIA   D000     19#28   58/60
CTIM   0002      9#28  253/58  253/59
CUP    F3E6    295#15  299/34  305/39  338/16
CUP1   F3EE    295#22  295/46
CUP2   F3F0    295/17  295#24  295/43
DATAER 009C      8#45   71/35   71/40   71/60   72/45   72/50
DAUX1  030A     16#53   62/20   64/ 5  221/15  224/51  237/32  361/57
               367/14
DAUX2  030B     16#54   62/18  221/13  237/34  248/31  248/55  249/19
               359/20
DAW    5387    117/46  118#32  124/10  124/13  130/44
DAW1   5387    118#34  118/41
DAW2   5389    118#36  118/38
DBE    C63E     62/51   64/12   65#41
DBL    E6BB    208/27  208/32  210/10  213#18
DBL1   E6C1    213/20  213#24
DBP    E6C8    208/60  213#42
DBP1   E6CE    213/44  213#48
DBS    F57A    276/10  276/45  305/56  307#32
DBS1   F58D    307/49  307#55
DBS2   F59B    307/59  308#10
DBS3   F59F    306/40  307/41  307/57  308/ 6  308#15
DBSECT 0241     14#11   63/57
```

```
  DBUFHI 0305    16#48   62/25   69/18   70/43  247/42  358/59  365/30
  DBUFLO 0304    16#47   62/23   69/16   70/41  247/38  358/61  365/29
  DBUFSZ 0014     9#31  366/56
  DBYTHI 0309    16#52   69/32   70/20  247/44  358/55  365/46
  DBYTLO 0308    16#51   69/31   70/18  247/40  358/57  365/44
  DCB    0300    16#42  221/ 7  224/44
  DCH    F4D5   302#15  338/55
  DCH1   F4D8   302#23  302/42
  DCH2   F4FE   302/33  302/37  302#44
  DCOMND 0302    16#45   61/52   66/29   68/44   68/56   69/42   69/53
                237/30  358/53  359/11  361/56  365/36  367/13
  DCSORG E000     4#57   52/38  191/ 6  191/ 6  191/ 6  191/ 6  191/ 6
                191/ 6  274/51
  DCT    C255    38/43   40/21   40/36   43#60
  DCT1   C262    44/ 5   44#12
  DCT2   C26F    44/ 8   44/13   44/16   44#21
  DDD    F565   277/27  305#54
  DDEVIC 0300    16#43   68/42  224/55  236/54  237/26  250/11  251/42
                352/35  359/ 6  365/32
  DELTAC 0077    12# 5  334/14  334/17  334/24  334/26  334/27  334/29
                334/30  334/33  334/50  334/53  334/58  335/56  335/59
  DELTAR 0076    11#61  333/53  333/60  334/ 7  334/59  335/ 5  335/31
  DERRF  03EC    17#38   50/26  280/ 9  280/11  280/13
  DERROR 0090     6#42  240/39
  DFLAGS 0240    14#10   63/22
  DFS    538E   113/28  113/38  116#39
  DIGRT  00F1    12#47  139/20  139/47  139/53  139/59  140/39  140/43
                142/12  142/18
  DINDEX 0057    11#35  274/40  275/56  277/43  278/35  278/58  284/18
                309/12  311/23  311/41  312/48  314/25  314/53  320/ 8
                332/22
  DINITV E450    24#32   60/54  196/ 9
  DIO    C6B3    68#37  196/12
  DIO1   C6C4    68/46   68#50
  DIO10  C736    69/55   70#24
  DIO2   C6D4    68/58   69# 6
  DIO3   C6D6    68/61   69#10
  DIO4   C6EA    69/11   69#25
  DIO5   C6F0    69/21   69#30
  DIO6   C6FF    69/34   69#42
  DIO7   C710    69/44   69#53
  DIO8   C71C    69#60   70/ 8
  DIO9   C71E    70# 6   70/14
n DIRECT 0002     5#35
n DISK   0044     5#48
  DISKID 0031     6#50   68/41
  DISL1  513A   104/31  104/32  109#19  109/39
  DISL2  51D1   110#49  113/10  113/11
  DISL3  51ED   111/ 6  111#14  111/58  112/26
  DISL4  5215   111#45  122/ 6  122/ 7
  DISL5  5231   112# 8  127/16  127/17
  DISPLY 0053     5#52   55/20
  DLISTH D403    22#45   39/29
  DLISTL D402    22#44   39/31
  DLN    F520   303#54  338/40
  DLN0   F527   304#17  304/43
  DLN1   F527   304#15  328/58
```

```
  DLN2   F529   304#19  304/30
  DLP    DCC1   143/59  145/52  170#42
  DLW    5385   115/41  116/19  118#15  122/34  129/48  130/14
  DMACTL D400    22#42   39/33
  DMASAV 02DD    16# 5  345/52  346/41
  DMASK  02A0    15#19  281/18  283/49  283/51  309/51
  DMW    5381   113/29  113/48  115/34  117#44  128/13
  DNACK  008B     6#37  240/45
  DOSINI 000C    10#23   63/34   63/36   65/22   67/22   67/24
  DOSVEC 000A    10#22   49/51   49/53   54/11
  DOUBLE 0044     7#20  366/53
  DQQ    F918   302/47  328#15
  DQQ1   F920   328/18  328#22
n DRAWLN 0011     5#25
  DRETRI 0001     9#27  237/10
  DRETRY 028D    15#30  237/11  239/10
  DRKMSK 004E    11#27   38/30   38/37   40/ 6  347/17
  DRS    53C3   114/26  115/36  115/46  118#58
  DSCTLN 02D5    15#56   68/19   68/21   69/25   69/26
  DSCTSZ 0080     4#49   68/18   68/20
  DSD    F578   277/19  307#15
n DSKFMS 0018    10#35
  DSKINV E453    24#33   61/53   66/32  196/12
  DSKTIM 0246    14#15   68/17   68/43   69/49
n DSKUTL 001A    10#36
  DSPFLG 02FE    16#33  289/38
  DSR    EC84   239/31  242/10  252#18  258/16
  DSR1   EC92   252#33  252/36
  DSS    5399   113/47  113/57  116#59
  DSTAT  004C    11#25  274/59  279/61  284/42  286/ 8  287/32  288/11
                288/26  292/58  306/39  307/40  308/11  315/24  315/39
  DSTATS 0303    16#46   69/30   70/24   83/51   83/52  238/13  238/34
                239/35  248/19  359/18  365/42
  DT1    00FA     7#54  354/21  356/19
  DTA    00FC     7#53  355/17
  DTIMLO 0306    16#49   68/50  252/54  359/10  365/48
  DUNIT  0301    16#44   61/50   66/31   83/14   83/47  237/27  359/ 8
                365/34
n DUNUSE 0307    16#50
  DVN    5685   127/57  128/20  128/30  128/40  128/44  128/48  128/58
                129/ 5  129/ 9  129/19  129/23  129/27  129/37  129/41
                131#15
  DVN1   5694   131#25  131/30  131/35
  DVNM   C28A    39/ 5   45#31   56/14  196/27
  DVSTAT 02EA    16#15   69/15   69/17  204/42  204/43  219/37  219/40
                220/ 7  220/ 9  220/11  221/28  222/23  222/27  222/30
                222/34  222/36  228/44  228/45  229/31  229/35  230/24
                230/25  268/60  361/30  366/19
  DWQ    F923   299/24  328#38
  DWQ1   F932   328#52  328/56
  EBL    C629    64/27   64#49
  ECL    F22E   194/14  194/34  286#43
  EDITRV E400    24#20   55/18   60/43  194/ 6
  EEXP   00ED    12#35  139/16  140/ 5  140/56  140/58  141/ 9  141/11
                141/14  141/15  141/18  141/19  141/31  141/35  141/41
                141/42  142/11  142/24  142/29  144/49  144/54  144/59
                144/60  145/11  155/58  171/47
```

```
  EGB    F24A   194/15  287#15
  EGB1   F25C   287#31  287/45
  EGB2   F277   287/41  287#45
  EGB3   F27A   287/35  287#49
  EGB4   F288   287/22  287#58
  EGB5   F28C   288# 6  288/12
  EGB6   F290   287/59  288/ 7  288#22
  EHC    E6EA   205/24  206/26  207/10  207/59  208/ 8  208/43  210/ 7
                210/41  214#60
  ELL    F7C2   303/33  321#51
  ELL1   F7C5   321#55  322/10
  ELL2   F7D9   322/ 7  322#12
  EMS    5003    52/40  103#41
  END    C795    71/46   72#43   80/41
  END1   C7C0    73#12   73/21
  END2   C7C6    72/56   73#15
  END3   C7C8    72/46   72/51   73#17
  END4   C7C9    72/58   73#19
  ENDPT  0074    11#60  313/53  313/56  334/55  335/11  335/13  335/38
                335/44  335/61  336/10
n ENTVEC 002C    10#55
  EOFERR 0088     6#34  292/54  354/40
  EOL    009B     6# 9   55/34   55/42  208/22  208/50  209/ 5  210/23
                210/40  282/17  287/34  288/23  293/ 6  310/33  312/26
                313/14  363/30  365/ 6
  EOP    EF94   194/13  274#15  280/10  286/60  314/30
  EOT    00FE     7#55  354/16  356/34
  EPB    F2B0   194/16  288#58
  EPC    C9DC    85/17   85/34   85/51   86/17   86/34   86/51   88#31
  EPC1   C9ED    88#48   89/31
  EPC2   CA03    88/49   89#10
  EPC3   CA05    89/ 6   89#14
  EPC4   CA1B    88/56   89#27
  ERRFLG 023F    14# 6   14/ 8  237/53  238/28  238/37  238/51  240/ 7
                240/57
  ERROR  0045     7#33  240/34
  ESC    F3E0   294#50  338/13
  ESCFLG 02A2    15#21  289/32  289/39  289/44  294/52
  ESIGN  00EF    12#41  141/26  141/53
  ESP    F22D   194/18  194/56  194/58  286#27
  ESR    EC40   244/38  250#45
  ESS    EC17   196/33  241/39  248/28  249#58
  ESS1   EC31   250/13  250#25
  EST    5009   103/26  103#58
  EXP    DDC0    23#45  181/17
  EXP1   DE03   182/35  182#60
  EXP10  DDCC    23#46  182/ 6
  EXP2   DE20   183/21  183#25
  EXP3   DE26   183#29  183/31
  EXP4   DE39   183/13  183#48
  EXP5   DE4A   183/49  183#59
  EXP6   DE4B   181/42  182/40  182/49  183/36  183/38  184# 6
  F1R    DD0B   172/22  173#29
  F2R    DD11   172/44  174#20
  FADD   DA66    23#32  152/ 6  177/ 8  185/11  188/24  188/45
  FADD1  DA66   152#31  152/52
  FADD10 DAC5   153/55  154# 7
```

```
  FADD11 DACE     154#14  154/18
  FADD2  DA77     152#44  152/50
  FADD3  DA85     152/38  152#56
  FADD4  DA8E     152/56  153#10
  FADD5  DA98     153#20  153/24
  FADD6  DAA4     152/59  153#31
  FADD7  DAA7     153/27  153#35
  FADD8  DAB3     153/13  153#46
  FADD9  DAB6     153#49  153/53
  FALSE  0000       4# 8    4/ 9    4/12    4/13    4/14
  FARR   DD13     173/31  174#42
  FARR1  DD19     174#53  174/59
  FASC   D8E6      23#26  143/ 6
  FASC1  D914     143/55  143#59
  FASC10 D988     143/57  145#31
  FASC11 D99C     143/60  145/33  145#47
  FASC12 D9A9     145/48  145#59
  FASC2  D91A     143/36  144# 7
  FASC3  D920     143/40  143/43  144#13
  FASC4  D94C     144/40  144#44
  FASC5  D94F     144/30  144#49
  FASC6  D969     144/55  145# 7
  FASC7  D96B     145/ 5  145# 9
  FASC8  D972     145#13  145/18
  FASC9  D97A     145/15  145#20
  FCHFLG 00F0      12#44  139/27  140/23
  FDH    E716      91/16  216#26
  FDH1   E71A     216#36  216/42
  FDH2   E724     216/30  216#46
  FDH3   E728     216/37  216#52
  FDIV   DB28      23#34  156/23  183/55  185/25  187/58  188/57
  FDIV1  DB43     157#12  158/14
  FDIV2  DB45     157#14  157/18
  FDIV3  DB4E     157/ 8  157#22  157/36
  FDIV4  DB52     157#26  157/30
  FDIV5  DB65     157/33  157#40
  FDIV6  DB70     157#51  158/ 8
  FDIV7  DB74     157#55  157/59
  FDIV8  DB87     158/ 5  158#12
  FDL    FCC4     275/15  275/17  347#11
  FEOF   003F      11#16  350/36  353/27  354/36
  FHALF  DF6C     188/21  188/22  189#21
  FILDAT 02FD      16#32  336/60
  FILFLG 02B7      15#25  333/34  336/42
n FILLIN 0012       5#26
  FINE   026E      14#30  275/12  277/61  278/40  286/47  286/55  306/16
                  323/11
  FIX    mac       28#36  100/ 6  103/ 6  138/11  138/37  143/ 6  146/ 6
                  147/11  149/14  149/37  151/29  152/ 6  154/29  156/23
                  176/11  177/25  178/ 6  178/39  179/ 6  179/39  180/11
                  180/45  181/17  182/ 6  185/32  186/ 6  191/ 6  194/ 6
                  194/26  194/46  195/ 6  195/26  196/ 9  196/12  196/15
                  196/18  196/21  196/24  196/27  196/30  196/33  196/36
                  196/39  196/42  196/45  196/48  196/51  196/54  196/57
                  196/60  197/ 6  197/ 9  197/12  198/ 6  199/ 6  232/ 6
                  235/ 6  272/ 6  285/ 6  348/ 6  371/ 6  372/ 6
  FKDEF  0060      11#41  273/32  273/34  293/42
```

```
  FLD01   DD8F     178#28  178/31
  FLD0P   DD8D      23#37  178/ 6
  FLD0R   DD89      23#36  176/45  177/25  182/55  183/54  185/17
  FLD11   DD9E     179#28  179/31
  FLD1P   DD9C      23#39  179/ 6
  FLD1R   DD98      23#38  177/ 7  177/16  178/39  181/37  185/10  185/20
                   185/24  188/19  188/23  188/56
  FLPTR   00FC      12#56  177/45  177/46  178/28  178/59  178/60  179/28
                   179/60  179/61  180/35
  FM01    DDB8     181# 6  181/ 9
  FMOVE   DDB6      23#42  176/42  180/45  182/52  183/ 7  183/51  188/11
                   188/25
  FMPREC  0005       7#44  139/42  139/43  152/58  153/17  153/46  154/12
                   157/17  163/34  163/48  165/40  166/16  167/14  171/56
                   183/26  187/51
n FMSZPG  0043      11#22
  FMUL    DADB      23#33  154/29  176/49  181/41  183/ 8  183/44  188/12
                   188/20
  FMUL1   DAF1     155#18  155/54
  FMUL2   DAF7     155#24  155/28
  FMUL3   DB01     155/25  155#32
  FMUL4   DB09     155#41  155/45
  FMUL5   DB13     155/42  155#49
  FMUL6   DB1A     155#58  158/22
  FMUL7   DB21     154/60  156# 7
  FMUL8   DB24     154/55  156#11  156/54
  FMUL9   DB26     155/10  156#16  156/49  156/61
  FNCNOT  0092       6#44   88/11  214/61  271/ 6
  FNZ     DCA4     143/50  144/15  169#24
  FNZ1    DCA6     169#32  169/42
  FNZ2    DCB4     169/34  169#46
  FNZ3    DCB8     169/37  169#51
  FOMAT   0021       7# 8   68/45   69/54
  FONE    DE8F     183/52  183/53  184#42
  FPB     FEED     363#56  365/ 8
  FPB1    FEED     363#60  364/ 6
  FPI     D9D2      23#28  147/11  182/45
  FPI1    D9EA     147#59  148/34
  FPI2    DA24     147/49  148#38
  FPI3    DA38     148/40  148#54
  FPI4    DA42     147/42  147/45  147/60  148/10  148/13  148/22  148/30
                   149# 6
  FPREC   0006       7#42    7/44  152/42  155/18  155/32  157/22  157/51
                   172/21  172/43  173/ 8  173/30  173/52  174/21  174/46
                   175/28  175/55  176/54  178/26  179/26  180/32  180/61
                   182/39  184/26  189/40
  FPSCR   05E6      18#35  182/42  182/43  182/53  182/54  185/12  185/13
                   185/22  185/23  188/ 8  188/ 9  188/17  188/18
n FPSCR1  05EC      18#36
  FPTR2   00FE      12#57  176/36  176/37  176/43  176/44  176/53  176/55
                   176/58  176/60  177/ 5  177/ 6  184/60  184/61  185/ 8
                   185/ 9  185/18  185/19
  FR0     00D4      12#22  142/32  142/44  142/46  143/35  144/21  145/47
                   146/28  146/30  146/44  146/45  146/46  146/58  147/41
                   148/55  148/57  149/30  152/34  152/44  152/48  153/11
                   154/ 7  154/ 9  154/54  155/59  156/53  157/14  157/15
                   163/35  163/53  164/ 7  164/13  164/52  166/18  166/19
```

```
                    166/24  171/21  171/27  171/49  172/21  172/43  175/57
                    178/29  180/34  181/ 6  182/26  182/29  182/46  182/48
                    186/55  187/40  188/27  188/29  188/35  188/42  188/43
                    190/28
  FROM    00D5       12#23  139/37  139/42  139/43  153/20  153/22  153/38
                    153/49  153/51  154/15  154/16  162/21  163/38  163/45
                    163/46  163/59  166/58  167/10
  FR1     00E0       12#28  151/56  151/58  152/31  152/45  152/46  153/12
                    154/59  155/ 9  156/48  156/58  157/56  165/17  171/22
                    171/25  171/26  171/50  173/30  175/30  179/29  181/ 7
                    183/19  183/33  183/40  190/29
  FR1M    00E1       12#29  153/21  153/50  183/25  183/29  187/50  187/54
  FR2     00E6       12#31  157/27  161/61  171/55  174/21  175/31
  FRA10   DD01      155/27  172#20
  FRA1E   DD09      158/12  173# 7
  FRA20   DD05      155/44  172#42
  FRA2E   DD0F      157/40  173#51
  FRE     00DA       12#26  155/18  155/32  157/26  157/28  157/55  157/57
                    163/17  173/ 8  173/52  175/58
  FREQ    0040       11#17  356/60  357/42
  FRMADR  0068       11#51  320/55  320/58  321/17  321/25  321/28  321/29
                    321/32
  FRMERR  008C        6#38  245/57
  FRSIGN  00EE       12#37  171/24  171/46
  FRX     00EC       12#33  140/49  142/ 5  162/49  170/21  171/53
  FST01   DDAD      180#34  180/37
  FSTOP   DDA8       23#41  180/11
  FSTOR   DDA7       23#40  176/41  179/39  182/44  185/ 7  185/14  188/10
  FSUB    DA60       23#31  151/29  182/56  185/21
  FTX     C851       74/49   76#30
  FTYPE   003E       11#15   67/13  349/58  359/19
  GBY     C7CF       71/34   71/39   71/59   72/44   72/49   73#36
  GBYTEA  02CF       15#52   73/37  222/39  222/41
  GCL     C996       85#32  198/14
  GDC     F18F      280/52  281#15  284/16  301/17  301/32  302/39  327/17
                    336/57
  GDC1    F197      281#20  281/24
  GDC2    F19E      281/21  281#26
n GETCHR  0007        5#10
  GETDAT  0040        9#23   68/54  359/12
n GETREC  0005        5# 9
  GGB     C998       85#49  198/15
  GIN     C90C       82#15  198/19
  GIN1    C911       82#24   82/38
  GIN2    C928       82/29   82/33   82#37
  GINTLK  03FA       17#45   30/23   39/19   46/27
  GNC     DB94      139/31  140/50  141/ 5  141/57  158#45  160/60  161/26
                    161/34
  GND     DCB9      148/24  148/38  170#19
  GNL     E833      223/57  224#37
  GNL1    E835      224#43  224/46
  GNLA    E851      224/43  225# 7  225/18
  GNLAL   000C      224/41  225#18
  GNS     C659       62/46   64/ 7   64/23   66#15
  GNS1    C661       66/20   66#28
  GOP     C991       85#15  198/13
  GPB     C9A0       86#15  198/16
```

```
   GPDVV  E48F    25# 8   61/ 6  198/ 6
   GPRIOR 026F    14#31   39/34  276/16  276/59
 n GRACTL D01D    20#47
 n GRAFM  D011    20#31
 n GRAFP0 D00D    20#26
 n GRAFP1 D00E    20#27
 n GRAFP2 D00F    20#28
 n GRAFP3 D010    20#29
   GSP    C9AA    86#49  198/18
   GST    C9A5    86#32  198/17
   GTO    EC9A   238/26  249/12  252#53
   HATABS 031A    17#12   52/ 9  212/35  212/37  216/36  266/54  267/12
                 267/41  267/43  267/45
 n HDR    00FB     7#52
   HELP   0011     6#11   41/34  345/25
   HELPFG 02DC    15#61  106/36  119/54  119/60  345/30
   HIBYTE 0288    14#61   71/38   71/44   74/60   75/51   78/43   80/ 9
                  80/19
   HIG    C8B5    79#43   80/38   80/39
   HIG1   C8D1    80#11   80/22
   HIG2   C8D2    79/54   80#15
 n HITCLR D01E    20#48
   HITONE 0005     9#13  250/20
   HIUSED 02CB    15#48   75/34   75/40   75/45   75/48
   HNDLOD 02E9    16#14  201/61  204/31  211/48  270/20
   HOLD1  0051    11#30  275/58  278/47  278/51  279/20  279/24  279/35
                 303/56  325/48  325/52  325/60  327/54
 n HOLD2  029F    15#18
   HOLD3  029D    15#16  313/ 5  313/26  313/31
   HOLD4  028C    15#29  336/49  337/ 8
   HOLDCH 007C    12# 8  291/19  293/36  293/61  294/18
 n HPOSM0 D004    20#14
 n HPOSM1 D005    20#15
 n HPOSM2 D006    20#16
 n HPOSM3 D007    20#17
 n HPOSP0 D000    20# 9
 n HPOSP1 D001    20#10
 n HPOSP2 D002    20#11
 n HPOSP3 D003    20#12
 n HSCROL D404    22#46
   IBP    E6D1   208/14  209/ 7  210/ 9  214#15
   IBP1   E6D7   214/17  214#21
   IBS    C63B    61/45   64/32   65#21
   ICAX1  034A    17#29   53/11
   ICAX1Z 002A    10#51  207/37  209/32  274/16  274/18  277/52  280/19
                 290/58  350/ 5
 n ICAX2  034B    17#30
   ICAX2Z 002B    10#52  273/55  349/57  366/43
   ICBAH  0345    17#24   53/ 9   65/51  211/42
   ICBAHZ 0025    10#46  211/43
   ICBAL  0344    17#23   53/ 7   65/49  211/40
   ICBALZ 0024    10#45  208/13  209/ 6  209/61  211/41  213/43  213/46
                 213/48  214/16  214/19  215/52  216/10  268/46  269/ 8
   ICBLH  0349    17#28  214/42
   ICBLHZ 0029    10#50  214/44
   ICBLL  0348    17#27   65/55  214/39
   ICBLLZ 0028    10#49  207/53  207/54  209/48  209/49  209/55  210/28
```

```
                  210/29  213/19  213/22  213/24  213/25  213/26  214/40
                  214/41  214/43
  ICCOM   0342     17#21   53/ 5   65/53
  ICCOMT  0017     10#34   91/29  203/33  205/29  212/39
  ICCOMZ  0022     10#43  201/55  203/20  207/36  208/15  208/37  209/31
                  210/16  210/34  333/22
  ICDNO   0341     17#20  363/ 8
  ICDNOZ  0021     10#42  215/61  216/ 5  268/47  363/ 9  365/33
  ICH     F49F    301#15  338/58
  ICH1    F4AC    301#22  301/35
  ICH2    F4C6    301/28  301#39
  ICH3    F4C9    301#41  301/45
  ICH4    F4D2    301/42  301#47
  ICHID   0340     17#19   91/23  200/22  207/15
  ICHIDZ  0020     10#41   91/24  201/49  204/19  206/31  206/55  207/16
                  212/24  216/53  268/55
  ICIDNO  002E     10#56   90/57   90/57   91/14   91/14   91/22   91/22
                  201/20  207/14  211/36  211/60  214/37  215/30  269/ 5
                  269/ 5  270/39  270/39
  ICIO    E4C1    196/39  200#15
  ICIO1   E4C3    200#21  200/32
  ICPTH   0347     17#26  200/26
  ICPTHZ  0027     10#48  205/34  206/33  268/59  270/58
  ICPTL   0346     17#25  200/24
  ICPTLZ  0026     10#47  205/32  206/35  268/57  270/60
  ICR     FD05    272/15  350#58
  ICR1    FD18    351#18  351/19
  ICS     C429     52/56   54#27
  ICSORG  CC00      4#56  100/ 6  100/ 6  100/ 6  100/ 6  100/ 6  100/ 6
                  274/53  346/23
  ICSPR   034C     17#31   90/58   91/15  269/ 6  269/ 9
  ICSPRZ  002C     10#53  201/44  205/31  205/33  211/54  212/36  212/38
                  212/42  212/45  212/46  212/47  215/25  215/27
n ICSTA   0343     17#22
  ICSTAZ  0023     10#44  206/20  208/56  211/16  211/61  215/ 6
  ICX     DB9D    141/24  158/48  159#19
  IDCPU   0002      4#26  371/14
  IDDAY   0010      4#23   29/14  371/13
  IDH     E6F4    215/ 5  215#23
  IDIO    C6A3     68#15  196/ 9
  IDMON   0005      4#24   29/14  371/13
  IDPN1   0042      4#27   29/16  371/15
  IDPN2   0042      4#28   29/16  371/15
  IDPN3   0000      4#29   29/16  371/15
  IDPN4   0000      4#30   29/16  371/15
  IDPN5   0001      4#31   29/16  371/15
  IDREV   0002      4#22   29/17  371/16
  IDYEAR  0083      4#25   29/14  371/13
  IFP     D9AA     23#27  146/ 6  182/51  188/37
  IFP1    D9B8    146#39  146/51
  IFP2    D9BE    146#44  146/48
  IGN     F2F8    290#28  291/37  291/54  292/ 9  292/23  292/34  292/45
                  293/23
  IHW     C4DA     56/55   58#52
  IHW1    C4E0     58#60   59/12
  IHW2    C4F0     59/ 7   59#11
  IIH     C00C     30#17  196/36
```

```
  IIN     E4DC   200/23  200/25  200#52  206/32  206/34
  IIO     E55C    91/30  205#15
  IIR     C030    32#18   56/ 7
  IIR1    C045    32/28   32#40
  IIR2    C052    32/53   32#61
  IIR3    C054    33# 6   33/17
  IIR4    C05F    33/ 8   33#13
  IIR5    C064    33/11   33#16
  IIR6    C06A    33/14   33#25
  IIR7    C070    33/33   33#42
  ILN     F50C   303#15  338/43
  ILN1    F50D   303#32  312/ 8
  ILP     DA51   143/29  150#44
n IMASK   028B    15# 7
  INBUFF  00F3    12#50  145/38  145/40  145/41  145/43  145/55  150/46
                 150/48  158/50  159/51  160/23  170/44  170/46  170/47
                 170/49
  IND     E510    91/34  202#24
  IND1    E512   202#27  202/44
  INIML   0700     4#54   60/36   60/38
n INSCLR  0020     5#39
  INSDAT  007D    12# 9  301/18  301/30  301/33  310/50  311/48  311/61
                 312/27  313/ 9
  INTABS  0200    13#16   33/44   33/46   37/14   37/15.  37/16   37/17
                  37/18   37/19   37/20   37/21   51/57
  INTEMP  022D    13#43   44/53   45/ 9
  INTINV  E46B    24#41   60/53  196/36
  INTZBS  0010    10#28   50/52
  INVFLG  02B6    15#24  292/ 6  292/ 8  294/27
  IOCB    0340    17#18  201/40  211/51  270/44
  IOCBAS  0020    10#40  201/41  201/44  211/50  211/54
  IOCBSZ  0010     4#44    4/46    4/47  200/29
  IOCFRE  00FF     9# 8  200/21  204/20  206/30  206/56
  IPH     C9CA    87#60   88/55
  IRA     F2AD   288#42  290/ 8
  IRI1    EB3A   245/53  246# 5
  IRI2    EB42   246/ 6  246#15
  IRI3    EB51   246/22  246#31  247/15
  IRI4    EB55   246#36  246/61  247/21
  IRI5    EB59   246/16  246#43
  IRI6    EB6D   246/51  246#57
  IRI7    EB81   247/ 9  247#19
  IRIR    EB2C    55/58  245#42  253/18
  IRQ     C02C    31#39  372/15
  IRQEN   D20E    22# 6   32/33   32/35   33/26   33/28  243/ 9  243/61
                 251/20  252/26  275/ 6
  IRQST   D20E    21#41   32/26   33/13
  ISIO    E95C   196/30  236#15
  IST     5086   103/42  103/59  105#58
  ISW     C535    52/15   60#15
  ITO     EB27   244/54  245#21  257/ 8
  IVN1    C0EE    38/11   38/15   38#21
  IVN10   C19D    40/34   40#39
  IVN11   C1B0    40/46   40/51   40#57
  IVN12   C1EE    41/ 5   41#45
  IVN13   C1F3    40/58   41/ 8   41/13   41/22   41/25   41/28   41/31
                  41/35   41/41   41#50
```

IVN14	C21F	42#26	42/32					
IVN15	C232	42#43	42/61					
IVN2	C0FC	38/24	38#30					
IVN3	C114	38/44	38#50					
IVN4	C120	38/51	39# 5					
IVN5	C123	38/58	39# 9					
IVN6	C162	39/40	39#53					
IVN7	C167	39#60	40/ 9					
IVN8	C188	40/22	40#28					
IVN9	C18D	40#30	40/40					
IVNM	C0E2	38# 6	56/13	196/24				
n JMPERS	030E	16#57						
JVECK	028C	15# 8	33/45	33/47	33/50	35/22	35/31	35/37
KBCODE	D209	21#38	41/17	41/39	344/27	345/ 7		
KBD	004B	5#50	55/23					
KBK	5527	123/27	125#42					
KEYBDV	E420	24#22	55/24	60/45	194/46	358/30	358/32	
KEYDEF	0079	12# 6	273/27	273/29	291/41			
KEYDEL	02F1	16#19	40/50	40/53	344/31	345/41		
KEYDIS	026D	14#29	33/37	41/12	344/36	344/44		
KEYREP	02DA	15#59	41/15	51/43				
KGB	F2FD	194/55	287/31	290#49				
KGB1	F2FD	290#53	291/15					
KGB10	F372	291/ 9	292#58					
KGB11	F376	290/60	293# 6					
KGB12	F378	292/50	293#11					
KGB13	F38C	293/19	293#23	293/31	293/51			
KGB14	F38F	293/12	293#27					
KGB15	F39F	293/37	293#41					
KGB16	F3A5	293/28	293#47					
KGB17	F3B4	291/49	293/48	293#61				
KGB18	F3CF	294/ 6	294/10	294/13	294/16	294#23		
KGB19	F3DA	293/ 7	293/57	294#31				
KGB2	F323	291/26	291#32	294/19				
KGB20	F3DD	294/24	294#35					
KGB3	F32A	291#45	293/43					
KGB4	F333	291/47	291#53					
KGB5	F345	291/59	292#13					
KGB6	F355	292/14	292#27					
KGB7	F359	292/19	292#32					
KGB8	F360	292/28	292#38					
KGB9	F36C	292/39	292#49					
KIR	FC19	55/57	344#18					
KIR1	FC2D	344/29	344#36					
KIR10	FC90	345/26	346#10					
KIR11	FCAE	346/24	346#31					
KIR12	FCB5	346/11	346#38					
KIR2	FC41	344/45	344#51					
KIR3	FC44	344/49	344#54					
KIR4	FC47	344/38	344#59					
KIR5	FC5C	345/13	345#24					
KIR6	FC67	345#35	346/14					
KIR7	FC6D	344/55	345/20	345/31	345#40	346/29	346/34	
KIR8	FC76	344/32	345#47					
KIR9	FC87	344/60	345/50	345#57	346/39	346/44		
KRK	5539	123/30	126#22					
KRPDEL	02D9	15#58	51/45	345/47				

```
  KSB     551F    123/21  125#24
  KTK     552F    123/24  125#60
n LBPR1   057E     18#30
  LBPR2   057F     18#31  143/31
  LBUFF   0580     18#32  143/52  143/53  144/ 8  144/28  144/34  144/35
                  144/36  144/37  145/31  150/45  150/47  168/44  169/32
                  169/47
  LCOUNT  0233     13#49   71/33   71/56   72/ 7   74/45   76/33   79/51
  LEDGE   0002      4#51   51/23
  LGCOEF  DF72    188/14  188/15  189#29  189/40
  LHO     E515    202/ 5  202#42
  LMARGN  0052     11#31   51/24  280/27  296/20  297/39  299/17  299/21
                  299/59  303/34  322/32  322/37  327/11  327/22  327/56
                  328/17  328/43  328/49  332/25
  LNBUG   0000      4#13   19/ 8   31/18   31/47   48/57
n LNFLG   0000     10# 8
  LOADAD  02D1     15#53   73/ 7   73/10   75/12   75/15   77/34   77/42
                   78/58   80/16   80/18  222/35  222/37
  LOG     DECD     23#43  185/32
  LOG10   DED1     23#44  186/ 6
  LOG10E  DE89    181/35  181/36  184#34  188/54  188/55
  LOGC    DEE0    187#33  190/31
  LOGC2   DEEF    187/42  187#47
  LOGC3   DEF3    187/45  187#50
  LOGC4   DEF9    187#54  187/56
  LOGC5   DF46    188/30  188#37
  LOGC6   DF53    188/39  188#45
  LOGC7   DF64    188/50  188#61
  LOGCOL  0063     11#43  287/39  298/48  299/16  300/11  300/33  300/50
                  301/23  301/27  302/28  302/32  311/16  311/44  325/46
                  325/56  325/59  326/ 9  326/11  326/36  327/10  327/21
                  327/36  328/16  328/42
  LOGMAP  02B2     15#23  298/24  304/25  304/32  304/34  311/57  313/27
                  317/19  317/20  317/21  319/16  321/60  322/14
  LOGQ    DFF6    187/ 7  190#27
  LOGS    DED3    185/50  186#47
  LOGS1   DEDE    186/56  186/58  187#11
  LOO     C892     78#41   80/32   80/33   80/34   80/35
  LOO1    C89D     78/45   78#49
  LOTONE  0007      9#14  250/18
  LPENH   0234     13#50   39/27
  LPENV   0235     13#51   39/25
  LPH     E7DE     90/59  220/12  222#11
  LPH1    E7F8    222/25  222/28  222#34
  LTEMP   0036     11# 8   76/37   76/40   76/43   77/27   77/30   77/32
                   77/35   77/36   77/39   77/41   77/43   78/51   78/54
                   78/56   78/59   79/60   80/ 6   80/ 8   80/21
  M0E     DD34    171/58  175#54
  M0E1    DD36    175#57  175/60
n M0PF    D000     19#36
n M0PL    D008     19#46
  M12     DD26    171/51  175#27
  M121    DD2A    175#30  175/33
n M1PF    D001     19#37
n M1PL    D009     19#47
n M2PF    D002     19#38
n M2PL    D00A     19#48
```

```
n M3PF    D003      19#39
n M3PL    D006      19#49
  MAXDEV  0021       4#43   212/25   216/34   266/60   267/18
  MAXIOC  0080       4#47   200/31   201/28   270/17
  MEMERR  009D       8#46    76/10
  MEMLO   02E7      16#13    60/37    60/39   219/36   219/39   220/ 8   220/10
                   230/51   230/53   230/55   230/57
  MEMTOP  02E5      16#12    60/30    60/32    75/57    76/ 5   219/43   219/45
                   279/49   279/51
n MINTLK  03F9      17#44
  MLN     F78E     303/37   320#49
  MLN1    F79F     321# 9   321/36
  MLN2    F7A7     321#17   321/20
  MLTTMP  0066      11#46   308/55   309/27   309/48
  MOTRGO  0034       9#19   248/38   249/ 5   351/ 5   352/37
  MOTRST  003C       9#20   236/17   249/22   258/17   356/ 9
  MXDMOD  0010       5#38   277/53   278/25
  NOE     DC04     155/60   163#33
  NOE1    DC0A     163#38   163/55
  NOE2    DC10     163#45   163/49
  NOE3    DC28     163/39   163/60   164#13
  NOE4    DC31     164/16   164#25
  NOE5    DC38     163/36   164/26   164#34
n NACK    004E       7#34
  NBUFSZ  0028       9#30   366/50
  NCOMHI  003C       9#17   236/20   244/39   258/19
  NCOMLO  0034       9#16   237/50
  NEWADR  028E      15# 9    74/52    74/58    75/13    75/16    75/19    75/23
                    76/36    76/39    77/26    77/29    78/50    78/53    79/59
                    80/ 5
  NEWCOL  02F6      16#25   333/41   333/43   334/12   334/15
  NEWROW  02F5      16#24   333/39   333/51
  NGFLAG  0001      10# 9    49/10    49/29    49/36    50/10    52/28
  NLCOEF  000A     188/13   189#40
  NMI     C018      30#46    30/48   372/13
  NMI1    C020      30/53    30#59
  NMIEN   D40E      22#51    30/20   104/28   104/61   106/ 7   275/11   275/21
                   286/53
  NMIRES  D40F      22#52    31/14
  NMIST   D40F      22#38    30/52
  NOCKSM  003C      11#13   240/20   247/ 8   247/14
  NOCLIK  02DB      15#60   291/25   293/16   293/18
n NODAT   0000       9#22
  NONDEV  0082       6#28    89/10   202/25   216/46   268/51   270/23
  NORM    DC00     142/34   147/ 5   153/31   153/42   153/60   154/23   163#15
  NORMAL  004E       7#21   366/47   367/19
  NOTOPN  0085       6#31   200/53   212/30
  NPCOEF  000A     182/60   184#26
  NSIGN   00EE      12#38   140/34   142/39
  NVALID  0084       6#30   203/19   333/29
  OCI     FCF7     196/54   350/ 8   350#30
  OCI1    EAFB     243/50   244# 8
  OCIR    EAEC      55/60   243#45   253/20
  OCO     FD34     350/11   352#15
  OCO1    FD6A     352#51   352/55
  OCO2    FD77     351/26   353# 7
  OLDADR  005E      11#40   309/39   309/43   316/17   316/21
```

```
  OLDCHR 005D     11#39  284/17  316/20
  OLDCOL 005B     11#38  334/13  334/16
n OLDPAR 02CF     15#51
  OLDROW 005A     11#37  282/52  333/52  334/43
  OPEN   0003      5# 8   52/60   91/28  203/21
  OPNIN  0004      5#36   53/10
  OPNOT  0008      5#37   53/10
  OPNTMP 0066     11#47  276/18  276/39  277/37  278/26  278/33
  ORI1   EAB5    242/36  242#42
  ORI2   EACE    242/51  243# 6
  ORI3   EAD7    242/59  243#13  243/27
  ORI4   EADB    242/46  243#20
  ORIR   EAAD     55/59  242#29  253/19
  OVRRUN 008E      6#40  246/10
n POPF   D004     19#41
n POPL   D00C     19#51
  P10COF DE4D    182/61  183/ 5  184#15  184/26
n P1PF   D005     19#42
n P1PL   D00D     19#52
n P2PF   D006     19#43
n P2PL   D00E     19#53
n P3PF   D007     19#44
n P3PL   D00F     19#54
  PACTL  D302     22#22   34/58   59/21   59/28  236/18  248/39  249/ 6
                 249/23  258/18  351/ 6  352/38  356/10
  PADDL0 0270     14#33   42/27
n PADDL1 0271     14#34
n PADDL2 0272     14#35
n PADDL3 0273     14#36
  PADDL4 0274     14#37   42/29
n PADDL5 0275     14#38
n PADDL6 0276     14#39
n PADDL7 0277     14#40
n PAI    C42C     54#49
  PAL    D014     20# 5   51/30
  PALNTS 0062     11#42   51/44  248/29  248/53  254/53  256/ 6  351/10
                 352/42  357/ 9  357/49
  PARMBL 02C9     15#46   71/26
  PBC    FD2A    351#41  352/52
  PBC1   FD2E    350/39  351#45  352/23
  PBCTL  D303     22#23   35/11   58/58   59/17   59/22   59/29  236/21
                 237/51  244/40  258/20
n PBI    D100     21# 8
n PBIRAM D600     23# 8
  PBK    EDC7    241/58  244/49  256/37  256/59  258#15
  PBPNT  02DE     16# 6  362/44  363/14  363/26  364/30  365/ 7
  PBUFSZ 02DF     16# 7  361/52  363/21  364/ 5  365/43  367/ 9
  PCC    E51A    201/51  203#15
  PCC1   E529    203/27  203#33
  PCH    F2BE    287/37  289#25
  PCH1   F2C6    289#32  289/40
  PCH2   F2CF    289/28  289#38
  PCH3   F2E5    289/50  289#61
  PCI    EB9D    236/60  248#15
  PCI1   EBBC    248/32  248#36
  PCI2   EBC6    248#41  248/42
  PCI3   EBD4    248/20  248#50
```

```
  PCI4    EBE6    248/56  248#60
  PCI5    EBF0    249# 8  249/ 9
  PCI6    EC04    248/46  249#19
  PCI7    EC0E    249/20  249#25
  PCL     FF07    195/14  364#55
  PCOLR0  02C0     15#34   39/61
n PCOLR1  02C1     15#35
n PCOLR2  02C2     15#36
n PCOLR3  02C3     15#37
  PCS     C2C8     46/28   46/38   46/43   47/38   47/42   48#18  196/48
  PDEVN   0040      6#51  365/31
  PDID1   D803     24# 8   82/27
  PDID2   D80B     24#11   82/31
  PDIMSK  0249     14#18   32/52
  PDIOV   D805     24# 9   83/28
  PDIRQV  D808     24#10   84/37
  PDVI    D1FF     21#12   32/51
  PDVMSK  0247     14#16   83/16   87/31
  PDVS    D1FF     21#16   82/25   82/44   83/37   84/36   84/43   87/26
                   87/38   89/16
  PDVV    D80D     24#12   82/35   87/61   88/ 7
  PENH    D40C     22#36   39/26
  PENV    D40D     22#37   39/24
  PHC     E89E     91/ 8  220/16  229#15
  PHE     EEBC    197/ 6  266#41
  PHE1    EEC2    266#54  266/61
  PHE2    EED3    267#12  267/19
  PHE3    EEE5    266/55  267#31
  PHE4    EEEB    267/13  267#40
  PHENTV  E486     24#50  197/ 6
  PHG     E816    222/38  222/40  223#37
  PHG1    E81F    223#48  223/58
  PHG2    E827    223/44  223#55
  PHI     E898    197/12  228#43
  PHINIV  E48C     24#52  197/12
  PHL     CA29     90#52  202/43  270/51
  PHL1    CA54     90/60   91/ 9   91/17   91#34
  PHO     EEF9    205/51  268#44
  PHO1    EF07    268/49  268#54
  PHP     E7BE    219/26  219/59  220#53  268/48
  PHP1    E7C1    221# 6  221/ 9
  PHPA    E7D4    221/ 6  221#24  221/33
  PHPAL   000A    220/61  221#33
  PHQ     E8C0    228/28  230#15
  PHQ1    E8CE    229/26  230#30
  PHQ2    E8D1    230/20  230#36
  PHQ3    E8DD    230/37  230#49
  PHR     E739     53/46  218#32
  PHR1    E745    218#48  219/12
  PHR2    E762    218/37  219#16
  PHR3    E76E    219#24  219/60  220/19
  PHR4    E776    218/54  219/ 5  219/ 8  219#31
  PHR5    E777    219/27  219#35
  PHR6    E799    219#51  220/13  220/17
  PHR7    E79B    219/20  219#58
  PHR8    E7A2    219/47  220# 7
  PHU     E915    197/ 9  230/26  233#44
```

```
  PHU1    E939    233/61  234#16
  PHU2    E953    234/11  234/14  234#37
  PHU3    E955    233/49  234#42
  PHUNLV  E489     24#51  197/ 9
  PHW     E894    219/ 7  228#25
  PHX     E900    230/19  231#29
  PIA     D300     22#15   59/ 9
  PIN     FE99    195/19  361#15
  PIO     C933     83# 7  196/18  235/15
  PIO1    C943     83#23   83/31
  PIO2    C958     83/17   83/24   83#42
  PIO3    C95E     83/38   83#46
  PIR     C96E     60/58   60/60   84#18
  PIR1    C970     84#24   84/28
  PIR2    C976     84/25   84#32
  PLO     F1CA    282/23  282#40  299/38  336/38  337/ 5
  PLO0    F1CA    282#44  282/45
  PLO1    F1D1    282#51  282/54
n PLOT    0050      7#22
  PLY1    DD58    176#49  177/17
  PLY2    DD6F    176/56  177# 5
  PLY3    DD88    176/47  176/50  177/ 9  177/12  177#19
  PLYARG  05E0     18#34  176/39  176/40  177/14  177/15  185/ 5  185/ 6
                  185/15  185/16
  PLYCNT  00EF     12#40  176/38  176/46  177/11
  PLYEVL  DD40     23#35  176/11  183/ 6  188/16
n PMBASE  D407     22#48
  PMD     50D0    104/34  104/36  107#24
  PMD1    50D3    107#34  109/11
  PMD2    50E9    107/42  107/49  107#55
  PMD3    50FA    107/37  108#14
  PMD4    5114    108/15  108#36
  PMD5    511F    108/21  108/32  108/38  108#43
  PMD6    5120    108/51  108#57
  PMD7    5137    108/59  109#11
  PMI     C471     49/ 5   56#35
  PMI1    C484     56/41   56/44   56/47   56#55
  PMI2    C49A     57/ 9   57#18
  PMI3    C4A1     57/14   57#24
  PMI4    C4A9     57/12   57/20   57#33
  PMI5    C4B2     57#40   57/52
  PMI6    C4C8     57/44   57/49   57#56
  POD     530A    106/37  106/39  119#34
  POD1    53FD    119/55  120#11
  POD10   5448    120/12  121#10
  POD2    540C    120/18  120#23
  POD3    540E    120/21  120#25
  POD4    541C    120/29  120#34
  POD5    541E    120/32  120#36
  POD6    542C    120/40  120#45
  POD7    542E    120/43  120#47
  POD8    5443    120/54  120#61
  POD9    5445    120/59  121# 6
  POKEY   D200     21#24   59/ 5
  POKMSK  0010     10#30   32/34   33/10   33/27  243/ 6  243/ 8  243/58
                  243/60  250/28  250/53  251/19  252/24  252/25  274/61
                  275/ 5
```

```
  POP     FEC2    195/13   362#41
  PORTA   D300     22#19    35/ 6    41/50    41/59    59/24    59/31
  PORTB   D301     22#20    35/16    49/57    49/59    50/12    50/14    50/61
                   52/33    52/35    56/59    56/61    57/24    57/26    59/ 6
                   59/19    59/26    59/30    81/26    81/28   117/25   117/28
                  344/26   345/57
  POT0    D200     21#28    42/26
n POT1    D201     21#29
n POT2    D202     21#30
n POT3    D203     21#31
n POT4    D204     21#32
n POT5    D205     21#33
n POT6    D206     21#34
n POT7    D207     21#35
  POTGO   D20B     21#61    42/36
  PPB     FECB    195/16   362#60
  PPB1    FEEB    363/31   363#40
  PPD     F223     49/50    49/52   196/42   196/57   285#15
  PPM     FF4B    363/10   364/59   366#38
  PPM1    FF4F    366#47   367/20
  PPM2    FF57    366/48   366#53
  PPM3    FF5F    366/54   366#59
  PPM4    FF65    366/51   366/57   367# 9
  PPM5    FF6F    366/60   367#19
  PPO     E576    204/32   204/37   205#50
  PPP     FEF6    363/22   364#25
  PPRB    FEA1    361#32   364/34   364/35
  PPTMPA  024C     14#20    88/ 9    88/35    88/60    89/20
  PPTMPX  024D     14#21    88/10    88/36    89/21    89/22
  PRINTR  0050      5#51    55/11
  PRINTV  E430     24#23    55/12    60/46   195/ 6
  PRIOR   D01B     20#45    39/35
  PRNBUF  03C0     17#33   361/32   363/16   363/60
  PRS     C2CA     46/48    48#42
  PRS10   C346     50/43    50#47
  PRS11   C34B     50#54    50/56
  PRS12   C350     50/20    50#60
  PRS13   C35A     51/ 6    51#10
  PRS14   C383     51/32    51#39
  PRS15   C389     51/37    51#43
  PRS17   C393     51#56    51/59
  PRS18   C39E     52# 8    52/11
  PRS21   C3C4     52/29    52#44
  PRS22   C3DC     52/49    52/52    52#60
  PRS23   C3FA     53/13    53#21    53/22    53/25
  PRS24   C40D     53/34    53#42
  PRS25   C413     53/38    53#50
  PRS26   C426     53/56    53/60    54#11
  PRS3    C2E4     49#24    49/39    49/46
  PRS4    C2EE     49/27    49#31
  PRS5    C2F8     49/34    49#38
  PRS6    C31D     50/ 5    50#10
  PRS7    C31F     50/ 8    50#12
  PRS8    C32E     49/15    50#24
  PRS9    C33C     50/27    50#40    50/48
  PRVOPN  0081      6#27   204/25
  PSP     FEC1    195/15   195/18   361/60   362#25
```

```
  PST      FEA3    195/17   361#47   362/42
  PSTB     FE9F    361#30   361/53   361/54
  PTE      EC11    249#40   258/43   258/45
  PTIMOT   0314     17# 5   361/17   365/47   366/20
  PTL      EF26    268/56   268/58   270#10
  PTL1     EF38    270#27   270/33   270/52
  PTL2     EF3D    270/15   270/18   270#32
  PTL3     EF41    270/21   270#37
  PTL4     EF46    270#44   270/49
  PTO      C24F     38/46    43#23
  PTRIG0   027C     14#47    42/54
  PTRIG1   027D     14#48    42/47
n PTRIG2   027E     14#49
n PTRIG3   027F     14#50
  PTRIG4   0280     14#51    42/56
  PTRIG5   0281     14#52    42/49
n PTRIG6   0282     14#53
n PTRIG7   0283     14#54
  PTT      C252     40/24    43#39
  PUPBT1   033D     17#14    47/36    51/15
  PUPBT2   033E     17#15    47/40    51/17
  PUPBT3   033F     17#16    47/44    51/19
  PUPDIV   E480     24#48   196/57
  PUPVL1   005C      8#35    47/37    51/14
  PUPVL2   0093      8#36    47/41    51/16
  PUPVL3   0025      8#37    47/45    51/18
  PUTCHR   0008      5#12   205/28
  PUTDAT   0080      9#24    69/ 6   359/16
  PUTREC   0009      5#11    65/52
  PUTSEC   0050      7# 9    68/57
  PWS      C290     46#18    47/46   196/45
  PWS1     C2A1     46/33    46#42
  QTEMP    00D9     12#24   157/35   157/44   157/45   157/46   157/47   158/ 7
  RAMLO    0004     10#14    49/21    49/22    49/25    49/26    49/32    49/33
                    49/43    49/44    63/29    63/31    63/41    63/48    63/50
                    63/51    63/53    64/56    64/59    65/ 6
  RAMSIZ   02E4     16#11    52/47    60/29   115/60   273/20
  RAMSYS   0000      4#12    57/30
  RAMTOP   006A     11#53   273/21   275/51   275/59   278/16   279/14   298/16
                   320/53   324/11   324/23   325/14
n RANDOM   D20A     21#39
  RBLOKV   E47A     24#46    66/24   196/51
  RCB      FD8D    196/51   353/34   353#57
  RCB1     FDA9    354/22   354#28
  RCB2     FDAF    354/17   354#36
  RCB3     FDB1    353/28   354#40
  RDONLY   0087      6#33   209/37
  READ     0052      7#10    66/28   359/13
  REC      EAFD    238/47   240/21   244#24   249/15
  REC1     EB06    244/30   244#34
  REC2     EB16    244#44   244/59
  REC3     EB1D    244/45   244#53
  RECLEN   0245     14#14    71/43    71/55    72/54    75/20
  RECVDN   0039     11#10   244/35   244/58   246/32
  REDGE    0027      4#52    51/25
  RELADR   024A     14#19    74/51    74/57    75/11    75/14
  RES      C2AA     47#18    53/17   372/14
```

```
RES1    C2AD     47#28   47/29   47/32
RET     F665    300/ 5  311/42  311/49  311/54  312#43
RET1    F67B    312/51  312#57
RET2    F67E    312/55  312#59
RET3    F691    313/15  313#19  313/29
RET4    F69D    313/22  313#26
RET5    F6AB    311/52  312/60  313/ 7  313/10  313#34
RIR     C0CE     36#18   55/53  286/56  286/58
RIRGHI  0000      8#27  248/60
RIRGLN  0078      8#11  260/11
RIRGLP  0064      8#20  260/12
RIRGLX  EE13    248/58  260#11
RLEADH  FE91    351/12  360#14
RLEADL  FE93    351/11  360#17
RLEADN  0240      8# 9  360/14  360/17
RLEADP  01E0      8#16  360/15  360/18
RLR     C745     71#19  222/47
RLR1    C747     71#25   71/28
RLR2    C74F     71#32   71/57
RLR3    C77D     71#55   72/ 8
RLR4    C794     71/36   71/41   71/61   72#10
RMARGN  0053     11#32   51/26  296/39  297/18  299/28  311/30  314/48
                314/51  322/36  326/59  327/27  328/47
ROO     F718    287/49  289/26  316#15
ROD1    F722    316/16  316#23
ROWAC   0070     11#58  313/52  313/54  313/55  313/57  334/42  335/17
                335/30  335/32  335/35  335/37  335/43
ROWCRS  0054     11#33  282/51  287/26  287/52  295/16  295/22  295/40
                295/41  298/51  299/31  301/44  302/35  303/57  304/17
                308/58  309/ 6  311/55  312/47  312/59  313/32  314/55
                318/42  321/ 5  322/ 9  322/12  325/47  326/39  326/55
                327/14  327/30  328/20  329/46  330/19  330/50  330/53
                333/38  334/44  335/48  335/50  336/51  336/55
ROWINC  02F8     16#26  333/48  333/59  335/49
RRC     F957    301/39  302/48  327/37  330#15  337/10
RRC1    F959    330#18  330/21
RSIRGN  000A      8#13  260/17
RSIRGP  0008      8#22  260/18
RSIRGX  EE17    248/54  260#17
RTCLOK  0012     10#32   38/10   38/14   38/17   38/27  256/51  257/23
                357/ 7  357/39  357/55
RUNADR  02C9     15#47   71/51   71/53   72/48   72/53   73/ 6   73/ 9
                 73/12   73/13   73/53
RWS     F661    282/20  288/22  311/51  312#25  338/37
S0ER    DC62    155/49  158/18  166#15
S0ER1   DC64    166#18  166/21
S0L     DBEB    139/40  142/27  162#20  170/20
S0R     DC3A    153/36  164#51
S1R     DC3E    153/ 6  165#16
S2L     DBE7    161#60  171/52
SAL     DC9F    144/52  145/ 9  167/45  168#43
SAS     5510    104/26  124/11  124/24  124#60  130/52
SAVADR  0068     11#50  277/29  277/31  279/54  279/57  302/25  302/27
                302/41  302/46
SAVIO   0316     17# 7  256/49  257/12  257/15
SAVMSC  0058     11#36  277/ 7  277/ 9  279/31  279/33  309/37  309/41
                332/44  332/46
```

SBA	C73A	69/46	69/57	70#40				
SBP	E887	238/20	238/46	247#36	248/44	249/11		
SBR	ED3D	249/14	256#32					
SBR1	ED3D	256#34	256/47					
SBR2	ED44	256/35	256#39					
SBR3	ED4C	256/41	256#45					
SBR4	ED68	256#58	257/13	257/17	257/26			
SBR5	ED71	256/43	257# 7					
SBR6	ED75	257/ 5	257#10					
SBR7	ED96	257/20	257#28					
SBS	F90C	295/24	327#52					
SBS1	F917	327#59	328/44	328/53				
SBT	5505	124/ 8	124/18	124#40				
SBUFSZ	001D	9#32	367/ 5					
SCB	FE3F	354/ 5	358#52	359/46				
SCB1	FE70	359/14	359#18					
SCC	F40C	296#55	297/40					
SCC1	F40E	296/21	296#58	297/19	297/21			
SCL	F997	298/47	312/44	332#15				
SCL1	F9A1	332/20	332#25					
SCL2	F9A3	332/23	332#27					
SCR	F7F7	313/19	322#60					
SCR1	F7FF	323#14	323/15					
SCR2	F809	323#22	323/24					
SCR3	F810	323#26	323/28					
SCR4	F822	323/33	323#37	323/38				
SCR5	F827	323/12	323#42					
SCREDT	0045	5#49	55/17					
SCRENV	E410	24#21	55/21	60/44	194/26			
SCRFLG	02BB	15#28	301/20	301/41	313/20			
SCRMEM	0093	6#45	308/10					
SDA	F610	310/17	310#49					
SDF	F569	278/10	278/23	306#15				
SDI	F570	277/36	278/19	278/21	278/52	279/17	279/19	279/22
		279/25	279/32	279/34	279/37	279/39	279/41	279/47
		306/17	306#35	308/34	308/36			
SDL	509E	104/33	106#27	113/13	122/ 9	127/19		
SDLSTH	0231	13#47	39/28	106/50				
SDLSTL	0230	13#46	39/30	106/48	279/43	279/45	279/53	279/56
SDMCTL	022F	13#45	39/32	106/ 5	106/35	106/53	274/57	280/33
		280/34	345/49	345/53	346/38	346/43		
SDP	FF14	361/58	364/36	365#28				
SDP1	FF2F	365/38	365#42					
SEA	F60E	282/24	310#32					
SEC	F20B	280/35	281/52	282/21	282/25	284#15	288/25	290/ 9
SEDS	C448	53/ 6	53/ 8	55#42				
SEIOCB	0000	4#46	52/61	65/48				
SEL	500C	104#15	120/ 7					
SEL1	5043	104#46	104/48					
SEL2	5049	104#52	104/54					
SEL3	5059	105# 8	105/10					
SEL4	5075	105#28	109/ 7					
SEL5	507D	105/18	105#35					
SEL6	5080	105/21	105#39					
SEL7	5083	105/24	105#43					
SEN	EA88	241#33	248/45	253/54				
SEN1	EA9E	241#53	242/ 6					

```
  SEN2    EAA5   241/54  242# 5
  SENDEV  E468    24#40  196/33  352/36
  SEP     F6AE   313#50  335/52  336/33
  SERIN   D20D    21#40  246/20  246/43
  SEROUT  D20D    22# 5   59/46  241/48  242/56  243/22
  SES     C8FC    81#21  285/16
  SETVBV  E45C    24#36  196/21  258/49  351/16  352/47
  SFL     E6D8   209/11  210/45  214#36
  SGB     F180   194/35  280#50  288/16
  SGNFLG  00F0    12#43  182/27  183/48  186/51  188/49
  SHC     E85D   226# 5  229/25  233/48
  SHC1    E86B   226#18  226/57
  SHC2    E87F   226/24  226/27  226#36
  SHC3    E887   226#44  226/53
  SHC4    E889   226/37  226/40  226#49
  SHFAMT  006F    11#57  281/20  283/41  309/52
  SHFLOK  02BE    15#31  273/24  292/18  292/22  292/33  292/44  294/15
  SHPDVS  0248    14#17   82/20   82/24   82/37   83/36   84/32   84/35
                  84/42   87/25   87/37   89/15
  SHT     E6FF   204/36  206/61  215#46
  SHT1    E70C   215/54  215#59
  SHT2    E70E   215/57  215#61
  SID     ECAF   237/52  238/21  253#38
  SID1    ECB1   253#44  253/50
  SID2    ECB3   253#46  253/47
  SIDWAY  0053     7#23  366/59
  SIL     DA5A   147/59  148/ 9  148/12  151#19
  SIN     EF6E   194/19  194/39  194/59  273#15
  SIO     E971    83/42  236#43
  SIO1    E983   236/56  237# 7
  SIO10   EA22   238/ 9  238/22  238/38  239#10
  SIO2    E98D   237#13  239/15
  SIO3    E992   237#18  238/ 5
  SIO4    E9D9   237/54  237#61
  SIO5    E9E1   237/57  238#13
  SIO6    E9F3   238/14  238#26
  SIO7    EA0C   238/35  238#46
  SIO8    EA12   238/30  238#51
  SIO9    EA1C   238/52  238#61
  SIOINV  E465    24#39   60/52  196/30
  SIOV    E459    24#35   69/33  196/18  221/19  224/59  359/21  361/59
                 364/40
n SIZEM   D00C    20#24
n SIZEP0  D008    20#19
n SIZEP1  D009    20#20
n SIZEP2  D00A    20#21
n SIZEP3  D00B    20#22
  SKC     F983   291/28  293/21  305/18  331#25  348/15
  SKC1    F986   331#34  331/47
  SKC2    F98C   331#40  331/41
  SKCTL   D20F    22# 7   59/36  104/25  236/26  250/26  250/50
  SKRES   D20A    21#60  245/49  250/51
  SKSTAT  D20F    21#42   40/44   40/60  124/20  245/48  256/45  257/10
                 257/33  257/35
  SLB     DBA1   139/ 7  159#42
  SLB1    DBA5   159#51  159/55
  SLB2    DBAC   159/52  159#59
```

SLC	F88E	296/58	299/40	300/17	301/47	302/49	303/39	303/55
		312/10	313/34	325#41	327/53	328/22		
SLC1	F896	325#52	325/61					
SLC2	F8A9	325/54	326# 8					
SLD	53A4	114/28	116/18	117#23	120/ 5			
SLFTSV	E483	24#49	81/30	196/60				
SMEM1	5161	109/21	109#47					
SMEM2	5171	109/28	110# 8					
SMEM3	51B1	109/37	110#29					
SMEM4	51F5	111/16	111#26					
SMEM5	5271	112/10	112#34					
SML	DBED	162/ 5	162#40					
SML2	DBEF	162#43	162/51					
SMS	F9A6	297/59	323/42	332#43				
SNL	DC9D	145/24	145/27	167/ 6	167/12	168#20		
SNP	C9AF	83/23	87#15	88/48				
SNP1	C9AF	87#19	87/33					
SNP2	C9BB	87/20	87#31					
SOP	EF8E	194/33	273#51					
SOUNDR	0041	11#18	236/25	251/28				
SPB	F1A4	194/36	281#43					
SPB1	F1B1	281/49	281#54					
SPECIL	000E	5#15	203/26	203/29				
SPQ	F1E9	283#27	284/25	301/22				
SPQ1	F1F2	283#41	283/45					
SPQ2	F1FA	283/42	283#49					
SQR10	DF66	188/ 5	188/ 6	189#13				
SRC	F94C	301/16	302/19	326/35	329#43	336/47		
SRC1	F94E	329#46	329/49					
SRR	DC40	164/53	165#35					
SRR1	DC46	165#40	165/52					
SRR2	DC48	165#42	165/46					
SRTIMR	022B	13#41	40/57	41/ 7	41/16	41/46	345/48	
SSC	E670	201/34	202/27	204/27	205/52	207/44	209/39	211#15
SSD	F82A	304/38	323#58					
SSD1	F82E	324#10	324/16					
SSD2	F839	324/12	324#18					
SSD3	F847	324/19	324#31					
SSD4	F855	324/37	324#43	325/ 7	325/10			
SSD5	F862	324#51	324/54					
SSD6	F87C	324/24	324/59	325#14				
SSD7	F889	325#22	325/24					
SSE	F5A0	278/ 7	278/45	308#32				
SSFLAG	02FF	16#34	34/24	282/44	345/17	345/19		
SSKCTL	0232	13#48	236/24	250/ 6	250/25	250/47	250/49	257/34
SSM	5759	113/17	113/19	122/14	122/16	122/57	123/12	123/14
		125/26	125/44	127/24	134#15			
SSM1	5766	134#23	134/28					
SSP	F9AF	194/38	333#17					
SSP1	F9BE	333/27	333#32					
SSP10	FA6F	335/41	335#47					
SSP11	FA7C	335/39	335/45	335#54				
SSP12	FA95	336/ 7	336#13					
SSP13	FAAA	336/14	336#27					
SSP14	FAB0	336/19	336/22	336/25	336/28	336#32		
SSP15	FAB5	336/ 5	336/11	336#37				
SSP16	FAC9	336#51	337/ 6					

```
SSP17   FAE6     336/58   337# 8
SSP18   FAEF     336/43   337#14
SSP19   FB01     335/25   337/24   337#30
SSP2    F9BF     333/24   333#34
SSP3    F9F1     333/54   334#11
SSP4    FA19     334/18   334/31   334#37
SSP5    FA1F     334#41   334/46
SSP6    FA40     334/56   334/60   335# 9
SSP7    FA4D     335#21   337/26
SSP8    FA56     335/23   335#29
SSP9    FA61     335/33   335#37
SSR     EC56     250/30   251#15
SSR1    EC6A     251/29   251#33   251/36
SSR2    EC83     251/44   251#49
SST     F21E     194/17   194/37   194/53   194/54   194/57   280/55   284/19
                 284/22   284#41   286/48   294/35   315/47   331/10   337/30
SST1    F226     284/43   286# 7
SSV     EDE2     248/37   248/61   249/13   254/16   258#42
STIK    008F      26#19   114/12   115/56   115/59
ST3000 3000       26#44   109/24   109/34   110/51   111/47   112/23   121/25
                 121/26   121/27   134/24   137/ 8   137/ 8   137/11   137/11
                 137/36   137/36   137/37   137/38   137/38   137/39   137/40
                 137/41   137/42   137/43   137/44   137/45
ST3002 3002       26#45   132/47
ST3004 3004       26#46   137/45
ST300B 300B       26#47   127/29
ST301C 301C       26#48   120/25
ST301E 301E       26#49   120/36
ST3020 3020       26#50   113/23   116/43   120/47   137/ 9   137/ 9
ST3021 3021       26#51   123/38   123/40
ST3022 3022       26#52   137/43
ST3024 3024       26#53   113/42   117/ 6   137/10   137/10
ST3028 3028       26#54   137/37
ST3038 3038       26#55   114/17   119/ 6   119/ 7   119/10   119/11
ST303C 303C       26#56   129/16   129/17
ST304C 304C       26#57   137/42
ST3052 3052       26#58   126/ 5   126/ 6
ST3062 3062       26#59   132/48
ST306D 306D       26#60   126/24   126/26
ST3072 3072       26#61   137/44
ST3092 3092       27# 5   137/40
ST309E 309E       27# 6   129/34   129/35
ST30AB 30AB       27# 7   137/41
ST30B7 30B7       27# 8   137/39
ST30C1 30C1       27# 9   132/43
ST30C2 30C2       27#10   132/51
ST30C7 30C7       27#11   128/41   128/42
ST30CA 30CA       27#12   128/55   128/56
ST30F8 30F8       27#13   128/37   128/38
ST3100 3100       27#14   112/14
ST3121 3121       27#15   132/44
ST3122 3122       27#16   132/49
ST313C 313C       27#17   129/24   129/25
ST3150 3150       27#18   127/36
ST3154 3154       27#19   128/17   128/18
ST3181 3181       27#20   132/45
ST3182 3182       27#21   132/50
```

```
  ST3186 3186    27#22  128/27  128/28
  ST318C 318C    27#23  129/20  129/21
  ST31B0 31B0    27#24  127/37
  ST31C2 31C2    27#25  132/52
  ST31CA 31CA    27#26  129/ 6  129/ 7
  ST31EE 31EE    27#27  129/38  129/39
  ST31F1 31F1    27#28  132/46
  ST3210 3210    27#29  127/38
  ST321A 321A    27#30  128/59  128/60
  ST3248 3248    27#31  128/45  128/46
  ST3270 3270    27#32  127/39
  ST32D0 32D0    27#33  127/40
  STACKP 0318    17# 9  236/48  258/23
  STADR1 009E    26#31  133/29  133/31  133/42  133/43  133/46  133/49
                133/53  134/19  134/27  369/57  370/11  370/18  370/21
                370/23  370/27
  STADR2 00A0    26#32  133/33  133/35  133/50  133/54  370/24  370/28
  STATC  0053     7#11   61/51   69/10   69/43  361/55  365/37
n STATIS 000D     5#14
  STATUS 0030    10#59  238/57  238/61  239/34  240/23  240/40  240/46
                240/50  241/ 9  241/38  244/37  245/23  245/58  246/11
                246/27  258/22
  STAUT  0082    26# 9  104/22  105/31  116/ 8  122/20  123/60  129/60
  STB    F495   300#32  338/49
  STBL   00A2    26#33  107/46  107/47  108/29
  STCDA  009A    26#28  127/49  127/56  127/59  127/61
  STCDI  0099    26#27  127/47  127/51  128/ 5  128/ 6  128/ 7
  STCHK  008B    26#16  368/24  368/25  368/53  368/56  369/27  369/28
                370/12  370/13  370/16
  STH    5000   103#25  196/60
  STICK0 0278    14#42   41/61   42/43
  STICK1 0279    14#43   41/55
  STICK2 027A    14#44   42/ 6
  STICK3 027B    14#45   41/57
n STIMER D209    21#59
  STJMP  0083    26#10  106/ 9  106/49  106/51  111/18
  STK    5450   105/43  121#55  130/15
  STK1   5462   122#13  124/14  124/25
  STK10  54F8   124#20  124/22
n STK2   5470   122#25
  STK3   5483   122/21  122#39  122/41  122/44
  STK4   5491   122/30  122#50
  STK5   54A0   122/54  122#61
  STK6   54B0   123/ 7  123#18
  STK7   54D3   123#48  123/50
  STK8   54DE   123#60  125/27  125/45  126/ 7  126/27
  STK9   54F5   123/61  124#18
  STKST  008A    26#15  106/29  120/11
  STLM   00A4    26#35  114/ 7  114/27  114/29  114/31
  STM    5291   103/43  105/35  113# 6  122/35
  STM1   52A9   113#23  116/13
  STM10  5326   114#51  115/28
  STM11  5328   114#55  115/22
  STM12  532A   114#57  114/61
  STM13  5335   115#11  115/17
  STM14  5348   115/13  115#34
  STM15  5356   114/39  115#41
```

```
STM16   5359    115/30  115#45
STM17   535E    114/20  114/23  115/37  115#48
STM18   5368    115#55  116/24
STM19   537D    116/ 9  116#17
STM2    52C2    113/31  113#36
STM20   5388    115/49  116#22
STM3    52C4    113/34  113#38
STM4    52C7    113/25  113#42
STM5    52E0    113/50  113#55
STM6    52E2    113/53  113#57
STM7    52E5    113/44  113#61
STM8    52F7    114#16  115/61
STM9    5318    114#37  114/42
STMVAL  0093     26#22  115/ 8  115/12  115/15
STNOT   0098     26#26  127/15  130/41  130/48
STPAG   0090     26#20  114/10  114/11  114/38  114/58  115/11  115/26
                115/58
STPASS  0087     26#12  104/59  107/35  107/58  108/46
STPC    0092     26#21  114/47  115/27
STRIG0  0284     14#56   42/12
STRIG1  0285     14#57   42/17
STRIG2  0286     14#58   42/14
STRIG3  0287     14#59   42/19
STS     FF44    362/ 7  366#18
STSEL   0086     26#11  105/15  105/29  105/60  108/25  108/27  108/43
STSKP   0094     26#23  121/60  122/25  122/27  122/28
STSMM   008D     26#17  114/ 5  114/18  115/48  115/52  116/23  118/61
                119/ 9
STSMP   008E     26#18  114/ 9  114/16  115/53  118/60
STSPP   0088     26#13  108/14  108/31  108/41
STTIME  0080     26# 8  104/20  104/21  108/50  108/53  108/57
STTMP1  0095     26#24  123/39  123/41  123/49  123/54  123/56
STTMP2  009B     26#29  131/16  131/17  131/26  131/27  131/54  131/56
                131/59
STTMP3  009D     26#30  131/21  131/29  131/32
STTMP4  00A3     26#34  131/23  131/34
STTMP5  00A5     26#36  117/24  117/27
STV     5557    105/39  116/20  126#57
STV1    555C    127# 9  130/ 6
STV2    5560    127#14  130/10
STV3    557D    127#35  127/42
STV4    5599    127#51  128/ 9
STV5    5663    129/56  130#10
STV6    5666    129/61  130#14
STVOC   0097     26#25  127/10  127/25  129/52  129/53  129/54  130/34
STW     ECC0    238/29  254#15
SUBTMP  029E     15#17  307/36  307/47
SUC     5773    104/30  113/15  122/ 5  127/ 5  134#58
SUCA    578C    134/60  135#18
SUCB    5790    135/ 6  135#23
SUCC    5794    135/ 9  135#28
SUCCES  0001      6#24   73/15  206/19  239/ 5  240/24  241/37  244/36
                269/10  274/58  286/ 7  315/23  353/ 7  353/40  355/ 6
                355/37  356/ 5  363/35  365/12
SUCD    5798    135/12  135#33
SUE     DCCF    155/ 5  156/56  171#20
SUP     DCE0    155/14  157/ 6  171#45
```

SUPERF	03E8	17#34	289/ 6	289/49	290/54	293/56		
SVN	566C	128/23	128/33	128/51	129/12	129/30	129/44	130#30
SVP	C272	44#48	196/21	232/15				
SVP1	C27C	44#61	45/ 5					
SVR	572A	104/40	106/43	116/41	116/61	133#20		
SVR1	5741	133#41	133/51	133/55				
SVR2	574A	133/44	133#48					
SWA	F962	287/19	287/38	288/18	288/27	288/60	289/34	290/10
		315/45	330#38					
SWA1	F968	330#50	330/57					
SWA2	F980	330/44	331#10					
SWPFLG	007B	12# 7	275/28	312/50	315/42	326/48	330/61	331/ 6
		332/19						
SYSVBV	E45F	24#37	196/24					
SZA	F60A	280/54	302/30	310#15	327/ 8			
TAB	F47A	299#55	338/34					
TAB1	F47A	299#57	300/13					
TAB2	F483	299/60	300#11					
TAB3	F492	300/ 7	300#17					
TABMAP	0243	15#22	275/36	304/25	311/57	317/59	317/61	318/24
		318/25	319/16	319/35	321/60	322/14		
TAGM	EE4D	262#11	275/57					
TAIC	F849	283/11	339#30					
TARS	57DC	133/28	133/30	133/32	133/34	137# 8		
TARV	FFD7	369/56	370#39					
TAVD	56BC	131/25	132#21					
TBTM	EEB4	265#19	316/44					
TCCR	F80D	289/56	289/61	290/ 6	329/19	338#12	338/60	
TCCRL	0030	329/17	338#60					
TCDA	5716	127/52	127/54	132#43	132/54			
TCDAL	0014	128/ 8	132#54					
TCKD	F851	273/26	273/28	339#54				
TCVO	E72D	203/34	212/40	217#10				
TDLE	EE20	261#42	278/28	278/34				
TDLV	EE5D	262#37	277/20					
TDP	DC93	166/51	167#35					
TDP1	DC9C	167/40	167#49					
TDSC	F808	275/44	337#49					
TDSM	EEAD	265#11	309/50					
TEMP	023E	13#59	13/61	240/12	240/16	240/27		
TEMP1	0312	16#60	219/38	219/41	219/44	219/46	222/17	224/50
		224/53	226/ 9	226/10	226/26	231/33	231/36	231/37
		254/48	255/ 7					
TEMP2	0313	16#61	222/15	224/54	226/23			
TEMP3	0315	17# 6	222/19	223/41	223/43	223/48	223/56	256/55
		257/19						
TEX	C7D5	74#44	60/30	80/31	80/40			
TEX1	C7E6	74/47	74#57					
TEX2	C7F9	74/61	75#10					
TEX3	C80C	75/ 6	75#18					
TEX4	C821	75/28	75#32					
TEX5	C830	75/38	75#45					
TEX6	C839	75/36	75/41	75#51				
TEX7	C84C	75/60	76# 8					
TEX8	C850	74/53	75/53	75/58	76/ 6	76#12	76/44	
TFKD	FC11	273/31	273/33	343#47				
THAV	C42E	52/ 8	55#11	55/26				

```
  THAVL  000F     52/ 6   55#26
  TIAC   FB4D    320/28  339#41
  TIHV   C44B     51/56   55#53   56/16
  TIHVL  0026     51/54   56#16
  TIMER1 030C     16#56  254/42  254/44  254/47  254/51  256/52  256/53
  TIMER2 0310     16#59  254/38  254/39  254/41  254/45  254/49
  TIMFLG 0317     17# 8  244/53  248/41  249/ 8  249/42  256/40  256/61
                 258/51
  TIMOUT 008A      6#36  240/51  245/22
  TINDEX 0293     15#12  275/26
  TIRQ   C0CF     33/ 6   36#50   36/59
  TIRQL  0008     32/61   36#59
  TLSC   EE6D    263#11  309/13
  TMCC   EE7D    263#36  311/24  315/17
  TMNT   544A    114/37  121#22  121/29
  TMNTL  0006    114/35  121#29
  TMPCHR 0050     11#29  283/50  283/54
n TMPCOL 02B9     15#27
n TMPLBT 02A1     15#20
  TMPROW 02B8     15#26  329/47  330/18
  TMRC   EE8D    264#10  312/57  314/54
  TMSK   FB04    309/46  337#38
  TNC    DBAF    158/46  160#21
  TNDD   56B6    130/42  132# 8
  TOADR  0066     11#48  320/56  320/60  321/18  321/26  321/30
  TOIH   C0D7     33/42   37#14
  TONE1  0002      7#57  352/21
  TONE2  0001      7#58  350/37
  TPOS   CA21     84/34   87/32   89#43   89/52
  TPOSL  0006     83/21   84/22   88/46   89#52
  TPFV   EDF9    255/34  255/37  259#25
  TRAMSZ 0006     10#15   49/45   52/45   52/54   53/33   53/55   57/35
                  57/38   57/40   57/42   57/43   57/47   57/48   57/51
                  60/28
  TRIG0  D010     19#56   42/11
  TRIG1  D011     19#57   42/16
n TRIG2  D012     19#59
  TRIG3  D013     19#60   30/22   39/18   46/26   56/39
  TRNRCD 0089      6#35  208/55
  TRPR   C8E4     71/50   71/52   80#30
  TRSC   EE9D    264#36  309/23  309/45
  TRUE   FFFF      4# 9    4/11
  TSFR   FB3D    289/56  339#12
n TSIH   ECA9    253#18
  TSKP   5545    122/26  126#35  126/39
  TSKPL  0012    122/29  126#39
  TSMA   EE1D    261#17  276/ 7
  TSMC   579C    123/35  135#46
  TSTAT  0319     17#10  238/56  241/10
  TSTD   57F6    134/20  137#36
  TSTL   57EC    134/18  137#19
  TSTO   CA57     92# 8  134/16
  TTXT   CA61     92/ 8   92/ 9   92/10   92/11   92/12   92/13   92/14
                  92/15   92/16   92/17   92#25  134/23
  TVN    DBB8    139/11  160#51
  TVN1   DBCF    161#17  161/30  161/37
  TVN2   DBD0    160/26  161#21
```

```
    TVN3    DBD2     161/10   161/13   161#26
    TVN4    DBD8     161/ 7   161#34
    TVN5    DBE2     160/61   161/27   161/35   161#41
    TXT0    CA61      92/ 8    92#33    92/40
    TXT0L   0013      92#40   137/19
    TXT1    CA74      92/ 9    92#48    92/50
    TXT1L   0003      92#50   137/20
    TXT2    CA77      92/10    93# 8    93/15
    TXT2L   0013      93#15   137/21
    TXT3    CB32      92/11    96#34    96/54
    TXT3L   0013      96#54   137/22
    TXT4    CB45      92/12    97# 8    97/12    97/20
    TXT4L   0004      97#12    97/22   137/23
    TXT5    CB45      92/13    97#20
    TXT5L   0004      97#22   137/24
    TXT6    CB49      92/14    97#30    97/34
    TXT6L   0003      97#34   137/25
    TXT7    CA8A      92/15    93#23    96/26
    TXT7L   00A8      96#26   137/26
    TXT8    CB4C      92/16    97#42    97/46
    TXT8L   0003      97#46   137/27
    TXT9    CB4F      92/17    98# 8    98/12
    TXT9L   0007      98#12   137/28
    TXTCOL  0291      15#11   280/28
    TXTMSC  0294      15#13   275/53   275/55
  n TXTOLD  0296      15#14
    TXTROW  0290      15#10   280/26   330/52   330/55
  n USAREA  0480      18#12
    VBREAK  0206      13#21    35/33
    VCOUNT  D40B      22#35   104/52   256/50   257/22   277/13   323/26   323/37
                     331/38   331/40
    VCS     FF86     368#52   369/34
    VCS1    FF90     368/54   368/57   369# 5
  n VDELAY  D01C      20#46
    VDSLST  0200      13#18    30/55   104/35   104/37   106/38   106/40   275/16
                     275/18   286/57   286/59
    VFR     FF73      49/61   113/30   368#19
    VFR1    FF79     368#29   368/31
    VGC     FFFF       4#11    41/56    42/ 5    42/13    42/18    42/28    42/48
                      42/55    51/49
    VIMIRQ  0216      13#29    31/49
    VINTER  0204      13#20    35/17
    VKEYBD  0208      13#22    37/15
    VPIRQ   0238      13#53    32/57    60/59    60/61
    VPRCED  0202      13#19    35/ 7
    VSCROL  D405      22#47    39/49
    VSERIN  020A      13#23    32/36    37/21
    VSEROC  020E      13#25    37/19
    VSEROR  020C      13#24    37/20
    VSFLAG  026C      14#28    39/39    39/44    39/47   323/14   323/18   323/22
    VSR     FF92      50/ 7   113/49   369#25
    VTIMR1  0210      13#26    37/18
    VTIMR2  0212      13#27    37/17
    VTIMR4  0214      13#28    37/16
    VVBLKD  0224      13#36    43/ 8
    VVBLKI  0222      13#35    31/20
    WARMST  0008      10#20    48/46    49/14    49/20    57/ 8    61/34    66/51
```

```
                218/36
WARMSV  E474     24#44   196/45
WCA     EA37    239#59   254/17
WCA1    EA6F    240/35   240#45
WCA2    EA73    240/25   240/41   240#50
WCA3    EA80    240/52   241# 5
WCA4    EA82    240/29   240/32   240/58   241# 9
WCB     FE7C    355/18   356/20   356/35   359#40
WFK     FE36    358/ 7   358#29
WFR     C0DF     37#41    39/20
WFR1    C0DF     37#45    37/45
WIRGHI  0000      8#26   248/36
WIRGLN  00B4      8#10   260/ 8
WIRGLP  0096      8#19   260/ 9
WIRGLX  EE11    248/34   260# 8
WLEADH  FE8D    352/44   360# 8
WLEADL  FE8F    352/43   360#11
WLEADN  0480      8# 8   360/ 8   360/11
WLEADP  03C0      8#17   360/ 9   360/12
WMODE   0289     15# 5   350/35   351/46   352/20   355/57
WOR     C86D     77#24    80/36    80/37
WOR1    C88A     77/37    77#41
WRITE   0057      7#12    68/60   366/42
WRONLY  0083      6#29   207/42
WSIRGN  000F      8#12   260/14
WSIRGP  000D      8#21   260/15
WSIRGX  EE15    248/30   260#14
WSYNC   D40A     22#50    44/59   104/46   107/55   118/36   347/18
XCL     E57C    201/57   203/40   206#15
XCL1    E588    206/22   206#30
XFMFLG  00F1     12#46   182/25   182/47   183/12   187/39   187/47   188/28
                188/38
XFORM   DE95    184#59   188/ 7
XGT     E5B2    203/46   207#32
XGT1    E5BA    207#44   207/49
XGT2    E5BD    207/38   207#48
XGT3    E5D0    207/55   208# 8   208/33
XGT4    E5F0    208/17   208/23   208#32
XGT5    E5FB    208#43   208/51
XGT6    E60C    208/45   208#60
XGT7    E618    208/10   208/28   208/39   209#11
XIR     C0CD     35#57    55/54    55/55    55/56    55/61    56/ 5    56/ 6
XITVBV  E462     24#38   196/27
XMTDON  003A     11#11   241/43   242/ 5   243/54
XOP     E53F    203/35   204#15
XOP1    E547    203/22   204#27   205/20   207/ 5
XOP2    E54A    204/21   204#31
XPT     E61E    203/48   209#27
XPT1    E626    209#39   209/44
XPT2    E629    209/33   209#43
XPT3    E63A    209/50   209#60   210/30
XPT4    E640    209/56   210# 7
XPT5    E659    210/18   210#28
XPT6    E66A    210/12   210/24   210/36   210#45
XSS     E597    203/43   206#51
XSS1    E5A2    206/57   207# 9
ZCHAIN  004A     11#24    99/31   218/42   218/44   218/50   218/53   218/58
```

			218/59	218/60	226/12	226/14	226/19	226/22	226/49
			226/50	229/32	229/36	229/37	229/38	229/40	229/42
			230/43	230/45	230/52	230/56	231/ 6	231/ 9	231/31
			231/34	233/54	233/56	233/58	233/59	234/ 8	234/10
			234/17	234/20	234/23	234/25	234/28	234/31	
	ZF1	DA46	23#30	149/37					
	ZFR0	DA44	23#29	139/21	146/32	149/14	156/ 7	164/30	171/59
n	ZHIUSE	02CD	15#49						
	ZIOCB	0020	10#39	270/45					
	ZLOADA	02D3	15#54	222/43					
	ZTEMP1	00F5	12#52	147/55	148/33	155/20	155/24	155/37	155/41
			155/53	157/ 7	158/13	171/57			
	ZTEMP3	00F9	12#54	148/ 6	148/ 8	148/17	148/20	165/36	165/50
	ZTEMP4	00F7	12#53	146/29	146/31	146/39	146/40	147/36	147/37
			148/ 5	148/ 7	148/16	148/18	148/19	148/21	148/26
			148/27	148/28	148/32	148/45	148/47	148/48	148/50
			148/54	148/56	151/21	151/22	152/33	152/37	165/37
			165/38	165/51	165/58	166/45	166/54	167/39	174/47
			174/58						
	ZXLY	DA48	139/18	150#17					
	ZXLY1	DA4A	150#21	150/24					

```
  $          $   $  $$$$$  $   $   $$$
 $ $         $   $    $    $$  $  $   $
$   $        $   $    $    $ $ $  $   $
$   $        $   $    $    $  $$  $   $
$$$$$        $   $    $    $   $  $   $
$   $   $$    $ $     $    $   $  $   $
$   $   $$     $    $$$$$  $   $   $$$
```

DEST=A.VINO USER=A.VINO QUEUE=LPT DEVICE=@p1

PAGES=408

PATH=:UDD:SYSTEMS:OS:REV2:OS.PRN

```
  $    $$$$$  $   $  $$$$     $
$ $ $  $      $$  $  $   $  $ $ $
 $$$   $      $ $ $  $   $   $$$
  $    $$$$   $  $$  $   $    $
 $$$   $      $   $  $   $   $$$
$ $ $  $      $   $  $   $  $ $ $
  $    $$$$$  $   $  $$$$     $
```

$$+

Atari S/W Development

AOS/VS REV 03.07
AOS/VS XLPT REV 03.00